

HOST CONTROLLER INTERFACE FUNCTIONAL SPECIFICATION

***This Part describes the functional specification for the Host Controller interface (HCI).
The HCI provides a uniform command interface to a Controller.***



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1 INTRODUCTION

This Part of the specification describes the functional specifications for the Host Controller interface (HCI). The HCI provides a uniform interface method of accessing a Bluetooth Controller's capabilities. The next three sections provide a brief overview of the lower layers of the Bluetooth software stack and of the Bluetooth hardware. [Section 2](#) provides an overview of the Lower HCI Device Driver Interface on the Host device. [Section 4](#) describes the flow control used between the Host and the Controller. [Section 7](#) describes each of the HCI commands in detail, identifies parameters for each of the commands, and lists events associated with each command.

The specification is applicable to the following types of Controllers:

- BR/EDR Controller
- BR/EDR/LE Controller
- LE Controller
- AMP Controller

A Host may drive a Primary Controller and zero or more AMP Controllers. A Primary Controller shall support one of the following sets of functionality:

- BR/EDR only
- LE only
- BR/EDR/LE

In the following sections, the term “BR/EDR Controller” is used to describe the BR/EDR functionality of a Primary Controller which may be either a BR/EDR only Controller or a BR/EDR/LE Controller. Similarly, the term “LE Controller” is used to describe the LE functionality of a Primary Controller which may be either an LE only Controller or a BR/EDR/LE Controller.

A BR/EDR Controller supports BR/EDR functionality only. A BR/EDR/LE Controller supports BR/EDR functionality and LE functionality. An LE Controller supports LE functionality only. An AMP Controller supports an AMP PAL. A Primary Controller is either a BR/EDR Controller, a BR/EDR/LE Controller, or an LE Controller.

1.1 LOWER LAYERS OF THE BLUETOOTH SOFTWARE STACK

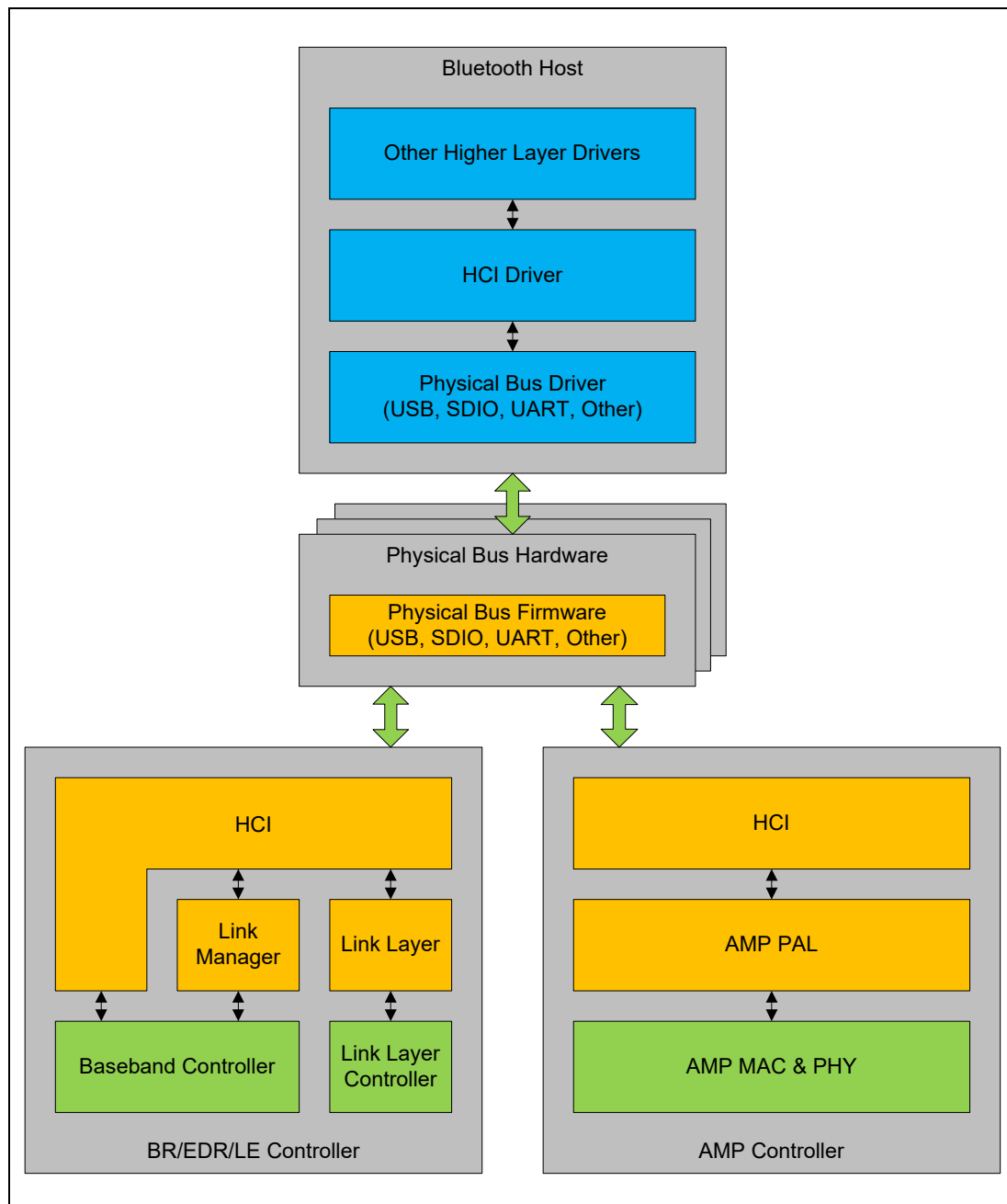


Figure 1.1: Overview of the lower software layers

Figure 1.1 provides an overview of the lower software layers.

Several layers may exist between the HCI driver on the Host system and the HCI layer in the Controller(s). These intermediate layers, the Host Controller Transport Layer, provide the ability to transfer data without intimate knowledge of the data.

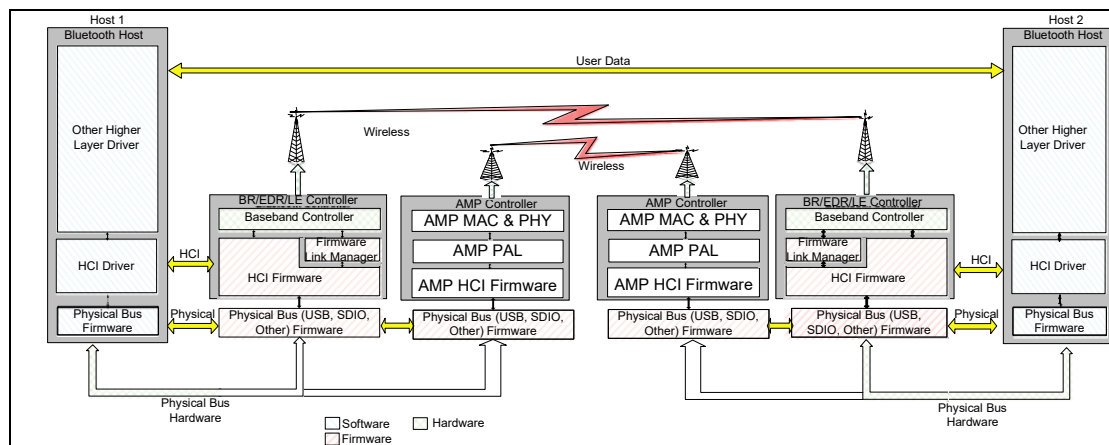


Figure 1.2: End to end overview of lower software layers to transfer data

Figure 1.2 illustrates the path of a data transfer from one device to another. The HCI driver on the Host exchanges data and commands with the HCI firmware on the Bluetooth hardware. The Host Control Transport Layer (i.e. physical bus) driver provides both HCI layers with the ability to exchange information with each other.

The Host will receive asynchronous notifications of HCI events independent of which Host Controller Transport Layer is used. HCI events are used for notifying the Host when something occurs. When the Host discovers that an event has occurred it will then parse the received event packet to determine which event occurred.

A BR/EDR/LE Controller uses one shared command buffer and flow control for BR/EDR and LE. Data buffers can be either shared between BR/EDR and LE or there may be separate data buffers for BR/EDR and LE. The configuration of a Controller is determined through HCI.

LE Controllers use a reduced set of HCI commands and events. Some commands and events are re-used for multiple Controller types.



2 OVERVIEW OF HOST CONTROLLER TRANSPORT LAYER

The Host driver stack has a transport layer between the Host Controller Interface driver and the Host.

The main goal of this transport layer is transparency. The Host Controller Interface driver (which interfaces to the Controller) should be independent of the underlying transport technology. In addition, the transport should not require any understanding of the data that the Host Controller Interface driver passes to the Controller. This allows the logical interface (HCI) or the Controller to be upgraded without affecting the transport layer.

The specified Host Controller Transport Layers are described in [Volume 4](#).

2.1 HOST CONTROLLER TRANSPORT LAYER AND AMPS

The logical Host Controller interface does not consider multiplexing/routing over the Host Controller transport layer(s). The Host designer must consider this in deciding which of the multiple Controller configurations it will support. Each logical AMP HCI instantiation is particular to a single AMP Controller. The Host shall direct each HCI command to a specific AMP, and be aware of the source of each HCI event, and manage flow control separately for each HCI. The Host's Controller_ID concept used for AMP enumeration may assist with this.

The logical HCI definition does not consider the initial discovery of what AMPs are present in the system.

The upper layers discover the AMP capabilities. As seen by the AMP Manager, each local AMP has a Controller_ID that is used to route the data and commands to the correct AMP Controller. This identifier is not used in AMP HCI.

The separate logical HCIs within a system may be on the same physical HCI transport, on separate physical HCI transports, or any combination. This may require a layer of multiplexing, which is not defined within this AMP HCI logical specification.

The logical HCI is not affected by this topology. The logical HCI definition describes commands and events as seen by a single AMP.



3 OVERVIEW OF COMMANDS AND EVENTS

The commands and events are sent between the Host and the Controller(s). The version information in this section denotes the version number of the specification in which this command or event is first specified. The Supported Controllers column denotes the Controller types that can support the command or event: All, BR/EDR, LE, or AMP, or a comma-separated list of Controller types (e.g. "BR/EDR, LE"). Not all Controllers of the named type(s) will necessarily support the command or event.

When more than one specification version number is shown for a command, the highest version listed is the one that introduced the current version of the command, while the lower specification versions listed introduced previous versions of the command.

If a Controller supports version v2 or higher of a command, it shall also support all lower-numbered versions of that command. This requirement overrides anything in [Section 3.1](#) allowing a command to be optional.

Note: A lower-numbered version of a command will always be a proper subset of all higher-numbered versions.

Name	Vers.	Summary Description	Supported Controllers
Accept Connection Request command	1.1	The HCI_Accept_Connection_ - Request command is used to accept a new incoming BR/EDR connection request.	BR/EDR
Accept Logical Link command	3.0 + HS	The HCI_Accept_Logical_Link command will cause the AMP Controller to create a logical link to the remote AMP Controller identified by the physical link handle.	AMP
Accept Physical Link command	3.0 + HS	The HCI_Accept_Physical_Link command will cause the AMP Controller to accept a Physical Link request from the initiating BR/EDR Controller's AMP Controller specified by the command parameters.	AMP
Accept Synchronous Connection Request command	1.2	The HCI_Accept_Synchronous_Connection_Request command is used to accept an incoming request for a synchronous connection and to inform the local Link Manager about the acceptable parameter values for the synchronous connection.	BR/EDR

Table 3.1: Alphabetical list of commands and events (Sheet 1 of 46)



Name	Vers.	Summary Description	Supported Controllers
AMP Receiver Report event	3.0 + HS	The HCI_AMP_Receiver_Report event shall be sent from the AMP to the tester via the AMP Test Manager at the interval configured by the HCI_Enable_AMP_Receiver_Reports command.	AMP
AMP Start Test event	3.0 + HS	The HCI_AMP_Start_Test event shall be generated when the HCI_AMP_Test command has completed and the first data is ready to be sent or received.	AMP
AMP Status Change event	3.0 + HS	The HCI_AMP_Status_Change event can occur at any time, after initial Read Local AMP_Info request, the Controller detects a change has occurred regarding status.	AMP
AMP Test command	3.0 + HS	The HCI_AMP_Test command is used to configure and start a test. This command shall be only valid in AMP Test Mode.	AMP
AMP Test End command	3.0 + HS	The HCI_AMP_Test_End command is used to stop any test scenario which is in progress. This command shall only be used in AMP Test Mode when a test is in progress.	AMP
AMP Test End event	3.0 + HS	The HCI_AMP_Test_End event shall be generated to indicate that the AMP has transmitted or received the number of frames/bursts configured.	AMP
Authenticated Payload Timeout Expired event	4.1	The HCI_Authenticated_Payload_Timeout_Expired event is used to indicate that a packet containing a valid MIC on the Handle was not received within the <i>authenticated-PayloadTO</i> .	BR/EDR, LE
Authentication Complete event	1.1	The HCI_Authentication_Complete event occurs when authentication has been completed for the specified connection.	BR/EDR
Authentication Requested command	1.1	The HCI_Authentication_Requested command is used to establish authentication between the two devices associated with the specified Connection_Handle.	BR/EDR

Table 3.1: Alphabetical list of commands and events (Sheet 2 of 46)



Name	Vers.	Summary Description	Supported Controllers
Change Connection Link Key command	1.1	The HCI_Change_Connection_Link_Key command is used to force both devices of a connection associated to the Connection_Handle, to generate a new link key.	BR/EDR
Change Connection Link Key Complete event	1.1	The HCI_Change_Connection_Link_Key_Complete event is used to indicate that the change in the Link Key for the Connection_Handle specified by the Connection_Handle event parameter had been completed.	BR/EDR
Change Connection Packet Type command	1.1	The HCI_Change_Connection_Packet_Type command is used to change which packet types can be used for a connection that is currently established.	BR/EDR
Channel Selected event	3.0 + HS	The HCI_Channel_Selected event indicates to the Host originating the physical link that the link information data is available to be read using the HCI_Read_Local_AMP_Assoc command.	AMP
Command Complete event	1.1	The HCI_Command_Complete event is used by the Controller to pass the return status of a command and the other event parameters for each HCI command.	All
Command Status event	1.1	The HCI_Command_Status event is used to indicate that the command described by the Command_Op-code parameter has been received and the Controller is currently performing the task for this command.	All
Connection Complete event	1.1	The HCI_Connection_Complete event indicates to both of the Hosts forming the connection that a new BR/EDR connection has been established.	BR/EDR
Connection Packet Type Changed event	1.1	The HCI_Connection_Packet_Type_Changed event is used to indicate the completion of the process of the Link Manager changing the packet type mask used for the specified Connection_Handle.	BR/EDR

Table 3.1: Alphabetical list of commands and events (Sheet 3 of 46)



Name	Vers.	Summary Description	Supported Controllers
Connection Request event	1.1	The HCI_Connection_Request event is used to indicate that a new incoming BR/EDR connection is trying to be established.	BR/EDR
Connectionless Slave Broadcast Channel Map Change event	CSA4	The HCI_Connectionless_Slave_Broadcast_Channel_Map_Change event indicates to the Host that the BR/EDR Controller has moved to a new AFH channel map for the PBD logical link.	BR/EDR
Connectionless Slave Broadcast Receive event	CSA4	The HCI_Connectionless_Slave_Broadcast_Receive event provides the Host with the data received from a Connectionless Slave Broadcast packet.	BR/EDR
Connectionless Slave Broadcast Timeout event	CSA4	On the HCI_Connectionless_Slave_Broadcast_Receiver, the HCI_Connectionless_Slave_Broadcast_Timeout event indicates to the Host that the BR/EDR Controller has lost synchronization with the Connectionless Slave Broadcast Transmitter. On the Connectionless Slave Broadcast Transmitter, the HCI_Connectionless_Slave_Broadcast_Timeout event indicates to the Host that the BR/EDR Controller has been unable to transmit a Connectionless Slave Broadcast packet for the timeout interval specified in the HCI_Set_Connectionless_Slave_Broadcast command.	BR/EDR
Create Connection Cancel command	1.2	The HCI_Create_Connection_Cancel command is used to cancel an ongoing Create Connection.	BR/EDR
Create Connection command	1.1	The HCI_Create_Connection command will cause the BR/EDR Link Manager to create an ACL connection to the BR/EDR Controller with the BD_ADDR specified by the command parameters.	BR/EDR

Table 3.1: Alphabetical list of commands and events (Sheet 4 of 46)



Name	Vers.	Summary Description	Supported Controllers
Create Logical Link command	3.0 + HS	The HCI_Create_Logical_Link command will cause the AMP Controller to create a logical link to the remote AMP Controller identified by the physical link handle.	AMP
Create Physical Link command	3.0 + HS	The HCI_Create_Physical_Link command will cause the AMP Controller to initiate a Physical Link creation to the remote BR/EDR Controller's AMP Controller specified by the command parameters.	AMP
Data Buffer Overflow event	1.1	The HCI_Data_Buffer_Overflow event is used to indicate that the Controller's data buffers have overflowed, because the Host has sent more packets than allowed.	All
Delete Reserved LT_ADDR command	CSA4	The HCI_Delete_Reserved_LT_ADDR command requests that the BR/EDR Controller cancel the reservation of a specific LT_ADDR reserved for the purposes of Connectionless Slave Broadcast.	BR/EDR
Delete Stored Link Key command	1.1	The HCI_Delete_Stored_Link_Key command provides the ability to remove one or more of the link keys stored in the Controller.	BR/EDR
Disconnect command	1.1	The HCI_Disconnect command is used to terminate an existing BR/EDR or LE connection.	BR/EDR, LE
Disconnect Logical Link command	3.0 + HS	The HCI_Disconnect_Logical_Link command is used to terminate the designated logical link.	AMP
Disconnect Physical Link command	3.0 + HS	The HCI_Disconnect_Physical_Link command is used to terminate the designated physical link (during or after creation).	AMP
Disconnection Complete event	1.1	The HCI_Disconnection_Complete event occurs when a connection has been terminated.	BR/EDR, LE
Disconnection Logical Link Complete event	3.0 + HS	The HCI_Disconnection_Logical_Link_Complete event occurs when an AMP Logical Link has been terminated.	AMP

Table 3.1: Alphabetical list of commands and events (Sheet 5 of 46)



Name	Vers.	Summary Description	Supported Controllers
Disconnection Physical Link Complete event	3.0 + HS	The HCI_Disconnection_Physical_Link_Complete event occurs when an AMP Physical Link has been terminated.	AMP
Enable AMP Receiver Reports command	3.0+HS	The HCI_Enable_AMP_Receiver_Reports command is used to enable and disable the reporting of frames received.	AMP
Enable Device Under Test Mode command	1.1	The HCI_Enable_Device_Under_Test_Mode command will allow the local Controller to enter test mode via LMP test commands. The Host issues this command when it wants the local device to be the DUT for the Testing scenarios as described in the Bluetooth Test Mode document.	BR/EDR, AMP
Encryption Change event	1.1	The HCI_Encryption_Change event is used to indicate that the change in the encryption has been completed for the Connection_Handle specified by the Connection_Handle event parameter.	BR/EDR, LE
Encryption Key Refresh Complete event	2.1 + EDR	The HCI_Encryption_Key_Refresh_Complete event is used to indicate to the Host that the encryption key was refreshed on the given Connection_Handle any time encryption is paused and then resumed.	BR/EDR, LE
Enhanced Accept Synchronous Connection Request command	CSA2	The HCI_Enhanced_Accept_Synchronous_Connection_Request command is used to accept an incoming request for a synchronous connection and to inform the local Link Manager about the acceptable parameter values for the synchronous connection.	BR/EDR
Enhanced Flush command	2.1 + EDR	The HCI_Enhanced_Flush command is used to discard specific packets currently pending for transmission in the Controller for the specified Handle. This command takes a parameter specifying the type of packets to be flushed.	BR/EDR, AMP
Enhanced Flush Complete event	2.1 + EDR	The HCI_Enhanced_Flush_Complete event is used to indicate that an Enhanced Flush is complete.	BR/EDR, AMP

Table 3.1: Alphabetical list of commands and events (Sheet 6 of 46)



Name	Vers.	Summary Description	Supported Controllers
Enhanced Setup Synchronous Connection command	CSA2	The HCI_Enhanced_Setup_Synchronous_Connection command adds a new or modifies an existing synchronous logical transport (SCO or eSCO) on a physical link depending on the Connection_Handle parameter specified.	BR/EDR
Exit Periodic Inquiry Mode command	1.1	The HCI_Exit_Periodic_Inquiry_Mode command is used to end the Periodic Inquiry mode when the local device is in Periodic Inquiry Mode.	BR/EDR
Exit Sniff Mode command	1.1	The HCI_Exit_Sniff_Mode command is used to end the Sniff mode for a Connection_Handle which is currently in Sniff mode.	BR/EDR
Extended Inquiry Result event	2.1 + EDR	The HCI_Extended_Inquiry_Result event indicates that a BR/EDR Controller has responded with an extended inquiry response during the current Inquiry process.	BR/EDR
Flow Spec Modify command	3.0 + HS	The HCI_Flow_Spec_Modify command modifies certain parameters of an existing Logical Link's Flow Spec on this AMP Controller.	AMP
Flow Spec Modify Complete event	3.0 + HS	The HCI_Flow_Spec_Modify_Complete event indicates to the Host that an HCI_Flow_Spec_Modify command has been completed.	AMP
Flow Specification command	1.2	The HCI_Flow_Specification command is used to specify the flow parameters for the traffic carried over the ACL connection identified by the Connection_Handle.	BR/EDR
Flow Specification Complete event	1.2	The HCI_Flow_Specification_Complete event is used to inform the Host about the Quality of Service for the ACL connection the Controller is able to support.	BR/EDR
Flush command	1.1	The HCI_Flush command is used to discard all data that is currently pending for transmission in the Controller for the specified Connection_Handle.	BR/EDR

Table 3.1: Alphabetical list of commands and events (Sheet 7 of 46)



Name	Vers.	Summary Description	Supported Controllers
Flush Occurred event	1.1	The HCI_Flush_Occurred event is used to indicate that, for the specified Handle, the data to be transmitted has been discarded.	BR/EDR, AMP
Get MWS Transport Layer Configuration command	CSA3	The HCI_Get_MWS_Transport_Layer_Configuration command reads the supported baud rates from the Controller.	BR/EDR, LE and AMP
Hardware Error event	1.1	The HCI_Hardware_Error event is used to indicate some type of hardware failure for the Controller.	All
Hold Mode command	1.1	The HCI_Hold_Mode command is used to initiate Hold mode.	BR/EDR
Host Buffer Size command	1.1	The HCI_Host_Buffer_Size command is used by the Host to notify the Controller about its buffer sizes for ACL and synchronous data. The Controller will segment the data to be transmitted from the Controller to the Host, so that data contained in HCI Data packets will not exceed these sizes.	BR/EDR, LE
Host Number Of Completed Packets command	1.1	The HCI_Host_Number_Of_Completed_Packets command is used by the Host to indicate to the Controller when the Host is ready to receive more HCI packets for any Connection_Handle.	All
Inquiry Cancel command	1.1	The HCI_Inquiry_Cancel command will cause the BR/EDR Controller to stop the current Inquiry if the BR/EDR Controller is in Inquiry Mode.	BR/EDR
Inquiry command	1.1	The HCI_Inquiry command will cause the BR/EDR Controller to enter Inquiry Mode. Inquiry Mode is used to discovery other nearby BR/EDR Controllers.	BR/EDR
Inquiry Complete event	1.1	The HCI_Inquiry_Complete event indicates that the Inquiry is finished.	BR/EDR
Inquiry Response Notification event	CSA4	The HCI_Inquiry_Response_Notification event indicates to the Host that the BR/EDR Controller responded to an inquiry message.	BR/EDR

Table 3.1: Alphabetical list of commands and events (Sheet 8 of 46)

Name	Vers.	Summary Description	Supported Controllers
Inquiry Result event	1.1	The HCI_Inquiry_Result event indicates that a BR/EDR Controller or multiple BR/EDR Controllers have responded so far during the current Inquiry process.	BR/EDR
Inquiry Result with RSSI event	1.2	The HCI_Inquiry_Result_with_RSSI event indicates that a BR/EDR Controller or multiple BR/EDR Controllers have responded so far during the current Inquiry process.	BR/EDR
IO Capability Request event	2.1 + EDR	The HCI_IO_Capability_Request event is used to indicate that the IO capabilities of the Host are required for a simple pairing process.	BR/EDR
IO Capability Request Negative Reply command	2.1 + EDR	The HCI_IO_Capability_Request_Negative_Reply command is used to reject a pairing attempt after an HCI_IO_Capability_Request event has been received by the Host.	BR/EDR
IO Capability Request Reply command	2.1 + EDR	The HCI_IO_Capability_Request_Reply command is used to reply to an HCI_IO_Capability_Request event from the Controller, and specifies the current I/O capabilities of the Host.	BR/EDR
IO Capability Response event	2.1 + EDR	The HCI_IO_Capability_Response event is used to indicate to the Host that IO capabilities from a remote device specified by BD_ADDR have been received during a simple pairing process.	BR/EDR
Keypress Notification event	2.1 + EDR	The HCI_Keypress_Notification event is sent to the Host after a pass-key notification has been received by the Link Manager on the given BD_ADDR.	BR/EDR
LE Add Device To Periodic Advertiser List command	5.0	The HCI_LE_Add_Device_To_Periodic_Advertiser_List command will add a device to the Periodic Advertiser List.	LE
LE Add Device To Resolving List command	4.2	The HCI_LE_Add_Device_To_Resolving_List command is used to add one device to the resolving list used to resolve Resolvable Private Addresses in the Controller.	LE

Table 3.1: Alphabetical list of commands and events (Sheet 9 of 46)



Name	Vers.	Summary Description	Supported Controllers
LE Add Device To White List command	4.0	The HCI_LE_Add_Device_To_White_List command will add a device to the White List.	LE
LE Advertising Report event	4.0	The HCI_LE_Advertising_Report event indicates that an advertising or scan response packet has been received.	LE
LE Advertising Set Terminated event	5.0	The HCI_LE_Advertising_Set_Terminated event indicates that advertising in a given advertising set has stopped.	LE
LE Channel Selection Algorithm event	5.0	The HCI_LE_Channel_Selection_Algorithm event indicates the channel selection algorithm used on a connection.	LE
LE Clear Advertising Sets command	5.0	The HCI_LE_Clear_Advertising_Sets command will remove all existing advertising sets from the Controller.	LE
LE Clear Periodic Advertiser List command	5.0	The HCI_LE_Clear_Periodic_Advertiser_List command will clear the Periodic Advertiser List.	LE
LE Clear Resolving List command	4.2	The HCI_LE_Clear_Resolving_List command is used to remove all devices from the resolving list used to resolve Resolvable Private Addresses in the Controller.	LE
LE Clear White List command	4.0	The HCI_LE_Clear_White_List command will clear the White List.	LE
LE Connection Complete event	4.0	The HCI_LE_Connection_Complete event indicates to the Host that a new connection has been created.	LE
LE Connection CTE Request Enable command	5.1	The HCI_LE_Connection_CTE_Request_Enable command will request the Controller to start or stop sending of LL_CTE_REQ PDUs on a connection.	LE
LE Connection CTE Response Enable command	5.1	The HCI_LE_Connection_CTE_Response_Enable command will command the Controller to respond to LL_CTE_REQ PDUs with LL_CTE_RSP PDUs.	LE

Table 3.1: Alphabetical list of commands and events (Sheet 10 of 46)



Name	Vers.	Summary Description	Supported Controllers
LE Connection IQ Report event	5.1	The HCI_LE_Connection_IQ_Report event is used to report IQ samples from the Constant Tone Extension field of a received packet containing an LL_CTE_RSP PDU.	LE
LE Connection Update command	4.0	The HCI_LE_Connection_Update command will be used to change the connection parameters of an existing connection.	LE
LE Connection Update Complete event	4.0	The HCI_LE_Connection_Update_Complete event indicates the completion of the process to change the connection parameters.	LE
LE Connectionless IQ Report event	5.1	The HCI_LE_Connectionless_IQ_Report event reports IQ information from the Constant Tone Extension of a received advertising packet.	LE
LE Create Connection Cancel command	4.0	The HCI_LE_Create_Connection_Cancel command is used to cancel an ongoing HCI_LE_Create_Connection command.	LE
LE Create Connection command	4.0	The HCI_LE_Create_Connection command is used to create a new connection.	LE
LE CTE Request Failed event	5.1	The HCI_LE_CTE_Request_Failed event indicates a problem with a request generated by an HCI_LE_Connection_CTE_Request_Enable command for a peer device to send Constant Tone Extensions.	LE
LE Data Length Change event	4.2	The HCI_LE_Data_Length_Change event is used to indicate a change in the maximum packet sizes by the Link Layer.	LE
LE Directed Advertising Report event	4.2	The HCI_LE_Directed_Advertising_Report event indicates that directed advertisements have been received where the advertiser is using a resolvable private address for the TargetA field in the ADV_DIRECT_IND PDU and the filter policy is set to send this event to the Host.	LE

Table 3.1: Alphabetical list of commands and events (Sheet 11 of 46)



Name	Vers.	Summary Description	Supported Controllers
LE Encrypt command	4.0	The HCI_LE_Encrypt command will encrypt a block of unencrypted data against a key and generate a block of encrypted data.	LE
LE Enhanced Connection Complete event	4.2	The HCI_LE_Enhanced_Connection_Complete event indicates to the Host that a new connection has been created. This event contains the additional parameters of the local and peer resolvable private addresses.	LE
LE Extended Advertising Report event	5.0	The HCI_LE_Extended_Advertising_Report event indicates that an advertising packet has been received.	LE
LE Generate DHKey command	4.2 5.1	The HCI_LE_Generate_DHKey command is used to initiate generation of a Diffie-Hellman key in the Controller for use over the LE transport.	LE
LE Generate DHKey Complete event	4.2	The HCI_LE_Generate_DHKey_Complete event indicates that LE Diffie-Hellman key generation has been completed by the Controller.	LE
LE Long Term Key Request event	4.0	The HCI_LE_Long_Term_Key_Request event indicates that a Long Term Key is required for a connection.	LE
LE Long Term Key Request Negative Reply command	4.0	The HCI_LE_Long_Term_Key_Request_Negative_Reply command is used to reply to an HCI_LE_Long_Term_Key_Request event and indicates that the Host does not have a Long Term Key for that connection.	LE
LE Long Term Key Request Reply command	4.0	The HCI_LE_Long_Term_Key_Request_Reply command is used to reply to an HCI_LE_Long_Term_Key_Request event and includes the Long Term Key stored in the Host for that connection.	LE
LE Modify Sleep Clock Accuracy command	5.1	The HCI_LE_Modify_Sleep_Clock_Accuracy command requests the Controller changes its sleep clock accuracy for testing purposes.	LE

Table 3.1: Alphabetical list of commands and events (Sheet 12 of 46)



Name	Vers.	Summary Description	Supported Controllers
LE Periodic Advertising Create Sync Cancel command	5.0	The HCI_LE_Periodic_Advertising_-Create_Sync_Cancel command is used to cancel a pending HCI_LE_Periodic_Advertising_Create_Sync command.	LE
LE Periodic Advertising Create Sync command	5.0	The HCI_LE_Periodic_Advertising_-Create_Sync command is used to start receiving periodic advertising packets from an advertiser.	LE
LE Periodic Advertising Report event	5.0	The HCI_LE_Periodic_Advertising_-Report event indicates that a periodic advertising packet has been received.	LE
LE Periodic Advertising Set Info Transfer command	5.1	The HCI_LE_Periodic_Advertising_-Set_Info_Transfer command is used to send periodic advertising synchronization information, describing periodic advertising events that the Controller is transmitting, to a connected Controller.	LE
LE Periodic Advertising Sync Established event	5.0	The HCI_LE_Periodic_Advertising_-Sync_Established event indicates that the Controller has started receiving periodic advertising packets from an advertiser.	LE
LE Periodic Advertising Sync Lost event	5.0	The HCI_LE_Periodic_Advertising_-Sync_Lost event indicates the Controller has ended receiving a periodic advertising train.	LE
LE Periodic Advertising Sync Transfer command	5.1	The HCI_LE_Periodic_Advertising_-Sync_Transfer command is used to send periodic advertising synchronization information to a connected Controller.	LE
LE Periodic Advertising Sync Transfer Received event	5.1	The HCI_LE_Periodic_Advertising_-Sync_Transfer_Received event reports reception of periodic advertising synchronization information from a connected Controller.	LE
LE Periodic Advertising Terminate Sync command	5.0	The HCI_LE_Periodic_Advertising_-Terminate_Sync command is used to end receiving of a periodic advertising train.	LE

Table 3.1: Alphabetical list of commands and events (Sheet 13 of 46)



Name	Vers.	Summary Description	Supported Controllers
LE PHY Update Complete event	5.0	The HCI_LE_PHY_Update_ - Complete event is used to inform the Host of the current PHY.	LE
LE Rand command	4.0	The HCI_LE_Rand command will generate a random number.	LE
LE Read Advertising Physical Channel Tx Power command	4.0	The HCI_LE_Read_Advertising_ - Physical_Channel_Tx_Power command will read the transmit power level that will be used for advertising.	LE
LE Read Antenna Information command	5.1	The HCI_LE_Read_Antenna_ - Information command allows the Host to read the switching rates, the sampling rates, the number of antennae, and the maximum length of the Constant Tone Extension supported by the Controller.	LE
LE Read Buffer Size command	4.0	The HCI_LE_Read_Buffer_Size command returns the size of the HCI buffers. These buffers are used by the LE Controller to buffer data that is to be transmitted.	LE
LE Read Channel Map command	4.0	The HCI_LE_Read_Channel_Map command will read the current state of the channel map for a connection.	LE
LE Read Local P-256 Public Key command	4.2	The HCI_LE_Read_Local_P-256_ - Public_Key command is used to return the local P-256 public key from the Controller.	LE
LE Read Local P-256 Public Key Complete event	4.2	The HCI_LE_Read_Local_P-256_ - Public_Key_Complete event is generated when local P-256 key generation is complete.	LE
LE Read Local Resolvable Address command	4.2	The HCI_LE_Read_Local_ - Resolvable_Address command is used to get the current local Resolvable Private Address being used for the corresponding peer Identity Address.	LE
LE Read Local Supported Features command	4.0	The HCI_LE_Read_Local_ - Supported_Features command will read the version information for the local LE Controller.	LE

Table 3.1: Alphabetical list of commands and events (Sheet 14 of 46)

Name	Vers.	Summary Description	Supported Controllers
LE Read Maximum Advertising Data Length command	5.0	The HCI_LE_Read_Maximum_Advertising_Data_Length command will read the maximum length of advertising data that the advertising Controller supports in a given advertising set.	LE
LE Read Maximum Data Length command	4.2	The HCI_LE_Read_Maximum_Data_Length command allows the Host to read the Controller's supportedMaxTxOctets, supportedMaxTxTime, supportedMaxRxOctets, and supportedMaxRxTime parameters.	LE
LE Read Number of Supported Advertising Sets command	5.0	The HCI_LE_Read_Number_of_Supported_Advertising_Sets command will read the maximum number of advertising sets supported by the advertising Controller at the same time.	LE
LE Read Peer Resolvable Address command	4.2	The HCI_LE_Read_Peer_Resolvable_Address command is used to get the current peer Resolvable Private Address being used for the corresponding peer Public and Random (static) Identity Address.	LE
LE Read Periodic Advertiser List Size command	5.0	The HCI_LE_Read_Periodic_Advertiser_List_Size command will read the maximum number of Periodic Advertiser List entries that the Controller supports.	LE
LE Read PHY command	5.0	The HCI_LE_Read_PHY command will read the current PHY.	LE
LE Read Remote Features command	4.0	The HCI_LE_Read_Remote_Features command is used to read the features used on a connection and the features supported by a remote LE device.	LE
LE Read Remote Features Complete event	4.0	The HCI_LE_Read_Remote_Features_Complete event indicates the completion of the process to read the features used on a connection and the features supported by a remote LE device.	LE

Table 3.1: Alphabetical list of commands and events (Sheet 15 of 46)



Name	Vers.	Summary Description	Supported Controllers
LE Read Resolving List Size command	4.2	The HCI_LE_Read_Resolving_List - Size command is used to read the total number of entries in the resolving list that can be stored in the Controller.	LE
LE Read RF Path Compensation command	5.0	The HCI_LE_Read_RF_Path - Compensation command is used to read the RF Path Compensation Value.	LE
LE Read Suggested Default Data Length command	4.2	The HCI_LE_Read_Suggested - Default_Data_Length command allows the Host to read the initial MaxTxOctets and MaxTxTime values for new connections it suggested to the Controller.	LE
LE Read Supported States command	4.0	The HCI_LE_Read_Supported - States command will read the current supported state and role combinations for the local LE Controllers.	LE
LE Read Transmit Power command	5.0	The HCI_LE_Read_Transmit - Power command will read the minimum and maximum transmit powers supported by the Controller for advertising packets.	LE
LE Read White List Size command	4.0	The HCI_LE_Read_White_List - Size command will read the maximum number of White List entries that this Controller supports.	LE
LE Receiver Test command	4.0 5.0 5.1	The HCI_LE_Receiver_Test command will run the LE receiver test.	LE
LE Remote Connection Parameter Request Negative Reply command	4.1	The HCI_LE_Remote_Connection - Parameter_Request_Negative - Reply command is used to reject the remote device's request to change the connection parameters of the LE connection.	LE
LE Remote Connection Parameter Request Reply command	4.1	The HCI_LE_Remote_Connection - Parameter_Request_Reply command is used to accept the remote device's request to change the connection parameters of the LE connection.	LE

Table 3.1: Alphabetical list of commands and events (Sheet 16 of 46)



Name	Vers.	Summary Description	Supported Controllers
LE Remove Advertising Set command	5.0	The HCI_LE_Remove_Advertising_Set command will remove an advertising set from the Controller.	LE
LE Remove Device From Periodic Advertiser List command	5.0	The HCI_LE_Remove_Device_From_Periodic_Advertiser_List command will remove a single device from the Periodic Advertiser List.	LE
LE Remove Device From Resolving List command	4.2	The HCI_LE_Remove_Device_From_Resolving_List command is used to remove one device from the resolving list used to resolve Resolvable Private Addresses in the Controller.	LE
LE Remove Device From White List command	4.0	The HCI_LE_Remove_Device_From_White_List command will remove a single device from the White List.	LE
LE Scan Request Received event	5.0	The HCI_LE_Scan_Request_Received event indicates that a scan request has been received.	LE
LE Scan Timeout event	5.0	The HCI_LE_Scan_Timeout event indicates that scanning has finished.	LE
LE Set Address Resolution Enable command	4.2	The HCI_LE_Set_Address_Resolution_Enable command is used to enable resolution of Resolvable Private Addresses in the Controller.	LE
LE Set Advertising Data command	4.0	The HCI_LE_Set_Advertising_Data command will set the data transmitted when advertising.	LE
LE Set Advertising Enable command	4.0	The HCI_LE_Set_Advertising_Enable command will enable or disable advertising.	LE
LE Set Advertising Parameters command	4.0	The HCI_LE_Set_Advertising_Parameters command will set the parameters used for advertising.	LE
LE Set Advertising Set Random Address command	5.0	The HCI_LE_Set_Advertising_Set_Random_Address command will set the random address used in advertising.	LE

Table 3.1: Alphabetical list of commands and events (Sheet 17 of 46)

Name	Vers.	Summary Description	Supported Controllers
LE Set Connection CTE Receive Parameters command	5.1	The HCI_LE_Set_Connection_-CTE_Receive_Parameters command will set the antenna-switching pattern, switching and sampling slot durations for receiving the Constant Tone Extension on a connection.	LE
LE Set Connection CTE Transmit Parameters command	5.1	The HCI_LE_Set_Connection_-CTE_Transmit_Parameters command will set the antenna-switching pattern, switching and sampling slot durations for transmitting the Constant Tone Extension on a connection.	LE
LE Set Connectionless CTE Transmit Enable command	5.1	The HCI_LE_Set_Connectionless_-CTE_Transmit_Enable command will request the Controller to enable or disable sending packets containing a Constant Tone Extension.	LE
LE Set Connectionless CTE Transmit Parameters command	5.1	The HCI_LE_Set_Connectionless_-CTE_Transmit_Parameters command will set the antenna-switching pattern and switching and sampling slot durations for the transmission of Constant Tone Extensions.	LE
LE Set Connectionless IQ Sampling Enable command	5.1	The HCI_LE_Set_Connectionless_-IQ_Sampling_Enable command will request the Controller to enable or disable taking IQ samples from the Constant Tone Extension of advertising packets.	LE
LE Set Data Length command	4.2	The HCI_LE_Set_Data_Length command is used to suggest maximum packet sizes to the Controller.	LE
LE Set Default Periodic Advertising Sync Transfer Parameters command	5.1	The HCI_LE_Set_Default_Periodic_-Advertising_Sync_Transfer_-Parameters command is used to specify the default behavior of the Controller when periodic advertising synchronization information is received from a connected Controller.	LE
LE Set Default PHY command	5.0	The HCI_LE_Set_Default_PHY command is used to configure preferred PHYs for new connections for the local device.	LE

Table 3.1: Alphabetical list of commands and events (Sheet 18 of 46)



Name	Vers.	Summary Description	Supported Controllers
LE Set Event Mask command	4.0	The HCI_LE_Set_Event_Mask command is used to control which events are generated by the HCI for the Host.	LE
LE Set Extended Advertising Data command	5.0	The HCI_LE_Set_Extended_Advertising_Data command will set the advertising data transmitted when advertising.	LE
LE Set Extended Advertising Enable command	5.0	The HCI_LE_Set_Extended_Advertising_Enable command will enable or disable advertising.	LE
LE Set Extended Advertising Parameters command	5.0	The HCI_LE_Set_Extended_Advertising_Parameters command will set the parameters used for advertising.	LE
LE Set Extended Scan Enable command	5.0	The HCI_LE_Set_Extended_Scan_Enable command will enable or disable scanning on the primary advertising physical channels.	LE
LE Set Extended Scan Parameters command	5.0	The HCI_LE_Set_Extended_Scan_Parameters command will set the parameters used for scanning on the primary advertising physical channel.	LE
LE Set Extended Scan Response Data command	5.0	The HCI_LE_Set_Extended_Scan_Response_Data command will set the data transmitted in a scan response.	LE
LE Set Host Channel Classification command	4.0	The HCI_LE_Set_Host_Channel_Classification command allows the Host to specify a channel classification based on its “local information”.	LE
LE Set Periodic Advertising Data command	5.0	The HCI_LE_Set_Periodic_Advertising_Data command will set the periodic advertising data transmitted when advertising.	LE
LE Set Periodic Advertising Enable command	5.0	The HCI_LE_Set_Periodic_Advertising_Enable command will enable or disable periodic advertising.	LE
LE Set Periodic Advertising Parameters command	5.0	The HCI_LE_Set_Periodic_Advertising_Parameters command will set the parameters used for periodic advertising.	LE

Table 3.1: Alphabetical list of commands and events (Sheet 19 of 46)



Name	Vers.	Summary Description	Supported Controllers
LE Set Periodic Advertising Receive Enable command	5.1	The HCI_LE_Set_Periodic_Advertising_Receive_Enable command will enable or disable periodic advertising reports once synchronized.	LE
LE Set Periodic Advertising Sync Transfer Parameters command	5.1	The HCI_LE_Set_Periodic_Advertising_Sync_Transfer_Parameters command is used to allow the Host to specify the behavior of the Controller when periodic advertising synchronization information is received from a connected Controller.	LE
LE Set PHY command	5.0	The HCI_LE_Set_PHY command is used to request a change of the PHY for a Connection_Handle.	LE
LE Set Privacy Mode command	5.0	The HCI_LE_Set_Privacy_Mode command is used to allow the Host to specify the privacy mode for an entry on the resolving list.	LE
LE Set Random Address command	4.0	The HCI_LE_Set_Random_Address command will set the Random Device Address that may be used in a packet sent on an advertising PHY channel.	LE
LE Set Resolvable Private Address Timeout command	4.2	The HCI_LE_Set_Resolvable_Private_Address_Timeout sets the length of time the Controller uses a random private address before a new random private address is generated and starts being used.	LE
LE Set Scan Enable command	4.0	The HCI_LE_Set_Scan_Enable command will enable or disable scanning.	LE
LE Set Scan Parameters command	4.0	The HCI_LE_Set_Scan_Parameters command will set the parameters used for scanning.	LE
LE Set Scan Response Data command	4.0	The HCI_LE_Set_Scan_Response_Data command will set the data transmitted in a scan response.	LE
LE Start Encryption command	4.0	The HCI_LE_Start_Encryption command is used to enable link level encryption.	LE

Table 3.1: Alphabetical list of commands and events (Sheet 20 of 46)



Name	Vers.	Summary Description	Supported Controllers
LE Test End command	4.0	The HCI_LE_Test_End command will end the current the receiver or transmitter test.	LE
LE Transmitter Test command	4.0 5.0 5.1	The HCI_LE_Transmitter_Test command will run the LE transmitter test.	LE
LE Write RF Path Compensation command	5.0	The HCI_LE_Write_RF_Path_Compensation command is used to indicate the RF path gain or loss from the RF transceiver output to the antenna output contributed by intermediate components.	LE
LE Write Suggested Default Data Length command	4.2	The HCI_LE_Write_Suggested_Default_Data_Length command allows the Host to suggest initial MaxTxOctets and MaxTxTime values for new connections.	LE
Link Key Notification event	1.1	The HCI_Link_Key_Notification event is used to indicate to the Host that a new Link Key has been created for the connection with the BR/EDR Controller specified in BD_ADDR.	BR/EDR
Link Key Request event	1.1	The HCI_Link_Key_Request event is used to indicate that a Link Key is required for the connection with the device specified in BD_ADDR.	BR/EDR
Link Key Request Negative Reply command	1.1	The HCI_Link_Key_Request_Negative_Reply command is used to reply to an HCI_Link_Key_Request event from the BR/EDR Controller if the Host does not have a stored Link Key for the connection with the other BR/EDR Controller specified by BD_ADDR.	BR/EDR
Link Key Request Reply command	1.1	The HCI_Link_Key_Request_Reply command is used to reply to an HCI_Link_Key_Request event from the BR/EDR Controller, and specifies the Link Key stored on the Host to be used as the link key for the connection with the other BR/EDR Controller specified by BD_ADDR.	BR/EDR

Table 3.1: Alphabetical list of commands and events (Sheet 21 of 46)



Name	Vers.	Summary Description	Supported Controllers
Link Supervision Timeout Changed event	2.1 + EDR	The HCI_Link_Supervision_Timeout_Changed event indicates that the remote device changed the Link Supervision Timeout.	BR/EDR
Logical Link Cancel command	3.0 + HS	The HCI_Logical_Link_Cancel command is used to request cancellation of the ongoing logical link creation process identified by the Physical_Link_Handle and the Tx_Flow_Spec_ID.	AMP
Logical Link Complete event	3.0 + HS	The HCI_Logical_Link_Complete event indicates to both of the Hosts forming the connection that a new AMP Logical Link has been established.	AMP
Loopback Command event	1.1	The HCI_Loopback_Command event is used to loop back all commands that the Host sends to the BR/EDR Controller with some exceptions.	BR/EDR
Master Link Key command	1.1	The HCI_Master_Link_Key command is used to force both BR/EDR Controllers of a connection associated to the Connection_Handle to use the temporary link key of the Master device or the regular link keys.	BR/EDR
Master Link Key Complete event	1.1	The HCI_Master_Link_Key_Complete event is used to indicate that the change in the temporary Link Key or in the semi-permanent link keys on the Bluetooth master side has been completed.	BR/EDR
Max Slots Change event	1.1	The HCI_Max_Slots_Change event is used to indicate a change in the max slots by the LM.	BR/EDR
Mode Change event	1.1	The HCI_Mode_Change event is used to indicate that the current mode has changed.	BR/EDR

Table 3.1: Alphabetical list of commands and events (Sheet 22 of 46)



Name	Vers.	Summary Description	Supported Controllers
Number Of Completed Data Blocks event	3.0 + HS	The HCI_Number_Of_Completed_Data_Blocks event is used by the Controller to indicate to the Host how many HCI ACL Data packets have been completed and how many data block buffers have been freed for each Handle since the previous HCI_Number_Of_Completed_Data_Blocks event was sent.	BR/EDR, AMP
Number Of Completed Packets event	1.1	The HCI_Number_Of_Completed_Packets event is used by the Controller to indicate to the Host how many HCI Data packets have been completed for each Connection_Handle since the previous HCI_Number_Of_Completed_Packets event was sent.	All
Page Scan Repetition Mode Change event	1.1	The HCI_Page_Scan_Repetition_Mode_Change event indicates that the connected remote BR/EDR Controller with the specified Connection_Handle has successfully changed the Page Scan Repetition Mode (SR).	BR/EDR
Periodic Inquiry Mode command	1.1	The HCI_Periodic_Inquiry_Mode command is used to configure the BR/EDR Controller to perform an automatic Inquiry based on a specified period range.	BR/EDR
Physical Link Complete event	3.0 + HS	The HCI_Physical_Link_Complete event indicates to both of the Hosts forming the connection that a new AMP Physical Link has been established.	AMP
Physical Link Loss Early Warning event	3.0 + HS	The HCI_Physical_Link_Loss_Early_Warning event occurs when a physical link has indication that the link may be disrupted.	AMP
Physical Link Recovery event	3.0 + HS	The HCI_Physical_Link_Recovery event indicates that whatever interruption caused an earlier HCI_Physical_Link_Loss_Early_Warning event has now been cleared.	AMP

Table 3.1: Alphabetical list of commands and events (Sheet 23 of 46)



Name	Vers.	Summary Description	Supported Controllers
PIN Code Request event	1.1	The HCI_PIN_Code_Request event is used to indicate that a PIN code is required to create a new link key for a connection.	BR/EDR
PIN Code Request Negative Reply command	1.1	The HCI_PIN_Code_Request_Negative_Reply command is used to reply to an HCI_PIN_Code_Request event from the Controller when the Host cannot specify a PIN code to use for a connection.	BR/EDR
PIN Code Request Reply command	1.1	The HCI_PIN_Code_Request_Reply command is used to reply to an HCI_PIN_Code_Request event from the Controller and specifies the PIN code to use for a connection.	BR/EDR
QoS Setup command	1.1	The HCI_QoS_Setup command is used to specify Quality of Service parameters for a Connection_Handle.	BR/EDR
QoS Setup Complete event	1.1	The HCI_QoS_Setup_Complete event is used to indicate that QoS is set up.	BR/EDR
QoS Violation event	1.1	The HCI_QoS_Violation event is used to indicate the Controller's Link Manager or PAL is unable to provide the current QoS requirement for the Handle.	BR/EDR, AMP
Read AFH Channel Assessment Mode command	1.2	The HCI_Read_AFH_Channel_Assessment_Mode command will read the value for the AFH Channel Classification Mode parameter. This value is used to enable or disable the Controller's channel assessment scheme.	BR/EDR
Read AFH Channel Map command	1.2	The HCI_Read_AFH_Channel_Map command will read the current state of the channel map for a connection.	BR/EDR
Read Authenticated Payload Timeout command	4.1	The HCI_Read_Authenticated_Payload_Timeout command is used to read the Authenticated Payload Timeout parameter, which is used to set the maximum time between packets being received from the remote device without a valid MIC.	BR/EDR, LE

Table 3.1: Alphabetical list of commands and events (Sheet 24 of 46)



Name	Vers.	Summary Description	Supported Controllers
Read Authentication Enable command	1.1	The HCI_Read_Authentication_ - Enable command will read the value for the Authentication Enable parameter, which controls whether the Bluetooth device will require authentication for each connection with other Bluetooth devices.	BR/EDR
Read Automatic Flush Timeout command	1.1	The HCI_Read_Automatic_Flush_ - Timeout command will read the value for the Flush Timeout configuration parameter for the specified Connection_Handle. The Flush Timeout parameter is only used for ACL connections.	BR/EDR
Read BD_ADDR command	1.1	The HCI_Read_BD_ADDR command will read the value for the BD_ADDR parameter.	BR/EDR, LE
Read Best Effort Flush Timeout command	3.0 + HS	The HCI_Read_Best_Effort_Flush_ - Timeout command returns the last value written with the HCI_Write_ - Best_Effort_Flush_Timeout command.	AMP
Read Buffer Size command	1.1	The HCI_Read_Buffer_Size command returns the size of the HCI buffers. These buffers are used by the Controller to buffer data that is to be transmitted.	BR/EDR, AMP
Read Class of Device command	1.1	The HCI_Read_Class_of_Device command will read the value for the Class of Device configuration parameter, which is used to indicate its capabilities to other devices.	BR/EDR
Read Clock command	1.2	The HCI_Read_Clock command will read an estimate of a piconet or the local Bluetooth Clock.	BR/EDR
Read Clock Offset command	1.1	The HCI_Read_Clock_Offset command allows the Host to read the clock offset of remote BR/EDR Controllers.	BR/EDR
Read Clock Offset Complete event	1.1	The HCI_Read_Clock_Offset_ - Complete event is used to indicate the completion of the process of the LM obtaining the Clock offset information.	BR/EDR

Table 3.1: Alphabetical list of commands and events (Sheet 25 of 46)



Name	Vers.	Summary Description	Supported Controllers
Read Connection Accept Timeout command	1.1	The HCI_Read_Connection_Accept_Timeout command will read the value for the Connection Accept Timeout configuration parameter, which allows the Controller to automatically deny a connection request after a specified period has occurred, and to refuse a new connection.	BR/EDR, AMP
Read Current IAC LAP command	1.1	The HCI_Read_Current_IAC_LAP command will read the LAP(s) used to create the Inquiry Access Codes (IAC) that the local BR/EDR Controller is simultaneously scanning for during Inquiry Scans.	BR/EDR
Read Data Block Size command	3.0 + HS	The HCI_Read_Data_Block_Size command returns the maximum size of the HCI buffers. These buffers are used by the Controller to buffer data that is to be transmitted.	BR/EDR, AMP
Read Default Erroneous Data Reporting command	2.1 + EDR	The HCI_Read_Default_Erroneous_Data_Reporting command will read the value for the Erroneous Data Reporting configuration parameter, which controls whether the BR/EDR Controller will provide data for every (e)SCO interval, with the Packet_Status_Flag in HCI Synchronous Data packets set according to HCI Synchronous Data packets.	BR/EDR
Read Default Link Policy Settings command	1.2	The HCI_Read_Default_Link_Policy_Settings command will read the Default Link Policy configuration parameter for all new connections.	BR/EDR
Read Encryption Key Size command	3.0 + HS	The HCI_Read_Encryption_Key_Size command is used to read the encryption key size on a given Connection_Handle.	BR/EDR
Read Enhanced Transmit Power Level command	3.0 + HS	The HCI_Read_Enhanced_Transmit_Power_Level command will read the values for the GFSK, $\pi/4$ -DQPSK and 8DPSK Transmit Power Level parameters for the specified Connection_Handle.	BR/EDR

Table 3.1: Alphabetical list of commands and events (Sheet 26 of 46)



Name	Vers.	Summary Description	Supported Controllers
Read Extended Inquiry Length command	4.1	The HCI_Read_Extended_Inquiry_Length command is used to read the Extended Inquiry Length parameter from the Controller.	BR/EDR
Read Extended Inquiry Response command	2.1 + EDR	The HCI_Read_Extended_Inquiry_Response command will read the data that the BR/EDR Controller sends in the extended inquiry response packet during inquiry response.	BR/EDR
Read Extended Page Timeout command	4.1	The HCI_Read_Extended_Page_Timeout command is used to read the Extended Page Timeout parameter from the Controller.	BR/EDR
Read Failed Contact Counter command	1.1	The HCI_Read_Failed_Contact_Counter command will read the value for the Failed Contact Counter configuration parameter for a particular connection to another device.	BR/EDR, AMP
Read Flow Control Mode command	3.0 + HS	The HCI_Read_Flow_Control_Mode command returns the value of the Flow_Control_Mode configuration parameter supported by this Controller.	BR/EDR, AMP
Read Hold Mode Activity command	1.1	The HCI_Read_Hold_Mode_Activity command is used to read which activities should be suspended when the BR/EDR Controller is in Hold mode.	BR/EDR
Read Inquiry Mode command	1.2	The HCI_Read_Inquiry_Mode command is used to read the Inquiry Mode configuration parameter of the local BR/EDR Controller.	BR/EDR
Read Inquiry Response Transmit Power Level command	2.1 + EDR	The HCI_Read_Inquiry_Response_Transmit_Power_Level command will read the inquiry response Transmit Power level used to transmit the FHS and EIR data packets. This can be used directly in the Tx Power Level EIR data type.	BR/EDR

Table 3.1: Alphabetical list of commands and events (Sheet 27 of 46)



Name	Vers.	Summary Description	Supported Controllers
Read Inquiry Scan Activity command	1.1	The HCI_Read_Inquiry_Scan_Activity command will read the value for Inquiry Scan Interval and Inquiry Scan Window configuration parameters. Inquiry Scan Interval defines the amount of time between consecutive inquiry scans. Inquiry Scan Window defines the amount of time for the duration of the inquiry scan.	BR/EDR
Read Inquiry Scan Type command	1.2	The HCI_Read_Inquiry_Scan_Type command is used to read the Inquiry Scan Type configuration parameter of the local BR/EDR Controller. The Inquiry Scan Type configuration parameter can set the inquiry scan to either normal or interlaced scan.	BR/EDR
Read LE Host Support command	4.0	The HCI_Read_LE_Host_Support command reads the LE Supported Host and Simultaneous LE Host settings from the BR/EDR Controller.	BR/EDR
Read Link Policy Settings command	1.1	The HCI_Read_Link_Policy_Settings command will read the Link Policy configuration parameter for the specified Connection_Handle. The Link Policy settings allow the Host to specify which Link Modes the Link Manager can use for the specified Connection_Handle.	BR/EDR
Read Link Quality command	1.1	The HCI_Read_Link_Quality command will read the value for the Link Quality for the specified Connection_Handle.	BR/EDR, AMP
Read Link Supervision Timeout command	1.1	The HCI_Read_Link_Supervision_Timeout command will read the value for the Link Supervision Timeout configuration parameter for the device. This parameter is used by the Controller to determine link loss.	BR/EDR, AMP
Read LMP Handle command	1.2	The HCI_Read_LMP_Handle command will read the current LMP Handle associated with the Connection_Handle.	BR/EDR
Read Local AMP ASSOC command	3.0 + HS	The HCI_Read_Local_AMP_ASSOC command will return a fragment of the AMP_ASSOC structure, which contains AMP specific information for this AMP Controller.	AMP

Table 3.1: Alphabetical list of commands and events (Sheet 28 of 46)



Name	Vers.	Summary Description	Supported Controllers
Read Local AMP Info command	3.0 + HS	The HCI_Read_Local_AMP_Info command returns information about this AMP Controller.	AMP
Read Local Extended Features command	1.2	The HCI_Read_Local_Extended_Features command requests a list of the supported extended features for the local device.	BR/EDR
Read Local Name command	1.1	The HCI_Read_Local_Name command provides the ability to read the stored user-friendly name for the BR/EDR Controller.	BR/EDR
Read Local OOB Data command	2.1 + EDR	The HCI_Read_Local_OOB_Data command is used to obtain a Simple Pairing Hash C and Simple Pairing Randomizer R which are intended to be transferred to a remote device using an OOB mechanism.	BR/EDR
Read Local OOB Extended Data command	4.1	The HCI_Read_Local_OOB_Extended_Data command is used to obtain Simple Pairing Hash C and Randomizer R associated with both P-192 and P-256 public keys, which are intended to be transferred to a remote device using an OOB mechanism.	BR/EDR
Read Local Simple Pairing Options command	Erratum 10734	The HCI_Read_Local_Simple_Pairing_Options command is used to read the simple pairing options and the maximum encryption key size supported.	BR/EDR
Read Local Supported Codecs command	CSA2	The HCI_Read_Local_Supported_Codecs command is used for a Host to query a Controller's supported codecs.	BR/EDR
Read Local Supported Commands command	1.2	The HCI_Read_Local_Supported_Commands command requests a list of the supported HCI commands for the local device.	All
Read Local Supported Features command	1.1	The HCI_Read_Local_Supported_Features command requests a list of the supported features for the local device.	All

Table 3.1: Alphabetical list of commands and events (Sheet 29 of 46)



Name	Vers.	Summary Description	Supported Controllers
Read Local Version Information command	1.1	The HCI_Read_Local_Version_Information command will read the version information for the local Controller.	All
Read Location Data command	3.0 + HS	The HCI_Read_Location_Data command provides the ability to read the Location Data parameters from any stored knowledge of environment or regulations or currently in use in the AMP Controller.	AMP
Read Logical Link Accept Timeout command	3.0 + HS	The HCI_Read_Logical_Link_Accept_Timeout command will read the value for the Logical Link Accept Timeout configuration parameter, which allows the AMP Controller to automatically deny a Logical Link request after a specified period has occurred, and to refuse a new connection.	AMP
Read Loopback Mode command	1.1	The HCI_Read_Loopback_Mode command will read the value for the setting of the BR/EDR Controller's Loopback Mode. The setting of the Loopback Mode will determine the path of information.	BR/EDR
Read Num Broadcast Retransmissions command	1.1	The HCI_Read_Num_Broadcast_Retransmissions command will read the parameter value for the Number of Broadcast Retransmissions for the BR/EDR Controller.	BR/EDR
Read Number Of Supported IAC command	1.1	The HCI_Read_Number_Of_Supported_IAC command will read the value for the number of Inquiry Access Codes (IAC) that the local BR/EDR Controller can simultaneously listen for during an Inquiry Scan.	BR/EDR
Read Page Scan Activity command	1.1	The HCI_Read_Page_Scan_Activity command will read the values for the Page Scan Interval and Page Scan Window configuration parameters. Page Scan Interval defines the amount of time between consecutive page scans. Page Scan Window defines the duration of the page scan.	BR/EDR

Table 3.1: Alphabetical list of commands and events (Sheet 30 of 46)



Name	Vers.	Summary Description	Supported Controllers
Read Page Scan Type command	1.2	The HCI_Read_Page_Scan_Type command is used to read the page scan type of the local BR/EDR Controller. The Page Scan Type configuration parameter can set the page scan to either normal or interlaced scan.	BR/EDR
Read Page Timeout command	1.1	The HCI_Read_Page_Timeout command will read the value for the Page Reply Timeout configuration parameter, which determines the time the BR/EDR Controller will wait for the remote device to respond to a connection request before the local device returns a connection failure.	BR/EDR
Read PIN Type command	1.1	The HCI_Read_PIN_Type command is used for the Host to read the value that is specified to indicate whether the Host supports variable PINs or only fixed PINs.	BR/EDR
Read Remote Extended Features command	1.2	The HCI_Read_Remote_Extended_Features command requests a list of the supported extended features of a remote device.	BR/EDR
Read Remote Extended Features Complete event	1.2	The HCI_Read_Remote_Extended_Features_Complete event is used to indicate the completion of the process of the Link Manager obtaining the supported Extended features of the remote BR/EDR Controller specified by the Connection_Handle event parameter.	BR/EDR
Read Remote Supported Features command	1.1	The HCI_Read_Remote_Supported_Features command requests a list of the supported features of a remote device.	BR/EDR
Read Remote Supported Features Complete event	1.1	The HCI_Read_Remote_Supported_Features_Complete event is used to indicate the completion of the process of the Link Manager obtaining the supported features of the remote BR/EDR Controller specified by the Connection_Handle event parameter.	BR/EDR

Table 3.1: Alphabetical list of commands and events (Sheet 31 of 46)



Name	Vers.	Summary Description	Supported Controllers
Read Remote Version Information command	1.1	The HCI_Read_Remote_Version_Information command will read the values for the version information for the remote device associated with the Connection_Handle.	BR/EDR, LE
Read Remote Version Information Complete event	1.1	The HCI_Read_Remote_Version_Information_Complete event is used to indicate the completion of the process of the Link Manager obtaining the version information of the remote device associated with the Connection_Handle event parameter.	BR/EDR, LE
Read RSSI command	1.1	The HCI_Read_RSSI command will read the value for the Received Signal Strength Indication (RSSI) for a Connection_Handle to another Controller.	All
Read Scan Enable command	1.1	The HCI_Read_Scan_Enable command will read the value for the Scan Enable configuration parameter, which controls whether or not the BR/EDR Controller will periodically scan for page attempts and/or inquiry requests from other BR/EDR Controllers.	BR/EDR
Read Secure Connections Host Support command	4.1	The HCI_Read_Secure_Connections_Host_Support command is used to read the Secure Connections Host Supports parameter from the Controller.	BR/EDR
Read Simple Pairing Mode command	2.1 + EDR	The HCI_Read_Simple_Pairing_Mode command reads the Simple Pairing mode setting in the BR/EDR Controller.	BR/EDR
Read Stored Link Key command	1.1	The HCI_Read_Stored_Link_Key command provides the ability to read whether one or more link keys are stored in the Controller.	BR/EDR
Read Synchronization Train Parameters command	CSA4	The HCI_Read_Synchronization_Train_Parameters command returns the currently configured values for the Synchronization Train functionality in the BR/EDR Controller.	BR/EDR

Table 3.1: Alphabetical list of commands and events (Sheet 32 of 46)



Name	Vers.	Summary Description	Supported Controllers
Read Synchronous Flow Control Enable command	1.1	The HCI_Read_Synchronous_Flow_Control_Enable command provides the ability to read the Synchronous Flow Control Enable setting. By using this setting, the Host can decide if the Controller will send HCI_Number_Of_Completed_Packets events for synchronous Connection_Handles.	BR/EDR
Read Transmit Power Level command	1.1	The HCI_Read_Transmit_Power_Level command will read the values for the Transmit Power Level parameter for the specified Connection_Handle.	BR/EDR, LE
Read Voice Setting command	1.1	The HCI_Read_Voice_Setting command will read the values for the Voice Setting configuration parameter, which controls all the various settings for the voice connections.	BR/EDR
Receive Synchronization Train command	CSA4	The HCI_Receive_Synchronization_Train command requests synchronization with the specified Connectionless Slave Broadcast transmitter.	BR/EDR
Refresh Encryption Key command	2.1 + EDR	The HCI_Refresh_Encryption_Key command is used by the Host to cause the Controller to refresh the encryption key by pausing and resuming encryption	BR/EDR
Reject Connection Request command	1.1	The HCI_Reject_Connection_Request command is used to decline a new incoming BR/EDR connection request.	BR/EDR
Reject Synchronous Connection Request command	1.2	The HCI_Reject_Synchronous_Connection_Request command is used to decline an incoming request for a synchronous link.	BR/EDR
Remote Host Supported Features Notification event	2.1 + EDR	The HCI_Remote_Host_Supported_Features_Notification event is used to return the LMP extended features page containing the Host features.	BR/EDR
Remote Name Request Cancel command	1.2	The HCI_Remote_Name_Request_Cancel command is used to cancel an ongoing Remote Name Request.	BR/EDR

Table 3.1: Alphabetical list of commands and events (Sheet 33 of 46)



Name	Vers.	Summary Description	Supported Controllers
Remote Name Request command	1.1	The HCI_Remote_Name_Request command is used to obtain the user-friendly name of another BR/EDR Controller.	BR/EDR
Remote Name Request Complete event	1.1	The HCI_Remote_Name_Request - Complete event is used to indicate a remote name request has been completed.	BR/EDR
Remote OOB Data Request event	2.1 + EDR	The HCI_Remote_OOB_Data - Request event is used to indicate that the Simple Pairing Hash C and the Simple Pairing Randomizer R is required for the Secure Simple Pairing process involving the device identified by BD_ADDR.	BR/EDR
Remote OOB Data Request Negative Reply command	2.1 + EDR	The HCI_Remote_OOB_Data - Request_Negative_Reply command is used to reply to an HCI_Remote_OOB_Data_Request event that the Host does not have the C and R	BR/EDR
Remote OOB Data Request Reply command	2.1 + EDR	The HCI_Remote_OOB_Data - Request_Reply command is used to reply to an HCI_Remote_OOB_Data_Request event with the C and R values received via an OOB transfer from a remote BR/EDR Controller identified by BD_ADDR.	BR/EDR
Remote OOB Extended Data Request Reply command	4.1	The HCI_Remote_OOB_Extended - Data_Request_Reply command is used to reply to an HCI_Remote_OOB_Data_Request event with the C and R values received via an OOB transfer from a remote BR/EDR Controller identified by the BD_ADDR.	BR/EDR
Reset command	1.1	For a BR/EDR Controller, the HCI_Reset command resets HCI, the Link Manager, and the Bluetooth radio. For an AMP Controller, the HCI_Reset command resets HCI and the AMP PAL. For an LE Controller, the HCI_Reset command resets HCI, the Link Layer, and LE PHY.	All

Table 3.1: Alphabetical list of commands and events (Sheet 34 of 46)



Name	Vers.	Summary Description	Supported Controllers
Reset Failed Contact Counter command	1.1	The HCI_Reset_Failed_Contact_-Counter command will reset the value for the Failed Contact Counter configuration parameter for a particular connection to another device.	BR/EDR, AMP
Return Link Keys event	1.1	The HCI_Return_Link_Keys event is used to return stored link keys after an HCI_Read_Stored_Link_Key command is used.	BR/EDR
Role Change event	1.1	The HCI_Role_Change event is used to indicate that the current BR/EDR Controller role related to the particular connection has been changed.	BR/EDR
Role Discovery command	1.1	The HCI_Role_Discovery command is used for a BR/EDR Controller to determine which role the device is performing for a particular Connection_Handle.	BR/EDR
SAM Status Change event	5.0	The HCI_SAM_Status_Change event is used to indicate that either the local or remote SAM slot map on a particular connection has been changed.	BR/EDR
Send Keypress Notification command	2.1 + EDR	The HCI_Send_Keypress_-Notification command is used during the Passkey Entry protocol by a device with KeyboardOnly IO capabilities. It is used by a Host to inform the remote device when keys have been entered or erased.	BR/EDR
Set AFH Host Channel Classification command	1.2	The HCI_Set_AFH_Host_Channel_-Classification command allows the Host to specify a channel classification based on its "local information".	BR/EDR
Set Connection Encryption command	1.1	The HCI_Set_Connection_-Encryption command is used to enable and disable the link level encryption.	BR/EDR
Set Connectionless Slave Broadcast command	CSA4	The HCI_Set_Connectionless_-Slave_Broadcast command controls Connectionless Slave Broadcast functionality (for transmission) in the BR/EDR Controller including enabling and disabling the broadcast.	BR/EDR

Table 3.1: Alphabetical list of commands and events (Sheet 35 of 46)



Name	Vers.	Summary Description	Supported Controllers
Set Connectionless Slave Broadcast Data command	CSA4	The HCI_Set_Connectionless_Slave_Broadcast_Data command is used by the Host to set Connectionless Slave Broadcast data in the BR/EDR Controller.	BR/EDR
Set Connectionless Slave Broadcast Receive command	CSA4	The HCI_Set_Connectionless_Slave_Broadcast_Receive command enables and disables Connectionless Slave Broadcast reception in the BR/EDR Controller.	BR/EDR
Set Controller To Host Flow Control command	1.1	The HCI_Set_Controller_To_Host_Flow_Control command is used by the Host to turn flow control on or off in the direction from the Controller to the Host.	BR/EDR, LE
Set Event Filter command	1.1	The HCI_Set_Event_Filter command is used by the Host to specify different event filters. The Host may issue this command multiple times to request various conditions for the same type of event filter and for different types of event filters.	BR/EDR
Set Event Mask command	1.1	The HCI_Set_Event_Mask command is used to control which events are generated by the HCI for the Host.	All
Set Event Mask Page 2 command	3.0 + HS	The HCI_Set_Event_Mask_Page 2 command is used to control which events are generated by the HCI for the Host.	All
Set External Frame Configuration command	CSA3	The HCI_Set_External_Frame_Configuration command enables an external device to describe a frame structure to the Controller.	BR/EDR, LE and AMP
Set MWS Channel Parameters command	CSA3	The HCI_Set_MWS_Channel_Parameters command enables an MWS device to inform the Controller about the MWS channel configuration.	BR/EDR, LE and AMP
Set MWS Scan Frequency Table command	CSA3	The HCI_Set_MWS_Scan_Frequency_Table command specifies the frequencies represented by the frequency index supplied by the MWS_SCAN_FREQUENCY signal.	BR/EDR, LE and AMP

Table 3.1: Alphabetical list of commands and events (Sheet 36 of 46)



Name	Vers.	Summary Description	Supported Controllers
Set MWS Signaling command	CSA3	The HCI_Set_MWS_Signaling command enables an MWS device to inform the Controller about the timing parameters for the MWS coexistence interface.	BR/EDR, LE and AMP
Set MWS Transport Layer command	CSA3	The HCI_Set_MWS_Transport_Layer command selects the MWS coexistence signaling transport layer in the Controller.	BR/EDR, LE and AMP
Set MWS_PATTERN Configuration command	CSA3	The HCI_Set_MWS_PATTERN_Configuration command specifies the configuration of the pattern indicated over the MWS Coexistence Transport Layer.	BR/EDR, LE and AMP
Set Reserved LT_ADDR command	CSA4	The HCI_Set_Reserved_LT_ADDR command requests that the BR/EDR Controller reserve a specific LT_ADDR for the purposes of Connectionless Slave Broadcast.	BR/EDR
Set Triggered Clock Capture command	CSA4	The HCI_Set_Triggered_Clock_Capture command is used to configure the Controller to return events containing an estimate of a piconet or the local Bluetooth clock.	BR/EDR
Setup Synchronous Connection command	1.2	The HCI_Setup_Synchronous_Connection command adds a new or modifies an existing synchronous logical transport (SCO or eSCO) on a physical link depending on the Connection_Handle parameter specified.	BR/EDR
Short Range Mode Change Complete event	3.0 + HS	The HCI_Short_Range_Mode_Change_Complete event is used to indicate completion of any tasks relating to establishment of short range mode.	802.11 PAL only
Short Range Mode command	3.0 + HS	The HCI_Short_Range_Mode command is used to notify the AMP that the peer may be at close enough range to require modification of AMP transmission power level.	802.11 PAL only
Simple Pairing Complete event	2.1 + EDR	The HCI_Simple_Pairing_Complete event is used to indicate that the simple pairing process has completed.	BR/EDR

Table 3.1: Alphabetical list of commands and events (Sheet 37 of 46)



Name	Vers.	Summary Description	Supported Controllers
Slave Page Response Timeout event	CSA4	The HCI_Slave_Page_Response_-Timeout event indicates to the Host that the <i>pagerespTO</i> has been exceeded on the BR/EDR Controller after the Controller responded to an ID packet.	BR/EDR
Sniff Mode command	1.1	The HCI_Sniff_Mode command is used to alter the behavior of the LM and have the LM place the local or remote device into the Sniff mode.	BR/EDR
Sniff Subrating command	2.1 + EDR	The HCI_Sniff_Subrating command is used to configure the sniff subrating parameters in the local device.	BR/EDR
Sniff Subrating event	2.1 + EDR	The HCI_Sniff_Subrating event is used to inform the Host of the local and remote transmit and receive latencies.	BR/EDR
Start Synchronization Train command	CSA4	The HCI_Start_Synchronization_Train command enables the Synchronization Train on the BR/EDR Controller using the currently configured Synchronization Train parameters.	BR/EDR
Switch Role command	1.1	The HCI_Switch_Role command is used to switch master and slave roles of the devices on either side of a connection.	BR/EDR
Synchronization Train Complete event	CSA4	The HCI_Synchronization_Train_-Complete event indicates that the Synchronization Train has completed.	BR/EDR
Synchronization Train Received event	CSA4	The HCI_Synchronization_Train_-Received event provides the status of Synchronization Train packets received from the device with the given BD_ADDR.	BR/EDR
Synchronous Connection Changed event	1.2	The HCI_Synchronous_-Connection_Changed event indicates to the Host that an existing synchronous connection has been reconfigured.	BR/EDR

Table 3.1: Alphabetical list of commands and events (Sheet 38 of 46)



Name	Vers.	Summary Description	Supported Controllers
Synchronous Connection Complete event	1.2	The HCI_Synchronous_Connection_Complete event indicates to both the Hosts that a new synchronous connection has been established.	BR/EDR
Triggered Clock Capture event	CSA4	The HCI_Triggered_Clock_Capture event reports the Bluetooth clock when an external trigger occurred.	BR/EDR
Truncated Page Cancel command	CSA4	The HCI_Truncated_Page_Cancel command is used to cancel an ongoing Truncated Page.	BR/EDR
Truncated Page command	CSA4	The HCI_Truncated_Page command will cause the BR/EDR Controller to page the BR/EDR Controller with the BD_ADDR specified by the command parameters and abort the page sequence after receiving the ID response packet.	BR/EDR
Truncated Page Complete event	CSA4	The HCI_Truncated_Page_Complete event indicates to the Host that a Truncated Page has completed.	BR/EDR
User Confirmation Request event	2.1 + EDR	The HCI_User_Confirmation_Request event is used to indicate that user confirmation of a numeric value is required.	BR/EDR
User Confirmation Request Negative Reply command	2.1 + EDR	The HCI_User_Confirmation_Request_Negative_Reply command is used to reply to an HCI_User_Confirmation_Request event and indicates that the user selected “no”. This command will terminate Secure Simple Pairing.	BR/EDR
User Confirmation Request Reply command	2.1 + EDR	The HCI_User_Confirmation_Request_Reply command is used to reply to an HCI_User_Confirmation_Request event and indicates that the user selected “yes”. It is also used when the Host has no input and no output capabilities.	BR/EDR
User Passkey Notification event	2.1 + EDR	The HCI_User_Passkey_Notification event is used to provide a passkey for the Host to display to the user as required as part of a Simple Pairing process.	BR/EDR

Table 3.1: Alphabetical list of commands and events (Sheet 39 of 46)



Name	Vers.	Summary Description	Supported Controllers
User Passkey Request event	2.1 + EDR	The HCI_User_Passkey_Request event is used to indicate that a passkey is required as part of a Simple Pairing process.	BR/EDR
User Passkey Request Negative Reply command	2.1 + EDR	The HCI_User_Passkey_Request_Negative_Reply command is used to reply to an HCI_User_Passkey_Request event and indicates the Host could not provide a passkey. This command will terminate Simple Pairing.	BR/EDR
User Passkey Request Reply command	2.1 + EDR	The HCI_User_Passkey_Request_Reply command is used to reply to an HCI_User_Passkey_Request event and specifies the Numeric_Value (passkey) entered by the user to be used in the Secure Simple Pairing process.	BR/EDR
Write AFH Channel Assessment Mode command	1.2	The HCI_Write_AFH_Channel_Assessment_Mode command will write the value for the Channel Classification Mode configuration parameter. This value is used to enable or disable the Controller's channel assessment scheme.	BR/EDR
Write Authenticated Payload Timeout command	4.1	The HCI_Write_Authenticated_Payload_Timeout command is used to write the Authenticated Payload Timeout parameter, which is used to set the maximum time between packets being received from the remote device without a valid MIC.	BR/EDR, LE
Write Authentication Enable command	1.1	The HCI_Write_Authentication_Enable command will write the value for the Authentication Enable parameter, which controls whether the Bluetooth device will require authentication for each connection with other Bluetooth devices.	BR/EDR
Write Automatic Flush Timeout command	1.1	The HCI_Write_Automatic_Flush_Timeout command will write the value for the Flush Timeout configuration parameter for the specified Connection_Handle. The Flush Timeout parameter is only used for ACL connections.	BR/EDR

Table 3.1: Alphabetical list of commands and events (Sheet 40 of 46)



Name	Vers.	Summary Description	Supported Controllers
Write Best Effort Flush Timeout command	3.0 + HS	The HCI_Write_Best_Effort_Flush_Timeout command configures the AMP device with a maximum time to attempt to transmit any given frame on the Best Effort logical link.	802.11 PAL only
Write Class of Device command	1.1	The HCI_Write_Class_of_Device command will write the value for the Class_of_Device configuration parameter, which is used to indicate its capabilities to other devices.	BR/EDR
Write Connection Accept Timeout command	1.1	The HCI_Write_Connection_Accept_Timeout command will write the value for the Connection Accept Timeout configuration parameter, which allows the Controller to automatically deny a connection request after a specified period has occurred, and to refuse a new connection.	BR/EDR, AMP
Write Current IAC LAP command	1.1	The HCI_Write_Current_IAC LAP command will write the LAP(s) used to create the Inquiry Access Codes (IAC) that the local BR/EDR Controller is simultaneously scanning for during Inquiry Scans.	BR/EDR
Write Default Erroneous Data Reporting command	2.1 + EDR	The HCI_Write_Default_Erroneous_Data_Reporting command will write the value for the Erroneous Data Reporting configuration parameter, which controls whether the Bluetooth Controller will provide data for every (e)SCO interval, with the Packet_Status_Flag in HCI Synchronous Data packets set according to HCI Synchronous Data packets.	BR/EDR
Write Default Link Policy Settings command	1.2	The HCI_Write_Default_Link_Policy_Settings command will write the Default Link Policy configuration parameter for all new connections.	BR/EDR
Write Extended Inquiry Length command	4.1	The HCI_Write_Extended_Inquiry_Length command is used to write the Extended Inquiry Length parameter to the Controller.	BR/EDR

Table 3.1: Alphabetical list of commands and events (Sheet 41 of 46)



Name	Vers.	Summary Description	Supported Controllers
Write Extended Inquiry Response command	2.1 + EDR	The HCI_Write_Extended_Inquiry_Response command will write the data that the BR/EDR Controller sends in the extended inquiry response packet during inquiry response.	BR/EDR
Write Extended Page Timeout command	4.1	The HCI_Write_Extended_Page_Timeout command is used to write the Extended Page Timeout parameter to the Controller.	BR/EDR
Write Flow Control Mode command	3.0 + HS	The HCI_Write_Flow_Control_Mode command sets the value of the Flow_Control_Mode configuration parameter for this Controller.	BR/EDR, AMP
Write Hold Mode Activity command	1.1	The HCI_Write_Hold_Mode_Activity command is used to write which activities should be suspended when the BR/EDR Controller is in Hold mode.	BR/EDR
Write Inquiry Mode command	1.2	The HCI_Write_Inquiry_Mode command is used to write the Inquiry Mode configuration parameter of the local BR/EDR Controller.	BR/EDR
Write Inquiry Scan Activity command	1.1	The HCI_Write_Inquiry_Scan_Activity command will write the value for Inquiry Scan Interval and Inquiry Scan Window configuration parameters. Inquiry Scan Interval defines the amount of time between consecutive inquiry scans. Inquiry Scan Window defines the amount of time for the duration of the inquiry scan.	BR/EDR
Write Inquiry Scan Type command	1.2	The HCI_Write_Inquiry_Scan_Type command is used to write the Inquiry Scan Type configuration parameter of the local BR/EDR Controller. The Inquiry Scan Type configuration parameter can set the inquiry scan to either normal or interlaced scan.	BR/EDR
Write Inquiry Transmit Power Level command	2.1 + EDR	The HCI_Write_Inquiry_Transmit_Power_Level command is used to write the transmit power level used to transmit the inquiry (ID) data packets.	BR/EDR

Table 3.1: Alphabetical list of commands and events (Sheet 42 of 46)



Name	Vers.	Summary Description	Supported Controllers
Write LE Host Support command	4.0	The HCI_Write_LE_Host_Support command writes the LE Supported Host and Simultaneous LE Host settings to the BR/EDR Controller.	BR/EDR
Write Link Policy Settings command	1.1	The HCI_Write_Link_Policy_Settings command will write the Link Policy configuration parameter for the specified Connection_Handle. The Link Policy settings allow the Host to specify which Link Modes the Link Manager can use for the specified Connection_Handle.	BR/EDR
Write Link Supervision Timeout command	1.1	The HCI_Write_Link_Supervision_Timeout command will write the value for the Link Supervision Timeout configuration parameter for the device. This parameter is used by the Controller to determine link loss.	BR/EDR, AMP
Write Local Name command	1.1	The HCI_Write_Local_Name command provides the ability to modify the user-friendly name for the BR/EDR Controller.	BR/EDR
Write Location Data command	3.0 + HS	The HCI_Write_Location_Data command is used to tell the Controller any knowledge of environment or regulations currently in force, which may affect the operation of the Controller.	AMP
Write Logical Link Accept Timeout command	3.0 + HS	The HCI_Write_Logical_Link_Accept_Timeout command will write the value for the Logical Link Accept Timeout configuration parameter, which allows the AMP Controller to automatically deny a connection request after a specified period has occurred, and to refuse a new connection.	AMP
Write Loopback Mode command	1.1	The HCI_Write_Loopback_Mode command will write the value for the setting of the BR/EDR Controllers Loopback Mode. The setting of the Loopback Mode will determine the path of information.	BR/EDR

Table 3.1: Alphabetical list of commands and events (Sheet 43 of 46)



Name	Vers.	Summary Description	Supported Controllers
Write Num Broadcast Retransmissions command	1.1	The HCI_Write_Num_Broadcast_Retransmissions command will write the parameter value for the Number of Broadcast Retransmissions for the BR/EDR Controller.	BR/EDR
Write Page Scan Activity command	1.1	The HCI_Write_Page_Scan_Activity command will write the value for Page Scan Interval and Page Scan Window configuration parameters. Page Scan Interval defines the amount of time between consecutive page scans. Page Scan Window defines the duration of the page scan.	BR/EDR
Write Page Scan Type command	1.2	The HCI_Write_Page_Scan_Type command is used to write the page scan type of the local BR/EDR Controller. The Page Scan Type configuration parameter can set the page scan to either normal or interlaced scan.	BR/EDR
Write Page Timeout command	1.1	The HCI_Write_Page_Timeout command will write the value for the Page Reply Timeout configuration parameter, which allows the BR/EDR Controller to define the amount of time a connection request will wait for the remote device to respond before the local device returns a connection failure.	BR/EDR
Write PIN Type command	1.1	The HCI_Write_PIN_Type command is used for the Host to specify whether the Host supports variable PIN or only fixed PINs.	BR/EDR
Write Remote AMP ASSOC command	3.0 + HS	The HCI_Write_Remote_-AMP_ASSOC command allows the Host to write a fragment of the AMP_ASSOC structure, which contains AMP specific information for the AMP Controller.	AMP

Table 3.1: Alphabetical list of commands and events (Sheet 44 of 46)



Name	Vers.	Summary Description	Supported Controllers
Write Scan Enable command	1.1	The HCI_Write_Scan_Enable command will write the value for the Scan Enable configuration parameter, which controls whether or not the BR/EDR Controller will periodically scan for page attempts and/or inquiry requests from other BR/EDR Controllers.	BR/EDR
Write Secure Connections Host Support command	4.1	The HCI_Write_Secure_Connections_Host_Support command is used to write the Secure Connections Host Supports parameter to the Controller.	BR/EDR
Write Secure Connections Test Mode command	4.1	The HCI_Write_Secure_Connections_Test_Mode command is used to put the Controller in a test mode where DM1 packets are not allowed to be used for ACL-U traffic and/or the contents of eSCO payloads can be looped back.	BR/EDR
Write Simple Pairing Debug Mode command	2.1 + EDR	The HCI_Write_Simple_Pairing_Debug_Mode command configures the BR/EDR Controller to use a pre-defined Diffie Hellman private key for Simple Pairing to enable debug equipment to monitor the encrypted connection.	BR/EDR
Write Simple Pairing Mode command	2.1 + EDR	The HCI_Write_Simple_Pairing_Mode command writes the Simple Pairing mode setting in the BR/EDR Controller.	BR/EDR
Write Stored Link Key command	1.1	The HCI_Write_Stored_Link_Key command provides the ability to write one or more link keys to be stored in the Controller.	BR/EDR
Write Synchronization Train Parameters command	CSA4	The HCI_Write_Synchronization_Train_Parameters command configures the Synchronization Train functionality in the BR/EDR Controller.	BR/EDR

Table 3.1: Alphabetical list of commands and events (Sheet 45 of 46)



Name	Vers.	Summary Description	Supported Controllers
Write Synchronous Flow Control Enable command	1.1	The HCI_Write_Synchronous_Flow_Control_Enable command provides the ability to write the Synchronous Flow Control Enable setting. By using this setting, the Host can decide if the Controller will send HCI_Number_Of_Completed_Packets events for synchronous Connection_Handles.	BR/EDR
Write Voice Setting command	1.1	The HCI_Write_Voice_Setting command will write the values for the Voice Setting configuration parameter, which controls all the various settings for the voice connections.	BR/EDR

Table 3.1: Alphabetical list of commands and events (Sheet 46 of 46)



3.1 LE CONTROLLER REQUIREMENTS

Table 3.2 lists the commands and events that a Controller supporting LE shall implement if HCI is claimed. The Link Layer features are defined in [Vol 6] Part B, Section 4.6.

Name	LE Feature Requirements
HCI_Authenticated_Payload_Timeout_Expired event	C7
HCI_Command_Complete event	Mandatory
HCI_Command_Status event	Mandatory
HCI_Data_Buffer_Overflow event	Optional
HCI_Disconnect command	C3
HCI_Disconnection_Complete event	C3
HCI_Encryption_Change event	C4
HCI_Encryption_Key_Refresh_Complete event	C4
HCI_Hardware_Error event	Optional
HCI_Host_Buffer_Size command	Optional
HCI_Host_Number_Of_Completed_Packets command	Optional
HCI_LE_Add_Device_To_Periodic_Advertiser_List command	C21
HCI_LE_Add_Device_To_Resolving_List command	C9
HCI_LE_Add_Device_To_White_List command	Mandatory
HCI_LE_Advertising_Report event	C2
HCI_LE_Advertising_Set_Terminated event	C17
HCI_LE_Channel_Selection_Algorithm event	C23
HCI_LE_Clear_Advertising_Sets command	C17
HCI_LE_Clear_Periodic_Advertiser_List command	C21
HCI_LE_Clear_Resolving_List command	C9
HCI_LE_Clear_White_List command	Mandatory
HCI_LE_Connection_Complete event	C3
HCI_LE_Connection_CTE_Request_Enable command	C25
HCI_LE_Connection_CTE_Response_Enable command	C26
HCI_LE_Connection_IQ_Report event	C25
HCI_LE_Connection_Update command	C3
HCI_LE_Connection_Update_Complete event	C3
HCI_LE_Connectionless_IQ_Report event	C28

Table 3.2: Bluetooth Controller supporting LE requirements



Name	LE Feature Requirements
HCI_LE_Create_Connection_Cancel command	C3
HCI_LE_Create_Connection command	C3
HCI_LE_CTE_Request_Failed event	C25
HCI_LE_Data_Length_Change event	C8
HCI_LE_Directed_Advertising_Report event	C9
HCI_LE_Encrypt command	C4
HCI_LE_Enhanced_Connection_Complete event	C24
HCI_LE_Extended_Advertising_Report event	C19
HCI_LE_Extended_Create_Connection command	C20
HCI_LE_Generate_DHKey command [v1]	Optional
HCI_LE_Generate_DHKey command [v2]	Optional
HCI_LE_Generate_DHKey_Complete event	Optional
HCI_LE_Long_Term_Key_Request event	C4
HCI_LE_Long_Term_Key_Request_Negative_Reply command	C4
HCI_LE_Long_Term_Key_Request_Reply command	C4
HCI_LE_Modify_Sleep_Clock_Accuracy command	C37
HCI_LE_Periodic_Advertising_Create_Sync_Cancel command	C16
HCI_LE_Periodic_Advertising_Create_Sync command	C16
HCI_LE_Periodic_Advertising_Report event	C21
HCI_LE_Periodic_Advertising_Set_Info_Transfer command	C34
HCI_LE_Periodic_Advertising_Sync_Established event	C16
HCI_LE_Periodic_Advertising_Sync_Lost event	C21
HCI_LE_Periodic_Advertising_Sync_Transfer command	C33
HCI_LE_Periodic_Advertising_Sync_Transfer_Received event	C35
HCI_LE_Periodic_Advertising_Terminate_Sync command	C21
HCI_LE_PHY_Update_Complete event	C11
HCI_LE_Rand command	C4
HCI_LE_Read_Advertising_Physical_Channel_Tx_Power command	C1
HCI_LE_Read_Antenna_Information command	C31

Table 3.2: Bluetooth Controller supporting LE requirements



Name	LE Feature Requirements
HCI_LE_Read_Buffer_Size command	Mandatory
HCI_LE_Read_Channel_Map command	C3
HCI_LE_Read_Local_P-256_Public_Key command	Optional
HCI_LE_Read_Local_P-256_Public_Key_Complete event	Optional
HCI_LE_Read_Local_Resolvable_Address command	C10
HCI_LE_Read_Local_Supported_Features command	Mandatory
HCI_LE_Read_Maximum_Advertising_Data_Length command	C17
HCI_LE_Read_Maximum_Data_Length command	C8
HCI_LE_Read_Number_of_Supported_Advertising_Sets command	C17
HCI_LE_Read_Peer_Resolvable_Address command	C10
HCI_LE_Read_Periodic_Advertiser_List_Size command	C21
HCI_LE_Read_PHY command	C11
HCI_LE_Read_Remote_Features command	C3
HCI_LE_Read_Remote_Features_Complete event	C3
HCI_LE_Read_Resolving_List_Size command	C9
HCI_LE_Read_RF_Path_Compensation command	C22
HCI_LE_Read_Suggested_Default_Data_Length command	C8
HCI_LE_Read_Supported_States command	Mandatory
HCI_LE_Read_Transmit_Power command	C1
HCI_LE_Read_White_List_Size command	Mandatory
HCI_LE_Receiver_Test command [v1]	C2
HCI_LE_Receiver_Test command [v2]	C13
HCI_LE_Receiver_Test command [v3]	C30
HCI_LE_Remote_Connection_Parameter_Request event	C6
HCI_LE_Remote_Connection_Parameter_Request - Negative_Reply command	C6
HCI_LE_Remote_Connection_Parameter_Request_Reply command	C6
HCI_LE_Remove_Advertising_Set command	C17
HCI_LE_Remove_Device_From_Periodic_Advertiser_List command	C21
HCI_LE_Remove_Device_From_Resolving_List command	C9

Table 3.2: Bluetooth Controller supporting LE requirements



Name	LE Feature Requirements
HCI_LE_Remove_Device_From_White_List command	Mandatory
HCI_LE_Scan_Request_Received event	C17
HCI_LE_Scan_Timeout event	C19
HCI_LE_Set_Address_Resolution_Enable command	C9
HCI_LE_Set_Advertising_Data command	C1
HCI_LE_Set_Advertising_Enable command	C1
HCI_LE_Set_Advertising_Parameters command	C1
HCI_LE_Set_Advertising_Set_Random_Address command	C17
HCI_LE_Set_Connection_CTE_Receive_Parameters command	C25
HCI_LE_Set_Connection_CTE_Transmit_Parameters command	C26
HCI_LE_Set_Connectionless_CTE_Transmit_Enable command	C27
HCI_LE_Set_Connectionless_CTE_Transmit_Parameters command	C27
HCI_LE_Set_Connectionless_IQ_Sampling_Enable command	C28
HCI_LE_Set_Data_Length command	C8
HCI_LE_Set_Default_Periodic_Advertising_Sync_Transfer_Parameters command	C35
HCI_LE_Set_Default_PHY command	C11
HCI_LE_Set_Event_Mask command	Mandatory
HCI_LE_Set_Extended_Advertising_Data command	C17
HCI_LE_Set_Extended_Advertising_Enable command	C17
HCI_LE_Set_Extended_Advertising_Parameters command	C17
HCI_LE_Set_Extended_Scan_Enable command	C19
HCI_LE_Set_Extended_Scan_Parameters command	C19
HCI_LE_Set_Extended_Scan_Response_Data command	C17
HCI_LE_Set_Host_Channel_Classification command	C36
HCI_LE_Set_Periodic_Advertising_Data command	C18
HCI_LE_Set_Periodic_Advertising_Enable command	C18
HCI_LE_Set_Periodic_Advertising_Parameters command	C18

Table 3.2: Bluetooth Controller supporting LE requirements



Name	LE Feature Requirements
HCI_LE_Set_Periodic_Advertising_Receive_Enable command	C32
HCI_LE_Set_Periodic_Advertising_Sync_Transfer_Parameters command	C35
HCI_LE_Set_PHY command	C11
HCI_LE_Set_Privacy_Mode command	C9
HCI_LE_Set_Random_Address command	C1
HCI_LE_Set_Resolvable_Private_Address_Timeout command	C9
HCI_LE_Set_Scan_Enable command	C2
HCI_LE_Set_Scan_Parameters command	C2
HCI_LE_Set_Scan_Response_Data command	C15
HCI_LE_Start_Encryption command	C4
HCI_LE_Test_End command	Mandatory
HCI_LE_Transmitter_Test command [v1]	C1
HCI_LE_Transmitter_Test command [v2]	C12
HCI_LE_Transmitter_Test command [v3]	C29
HCI_LE_Write_RF_Path_Compensation command	C22
HCI_LE_Write_Suggested_Default_Data_Length command	C8
HCI_Number_Of_Completed_Packets event	C3
HCI_Read_Authenticated_Payload_Timeout command	C7
HCI_Read_BD_ADDR command	Mandatory
HCI_Read_Buffer_Size command	C5
HCI_Read_LE_Host_Support command	C5
HCI_Read_Local_Supported_Commands command	Mandatory
HCI_Read_Local_Supported_Features command	Mandatory
HCI_Read_Local_Version_Information command	Mandatory
HCI_Read_Remote_Version_Information command	C3
HCI_Read_Remote_Version_Information_Complete event	C3
HCI_Read_RSSI command	C3
HCI_Read_Transmit_Power_Level command	C3
HCI_Reset command	Mandatory
HCI_Set_Controller_To_Host_Flow_Control command	Optional

Table 3.2: Bluetooth Controller supporting LE requirements



Name	LE Feature Requirements
HCI_Set_Event_Mask command	Mandatory
HCI_Set_Event_Mask_Page_2 command	C14
HCI_Write_Authenticated_Payload_Timeout command	C7
HCI_Write_LE_Host_Support command	C5

Table 3.2: Bluetooth Controller supporting LE requirements

- C1: Mandatory if the Controller supports transmitting packets, otherwise excluded.
- C2: Mandatory if the Controller supports receiving packets, otherwise excluded.
- C3: Mandatory if the Controller supports Connection State, otherwise excluded.
- C4: Mandatory if LE Feature (LE Encryption) is supported, otherwise excluded.
- C5: Mandatory if BR/EDR is supported, otherwise optional.
- C6: Mandatory if LE Feature (Connection Parameters Request procedure) is supported, otherwise excluded.
- C7: Mandatory if LE Feature (LE Ping) is supported, otherwise excluded.
- C8: Mandatory if LE Feature (LE Data Packet Length Extension) is supported, otherwise optional.
- C9: Mandatory if LE Feature (LL Privacy) is supported, otherwise excluded.
- C10: Optional if LE Feature (LL Privacy) is supported, otherwise excluded.
- C11: Mandatory if LE Feature (LE 2M PHY) or LE Feature (LE Coded PHY) is supported, otherwise optional.
- C12: Mandatory if LE Feature (LE 2M PHY) or LE Feature (LE Coded PHY) or LE Feature (Stable Modulation Index - Transmitter) is supported, otherwise optional if the Controller supports transmitting packets, otherwise excluded.
- C13: Mandatory if LE Feature (LE 2M PHY) or LE Feature (LE Coded PHY) or LE Feature (Stable Modulation Index - Receiver) is supported, otherwise optional if the Controller supports receiving packets, otherwise excluded.
- C14: Mandatory if LE Feature (LE Ping) is supported, otherwise optional.
- C15: Mandatory if Controller supports transmitting scannable advertisements, otherwise excluded.
- C16: Mandatory if LE Feature (Periodic Advertising) is supported and the Controller supports both Scanning State and Synchronization State, otherwise excluded.
- C17: Mandatory if LE Feature (Extended Advertising) is supported and the Controller supports Advertising State, otherwise excluded.
- C18: Mandatory if LE Feature (Periodic Advertising) is supported and the Controller supports Advertising State, otherwise excluded.
- C19: Mandatory if LE Feature (Extended Advertising) is supported and the Controller supports Scanning State, otherwise excluded.
- C20: Mandatory if LE Feature (Extended Advertising) is supported and the Controller supports Initiating State, otherwise excluded.
- C21: Mandatory if LE Feature (Periodic Advertising) is supported and the Controller supports Synchronization State, otherwise excluded.
- C22: Mandatory if the Controller supports sending Transmit Power in advertisements, otherwise optional.
- C23: Mandatory if LE Feature (LE Channel Selection Algorithm #2) is supported, otherwise excluded.



- C24: Mandatory if the Controller supports Connection State and either LE Feature (LL Privacy) or LE Feature (Extended Advertising) is supported, otherwise optional if the Controller supports Connection State, otherwise excluded.
- C25: Mandatory if LE Feature (Connection CTE Request) is supported, otherwise excluded.
- C26: Mandatory if LE Feature (Connection CTE Response) is supported, otherwise excluded.
- C27: Mandatory if LE Feature (Connectionless CTE Transmitter) is supported, otherwise excluded.
- C28: Mandatory if LE Feature (Connectionless CTE Receiver) is supported, otherwise excluded.
- C29: Mandatory if LE Feature (Connection CTE Response) or LE Feature (Connectionless CTE Transmitter) is supported, otherwise optional if the Controller supports transmitting packets, otherwise excluded.
- C30: Mandatory if LE Feature (Connection CTE Request) or LE Feature (Connectionless CTE Receiver) is supported, otherwise optional if the Controller supports receiving packets, otherwise excluded.
- C31: Mandatory if LE Feature (Connection CTE Request) or LE Feature (Connection CTE Response) or LE Feature (Connectionless CTE Transmitter) or LE Feature (Connectionless CTE Receiver) is supported, otherwise excluded.
- C32: Mandatory if LE Feature (Periodic Advertising Sync Transfer – Recipient) is supported, otherwise optional if LE Feature (Periodic Advertising) is supported and the Controller supports Synchronization State, otherwise excluded.
- C33: Mandatory if LE Feature (Periodic Advertising Sync Transfer – Sender) is supported and the Controller supports Scanning State, otherwise excluded.
- C34: Mandatory if LE Feature (Periodic Advertising Sync Transfer – Sender) is supported and the Controller supports Advertising State, otherwise excluded.
- C35: Mandatory if LE Feature (Periodic Advertising Sync Transfer – Recipient) is supported, otherwise excluded.
- C36: Mandatory if the Controller supports Connection State, otherwise optional if LE Feature (Extended Advertising) is supported and the Controller supports Advertising State, otherwise excluded.
- C37: Mandatory if the Controller can change its sleep clock accuracy, otherwise excluded.



3.1.1 Legacy and extended advertising

Table 3.3 lists the legacy and extended advertising commands and events.

If a Controller supports any legacy advertising command or event listed in the table and also supports the LE Feature (Extended Advertising), it shall support the corresponding extended advertising command or event in the same row of the table.

If, since the last power-on or reset, the Host has ever issued a legacy advertising command and then issues an extended advertising command, or has ever issued an extended advertising command and then issues a legacy advertising command, the Controller shall return the error code *Command Disallowed* (0x0C).

A Host should not issue legacy commands to a Controller that supports the LE Feature (Extended Advertising).

Legacy advertising command or event	Extended advertising command or event
HCI_LE_Advertising_Report event	HCI_LE_Extended_Advertising_Report event
HCI_LE_Directed_Advertising_Report event	HCI_LE_Extended_Advertising_Report event
HCI_LE_Set_Advertising_Parameters command	HCI_LE_Set_Extended_Advertising_Parameters command
HCI_LE_Read_Advertising_Physical_Channel_Tx_Power command	<i>none</i>
HCI_LE_Set_Advertising_Data command	HCI_LE_Set_Extended_Advertising_Data command
HCI_LE_Set_Scan_Response_Data command	HCI_LE_Set_Extended_Scan_Response_Data command
HCI_LE_Set_Advertising_Enable command	HCI_LE_Set_Extended_Advertising_Enable command
<i>none</i>	HCI_LE_Read_Maximum_Advertising_Data_Length command
<i>none</i>	HCI_LE_Read_Number_of_Supported_Advertising_Sets command
<i>none</i>	HCI_LE_Remove_Advertising_Set command
<i>none</i>	HCI_LE_Clear_Advertising_Sets command
<i>none</i>	HCI_LE_Set_Periodic_Advertising_Parameters command
<i>none</i>	HCI_LE_Set_Periodic_Advertising_Data command

Table 3.3: Legacy and extended advertising commands and events



Legacy advertising command or event	Extended advertising command or event
<i>none</i>	HCI_LE_Set_Periodic_Advertising_Enable command
HCI_LE_Set_Scan_Parameters command	HCI_LE_Set_Extended_Scan_Parameters command
HCI_LE_Set_Scan_Enable command	HCI_LE_Set_Extended_Scan_Enable command
HCI_LE_Create_Connection command	HCI_LE_Extended_Create_Connection command
<i>none</i>	HCI_LE_Periodic_Advertising_Create_Sync command
<i>none</i>	HCI_LE_Periodic_Advertising_Create_Sync - Cancel command
<i>none</i>	HCI_LE_Periodic_Advertising_Terminate_Sync command
<i>none</i>	HCI_LE_Add_Device_To_Periodic_Advertiser_List command
<i>none</i>	HCI_LE_Remove_Device_From_Periodic_Advertiser_List command
<i>none</i>	HCI_LE_Clear_Periodic_Advertiser_List command
<i>none</i>	HCI_LE_Read_Periodic_Advertiser_List_Size command
<i>none</i>	HCI_LE_Set_Periodic_Advertising_Sync_Transfer_Parameters command
<i>none</i>	HCI_LE_Set_Default_Periodic_Advertising_Sync_Transfer_Parameters command
<i>none</i>	HCI_LE_Periodic_Advertising_Sync_Transfer_Received event

Table 3.3: Legacy and extended advertising commands and events



4 HCI FLOW CONTROL

Flow control for data shall be used in the direction from the Host to the Controller to avoid overflowing the Controller data buffers with ACL data destined for a remote device (using a `Connection_Handle`) that is not responding. The Host manages the data buffers of the Controller. For Primary Controller, packet based flow control is the default. For AMP Controllers, data block based data flow control is the default for ACL traffic. Flow control for data moving from the Controller to the Host may be used in the Primary Controller in accordance with [Section 4.2](#).

Command flow control is covered in [Section 4.4](#) and [Section 4.5](#).

4.1 HOST TO CONTROLLER DATA FLOW CONTROL

Two methods of data flow control are defined: 'packet-based' flow control and 'data-block-based' flow control, known as buffer management. Selection of the data flow control mechanism is performed with the `HCI_Write_Flow_Control_Mode` command (see [Section 7.3.73](#)).

If a BR/EDR/LE Controller implements separate buffers for ACL Data:

1. The Host shall use the `HCI_LE_Read_Buffer_Size` command to determine the buffers that are used for ACL Data on an LE-U logical link.
2. The Host shall use separate packet based flow control for each set of buffers.
3. The `Connection_Handle` contained in the ACL Data packet shall be used by the Controller to determine which set of buffers to use and the logical link (ACL-U, ASB-U, or LE-U) over which the data is to be sent.

If a BR/EDR/LE Controller does not implement separate buffers, then all ACL Data shall use the BR/EDR buffer management as described below, and only the logical link (ACL-U, ASB-U, or LE-U) shall be determined by the `Connection_Handle`.

4.1.1 Packet-based data flow control

When the packet based flow control mechanism is enabled, on initialization, a Host that supports LE shall issue the `HCI_LE_Read_Buffer_Size` command. Two of the return parameters of this command determine the maximum size of HCI ACL (excluding header) Data packets that can be used to transmit ACL data for an LE transport sent from the Host to the Controller. There is an additional return parameter that specifies the total number of HCI ACL Data packets that the Controller may have waiting for transmission in those buffers. A Primary Controller that supports BR/EDR and LE may return zero for the total number of HCI ACL packets used to transmit ACL data for an LE transport. In this case the Host shall then send all BR/EDR and LE data using the HCI ACL



Data packets into the buffers identified using the `HCI_Read_Buffer_Size` command.

In a BR/EDR Controller, when the packet based flow control mechanism is enabled, on initialization, the Host shall issue the `HCI_Read_Buffer_Size` command. Two of the return parameters of this command determine the maximum size of HCI ACL and synchronous Data packets (excluding header) sent from the Host to the Controller. There are also two additional return parameters that specify the total number of HCI ACL and synchronous Data packets that the Controller may have waiting for transmission in its buffers.

When there is at least one connection to another device, or when in local loopback mode on a BR/EDR Controller, the Controller shall use the `HCI_Number_Of_Completed_Packets` event to control the flow of data from the Host. This event contains a list of `Connection_Handles` and a corresponding number of HCI Data packets that have been completed (transmitted, flushed, or looped back to the Host) since the previous time the event was returned (or since the connection was established, if the event has not been returned before for a particular `Connection_Handle`).

The Host chooses the `Connection_Handles` for the following HCI Data packets based on the information returned in this event, and/or the `HCI_LE_Read_Buffer_Size` commands.

Every time it has sent an HCI Data packet, the Host shall assume that the free buffer space for the corresponding link type (ACL, SCO, or eSCO) in the Controller has decreased by one HCI Data packet.

Each `HCI_Number_Of_Completed_Packets` event received by the Host provides information about how many HCI Data packets have been completed (transmitted or flushed) for each `Connection_Handle` since the previous `HCI_Number_Of_Completed_Packets` event was sent to the Host. It can then calculate the actual current buffer usage.

When the Controller has completed one or more HCI Data packet(s) it shall send an `HCI_Number_Of_Completed_Packets` event to the Host, until it finally reports that all the pending HCI Data packets have been completed. The frequency at which this event is sent is manufacturer specific.

Note: The `HCI_Number_Of_Completed_Packets` events will not report on synchronous `Connection_Handles` if Synchronous Flow Control is disabled. (See [Read Synchronous Flow Control Enable command](#), and [Write Synchronous Flow Control Enable command](#).)

For each individual `Handle`, the data shall be sent to the Controller in HCI Data packets in the order in which it was created in the Host. The Controller shall also transmit data on the air that is received from the Host for a given `Handle` in the same order as it is received from the Host.



Data that is received on the air from another device shall, for the corresponding Connection_Handle, be sent in HCI Data packets to the Host in the same order as it is received. This means the scheduling shall be decided separately for each Handle basis. For each individual Handle, the order of the data shall not be changed from the order in which the data has been created.

HCI shall not fragment an L2CAP message on an LE-U logical link that is 27 octets or less in length.

4.1.2 Data-block-based data flow control

When the data-block-based flow control mechanism is enabled, on initialization the Host shall issue the [Read Data Block Size command](#). Two of the return parameters of this command determine the maximum size of HCI ACL Data packets (excluding header) sent from the Host to the Controller. A further return parameter specifies the total number of HCI ACL Data packets that the Controller may have waiting for transmission in its buffers.

When there is at least one Logical Link to another AMP the Controller shall use the HCI_Number_Of_Completed_Data_Blocks event to control the flow of data from the Host. This event contains a list of Handles and a corresponding number of HCI Data packets that have been completed (transmitted or flushed) since the previous time the event was returned (or since the link was established, if the event has not been returned before for a particular Handle).

Based on the information returned in this event, and the return parameters of the HCI_Read_Data_Block_Size command that specify the total number of HCI ACL Data packets that can be stored in the Controller, the Host decides for which Handles the following HCI Data packets should be sent.

Every time it has sent an HCI Data packet, the Host shall assume that the free buffer space for the corresponding ACL link type in the Controller has decreased by one HCI Data packet.

Each HCI_Number_Of_Completed_Data_Blocks event received by the Host provides information about how many HCI Data packets have been completed (transmitted or flushed) for each Handle since the previous HCI_Number_Of_Completed_Data_Blocks event was sent to the Host. It can then calculate the actual current buffer usage.

When the Controller has completed one or more HCI Data packet(s) it shall send an HCI_Number_Of_Completed_Data_Blocks event to the Host until it finally reports that all the pending HCI Data packets have been completed. The frequency at which this event is sent is manufacturer specific.

For each individual Handle, the data shall be sent to the Controller in HCI Data packets in the order in which it was created in the Host. The Controller shall also transmit data for a given Handle in the same order as it is received from the Host.



Data that is received shall be sent in HCI Data packets to the Host in the same order as it is received. This means the scheduling shall be decided separately for each Handle basis.

4.2 CONTROLLER TO HOST DATA FLOW CONTROL

In some implementations, flow control may also be necessary in the direction from the Controller to the Host. The `HCI_Set_Host_Controller_To_Host_Flow_Control` command can be used to turn flow control on or off in that direction. For AMP Controllers, Controller to Host data flow control is provided by the HCI physical transport or software equivalent.

On initialization, the Host uses the `HCI_Host_Buffer_Size` command to notify the Controller about the maximum size of HCI ACL and Synchronous Data packets sent from the Controller to the Host. The command also contains two additional command parameters to notify the Primary Controller about the total number of ACL and Synchronous Data packets that can be stored in the data buffers of the Host.

The Host uses the `HCI_Host_Number_Of_Completed_Packets` command in exactly the same way as the Primary Controller uses the `HCI_Number_Of_Completed_Packets` event as was previously described in this section.

The `HCI_Host_Number_Of_Completed_Packets` command is a special command for which no command flow control is used, and which can be sent anytime there is a connection or when in local loopback mode. The command also has no event after the command has completed. This makes it possible for the flow control to work in exactly the same way in both directions, and the flow of normal commands will not be disturbed.

4.3 DISCONNECTION BEHAVIOR

When the Host receives an `HCI_Disconnection_Complete`, `HCI_Disconnection_Physical_Link_Complete` or `HCI_Disconnection_Logical_Link_Complete` event, the Host shall assume that all unacknowledged HCI Data packets that have been sent to the Controller for the returned Handle have been flushed, and that the corresponding data buffers have been freed. A Primary Controller does not have to notify the Host about this in an `HCI_Number_Of_Completed_Packets` event, nor does the AMP Controller have to notify the Host about this in an `HCI_Number_Of_Completed_Data_Blocks` event, before the disconnection event (and shall not do so afterwards because the `Connection_Handle` is no longer valid).

If flow control is also enabled in the direction from the Controller to the Host, the Controller may, after it has sent an `HCI_Disconnection_Complete` event,



assume that the Host will flush its data buffers for the sent Handle when it receives the HCI_Disconnection_Complete event. The Host does not have to notify the Primary Controller about this in an HCI_Host_Number_Of_Completed_Packets command.

4.4 COMMAND FLOW CONTROL

On initial power-on, and after a reset, the Host shall send a maximum of one outstanding HCI Command packet until an HCI_Command_Complete or HCI_Command_Status event has been received.

The HCI_Command_Complete and HCI_Command_Status events contain a parameter called Num HCI Command Packets, which indicates the number of HCI Command packets the Host is currently allowed to send to the Controller. The Controller may buffer one or more HCI Command packets, but the Controller shall start performing the commands in the order in which they are received. The Controller can start performing a command before it completes previous commands. Therefore, the commands do not always complete in the order they are started.

To indicate to the Host that the Controller is ready to receive HCI command packets, the Controller may generate an HCI_Command_Complete or HCI_Command_Status event with the Command Opcode set to 0x0000 and the HCI_Num_HCI_Command_Packets event parameter set to 1 or more. Command Opcode 0x0000 is a special value indicating that this event is not associated with a command sent by the Host. The Controller can send an HCI_Command_Complete or HCI_Command_Status event with Command Opcode 0x0000 at any time to change the number of outstanding HCI Command packets that the Host can send before waiting. If the Controller generates an HCI_Command_Complete or HCI_Command_Status event with the Num HCI Command Packets event parameter set to zero, the Host shall stop sending commands.

For most commands, an HCI_Command_Complete event shall be sent to the Host when the Controller completes the command. Some commands are executed in the background and do not return an HCI_Command_Complete event when they have been completed. Instead, the Controller shall send an HCI_Command_Status event back to the Host when it has begun to execute the command. When the actions associated with the command have finished, an event that is associated with the command shall be sent by the Controller to the Host.

If the command does not begin to execute (for example, if there was a parameter error or the command is currently not allowed), the HCI_Command_Status event shall be returned with the appropriate error code in the Status parameter, and the event associated with the sent command shall not be returned.



When a connection is disconnected and there are outstanding commands relating to that connection, the Controller may return the event associated with each command to the Host. Each such event shall have a non-zero status and shall precede the HCI_Disconnection_Complete event. No events for that Connection_Handle shall be sent after the HCI_Disconnection_Complete event.

4.5 COMMAND ERROR HANDLING

Unless explicitly stated otherwise in the description of a command, any error in any parameter means that the command will not begin to execute; it will only return an error code. If more than one error code is applicable, the implementation shall choose one of them.

If an error occurs for a command for which an HCI_Command_Complete event is returned, the Return Parameters field may only contain some of the return parameters specified for the command. The Status parameter, which explains the error reason and which is the first return parameter, shall always be returned. If there is a Handle parameter or a BD_ADDR parameter right after the Status parameter, this parameter shall also be returned so that the Host can identify to which instance of a command the HCI_Command_Complete event belongs. In this case, the Handle or BD_ADDR parameter shall have exactly the same value as that in the corresponding command parameter. It is implementation specific whether more parameters will be returned in case of an error.

The above also applies to commands that have associated command specific completion events with a status parameter in their completion event, with the exceptions shown in [Table 4.1](#), which indicates the only parameters (other than Status) that are valid. The validity of other parameters is likewise implementation specific for failed commands in this group, but they shall be sent in any case.

Event	Valid parameters
Connection Complete	BD_ADDR
Synchronous Connection Complete	BD_ADDR
LE Connection Complete	<i>none</i>
LE Enhanced Connection Complete	<i>none</i>
Logical Link Complete	Physical_Link_Handle
LE Periodic Advertising Sync Established	<i>none</i>

Table 4.1: Valid parameters for command completion events reporting an error

For the purposes of this section, the Subevent_Code parameter of the HCI_LE_Meta event (see [\[Vol 2\] Part E, Section 7.7.65](#)) is not treated as a parameter and is always valid.



Note: The BD_ADDR return parameter of the command HCI_Read_BD_ADDR is not used to identify to which instance of the HCI_Read_BD_ADDR command the HCI_Command_Complete event belongs. It is optional for the Controller to return this parameter in case of an error.

4.6 LMP TRANSACTION AND LL PROCEDURE COLLISIONS

If the Host issues a command that returns a successful HCI_Command_Status event but then triggers an LMP transaction or Link Layer procedure that terminates with the error code *LMP Error Transaction Collision / LL Procedure Collision* (0x23) because the peer has also initiated the same transaction or procedure, then the Controller shall take one of the following actions:

- The Controller shall wait until the peer-initiated transaction or procedure has completed and use the result to generate the command-specific completion event. In this case, the collision is not reported to the Host.
- The Controller shall generate the command-specific completion event with the error code *LMP Error Transaction Collision / LL Procedure Collision* (0x23).



5 HCI DATA FORMATS

5.1 INTRODUCTION

The HCI provides a uniform command method of accessing Controller capabilities. The HCI Link commands provide the Host with the ability to control connections to other BR/EDR Controllers. For the BR/EDR Controller, these commands typically involve the Link Manager (LM) to exchange LMP commands or the Link Layer (LL) to exchange LL Control packets with remote Bluetooth devices. For details see [\[Vol 2\] Part C, Link Manager Protocol Specification](#) and [\[Vol 6\] Part B, Link Layer Specification](#). For AMP Controllers, these commands typically involve the AMP PAL. For details see the appropriate PAL specification (see [\[Vol 5\] Part A, 802.11 Protocol Adaptation Layer Functional Specification](#)).

The HCI Policy commands are used to affect the behavior of the local and remote LM or LL. These Policy commands provide the Host with methods of influencing how the LM or LL manages the piconet. The Controller & Baseband, Informational, and Status commands provide the Host access to various registers in the Controller.

HCI commands may take different amounts of time to be completed. Therefore, the results of commands will be reported back to the Host in the form of an event. For example, for most HCI commands the Controller will generate the HCI_Command_Complete event when a command is completed. This event contains the return parameters for the completed HCI command. For enabling the Host to detect errors on the HCI-Transport Layer, there needs to be a timeout between the transmission of the Host's command and the reception of the Controller's response (e.g. an HCI_Command_Complete or HCI_Command_Status event). Since the maximum response timeout is strongly dependent on the HCI-Transport Layer used, it is recommended to use a default value of one second for this timer. This amount of time is also dependent on the number of commands unprocessed in the command queue.

5.2 DATA AND PARAMETER FORMATS

- All values are in Binary and Hexadecimal little-endian formats unless otherwise noted.
- In addition, all parameters which can have negative values shall use 2's complement when specifying values.
- Unless noted otherwise, the order of parameters in an HCI Command packet or HCI Event packet is the order the parameters are listed in the command or event.
- Arrayed parameters are specified using the following notation: ParameterA[i]. If more than one set of arrayed parameters are specified (e.g. ParameterA[i], ParameterB[i]), then, unless noted otherwise, the order



of the parameters are as follows: ParameterA[0], ParameterB[0], ParameterA[1], ParameterB[1], ParameterA[2], ParameterB[2], ... ParameterA[n], ParameterB[n].

- Unless noted otherwise, all parameter values are sent and received in little-endian format (i.e. for multi-octet parameters the rightmost (Least Significant Octet) is transmitted first).
- Most command and event parameters that are not-arrayed and all elements in an arrayed parameter have fixed sizes (an integer number of octets). Where a parameter, or an element of an arrayed parameter, has a variable length, this will be noted in the specific command; the length will then be specified in another parameter. The parameters and the size of each not-arrayed parameter (or of each element in an arrayed parameter) contained in a command or an event is specified for each command or event. The number of elements in an arrayed parameter is not fixed.
- Where bit strings are specified, the low order bit is the right hand bit, e.g. 0 is the low order bit in 10b.
- Where a parameter value is described as "0xXX" or "N = 0xXX" (with an appropriate number of "X"s), then the description applies to all possible values other than any for which a separate description is given. Where the description specifies a range, values outside that range (if any) are reserved for future use.
- Parameter values or opcodes that an implementation does not know how to interpret shall be ignored and the operation that is being attempted shall be completed with the correct signaling. The Host or Controller shall not stop functioning because of receiving a reserved value.
- Unless noted otherwise, such as in a "Mandatory Range" statement or because a value relates to an optional feature that is not supported, the Controller shall support all valid values of all parameters (values reserved for future use are not valid).

5.3 HANDLES

Three types of handles are used to identify logical channels between the Host and a Controller: Connection_Handles, Logical Link Handles, and Physical Link Handles.

5.3.1 Primary Controller handles

Connection_Handles are used to identify logical channels between the Host and the Primary Controller. Connection_Handles are assigned by the Primary Controller when a new logical link is created, using the Connection Complete, Synchronous Connection Complete, LE Connection Complete, or LE Enhanced Connection Complete events. Broadcast Connection_Handles are handled differently, and are described in [Section 5.3.1.1](#).



Note: There are no broadcast or synchronous Connection_Handles in a Controller that only supports LE.

5.3.1.1 Broadcast Connection_Handles

The first time the Host sends an HCI Data packet with Broadcast_Flag set to 01b (active slave broadcast) after a power-on or a reset, the value of the Connection_Handle parameter shall be a value which is not currently assigned by the Controller.

The BR/EDR Controller shall then continue to use the same Connection_Handle for broadcast until a reset is made. The Controller shall not send a Connection Complete event containing a new Connection_Handle that it knows is used for broadcast.

In some situations, it may happen that the Controller sends a Connection Complete event before having interpreted a Broadcast packet received from the Host, and that the Connection_Handles of both Connection Complete event and HCI Data packet are the same. This conflict is avoided as follows:

If a Connection Complete event is received containing the Connection_Handle used for broadcast, the Host shall wait before sending any packets for the new connection until it receives a Number Of Completed Packets event indicating that there are no pending broadcast packets belonging to the Connection_Handle. The Host shall also change the Connection_Handle used for broadcast to a Connection_Handle which is currently not assigned by the Controller. This Connection_Handle shall then be used for all following broadcasts until a reset is performed or the same conflict situation happens again. However, this will occur very rarely.

By following this procedure, the Controller can distinguish between the Broadcast message sent by the Host and the new connection made (this could be even a new synchronous link) even though the Connection_Handles are the same.

For an HCI Data packet sent from the Controller to the Host where the Broadcast_Flag is 01, the Connection_Handle parameter should contain the Connection_Handle for the ACL connection to the master that sent the broadcast.

For Connectionless Slave Broadcast, no Connection_Handle is assigned.

5.3.2 AMP Controller handles

AMP Controllers have two types of handles: Physical Link Handles and Logical Link Handles. For data, command and event operations between the Host and an AMP Controller, a Logical Link Handle is used where a Connection_Handle is specified, unless a Physical Link Handle is explicitly specified.



A Physical Link is an association between the local Controller and a remote Controller. For any AMP Controller, there can be at most one Physical Link. The Host is responsible for allocating Physical Link Handles. A value of 0 shall not be used for a Physical Link Handle. Physical Links are created and removed explicitly, and encryption keys are established on a per Physical Link basis. The AMP may support one or more Physical Links (representing different remote devices) at any one time.

For each Physical Link, an AMP Controller provides zero or more Logical Links.

A Logical Link is an allocation of bandwidth or other resources for application level data transfer between the local AMP Controller and a remote AMP Controller. A Logical Link exists with respect to a specific Physical Link and the Controller ensures this Handle's uniqueness. The AMP shall support one or more Logical Links (representing distinct resource allocations) over any one Physical Link.

5.4 EXCHANGE OF HCI-SPECIFIC INFORMATION

The Host Controller Transport Layer provides transparent exchange of HCI specific information. These transporting mechanisms provide the ability for the Host to send HCI commands, ACL data and synchronous data to the BR/EDR Controller, and HCI commands and ACL data to the LE Controller or AMP Controller. These transport mechanisms also provide the ability for the Host to receive HCI events, ACL data and synchronous data from the BR/EDR Controller, and HCI events and ACL data from the LE Controller or AMP Controller. Since the Host Controller Transport Layer provides transparent exchange of HCI-specific information, the HCI specification specifies the format of the commands, events, and data exchange between the Host and the Controller(s). The next sections specify the HCI packet formats.

5.4.1 HCI Command packet

The HCI Command packet is used to send commands to the Controller from the Host. The format of the HCI Command packet is shown in [Figure 5.1](#), and the definition of each field is explained below.

Controllers shall be able to accept HCI Command packets with up to 255 bytes of data excluding the HCI Command packet header. The HCI Command packet header is the first 3 octets of the packet.

Each command is assigned a 2 byte Opcode used to uniquely identify different types of commands. The Opcode parameter is divided into two fields, called the OpCode Group Field (OGF) and OpCode Command Field (OCF). The OGF occupies the upper 6 bits of the Opcode, while the OCF occupies the remaining 10 bits. The OGF of 0x3F is reserved for vendor-specific debug commands. The organization of the opcodes allows additional information to be inferred without fully decoding the entire Opcode.



Note: The OGF composed of all ‘ones’ has been reserved for vendor-specific debug commands. These commands are vendor-specific and are used during manufacturing, for a possible method for updating firmware, and for debugging.

On receipt of a Vendor Specific Debug command the Controller should respond with either:

1. An HCI_Command_Status event. If the status indicates success ([Section 7.7.15](#)) then this event shall be followed by an HCI event with Event Code field of 0xFF ([Section 5.4.4](#)).
2. An HCI_Command_Complete event specifying the corresponding Vendor Specific Debug command opcode.

The Host shall assume that sending of an HCI_Vendor_Specific_Debug command will consume an HCI command credit.

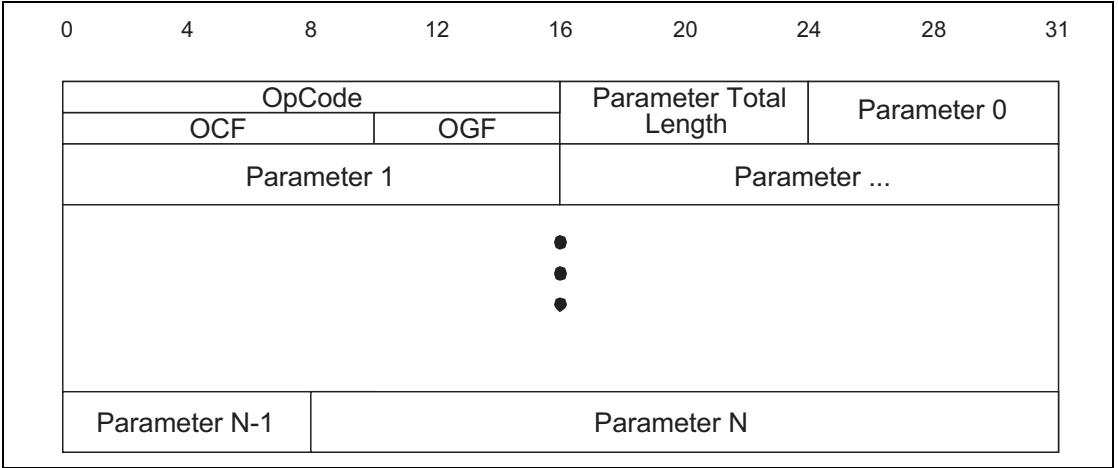


Figure 5.1: HCI Command packet

Op_Code: Size: 2 octets

Value	Parameter Description
0xFFFF	OGF Range (6 bits): 0x00 to 0x3F (0x3F reserved for vendor-specific debug commands) OCF Range (10 bits): 0x0000 to 0x3FFF

Parameter_Total_Length: Size: 1 octet

Value	Parameter Description
0xFF	Lengths of all of the parameters contained in this packet measured in octets. (N.B.: total length of parameters, not number of parameters)



Parameter 0 - N:

Size: Parameter Total Length

Value	Parameter Description
0xXX	Each command has a specific number of parameters associated with it. These parameters and the size of each of the parameters are defined for each command. Each parameter is an integer number of octets in size.

5.4.2 HCI ACL Data packets

HCI ACL Data packets are used to exchange data between the Host and Controller. There are two types of HCI ACL Data packets:

- Automatically-Flushable
- Non-Automatically-Flushable

Automatically-Flushable HCI Data packets are flushed based on the setting of an automatic flush timer (see [Read Automatic Flush Timeout command](#)). Non-Automatically-Flushable HCI Data packets are not controlled by the automatic flush timeout and shall not be automatically flushed. The format of the HCI ACL Data packet is shown in [Figure 5.2](#). The definition for each of the fields in the data packets is explained below.

Hosts and Controllers shall be able to accept HCI ACL Data packets with up to 27 bytes of data excluding the HCI ACL Data packet header on Connection_Handles associated with an LE-U logical link. The HCI ACL Data packet header is the first 4 octets of the packet.

Note: HCI ACL Data packets with a Connection_Handle associated with an LE-U logical link will not be affected by the automatic flush timer because only non-flushable packet boundary flags are allowed.

All packets on an AMP logical link are affected by the automatic flush timer.

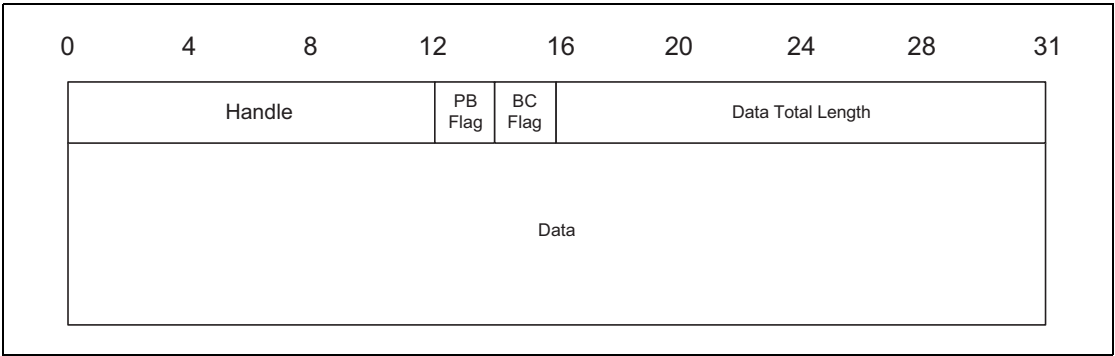


Figure 5.2: HCI ACL Data packet

**Handle:****Size: 12 Bits**

Value	Parameter Description
0xXXX	Connection_Handle to be used for transmitting a data packet or segment over a Primary Controller. Logical_Link_Handle to be used for transmitting a data packet over an AMP Controller. On the receiving side (AMP Controller to Host), the least significant 8 bits of the Handle are the Physical_Link_Handle and the most significant 4 bits are reserved for future use. (This is because certain AMP Controllers might not be able to provide the logical link context on the receiving side, and the Host does not require it.) Range: 0x000 to 0xEFF (all other values reserved for future use)

The Flag Bits consist of the Packet_Boundary_Flag and Broadcast_Flag. The Packet_Boundary_Flag is located in bit 4 and bit 5, and the Broadcast_Flag is located in bit 6 and bit 7 in the second octet of the HCI ACL Data packet.

Packet_Boundary_Flag:**Size: 2 Bits**

Value	Parameter Description		ASB-U	ACL-U	AMP-U	LE-U
00b	First non-automatically-flushable packet of a higher layer message (start of a non-automatically-flushable L2CAP PDU) from Host to Controller.	Host to Controller	Not allowed	Allowed	Not allowed	Allowed
		Controller to Host	Not allowed	Not allowed (except during loopback)	Not allowed	Not allowed
01b	Continuing fragment of a higher layer message	Host to Controller	Allowed	Allowed	Not allowed	Allowed
		Controller to Host	Allowed	Allowed	Not allowed	Allowed
10b	First automatically flushable packet of a higher layer message (start of an automatically-flushable L2CAP PDU).	Host to Controller	Allowed	Allowed	Not allowed	Not Allowed
		Controller to Host	Allowed	Allowed	Not allowed	Allowed
11b	A complete L2CAP PDU. Automatically flushable.	Host to Controller	Not allowed	Not Allowed	Allowed	Not Allowed
		Controller to Host	Not allowed	Not Allowed	Allowed	Not Allowed



The start of a non-flushable packet of a higher layer message (start of a non-automatically-flushable L2CAP PDU) with the PBF of 00b shall be transmitted with an LLID of 10b. All continuing fragment packets of a higher layer message shall be transmitted with an LLID of 01b.

Broadcast_Flag (in packet from Host to Controller):

Size: 2 Bits

Value	Parameter Description
00b	No broadcast. Only point-to-point (this is the only valid option for AMPs).
01b	Active Slave Broadcast: packet is sent to all active slaves and may be received by slaves in Sniff mode.
10b	Reserved for future use.
11b	Reserved for future use.

Broadcast_Flag (in packet from Controller to Host):

Size: 2 Bits

Value	Parameter Description
00b	Point-to-point
01b	Active Slave Broadcast
10b	Reserved for future use.
11b	Reserved for future use.

Note: Slaves in Sniff mode will only receive a broadcast packet if it happens to be sent in a sniff slot when the slave is listening.

Data_Total_Length:

Size: 2 octets

Value	Parameter Description
0xFFFF	Length of data measured in octets.



5.4.3 HCI Synchronous Data packets

HCI synchronous (SCO and eSCO) Data packets are used to exchange synchronous data between the Host and Controller. The format of the synchronous Data packet is shown in Figure 5.3. The definition for each of the fields in the data packets is explained below. The HCI SCO Data packet header is the first 3 octets of the packet.

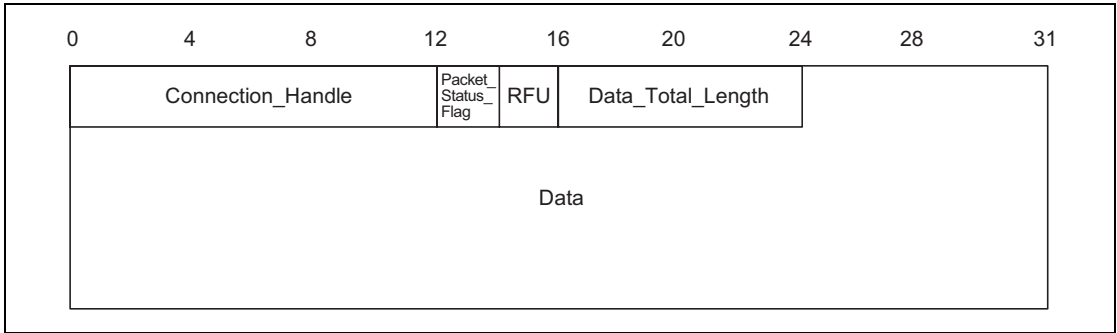


Figure 5.3: HCI Synchronous Data packet

Connection_Handle: Size: 12 Bits

Value	Parameter Description
0xXXX	Connection_Handle to be used to for transmitting a synchronous data packet or segment. Range: 0x0000 to 0x0EFF

The Packet_Status_Flag bits consist of two bits, which are located from bit 4 to 5 in the second octet of the HCI Synchronous Data packet.

The Host shall set the Packet_Status_Flag bits to 00b.

If the Erroneous_Data_Reporting parameter was set to disabled when the synchronous connection was created, the Controller shall set the Packet_Status_Flag bits to 00b and whether or not data is provided for cases when a valid (e)SCO packet was not received is unspecified.

If the Erroneous_Data_Reporting parameter was set to enabled when the synchronous connection was created, the Controller shall set the Packet_Status_Flag according to the following table.

**Packet_Status_Flag** (in packets sent by the Controller)

Size: 2 Bits

Value	Parameter Description
00b	Correctly received data. The payload data belongs to received eSCO or SCO packets that the baseband marked as "good data".
01b	Possibly invalid data. At least one eSCO packet has been marked by the baseband as "data with possible errors" and all others have been marked as "good data" in the eSCO interval(s) corresponding to the HCI Synchronous Data packet.
10b	No data received. All data from the baseband received during the (e)SCO interval(s) corresponding to the HCI Synchronous Data packet have been marked as "lost data" by the baseband. The Payload data octets shall be set to 0.
11b	Data partially lost. Not all, but at least one (e)SCO packet has been marked as "lost data" by the baseband in the (e)SCO intervals corresponding to the HCI Synchronous Data packet. The payload data octets corresponding to the missing (e)SCO packets shall be set to 0.

Note: Some HCI transports and/or Controller implementations will align the HCI Synchronous Data packets with the (e)SCO Baseband packets such that data integrity can be explicitly marked in the Packet_Status_Flag. For HCI transports or Controller implementations that do not preserve this alignment, information in the Packet_Status_Flag may be ambiguous.

Data_Total_Length:

Size: 1 octet

Value	Parameter Description
0xXX	Length of synchronous data measured in octets



5.4.4 HCI Event packet

The HCI Event packet is used by the Controller to notify the Host when events occur. If the Controller sends an HCI Event Packet containing an Event Code or an LE subevent code that the Host has not masked out and does not support, the Host shall ignore that packet. The Host shall be able to accept HCI Event packets with up to 255 octets of data excluding the HCI Event packet header. The format of the HCI Event packet is shown in [Figure 5.4](#), and the definition of each field is explained below. The HCI Event packet header is the first 2 octets of the packet.

Note: An LE Controller uses a single Event Code (see [Section 7.7.65](#)) to transmit all LE specific events from the Controller to the Host. The first Event Parameter is always a subevent code identifying the specific event.

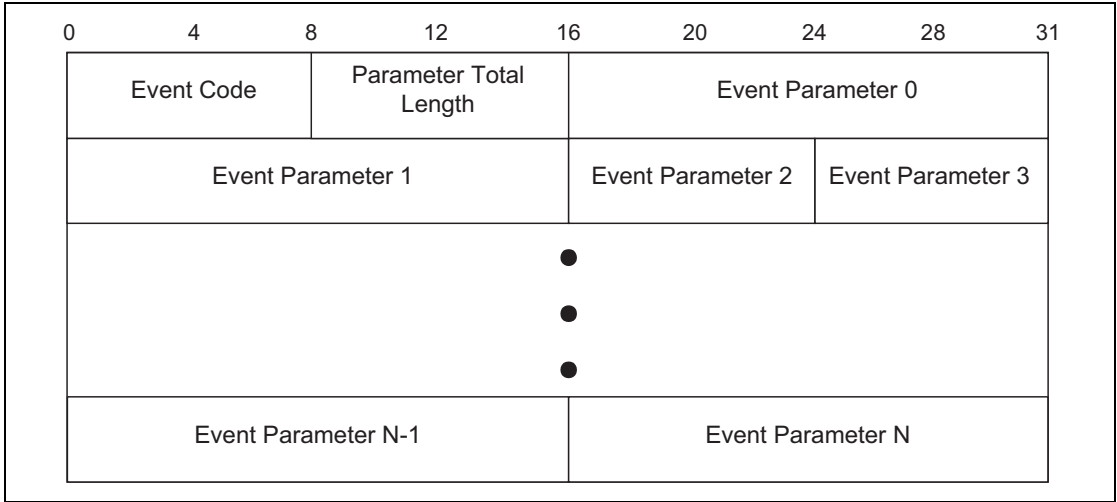


Figure 5.4: HCI Event packet

Event_Code: **Size: 1 octet**

Value	Parameter Description
0xXX	Each event is assigned a 1-Octet event code used to uniquely identify different types of events. Range: 0x00 to 0xFF (The event code 0xFF is reserved for the event code used for vendor-specific debug events.)

Parameter_Total_Length: **Size: 1 octet**

Value	Parameter Description
0xXX	Length of all of the parameters contained in this packet, measured in octets

*Event_Parameter 0 - N:**Size: Parameter Total Length*

Value	Parameter Description
0xXX	Each event has a specific number of parameters associated with it. These parameters and the size of each of the parameters are defined for each event. Each parameter is an integer number of octets in size.

5.5 IGNORED PARAMETERS

When a parameter of a command or event is described as “ignored” (this is usually only in a specific circumstance), then the value of this parameter shall not affect the behavior of the Controller or Host (as appropriate) in any way. In particular, the Controller shall not report an error because the value is out of range, reserved for future use, or represents an inappropriate entity (e.g. a `Connection_Handle` may be unassigned or for the wrong logical channel), or because two related values have the wrong relationship (e.g. minimum > maximum).



6 HCI CONFIGURATION PARAMETERS

6.1 SCAN ENABLE

The Scan_Enable parameter controls whether or not the BR/EDR Controller will periodically scan for page attempts and/or inquiry requests from other BR/EDR Controllers. If Page_Scan is enabled, then the device will enter page scan mode based on the value of the Page_Scan_Interval and Page_Scan_Window parameters. If Inquiry_Scan is enabled, then the BR/EDR Controller will enter Inquiry Scan mode based on the value of the Inquiry_Scan_Interval and Inquiry_Scan_Window parameters.

Value	Parameter Description
0x00	No Scans enabled.
0x01	Inquiry Scan enabled. Page Scan always disabled.
0x02	Inquiry Scan disabled. Page Scan enabled.
0x03	Inquiry Scan enabled. Page Scan enabled.
All other values	Reserved for future use

6.2 INQUIRY SCAN INTERVAL

The Inquiry_Scan_Interval configuration parameter defines the amount of time between consecutive inquiry scans. This is defined as the time interval from when the BR/EDR Controller started its last inquiry scan until it begins the next inquiry scan.

Value	Parameter Description
N = 0xXXXX	Size: 2 octets Range: 0x0012 to 0x1000; only even values are valid Default: 0x1000 Time = N * 0.625 ms Time Range: 11.25 to 2560 ms Time Default: 2.56 s



6.3 INQUIRY SCAN WINDOW

The Inquiry_Scan_Window configuration parameter defines the amount of time for the duration of the inquiry scan. The Inquiry_Scan_Window can only be less than or equal to the Inquiry_Scan_Interval.

Value	Parameter Description
N = 0xXXXX	Size: 2 octets Range: 0x0011 to 0x1000 Default: 0x0012 Mandatory Range: 0x0011 to Inquiry Scan Interval Time = N * 0.625 ms Time Range: 10.625 ms to 2560 ms Time Default: 11.25 ms

6.4 INQUIRY SCAN TYPE

The Inquiry_Scan_Type configuration parameter indicates whether inquiry scanning will be done using non-interlaced scan or interlaced scan. Currently, one mandatory inquiry scan type and one optional inquiry scan type are defined. For details, see the Baseband Specification, [\[Vol 2\] Part B, Section 8.4.1](#).

Value	Parameter Description
0x00	Mandatory: Standard Scan (default)
0x01	Optional: Interlaced Scan
All other values	Reserved for future use

6.5 INQUIRY MODE

The Inquiry_Mode configuration parameter indicates whether inquiry returns Inquiry Result events in the standard format, with RSSI, or with RSSI and extended inquiry response information.

Value	Parameter Description
0x00	Standard Inquiry Result event format
0x01	Inquiry Result format with RSSI
0x02	Inquiry Result with RSSI format or Extended Inquiry Result format
All other values	Reserved for future use



6.6 PAGE TIMEOUT

The Page_Timeout configuration parameter defines the maximum time the local Link Manager will wait for a baseband page response from the remote device at a locally initiated connection attempt. If this time expires and the remote BR/EDR Controller has not responded to the page at baseband level, the connection attempt will be considered to have failed.

Value	Parameter Description
N = 0xFFFF	Size: 2 octets Range: 0x0001 to 0xFFFF Default: 0x2000 Mandatory Range: 0x0016 to 0xFFFF Time = $N * 0.625 \text{ ms}$ Time Range: 0.625 ms to 40.9 s Time Default: 5.12 s

6.7 CONNECTION ACCEPT TIMEOUT

The Connection_Accept_Timeout configuration parameter allows the BR/EDR Controller to automatically deny a connection request after a specified time period has occurred and the new connection is not accepted. The parameter defines the time duration from when the BR/EDR Controller sends a Connection Request event until the BR/EDR Controller will automatically reject an incoming connection.

The Connection_Accept_Timeout configuration parameter allows each AMP Controller to limit the duration of the HCI_Create_Physical_Link command or HCI_Accept_Physical_Link command it is locally serving. The parameter defines the maximum time duration for all physical connection setup activities that occur between the HCI_Create_Physical_Link command or HCI_Accept_Physical_Link command and the corresponding HCI_Physical_Link_Complete event (including all authentication activity) before the AMP Controller will automatically reject the physical link setup.

Value	Parameter Description
N = 0xFFFF	Size: 2 octets Range: 0x0001 to 0xB540 Default: 0x1F40 Mandatory Range: 0x00A0 to 0xB540 Time = $N * 0.625 \text{ ms}$ Time Range: 0.625 ms to 29 s Time Default: BR/EDR 5 s AMP type dependent



6.8 PAGE SCAN INTERVAL

The Page_Scan_Interval configuration parameter defines the amount of time between consecutive page scans. This time interval is defined from when the Controller started its last page scan until it begins the next page scan.

Value	Parameter Description
N = 0xXXXX	Size: 2 octets Range: 0x0012 to 0x1000; only even values are valid Default: 0x0800 $\text{Time} = N * 0.625 \text{ ms}$ Time Range: 11.25 ms to 2560 ms Time Default: 1.28 s

6.9 PAGE SCAN WINDOW

The Page_Scan_Window configuration parameter defines the amount of time for the duration of the page scan. The Page_Scan_Window can only be less than or equal to the Page_Scan_Interval.

Value	Parameter Description
N = 0xXXXX	Size: 2 octets Range: 0x0011 to 0x1000 Default: 0x0012 Mandatory Range: 0x0011 to Page Scan Interval $\text{Time} = N * 0.625 \text{ ms}$ Time Range: 10.625 ms to Page Scan Interval Time Default: 11.25 ms

6.10 THIS SECTION NO LONGER USED

6.11 PAGE SCAN TYPE

The Page_Scan_Type parameter indicates whether page scanning will be done using non-interlaced scan or interlaced scan. For details, see the Baseband Specification, [\[Vol 2\] Part B, Section 8.3.1](#).

Value	Parameter Description
0x00	Mandatory: Standard Scan (default)
0x01	Optional: Interlaced Scan
All other values	Reserved for future use



6.12 VOICE SETTING

The Voice_Setting parameter controls all the various settings for voice connections. The Voice_Setting parameter controls the configuration for voice connections: Input Coding, Air coding format, input data format, Input sample size, and linear PCM parameter. The air coding format bits in the Voice_Setting command parameter specify which air coding format the local device requests. The air coding format bits do not specify which air coding format(s) the local device accepts when a remote device requests an air coding format. This is determined by the hardware capabilities of the local BR/EDR Controller.

Bit number	Parameter description	
0 – 1	Air coding format	0: CVSD 1: μ -law 2: A-law 3: transparent data
2 – 4	Linear PCM bit position - number of bit positions that the MSB of the sample is from the MSB of the value (only for linear PCM)	
5	Input sample size (only for linear PCM)	0: 8 bits 1: 16 bits
6 – 7	Input data format	0: 1's complement 1: 2's complement 2: sign and magnitude 3: unsigned
8 – 9	Input coding format	0: linear 1: μ -law 2: A-law 3: Reserved for future use
10 – 15	Reserved for future use	

6.13 PIN TYPE

The PIN Type configuration parameter determines whether the Link Manager assumes that the Host supports variable PIN codes or a fixed PIN code. The Controller uses the PIN-type information during pairing.

Value	Parameter Description
0x00	Variable PIN.
0x01	Fixed PIN.



6.14 LINK KEY

The Controller can store a limited number of link keys for other BR/EDR Controllers. Link keys are shared between two BR/EDR Controllers, and are used for all security transactions between the two BR/EDR Controllers. A Host device may have additional storage capabilities, which can be used to save additional link keys to be reloaded to the BR/EDR Controller when needed. A Link Key is associated with a BD_ADDR.

Value	Parameter Description
0xFFFFFFFFXX XXXXXXXXXXXX XXXXXXXXXXXX	Link Key for an associated BD_ADDR.

6.15 FAILED CONTACT COUNTER

The Failed_Contact_Counter records the number of consecutive incidents in which either the local or remote device did not respond after the flush timeout had expired, and the L2CAP PDU that was currently being transmitted was automatically 'flushed'. When this occurs, the Failed_Contact_Counter is incremented by 1.

The Failed Contact Counter is maintained for each Connection_Handle and Logical_Link_Handle.

The Failed_Contact_Counter for a connection is reset to zero on the following conditions:

1. When a new connection is established
2. When the Failed_Contact_Counter is > zero and an L2CAP PDU is acknowledged for that connection
3. When the HCI_Reset_Failed_Contact_Counter command has been issued

Value	Parameter Description
0xFFFF	Number of consecutive failed contacts for a connection corresponding to the Connection or Logical Link Handle.

6.16 AUTHENTICATION ENABLE

The Authentication_Enable parameter controls if the local device requires to authenticate the remote device at connection setup (between the HCI_Create_Connection command or acceptance of an incoming ACL connection and the corresponding Connection Complete event). At connection setup, only the device(s) with the Authentication_Enable parameter enabled will try to authenticate the other device.



Note: Changing this parameter does not affect existing connections.

Value	Parameter Description
0x00	Authentication not required.
0x01	Authentication required for all connections.
All other values	Reserved for future use

6.17 HOLD MODE ACTIVITY

The Hold_Mode_Activity value is used to determine what activities should be suspended when the BR/EDR Controller is in Hold mode. After the hold period has expired, the device will return to the previous mode of operation. Multiple Hold mode activities may be specified for the Hold_Mode_Activity parameter by performing a bitwise OR operation of the different activity types. If no activities are suspended, then all of the current Periodic Inquiry, Inquiry Scan, and Page Scan settings remain valid during the Hold mode. If the Hold_Mode_Activity parameter is set to Suspend Page Scan, Suspend Inquiry Scan, and Suspend Periodic Inquiries, then the device can enter a low-power state during the Hold mode period, and all activities are suspended. Suspending multiple activities can be specified for the Hold_Mode_Activity parameter by performing a bitwise OR operation of the different activity types. The Hold Mode Activity is only valid if all connections are in Hold mode.

Value	Parameter Description
0x00	Maintain current Power State.
0x01	Suspend Page Scan.
0x02	Suspend Inquiry Scan.
0x04	Suspend Periodic Inquiries.
All other values	Reserved for future use.



6.18 LINK POLICY SETTINGS

The Link_Policy_Settings parameter determines the behavior of the local Link Manager when it receives a request from a remote Link Manager or it determines itself to change the master-slave role or to enter Hold or Sniff mode. The local Link Manager will automatically accept or reject such a request from the remote device, and may even autonomously request itself, depending on the value of the Link_Policy_Settings parameter for the corresponding Connection_Handle. When the value of the Link_Policy_Settings parameter is changed for a certain Connection_Handle, the new value will only be used for requests from a remote device or from the local Link Manager itself made after this command has been completed. By enabling each mode individually, the Host can choose any combination needed to support various modes of operation. Multiple LM policies may be specified for the Link_Policy_Settings parameter by performing a bitwise OR operation of the different activity types.

Note: The local BR/EDR Controller may be forced into Hold mode (regardless of whether the local device is master or slave) by the remote device regardless of the value of the Link_Policy_Settings parameter. The forcing of Hold mode can however only be done once the connection has already been placed into Hold mode through an LMP request (the Link_Policy_Settings determine if requests from a remote device should be accepted or rejected). The forcing of Hold mode can after that be done as long as the connection lasts regardless of the setting for Hold mode in the Link_Policy_Settings parameter.

Note: If the implementation in the remote device is a “polite” implementation that does not force another device into Hold mode via LMP PDUs, then the Link_Policy_Settings will never be overruled.

Bit Number	Parameter Description
0	Enable Role switch.
1	Enable Hold mode.
2	Enable Sniff mode.
All other bits	Reserved for future use.



6.19 FLUSH TIMEOUT

The Flush_Timeout configuration parameter is used for ACL connections on a BR/EDR Controller only. The Flush Timeout is defined in the Baseband specification [Vol 2] Part B, Section 7.6.3. This parameter allows automatically-flushable ACL packets to be automatically flushed without the Host device issuing an HCI_Flush command. This provides support for isochronous data, such as audio. Non-automatically-flushable ACL packets shall not be affected by this parameter (see Section 5.4.2) providing support for both asynchronous and isochronous data on the same ACL connection. When the L2CAP packet that is currently being transmitted is automatically ‘flushed’, the Failed Contact Counter is incremented by one.

Note: Since for AMP Controllers a Flush Timeout value is maintained as part of the Extended Flow Specification for each logical link, the separate Flush Timeout configuration parameter is not used.

Value	Parameter Description
0x0000	Timeout = ∞ ; No Automatic Flush
N = 0xXXXX	Size: 2 octets Range: 0x0001 to 0x07FF Mandatory Range: 0x0002 to 0x07FF Time = $N * 0.625$ ms Time Range: 0.625 ms to 1279.375 ms

6.20 NUM BROADCAST RETRANSMISSIONS

Broadcast packets are not acknowledged and are unreliable. The Number of Broadcast Retransmissions parameter, N, is used to increase the reliability of a broadcast message by retransmitting the broadcast message multiple times. This sets the value N_{BC} in the baseband to one greater than the Num Broadcast Retransmissions value. (See [Vol 2] Part B, Section 7.6.5) This parameter should be adjusted as the link quality measurement changes.

Value	Parameter Description
N = 0xXX	$N_{BC} = N + 1$ Range: 0x00 to 0xFE



6.21 LINK SUPERVISION TIMEOUT

The Link_Supervision_Timeout parameter is used by the Controller to monitor link loss. If, for any reason, no packets are received from that Handle (a Connection_Handle for the BR/EDR Controller or the Physical_Link_Handle for an AMP Controller) for a duration longer than the Link_Supervision_Timeout, the connection shall be disconnected. For the BR/EDR Controller, the same timeout value is used for both synchronous and ACL connections for the device specified by the Connection_Handle. For an AMP Controller, the timeout value is maintained for each physical link.

Note: Some AMP Controllers may also have an early notification event (See [Physical Link Loss Early Warning event](#)).

Note: Setting the Link_Supervision_Timeout to No Link_Supervision_Timeout (0x0000) on a BR/EDR Controller will disable the Link_Supervision_Timeout check for the specified Connection_Handle.

Value	Parameter Description
0x0000	No Link_Supervision_Timeout. Shall not be used with AMP Controllers.
N = 0xXXXX	Size: 2 octets Range: 0x0001 to 0xFFFF Default: 0x7D00 Mandatory Range: 0x0190 to 0xFFFF Time = N * 0.625 ms Time Range: 0.625 ms to 40.9 s Time Default: BR/EDR 20 s AMP 10 s

6.22 SYNCHRONOUS FLOW CONTROL ENABLE

The Synchronous Flow Control Enable configuration parameter allows the Host to decide if the BR/EDR Controller will send Number Of Completed Packets events for synchronous Connection_Handles. This setting allows the Host to enable and disable synchronous flow control.

Value	Parameter Description
0x00	Synchronous Flow Control is disabled. No Number of Completed Packets events will be sent from the BR/EDR Controller for synchronous Connection_Handles.
0x01	Synchronous Flow Control is enabled. Number of Completed Packets events will be sent from the BR/EDR Controller for synchronous Connection_Handles.



6.23 LOCAL NAME

The user-friendly Local Name provides the user the ability to distinguish one BR/EDR Controller from another. The Local Name configuration parameter is a UTF-8 encoded string with up to 248 octets in length. The Local Name configuration parameter will be null terminated (0x00) if the UTF-8 encoded string is less than 248 octets.

Note: The Local Name configuration parameter is a string parameter. Endianness does therefore not apply to the Local Name configuration parameter. The first octet of the name is received first.

Value	Parameter Description
	A UTF-8 encoded User Friendly Descriptive Name for the device. If the name contained in the parameter is shorter than 248 octets, the end of the name is indicated by a NULL octet (0x00), and the following octets (to fill up 248 octets, which is the length of the parameter) do not have valid values.

6.24 EXTENDED INQUIRY RESPONSE

The Extended Inquiry Response provides information about the local device in response to inquiry from remote devices. The configuration parameter has two parts, a significant part followed by a non-significant part. The non-significant part contains only zero octets. The length of the extended inquiry response configuration parameter is 240 octets. The data format of the significant part is defined in [\[Vol 3\] Part C, Section 8](#).

Value	Parameter Description
	Information about the local device that will be sent in an extended inquiry response packet to remote devices during inquiry response.

6.25 ERRONEOUS DATA REPORTING

The Erroneous_Data_Reporting configuration parameter shall be used for SCO and eSCO connections only. This parameter determines if the BR/EDR Controller is required to provide data to the Host for every (e)SCO interval, with the Packet_Status_Flag in HCI Synchronous Data packets set according to [Section 5.4.3](#).

Value	Parameter Description
0x00	Erroneous data reporting disabled
0x01	Erroneous data reporting enabled.
All other values	Reserved for future use



6.26 CLASS OF DEVICE

The Class_of_Device parameter is used to indicate the capabilities of the local device to other devices.

Value	Parameter Description
0xXXXXXX	Class of Device for the device.

6.27 SUPPORTED COMMANDS

The Supported Commands configuration parameter lists which HCI commands the local Controller supports. If a command is listed as supported, the feature underlying that command shall also be supported.

The Supported Commands is a 64 octet bit field. If a bit is set to 1, then this command is supported.

Octet	Bit	Command Supported
0	0	HCI_Inquiry
	1	HCI_Inquiry_Cancel
	2	HCI_Periodic_Inquiry_Mode
	3	HCI_Exit_Periodic_Inquiry_Mode
	4	HCI_Create_Connection
	5	HCI_Disconnect
	6	HCI_Add_SCO_Connection (deprecated)
	7	HCI_Create_Connection_Cancel
1	0	HCI_Accept_Connection_Request
	1	HCI_Reject_Connection_Request
	2	HCI_Link_Key_Request_Reply
	3	HCI_Link_Key_Request_Negative_Reply
	4	HCI_PIN_Code_Request_Reply
	5	HCI_PIN_Code_Request_Negative_Reply
	6	HCI_Change_Connection_Packet_Type
	7	HCI_Authentication_Requested



Octet	Bit	Command Supported
2	0	HCI_Set_Connection_Encryption
	1	HCI_Change_Connection_Link_Key
	2	HCI_Master_Link_Key
	3	HCI_Remote_Name_Request
	4	HCI_Remote_Name_Request_Cancel
	5	HCI_Read_Remote_Supported_Features
	6	HCI_Read_Remote_Extended_Features
	7	HCI_Read_Remote_Version_Information
3	0	HCI_Read_Clock_Offset
	1	HCI_Read_LMP_Handle
	2	Reserved for future use
	3	Reserved for future use
	4	Reserved for future use
	5	Reserved for future use
	6	Reserved for future use
	7	Reserved for future use
4	0	Reserved for future use
	1	HCI_Hold_Mode
	2	HCI_Sniff_Mode
	3	HCI_Exit_Sniff_Mode
	4	Reserved for future use
	5	Reserved for future use
	6	HCI_QoS_Setup
	7	HCI_Role_Discovery
5	0	HCI_Switch_Role
	1	HCI_Read_Link_Policy_Settings
	2	HCI_Write_Link_Policy_Settings
	3	HCI_Read_Default_Link_Policy_Settings
	4	HCI_Write_Default_Link_Policy_Settings
	5	HCI_Flow_Specification
	6	HCI_Set_Event_Mask
	7	HCI_Reset



Octet	Bit	Command Supported
6	0	HCI_Set_Event_Filter
	1	HCI_Flush
	2	HCI_Read_PIN_Type
	3	HCI_Write_PIN_Type
	4	Reserved for future use
	5	HCI_Read_Stored_Link_Key
	6	HCI_Write_Stored_Link_Key
	7	HCI_Delete_Stored_Link_Key
7	0	HCI_Write_Local_Name
	1	HCI_Read_Local_Name
	2	HCI_Read_Connection_Accept_Timeout
	3	HCI_Write_Connection_Accept_Timeout
	4	HCI_Read_Page_Timeout
	5	HCI_Write_Page_Timeout
	6	HCI_Read_Scan_Enable
	7	HCI_Write_Scan_Enable
8	0	HCI_Read_Page_Scan_Activity
	1	HCI_Write_Page_Scan_Activity
	2	HCI_Read_Inquiry_Scan_Activity
	3	HCI_Write_Inquiry_Scan_Activity
	4	HCI_Read_Authentication_Enable
	5	HCI_Write_Authentication_Enable
	6	HCI_Read_Encryption_Mode (deprecated)
	7	HCI_Write_Encryption_Mode (deprecated)
9	0	HCI_Read_Class_Of_Device
	1	HCI_Write_Class_Of_Device
	2	HCI_Read_Voice_Setting
	3	HCI_Write_Voice_Setting
	4	HCI_Read_Automatic_Flush_Timeout
	5	HCI_Write_Automatic_Flush_Timeout
	6	HCI_Read_Num_Broadcast_Retransmissions
	7	HCI_Write_Num_Broadcast_Retransmissions



Octet	Bit	Command Supported
10	0	HCI_Read_Hold_Mode_Activity
	1	HCI_Write_Hold_Mode_Activity
	2	HCI_Read_Transmit_Power_Level
	3	HCI_Read_Synchronous_Flow_Control_Enable
	4	HCI_Write_Synchronous_Flow_Control_Enable
	5	HCI_Set_Controller_To_Host_Flow_Control
	6	HCI_Host_Buffer_Size
	7	HCI_Host_Number_Of_Completed_Packets
11	0	HCI_Read_Link_Supervision_Timeout
	1	HCI_Write_Link_Supervision_Timeout
	2	HCI_Read_Number_Of_Supported_IAC
	3	HCI_Read_Current_IAC_LAP
	4	HCI_Write_Current_IAC_LAP
	5	HCI_Read_Page_Scan_Mode_Period (deprecated)
	6	HCI_Write_Page_Scan_Mode_Period (deprecated)
	7	HCI_Read_Page_Scan_Mode (deprecated)
12	0	HCI_Write_Page_Scan_Mode (deprecated)
	1	HCI_Set_AFH_Host_Channel_Classification
	2	Reserved for future use
	3	Reserved for future use
	4	HCI_Read_Inquiry_Scan_Type
	5	HCI_Write_Inquiry_Scan_Type
	6	HCI_Read_Inquiry_Mode
	7	HCI_Write_Inquiry_Mode
13	0	HCI_Read_Page_Scan_Type
	1	HCI_Write_Page_Scan_Type
	2	HCI_Read_AFH_Channel_Assessment_Mode
	3	HCI_Write_AFH_Channel_Assessment_Mode
	4	Reserved for future use
	5	Reserved for future use
	6	Reserved for future use
	7	Reserved for future use



Octet	Bit	Command Supported
14	0	Reserved for future use
	1	Reserved for future use
	2	Reserved for future use
	3	HCI_Read_Local_Version_Information
	4	Reserved for future use
	5	HCI_Read_Local_Supported_Features
	6	HCI_Read_Local_Extended_Features
	7	HCI_Read_Buffer_Size
15	0	HCI_Read_Country_Code (deprecated)
	1	HCI_Read_BD_ADDR
	2	HCI_Read_Failed_Contact_Counter
	3	HCI_Reset_Failed_Contact_Counter
	4	HCI_Read_Link_Quality
	5	HCI_Read_RSSI
	6	HCI_Read_AFH_Channel_Map
	7	HCI_Read_Clock
16	0	HCI_Read_Loopback_Mode
	1	HCI_Write_Loopback_Mode
	2	HCI_Enable_Device_Under_Test_Mode
	3	HCI_Setup_Synchronous_Connection_Request
	4	HCI_Accept_Synchronous_Connection_Request
	5	HCI_Reject_Synchronous_Connection_Request
	6	Reserved for future use
	7	Reserved for future use
17	0	HCI_Read_Extended_Inquiry_Response
	1	HCI_Write_Extended_Inquiry_Response
	2	HCI_Refresh_Encryption_Key
	3	Reserved for future use
	4	HCI_Sniff_Subrating
	5	HCI_Read_Simple_Pairing_Mode
	6	HCI_Write_Simple_Pairing_Mode
	7	HCI_Read_Local_OOB_Data



Octet	Bit	Command Supported
18	0	HCI_Read_Inquiry_Response_Transmit_Power_Level
	1	HCI_Write_Inquiry_Transmit_Power_Level
	2	HCI_Read_Default_Erroneous_Data_Reporting
	3	HCI_Write_Default_Erroneous_Data_Reporting
	4	Reserved for future use
	5	Reserved for future use
	6	Reserved for future use
	7	HCI_IO_Capability_Request_Reply
19	0	HCI_User_Confirmation_Request_Reply
	1	HCI_User_Confirmation_Request_Negative_Reply
	2	HCI_User_Passkey_Request_Reply
	3	HCI_User_Passkey_Request_Negative_Reply
	4	HCI_Remote_OOB_Data_Request_Reply
	5	HCI_Write_Simple_Pairing_Debug_Mode
	6	HCI_Enhanced_Flush
	7	HCI_Remote_OOB_Data_Request_Negative_Reply
20	0	Reserved for future use
	1	Reserved for future use
	2	HCI_Send_Keypress_Notification
	3	HCI_IO_Capability_Request_Negative_Reply
	4	HCI_Read_Encryption_Key_Size
	5	Reserved for future use
	6	Reserved for future use
	7	Reserved for future use
21	0	HCI_Create_Physical_Link
	1	HCI_Accept_Physical_Link
	2	HCI_Disconnect_Physical_Link
	3	HCI_Create_Logical_Link
	4	HCI_Accept_Logical_Link
	5	HCI_Disconnect_Logical_Link
	6	HCI_Logical_Link_Cancel
	7	HCI_Flow_Spec_Modify



Octet	Bit	Command Supported
22	0	HCI_Read_Logical_Link_Accept_Timeout
	1	HCI_Write_Logical_Link_Accept_Timeout
	2	HCI_Set_Event_Mask_Page_2
	3	HCI_Read_Location_Data
	4	HCI_Write_Location_Data
	5	HCI_Read_Local_AMP_Info
	6	HCI_Read_Local_AMP_ASSOC
	7	HCI_Write_Remote_AMP_ASSOC
23	0	HCI_Read_Flow_Control_Mode
	1	HCI_Write_Flow_Control_Mode
	2	HCI_Read_Data_Block_Size
	3	Reserved for future use
	4	Reserved for future use
	5	HCI_Enable_AMP_Receiver_Reports
	6	HCI_AMP_Test_End
	7	HCI_AMP_Test
24	0	HCI_Read_Enhanced_Transmit_Power_Level
	1	Reserved for future use
	2	HCI_Read_Best_Effort_Flush_Timeout
	3	HCI_Write_Best_Effort_Flush_Timeout
	4	HCI_Short_Range_Mode
	5	HCI_Read_LE_Host_Support
	6	HCI_Write_LE_Host_Support
	7	Reserved for future use
25	0	HCI_LE_Set_Event_Mask
	1	HCI_LE_Read_Buffer_Size
	2	HCI_LE_Read_Local_Supported_Features
	3	Reserved for future use
	4	HCI_LE_Set_Random_Address
	5	HCI_LE_Set_Advertising_Parameters
	6	HCI_LE_Read_Advertising_Physical_Channel_Tx_Power
	7	HCI_LE_Set_Advertising_Data



Octet	Bit	Command Supported
26	0	HCI_LE_Set_Scan_Response_Data
	1	HCI_LE_Set_Advertising_Enable
	2	HCI_LE_Set_Scan_Parameters
	3	HCI_LE_Set_Scan_Enable
	4	HCI_LE_Create_Connection
	5	HCI_LE_Create_Connection_Cancel
	6	HCI_LE_Read_White_List_Size
	7	HCI_LE_Clear_White_List
27	0	HCI_LE_Add_Device_To_White_List
	1	HCI_LE_Remove_Device_From_White_List
	2	HCI_LE_Connection_Update
	3	HCI_LE_Set_Host_Channel_Classification
	4	HCI_LE_Read_Channel_Map
	5	HCI_LE_Read_Remote_Features
	6	HCI_LE_Encrypt
	7	HCI_LE_Rand
28	0	HCI_LE_Start_Encryption
	1	HCI_LE_Long_Term_Key_Request_Reply
	2	HCI_LE_Long_Term_Key_Request_Negative_Reply
	3	HCI_LE_Read_Supported_States
	4	HCI_LE_Receiver_Test [v1]
	5	HCI_LE_Transmitter_Test [v1]
	6	HCI_LE_Test_End
	7	Reserved for future use
29	0	Reserved for future use
	1	Reserved for future use
	2	Reserved for future use
	3	HCI_Enhanced_Setup_Synchronous_Connection
	4	HCI_Enhanced_Accept_Synchronous_Connection
	5	HCI_Read_Local_Supported_Codecs
	6	HCI_Set_MWS_Channel_Parameters
	7	HCI_Set_External_Frame_Configuration



Octet	Bit	Command Supported
30	0	HCI_Set_MWS_Signaling
	1	HCI_Set_MWS_Transport_Layer
	2	HCI_Set_MWS_Scan_Frequency_Table
	3	HCI_Get_MWS_Transport_Layer_Configuration
	4	HCI_Set_MWS_PATTERN_Configuration
	5	HCI_Set_Triggered_Clock_Capture
	6	HCI_Truncated_Page
	7	HCI_Truncated_Page_Cancel
31	0	HCI_Set_Connectionless_Slave_Broadcast
	1	HCI_Set_Connectionless_Slave_Broadcast_Receive
	2	HCI_Start_Synchronization_Train
	3	HCI_Receive_Synchronization_Train
	4	HCI_Set_Reserved_LT_ADDR
	5	HCI_Delete_Reserved_LT_ADDR
	6	HCI_Set_Connectionless_Slave_Broadcast_Data
	7	HCI_Read_Synchronization_Train_Parameters
32	0	HCI_Write_Synchronization_Train_Parameters
	1	HCI_Remote_OOB_Extended_Data_Request_Reply
	2	HCI_Read_Secure_Connections_Host_Support
	3	HCI_Write_Secure_Connections_Host_Support
	4	HCI_Read_Authenticated_Payload_Timeout
	5	HCI_Write_Authenticated_Payload_Timeout
	6	HCI_Read_Local_OOB_Extended_Data
	7	HCI_Write_Secure_Connections_Test_Mode
33	0	HCI_Read_Extended_Page_Timeout
	1	HCI_Write_Extended_Page_Timeout
	2	HCI_Read_Extended_Inquiry_Length
	3	HCI_Write_Extended_Inquiry_Length
	4	HCI_LE_Remote_Connection_Parameter_Request_Reply
	5	HCI_LE_Remote_Connection_Parameter_Request_Negative_Reply
	6	HCI_LE_Set_Data_Length
	7	HCI_LE_Read_Suggested_Default_Data_Length



Octet	Bit	Command Supported
34	0	HCI_LE_Write_Suggested_Default_Data_Length
	1	HCI_LE_Read_Local_P-256_Public_Key
	2	HCI_LE_Generate_DHKey [v1]
	3	HCI_LE_Add_Device_To_Resolving_List
	4	HCI_LE_Remove_Device_From_Resolving_List
	5	HCI_LE_Clear_Resolving_List
	6	HCI_LE_Read_Resolving_List_Size
	7	HCI_LE_Read_Peer_Resolvable_Address
35	0	HCI_LE_Read_Local_Resolvable_Address
	1	HCI_LE_Set_Address_Resolution_Enable
	2	HCI_LE_Set_Resolvable_Private_Address_Timeout
	3	HCI_LE_Read_Maximum_Data_Length
	4	HCI_LE_Read_PHY
	5	HCI_LE_Set_Default_PHY
	6	HCI_LE_Set_PHY
	7	HCI_LE_Receiver_Test [v2]
36	0	HCI_LE_Transmitter_Test [v2]
	1	HCI_LE_Set_Advertising_Set_Random_Address
	2	HCI_LE_Set_Extended_Advertising_Parameters
	3	HCI_LE_Set_Extended_Advertising_Data
	4	HCI_LE_Set_Extended_Scan_Response_Data
	5	HCI_LE_Set_Extended_Advertising_Enable
	6	HCI_LE_Read_Maximum_Advertising_Data_Length
	7	HCI_LE_Read_Number_of_Supported_Advertising_Sets
37	0	HCI_LE_Remove_Advertising_Set
	1	HCI_LE_Clear_Advertising_Sets
	2	HCI_LE_Set_Periodic_Advertising_Parameters
	3	HCI_LE_Set_Periodic_Advertising_Data
	4	HCI_LE_Set_Periodic_Advertising_Enable
	5	HCI_LE_Set_Extended_Scan_Parameters
	6	HCI_LE_Set_Extended_Scan_Enable
	7	HCI_LE_Extended_Create_Connection



Octet	Bit	Command Supported
38	0	HCI_LE_Periodic_Advertising_Create_Sync
	1	HCI_LE_Periodic_Advertising_Create_Sync_Cancel
	2	HCI_LE_Periodic_Advertising_Terminate_Sync
	3	HCI_LE_Add_Device_To_Periodic_Advertiser_List
	4	HCI_LE_Remove_Device_From_Periodic_Advertiser_List
	5	HCI_LE_Clear_Periodic_Advertiser_List
	6	HCI_LE_Read_Periodic_Advertiser_List_Size
	7	HCI_LE_Read_Transmit_Power
39	0	HCI_LE_Read_RF_Path_Compensation
	1	HCI_LE_Write_RF_Path_Compensation
	2	HCI_LE_Set_Privacy_Mode
	3	HCI_LE_Receiver_Test [v3]
	4	HCI_LE_Transmitter_Test [v3]
	5	HCI_LE_Set_Connectionless_CTE_Transmit_Parameters
	6	HCI_LE_Set_Connectionless_CTE_Transmit_Enable
	7	HCI_LE_Set_Connectionless_IQ_Sampling_Enable
40	0	HCI_LE_Set_Connection_CTE_Receive_Parameters
	1	HCI_LE_Set_Connection_CTE_Transmit_Parameters
	2	HCI_LE_Connection_CTE_Request_Enable
	3	HCI_LE_Connection_CTE_Response_Enable
	4	HCI_LE_Read_Antenna_Information
	5	HCI_LE_Set_Periodic_Advertising_Receive_Enable
	6	HCI_LE_Periodic_Advertising_Sync_Transfer
	7	HCI_LE_Periodic_Advertising_Set_Info_Transfer
41	0	HCI_LE_Set_Periodic_Advertising_Sync_Transfer_Parameters
	1	HCI_LE_Set_Default_Periodic_Advertising_Sync_Transfer_Parameters
	2	HCI_LE_Generate_DHKey [v2]
	3	HCI_Read_Local_Simple_Pairing_Options
	4	HCI_LE_Modify_Sleep_Clock_Accuracy
	5	Reserved for future use
	6	Reserved for future use
	7	Reserved for future use



6.28 LOGICAL LINK ACCEPT TIMEOUT

The Logical_Link_Accept_Timeout configuration parameter allows the AMP Controller to automatically deny a logical link request after a specified time period has elapsed and the new logical link has not completed. The parameter defines the timeout duration, which includes all logical link setup activities that occur between the HCI_Create_Logical_Link or HCI_Accept_Logical_Link commands and the HCI_Logical_Link_Complete event. If these activities are not completed within the specified time, the AMP Controller will automatically reject the logical link setup.

Value	Parameter Description
N = 0xXXXX	Size: 2 octets Range: 0x0001 to 0xB540 Default: 0x1F40 Mandatory Range: 0x00A0 to 0xB540 Time = N * 0.625 ms Time Range: 0.625 ms to 29 s Time Default: AMP Type dependent

6.29 LOCATION DOMAIN AWARE

The Location_Domain_Aware parameter indicates if the transmitter has information about the current regulatory domain.

Location_Domain_Aware:

Size: 1 octet

Value	Parameter Description
0x00	Regulatory domain unknown
0x01	Regulatory domain known
All other values	Reserved for future use



6.30 LOCATION DOMAIN

The Location_Domain parameter specifies the country or other location code if known.

Location_Domain:

Size: 2 octets

Value	Parameter Description
0xXXXX	Shall be set to the character string "XX" (0x5858) for a non-country entity or when Location_Domain_Aware is set to regulatory domain unknown. When the regulatory domain is known, it shall be set to the relevant ISO 3166-1 ¹ alpha-2 two character country code, with the first letter in the least significant octet (for example, "BT" is encoded as 0x5442).

1. ISO 3166-1-alpha-2 code elements (English),
http://www.iso.org/iso/country_names_and_code_elements



6.31 LOCATION DOMAIN OPTIONS

The Location_Domain_Options parameter may specify additional information specific to that country, or that the location is not a single country.

Location_Domain_Options:

Size: 1 octet

Values	Parameter Descriptions
0x20	Single octet suffix:
0x4F	"space" indicates that the code applies to the entire country
0x49	"O" indicates for use outdoors only
0x58	"I" indicates for use indoors only
	"X" indicates a non-country entity
All other values	Reserved for future use

The format of the Location_Domain_Options parameter is derived from IEEE 802.11-2007 Annex D "ASN.1 encoding of the MAC and PHY MIB."

6.32 LOCATION OPTIONS

The Location_Options parameter indicates if the transmitter has information about any options that may affect operation in different locations.

Location_Options:

Size: 1 octet

Bit	Value	Parameter Description
0	0	Not mains-powered
0	1	Mains powered
1-7		Reserved for future use

6.33 FLOW CONTROL MODE

The Flow_Control_Mode configuration parameter allows the Host to select the HCI Data flow control mode used by the Controller for ACL Data traffic.

Flow_Control_Mode:

Size: 1 octet

Value	Parameter Description
0x00	Packet based data flow control mode (default for a Primary Controller)
0x01	Data block based data flow control mode (default for an AMP Controller)
All other values	Reserved for future use



6.34 LE SUPPORTED HOST

The LE_Supported_Host parameter allows the Host to read and set the Link Manager Protocol feature bit LE Supported (Host). See [\[Vol 2\] Part C, Section 3.2](#).

Value	Parameter Description
0x00	LE Supported (Host) disabled (default)
0x01	LE Supported (Host) enabled
All other values	Reserved for future use

6.35 SIMULTANEOUS LE HOST

The Simultaneous_LE_Host parameter allows the Host to read and set the Link Manager Protocol feature bit Simultaneous LE and BR/EDR to Same Device Capable (Host). See [\[Vol 2\] Part C, Section 3.2](#).

Value	Parameter Description
0x00	Simultaneous LE and BR/EDR to Same Device Capable (Host) disabled (default). This value shall be ignored.
All other values	Reserved for future use

6.36 SYNCHRONIZATION TRAIN INTERVAL

The Sync_Train_Interval configuration parameter defines the time between Synchronization Train transmit events on a single transmit RF channel.

Sync_Train_Interval:

Size: 2 octets

Value	Parameter Description
N = 0xXXXX	Range: 0x0020 to 0xFFFFE; only even values are valid Default: 0x0080 Mandatory Range: 0x0020 to 0x1000 Time = N * 0.625 ms Time Range: 20 ms to 40.9 s Time Default: 80 ms



6.37 SYNCHRONIZATION TRAIN TIMEOUT

The *synchronization_trainTO* configuration parameter is used by the Controller to terminate the Synchronization Train after it has been started via the *HCI_Start_Synchronization_Train* command.

synchronization_trainTO:

Size: 4 octets

Value	Parameter Description
N = 0XXXXXXXX	Range: 0x00000002 to 0x07FFFFFFE; only even values are valid Default: 0x0002EE00 Time = N * 0.625 ms Time Range: 1.25 ms to 23.3 hours Time Default: 120 s

6.38 SERVICE DATA

The *Service_Data* configuration parameter defines the value of the service data field in the Synchronization Train.

Service_Data:

Size: 1 octet

Value	Parameter Description
N = 0xXX	Range: 0x00 to 0xFF Default: 0x00

6.39 SECURE CONNECTIONS HOST SUPPORT

The *Secure_Connections_Host_Support* configuration parameter allows the Host to indicate whether it supports Secure Connections or not. When Secure Connections Host Support is set to 'enabled' the Controller shall use the enhanced reporting mechanisms for the *Encryption_Enabled* parameter in the *HCI_Encryption_Change* event (see [Section 7.7.8](#)) and the *Key_Type* parameter in the *HCI_Link_Key_Notification* event (see [Section 7.7.24](#)).

Secure_Connections_Host_Support:

Size: 1 octet

Value	Parameter Description
0x00	<i>Secure_Connections_Host_Support</i> is 'disabled'. Host does not support Secure Connections (default)
0x01	<i>Secure_Connections_Host_Support</i> is 'enabled'. Host supports Secure Connections
All other values	Reserved for future use



6.40 AUTHENTICATED PAYLOAD TIMEOUT

The `Authenticated_Payload_Timeout` configuration parameter allows the Host to configure the maximum interval between packets containing a MIC received from the remote device when AES-CCM encryption is enabled.

Authenticated_Payload_Timeout:

Size: 2 octets

Value	Parameter Description
N = 0xXXXX	Maximum amount of time specified between packets authenticated by a MIC. Range: 0x0001 to 0xFFFF Time = N * 10 ms Time Range: 10 ms to 655,350 ms

6.41 EXTENDED PAGE TIMEOUT

The `Extended_Page_Timeout` configuration parameter together with `Page_Timeout` defines the maximum time the local Link Manager will wait for a baseband page response from the remote device at a locally initiated connection attempt. If this time expires and the remote device has not responded to the page at baseband level, the connection attempt will be considered to have failed.

Extended_Page_Timeout:

Size: 2 octets

Value	Parameter Description
N = 0xXXXX	Range: 0x0000 to 0xFFFF Default: 0x0000 Time = N * 0.625 ms Time Range: 0 to 40.9 s Time Default: 0 s

6.42 EXTENDED INQUIRY LENGTH

The `Extended_Inquiry_Length` configuration parameter together with `Inquiry_Length` defines the maximum time the local Link Manager will wait for a baseband inquiry response messages from the remote device at a locally initiated inquiry or periodic inquiry.

Extended_Inquiry_Length:

Size: 2 octets

Value	Parameter Description
N = 0xXXXX	Range: 0x0000 to 0xFFFF Default: 0x0000 Time = N * 0.625 ms Time Range: 0 to 40.9 s Time Default: 0 s



7 HCI COMMANDS AND EVENTS

7.1 LINK CONTROL COMMANDS

The Link Control commands allow a Controller to control connections to other BR/EDR Controllers. Some Link Control commands are used only with a BR/EDR Controller whereas other Link Control commands are used only with an AMP Controller.

In the BR/EDR Controller, when the Link Control commands are used, the Link Manager (LM) controls how the Bluetooth piconets and scatternets are established and maintained. These commands instruct the LM to create and modify Link Layer connections with Bluetooth remote devices, perform Inquiries of other BR/EDR Controllers in range, and other LMP commands.

In the AMP Controller, Link Control commands are used to create, modify and disconnect physical and logical links.

In the LE Controller, Link Control commands are used to disconnect physical links.

For the Link Control commands, the OGF is defined as 0x01.

7.1.1 Inquiry command

Command	OCF	Command Parameters	Return Parameters
HCI_Inquiry	0x0001	LAP, Inquiry_Length, Num_Responses	

Description:

This command causes the BR/EDR Controller to enter Inquiry Mode. Inquiry Mode is used to discover other nearby BR/EDR Controllers. The LAP input parameter contains the LAP from which the inquiry access code shall be derived when the inquiry procedure is made. The Inquiry_Length parameter specifies the total duration of the Inquiry Mode and, when this time expires, Inquiry will be halted. When Extended_Inquiry_Length is greater than zero, the duration of the Inquiry Mode may be changed to (Inquiry_Length + Extended_Inquiry_Length). The Num_Responses parameter specifies the number of responses that can be received before the Inquiry is halted. HCI_Inquiry_Result, HCI_Inquiry_Result_with_RSSI, or HCI_Extended_Inquiry_Result events will be sent to report the details of nearby BR/EDR Controllers that have responded to this inquiry. The HCI_Inquiry_Complete event is sent to report that Inquiry Mode has ended.



A device which responds during an inquiry or inquiry period should always be reported to the Host in an HCI_Inquiry_Result, HCI_Inquiry_Result_with_RSSI, or HCI_Extended_Inquiry_Result event if the device has not been reported earlier during the current inquiry or inquiry period and the device has not been filtered out using the command HCI_Set_Event_Filter. If the device has been reported earlier during the current inquiry or inquiry period, whether it is reported again will depend on the implementation (e.g. whether earlier results have been saved in the BR/EDR Controller and in that case how many responses have been saved). It is recommended that the BR/EDR Controller tries to report a particular device only once during an inquiry or inquiry period.

Command parameters:*LAP:**Size: 3 octets*

Value	Parameter Description
0xxxxxxx	The LAP from which the inquiry access code should be derived when the inquiry procedure is made; see Assigned Numbers . Range: 0x9E8B00 to 0x9E8B3F

*Inquiry_Length:**Size: 1 octet*

Value	Parameter Description
N = 0xXX	Maximum amount of time specified before the Inquiry is halted. Size: 1 octet Range: 0x01 to 0x30 Time = N * 1.28 s Range: 1.28 to 61.44 s

*Num_Responses:**Size: 1 octet*

Value	Parameter Description
0x00	Unlimited number of responses.
0xXX	Maximum number of responses from the Inquiry before the Inquiry is halted. Range: 0x01 to 0xFF

Return parameters:

None.

Event(s) generated (unless masked away):

An HCI_Command_Status event shall be sent from the BR/EDR Controller to the Host when the BR/EDR Controller has started the Inquiry process. Unless filtered, an HCI_Inquiry_Result, HCI_Inquiry_Result_with_RSSI, or HCI_Extended_Inquiry_Result event shall be created for each BR/EDR



Controller which responds to the Inquiry message. In addition, multiple BR/EDR Controllers which respond to the Inquire message may be combined into the same event. An HCI_Inquiry_Complete event shall be generated when the Inquiry process has completed.

Note: No HCI_Command_Complete event will be sent by the BR/EDR Controller to indicate that this command has been completed. Instead, the HCI_Inquiry_Complete event will indicate that this command has been completed. No HCI_Inquiry_Complete event will be generated for the cancelled Inquiry process.



7.1.2 Inquiry Cancel command

Command	OCF	Command Parameters	Return Parameters
HCI_Inquiry_Cancel	0x0002		Status

Description:

This command shall cause the BR/EDR Controller to stop the current Inquiry if the BR/EDR Controller is in Inquiry Mode. This command allows the Host to interrupt the BR/EDR Controller and request the BR/EDR Controller to perform a different task. The command should only be issued after the HCI_Inquiry command has been issued, an HCI_Command_Status event has been received for the Inquiry command, and before the HCI_Inquiry_Complete event occurs.

Command parameters:

None.

Return parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	HCI_Inquiry_Cancel command succeeded.
0x01-0xFF	HCI_Inquiry_Cancel command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Event(s) generated (unless masked away):

When the HCI_Inquiry_Cancel command has completed, an HCI_Command_Complete event shall be generated. No HCI_Inquiry_Complete event will be generated for the cancelled Inquiry process.



7.1.3 Periodic Inquiry Mode command

Command	OCF	Command Parameters	Return Parameters
HCI_Periodic_Inquiry_Mode	0x0003	Max_Period_Length, Min_Period_Length, LAP, Inquiry_Length, Num_Responses	Status

Description:

The HCI_Periodic_Inquiry_Mode command is used to configure the BR/EDR Controller to enter the Periodic Inquiry Mode that performs an automatic Inquiry. Max_Period_Length and Min_Period_Length define the time range between two consecutive inquiries, from the beginning of an inquiry until the start of the next inquiry. The BR/EDR Controller shall use this range to determine a new random time between two consecutive inquiries for each Inquiry. The LAP input parameter contains the LAP from which the inquiry access code shall be derived when the inquiry procedure is made. The Inquiry_Length parameter specifies the total duration of the Inquiry Mode and, when time expires, Inquiry will be halted. When Extended_Inquiry_Length is greater than zero, the duration of the Inquiry Mode may be changed to (Inquiry_Length + Extended_Inquiry_Length). The Num_Responses parameter specifies the number of responses that can be received before the Inquiry is halted. This command is completed when the Inquiry process has been started by the BR/EDR Controller, and an HCI_Command_Complete event is sent from the Controller to the Host. When each of the periodic Inquiry processes are completed, the Controller will send an HCI_Inquiry_Complete event to the Host indicating that the latest periodic Inquiry process has finished. When a BR/EDR Controller responds to the Inquiry message, an HCI_Inquiry_Result, HCI_Inquiry_Result_with_RSSI, or HCI_Extended_Inquiry_Result event will occur to notify the Host of the discovery.

Max_Period_Length shall be greater than Min_Period_Length.

Min_Period_Length shall be greater than (Inquiry_Length + Extended_Inquiry_Length).

A device which responds during an inquiry or inquiry period should always be reported to the Host in an HCI_Inquiry_Result, HCI_Inquiry_Result_with_RSSI, or HCI_Extended_Inquiry_Result event if the device has not been reported earlier during the current inquiry or inquiry period and the device has not been filtered out using the command HCI_Set_Event_Filter. If the device has been reported earlier during the current inquiry or inquiry period, whether it is reported again will depend on the implementation (e.g. whether earlier results have been saved in the BR/EDR Controller and in that case how many responses have been saved). It is



recommended that the BR/EDR Controller tries to report a particular device only once during an inquiry or inquiry period.

Command parameters:

Max_Period_Length:

Size: 2 octets

Value	Parameter Description
N = 0xXXXX	Maximum amount of time specified between consecutive inquiries. Size: 2 octets Range: 0x0003 to 0xFFFF Time = N * 1.28 s Range: 3.84 to 83884.8 s 0.0 to 23.3 hours

Min_Period_Length:

Size: 2 octets

Value	Parameter Description
N = 0xXXXX	Minimum amount of time specified between consecutive inquiries. Size: 2 octets Range: 0x0002 to 0xFFFE Time = N * 1.28 s Range: 2.56 to 83883.52 s 0.0 to 23.3 hours

LAP:

Size: 3 octets

Value	Parameter Description
0xXXXXXX	The LAP from which the inquiry access code should be derived when the inquiry procedure is made; see Assigned Numbers . Range: 0x9E8B00 to 0x9E8B3F

Inquiry_Length:

Size: 1 octet

Value	Parameter Description
N = 0xXX	Maximum amount of time specified before the Inquiry is halted. Size: 1 octet Range: 0x01 to 0x30 Time = N * 1.28 s Range: 1.28 to 61.44 s

Num_Responses:

Size: 1 octet

Value	Parameter Description
0x00	Unlimited number of responses.



Value	Parameter Description
0xXX	Maximum number of responses from the Inquiry before the Inquiry is halted. Range: 0x01 to 0xFF

Return parameters:*Status:**Size: 1 octet*

Value	Parameter Description
0x00	HCI_Periodic_Inquiry_Mode command succeeded.
0x01 to 0xFF	HCI_Periodic_Inquiry_Mode command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Event(s) generated (unless masked away):

The HCI Periodic Inquiry Mode begins when the BR/EDR Controller sends the HCI_Command_Complete event for this command to the Host. Unless filtered, an HCI_Inquiry_Result, HCI_Inquiry_Result_with_RSSI, or HCI_Extended_Inquiry_Result event shall be created for each remote device that has responded to the Inquiry message. In addition, multiple BR/EDR Controllers which response to the Inquiry message may be combined into the same event. An HCI_Inquiry_Complete event shall be generated when each of the periodic Inquiry processes has completed. No HCI_Inquiry_Complete event will be generated for the cancelled Inquiry process.



7.1.4 Exit Periodic Inquiry Mode command

Command	OCF	Command Parameters	Return Parameters
HCI_Exit_Periodic_Inquiry_Mode	0x0004		Status

Description:

The HCI_Exit_Periodic_Inquiry_Mode command is used to end the Periodic Inquiry mode when the local device is in Periodic Inquiry Mode. If the BR/EDR Controller is currently in an Inquiry process, the Inquiry process shall be stopped directly and the BR/EDR Controller shall no longer perform periodic inquiries until the HCI_Periodic_Inquiry_Mode command is reissued.

Command parameters:

None.

Return parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	HCI_Exit_Periodic_Inquiry_Mode command succeeded.
0x01-0xFF	HCI_Exit_Periodic_Inquiry_Mode command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Event(s) generated (unless masked away):

An HCI_Command_Complete event for this command shall occur when the local device is no longer in Periodic Inquiry Mode. No HCI_Inquiry_Complete event will be generated for the cancelled Inquiry process.



7.1.5 Create Connection command

Command	OCF	Command Parameters	Return Parameters
HCI_Create_Connection	0x0005	BD_ADDR, Packet_Type, Page_Scan_Repetition_Mode, Reserved, Clock_Offset, Allow_Role_Switch	

Description:

This command causes the Link Manager to create a connection to the remote device with the BD_ADDR specified by the command parameters. This command causes the local BR/EDR Controller to begin the Page process to create a link level connection. The Link Manager will determine how the new ACL connection is established. This ACL connection is determined by the current state of the device, its piconet, and the state of the device to be connected. The Packet_Type command parameter specifies which packet types the Link Manager shall use for the ACL connection. When sending HCI ACL Data packets the Link Manager shall only use the packet type(s) specified by the Packet_Type command parameter or the always-allowed DM1 packet type. Multiple packet types may be specified for the Packet Type parameter by performing a bit-wise OR operation of the different packet types. The Link Manager may choose which packet type to be used from the list of acceptable packet types. The Page_Scan_Repetition_Mode parameter specifies the Page Scan Repetition mode supported by the remote device with the BD_ADDR. This is the most recent version of the information that was acquired either during the inquiry process or from an HCI_Page_Scan_Repetition_Mode_Change event (see [Section 7.7.31](#)). The Clock_Offset parameter is the difference between its own clock and the clock of the remote device with BD_ADDR. Only bits 2 through 16 of the difference are used, and they are mapped to this parameter as bits 0 through 14 respectively. A Clock_Offset_Valid_Flag, located in bit 15 of the Clock_Offset parameter, is used to indicate if the Clock Offset is valid or not. A Connection_Handle for this connection is returned in the HCI_Connection_Complete event (see below). The Allow_Role_Switch parameter specifies if the local device accepts or rejects the request of a master-slave role switch when the remote device requests it at the connection setup (in the Role parameter of the HCI_Accept_Connection_Request command) (before the local Controller returns an HCI_Connection_Complete event). For a definition of the different packet types see the [\[Vol 2\] Part B, Baseband Specification](#).



Note: The Host should enable as many packet types as possible for the Link Manager to perform efficiently. However, the Host must not enable packet types that the local device does not support.

Command parameters:

BD_ADDR:

Size: 6 octets

Value	Parameter Description
0XXXXXXXXXXXXX	BD_ADDR of the Device to be connected.

Packet_Type:

Size: 2 octets

Bit Number	Parameter Description
1	2-DH1 shall not be used.
2	3-DH1 shall not be used.
3 ¹	DM1 may be used.
4	DH1 may be used.
8	2-DH3 shall not be used.
9	3-DH3 shall not be used.
10	DM3 may be used.
11	DH3 may be used.
12	2-DH5 shall not be used.
13	3-DH5 shall not be used.
14	DM5 may be used.
15	DH5 may be used.
All other bits	Reserved for future use.

1. This bit will be interpreted as set to 1 by Bluetooth V1.2 or higher Controllers.

Page_Scan_Repetition_Mode:

Size: 1 octet

Value	Parameter Description
0x00	R0
0x01	R1
0x02	R2
All other values	Reserved for future use.

**Reserved:****Size: 1 octet**

Value	Parameter Description
0x00	Reserved, must be set to 0x00. See Section A.10.2 .

Clock_Offset:**Size: 2 octets**

Bit format	Parameter Description
Bits 14-0	Bits 16-2 of CLKNslave-CLK
Bit 15	Clock_Offset_Valid_Flag Invalid Clock Offset = 0 Valid Clock Offset = 1

Allow_Role_Switch:**Size: 1 octet**

Value	Parameter Description
0x00	The local device will be a master, and will not accept a role switch requested by the remote device at the connection setup.
0x01	The local device may be a master, or may become a slave after accepting a role switch requested by the remote device at the connection setup.
All other values	Reserved for future use.

Return parameters:

None.

Event(s) generated (unless masked away):

When the BR/EDR Controller receives the HCI_Create_Connection command, the BR/EDR Controller shall send the HCI_Command_Status event to the Host. In addition, when the Link Manager determines the connection is established, the BR/EDR Controller, on both BR/EDR Controllers that form the connection, shall send an HCI_Connection_Complete event to each Host. The HCI_Connection_Complete event contains the Connection_Handle if this command is successful.

Note: No HCI_Command_Complete event will be sent by the Controller to indicate that this command has been completed. Instead, the HCI_Connection_Complete event will indicate that this command has been completed.



7.1.6 Disconnect command

Command	OCF	Command Parameters	Return Parameters
HCI_Disconnect	0x0006	Connection_Handle, Reason	

Description:

The HCI_Disconnect command is used to terminate an existing connection. The Connection_Handle command parameter indicates which connection is to be disconnected. The Reason command parameter indicates the reason for ending the connection and is copied into the error code field of the LMP_detach PDU on a BR/EDR connection or the error code field of the LL_TERMINATE_IND PDU on an LE connection. All synchronous connections on a physical link should be disconnected before the ACL connection on the same physical connection is disconnected.

Command parameters:

Connection_Handle:

Size: 2 octets (12 bits meaningful)

Value	Parameter Description
0xXXXX	Connection_Handle Range: 0x0000 to 0x0EFF

Reason:

Size: 1 octet

Value	Parameter Description
0x05, 0x13 to 0x15, 0x1A, 0x29, 0x3B	<i>Authentication Failure</i> error code (0x05), <i>Other End Terminated Connection</i> error codes (0x13 to 0x15), <i>Unsupported Remote Feature</i> error code (0x1A), <i>Pairing with Unit Key Not Supported</i> error code (0x29) and <i>Unacceptable Connection Parameters</i> error code (0x3B), see [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Return parameters:

None.

Event(s) generated (unless masked away):

When the Controller receives the HCI_Disconnect command, it shall send the HCI_Command_Status event to the Host. The HCI_Disconnection_Complete event will occur at each Host when the termination of the connection has completed, and on the local Host also indicates that this command has been completed. The Reason event parameter in the event on the local Host shall be set to the value *Connection Terminated by Local Host* (0x16), while that on the remote Host shall be set to the value of the Reason command parameter. However, if the termination procedure completes because a timer expires and,



therefore, the local Controller cannot determine whether or not the Reason command parameter was received by the remote Controller, the Reason event parameter on the local Host should instead be set to the value *LMP Response Timeout / LL Response Timeout* (0x22).

Note: No HCI_Command_Complete event will be sent by the Controller to indicate that this command has been completed. Instead, the HCI_Disconnection_Complete event will indicate that this command has been completed.



7.1.7 Create Connection Cancel command

Command	OCF	Command Parameters	Return Parameters
HCI_Create_ - Connection_Cancel	0x0008	BD_ADDR	Status, BD_ADDR

Description:

This command is used to request cancellation of the ongoing connection creation process, which was started by an HCI_Create_Connection command of the local BR/EDR Controller.

Command parameters:

BD_ADDR:

Size: 6 octets

Value	Parameter Description
0xFFFFFFFFXXXX	BD_ADDR of the HCI_Create_Connection command request that was issued before and is subject of this cancellation request

Return parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	HCI_Create_Connection_Cancel command succeeded
0x01-0xff	HCI_Create_Connection_Cancel command failed. See [Vol 2] Part D, Error Codes for list of error codes

BD_ADDR:

Size: 6 octets

Value	Parameter Description
0xFFFFFFFFXXXX	BD_ADDR of the HCI_Create_Connection command that was issued before and is the subject of this cancellation request.

Event(s) generated (unless masked away):

When the HCI_Create_Connection_Cancel command has completed, an HCI_Command_Complete event shall be generated.

If the connection is already established by the baseband, but the BR/EDR Controller has not yet sent the HCI_Connection_Complete event, then the local device shall detach the link and return an HCI_Command_Complete event with the status "Success".

If the connection is already established, and the HCI_Connection_Complete event has been sent, then the Controller shall return an



HCI_Command_Complete event with the error code *Connection Already Exists* (0x0B).

If the HCI_Create_Connection_Cancel command is sent to the Controller without a preceding HCI_Create_Connection command to the same device, the BR/EDR Controller shall return an HCI_Command_Complete event with the error code *Unknown Connection Identifier* (0x02).

The HCI_Connection_Complete event for the corresponding HCI_Create_Connection command shall always be sent. The HCI_Connection_Complete event shall be sent after the HCI_Command_Complete event for the HCI_Create_Connection_Cancel command. If the cancellation was successful, the HCI_Connection_Complete event will be generated with the error code *Unknown Connection Identifier* (0x02).



7.1.8 Accept Connection Request command

Command	OCF	Command Parameters	Return Parameters
HCI_Accept_Connection_Request	0x0009	BD_ADDR, Role	

Description:

The HCI_Accept_Connection_Request command is used to accept a new incoming connection request. The HCI_Accept_Connection_Request command shall only be issued after an HCI_Connection_Request event has occurred. The HCI_Connection_Request event will return the BD_ADDR of the device which is requesting the connection. This command will cause the Link Manager to create a connection to the BR/EDR Controller, with the BD_ADDR specified by the command parameters. The Link Manager will determine how the new connection will be established. This will be determined by the current state of the device, its piconet, and the state of the device to be connected. The Role command parameter allows the Host to specify if the Link Manager shall request a role switch and become the Master for this connection. This is a preference and not a requirement. If the Role Switch fails then the connection will still be accepted, and the HCI_Role_Discovery command will reflect the current role.

Note: The Link Manager may terminate the connection if it would be low on resources if the role switch fails. The decision to accept a connection must be completed before the connection accept timeout expires on the local Bluetooth Module.

Note: When accepting synchronous connection request, the Role parameter is not used and will be ignored by the BR/EDR Controller.

Note: See [Section 7.3.3](#) for the behavior when the HCI_Connection_Request event is masked or the connection is auto accepted.

Command parameters:

BD_ADDR: Size: 6 octets

Value	Parameter Description
0xFFFFFFFFXXXX	BD_ADDR of the Device to be connected

Role: Size: 1 octet

Value	Parameter Description
0x00	Become the Master for this connection. The LM will perform the role switch.



Value	Parameter Description
0x01	Remain the Slave for this connection. The LM will NOT perform the role switch.

Return parameters:

None.

Event(s) generated (unless masked away):

The HCI_Accept_Connection_Request command shall cause the HCI_Command_Status event to be sent from the BR/EDR Controller when the BR/EDR Controller begins setting up the connection. In addition, when the Link Manager determines the connection is established, the local BR/EDR Controller shall send an HCI_Connection_Complete event to its Host, and the remote Controller will send an HCI_Connection_Complete event or an HCI_Synchronous_Connection_Complete event to the Host. The HCI_Connection_Complete event contains the Connection_Handle if this command is successful.

Note: No HCI_Command_Complete event will be sent by the BR/EDR Controller to indicate that this command has been completed. Instead, the HCI_Connection_Complete event will indicate that this command has been completed.



7.1.9 Reject Connection Request command

Command	OCF	Command Parameters	Return Parameters
HCI_Reject_Connection_Request	0x000A	BD_ADDR, Reason	

Description:

The HCI_Reject_Connection_Request command is used to decline a new incoming connection request. The HCI_Reject_Connection_Request command shall only be called after an HCI_Connection_Request event has occurred. The HCI_Connection_Request event will return the BD_ADDR of the device that is requesting the connection. The Reason command parameter will be returned to the connecting device in the Status parameter of the HCI_Connection_Complete event returned to the Host of the connection device, to indicate why the connection was declined.

Command parameters:

BD_ADDR:

Size: 6 octets

Value	Parameter Description
0XXXXXXXXXXXXX	BD_ADDR of the Device to reject the connection from.

Reason:

Size: 1 octet

Value	Parameter Description
0x0D-0x0F	Host Reject error code. See [Vol 2] Part D, Error Codes for list of error codes and descriptions.

Return parameters:

None.

Event(s) generated (unless masked away):

When the Controller receives the HCI_Reject_Connection_Request command, the Controller shall send the HCI_Command_Status event to the Host. Then, the local BR/EDR Controller will send an HCI_Connection_Complete event to its Host, and the remote device shall send an HCI_Connection_Complete event or an HCI_Synchronous_Connection_Complete event to the Host. The Status parameter of the HCI_Connection_Complete event, which is sent to the Host of the device attempting to make the connection, will contain the Reason command parameter from this command.

Note: No HCI_Command_Complete event will be sent by the BR/EDR Controller to indicate that this command has been completed. Instead, the



HCI_Connection_Complete event will indicate that this command has been completed.



7.1.10 Link Key Request Reply command

Command	OCF	Command Parameters	Return Parameters
HCI_Link_Key_Request_Reply	0x000B	BD_ADDR, Link_Key	Status, BD_ADDR

Description:

The HCI_Link_Key_Request_Reply command is used to reply to an HCI_Link_Key_Request event from the Controller, and specifies the Link Key stored on the Host to be used as the link key for the connection with the other BR/EDR Controller specified by BD_ADDR. The HCI_Link_Key_Request event will be generated when the BR/EDR Controller needs a Link Key for a connection.

When the BR/EDR Controller generates an HCI_Link_Key_Request event in order for the local Link Manager to respond to the request from the remote Link Manager (as a result of an HCI_Create_Connection or HCI_Authentication_Requested command from the remote Host), the local Host shall respond with either an HCI_Link_Key_Request_Reply or HCI_Link_Key_Request_Negative_Reply command before the remote Link Manager detects LMP response timeout. (See [\[Vol 2\] Part C, Link Manager Protocol Specification](#).)

When the BR/EDR Controller supports the Secure Connections (Controller Support) feature, it shall discard the Link Key once the connection has been disconnected.

Command parameters:

BD_ADDR:

Size: 6 octets

Value	Parameter Description
0XXXXXXXXXXXXX	BD_ADDR of the Device of which the Link Key is for.

Link_Key:

Size: 16 octets

Value	Parameter Description
0XXXXXXXXXXXXX XXXXXXXXXXXXX XXXXXXXXXXXXX	Link Key for the associated BD_ADDR.

**Return parameters:***Status:**Size: 1 octet*

Value	Parameter Description
0x00	HCI_Link_Key_Request_Reply command succeeded.
0x01-0xFF	HCI_Link_Key_Request_Reply command failed. See [Vol 2] Part D, Error Codes for error codes and descriptions.

*BD_ADDR:**Size: 6 octets*

Value	Parameter Description
0XXXXXXXXX XXXX	BD_ADDR of the Device of which the Link Key request reply has completed.

Event(s) generated (unless masked away):

When the HCI_Link_Key_Request_Reply command has completed, an HCI_Command_Complete event shall be generated.



7.1.11 Link Key Request Negative Reply command

Command	OCF	Command Parameters	Return Parameters
HCI_Link_Key_Request_Negative_Reply	0x000C	BD_ADDR	Status, BD_ADDR

Description:

The HCI_Link_Key_Request_Negative_Reply command is used to reply to an HCI_Link_Key_Request event from the BR/EDR Controller if the Host does not have a stored Link Key for the connection with the other BR/EDR Controller specified by BD_ADDR. The HCI_Link_Key_Request event will be generated when the BR/EDR Controller needs a Link Key for a connection.

When the Controller generates an HCI_Link_Key_Request event in order for the local Link Manager to respond to the request from the remote Link Manager (as a result of an HCI_Create_Connection or HCI_Authentication_Requested command from the remote Host), the local Host shall respond with either an HCI_Link_Key_Request_Reply or HCI_Link_Key_Request_Negative_Reply command before the remote Link Manager detects LMP response timeout. (See [\[Vol 2\] Part C, Link Manager Protocol Specification](#).)

Command parameters:

BD_ADDR:

Size: 6 octets

Value	Parameter Description
0XXXXXXXXX XX	BD_ADDR of the Device which the Link Key is for.

Return parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	HCI_Link_Key_Request_Negative_Reply command succeeded.
0x01-0xFF	HCI_Link_Key_Request_Negative_Reply command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

BD_ADDR:

Size: 6 octets

Value	Parameter Description
0XXXXXXXXX XXXX	BD_ADDR of the Device for which the HCI_Link_Key_Request_Negative_Reply command has completed.

**Event(s) generated (unless masked away):**

When the HCI_Link_Key_Request_Negative_Reply command has completed, an HCI_Command_Complete event shall be generated.



7.1.12 PIN Code Request Reply command

Command	OCF	Command Parameters	Return Parameters
HCI_PIN_Code_Request_Reply	0x000D	BD_ADDR, PIN_Code_Length, PIN_Code	Status, BD_ADDR

Description:

The HCI_PIN_Code_Request_Reply command is used to reply to an HCI_PIN_Code_Request event from the BR/EDR Controller, and specifies the PIN code to use for a connection. The HCI_PIN_Code_Request event will be generated when a connection with remote initiating device has requested pairing.

When the BR/EDR Controller generates an HCI_PIN_Code_Request event in order for the local Link Manager to respond to the request from the remote Link Manager (as a result of an HCI_Create_Connection or HCI_Authentication_Requested command from the remote Host), the local Host shall respond with either an HCI_PIN_Code_Request_Reply or HCI_PIN_Code_Request_Negative_Reply command before the remote Link Manager detects LMP response timeout. (See [\[Vol 2\] Part C, Link Manager Protocol Specification](#).)

Command parameters:

BD_ADDR: Size: 6 octets

Value	Parameter Description
0XXXXXXXXXX XX	BD_ADDR of the Device which the PIN code is for.

PIN_Code_Length: Size: 1 octet

Value	Parameter Description
0xXX	The PIN code length specifies the length, in octets, of the PIN code to be used. Range: 0x01 to 0x10

**PIN_Code:****Size: 16 octets**

Value	Parameter Description
0xFFFFFFFFXXXXXXX	PIN code for the device that is to be connected. The Host should ensure that strong PIN Codes are used. PIN Codes can be up to a maximum of 128 bits. The first octet of the PIN code should be transmitted first. Note: The PIN_Code Parameter is a string parameter. Endianness does therefore not apply to the PIN_Code Parameter.

Return parameters:**Status:****Size: 1 octet**

Value	Parameter Description
0x00	HCI_PIN_Code_Request_Reply command succeeded.
0x01-0xFF	HCI_PIN_Code_Request_Reply command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

BD_ADDR:**Size: 6 octets**

Value	Parameter Description
0xFFFFFFFFXXXX	BD_ADDR of the Device for which the HCI_PIN_Code_Request_Reply command has completed.

Event(s) generated (unless masked away):

When the HCI_PIN_Code_Request_Reply command has completed, an HCI_Command_Complete event shall be generated.



7.1.13 PIN Code Request Negative Reply command

Command	OCF	Command Parameters	Return Parameters
HCI_PIN_Code_Request_Negative_Reply	0x000E	BD_ADDR	Status, BD_ADDR

Description:

The HCI_PIN_Code_Request_Negative_Reply command is used to reply to a PIN Code request event from the BR/EDR Controller when the Host cannot specify a PIN code to use for a connection. This command will cause the pair request with remote device to fail.

When the BR/EDR Controller generates an HCI_PIN_Code_Request event in order for the local Link Manager to respond to the request from the remote Link Manager (as a result of an HCI_Create_Connection or HCI_Authentication_Requested command from the remote Host), the local Host shall respond with either an HCI_PIN_Code_Request_Reply or HCI_PIN_Code_Request_Negative_Reply command before the remote Link Manager detects LMP response timeout. (See [\[Vol 2\] Part C, Link Manager Protocol Specification](#).)

Command parameters:

BD_ADDR:

Size: 6 octets

Value	Parameter Description
0XXXXXXXXXXXXX	BD_ADDR of the Device which this command is responding to.

Return parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	HCI_PIN_Code_Request_Negative_Reply command succeeded.
0x01-0xFF	HCI_PIN_Code_Request_Negative_Reply command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

BD_ADDR:

Size: 6 octets

Value	Parameter Description
0XXXXXXXXXXXXX	BD_ADDR of the Device for which the HCI_PIN_Code_Request_Negative_Reply command has completed.

**Event(s) generated (unless masked away):**

When the HCI_PIN_Code_Request_Negative_Reply command has completed, an HCI_Command_Complete event shall be generated.



7.1.14 Change Connection Packet Type command

Command	OCF	Command Parameters	Return Parameters
HCI_Change_ Connection_Packet_Type	0x000F	Connection_Handle, Packet_Type	

Description:

The HCI_Change_Connection_Packet_Type command is used to change which packet types can be used for a connection that is currently established. This allows current connections to be dynamically modified to support different types of user data. The Packet_Type command parameter specifies which packet types the Link Manager can use for the connection. When sending HCI ACL Data packets the Link Manager shall only use the packet type(s) specified by the Packet_Type command parameter or the always-allowed DM1 packet type. The interpretation of the value for the Packet_Type command parameter will depend on the Link_Type command parameter returned in the HCI_Connection_Complete event at the connection setup. Multiple packet types may be specified for the Packet_Type command parameter by bitwise OR operation of the different packet types. For a definition of the different packet types see the [\[Vol 2\] Part B, Baseband Specification](#).

Note: The Host should enable as many packet types as possible for the Link Manager to perform efficiently. However, the Host must not enable packet types that the local device does not support.

Note: To change an eSCO connection, use the HCI_Setup_Synchronous_Connection command.

Command parameters:

Connection_Handle:

Size: 2 octets (12 bits meaningful)

Value	Parameter Description
0xFFFF	Connection_Handle Range: 0x0000 to 0x0EFF

*Packet_Type:**Size: 2 octets**For ACL Link_Type*

Bit Number	Parameter Description
1	2-DH1 shall not be used.
2	3-DH1 shall not be used.
3 ¹	DM1 may be used.
4	DH1 may be used.
8	2-DH3 shall not be used.
9	3-DH3 shall not be used.
10	DM3 may be used.
11	DH3 may be used.
12	2-DH5 shall not be used.
13	3-DH5 shall not be used.
14	DM5 may be used.
15	DH5 may be used.
All other bits	Reserved for future use.

1. This bit will be set to 1 by Bluetooth V1.2 or higher Controllers.

For SCO Link_Type

Bit Number	Parameter Description
5	HV1 may be used.
6	HV2 may be used.
7	HV3 may be used.
All other bits	Reserved for future use.

Return parameters:

None.

Event(s) generated (unless masked away):

When the BR/EDR Controller receives the HCI_Change_Connection_Packet_Type command, BR/EDR the Controller shall send the HCI_Command_Status event to the Host. In addition, when the Link Manager determines the packet type has been changed for the connection, the Controller on the local device will send an HCI_Connection_Packet_Type_Changed event to the Host. This will be done at the local side only.



Note: No HCI_Command_Complete event will be sent by the BR/EDR Controller to indicate that this command has been completed. Instead, the HCI_Connection_Packet_Type_Changed event will indicate that this command has been completed.



7.1.15 Authentication Requested command

Command	OCF	Command Parameters	Return Parameters
HCI_Authentication_Requested	0x0011	Connection_Handle	

Description:

The HCI_Authentication_Requested command is used to try to authenticate the remote device associated with the specified Connection_Handle. On an authentication failure, the BR/EDR Controller or Link Manager shall not automatically detach the link. The Host is responsible for issuing an HCI_Disconnect command to terminate the link if the action is appropriate.

Note: The Connection_Handle command parameter is used to identify the other BR/EDR Controller which forms the connection. The Connection_Handle should be a Connection_Handle for an ACL connection. The authentication will apply to all Connection_Handles with the same remote BR/EDR Controller.

Command parameters:

Connection_Handle:

Size: 2 octets (12 bits meaningful)

Value	Parameter Description
0xXXXX	Connection_Handle Range: 0x0000 to 0x0EFF

Return parameters:

None.

Event(s) generated (unless masked away):

When the Controller receives the HCI_Authentication_Requested command, it shall send the HCI_Command_Status event to the Host. If Simple Pairing Mode is enabled, the HCI_Link_Key_Request event shall be generated, and Simple Pairing shall be started only if Host replies to it with HCI_Link_Key_Request_Negative_Reply command; if Host replies to it with HCI_Link_Key_Request_Reply command, only authentication procedure (see [\[Vol 2\] Part C, Section 4.2.1](#)) shall be performed and no Simple Pairing shall be started. If Simple Pairing Mode is not enabled and authentication fails, the HCI_PIN_Code_Request event may be generated. The HCI_Authentication_Complete event will occur when the authentication has been completed for the connection and is indication that this command has been completed.

Note: No HCI_Command_Complete event will be sent by the BR/EDR Controller to indicate that this command has been completed. Instead, the



HCI_Authentication_Complete event will indicate that this command has been completed.

On receipt of an HCI_Authentication_Requested command, a local BR/EDR Controller that is not in Simple Pairing Mode may use an existing stored link key and respond immediately to the Host with an HCI_Authentication_Complete event.

This behavior is not recommended for Controllers as it offers the Host no method for enhancing the security of an existing link (e.g., in the case where a profile mandating a minimum passkey length is started over a link that is already authenticated with shorter passkey than the new service requires).



7.1.16 Set Connection Encryption command

Command	OCF	Command Parameters	Return Parameters
HCI_Set_Connection_Encryption	0x0013	Connection_Handle, Encryption_Enable	

Description:

The HCI_Set_Connection_Encryption command is used to enable and disable the link level encryption. The Connection_Handle command parameter is used to identify the other BR/EDR Controller which forms the connection. The Connection_Handle should be a Connection_Handle for an ACL connection. The encryption setting will apply to all Connection_Handles with the same remote BR/EDR Controller. While the encryption is being changed, the Link Manager will suspend all ACL-U traffic on the connection.

If both devices support both the Secure Connections (Controller Support) and Secure Connections (Host Support) features, and Encryption_Enable is set to Turn Link Level Encryption OFF when encryption is currently enabled on the specified Connection_Handle, the Controller shall return the error code *Encryption Mode Not Acceptable* (0x25).

Command parameters:

Connection_Handle: *Size: 2 octets (12 bits meaningful)*

Value	Parameter Description
0xXXXX	Connection_Handle Range: 0x0000 to 0x0EFF

Encryption_Enable: *Size: 1 octet*

Value	Parameter Description
0x00	Turn Link Level Encryption OFF.
0x01	Turn Link Level Encryption ON.

Return parameters:

None.

Event(s) generated (unless masked away):

When the BR/EDR Controller receives the HCI_Set_Connection_Encryption command, the BR/EDR Controller shall send the HCI_Command_Status event to the Host. When the Link Manager has completed enabling/disabling encryption for the connection, the local BR/EDR Controller shall send an



HCI_Encryption_Change event to the Host, and the BR/EDR Controller on the remote device will also generate an HCI_Encryption_Change event.

Note: No HCI_Command_Complete event will be sent by the Controller to indicate that this command has been completed. Instead, the HCI_Encryption_Change event will indicate that this command has been completed.



7.1.17 Change Connection Link Key command

Command	OCF	Command Parameters	Return Parameters
HCI_Change_Connection_Link_Key	0x0015	Connection_Handle	

Description:

The HCI_Change_Connection_Link_Key command is used to force both devices of a connection associated with the Connection_Handle to generate a new link key. The link key is used for authentication and encryption of connections.

Note: The Connection_Handle command parameter is used to identify the other BR/EDR Controller forming the connection. The Connection_Handle should be a Connection_Handle for an ACL connection.

Note: The resulting link key, generated as a result of HCI_Change_Connection_Link_Key command, will be of equal link key strength to the previously used link key.

Command parameters:

Connection_Handle:

Size: 2 octets (12 bits meaningful)

Value	Parameter Description
0xXXXX	Connection_Handle Range: 0x0000 to 0x0EFF

Return parameters:

None.

Event(s) generated (unless masked away):

When the Controller receives the HCI_Change_Connection_Link_Key command, the BR/EDR Controller shall send the HCI_Command_Status event to the Host. When the Link Manager has changed the Link Key for the connection, the local BR/EDR Controller shall send an HCI_Link_Key_Notification event and an HCI_Change_Connection_Link_Key_Complete event to the Host, and the remote BR/EDR Controller will also generate an HCI_Link_Key_Notification event. The HCI_Link_Key_Notification event indicates that a new connection link key is valid for the connection.

Note: No HCI_Command_Complete event will be sent by the BR/EDR Controller to indicate that this command has been completed. Instead, the



HCI_Change_Connection_Link_Key_Complete event will indicate that this command has been completed.



7.1.18 Master Link Key command

Command	OCF	Command Parameters	Return Parameters
HCI_Master_Link_Key	0x0017	Key_Flag	

Description:

The HCI_Master_Link_Key command is used to force the device that is master of the piconet to use the temporary link key of the master device, or the semi-permanent link keys. The temporary link key is used for encryption of broadcast messages within a piconet, and the semi-permanent link keys are used for private encrypted point-to-point communication. The Key_Flag command parameter is used to indicate which Link Key (temporary link key of the Master, or the semi-permanent link keys) shall be used.

The HCI_Master_Link_Key command shall be rejected with error code *Command Disallowed* (0x0C) when all slaves in the piconet support AES-CCM encryption and Key_Flag is set to “Use Temporary Link Key”.

Note: When at least one slave in the piconet cannot support AES-CCM encryption, encrypted broadcast packets will not be received by slave devices where both the Controller and Host support Secure Connections. When all slaves in the piconet support AES-CCM encryption, broadcast packets will not be encrypted and may be received by slaves that have AES-CCM encryption enabled.

Command parameters:

Key_Flag:

Size: 1 octet

Value	Parameter Description
0x00	Use semi-permanent Link Keys.
0x01	Use Temporary Link Key.

Return parameters:

None.

Event(s) generated (unless masked away):

When the BR/EDR Controller receives the HCI_Master_Link_Key command, the BR/EDR Controller shall send the HCI_Command_Status event to the Host. When the Link Manager has changed link key, the BR/EDR Controller on both the local and the remote device shall send an HCI_Master_Link_Key_Complete event to the Host. The Connection_Handle on the master side shall be a Connection_Handle for one of the existing connections to a slave. On the slave side, the Connection_Handle shall be a Connection_Handle to the initiating master.



The HCI_Master_Link_Key_Complete event contains the status of this command.

Note: No HCI_Command_Complete event will be sent by the BR/EDR Controller to indicate that this command has been completed. Instead, the HCI_Master_Link_Key_Complete event will indicate that this command has been completed.



7.1.19 Remote Name Request command

Command	OCF	Command Parameters	Return Parameters
HCI_Remote_Name_Request	0x0019	BD_ADDR, Page_Scan_Repetition_Mode, Reserved, Clock_Offset	

Description:

The HCI_Remote_Name_Request command is used to obtain the user-friendly name of another BR/EDR Controller. The user-friendly name is used to enable the user to distinguish one BR/EDR Controller from another. The BD_ADDR command parameter is used to identify the device for which the user-friendly name is to be obtained. The Page_Scan_Repetition_Mode parameter specifies the Page Scan Repetition mode supported by the remote device with the BD_ADDR. This is the most recent version of the information that was acquired either during the inquiry process or from an HCI_Page_Scan_Repetition_Mode_Change event (see [Section 7.7.31](#)). The Clock_Offset parameter is the difference between its own clock and the clock of the remote device with BD_ADDR. Only bits 2 through 16 of the difference are used and they are mapped to this parameter as bits 0 through 14 respectively. A Clock_Offset_Valid_Flag, located in bit 15 of the Clock_Offset command parameter, is used to indicate if the Clock Offset is valid or not.

When the HCI_Remote_Supported_Host_Features_Notification event is unmasked and when the HCI_Remote_Name_Request command initiates a connection, the Link Manager shall read the remote LMP features mask pages 0 and 1.

Note: If no connection exists between the local device and the device corresponding to the BD_ADDR, a temporary Link Layer connection will be established to obtain the LMP features and name of the remote device.

Command parameters:

BD_ADDR:

Size: 6 octets

Value	Parameter Description
0XXXXXXXXXX XX	BD_ADDR for the device whose name is requested.

**Page_Scan_Repetition_Mode:****Size: 1 octet**

Value	Parameter Description
0x00	R0
0x01	R1
0x02	R2
All other values	Reserved for future use.

Reserved:**Size: 1 octet**

Value	Parameter Description
0x00	Reserved, must be set to 0x00. See Section A.10.2 .

Clock_Offset:**Size: 2 octets**

Bit format	Parameter Description
Bits 14-0	Bits 16-2 of CLKNSlave-CLK
Bit 15	Clock_Offset_Valid_Flag Invalid Clock Offset = 0 Valid Clock Offset = 1

Return parameters:

None.

Event(s) generated (unless masked away):

When the BR/EDR Controller receives the HCI_Remote_Name_Request command, the BR/EDR Controller shall send the HCI_Command_Status event to the Host. If a temporary Link Layer connection was established, then when the Link Manager has completed the LMP sequence to obtain the remote Host supported features, if present, the BR/EDR Controller on the local device shall send an HCI_Remote_Host_Supported_Features_Notification event. When the Link Manager has completed the LMP messages to obtain the remote name, the local BR/EDR Controller shall send an HCI_Remote_Name_Request_Complete event to the Host. If the remote Host supported features page is present, the HCI_Remote_Host_Supported_Features_Notification event shall be sent before the HCI_Remote_Name_Request_Complete event. If not, only the HCI_Remote_Name_Request_Complete event shall be sent.

Note: No HCI_Command_Complete event shall be sent by the BR/EDR Controller to indicate that this command has been completed. Instead, only the HCI_Remote_Name_Request_Complete event will indicate that this command has been completed.



7.1.20 Remote Name Request Cancel command

Command	OCF	Command Parameters	Return Parameters
HCI_Remote_Name_Request_Cancel	0x001A	BD_ADDR	Status, BD_ADDR

Description:

This command is used to request cancellation of the ongoing remote name request process, which was started by the HCI_Remote_Name_Request command.

Command parameters:

BD_ADDR:

Size: 6 octets

Value	Parameter Description
0xFFFFFFFFXXXX	BD_ADDR of the HCI_Remote_Name_Request command that was issued before and that is subject of this cancellation request

Return parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	HCI_Remote_Name_Request_Cancel command succeeded
0x01-0xff	HCI_Remote_Name_Request_Cancel command failed. See [Vol 2] Part D, Error Codes for list of error codes

BD_ADDR:

Size: 6 octets

Value	Parameter Description
0xFFFFFFFFXXXX	BD_ADDR of the HCI_Remote_Name_Request_Cancel command that was issued before and that was subject of this cancellation request

Event(s) generated (unless masked away):

When the HCI_Remote_Name_Request_Cancel command has completed, an HCI_Command_Complete event shall be generated.

If the HCI_Remote_Name_Request_Cancel command is sent to the BR/EDR Controller without a preceding HCI_Remote_Name_Request command to the same device, the Controller shall return an HCI_Command_Complete event with the error code *Invalid HCI Command Parameters* (0x12).



The HCI_Remote_Name_Request_Complete event for the corresponding HCI_Remote_Name_Request command shall always be sent. The HCI_Remote_Name_Request_Complete event shall be sent after the HCI_Command_Complete event for the HCI_Remote_Name_Request_Cancel command. If the cancellation was successful, the HCI_Remote_Name_Request_Complete event shall be generated with the error code *Unknown Connection Identifier* (0x02).



7.1.21 Read Remote Supported Features command

Command	OCF	Command Parameters	Return Parameters
HCI_Read_Remote_Supported_Features	0x001B	Connection_Handle	

Description:

This command requests a list of the supported features for the remote device identified by the Connection_Handle parameter. The Connection_Handle shall be a Connection_Handle for an ACL-U logical link. The HCI_Read_Remote_Supported_Features_Complete event will return a list of the LMP features. For details see [\[Vol 2\] Part C, Link Manager Protocol Specification](#).

Command parameters:

Connection_Handle: *Size: 2 octets (12 bits meaningful)*

Value	Parameter Description
0xXXXX	Connection_Handle Range: 0x0000 to 0x0EFF

Return parameters:

None.

Event(s) generated (unless masked away):

When the BR/EDR Controller receives the HCI_Read_Remote_Supported_Features command, the BR/EDR Controller shall send the HCI_Command_Status event to the Host. When the Link Manager has completed the LMP sequence to determine the remote features or has determined that it will be using a cached copy, the BR/EDR Controller on the local device shall send an HCI_Read_Remote_Supported_Features_Complete event to the Host. The HCI_Read_Remote_Supported_Features_Complete event contains the status of this command, and parameters describing the supported features of the remote device.

Note: No HCI_Command_Complete event will be sent by the BR/EDR Controller to indicate that this command has been completed. Instead, the HCI_Read_Remote_Supported_Features_Complete event will indicate that this command has been completed.



7.1.22 Read Remote Extended Features command

Command	OCF	Command Parameters	Return Parameters
HCI_Read_Remote_Extended_Features	0x001C	Connection_Handle, Page Number	

Description:

The HCI_Read_Remote_Extended_Features command returns the requested page of the extended LMP features for the remote device identified by the specified Connection_Handle. The Connection_Handle shall be the Connection_Handle for an ACL-U logical link. This command is only available if the extended features feature is implemented by the remote device. The HCI_Read_Remote_Extended_Features_Complete event will return the requested information. For details see [\[Vol 2\] Part C, Link Manager Protocol Specification](#).

Command parameters:

Connection_Handle:

Size: 2 octets (12 bits meaningful)

Value	Parameter Description
0xFFFF	Connection_Handle Range: 0x0000 to 0x0EFF

Page Number:

Size: 1 octet

Value	Parameter Description
0x00	Requests the normal LMP features as returned by the HCI_Read_Remote_Supported_Features command
0x01-0xFF	Return the corresponding page of features

Return parameters:

None.

Event(s) generated (unless masked away):

When the BR/EDR Controller receives the HCI_Read_Remote_Extended_Features command the BR/EDR Controller shall send the HCI_Command_Status event to the Host. When the Link Manager has completed the LMP sequence to determine the remote extended features or has determined that it will be using a cached copy, the Controller on the local device shall generate an HCI_Read_Remote_Extended_Features_Complete event to the Host. The



HCI_Read_Remote_Extended_Features_Complete event contains the page number and the remote features returned by the remote device.

Note: No HCI_Command_Complete event will ever be sent by the BR/EDR Controller to indicate that this command has been completed. Instead the HCI_Read_Remote_Extended_Features_Complete event will indicate that this command has been completed.



7.1.23 Read Remote Version Information command

Command	OCF	Command Parameters	Return Parameters
HCI_Read_Remote_Version_Information	0x001D	Connection_Handle	

Description:

This command will obtain the values for the version information for the remote device identified by the Connection_Handle parameter. The Connection_Handle shall be a Connection_Handle for an ACL-U or LE-U logical link.

Command parameters:

Connection_Handle:

Size: 2 octets (12 bits meaningful)

Value	Parameter Description
0xXXXX	Connection_Handle Range: 0x0000 to 0x0EFF

Return parameters:

None.

Event(s) generated (unless masked away):

When the Controller receives the HCI_Read_Remote_Version_Information command, the Controller shall send the HCI_Command_Status event to the Host. When the Link Manager or Link Layer has completed the sequence to determine the remote version information, the local Controller shall send an HCI_Read_Remote_Version_Information_Complete event to the Host. The HCI_Read_Remote_Version_Information_Complete event contains the status of this command, and parameters describing the version and subversion of the LMP or Link Layer used by the remote device.

Note: No HCI_Command_Complete event will be sent by the Controller to indicate that this command has been completed. Instead, the HCI_Read_Remote_Version_Information_Complete event will indicate that this command has been completed.



7.1.24 Read Clock Offset command

Command	OCF	Command Parameters	Return Parameters
HCI_Read_Clock_Offset	0x001F	Connection_Handle	

Description:

Both the System Clock and the clock offset to a remote device are used to determine what hopping frequency is used by a remote device for page scan. This command allows the Host to read the clock offset of remote devices. The clock offset can be used to speed up the paging procedure when the local device tries to establish a connection to a remote device, for example, when the local Host has issued an HCI_Create_Connection or HCI_Remote_Name_Request command. The Connection_Handle shall be a Connection_Handle for an ACL-U logical link.

Command parameters:

Connection_Handle:

Size: 2 octets (12 bits meaningful)

Value	Parameter Description
0xXXXX	Connection_Handle Range: 0x0000 to 0x0EFF

Return parameters:

None.

Event(s) generated (unless masked away):

When the BR/EDR Controller receives the HCI_Read_Clock_Offset command, the BR/EDR Controller shall send the HCI_Command_Status event to the Host. If this command was requested at the master and the Link Manager has completed the LMP messages to obtain the Clock Offset information, the BR/EDR Controller on the local BR/EDR Controller shall send an HCI_Read_Clock_Offset_Complete event to the Host.

Note: No HCI_Command_Complete event will be sent by the BR/EDR Controller to indicate that this command has been completed. Instead, only the HCI_Read_Clock_Offset_Complete event will indicate that this command has been completed. If the command is requested at the slave, the LM will immediately send an HCI_Command_Status event and an HCI_Read_Clock_Offset_Complete event to the Host, without an exchange of LMP PDU.



7.1.25 Read LMP Handle command

Command	OCF	Command Parameters	Return Parameters
HCI_Read_LMP_Handle	0x0020	Connection_Handle	Status, Connection_Handle, LMP_Handle, Reserved

Description:

This command reads the current LMP Handle associated with the Connection_Handle. The Connection_Handle shall be a SCO or eSCO Handle. If the Connection_Handle is a SCO Connection_Handle, then this command shall read the LMP SCO Handle for this connection. If the Connection_Handle is an eSCO Connection_Handle, then this command shall read the LMP eSCO Handle for this connection.

Command parameters:

Connection_Handle:

Size: 2 octets (12 bits meaningful)

Value	Parameter Description
0xFFFF	Connection_Handle Range: 0x0000 to 0x0EFF

Return parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	HCI_Read_LMP_Handle command succeeded.
0x01 – 0xFF	HCI_Read_LMP_Handle command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Connection_Handle:

Size: 2 octets (12 bits meaningful)

Value	Parameter Description
0xFFFF	Connection_Handle Range: 0x0000 to 0x0EFF

**LMP_Handle:****Size: 1 octet**

Value	Parameter Description
0xXX	The LMP Handle is the LMP Handle that is associated with this Connection_Handle. For a synchronous handle, this would be the LMP Synchronous Handle used when negotiating the synchronous connection in the link manager.

Reserved:**Size: 4 octets**

Value	Parameter Description
0xFFFFFFFF	This parameter is reserved, must be set to zero.

Events(s) generated (unless masked away):

When the HCI_Read_LMP_Handle command has completed, an HCI_Command_Complete event shall be generated.



7.1.26 Setup Synchronous Connection command

Command	OCF	Command Parameters	Return Parameters
HCI_Setup_ Synchronous_Connection	0x0028	Connection_Handle, Transmit_Bandwidth, Receive_Bandwidth, Max_Latency, Voice_Setting, Retransmission_Effort, Packet_Type	

Description:

The HCI_Setup_Synchronous_Connection command adds a new or modifies an existing synchronous logical transport (SCO or eSCO) on a physical link depending on the Connection_Handle parameter specified. If the Connection_Handle refers to an ACL link a new synchronous logical transport will be added. If the ACL link has encryption enabled using AES-CCM and the Controller cannot establish an eSCO transport (e.g. the Host parameters restricting the packet types to only SCO packet types), the Controller shall return the error code *Connection Rejected Due to Security Reasons* (0x0E) and a SCO transport will not be established. If the Connection_Handle refers to an already existing synchronous logical transport (eSCO only) this link will be modified. The parameters are specified per connection. This synchronous connection can be used to transfer synchronous voice at 64kb/s or transparent synchronous data.

When used to setup a new synchronous logical transport, the Connection_Handle parameter shall specify an ACL connection with which the new synchronous connection will be associated. The other parameters relate to the negotiation of the link, and may be reconfigured during the lifetime of the link. The transmit and receive bandwidth specify how much bandwidth shall be available for transmitting and for receiving data. While in many cases the receive and transmit bandwidth parameters may be equal, they may be different. The latency specifies an upper limit to the time in milliseconds between the eSCO (or SCO) instants, plus the size of the retransmission window, plus the length of the reserved synchronous slots for this logical transport. The content format specifies the settings for voice or transparent data on this connection. The retransmission effort specifies the extra resources that are allocated to this connection if a packet may need to be retransmitted. The Retransmission_Effort parameter shall be set to indicate the required behavior, or to don't care.

When used to modify an existing synchronous logical transport, the Transmit_Bandwidth, Receive_Bandwidth and Voice_Setting shall be set to the same values as were used during the initial setup. The Packet_Type, Retransmission_Effort and Max_Latency parameters may be modified.



The `Packet_Type` field is a bitmap specifying which packet types the LM shall accept in the negotiation of the link parameters. Multiple packet types are specified by bitwise OR of the packet type codes in the table. At least one packet type must be specified for each negotiation. It is recommended to enable as many packet types as possible.

Note: It is allowed to enable packet types that are not supported by the local device.

A `Connection_Handle` for the new synchronous connection will be returned in an `HCI_Synchronous_Connection_Complete` event.

Note: The link manager may choose any combination of packet types, timing, and retransmission window sizes that satisfy the parameters given. This may be achieved by using more frequent transmissions of smaller packets. The link manager may choose to set up either a SCO or an eSCO connection, if the parameters allow, using the corresponding LMP sequences.

Note: To modify a SCO connection, use the `HCI_Change_Connection_Packet_Type` command.

If the lower layers cannot achieve the exact transmit and receive bandwidth requested subject to the other parameters, then the link shall be rejected.

A synchronous connection may only be created when an ACL connection already exists.

Command parameters:

Connection_Handle: 2 octets (12 bits meaningful)

Value	Parameter Description
0xXXXX	Connection_Handle Range: 0x0000 to 0x0EFF

Transmit_Bandwidth: 4 octets

Value	Parameter Description
0XXXXXXXX	Transmit bandwidth in octets per second.

Receive_Bandwidth: 4 octets

Value	Parameter Description
0XXXXXXXX	Receive bandwidth in octets per second.

**Max_Latency:****2 octets**

Value	Parameter Description
0x0000-0x0003	Reserved for future use
0x0004-0xFFFE	This is a value in milliseconds representing the upper limit of the sum of the synchronous interval, and the size of the eSCO window, where the eSCO window is the reserved slots plus the retransmission window. (See Figure 8.9 in Part B, Baseband Specification)
0xFFFF	Don't care.

Voice_Setting:**2 octets (10 bits meaningful)**

Value	Parameter Description
See Section 6.12 .	

Retransmission_Effort:**1 octet**

Value	Parameter Description
0x00	No retransmissions (SCO or eSCO connection allowed)
0x01	At least one retransmission, optimize for power consumption (eSCO connection required).
0x02	At least one retransmission, optimize for link quality (eSCO connection required)
0xFF	Don't care (SCO or eSCO connection allowed)
All other values	Reserved for future use

Packet_Type:**2 octets**

Bit Number	Parameter Description
0	HV1 may be used.
1	HV2 may be used.
2	HV3 may be used.
3	EV3 may be used.
4	EV4 may be used.
5	EV5 may be used.
6	2-EV3 shall not be used.
7	3-EV3 shall not be used.
8	2-EV5 shall not be used.
9	3-EV5 shall not be used.
All other bits	Reserved for future use

**Return parameters:**

None.

Event(s) generated (unless masked away)

When the BR/EDR Controller receives the HCI_Setup_Synchronous_Connection command, it shall send the HCI_Command_Status event to the Host. In addition, when the LM determines the connection is established, the local BR/EDR Controller shall send an HCI_Synchronous_Connection_Complete event to the local Host, and the remote Controller will send an HCI_Synchronous_Connection_Complete event or an HCI_Connection_Complete event to the remote Host. The HCI_Synchronous_Connection_Complete event contains the Connection_Handle if this command is successful.

If this command is used to change the parameters of an existing eSCO link, the HCI_Synchronous_Connection_Changed event is sent to both Hosts. In this case no HCI_Connection_Setup_Complete event or HCI_Connection_Request event will be sent to either Host. This command cannot be used to change the parameters of a SCO link.

Note: No HCI_Command_Complete event will be sent by the BR/EDR Controller to indicate that this command has been completed. Instead, the HCI_Synchronous_Connection_Complete event will indicate that this command has been completed.



7.1.27 Accept Synchronous Connection Request command

Command	OCF	Command Parameters	Return Parameters
HCI_Accept_Synchronous_Connection_Request	0x0029	BD_ADDR, Transmit_Bandwidth, Receive_Bandwidth, Max_Latency, Voice_Setting, Retransmission_Effort, Packet_Type	

Description:

The HCI_Accept_Synchronous_Connection_Request command is used to accept an incoming request for a synchronous connection and to inform the local Link Manager about the acceptable parameter values for the synchronous connection.

The command shall only be issued after an HCI_Connection_Request event with link type SCO or eSCO has occurred. The HCI_Connection_Request event contains the BD_ADDR of the device requesting the connection. The decision to accept a connection must be taken before the connection accept timeout expires on the local device.

If the ACL link has encryption enabled using AES-CCM and the Controller cannot establish an eSCO transport (e.g. the Host parameters restricting the packet types to only SCO packet types), the Controller shall return the error code *Connection Rejected Due to Security Reasons* (0x0E) and a SCO transport will not be established.

The parameter set of the HCI_Accept_Synchronous_Connection_Request command is the same as for the HCI_Setup_Synchronous_Connection command. The Transmit_Bandwidth and Receive_Bandwidth values are required values for the new link and shall be met. The Max_Latency is an upper bound to the acceptable latency for the Link, as defined in [Section 7.1.26](#) Setup_Synchronous_Connection and shall not be exceeded. Voice_Setting specifies the encoding in the same way as in the HCI_Setup_Synchronous_Connection command and shall be met. The Retransmission_Effort parameter shall be set to indicate the required behavior, or to don't care. The Packet_Type parameter is a bit mask specifying the synchronous packet types that are allowed on the link.

If the Link Type of the incoming request is SCO, then only the Transmit_Bandwidth, Max_Latency, Voice_Settings, and Packet_Type fields are valid.



Note: See [Section 7.3.3](#) for the behavior when the HCI_Connection_Request event is masked or the connection is auto accepted.

Command parameters:

BD_ADDR: 6 octets

Value	Parameter Description
0xFFFFFFFFXXXXXX	BD_ADDR of the device requesting the connection

Transmit_Bandwidth: 4 octets

Value	Parameter Description
0x00000000-0xFFFFFFFF	Maximum possible transmit bandwidth in octets per second.
0xFFFFFFFF	Don't care

Default: Don't care

Receive_Bandwidth: 4 octets

Value	Parameter Description
0x00000000-0xFFFFFFFF	Maximum possible receive bandwidth in octets per second.
0xFFFFFFFF	Don't care

Default: Don't care

Max_Latency: 2 octets

Value	Parameter Description
0x0000-0x0003	Reserved for future use
0x0004-0xFFFF	This is a value in milliseconds representing the upper limit of the sum of the synchronous interval and the size of the eSCO window, where the eSCO window is the reserved slots plus the retransmission window (See Figure 8.9 in Part B, Baseband Specification).
0xFFFF	Don't care.

Default: Don't care

**Voice_Settings:****2 octets (10 bits meaningful)**

Value	Parameter Description
See Section 6.12 .	

Default: When links are auto-accepted, the values written by the HCI_Write_Voice_Settings command are used.

Retransmission_Effort:**1 octet**

Value	Parameter Description
0x00	No retransmissions (SCO or eSCO connection allowed)
0x01	At least one retransmission, optimize for power consumption (eSCO connection required).
0x02	At least one retransmission, optimize for link quality (eSCO connection required).
0xFF	Don't care (SCO or eSCO connection allowed)
All other values	Reserved for future use

Default: Don't care

Packet_Type:**2 octets**

Bit Number	Parameter Description
0	HV1 may be used.
1	HV2 may be used.
2	HV3 may be used.
3	EV3 may be used.
4	EV4 may be used.
5	EV5 may be used.
6	2-EV3 shall not be used.
7	3-EV3 shall not be used
8	2-EV5 shall not be used.
9	3-EV5 shall not be used.
All other bits	Reserved for future use

Default: 0x003F - means all defined packet types may be used.

Return parameters:

None.

Event(s) generated (unless masked away):



The `HCI_Accept_Synchronous_Request` command shall cause the `HCI_Command_Status` event to be sent from the BR/EDR Controller when the BR/EDR Controller starts setting up the connection. When the link setup is complete, the local BR/EDR Controller shall send an `HCI_Synchronous_Connection_Complete` event to its Host, and the remote BR/EDR Controller will send an `HCI_Synchronous_Connection_Complete` event to the Host. The `HCI_Synchronous_Connection_Complete` will contain the `Connection_Handle` and the link parameters if the setup is successful.

Note: No `HCI_Command_Complete` event will be sent by the Controller as the result of this command. Instead, the `HCI_Synchronous_Connection_Complete` event will indicate that this command has been completed.



7.1.28 Reject Synchronous Connection Request command

Command	OCF	Command Parameters	Return Parameters
HCI_Reject_Synchronous_Connection_Request	0x002A	BD_ADDR, Reason	

Description:

The HCI_Reject_Synchronous_Connection_Request command is used to decline an incoming request for a synchronous link. It shall only be issued after an HCI_Connection_Request event with Link Type equal to SCO or eSCO has occurred. The HCI_Connection_Request event contains the BD_ADDR of the device requesting the connection. The Reason parameter will be returned to the initiating Host in the Status parameter of the HCI_Synchronous_Connection_Complete event on the remote side.

Command parameters:

BD_ADDR: 6 octets

Value	Parameter Description
0xFFFFFFFFXXXX	BD_ADDR of the device requesting the connection

Reason: 1 octet

Value	Parameter Description
0x0D-0x0F	Host Reject error code. See [Vol 2] Part D, Error Codes for error codes and descriptions.

Return parameters:

None.

Event(s) generated (unless masked away):

When the Controller receives the HCI_Reject_Synchronous_Connection_Request, it shall send an HCI_Command_Status event to the Host. When the setup is terminated, the local Controller shall send an HCI_Synchronous_Connection_Complete event to its Host, and the remote Controller will send an HCI_Synchronous_Connection_Complete event to the Host with the Reason code from this command.

Note: No HCI_Command_Complete event will be sent by the Controller to indicate that this command has been completed. Instead, the HCI_Synchronous_Connection_Complete event will indicate that this command has been completed.



7.1.29 IO Capability Request Reply command

Command	OCF	Command Parameters	Return Parameters
HCI_IO_Capability_Request_Reply	0x002B	BD_ADDR, IO_Capability, OOB_Data_Present, Authentication_Requirements	Status, BD_ADDR

Description:

The HCI_IO_Capability_Request_Reply command is used to reply to an HCI_IO_Capability_Request event from the Controller, and specifies the current I/O capabilities of the Host. This includes the Host input, output and out-of-band (OOB) capabilities.

If an authenticated link key is not required by the Host, the Authentication Requirements parameter may be set to one of the following:

- MITM Protection Not Required – No Bonding
- MITM Protection Not Required – Dedicated Bonding
- MITM Protection Not Required – General Bonding

If both Hosts set the Authentication_Requirements parameter to one of the above values, the Link Managers shall use the numeric comparison authentication procedure and the Hosts shall use the Just Works association model.

If an authenticated link key is required by the Host, the Authentication Requirements parameter shall be set to one of the following:

- MITM Protection Required – No Bonding
- MITM Protection Required – Dedicated Bonding
- MITM Protection Required – General Bonding

In addition, the following requirements apply:

- If one or both Hosts set the Authentication Requirements parameter to one of the above values, the Link Managers shall use the IO_Capability parameter to determine the authentication procedure.
- A Host that sets the Authentication_Requirements parameter to one of the above values shall verify that the resulting Link Key type meets the security requirements requested.

If the Host has received OOB authentication data from a device with the same BD_ADDR sent in the HCI_IO_Capability_Request event, then the OOB_Data_Present parameter shall be set to:

- "P-192 OOB authentication data from remote device present" when the Host has received only P-192 OOB data,



- "P-256 OOB authentication data from remote device present" when the Host has received only P-256 OOB data, or
- "P-192 and P-256 OOB authentication data from remote device present" when the Host has received both P-192 and P-256 OOB data.

Otherwise OOB_Data_Present shall be set to "OOB authentication data not present".

Command parameters:

BD_ADDR:

Size: 6 octets

Value	Parameter Description
0XXXXXXXXXX XX	BD_ADDR of remote device involved in simple pairing process

IO_Capability:

Size: 1 octet

Value	Parameter Description
0x00	DisplayOnly
0x01	DisplayYesNo
0x02	KeyboardOnly
0x03	NoInputNoOutput
All other values	Reserved for future use

OOB_Data_Present:

Size: 1 octet

Value	Parameter Description
0x00	OOB authentication data not present
0x01	P-192 OOB authentication data from remote device present
0x02	P-256 OOB authentication data from remote device present
0x03	P-192 and P-256 OOB authentication data from remote device present
All other values	Reserved for future use

**Authentication_Requirements:****Size: 1 octet**

Value	Parameter Description
0x00	MITM Protection Not Required – No Bonding. Numeric comparison with automatic accept allowed.
0x01	MITM Protection Required – No Bonding. Use IO Capabilities to determine authentication procedure
0x02	MITM Protection Not Required – Dedicated Bonding. Numeric comparison with automatic accept allowed.
0x03	MITM Protection Required – Dedicated Bonding. Use IO Capabilities to determine authentication procedure
0x04	MITM Protection Not Required – General Bonding. Numeric Comparison with automatic accept allowed.
0x05	MITM Protection Required – General Bonding. Use IO capabilities to determine authentication procedure.
All other values	Reserved for future use

Return parameters:**Status:****Size: 1 octet**

Value	Parameter Description
0x00	HCI_IO_Capability_Request_Reply command succeeded
0x01 - 0xFF	HCI_IO_Capability_Request_Reply command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

BD_ADDR:**Size: 6 octets**

Value	Parameter Description
0XXXXXXXXXX XX	BD_ADDR of remote device involved in simple pairing process

Event(s) generated (unless masked away):

When the HCI_IO_Capability_Request_Reply command has completed, an HCI_Command_Complete event shall be generated. When the device is the initiator of Secure Simple Pairing, an HCI_IO_Capability_Response event shall be generated. Additionally, when the OOB_Data_Present parameter indicates that OOB authentication data from the remote device is present, the HCI_Remote_OOB_Data_Request event shall be generated.



7.1.30 User Confirmation Request Reply command

Command	OCF	Command Parameters	Return Parameters
HCI_User_Confirmation_Request_Reply	0x002C	BD_ADDR	Status, BD_ADDR

Description:

The HCI_User_Confirmation_Request_Reply command is used to reply to an HCI_User_Confirmation_Request event and indicates that the user selected "yes". It is also used when the Host has no input and no output capabilities.

Command parameters:

BD_ADDR:

Size: 6 octets

Value	Parameter Description
0XXXXXXXXXX X	BD_ADDR of remote device involved in simple pairing process

Return parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	HCI_User_Confirmation_Request_Reply command succeeded
0x01 - 0xFF	HCI_User_Confirmation_Request_Reply command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

BD_ADDR:

Size: 6 octets

Value	Parameter Description
0XXXXXXXXXX X	BD_ADDR of remote device involved in simple pairing process

Event(s) generated (unless masked away):

When the HCI_User_Confirmation_Request_Reply command has completed, an HCI_Command_Complete event shall be generated.



7.1.31 User Confirmation Request Negative Reply command

Command	OCF	Command Parameters	Return Parameters
HCI_User_Confirmation_Request_Negative_Reply	0x002D	BD_ADDR	Status, BD_ADDR

Description:

The HCI_User_Confirmation_Request_Negative_Reply command is used to reply to an HCI_User_Confirmation_Request event and indicates that the user selected "no". This command shall cause the initiating Link Manager to transmit an LMP_Numeric_Comparison_Failed PDU and terminate Secure Simple Pairing.

Command parameters:

BD_ADDR:

Size: 6 octets

Value	Parameter Description
0XXXXXXXXX XX	BD_ADDR of remote device involved in simple pairing process

Return parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	HCI_User_Confirmation_Request_Negative_Reply command succeeded
0x01 - 0xFF	HCI_User_Confirmation_Request_Negative_Reply command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

BD_ADDR:

Size: 6 octets

Value	Parameter Description
0XXXXXXXXX XX	BD_ADDR of remote device involved in simple pairing process

Event(s) generated (unless masked away):

When the HCI_User_Confirmation_Request_Negative_Reply command has completed, an HCI_Command_Complete event shall be generated.



7.1.32 User Passkey Request Reply command

Command	OCF	Command Parameters	Return Parameters
HCI_User_Passkey_Request_Reply	0x002E	BD_ADDR, Numeric_Value	Status, BD_ADDR

Description:

The HCI_User_Passkey_Request_Reply command is used to reply to an HCI_User_Passkey_Request event and specifies the Numeric_Value (passkey) entered by the user to be used in the Secure Simple Pairing process.

Command parameters:

BD_ADDR:

Size: 6 octets

Value	Parameter Description
0xFFFFFFFFXXXX	BD_ADDR of remote device involved in simple pairing process

Numeric_Value:

Size: 4 octets

Value	Parameter Description
0x00000000 to 0x000F423F	Numeric value (passkey) entered by user. Valid values are decimal 000000 to 999999.

Return parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	HCI_User_Passkey_Request_Reply command succeeded
0x01 to 0xFF	HCI_User_Passkey_Request_Reply command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

BD_ADDR:

Size: 6 octets

Value	Parameter Description
0xFFFFFFFFXXXX	BD_ADDR of remote device involved in simple pairing process

Event(s) generated (unless masked away):

When the HCI_User_Passkey_Request_Reply command has completed, an HCI_Command_Complete event shall be generated.



7.1.33 User Passkey Request Negative Reply command

Command	OCF	Command Parameters	Return Parameters
HCI_User_Passkey_Request_Negative_Reply	0x002F	BD_ADDR	Status, BD_ADDR

Description:

The HCI_User_Passkey_Request_Negative_Reply command is used to reply to an HCI_User_Passkey_Request event and indicates the Host could not provide a passkey. This command shall cause the initiating Link Manager to transmit an LMP_Passkey_Entry_Failed PDU and terminate Simple Pairing.

Command parameters:

BD_ADDR:

Size: 6 octets

Value	Parameter Description
0xFFFFFFFFXX	BD_ADDR of remote device involved in simple pairing process

Return parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	HCI_User_Passkey_Request_Negative_Reply command succeeded
0x01 to 0xFF	HCI_User_Passkey_Request_Negative_Reply command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

BD_ADDR:

Size: 6 octets

Value	Parameter Description
0xFFFFFFFFXX	BD_ADDR of remote device involved in simple pairing process

Event(s) generated (unless masked away):

When the HCI_User_Passkey_Request_Negative_Reply command has completed, an HCI_Command_Complete event shall be generated.



7.1.34 Remote OOB Data Request Reply command

Command	OCF	Command Parameters	Return Parameters
HCI_Remote_OOB_Data_Request_Reply	0x0030	BD_ADDR, C, R	Status, BD_ADDR

Description:

The HCI_Remote_OOB_Data_Request_Reply command is used to reply to an HCI_Remote_OOB_Data_Request event with the C and R values received via an OOB transfer from a remote device identified by BD_ADDR. If the R value is not present in the received OOB data from the remote device, the Host shall set R to zeros.

Command parameters:

BD_ADDR:

Size: 6 octets

Value	Parameter Description
0XXXXXXXXXX XX	BD_ADDR of remote device from which the C and R values were received

C:

Size: 16 octets

Value	Parameter Description
0XXXXXXXXXX XXXXXXXXXXXX XXXXXXXXXXXX	Simple Pairing Hash C

R:

Size: 16 octets

Value	Parameter Description
0XXXXXXXXXX XXXXXXXXXXXX XXXXXXXXXXXX	Simple Pairing Randomizer R

Return parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	HCI_Remote_OOB_Data_Request_Reply command succeeded
0x01 - 0xFF	HCI_Remote_OOB_Data_Request_Reply command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

**BD_ADDR:****Size: 6 octets**

Value	Parameter Description
0XXXXXXXXXX XX	BD_ADDR of remote device involved in simple pairing process

Event(s) generated (unless masked away):

When the HCI_Remote_OOB_Data_Request_Reply command has completed, an HCI_Command_Complete event shall be generated.



7.1.35 Remote OOB Data Request Negative Reply command

Command	OCF	Command Parameters	Return Parameters
HCI_Remote_OOB_Data_Request_Negative_Reply	0x0033	BD_ADDR	Status, BD_ADDR

Description:

The HCI_Remote_OOB_Data_Request_Negative_Reply command is used to reply to an HCI_Remote_OOB_Data_Request event that the Host does not have the C and R values associated with the remote device identified by BD_ADDR.

Command parameters:

BD_ADDR:

Size: 6 octets

Value	Parameter Description
0xFFFFFFFFXXXX	BD_ADDR of remote device

Return parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	HCI_Remote_OOB_Data_Request_Negative_Reply command succeeded
0x01 - 0xFF	HCI_Remote_OOB_Data_Request_Negative_Reply command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

BD_ADDR:

Size: 6 octets

Value	Parameter Description
0xFFFFFFFFXXXX X	BD_ADDR of remote device involved in simple pairing process

Event(s) generated (unless masked away):

When the HCI_Remote_OOB_Data_Request_Negative_Reply command has completed, an HCI_Command_Complete event shall be generated.



7.1.36 IO Capability Request Negative Reply command

Command	OCF	Command Parameters	Return Parameters
HCI_IO_Capability_Request_Negative_Reply	0x0034	BD_ADDR, Reason	Status, BD_ADDR

Description:

The HCI_IO_Capability_Request_Negative_Reply command shall be used to reject a pairing attempt after an HCI_IO_Capability_Request event has been received by the Host. The reason for the rejection is given in the Reason parameter. The error code *Simple Pairing not Supported by Host* (0x37) shall not be used in the Reason parameter.

Command parameters:

BD_ADDR:

Size: 6 octets

Value	Parameter Description
0xFFFFFFFFXXXX	BD_ADDR of remote device involved in the Simple Pairing process

Reason:

Size: 1 octet

Value	Parameter Description
0xXX	Reason that Simple Pairing rejected. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Return parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	HCI_IO_Capability_Request_Negative_Reply command succeeded.
0x01 - 0xFF	HCI_IO_Capability_Request_Negative_Reply command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

BD_ADDR:

Size: 6 octets

Value	Parameter Description
0xFFFFFFFFXXXXX	BD_ADDR of remote device involved in Simple Pairing process

Event(s) generated (unless masked away):

When the HCI_IO_Capability_Request_Negative_Reply command has completed, an HCI_Command_Complete event shall be generated.



7.1.37 Create Physical Link command

Command	OCF	Command Parameters	Return Parameters
HCI_Create_Physical_Link	0x0035	Physical_Link_Handle, Dedicated_AMP_Key_Length, Dedicated_AMP_Key_Type, Dedicated_AMP_Key	

Description:

When this command is received by an AMP Controller, the AMP Controller shall initiate a connection to another AMP Controller. The AMP_ASSOC used to create the physical link shall be written to the AMP Controller using the HCI_Write_Remote_AMP_ASSOC command following the HCI_Create_Physical_Link command.

The Dedicated_AMP_Key_Length parameter is used by the Host to indicate the number of valid octets of the Dedicated_AMP_Key. The AMP shall use the supplied key to generate a session key in order to encrypt all data on the indicated physical link. The method by which to do this is AMP type specific. The Dedicated_AMP_Key_Type parameter indicates whether the key is an authenticated combination key, an unauthenticated combination key, or a debug combination key.

Command parameters:

Physical_Link_Handle:

Size: 1 octet

Value	Parameter Description
0xXX	Physical Link Handle identifying the physical link to be created (see Section 5.3.2).
0x00	Reserved for future use

Dedicated_AMP_Key_Length:

Size: 1 octet

Value	Parameter Description
0	Reserved for future use
0xXX	See Assigned Numbers .
33 to 255	Reserved for future use

**Dedicated_AMP_Key_Type:****Size: 1 octet**

Value	Parameter Description
0x00	Reserved for future use
0x01	Reserved for future use
0x02	Reserved for future use
0x03	Debug combination keys
0x04	Unauthenticated combination key
0x05	Authenticated combination key
0x06	Reserved for future use
All other values	Reserved for future use

Dedicated_AMP_Key:**Size: AMP_Key_Length octets**

Value	Parameter Description
0xXX...XX	Dedicated AMP Key for the associated AMP. The Dedicated AMP Key is a multi-octet integer.

Return parameters:

None.

Event(s) generated (unless masked away):

When the AMP Controller receives the `HCI_Create_Physical_Link` command, the AMP Controller shall send the `HCI_Command_Status` event to the Host. When the AMP Controller has determined underlying connection details to populate the `AMP_ASSOC` the AMP Controller shall generate an `HCI_Channel_Selected` event to the Host. After the `HCI_Channel_Selected` event is received the Host uses the `HCI_Read_Local_AMP_ASSOC` command to obtain the capabilities and status of the local AMP. The `AMP_ASSOC` is sent to the remote AMP to be used to perform a matching `HCI_Accept_Physical_Link` command. When the connection has been completed the AMP Controller on both devices that form the connection shall send an `HCI_Physical_Link_Complete` event to its Host. The `HCI_Physical_Link_Complete` event indicates to the Host whether the physical link creation was successful. This sequence must be completed before any Logical Links using this Controller can be created.

If the AMP physical link security establishment process fails (for instance, if the two AMP Controllers do not have matching keys) then the physical link shall be disconnected and the `HCI_Physical_Link_Complete` event shall be sent to the Host with the *Authentication Failure* error code see [\[Vol 2\] Part D, Error Codes](#).

Note: No `HCI_Command_Complete` event will be sent by the AMP Controller to indicate that this command has been completed. Instead, the



HCI_Physical_Link_Complete event will indicate that this command has been completed.



7.1.38 Accept Physical Link command

Command	OCF	Command Parameters	Return Parameters
HCI_Accept_Physical_Link	0x0036	Physical_Link_Handle, Dedicated_AMP_Key_Length, Dedicated_AMP_Key_Type, Dedicated_AMP_Key	

Description:

When this command is received by an AMP Controller, the AMP Controller shall attempt to accept a connection request from the initiating BR/EDR Controller's AMP Controller. The AMP_ASSOC used to create the physical link shall be written to the AMP Controller using the HCI_Write_Remote_AMP_ASSOC command after issuing the HCI_Accept_Physical_Link command.

The Dedicated_AMP_Key_Length parameter is used by the Host to indicate the number of valid octets of the Dedicated_AMP_Key. The AMP shall use the Dedicated_AMP_Key for AMP type-specific security functions. The Dedicated_AMP_Key_Type parameter indicates whether the key is an authenticated combination key, an unauthenticated combination key, or a debug combination key.

Command parameters:

Physical_Link_Handle:

Size: 1 octet

Value	Parameter Description
0xXX	Physical Link Handle identifying the physical link to be created.

Dedicated_AMP_Key_Length:

Size: 1 octet

Value	Parameter Description
0x00	Reserved for future use
0xXX	See Assigned Numbers .
33 to 255	Reserved for future use

**Dedicated_AMP_Key_Type****Size: 1 octet**

Value	Parameter Description
0x00	Reserved for future use
0x01	Reserved for future use
0x02	Reserved for future use
0x03	Debug combination key
0x04	Authenticated combination key
0x05	Unauthenticated combination key
0x06	Reserved for future use
All other values	Reserved for future use

Dedicated_AMP_Key**Size: Dedicated_AMP_Key_Length octets**

Value	Parameter Description
0xXX..XX	Dedicated AMP Key for the associated AMP. The Dedicated AMP Key is a multi-octet integer.

Return parameters:

None.

Event(s) generated (unless masked away):

When the AMP Controller receives the `HCI_Accept_Physical_Link` command, the AMP Controller shall send the `HCI_Command_Status` event to the Host. When the connection has been completed, the AMP Controller shall send an `HCI_Physical_Link_Complete` event to the Host. The `HCI_Physical_Link_Complete` event indicates to the Host whether the link creation was successful. This sequence must be completed before any Logical Links can be created.

If communication is not established within the Connection Accept Timeout then an `HCI_Physical_Link_Complete` event shall be returned with a Status value of *Connection Accept Timeout Exceeded* (0x10).

If the AMP-specific security establishment process fails (for instance, if the two AMP Controllers do not have matching keys) then the physical link shall be deleted and the `HCI_Physical_Link_Complete` event sent to the Host with the *Authentication Failure* error code.

No `HCI_Physical_Link_Complete` event shall be sent before the security is established.

Note: No `HCI_Command_Complete` event will be sent by the AMP Controller to indicate that this command has been completed. Instead, the



HCI_Physical_Link_Complete event will indicate that this command has been completed.



7.1.39 Disconnect Physical Link command

Command	OCF	Command Parameters	Return Parameters
HCI_Disconnect_Physical_Link	0x0037	Physical_Link_Handle, Reason	

Description:

When the HCI_Disconnect_Physical_Link command is received by an AMP Controller, the AMP Controller shall terminate the AMP physical link indicated by the Physical_Link_Handle parameter. The Reason parameter indicates the reason for ending the connection. The remote AMP Controller will receive the Reason parameter in the HCI_Disconnection_Physical_Link_Complete event.

All logical links on a physical link should be disconnected before the physical link is disconnected.

The HCI_Disconnect_Physical_Link command may be used to cancel the creation of a physical link, before the HCI_Physical_Link_Complete event for this link has been generated.

Command parameters:

Physical_Link_Handle:

Size: 1 octet

Value	Parameter Description
0xXX	Physical_Link_Handle identifying the physical link which has been created.

Reason:

Size: 1 octet

Value	Parameter Description
0x05, 0x13 to 0x16, 0x1A	<i>Authentication Failure</i> error code (0x05), <i>Other End Terminated Connection</i> error codes (0x13 to 0x16), <i>Unsupported Remote Feature</i> error code (0x1A), see [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Return parameters:

None.

Event(s) generated (unless masked away):

When the AMP Controller receives the HCI_Disconnect_Physical_Link command, an HCI_Command_Status event shall be sent to the Host. The HCI_Disconnection_Physical_Link_Complete event occurs at both Hosts when the termination of the connection has completed, and indicates that this command has been completed.



The HCI_Disconnect_Physical_Link command may be used to cancel the creation of a physical link. In such a case, the HCI_Physical_Link_Complete event for the corresponding HCI_Create_Physical_Link or HCI_Accept_Physical_Link command shall always be sent. The HCI_Physical_Link_Complete event shall be sent after the HCI_Disconnection_Physical_Link_Complete event, in the case the Controller has already issued an HCI_Command_Status event for the HCI_Disconnect_Physical_Link command, thus treating it as a creation cancel. If the cancellation was successful, the HCI_Physical_Link_Complete event shall be generated with the error code *Unknown Connection Identifier* (0x02).

Note: No HCI_Command_Complete event will be sent by the AMP Controller to indicate that this command has been completed. Instead, the HCI_Disconnection_Physical_Link_Complete event will indicate that this command has been completed.



7.1.40 Create Logical Link command

Command	OCF	Command Parameters	Return Parameters
HCI_Create_Logical_Link	0x0038	Physical_Link_Handle, Tx_Flow_Spec, Rx_Flow_Spec	

Description:

When the HCI_Create_Logical_Link command is received by an AMP Controller, the AMP Controller shall create a logical connection to the remote AMP Controller identified by the physical link handle.

The Flow_Spec structures define the traffic requirements of the link. The Flow Spec ID values in the Tx/Rx parameters identify the logical link.

Note: The Tx_Flow_Spec and Rx_Flow_Spec parameters only use 16 out of the 18 octets of the L2CAP Extended Flow Spec. The Type and Length fields from the L2CAP format are not included in the Tx_Flow_Spec and Rx_Flow_Spec parameters.

A logical link can be created between two devices that are already joined by a physical link. In order to create a logical link the HCI_Create_Logical_Link command is called on one device (the initiating device) and the HCI_Accept_Logical_Link command is called on the other device (the responding device). The Flow_Spec information shall match for the two calls (see [Vol 3] Part A, Section 5.6 and Section 9.1).

The AMP Controller may refuse to create the logical link if it has insufficient resources. If the AMP Controller is already in the process of creating one or more logical links, it may reject an HCI_Create_Logical_Link command with the error code *Controller Busy* (0x3A). When the Host receives this error code, it shall wait until a previous logical link creation completes before requesting another logical link.

The logical link must be completed before the Logical Link Accept Timeout expires on the local AMP.

Command parameters:

Physical_Link_Handle:

Size: 1 octet

Value	Parameter Description
0xXX	Physical Link Handle over which the logical link is to be established.



Tx_Flow_Spec:

Size: 16 octets

Parameter Description
Extended Flow Specification value defining transmitted traffic. See [Vol 3] Part A, Section 5.6 .



Rx_Flow_Spec:

Size: 16 octets

Parameter Description
Extended Flow Specification value defining received traffic. (See [Vol 3] Part A, Section 5.6).

Return parameters:

None.

Event(s) generated (unless masked away):

When the AMP Controller receives the HCI_Create_Logical_Link command, the AMP Controller shall send the HCI_Command_Status event to the Host. In addition, when the AMP Controller determines the logical link is established, the AMP Controller shall send an HCI_Logical_Link_Complete event to each Host. The HCI_Logical_Link_Complete event contains the Logical Link Handle, Physical Link Handle, and the corresponding flow spec ID if this command is successful.

Note: No HCI_Command_Complete event will be sent by the AMP Controller to indicate that this command has been completed. Instead, the HCI_Logical_Link_Complete event will indicate that this command has been completed.



7.1.41 Accept Logical Link command

Command	OCF	Command Parameters	Return Parameters
HCI_Accept_Logical_Link	0x0039	Physical_Link_Handle, Tx_Flow_Spec, Rx_Flow_Spec	

Description:

When the HCI_Accept_Logical_Link command is received by an AMP Controller, the AMP Controller shall attempt to accept a logical link with the remote AMP Controller identified by the Physical_Link_Handle.

The Flow_Spec structures define the traffic requirements of the link. The Flow Spec ID in the Tx/Rx parameter values identify the logical link.

A logical link may be created between two devices which are already joined by a physical link. In order to create a logical link the HCI_Create_Logical_Link command is called on one device (the initiating device) and the HCI_Accept_Logical_Link command is called on the other device (the responding device). The Flow_Spec information shall match for the two calls. See [\[Vol 3\] Part A, Section 5.6](#) and [Section 9.1](#) for how this is achieved.

The AMP Controller may refuse to create the logical link if it has insufficient resources.

The logical link must be completed before the Logical Link Accept Timeout expires on the local AMP.

Command parameters:

Physical_Link_Handle:

Size: 1 octet

Value	Parameter Description
0xXX	Physical Link Handle over which the logical link is to be established.

Tx_Flow_Spec:

Size: 16 octets

Parameter Description
Extended Flow Specification value defining transmitted traffic (see [Vol 3] Part A, Section 5.6).

Rx_Flow_Spec:

Size: 16 octets

Parameter Description
Extended Flow Specification value defining received traffic (see [Vol 3] Part A, Section 5.6).

**Return parameters:**

None.

Event(s) generated (unless masked away):

When the AMP Controller receives the `HCI_Accept_Logical_Link` command, the AMP Controller shall send the `HCI_Command_Status` event to the Host. In addition, when the AMP Controller determines the logical link is established, it shall send an `HCI_Logical_Link_Complete` event to each Host. This action shall occur on both Controllers that form the connection. The `HCI_Logical_Link_Complete` event contains the Logical Link Handle if this command is successful.

Note: No `HCI_Command_Complete` event will be sent by the AMP Controller to indicate that this command has been completed. Instead, the `HCI_Logical_Link_Complete` event will indicate that this command has been completed.



7.1.42 Disconnect Logical Link command

Command	OCF	Command Parameters	Return Parameters
HCI_Disconnect_Logical_Link	0x003A	Logical_Link_Handle	

Description:

When the HCI_Disconnect_Logical_Link command is received by an AMP Controller it shall terminate the designated logical link on the local Controller. The Logical_Link_Handle command parameter indicates which logical link is to be disconnected.

Command parameters:

Logical_Link_Handle:

Size: 2 octets (12 bits meaningful)

Value	Parameter Description
0xXXXX	Logical_Link_Handle for the logical link being disconnected. Range: 0x0000 to 0x0EFF

Return parameters:

None.

Event(s) generated (unless masked away):

When the AMP Controller receives the HCI_Disconnect_Logical_Link command it shall send the HCI_Command_Status event to the Host. The HCI_Disconnection_Logical_Link_Complete event shall be sent to the Host when the termination of the logical link has completed, and indicates that this command has been completed.

Note: No HCI_Command_Complete event will be sent by the AMP Controller to indicate that this command has been completed. Instead, the HCI_Disconnection_Logical_Link_Complete event will indicate that this command has been completed.



7.1.43 Logical Link Cancel command

Command	OCF	Command Parameters	Return Parameters
HCI_Logical_Link_Cancel	0x003B	Physical_Link_Handle, Tx_Flow_Spec_ID	Status, Physical_Link_Handle, Tx_Flow_Spec_ID

Description:

When the HCI_Logical_Link_Cancel command is received by an AMP Controller, the AMP Controller shall cancel the ongoing logical link creation process identified by the Physical_Link_Handle and the Tx_Flow_Spec_ID.

Command parameters:

Physical_Link_Handle:

Size: 1 octet

Value	Parameter Description
0xXX	Physical_Link_Handle over which the logical link is to be established.

Tx_Flow_Spec_ID:

Size: 1 octet

Value	Parameter Description
0xXX	Flow Spec ID identifying the logical link whose creation is being cancelled. This matches the ID field of the Tx_Flow_Spec in the matching HCI_Create_Logical_Link or HCI_Accept_Logical_Link command.

Return parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	HCI_Logical_Link_Cancel command succeeded
0x01 to 0xFF	HCI_Logical_Link_Cancel command failed. See [Vol 2] Part D, Error Codes for list of error codes.

Physical_Link_Handle:

Size: 1 octet

Value	Parameter Description
0xXX	Physical Link Handle identifying the logical link creation being cancelled.



Tx_Flow_Spec_ID:

Size: 1 octet

Value	Parameter Description
0xXX	Flow Spec ID identifying the logical link creation being cancelled. This matches the ID field of the Tx_Flow_Spec in the matching Create or HCI_Accept_Logical_Link command.

Event(s) generated (unless masked away):

When the HCI_Logical_Link_Cancel command has completed, an HCI_Command_Complete event shall be generated.

If the logical link is already established by the Controller, but the AMP Controller has not yet sent the HCI_Logical_Link_Complete event, then the local Controller shall detach the logical link and return an HCI_Command_Complete event with the status "Success".

If the logical link is already established, and the HCI_Logical_Link_Complete event has been sent, then the AMP Controller shall return an HCI_Command_Complete event with the error code *Connection Already Exists* (0x0B).

If the HCI_Logical_Link_Cancel command is sent to the AMP Controller without a preceding Create or HCI_Accept_Logical_Link command to the same device, the Controller shall return an HCI_Command_Complete event with the error code *Unknown Connection Identifier* (0x02).

The HCI_Logical_Link_Complete event for the corresponding HCI_Create_Logical_Link command shall always be sent. The HCI_Logical_Link_Complete event shall be sent after the HCI_Command_Complete event for the HCI_Logical_Link_Cancel command. If the cancellation was successful, the HCI_Logical_Link_Complete event shall be generated with the error code *Unknown Connection Identifier* (0x02).



7.1.44 Flow Spec Modify command

Command	OCF	Command Parameters	Return Parameters
HCI_Flow_Spec_Modify	0x003C	Handle, Tx_Flow_Spec, Rx_Flow_Spec	

Description:

When the HCI_Flow_Spec_Modify command is received by an AMP Controller, the AMP Controller shall modify the Flow Spec of the logical connection identified by the Handle.

Matching commands will be issued on both devices. This ensures that both Controllers have full knowledge of the traffic requirements on this logical link, in both directions. Each Controller is responsible for meeting its own Tx Flow specification.

The HCI_Flow_Spec_Modify command cannot be used to change the Identifier or the Service Type fields of an existing Handle's Flow Spec.

Command parameters:

Handle: *Size: 2 octets (12 bits meaningful)*

Value	Parameter Description
0xFFFF	Connection_Handle if the receiving Controller is a BR/EDR Controller. Logical_Link_Handle, if the receiving Controller is an AMP Controller, identifying the logical link to be changed. Range: 0x0000 to 0x0EFF (0x0F00 to 0x0FFF Reserved for future use)

Tx_Flow_Spec: *Size: 16 octets*

Parameter Description
Extended Flow Specification value defining transmitted traffic (see [Vol 3] Part A, Section 5.6).

Rx_Flow_Spec: *Size: 16 octets*

Parameter Description
Extended Flow Specification value defining received traffic (see [Vol 3] Part A, Section 5.6).

Return parameters:

None.

**Event(s) generated (unless masked away):**

When the AMP Controller receives the HCI_Flow_Spec_Modify command, the AMP Controller shall send the Command Status event to the Host. When each AMP Controller has completed the change a Flow Spec Modify Complete event shall be sent to the Host. If the modify fails, the Controllers shall revert to previously specified values.

Note: No HCI_Command_Complete event will be sent by the AMP Controller to indicate that this command has been completed. Instead, the HCI_Flow_Spec_Modify_Complete event will indicate that this command has been completed.



7.1.45 Enhanced Setup Synchronous Connection command

Command	OCF	Command Parameters	Return Parameters
HCI_Enhanced_Setup_Synchronous_Connection	0x003D	Connection_Handle, Transmit_Bandwidth, Receive_Bandwidth, Transmit_Coding_Format, Receive_Coding_Format, Transmit_Codec_Frame_Size, Receive_Codec_Frame_Size, Input_Bandwidth, Output_Bandwidth, Input_Coding_Format, Output_Coding_Format, Input_Coded_Data_Size, Output_Coded_Data_Size, Input_PCM_Data_Format, Output_PCM_Data_Format, Input_PCM_Sample_Payload_MSB_Position, Output_PCM_Sample_Payload_MSB_Position, Input_Data_Path, Output_Data_Path, Input_Transport_Unit_Size, Output_Transport_Unit_Size, Max_Latency, Packet_Type, Retransmission_Effort	

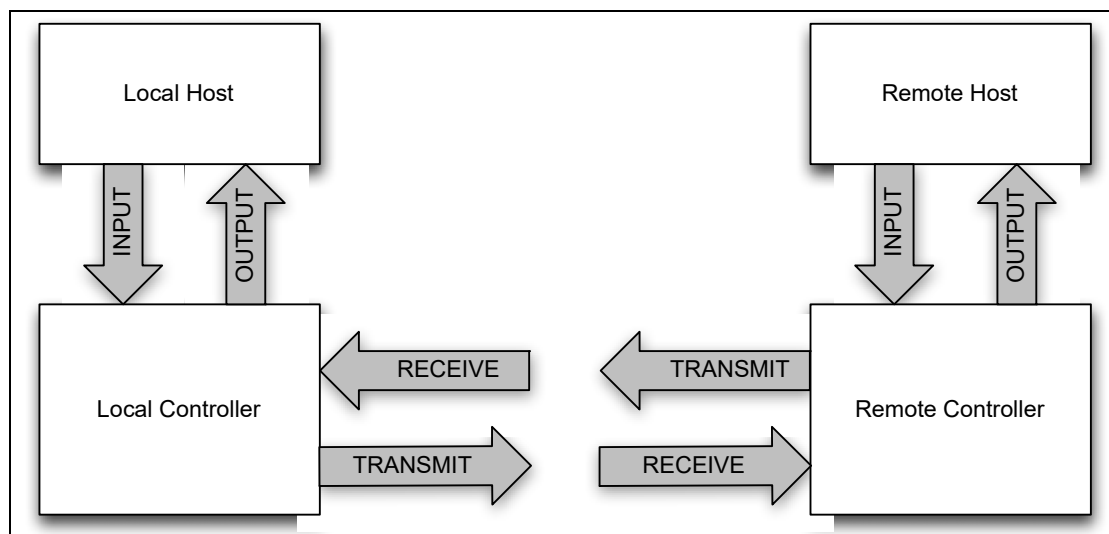
Description:

The HCI_Enhanced_Setup_Synchronous_Connection command adds a new, or modifies an existing, synchronous logical transport (SCO or eSCO) on a physical link depending on the Connection_Handle parameter specified. If the Connection_Handle refers to an ACL link, then a new synchronous logical transport shall be added. If the Connection_Handle refers to an existing synchronous logical transport (eSCO only), then this link shall be modified. The parameters are specified per connection. This synchronous connection can be used to transfer synchronous voice data or transparent synchronous data. If the ACL link has encryption enabled using AES-CCM and the Controller cannot establish an eSCO transport (e.g. the Host parameters restricting the packet types to SCO packet types), the Controller shall return the error code *Connection Rejected Due to Security Reasons* (0x0E) and a SCO transport will not be established.



When used to setup a new synchronous logical transport, the `Connection_Handle` parameter shall specify an ACL connection with which the new synchronous connection shall be associated. The other parameters relate to the negotiation of the link, and may be reconfigured during the lifetime of the link.

The following terms are used to describe the four different audio paths: Transmit, Receive, Input and Output. The Transmit and Receive paths are from the perspective of the local Controller's radio. The Input and Output paths are from the perspective of the Controller.



The following parameters are used to describe the transmit and receive paths over the air:

- The `Transmit_Bandwidth` and `Receive_Bandwidth` parameters specify how much bandwidth shall be available for transmitting and for receiving data. The Host shall set the `Transmit_Bandwidth` and `Receive_Bandwidth` parameters to be equal or shall set one of them to be zero and the other non-zero.
- The `Transmit_Coding_Format` and `Receive_Coding_Format` parameters specify the coding format used for transmitted or received data. The Host shall set the `Transmit_Coding_Format` and `Receive_Coding_Formats` to be equal. When the `Transmit_Coding_Format` and `Receive_Coding_Format` parameters are not equal to CVSD, A-law or μ -law, the Link Manager shall map these to Transparent air mode.
- The `Transmit_Codec_Frame_Size` and `Receive_Codec_Frame_Size` parameters specify the frame size produced by the codecs in the context of over-the-air coding. The over-the-air packet size should have the following relationship with the codec frame size:

$$\text{Packet_Size} = \text{Frame_Size} * N, \text{ or}$$

$$\text{Packet_Size} = \text{Frame_Size} / N$$

where N is an integer.



The following parameters are used to describe the coding format used prior to encapsulating over the audio data transport path:

- The `Input_Bandwidth` and `Output_Bandwidth` specify the nominal rate at which the Host or Controller transfers data (for HCI transports this excludes the HCI header). The Host shall either set the `Input_Bandwidth` and `Output_Bandwidth` to be equal, or shall set one of them to be zero and the other non-zero.
- The `Input_Coding_Format` and `Output_Coding_Format` parameters specify the coding format used over the transport. The Host shall set the `Input_Coding_Format` and `Output_Coding_Format` to be equal.
- The `Input_Coded_Data_Size` and `Output_Coded_Data_Size` specify the number of bits in each sample or frame of data. For CVSD, a frame of data shall be 8 bits.
- The `Input_PCM_Data_Format` and `Output_PCM_Data_Format` parameters specify the data format over the transport for linear samples. They shall be ignored when the data is encoded in any other way.
- The `Input_PCM_Sample_Payload_MSB_Position` and `Output_PCM_Sample_Payload_MSB_Position` parameters indicate, for linear samples, how many bit positions that the MSB of the sample is away from starting at the MSB of the data. They shall be ignored when the data is encoded in any other way. For example, if `Input_Coded_Data_Size` = 16 and `Input_PCM_Sample_Payload_MSB_Position` = 3, then each sample is actually only 13 bits, the MSB (which is the sign bit for signed formats) is bit 12 (counting from the LSB at bit 0), and the contents of bits 13, 14, and 15 of each sample shall be ignored.

The following parameters describe the audio data transport path characteristics:

- The `Input_Data_Path` and `Output_Data_Path` parameters specify the audio data transport path. When set to 0x00, the audio data path shall be over the HCI transport. When set to 0xFF, audio test mode (see [Section 7.6.2](#)) is selected (this is only applicable during test mode). When set to 0x01 to 0xFE, the audio data path shall use non-HCI transport data paths (e.g. PCM interface) with logical transport channel numbers. The meanings of these logical transport channel numbers are vendor specific.
- The `Input_Transport_Unit_Size` and `Output_Transport_Unit_Size` indicate how many bits are in each unit of data delivered by the audio data transport. Except for HCI, the meaning of “unit” depends on the Host transport used and, therefore, is vendor specific (for example, on a PCM transport this should indicate the number of bits transported per sync pulse, and would normally be 8 or 16). The Host shall set the `Input_Transport_Unit_Size` and `Output_Transport_Unit_Size` to be equal. For HCI Host transport the Host shall set them to 0.



The following parameters are used by the Link Manager to negotiate the synchronous transport:

- The `Max_Latency` parameter specifies an upper limit to the time in milliseconds between the eSCO (or SCO) instants, plus the size of the retransmission window, plus the length of the reserved synchronous slots for this logical transport.
- The `Packet_Type` parameter is a bitmap specifying which synchronous packet types may be used by the Link Manager in the negotiation of the link parameters. Multiple packet types are specified by bitwise OR of the packet type codes in the table. At least one packet type shall be specified for each negotiation. It is recommended to enable as many packet types as possible.
Note: It is allowed to enable packet types that are not supported by the local device.
- The `Retransmission_Effort` parameter specifies the extra resources that are allocated to this connection if a packet may need to be retransmitted. The `Retransmission_Effort` parameter shall be set to indicate the required behavior, or to “Don’t care”.

The following restrictions shall apply:

- Either both the `Transmit_Coding_Format` and `Input_Coding_Format` shall be “transparent” or neither shall be. If both are “transparent”, the `Transmit_Bandwidth` and the `Input_Bandwidth` shall be the same and the Controller shall not modify the data sent to the remote device.
- Either both the `Receive_Coding_Format` and `Output_Coding_Format` shall be “transparent” or neither shall be. If both are “transparent”, the `Receive_Bandwidth` and the `Output_Bandwidth` shall be the same and the Controller shall not modify the data sent to the Host.

A `Connection_Handle` for the new synchronous connection will be returned in the `HCI_Synchronous_Connection_Complete` event if the command is used to set up a new synchronous connection.

When used to modify an existing synchronous logical transport, only the `Packet_Type`, `Retransmission_Effort` and `Max_Latency` parameters may be modified.

Note: The Link Manager may choose any combination of packet types, timing, and retransmission window sizes that satisfy the parameters given. This may be achieved by using more frequent transmissions of smaller packets. The link manager may choose to set up either a SCO or an eSCO connection, if the parameters allow, using the corresponding LMP sequences.

Note: To modify a SCO connection, use the `HCI_Change_Connection_Packet_Type` command.

If the lower layers cannot achieve the exact transmit and receive bandwidth requested subject to the other parameters, or cannot achieve the transcoding or resampling implied by the parameters, then the link creation or link



modification shall be rejected. A synchronous connection may only be created when an ACL connection already exists.

The data at the audio data transport interface shall be treated as a stream of bits. The bits in each unit of data delivered by the transport shall be taken LSB first, and the units shall be taken in the order of delivery. The samples, encoded samples, frames, or other entity to be transcoded for transmission, or that has been transcoded after reception, shall be taken in the order of transmission or reception, with each entity taken LSB first.

For example, if the audio data transport uses 16 bit units and the Input or Output coding format is A-law, each unit represents two samples with the first in the 8 least significant bits and the second in the 8 most significant bits. Similarly, if the audio data transport uses 8 bit units and the Input or Output coding format is linear PCM with a size of 16 bits, the 8 least significant bits of each sample are transmitted first.

Command parameters:

Connection_Handle: *Size: 2 octets (12 bits meaningful)*

Value	Parameter Description
0xFFFF	Connection_Handle Range: 0x0000 to 0xFFFF

Transmit_Bandwidth: *Size: 4 octets*

Value	Parameter Description
0x00000000 to 0xFFFFFFFF	Transmit bandwidth in octets per second.
0xFFFFFFFF	Don't care

Receive_Bandwidth: *Size: 4 octets*

Value	Parameter Description
0x00000000 to 0xFFFFFFFF	Receive bandwidth in octets per second.
0xFFFFFFFF	Don't care

Transmit_Coding_Format: *Size: 5 octets*

Value	Parameter Description
Octet 0	See Assigned Numbers for Coding_Format
Octets 1 to 4	Octet 1 to 2: Company ID, see Assigned Numbers for Company Identifier Octet 3 to 4: Vendor specific codec ID. Shall be ignored if octet 0 of Transmit_Coding_Format is not 0xFF.

**Receive_Coding_Format:****Size: 5 octets**

Value	Parameter Description
Octet 0	See Assigned Numbers for Coding_Format
Octets 1 to 4	Octet 1 to 2: Company ID, see Assigned Numbers for Company Identifier. Octet 3 to 4: Vendor specific codec ID. Shall be ignored if octet 0 of Receive_Coding_Format is not 0xFF.

Transmit_Codec_Frame_Size:**Size: 2 octets**

Value	Parameter Description
0xFFFF	Range: 0x0001 to 0xFFFF, the actual size of the over-the-air encoded frame in octets.

Receive_Codec_Frame_Size:**Size: 2 octets**

Value	Parameter Description
0xFFFF	Range: 0x0001 to 0xFFFF, the actual size of the over-the-air encoded frame in octets.

Input_Bandwidth:**Size: 4 octets**

Value	Parameter Description
0xFFFFFFFF	Host to Controller nominal data rate in octets per second.

Output_Bandwidth:**Size: 4 octets**

Value	Parameter Description
0xFFFFFFFF	Controller to Host nominal data rate in octets per second.

Input_Coding_Format:**Size: 5 octets**

Value	Parameter Description
Octet 0	See Assigned Numbers for Coding_Format
Octets 1 to 4	Octet 1 to 2: Company ID, see Assigned Numbers for Company Identifier. Octet 3 to 4: Vendor specific codec ID. Shall be ignored if octet 0 of Input_Coding_Format is not 0xFF.

Output_Coding_Format:**Size: 5 octets**

Value	Parameter Description
Octet 0	See Assigned Numbers for Coding_Format



Value	Parameter Description
Octets 1 to 4	Octet 1 to 2: Company ID, see Assigned Numbers for Company Identifier. Octet 3 to 4: Vendor specific codec ID. Shall be ignored if octet 0 of Output_Coding_Format is not 0xFF.

*Input_Coded_Data_Size:**Size: 2 octets*

Value	Parameter Description
0xFFFF	Size, in bits, of the sample or framed data

*Output_Coded_Data_Size:**Size: 2 octets*

Value	Parameter Description
0xFFFF	Size, in bits, of the sample or framed data

*Input_PCM_Data_Format:**Size: 1 octet*

Value	Parameter Description
0xFF	See Assigned Numbers for PCM_Data_Format

*Output_PCM_Data_Format:**Size: 1 octet*

Value	Parameter Description
0xFF	See Assigned Numbers for PCM_Data_Format

*Input_PCM_Sample_Payload_MSB_Position:**Size: 1 octet*

Value	Parameter Description
0xFF	The number of bit positions within an audio sample that the MSB of the sample is away from starting at the MSB of the data.

*Output_PCM_Sample_Payload_MSB_Position:**Size: 1 octet*

Value	Parameter Description
0xFF	The number of bit positions within an audio sample that the MSB of the sample is away from starting at the MSB of the data.

*Input_Data_Path:**Size: 1 octet*

Value	Parameter Description
0x00	HCI
0x01 to 0xFE	Logical_Channel_Number. The meaning of the logical channels will be vendor specific.
0xFF	Audio test mode

*Output_Data_Path:**Size: 1 octet*

Value	Parameter Description
0x00	HCI
0x01 to 0xFE	Logical_Channel_Number. The meaning of the logical channels will be vendor specific.
0xFF	Audio test mode

*Input_Transport_Unit_Size:**Size: 1 octet*

Value	Parameter Description
1 to 255	The number of bits in each unit of data received from the Host over the audio data transport.
0	Not applicable (implied by the choice of audio data transport)

*Output_Transport_Unit_Size:**Size: 1 octet*

Value	Parameter Description
1 to 255	The number of bits in each unit of data sent to the Host over the audio data transport.
0	Not applicable (implied by the choice of audio data transport)

*Max_Latency:**Size: 2 octets*

Value	Parameter Description
0x0000 to 0x0003	Reserved for future use
0x0004 to 0xFFFFE	The value in milliseconds representing the upper limit of the sum of the synchronous interval, and the size of the eSCO window, where the eSCO window is the reserved slots plus the retransmission window. (See Figure 8.9 in [Vol 2] Part B, Baseband Specification)
0xFFFF	Don't care.

*Packet_Type:**Size: 2 octets*

Bit Number	Parameter Description
0	HV1 may be used
1	HV2 may be used
2	HV3 may be used
3	EV3 may be used
4	EV4 may be used
5	EV5 may be used
6	2-EV3 shall not be used
7	3-EV3 shall not be used
8	2-EV5 shall not be used
9	3-EV5 shall not be used
All other bits	Reserved for future use

Note: 0x003F means all packet types may be used.

*Retransmission_Effort:**Size: 1 octet*

Value	Parameter Description
0x00	No retransmission (SCO or eSCO connection allowed)
0x01	At least one retransmission, optimize for power consumption (eSCO connection required)
0x02	At least one retransmission, optimize for link quality (eSCO connection required)
0xFF	Don't care (SCO or eSCO connection allowed)
All other values	Reserved for future use

Return parameters:

None.

Event(s) generated (unless masked away):

When the BR/EDR Controller receives the HCI_Enhanced_Setup_Synchronous_Connection command, it shall send the HCI_Command_Status event to the Host. In addition, when the LM determines that the connection is established, the local BR/EDR Controller shall send an HCI_Synchronous_Connection_Complete event to the local Host, and the remote Controller will send an HCI_Synchronous_Connection_Complete event or an HCI_Connection_Complete event to the remote Host. The HCI_Synchronous_Connection_Complete event contains the Connection_Handle if this command is successful.



This command cannot be used to change the parameters of a SCO link.

Note: No HCI_Command_Complete event will be sent by the local BR/EDR Controller to indicate that this command has been completed. Instead, the HCI_Synchronous_Connection_Complete event (for a new connection setup) or HCI_Synchronous_Connection_Changed event (for modifying an existing synchronous link) will indicate that this command has been completed.



7.1.46 Enhanced Accept Synchronous Connection Request command

Command	OCF	Command Parameters	Return Parameters
HCI_Enhanced_Accept_Synchronous_Connection_Request	0x003E	BD_ADDR, Transmit_Bandwidth, Receive_Bandwidth, Transmit_Coding_Format, Receive_Coding_Format, Transmit_Codec_Frame_Size, Receive_Codec_Frame_Size, Input_Bandwidth, Output_Bandwidth, Input_Coding_Format, Output_Coding_Format, Input_Coded_Data_Size, Output_Coded_Data_Size, Input_PCM_Data_Format, Output_PCM_Data_Format, Input_PCM_Sample_Payload_MSB_Position, Output_PCM_Sample_Payload_MSB_Position, Input_Data_Path, Output_Data_Path, Input_Transport_Unit_Size, Output_Transport_Unit_Size, Max_Latency, Packet_Type, Retransmission_Effort	

Description:

The HCI_Enhanced_Accept_Synchronous_Connection_Request command is used to accept an incoming request for a synchronous connection and to present the local Link Manager with the acceptable parameter values for the synchronous connection. This command shall only be issued after an HCI_Connection_Request event, with link type SCO or eSCO, has occurred. An HCI_Connection_Request event contains the BD_ADDR of the device requesting the connection. The command to accept a connection must be received by the Controller before the connection accept timeout expires on the local device.

The parameter set of the HCI_Enhanced_Accept_Synchronous_Connection_Request command is the



same as for the HCI_Enhanced_Setup_Synchronous_Connection command except for the Connection_Handle in the HCI_Enhanced_Setup_Synchronous_Connection command, which is replaced by the BD_ADDR in the HCI_Enhanced_Accept_Synchronous_Connection_Request command. See [Section 7.1.45](#) for the descriptions of these parameters.

If the Link Type of the incoming request is SCO, then the Retransmission Effort parameter shall be ignored.

Note: See [Section 7.3.3](#) for the behavior when the HCI_Connection_Request event is masked or the connection is auto accepted.

If the ACL link has encryption enabled using AES-CCM and the Controller cannot establish an eSCO transport (e.g. the Host parameters restricting the packet types to SCO packet types), the Controller shall return the error code *Connection Rejected Due to Security Reasons* (0x0E) and a SCO transport will not be established.

Command parameters:

BD_ADDR:

Size: 6 octets

Value	Parameter Description
0XXXXXXXXXX X	BD_ADDR of the device requesting the connection

Transmit_Bandwidth:

Size: 4 octets

Value	Parameter Description
0x00000000 to 0xFFFFFFFF	Transmit bandwidth in octets per second.
0xFFFFFFFF	Don't care

Receive_Bandwidth:

Size: 4 octets

Value	Parameter Description
0x00000000 to 0xFFFFFFFF	Receive bandwidth in octets per second.
0xFFFFFFFF	Don't care

Transmit_Coding_Format:

Size: 5 octets

Value	Parameter Description
Octet 0	See Assigned Numbers for Coding_Format



Value	Parameter Description
Octets 1 to 4	Octet 1 to 2: Company ID, see Assigned Numbers for Company Identifier. Octet 3 to 4: Vendor specific codec ID. Shall be ignored if octet 0 of Transmit_Coding_Format is not 0xFF.

*Receive_Coding_Format:**Size: 5 octets*

Value	Parameter Description
Octet 0	See Assigned Numbers for Coding_Format
Octets 1 to 4	Octet 1 to 2: Company ID, see Assigned Numbers for Company Identifier. Octet 3 to 4: Vendor specific codec ID. Shall be ignored if octet 0 of Receive_Coding_Format is not 0xFF.

*Transmit_Codec_Frame_Size:**Size: 2 octets*

Value	Parameter Description
0xFFFF	Range: 0x0001 to 0xFFFF, the actual size of the over-the-air encoded frame in octets.

*Receive_Codec_Frame_Size:**Size: 2 octets*

Value	Parameter Description
0xFFFF	Range: 0x0001 to 0xFFFF, the actual size of the over-the-air encoded frame in octets.

*Input_Bandwidth:**Size: 4 octets*

Value	Parameter Description
0xFFFFFFFF	Host to Controller nominal data rate in octets per second.

*Output_Bandwidth:**Size: 4 octets*

Value	Parameter Description
0xFFFFFFFF	Controller to Host nominal data rate in octets per second.

*Input_Coding_Format:**Size: 5 octets*

Value	Parameter Description
Octet 0	See Assigned Numbers for Coding_Format



Value	Parameter Description
Octets 1 to 4	Octet 1 to 2: Company ID, see Assigned Numbers for Company Identifier. Octet 3 to 4: Vendor specific codec ID. Shall be ignored if octet 0 of Input_Coding_Format is not 0xFF.

*Output_Coding_Format:**Size: 5 octets*

Value	Parameter Description
Octet 0	See Assigned Numbers for Coding_Format
Octets 1 to 4	Octet 1 to 2: Company ID, see Assigned Numbers for Company Identifier. Octet 3 to 4: Vendor specific codec ID. Shall be ignored if octet 0 of Output_Coding_Format is not 0xFF.

*Input_Coded_Data_Size:**Size: 2 octets*

Value	Parameter Description
0xFFFF	Size, in bits, of the sample or framed data

*Output_Coded_Data_Size:**Size: 2 octets*

Value	Parameter Description
0xFFFF	Size, in bits, of the sample or framed data

*Input_PCM_Data_Format:**Size: 1 octet*

Value	Parameter Description
0xFF	See Assigned Numbers for PCM_Data_Format

*Output_PCM_Data_Format:**Size: 1 octet*

Value	Parameter Description
0xFF	See Assigned Numbers for PCM_Data_Format

*Input_PCM_Sample_Payload_MSB_Position:**Size: 1 octet*

Value	Parameter Description
0xFF	The number of bit positions within an audio sample that the MSB of the sample is away from starting at the MSB of the data.

*Output_PCM_Sample_Payload_MSB_Position:**Size: 1 octet*

Value	Parameter Description
0xXX	The number of bit positions within an audio sample that the MSB of the sample is away from starting at the MSB of the data.

*InputData_Path:**Size: 1 octet*

Value	Parameter Description
0x00	HCI
0x01 to 0xFE	Logical_Channel_Number. The meaning of the logical channels will be vendor specific.
0xFF	Audio test mode

*Output_Data_Path:**Size: 1 octet*

Value	Parameter Description
0x00	HCI
0x01 to 0xFE	Logical_Channel_Number. The meaning of the logical channels will be vendor specific.
0xFF	Audio test mode

*Input_Transport_Unit_Size:**Size: 1 octet*

Value	Parameter Description
1 to 255	The number of bits in each unit of data received from the Host over the audio data transport.
0	Not applicable (implied by the choice of audio data transport)

*Output_Transport_Unit_Size:**Size: 1 octet*

Value	Parameter Description
1 to 255	The number of bits in each unit of data sent to the Host over the audio data transport.
0	Not applicable (implied by the choice of audio data transport)

*Max_Latency:**Size: 2 octets*

Value	Parameter Description
0x0000 to 0x0003	Reserved for future use
0x0004 to 0xFFFFE	The value in milliseconds representing the upper limit of the sum of the synchronous interval, and the size of the eSCO window, where the eSCO window is the reserved slots plus the retransmission window. (See Figure 8.9 in Part B, Baseband Specification)
0xFFFF	Don't care.

*Packet_Type:**Size: 2 octets*

Bit Number	Parameter Description
0	HV1 may be used
1	HV2 may be used
2	HV3 may be used
3	EV3 may be used
4	EV4 may be used
5	EV5 may be used
6	2-EV3 shall not be used
7	3-EV3 shall not be used
8	2-EV5 shall not be used
9	3-EV5 shall not be used
All other bits	Reserved for future use

Note: 0x003F means all packet types may be used.

*Retransmission_Effort:**Size: 1 octet*

Value	Parameter Description
0x00	No retransmission (SCO or eSCO connection allowed)
0x01	At least one retransmission, optimize for power consumption (eSCO connection required)
0x02	At least one retransmission, optimize for link quality (eSCO connection required)
0xFF	Don't care (SCO or eSCO connection allowed)
All other values	Reserved for future use

Return parameters:

None.

Event(s) generated (unless masked away):

The HCI_Enhanced_Accept_Synchronous_Request command requests the local BR/EDR Controller to start setting up the connection. When this action commences, the HCI_Command_Status event shall be sent by the BR/EDR Controller. When the link setup is complete, the BR/EDR Controller shall send an HCI_Synchronous_Connection_Complete event to its Host, and the remote BR/EDR Controller will send an HCI_Synchronous_Connection_Complete event to its



Host. The HCI_Synchronous_Connection_Complete will contain the Connection_Handle and the link parameters if the setup is successful.

Note: No HCI_Command_Complete event will be sent by the local Controller as the result of this command. Instead, the HCI_Synchronous_Connection_Complete event will indicate that this command has been completed.



7.1.47 Truncated Page command

Command	OCF	Command Parameters	Return Parameters
HCI_Truncated_Page	0x003F	BD_ADDR, Page_Scan_Repetition_Mode, Clock_Offset	

Description:

The HCI_Truncated_Page command is used to page the BR/EDR Controller with the specified BD_ADDR and then abort the paging sequence after an ID response has been received. See [\[Vol 2\] Part B, Section 8.3.3](#) for additional information.

The Page_Scan_Repetition_Mode parameter specifies the Page Scan Repetition mode supported by the remote BR/EDR Controller with the BD_ADDR. This is the most recent version of the information that was acquired either during the inquiry process or from an HCI_Page_Scan_Repetition_Mode_Change event (see [Section 7.7.31](#)).

The Clock_Offset parameter is the difference between the local BR/EDR Controller's own clock and the clock of the remote BR/EDR Controller with BD_ADDR. Only bits 2 through 16 of the difference are used, and they are mapped to this parameter as bits 0 through 14 respectively. A Clock_Offset_Valid_Flag, located in bit 15 of the Clock_Offset parameter, indicates if the Clock Offset is valid or not.

Command parameters:

BD_ADDR:

Size: 6 octets

Value	Parameter Description
0xFFFFFFFFXXXX	BD_ADDR of the Device to page

Page_Scan_Repetition_Mode:

Size: 1 octet

Value	Parameter Description
0x00	R0
0x01	R1
0x02	R2
All other values	Reserved for future use.

**Clock_Offset:****Size: 2 octets**

Value	Parameter Description
Bits 14-0	Bits 16-2 of CLKNSlave-CLK
Bit 15	Clock Offset is valid

Return parameters:

None.

Event(s) generated (unless masked away):

When the BR/EDR Controller receives the HCI_Truncated_Page command the BR/EDR Controller shall send the HCI_Command_Status event to the Host. In addition, when the Truncated Page procedure has completed, the BR/EDR Controller shall send an HCI_Truncated_Page_Complete event to the Host.

Note: No HCI_Command_Complete event will be sent by the BR/EDR Controller to indicate that this command has been completed. Instead, the HCI_Truncated_Page_Complete event will indicate that this command has been completed.



7.1.48 Truncated Page Cancel command

Command	OCF	Command Parameters	Return Parameters
HCI_Truncated_Page_Cancel	0x0040	BD_ADDR	Status, BD_ADDR

Description:

The HCI_Truncated_Page_Cancel command is used to request cancellation of an ongoing Truncated_Page process previously started by an HCI_Truncated_Page command.

Command parameters:

BD_ADDR: *Size: 6 octets*

Value	Parameter Description
0xFFFFFFFFXXXX	BD_ADDR of the device to which the HCI_Truncated_Page command was previously issued and that is the subject of the cancellation request.

Return parameters:

Status: *Size: 1 octet*

Value	Parameter Description
0x00	HCI_Truncated_Page_Cancel command succeeded.
0x01 to 0xFF	HCI_Truncated_Page_Cancel command failed. See [Vol 2] Part D, Error Codes , for error codes and descriptions

BD_ADDR: *Size: 6 octets*

Value	Parameter Description
0xFFFFFFFFXXXX	BD_ADDR of the device to which the HCI_Truncated_Page command was previously issued before and that is the subject of the cancellation request.

Event(s) generated (unless masked away):

When the HCI_Truncated_Page_Cancel command has completed, an HCI_Command_Complete event shall be generated.

If the truncated page procedure has already completed, but the BR/EDR Controller has not yet sent the HCI_Truncated_Page_Complete event, then the local device shall return an HCI_Command_Complete event with status "Success".



If the HCI_Truncated_Page_Cancel command is sent to the BR/EDR Controller without an outstanding HCI_Truncated_Page command to the same device, the BR/EDR Controller shall return an HCI_Command_Complete event with the error code *Unknown Connection Identifier* (0x02).

Note: From the BR/EDR Controller perspective this is identical to the situation where the HCI_Truncated_Page command has already completed and the HCI_Truncated_Page_Complete event already sent.



7.1.49 Set Connectionless Slave Broadcast command

Command	OCF	Command Parameters	Return Parameters
HCI_Set_Connectionless_Slave_Broadcast	0x0041	Enable, LT_ADDR, LPO_Allowed, Packet_Type, Interval_Min, Interval_Max, CSB_supervisionTO	Status, LT_ADDR, Interval

Description:

The HCI_Set_Connectionless_Slave_Broadcast command controls the Connectionless Slave Broadcast functionality in the BR/EDR Controller. Connectionless Slave Broadcast mode may be enabled or disabled by the Enable parameter. If Enable is set to 0x00 and the **synchronization train** substate is active, then the Controller shall also exit the **synchronization train** substate. If Enable is set to 0x01, the remaining parameters shall be ignored.

The LT_ADDR indicated in the Set_Connectionless_Slave_Broadcast shall be pre-allocated using the HCI_Set_Reserved_LT_ADDR command. If the LT_ADDR has not been reserved, the *Unknown Connection Identifier* (0x02) error code shall be returned. If the Controller is unable to reserve sufficient bandwidth for the requested activity, the *Connection Rejected Due to Limited Resources* (0x0D) error code shall be returned.

The LPO_Allowed parameter informs the BR/EDR Controller whether it is allowed to sleep.

The Packet_Type parameter specifies which packet types are allowed. The Host shall either enable BR packet types only, or shall enable EDR and DM1 packet types only.

The Interval_Min and Interval_Max parameters specify the range from which the BR/EDR Controller shall select the Connectionless Slave Broadcast Interval. The selected Interval is returned.

Command parameters:

Enable:

Size: 1 octet

Value	Parameter Description
0x00	Disabled
0x01	Enabled
All other values	Reserved for future use

**LT_ADDR:****Size: 1 octet**

Value	Parameter Description
0x01 to 0x07	LT_ADDR used for Connectionless Slave Broadcast
All other values	Reserved for future use

LPO_Allowed:**Size: 1 octet**

Value	Parameter Description
0x00	BR/EDR Controller shall not sleep (that is, clock accuracy shall be equal to or better than ± 20 ppm)
0x01	BR/EDR Controller may sleep (that is, clock accuracy shall be equal to or better than ± 250 ppm)
All other values	Reserved for future use

Packet_Type:**Size: 2 octets**

Bit Number	Parameter Description
1	2-DH1 shall not be used
2	3-DH1 shall not be used
3	DM1 may be used
4	DH1 may be used
8	2-DH3 shall not be used
9	3-DH3 shall not be used
10	DM3 may be used
11	DH3 may be used
12	2-DH5 shall not be used
13	3-DH5 shall not be used
14	DM5 may be used
15	DH5 may be used
All other bits	Reserved for future use.

*Interval_Min:**Size: 2 octets*

Value	Parameter Description
0xFFFF	Minimum interval between Connectionless Slave Broadcast packets in slots. Range: 0x0002 to 0xFFFF; only even values are valid

*Interval_Max:**Size: 2 octets*

Value	Parameter Description
0xFFFF	Maximum interval between Connectionless Slave Broadcast packets in slots. Range: 0x0002 to 0xFFFF; only even values are valid

*CSB_supervisionTO:**Size: 2 octets*

Value	Parameter Description
0xFFFF	Duration in slots after which the BR/EDR Controller reports a Connectionless HCI_Slave_Broadcast_Timeout event if it is unable to transmit a Connectionless Slave Broadcast packet. Range: 0x0002 to 0xFFFF; only even values are valid

Return parameters:*Status:**Size: 1 octet*

Value	Parameter Description
0x00	HCI_Set_Connectionless_Slave_Broadcast command succeeded.
0x01 to 0xFF	HCI_Set_Connectionless_Slave_Broadcast command failed. See [Vol 2] Part D, Error Codes , for error codes and descriptions.

*LT_ADDR:**Size: 1 octet*

Value	Parameter Description
0x01 to 0x07	LT_ADDR used for Connectionless Slave Broadcast
All other values	Reserved for future use

*Interval:**Size: 2 octets*

Value	Parameter Description
0xFFFF	Actual interval between Connectionless Slave Broadcast packets in slots. Range: 0x0002 to 0xFFFF; only even values are valid

**Event(s) generated (unless masked away):**

When the HCI_Set_Connectionless_Slave_Broadcast command has completed, an HCI_Command_Complete event shall be generated.

If the BR/EDR Controller is unable to transmit a Connectionless Slave Broadcast packet for *CSB_supervisionTO slots*, it shall generate an HCI_Connectionless_Slave_Broadcast_Timeout event.



7.1.50 Set Connectionless Slave Broadcast Receive command

Command	OCF	Command Parameters	Return Parameters
HCI_Set_Connectionless_Slave_Broadcast_Receive	0x0042	Enable, BD_ADDR, LT_ADDR, Interval, Clock_Offset, Next_Connectionless_Slave_Broadcast_Clock, CSB_supervisionTO, Remote_Timing_Accuracy, Skip, Packet_Type, AFH_Channel_Map	Status, BD_ADDR, LT_ADDR

Description:

The HCI_Set_Connectionless_Slave_Broadcast_Receive command controls the reception of Connectionless Slave Broadcast packets in the BR/EDR Controller of a Connectionless Slave Broadcast slave. If the Enable parameter is set to Disabled, the BR/EDR Controller does not attempt to receive Connectionless Slave Broadcast packets and the remaining parameters shall be ignored. If the Enable parameter is set to Enabled, the BR/EDR Controller starts attempting to receive Connectionless Slave Broadcast packets on the specified LT_ADDR.

The Interval parameter specifies the interval of the Connectionless Slave Broadcast to be used by the BR/EDR Controller.

The Skip parameter specifies the number of consecutive Connectionless Slave Broadcast instants which the receiver may skip after successfully receiving a Connectionless Slave Broadcast packet.

The Packet_Type parameter specifies which packet types are allowed. The Host shall either enable BR packet types only or shall enable EDR and DM1 packet types only.

The AFH_Channel_Map parameter is the AFH channel map used by the master for the PBD logical link, and is obtained by the slave's Host from the HCI_Synchronization_Train_Received event.

**Command parameters:***Enable:**Size: 1 octet*

Value	Parameter Description
0x00	Disabled
0x01	Enabled
All other values	Reserved for future use

*BD_ADDR:**Size: 6 octets*

Value	Parameter Description
0XXXXXXXXXXXXX	BD_ADDR of the Connectionless Slave Broadcast transmitter

*LT_ADDR:**Size: 1 octet*

Value	Parameter Description
0x01 to 0x07	LT_ADDR to use for receiving Connectionless Slave Broadcast messages
All other values	Reserved for future use

*Interval:**Size: 2 octets*

Value	Parameter Description
0XXXXX	Interval between Connectionless Slave Broadcast packets instants in slots. Range: 0x0002 to 0xFFFE; only even values are valid

*Clock_Offset:**Size: 4 octets (28 bits meaningful)*

Value	Parameter Description
0XXXXXXXX	$(\text{CLKN}_{\text{slave}} - \text{CLK}) \bmod 2^{28}$

Next_Connectionless_Slave_Broadcast_Clock: Size: 4 octets (28 bits meaningful)

Value	Parameter Description
0XXXXXXXX	CLK for next Connectionless Slave Broadcast instant

*CSB_supervisionTO:**Size: 2 octets*

Value	Parameter Description
0XXXXX	Duration in slots to continue listening for Connectionless Slave Broadcast packets after the last successfully received Connectionless Slave Broadcast packet. Range: 0x0002 to 0xFFFE; only even values are valid

**Remote_Timing_Accuracy:****Size: 1 octet**

Value	Parameter Description
0xXX	Timing accuracy of the master in ppm. Typical values are 20ppm and 250ppm.

Skip:**Size: 1 octet**

Value	Parameter Description
0xXX	Number of Connectionless Slave Broadcast instants to skip after successfully receiving a Broadcast packet.

Packet_Type:**Size: 2 octets**

Bit Number	Parameter Description
1	2-DH1 shall not be used
2	3-DH1 shall not be used
3	DM1 may be used
4	DH1 may be used
8	2-DH3 shall not be used
9	3-DH3 shall not be used
10	DM3 may be used
11	DH3 may be used
12	2-DH5 shall not be used
13	3-DH5 shall not be used
14	DM5 may be used
15	DH5 may be used
All other bits	Reserved for future use.

AFH_Channel_Map:**Size: 10 octets (79 bits meaningful)**

Value	Parameter Description
0XXXXXXXXX XXXXXXXXXX XX	This parameter contains 80 1-bit fields. The n^{th} such field (in the range 0 to 78) contains the value for channel n: 0: channel n is unused 1: channel n is used The most significant bit (bit 79) is reserved for future use At least N_{min} channels shall be marked as used (see [Vol 2] Part B, Section 2.3.1)

Return parameters:

**Status:****Size: 1 octet**

Value	Parameter Description
0x00	HCI_Set_Connectionless_Slave_Broadcast_Receive command succeeded.
0x01 to 0xFF	HCI_Set_Connectionless_Slave_Broadcast_Receive command failed. See [Vol 2] Part D, Error Codes, for error codes and descriptions.

BD_ADDR:**Size: 6 octets**

Value	Parameter Description
0XXXXXXXXXXXXX	BD_ADDR of the Connectionless Slave Broadcast transmitter

LT_ADDR:**Size: 1 octet**

Value	Parameter Description
0x01 to 0x07	LT_ADDR used for receiving Connectionless Slave Broadcast messages
All other values	Reserved for future use

Event(s) generated (unless masked away):

When the HCI_Set_Connectionless_Slave_Broadcast_Receive command has been received, an HCI_Command_Complete event shall be generated. Completion of the HCI_Set_Connectionless_Slave_Broadcast_Receive command does not require reception of a Connectionless Slave Broadcast packet. If the BR/EDR Controller does not receive a Connectionless Slave Broadcast packet for *CSB_supervisionTO* slots, it shall generate an HCI_Connectionless_Slave_Broadcast_Timeout event.



7.1.51 Start Synchronization Train command

Command	OCF	Command Parameters	Return Parameters
HCI_Start_Synchronization_Train	0x0043		

Description:

The HCI_Start_Synchronization_Train command controls the Synchronization Train functionality in the BR/EDR Controller. Connectionless Slave Broadcast mode shall be enabled on the BR/EDR Controller before this command may be used. If Connectionless Slave Broadcast mode is not enabled, the *Command Disallowed* (0x0C) error code shall be returned. After receiving this command and returning an HCI_Command_Status event, the Baseband starts attempting to send synchronization train packets containing information related to the enabled Connectionless Slave Broadcast packet timing.

Note: The AFH_Channel_Map used in the synchronization train packets is configured by the HCI_Set_AFH_Channel_Classification command and the local channel classification in the BR/EDR Controller.

The synchronization train packets will be sent using the parameters specified by the latest HCI_Write_Synchronization_Train_Parameters command. The Synchronization Train will continue until *synchronization_trainTO* slots (as specified in the last HCI_Write_Synchronization_Train command) have passed or until the Host disables the Connectionless Slave Broadcast logical transport.

Command parameters:

None.

Return parameters:

None

Event(s) generated (unless masked away):

When the BR/EDR Controller receives the HCI_Start_Synchronization_Train command, it shall send an HCI_Command_Status event to the Host.

Note: No HCI_Command_Complete event shall be sent by the BR/EDR Controller to indicate that this command has been completed. Instead the HCI_Synchronization_Train_Complete event will indicate that this command has been completed.



7.1.52 Receive Synchronization Train command

Command	OCF	Command Parameters	Return Parameters
HCI_Receive_ Synchronization_Train	0x0044	BD_ADDR, synchronization_scanTO, Sync_Scan_Window, Sync_Scan_Interval	

Description:

The HCI_Receive_Synchronization_Train command requests synchronization with the specified Connectionless Slave Broadcast Transmitter. The Sync_Scan_Window parameter specifies the duration of each scan and the Sync_Scan_Interval parameter specifies the interval between the start of consecutive scan windows. An HCI_Synchronization_Train_Received event shall be sent if a synchronization train packet is received with a non-zero Connectionless Slave Broadcast LT_ADDR or if the BR/EDR Controller fails to receive a synchronization train packet for *synchronization_scanTO* slots.

Command parameters:

BD_ADDR:

Size: 6 octets

Value	Parameter Description
0XXXXXXXXXXXXX	BD_ADDR of the Connectionless Slave Broadcast transmitter

synchronization_scanTO:

Size: 2 octets

Value	Parameter Description
0XXXXX	Duration in slots to search for the synchronization train Shall be greater than or equal to Sync_Scan_Window; only even values are valid

Sync_Scan_Window:

Size: 2 octets

Value	Parameter Description
0XXXXX	Duration in slots to listen for a synchronization train packet on a single frequency Range: 0x0022 to 0xFFFE; only even values are valid

**Sync_Scan_Interval:****Size: 2 octets**

Value	Parameter Description
0xXXXX	Duration in slots between the start of consecutive scan windows Shall be greater than or equal to Sync_Scan_Window+0x0002; only even values are valid

Return parameters:

None.

Event(s) generated (unless masked away):

When the BR/EDR Controller receives the HCI_Receive_Synchronization_Train command, it shall send an HCI_Command_Status event to the Host. In addition, when the BR/EDR Controller receives, or fails to receive within the duration specified by *synchronization_scanTO*, the synchronization train from the Connectionless Slave Broadcast transmitter, it shall send an HCI_Synchronization_Train_Received event to the Host.

Note: No HCI_Command_Complete event shall be sent by the BR/EDR Controller to indicate that this command has been completed. Instead the HCI_Synchronization_Train_Received event will indicate that this command has been completed.



7.1.53 Remote OOB Extended Data Request Reply command

Command	OCF	Command Parameters	Return Parameters
HCI_Remote_OOB_Extended_Data_Request_Reply	0x0045	BD_ADDR, C_192, R_192, C_256, R_256	Status, BD_ADDR

Description:

The HCI_Remote_OOB_Extended_Data_Request_Reply command is used to reply to an HCI_Remote_OOB_Data_Request event with the C and R values derived with the P-192 public key and the C and R values associated with the P-256 public key received via an OOB transfer from a remote device identified by BD_ADDR. If the C_192 and R_192 values are not present in the received OOB data from the remote device, the Host shall set C_192 and R_192 to zeros. If the C_256 and R_256 values are not present in the received OOB data from the remote device, the Host shall set C_256 and R_256 to zeros.

Command parameters:

BD_ADDR:

Size: 6 octets

Value	Parameter Description
0XXXXXXXXXXXXX X	BD_ADDR of remote device from which the C and R values were received

C_192:

Size: 16 octets

Value	Parameter Description
0XXXXXXXXXXXXX XXXXXXXXXXXXX XXXXXXXXXXXXX	Simple Pairing Hash C derived from the P-192 public key.

R_192:

Size: 16 octets

Value	Parameter Description
0XXXXXXXXXXXXX XXXXXXXXXXXXX XXXXXXXXXXXXX	Simple Pairing Randomizer associated with the P-192 public key.

**C_256:****Size: 16 octets**

Value	Parameter Description
0XXXXXXXXXXXXX XXXXXXXXXXXXX XXXXXXXXXXXXX	Simple Pairing Hash C derived from the P-256 public key.

R_256:**Size: 16 octets**

Value	Parameter Description
0XXXXXXXXXXXXX XXXXXXXXXXXXX XXXXXXXXXXXXX	Simple Pairing Randomizer associated with the P-256 public key.

Return parameters:**Status:****Size: 1 octet**

Value	Parameter Description
0x00	HCI_Remote_OOB_Extended_Data_Request_Reply command succeeded.
0x01-0xFF	HCI_Remote_OOB_Extended_Data_Request_Reply command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

BD_ADDR:**Size: 6 octets**

Value	Parameter Description
0XXXXXXXXXXXXX X	BD_ADDR of remote device from which the C and R values were received

Event(s) generated (unless masked away):

When the HCI_Remote_OOB_Extended_Data_Request_Reply command has completed, an HCI_Command_Complete event shall be generated.



7.2 LINK POLICY COMMANDS

The Link Policy commands provide methods for the Host to affect how the Link Manager manages the piconet. When Link Policy commands are used, the LM still controls how Bluetooth piconets and scatternets are established and maintained, depending on adjustable policy parameters. These policy commands modify the Link Manager behavior that can result in changes to the Link Layer connections with Bluetooth remote devices.

Note: Only one ACL connection can exist between two BR/EDR Controllers, and therefore there can only be one ACL HCI Connection_Handle for each physical Link Layer Connection. The BR/EDR Controller provides policy adjustment mechanisms to provide support for a number of different policies. This capability allows one Bluetooth module to be used to support many different usage models, and the same Bluetooth module can be incorporated in many different types of BR/EDR Controllers.

For the Link Policy commands, the OGF is defined as 0x02.

7.2.1 Hold Mode command

Command	OCF	Command Parameters	Return Parameters
HCI_Hold_Mode	0x0001	Connection_Handle, Hold_Mode_Max_Interval, Hold_Mode_Min_Interval	

Description:

The HCI_Hold_Mode command is used to alter the behavior of the Link Manager, and have it place the ACL baseband connection associated by the specified Connection_Handle into the Hold mode. The Connection_Handle shall identify an ACL connection and not a SCO, eSCO, or AMP connection. The Hold_Mode_Max_Interval and Hold_Mode_Min_Interval command parameters specify the length of time the Host wants to put the connection into the Hold mode. The local and remote devices will negotiate the length in the Hold mode. The Hold_Mode_Max_Interval parameter is used to specify the maximum length of the Hold interval for which the Host may actually enter into the Hold mode after negotiation with the remote device. The Hold interval defines the amount of time between when the Hold mode begins and when the Hold mode is completed. The Hold_Mode_Min_Interval parameter is used to specify the minimum length of the Hold interval for which the Host may actually enter into the Hold mode after the negotiation with the remote device. Therefore the Hold_Mode_Min_Interval shall not be greater than the Hold_Mode_Max_Interval. The BR/EDR Controller will return the actual Hold interval in the Interval parameter of the HCI_Mode_Change event, if the command is successful. This command enables the Host to support a low-power policy for itself or several other BR/EDR Controllers, and allows the



devices to enter Inquiry Scan, Page Scan, and a number of other possible actions.

Note: If the Host sends data to the BR/EDR Controller with a Connection_Handle corresponding to a connection in Hold mode, the BR/EDR Controller will keep the data in its buffers until either the data can be transmitted (the Hold mode has ended) or a flush, a flush timeout or a disconnection occurs. This is valid even if the Host has not yet been notified of the Hold mode through an HCI_Mode_Change event when it sends the data.

Note: The above is not valid for an HCI Data packet sent from the Host to the BR/EDR Controller on the master side where the Connection_Handle is a Connection_Handle used for broadcast and the Broadcast_Flag is set to Active Broadcast. The broadcast data will then never be received by slaves in Hold mode.

The Hold_Mode_Max_Interval shall be less than the Link Supervision Timeout configuration parameter.

Command parameters:

Connection_Handle:

Size: 2 octets (12 bits meaningful)

Value	Parameter Description
0xXXXX	Connection_Handle Range: 0x0000 to 0x0EFF

Hold_Mode_Max_Interval:

Size: 2 octets

Value	Parameter Description
N = 0xXXXX	Maximum acceptable number of Baseband slots to wait in Hold mode. Time Length of the Hold = $N * 0.625$ ms (1 Baseband slot) Range: 0x0002 to 0xFFFE; only even values are valid. Time Range: 1.25ms to 40.9 s Mandatory Range: 0x0014 to 0x8000

Hold_Mode_Min_Interval:

Size: 2 octets

Value	Parameter Description
N = 0xXXXX	Minimum acceptable number of Baseband slots to wait in Hold mode. Time Length of the Hold = $N * 0.625$ ms (1 Baseband slot) Range: 0x0002 to 0xFF00; only even values are valid Time Range: 1.25 ms to 40.9 s Mandatory Range: 0x0014 to 0x8000

**Return parameters:**

None.

Event(s) generated (unless masked away):

When the BR/EDR Controller receives the HCI_Hold_Mode command, the BR/EDR Controller shall send the HCI_Command_Status event to the Host. The HCI_Mode_Change event shall occur when the Hold mode has started and the HCI_Mode_Change event shall occur again when the Hold mode has completed for the specified Connection_Handle. The HCI_Mode_Change event signaling the end of the Hold mode is an estimation of the Hold mode ending if the event is for a remote BR/EDR Controller.

Note: No HCI_Command_Complete event will be sent by the Controller to indicate that this command has been completed. Instead, only the HCI_Mode_Change event will indicate that this command has been completed. If an error occurs after the HCI_Command_Status event has occurred, then the status in the HCI_Mode_Change event shall indicate the error.



7.2.2 Sniff Mode command

Command	OCF	Command Parameters	Return Parameters
HCI_Sniff_Mode	0x0003	Connection_Handle, Sniff_Max_Interval, Sniff_Min_Interval, Sniff_Attempt, Sniff_Timeout	

Description:

The HCI_Sniff_Mode command is used to alter the behavior of the Link Manager and have it place the ACL baseband connection associated with the specified Connection_Handle into the Sniff mode. The Connection_Handle command parameter is used to identify which ACL link connection is to be placed in Sniff mode and shall identify an ACL connection, not a SCO, eSCO, or AMP connection. The Sniff_Max_Interval and Sniff_Min_Interval command parameters are used to specify the requested acceptable maximum and minimum periods in the Sniff mode. The Sniff_Min_Interval shall not be greater than the Sniff_Max_Interval. The sniff interval defines the amount of time between each consecutive sniff period. The BR/EDR Controller will return the actual sniff interval in the Interval parameter of the HCI_Mode_Change event, if the command is successful. For a description of the meaning of the Sniff_Attempt and Sniff_Timeout parameters, see [\[Vol 2\] Part B, Section 8.7](#). Sniff_Attempt is there called $N_{\text{sniff attempt}}$ and Sniff_Timeout is called $N_{\text{sniff timeout}}$. This command enables the Host to support a low-power policy for itself or several other BR/EDR Controllers, and allows the devices to enter Inquiry Scan, Page Scan, and a number of other possible actions.

Note: If the Host sends data to the BR/EDR Controller with a Connection_Handle corresponding to a connection in Sniff mode, the BR/EDR Controller will keep the data in its buffers until either the data can be transmitted or a flush, a flush timeout or a disconnection occurs. This is valid even if the Host has not yet been notified of the Sniff mode through an HCI_Mode_Change event when it sends the data.

Note: It is possible for the master to transmit data to a slave without exiting Sniff mode (see description in [\[Vol 2\] Part B, Section 8.7](#)).

Note: The above is not valid for an HCI Data packet sent from the Host to the BR/EDR Controller on the master side where the Connection_Handle is a Connection_Handle used for broadcast and the Broadcast_Flag is set to Active Broadcast. In that case, the broadcast data will only be received by a slave in Sniff mode if that slave happens to listen to the master when the broadcast is made.

The Sniff_Max_Interval shall be less than the Link Supervision Timeout configuration parameter.

**Command parameters:***Connection_Handle:**Size: 2 octets (12 bits meaningful)*

Value	Parameter Description
0xXXXX	Connection_Handle Range: 0x0000 to 0x0EFF

*Sniff_Max_Interval:**Size: 2 octets*

Value	Parameter Description
N = 0xXXXX	Range: 0x0002 to 0xFFFFE; only even values are valid Mandatory Range: 0x0006 to 0x0540 Time = N * 0.625 ms Time Range: 1.25 ms to 40.9 s

*Sniff_Min_Interval:**Size: 2 octets*

Value	Parameter Description
N = 0xXXXX	Range: 0x0002 to 0xFFFFE; only even values are valid Mandatory Range: 0x0006 to 0x0540 Time = N * 0.625 ms Time Range: 1.25 ms to 40.9 s

*Sniff_Attempt:**Size: 2 octets*

Value	Parameter Description
N = 0xXXXX	Number of Baseband receive slots for sniff attempt. Length = N * 1.25 ms Range: 0x0001 to 0x7FFF Time Range: 1.25ms to 40.9 s Mandatory Range for Controller: 1 to $T_{sniff}/2$

*Sniff_Timeout:**Size: 2 octets*

Value	Parameter Description
N = 0xXXXX	Number of Baseband receive slots for sniff timeout. Length = N * 1.25 ms Range: 0x0000 to 0x7FFF Time Range: 0 ms to 40.9 s Mandatory Range for Controller: 0 to 0x0028

Return parameters:

None.

**Event(s) generated (unless masked away):**

When the BR/EDR Controller receives the HCI_Sniff_Mode command, the BR/EDR Controller shall send the HCI_Command_Status event to the Host. The HCI_Mode_Change event shall occur when the Sniff mode has started for the specified Connection_Handle.

Note: No HCI_Command_Complete event will be sent by the BR/EDR Controller to indicate that this command has been completed. Instead only the HCI_Mode_Change event will indicate that this command has been completed. If an error occurs after the HCI_Command_Status event has occurred, then the status in the HCI_Mode_Change event shall indicate the error.



7.2.3 Exit Sniff Mode command

Command	OCF	Command Parameters	Return Parameters
HCI_Exit_Sniff_Mode	0x0004	Connection_Handle	

Description:

The HCI_Exit_Sniff_Mode command is used to end the Sniff mode for a Connection_Handle, which is currently in Sniff mode. The Connection_Handle shall identify an ACL connection and not a SCO, eSCO, or AMP connection. The Link Manager will determine and issue the appropriate LMP commands to remove the Sniff mode for the associated Connection_Handle.

Command parameters:

Connection_Handle: Size: 2 octets (12 bits meaningful)

Value	Parameter Description
0xXXXX	Connection_Handle Range: 0x0000 to 0x0EFF

Return parameters:

None.

Event(s) generated (unless masked away):

When BR/EDR Controller receives the HCI_Exit_Sniff_Mode command, the BR/EDR Controller shall send the HCI_Command_Status event to the Host. The HCI_Mode_Change event shall occur when the Sniff mode has ended for the specified Connection_Handle.

Note: No HCI_Command_Complete event will be sent by the BR/EDR Controller to indicate that this command has been completed. Instead, only the HCI_Mode_Change event will indicate that this command has been completed.

7.2.4 This section no longer used

7.2.5 This section no longer used



7.2.6 QoS Setup command

Command	OCF	Command Parameters	Return Parameters
HCI_QoS_Setup	0x0007	Connection_Handle, Unused, Service_Type, Token_Rate, Peak_Bandwidth, Latency, Delay_Variation	

Description:

The HCI_QoS_Setup command is used to specify Quality of Service parameters for a Connection_Handle. The Connection_Handle shall be a Connection_Handle for an ACL connection. These QoS parameter are the same parameters as L2CAP QoS. For more detail see [\[Vol 3\] Part A, Logical Link Control and Adaptation Protocol Specification](#). This allows the Link Manager to have all of the information about what the Host is requesting for each connection. The LM will determine if the QoS parameters can be met. BR/EDR Controllers that are both slaves and masters can use this command. When a device is a slave, this command will trigger an LMP request to the master to provide the slave with the specified QoS as determined by the LM. When a device is a master, this command is used to request a slave device to accept the specified QoS as determined by the LM of the master. The Connection_Handle command parameter is used to identify for which connection the QoS request is requested.

The Unused parameter is reserved for future use.

Command parameters:

Connection_Handle: *Size: 2 octets (12 bits meaningful)*

Value	Parameter Description
0xFFFF	Connection_Handle Range: 0x0000 to 0x0EFF

Unused: *Size: 1 octet*

Value	Parameter Description
0x00	This value shall be used by the Host.
All other values	Reserved for future use.

*Service_Type:**Size: 1 octet*

Value	Parameter Description
0x00	No Traffic.
0x01	Best Effort.
0x02	Guaranteed.
All other values	Reserved for future use.

*Token_Rate:**Size: 4 octets*

Value	Parameter Description
0xFFFFFFFF	Token Rate in octets per second.

*Peak_Bandwidth:**Size: 4 octets*

Value	Parameter Description
0xFFFFFFFF	Peak Bandwidth in octets per second.

*Latency:**Size: 4 octets*

Value	Parameter Description
0xFFFFFFFF	Latency in microseconds.

*Delay_Variation:**Size: 4 octets*

Value	Parameter Description
0xFFFFFFFF	Delay Variation in microseconds.

Return parameters:

None.

Event(s) generated (unless masked away):

When the BR/EDR Controller receives the HCI_QoS_Setup command, the BR/EDR Controller shall send the HCI_Command_Status event to the Host. When the Link Manager has completed the LMP messages to establish the requested QoS parameters, the BR/EDR Controller shall send an HCI_QoS_Setup_Complete event to the Host, and the event may also be generated on the remote side if there was LMP negotiation. The values of the parameters of the HCI_QoS_Setup_Complete event may, however, be different on the initiating and the remote side. The HCI_QoS_Setup_Complete event returned by the BR/EDR Controller on the local side contains the status of this command, and returned QoS parameters describing the supported QoS for the connection.

Note: If the Link Manager performs an LMP transaction that involves the flow parameter values on the remote side, the remote Controller can send an



HCI_Flow_Specification_Complete event or HCI_QoS_Setup_Complete event to the remote Host.

Note: No HCI_Command_Complete event will be sent by the BR/EDR Controller to indicate that this command has been completed. Instead, the HCI_QoS_Setup_Complete event will indicate that this command has been completed.



7.2.7 Role Discovery command

Command	OCF	Command Parameters	Return Parameters
HCI_Role_Discovery	0x0009	Connection_Handle	Status, Connection_Handle, Current_Role

Description:

The HCI_Role_Discovery command is used for a Host to determine which role the device is performing for a particular Connection_Handle. The Connection_Handle shall be a Connection_Handle for an ACL connection.

Command parameters:

Connection_Handle: *Size: 2 octets (12 bits meaningful)*

Value	Parameter Description
0xFFFF	Connection_Handle Range: 0x0000 to 0x0EFF

Return parameters:

Status: *Size: 1 octet*

Value	Parameter Description
0x00	HCI_Role_Discovery command succeeded
0x01 to 0xFF	HCI_Role_Discovery command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Connection_Handle: *Size: 2 octets (12 bits meaningful)*

Value	Parameter Description
0xFFFF	Connection_Handle Range: 0x0000 to 0x0EFF

Current_Role: *Size: 1 octet*

Value	Parameter Description
0x00	Current Role is Master for this Connection_Handle.
0x01	Current Role is Slave for this Connection_Handle.

Event(s) generated (unless masked away):

When the HCI_Role_Discovery command has completed, an HCI_Command_Complete event shall be generated.



7.2.8 Switch Role command

Command	OCF	Command Parameters	Return Parameters
HCI_Switch_Role	0x000B	BD_ADDR, Role	

Description:

The HCI_Switch_Role command is used to switch the current BR/EDR role the device is performing for a particular connection with another specified BR/EDR Controller. The BD_ADDR command parameter indicates for which connection the role switch is to be performed and shall specify a BR/EDR Controller for which a connection already exists. The Role parameter indicates the requested new role that the local device performs.

If there is an (e)SCO connection between the local device and the device identified by the BD_ADDR parameter, an attempt to perform a role switch shall be rejected by the local device.

If the connection between the local device and the device identified by the BD_ADDR parameter is placed in Sniff mode, an attempt to perform a role switch shall be rejected by the local device.

Command parameters:

BD_ADDR:

Size: 6 octets

Value	Parameter Description
0XXXXXXXXXX XX	BD_ADDR for the connected device with which a role switch is to be performed.

Role:

Size: 1 octet

Value	Parameter Description
0x00	Change own Role to Master for this BD_ADDR.
0x01	Change own Role to Slave for this BD_ADDR.

Return parameters:

None.

Event(s) generated (unless masked away):

When the BR/EDR Controller receives the HCI_Switch_Role command, the BR/EDR Controller shall send the HCI_Command_Status event to the Host. When the role switch is performed, an HCI_Role_Change event shall occur to indicate that the roles have been changed, and will be communicated to both Hosts.



Note: No HCI_Command_Complete event will be sent by the BR/EDR Controller to indicate that this command has been completed. Instead, only the HCI_Role_Change event will indicate that this command has been completed.



7.2.9 Read Link Policy Settings command

Command	OCF	Command Parameters	Return Parameters
HCI_Read_Link_Policy_Settings	0x000C	Connection_Handle	Status, Connection_Handle, Link_Policy_Settings

Description:

This command will read the Link Policy setting for the specified Connection_Handle. The Connection_Handle shall be a Connection_Handle for an ACL connection. [Section 6.18](#).

Command parameters:

Connection_Handle: *Size: 2 octets (12 bits meaningful)*

Value	Parameter Description
0xFFFF	Connection_Handle Range: 0x0000 to 0x0EFF

Return parameters:

Status: *Size: 1 octet*

Value	Parameter Description
0x00	HCI_Read_Link_Policy_Settings command succeeded.
0x01 to 0xFF	HCI_Read_Link_Policy_Settings command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Connection_Handle: *Size: 2 octets (12 bits meaningful)*

Value	Parameter Description
0xFFFF	Connection_Handle Range: 0x0000 to 0x0EFF

**Link_Policy_Settings**

Size: 2 octets

Bit Number	Parameter Description
0	Enable Role Switch.
1	Enable Hold mode.
2	Enable Sniff mode.
All other bits	Reserved for future use.

Event(s) generated (unless masked away):

When the HCI_Read_Link_Policy_Settings command has completed, an HCI_Command_Complete event shall be generated.



7.2.10 Write Link Policy Settings command

Command	OCF	Command Parameters	Return Parameters
HCI_Write_Link_Policy_Settings	0x000D	Connection_Handle, Link_Policy_Settings	Status, Connection_Handle

Description:

This command writes the Link Policy setting for the specified Connection_Handle. The Connection_Handle shall be a Connection_Handle for an ACL connection. See [Section 6.18](#).

The default value is the value set by the HCI_Write_Default_Link_Policy_Settings command.

Command parameters:

Connection_Handle: *Size: 2 octets (12 bits meaningful)*

Value	Parameter Description
0xFFFF	Connection_Handle Range: 0x0000 to 0x0EFF

Link_Policy_Settings *Size: 2 octets*

Bit Number	Parameter Description
0	Enable Role Switch.
1	Enable Hold mode.
2	Enable Sniff mode.
All other bits	Reserved for future use.

Return parameters:

Status: *Size: 1 octet*

Value	Parameter Description
0x00	HCI_Write_Link_Policy_Settings command succeeded.
0x01 to 0xFF	HCI_Write_Link_Policy_Settings command failed. See [Vol 2] Part D, Error Codes for error codes and descriptions.

**Connection_Handle:****Size: 2 octets (12 bits meaningful)**

Value	Parameter Description
0xXXXX	Connection_Handle Range: 0x0000 to 0x0EFF

Event(s) generated (unless masked away):

When the HCI_Write_Link_Policy_Settings command has completed, an HCI_Command_Complete event shall be generated.



7.2.11 Read Default Link Policy Settings command

Command	OCF	Command Parameters	Return Parameters
HCI_Read_Default_Link_Policy_Settings	0x000E		Status, Default_Link_Policy_Settings

Description:

This command reads the Default Link Policy setting for all new BR/EDR connections.

Note: See the Link Policy Settings configuration parameter for more information. See [Section 6.18](#).

Command parameters:

None.

Return parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	HCI_Read_Link_Policy_Settings command succeeded
0x01 to 0xFF	HCI_Read_Link_Policy_Settings command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Default_Link_Policy_Settings

Size: 2 octets

Bit Number	Parameter Description
0	Enable Role Switch
1	Enable Hold mode
2	Enable Sniff mode
All other bits	Reserved for future use

Event(s) generated (unless masked away):

When the HCI_Read_Default_Link_Policy_Settings command has completed, an HCI_Command_Complete event shall be generated.



7.2.12 Write Default Link Policy Settings command

Command	OCF	Command Parameters	Return Parameters
HCI_Write_Default_Link_Policy_Settings	0x000F	Default_Link_Policy_Settings	Status

Description:

This command writes the Default Link Policy configuration value. The Default_Link_Policy_Settings parameter determines the initial value of the Link_Policy_Settings for all new BR/EDR connections.

Note: See the Link Policy Settings configuration parameter for more information. See [Section 6.18](#).

Command parameters:

Default_Link_Policy_Settings

Size: 2 octets

Bit Number	Parameter Description
0	Enable Role Switch
1	Enable Hold mode
2	Enable Sniff mode
All other bits	Reserved for future use

The default value is 0x0000.

Return parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	HCI_Write_Link_Policy_Settings command succeeded
0x01-0xFF	HCI_Write_Link_Policy_Settings command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Event(s) generated (unless masked away):

When the HCI_Write_Default_Link_Policy_Settings command has completed, an HCI_Command_Complete event will be generated.



7.2.13 Flow Specification command

Command	OCF	Command Parameters	Return Parameters
HCI_Flow_Specification	0x0010	Connection_Handle, Unused, Flow_direction, Service_Type, Token_Rate, Token_Bucket_Size, Peak_Bandwidth, Access_Latency	

Description:

The HCI_Flow_Specification command is used to specify the flow parameters for the traffic carried over the ACL connection identified by the Connection_Handle. The Connection_Handle shall be a Connection_Handle for an ACL connection. The Connection_Handle command parameter is used to identify for which connection the Flow Specification is requested. The flow parameters refer to the outgoing or incoming traffic of the ACL link, as indicated by the Flow_direction field. The HCI_Flow_Specification command allows the Link Manager to have the parameters of the outgoing as well as the incoming flow for the ACL connection. The flow parameters are defined in the L2CAP specification [\[Vol 3 Part A, Section 5.3, Quality of Service \(QoS\) option\]](#). The Link Manager will determine if the flow parameters can be supported. BR/EDR Controllers that are both master and slave can use this command.

The Unused parameter is reserved for future use.

Command parameters:

Connection_Handle: *Size: 2 octets (12 bits meaningful)*

Value	Parameter Description
0xFFFF	Connection_Handle Range: 0x0000 to 0x0EFF

Unused: *Size: 1 octet*

Value	Parameter Description
0x00	This value shall be used by the Host.
All other values	Reserved for future use.

Flow_direction: *Size: 1 octet*

Value	Parameter Description
0x00	Outgoing Flow i.e., traffic sent over the ACL connection
0x01	Incoming Flow i.e., traffic received over the ACL connection



Value	Parameter Description
All other values	Reserved for future use.

Service_Type:**Size: 1 octet**

Value	Parameter Description
0x00	No Traffic
0x01	Best Effort
0x02	Guaranteed
All other values	Reserved for future use

Token Rate:**Size: 4 octets**

Value	Parameter Description
0XXXXXXXX	Token Rate in octets per second

Token Bucket Size:**Size: 4 octets**

Value	Parameter Description
0XXXXXXXX	Token Bucket Size in octets

Peak_Bandwidth:**Size: 4 octets**

Value	Parameter Description
0XXXXXXXX	Peak Bandwidth in octets per second

Access Latency:**Size: 4 octets**

Value	Parameter Description
0XXXXXXXX	Latency in microseconds

Return parameters:

None.

Event(s) generated (unless masked away):

When the Controller receives the HCI_Flow_Specification command, the BR/EDR Controller shall send the HCI_Command_Status event to the Host. When the Link Manager has determined if the Flow specification can be supported, the BR/EDR Controller on the local BR/EDR Controller shall send an HCI_Flow_Specification_Complete event to the Host. The HCI_Flow_Specification_Complete event returned by the Controller on the



local side contains the status of this command, and returned Flow parameters describing the supported QoS for the ACL connection.

Note: If the Link Manager performs an LMP transaction that involves the flow parameter values on the remote side, the remote Controller can send an HCI_Flow_Specification_Complete event or HCI_QoS_Setup_Complete event to the remote Host.

Note: No HCI_Command_Complete event will be sent by the BR/EDR Controller to indicate that this command has been completed. Instead, the HCI_Flow_Specification_Complete event will indicate that this command has been completed.



7.2.14 Sniff Subrating command

Command	OCF	Command Parameters	Return Parameters
HCI_Sniff_Subrating	0x0011	Connection_Handle, Maximum_Latency, Minimum_Remote_Timeout, Minimum_Local_Timeout	Status, Connection_Handle

Description:

The HCI_Sniff_Subrating command specifies the parameters for sniff subrating for a given link. The interval shall be determined from the sniff interval and the maximum subrate latency parameters from the command. The link may have smaller subrates and therefore lower latencies and longer timeouts than those specified. When the sniff subrate has been exchanged an HCI_Sniff_Subrating event shall be generated. If this command is used on a link in Sniff mode this shall cause sniff subrating to be negotiated at the Link Manager, otherwise sniff subrating shall be negotiated only after the device has entered the Sniff mode.

The Connection_Handle shall be the primary Connection_Handle between the two devices.

The Maximum Latency parameter shall define the maximum allowed sniff subrate of the remote device.

If the Host does not write the sniff subrating parameters prior to sniff subrating being initiated by the Link Manager the default values shall be used.

Command parameters:

Connection_Handle: *Size: 2 octets (12 bits meaningful)*

Value	Parameter Description
0xXXXX	Connection_Handle Range: 0x0000 to 0x0EFF

Maximum_Latency: *Size: 2 octets*

Value	Parameter Description
N = 0xXXXX	The Maximum Latency parameter shall be used to calculate the maximum_sniff subrate that the remote device may use. Default: T_{sniff} Latency = $N * 0.625 \text{ ms}$ (1 Baseband slot) Range: 0x0002 to 0xFFFF Time Range: 1.25ms to 40.9 s

**Minimum_Remote_Timeout:****Size: 2 octets**

Value	Parameter Description
N = 0xFFFF	Minimum sniff mode timeout ($T_{\text{sniff_mode_timeout}}$) that the remote device may use Default: 0x0000 Timeout = $N * 0.625 \text{ ms}$ (1 Baseband slot) Range: 0x0000 to 0xFFFF Time Range: 0 s to 40.9 s

Minimum_Local_Timeout:**Size: 2 octets**

Value	Parameter Description
N = 0xFFFF	Minimum sniff mode timeout ($T_{\text{sniff_mode_timeout}}$) that the local device may use. Default: 0x0000 Timeout = $N * 0.625 \text{ ms}$ (1 Baseband slot) Range: 0x0000 to 0xFFFF Time Range: 0 s to 40.9 s

Return parameters:**Status:****Size: 1 octet**

Value	Parameter Description
0x00	The HCI_Sniff_Subrating command succeeded.
0x01 to 0xFF	HCI_Sniff_Subrating command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Connection_Handle:**Size: 2 octets (12 bits meaningful)**

Value	Parameter Description
0xFFFF	Connection_Handle Range: 0x0000 to 0x0EFF

Event(s) generated (unless masked away)

When the HCI_Sniff_Subrating command has been received by the BR/EDR Controller, an HCI_Command_Complete event shall be generated.

An HCI_Sniff_Subrating event shall occur when the sniff subrating has been negotiated for the specified Connection_Handle.



7.3 CONTROLLER & BASEBAND COMMANDS

The Controller & Baseband commands provide access and control to various capabilities of the Bluetooth hardware. These parameters provide control of BR/EDR Controllers and of the capabilities of the Link Manager and Baseband in the BR/EDR Controller, the PAL in an AMP Controller, and the Link Layer in an LE Controller. The Host can use these commands to modify the behavior of the local Controller.

For the HCI Control and Baseband commands, the OGF is defined as 0x03.

7.3.1 Set Event Mask command

Command	OCF	Command Parameters	Return Parameters
HCI_Set_Event_Mask	0x0001	Event_Mask	Status

Description:

The HCI_Set_Event_Mask command is used to control which events are generated by the HCI for the Host. If the bit in the Event_Mask is set to a one, then the event associated with that bit will be enabled. For an LE Controller, the “LE Meta event” bit in the event_mask shall enable or disable all LE events in the LE Meta event (see [Section 7.7.65](#)). The event mask allows the Host to control how much it is interrupted.

The Controller shall ignore those bits which are reserved for future use or represent events which it does not support. If the Host sets any of these bits to 1, the Controller shall act as if they were set to 0.

Command parameters:

Event_Mask:

Size: 8 octets

Bit	Parameter Description
0	Inquiry Complete event
1	Inquiry Result event
2	Connection Complete event
3	Connection Request event
4	Disconnection Complete event
5	Authentication Complete event
6	Remote Name Request Complete event
7	Encryption Change event
8	Change Connection Link Key Complete event



Bit	Parameter Description
9	Master Link Key Complete event
10	Read Remote Supported Features Complete event
11	Read Remote Version Information Complete event
12	QoS Setup Complete event
15	Hardware Error event
16	Flush Occurred event
17	Role Change event
19	Mode Change event
20	Return Link Keys event
21	PIN Code Request event
22	Link Key Request event
23	Link Key Notification event
24	Loopback Command event
25	Data Buffer Overflow event
26	Max Slots Change event
27	Read Clock Offset Complete event
28	Connection Packet Type Changed event
29	QoS Violation event
30	Page Scan Mode Change event [deprecated]
31	Page Scan Repetition Mode Change event
32	Flow Specification Complete event
33	Inquiry Result with RSSI event
34	Read Remote Extended Features Complete event
43	Synchronous Connection Complete event
44	Synchronous Connection Changed event
45	Sniff Subrating event
46	Extended Inquiry Result event
47	Encryption Key Refresh Complete event
48	IO Capability Request event
49	IO Capability Response event
50	User Confirmation Request event
51	User Passkey Request event



Bit	Parameter Description
52	Remote OOB Data Request event
53	Simple Pairing Complete event
55	Link Supervision Timeout Changed event
56	Enhanced Flush Complete event
58	User Passkey Notification event
59	Keypress Notification event
60	Remote Host Supported Features Notification event
61	LE Meta event
All other bits	Reserved for future use

The value with all bits set to 0 indicates that no events are specified.
The default is for bits 0 to 44 inclusive (the value 0x0000 1FFF FFFF FFFF) to be set.

Return parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	HCI_Set_Event_Mask command succeeded.
0x01 to 0xFF	HCI_Set_Event_Mask command failed. See [Vol 2] Part D, Error Codes for error codes and descriptions.

Event(s) generated (unless masked away):

When the HCI_Set_Event_Mask command has completed, an HCI_Command_Complete event shall be generated.



7.3.2 Reset command

Command	OCF	Command Parameters	Return Parameters
HCI_Reset	0x0003		Status

Description:

The HCI_Reset command will reset the Controller and the Link Manager on the BR/EDR Controller, the PAL on an AMP Controller, or the Link Layer on an LE Controller. If the Controller supports both BR/EDR and LE then the HCI_Reset command shall reset the Link Manager, Baseband and Link Layer. The HCI_Reset command shall not affect the used HCI transport layer since the HCI transport layers may have reset mechanisms of their own. After the reset is completed, the current operational state will be lost, the Controller will enter standby mode and the Controller will automatically revert to the default values for the parameters for which default values are defined in the specification.

Note: The HCI_Reset command will not necessarily perform a hardware reset. This is implementation defined.

On an AMP Controller, the HCI_Reset command shall reset the service provided at the logical HCI to its initial state, but beyond this the exact effect on the Controller device is implementation defined and should not interrupt the service provided to other protocol stacks.

The Host shall not send additional HCI commands before the HCI_Command_Complete event related to the HCI_Reset command has been received.

Command parameters:

None.

Return parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	HCI_Reset command succeeded, was received and will be executed.
0x01 to 0xFF	HCI_Reset command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Event(s) generated (unless masked away):

When the reset has been performed, an HCI_Command_Complete event shall be generated.



7.3.3 Set Event Filter command

Command	OCF	Command Parameters	Return Parameters
HCI_Set_Event_Filter	0x0005	Filter_Type, Filter_Condition_Type, Condition	Status

Description:

The HCI_Set_Event_Filter command is used by the Host to specify different event filters. The Host may issue this command multiple times to request various conditions for the same type of event filter and for different types of event filters. The event filters are used by the Host to specify items of interest, which allow the BR/EDR Controller to send only events which interest the Host. Only some of the events have event filters. By default (before this command has been issued after power-on or Reset) no filters are set, and the Auto_Accept_Flag is off (incoming connections are not automatically accepted). An event filter is added each time this command is sent from the Host and the Filter_Condition_Type is not equal to 0x00. (The old event filters will not be overwritten). To clear all event filters, the Filter_Type = 0x00 is used. The Auto_Accept_Flag will then be set to off. To clear event filters for only a certain Filter_Type, the Filter_Condition_Type = 0x00 is used.

The Inquiry Result filter allows the BR/EDR Controller to filter out HCI_Inquiry_Result, HCI_Inquiry_Result_with_RSSI, or HCI_Extended_Inquiry_Result events. The Inquiry Result filter allows the Host to specify that the BR/EDR Controller only sends Inquiry Results to the Host if the report meets one of the specified conditions set by the Host. For the Inquiry Result filter, the Host can specify one or more of the following Filter Condition Types:

1. Return responses from all devices during the Inquiry process
2. A device with a specific Class of Device responded to the Inquiry process
3. A device with a specific BD_ADDR responded to the Inquiry process

The Inquiry Result filter is used in conjunction with the HCI_Inquiry and HCI_Periodic_Inquiry_Mode commands.

The Connection Setup filter allows the Host to specify that the Controller only sends an HCI_Connection_Complete, HCI_Synchronous_Connection_Complete, or HCI_Connection_Request event to the Host if the event meets one of the specified conditions set by the Host. For the Connection Setup filter, the Host can specify one or more of the following Filter Condition Types:

1. Allow Connections from all devices
2. Allow Connections from a device with a specific Class of Device



3. Allow Connections from a device with a specific BD_ADDR

For each of these conditions, an `Auto_Accept_Flag` parameter allows the Host to specify what action should be done when the condition is met. The `Auto_Accept_Flag` allows the Host to specify if the incoming connection should be auto accepted (in which case the BR/EDR Controller will send a `HCI_Connection_Complete` event to the Host when an ACL or SCO connection is completed or an `HCI_Synchronous_Connection_Complete` event when an eSCO connection is completed) or if the Host should make the decision (in which case the BR/EDR Controller will send the `HCI_Connection_Request` event to the Host, to elicit a decision on the connection). If the `Auto_Accept_Flag` is off and the Host has masked the `HCI_Connection_Request` event, the Controller shall reject the connection attempt.

If a synchronous connection is auto-accepted, then the default parameter settings of the `Accept_Synchronous_Connection_Request` command (see [Section 7.1.27](#)) should be used by the local Link Manager when negotiating the SCO or eSCO link parameters.

The Connection Setup filter is used in conjunction with the `HCI_Read/Write_Scan_Enable` commands. If the local device is in the process of a page scan, and is paged by another device which meets one of the conditions set by the Host, and the `Auto_Accept_Flag` is off for this device, then an `HCI_Connection_Request` event will be sent to the Host by the BR/EDR Controller. An `HCI_Connection_Complete` event will be sent later on after the Host has responded to the incoming connection attempt. In this same example, if the `Auto_Accept_Flag` is on, then an `HCI_Connection_Complete` event will be sent to the Host by the Controller. (No `HCI_Connection_Request` event will be sent in that case.)

The BR/EDR Controller will store these filters in volatile memory until the Host clears the event filters using the `HCI_Set_Event_Filter` command or until the `HCI_Reset` command is issued. The number of event filters the BR/EDR Controller can store is implementation dependent. If the Host tries to set more filters than the BR/EDR Controller can store, the BR/EDR Controller will return the *Memory Full* error code and the filter will not be installed.

Note: The Clear All Filters has no Filter Condition Types or Conditions.

Note: In the condition that a connection is auto accepted, an `HCI_Link_Key_Request` event and possibly also an `HCI_PIN_Code_Request` event and an `HCI_Link_Key_Notification` event could be sent to the Host by the Controller before the `HCI_Connection_Complete` event is sent.

If there is a contradiction between event filters, the latest set event filter will override older ones. An example is an incoming connection attempt where more than one Connection Setup filter matches the incoming connection attempt, but the `Auto-Accept_Flag` has different values in the different filters.

**Command parameters:***Filter_Type:**Size: 1 octet*

Value	Parameter Description
0x00	Clear All Filters Note: In this case, the Filter_Condition_Type and Condition parameters should not be given; they should have a length of 0 octets. Filter_Type should be the only parameter.
0x01	Inquiry Result.
0x02	Connection Setup.
All other values	Reserved for future use.

Filter Condition Types: For each Filter Type one or more Filter Condition types exists.

*Inquiry_Result_Filter_Condition_Type:**Size: 1 octet*

Value	Parameter Description
0x00	Return responses from all devices during the Inquiry process. Note: A device may be reported to the Host in an HCI_Inquiry_Result, HCI_Inquiry_Result_with_RSSI, or HCI_Extended_Inquiry_Result event more than once during an inquiry or inquiry period depending on the implementation; see description in Section 7.1.1 and Section 7.1.3 .
0x01	A device with a specific Class of Device responded to the Inquiry process.
0x02	A device with a specific BD_ADDR responded to the Inquiry process.
All other values	Reserved for future use

*Connection_Setup_Filter_Condition_Type:**Size: 1 octet*

Value	Parameter Description
0x00	Allow Connections from all devices.
0x01	Allow Connections from a device with a specific Class of Device.
0x02	Allow Connections from a device with a specific BD_ADDR.
All other values	Reserved for future use.



Condition: For each Filter Condition Type defined for the Inquiry Result Filter and the Connection Setup Filter, zero or more Condition parameters are required – depending on the filter condition type and filter type.

Condition for Inquiry_Result_Filter_Condition_Type = 0x00

Condition:

Size: 0 octet

Value	Parameter Description
	The Condition parameter is not used.

Condition for Inquiry_Result_Filter_Condition_Type = 0x01

Condition:

Size: 6 octets

Class_of_Device:

Size: 3 octets

Value	Parameter Description
0x000000	Default, Return All Devices.
0xxxxxxx	Class of Device of Interest.

Class_of_Device_Mask:

Size: 3 octets

Value	Parameter Description
0xxxxxxx	Bit Mask used to determine which bits of the Class of Device parameter are 'don't care'. Zero-value bits in the mask indicate the 'don't care' bits of the Class of Device.

Condition for Inquiry_Result_Filter_Condition_Type = 0x02

Condition:

Size: 6 octets

BD_ADDR:

Size: 6 octets

Value	Parameter Description
0xxxxxxxxxxx xx	BD_ADDR of the Device of Interest

Condition for Connection_Setup_Filter_Condition_Type = 0x00

Condition:

Size: 1 octet

Auto_Accept_Flag:

Size: 1 octet

Value	Parameter Description
0x01	Do NOT Auto accept the connection. (Auto accept is off)
0x02	Do Auto accept the connection with role switch disabled. (Auto accept is on).



Value	Parameter Description
0x03	Do Auto accept the connection with role switch enabled. (Auto accept is on). Note: When auto accepting an incoming synchronous connection, no role switch will be performed. The value 0x03 of the Auto_Accept_Flag will then get the same effect as if the value had been 0x02.
All other values	Reserved for future use.

Condition for Connection_Setup_Filter_Condition_Type = 0x01

Condition:

Size: 7 octets

Class_of_Device:

Size: 3 octets

Value	Parameter Description
0x000000	Default, Return All Devices.
0xXXXXXX	<i>Class of Device</i> of Interest.

Class_of_Device_Mask:

Size: 3 octets

Value	Parameter Description
0xXXXXXX	Bit Mask used to determine which bits of the Class of Device parameter are 'don't care'. Zero-value bits in the mask indicate the 'don't care' bits of the Class of Device. Note: For an incoming SCO connection, if the class of device is unknown then the connection will be accepted.

Auto_Accept_Flag:

Size: 1 octet

Value	Parameter Description
0x01	Do NOT Auto accept the connection. (Auto accept is off)
0x02	Do Auto accept the connection with role switch disabled. (Auto accept is on).
0x03	Do Auto accept the connection with role switch enabled. (Auto accept is on). Note: When auto accepting an incoming synchronous connection, no role switch will be performed. The value 0x03 of the Auto_Accept_Flag will then get the same effect as if the value had been 0x02.
All other values	Reserved for future use.



Condition for Connection_Setup_Filter_Condition_Type = 0x02

Condition:

Size: 7 octets

BD_ADDR:

Size: 6 octets

Value	Parameter Description
0XXXXXXXXXX XX	BD_ADDR of the Device of Interest.

Auto_Accept_Flag:

Size: 1 octet

Value	Parameter Description
0x01	Do NOT Auto accept the connection. (Auto accept is off)
0x02	Do Auto accept the connection with role switch disabled. (Auto accept is on).
0x03	Do Auto accept the connection with role switch enabled. (Auto accept is on). Note: When auto accepting an incoming synchronous connection, no role switch will be performed. The value 0x03 of the Auto_Accept_Flag will then get the same effect as if the value had been 0x02.
All other values	Reserved for future use.

Return parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	HCI_Set_Event_Filter command succeeded.
0x01-0xFF	HCI_Set_Event_Filter command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Event(s) generated (unless masked away):

An HCI_Command_Complete event for this command shall occur when the Controller has enabled the filtering of events. When one of the conditions are met, a specific event shall occur.



7.3.4 Flush command

Command	OCF	Command Parameters	Return Parameters
HCI_Flush	0x0008	Connection_Handle	Status, Connection_Handle

Description:

The HCI_Flush command is used to discard all data that is currently pending for transmission in the Controller for the specified Connection_Handle, even if there currently are chunks of data that belong to more than one L2CAP packet in the Controller. Both automatically-flushable and non-automatically-flushable packets shall be discarded (see [Section 5.4.2](#)). After this, all data that is sent to the Controller for the same Connection_Handle will be discarded by the Controller until an HCI Data packet with one of the start Packet_Boundary_Flag values (0x00 or 0x02) is received. When this happens, a new transmission attempt can be made.

This command, when used on a BR/EDR Controller or an AMP Controller, will allow higher-level software to control how long the baseband should try to retransmit a Baseband packet for a Connection_Handle before all data that is currently pending for transmission in the Controller should be flushed.

Note: The HCI_Flush command is used for ACL connections only. In addition to the HCI_Flush command, the automatic flush timers (see [Section 7.3.29](#)) can be used to automatically flush an automatically-flushable L2CAP packet that is currently being transmitted after the specified flush timer has expired.

Command parameters:

Connection_Handle: *Size: 2 octets (12 bits meaningful)*

Value	Parameter Description
0xFFFF	Connection_Handle Range: 0x0000 to 0x0EFF

Return parameters:

Status: *Size: 1 octet*

Value	Parameter Description
0x00	HCI_Flush command succeeded.
0x01 to 0xFF	HCI_Flush command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.



Connection_Handle: *Size: 2 octets (12 bits meaningful)*

Value	Parameter Description
0xXXXX	Connection_Handle Range: 0x0000 to 0x0EFF

Event(s) generated (unless masked away):

The HCI_Flush_Occurred event shall occur once the flush is completed. An HCI_Flush_Occurred event could be from an automatic Flush or could be caused by the Host issuing the HCI_Flush command. When the HCI_Flush command has completed, an HCI_Command_Complete event shall be generated, to indicate that the Host caused the Flush.



7.3.5 Read PIN Type command

Command	OCF	Command Parameters	Return Parameters
HCI_Read_PIN_Type	0x0009		Status, PIN_Type

Description:

The HCI_Read_PIN_Type command is used to read the PIN_Type configuration parameter. See [Section 6.13](#).

Command parameters:

None.

Return parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	HCI_Read_PIN_Type command succeeded.
0x01 to 0xFF	HCI_Read_PIN_Type command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

PIN_Type:

Size: 1 octet

Value	Parameter Description
0x00	Variable PIN.
0x01	Fixed PIN.

Event(s) generated (unless masked away):

When the HCI_Read_PIN_Type command has completed, an HCI_Command_Complete event will be generated.



7.3.6 Write PIN Type command

Command	OCF	Command Parameters	Return Parameters
HCI_Write_PIN_Type	0x000A	PIN_Type	Status

Description:

The HCI_Write_PIN_Type command is used to write the PIN Type configuration parameter. See [Section 6.13](#).

Command parameters:

PIN_Type:

Size: 1 octet

Value	Parameter Description
0x00	Variable PIN.
0x01	Fixed PIN.

Return parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	HCI_Write_PIN_Type command succeeded.
0x01 to 0xFF	HCI_Write_PIN_Type command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Event(s) generated (unless masked away):

When the HCI_Write_PIN_Type command has completed, an HCI_Command_Complete event shall be generated.



7.3.7 This section is not currently used



7.3.8 Read Stored Link Key command

Command	OCF	Command Parameters	Return Parameters
HCI_Read_Stored_Link_Key	0x000D	BD_ADDR, Read_All_Flag	Status, Max_Num_Keys, Num_Keys_Read

Description:

The HCI_Read_Stored_Link_Key command provides the ability to read whether one or more link keys are stored in the BR/EDR Controller. The BR/EDR Controller can store a limited number of link keys for other BR/EDR Controllers. Link keys are shared between two BR/EDR Controllers, and are used for all security transactions between the two devices. The HCI_Read_Stored_Link_Key command shall not return the link key's value. A Host device may have additional storage capabilities, which can be used to save additional link keys to be reloaded to the BR/EDR Controller when needed. The Read_All_Flag parameter is used to indicate if all of the stored Link Keys should be returned. If Read_All_Flag indicates that all Link Keys are to be returned, then the BD_ADDR command parameter shall be ignored. The BD_ADDR command parameter is used to identify which link key to read. The stored Link Keys are returned by one or more HCI_Return_Link_Keys events. See [Section 6.14](#).

Command parameters:

BD_ADDR:

Size: 6 octets

Value	Parameter Description
0xFFFFFFFFXXXX	BD_ADDR for the stored link key to be read.

Read_All_Flag:

Size: 1 octet

Value	Parameter Description
0x00	Return Link Key for specified BD_ADDR.
0x01	Return all stored Link Keys.
All other values	Reserved for future use.

Return parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	HCI_Read_Stored_Link_Key command succeeded.
0x01 to 0xFF	HCI_Read_Stored_Link_Key command failed. See [Vol 2] Part D, Error Codes for error codes and descriptions.

**Max_Num_Keys:****Size: 2 octets**

Value	Parameter Description
0xFFFF	Total Number of Link Keys that the Controller can store. Range: 0x0000 to 0xFFFF

Num_Keys_Read:**Size: 2 octets**

Value	Parameter Description
0xFFFF	Number of Link Keys Read. Range: 0x0000 to 0xFFFF

Event(s) generated (unless masked away):

Zero or more instances of the HCI_Return_Link_Keys event shall occur after the command is issued. When there are no link keys stored, no HCI_Return_Link_Keys events will be returned. When there are link keys stored, the number of link keys returned in each HCI_Return_Link_Keys event is implementation specific. When the HCI_Read_Stored_Link_Key command has completed an HCI_Command_Complete event shall be generated.



7.3.9 Write Stored Link Key command

Command	OCF	Command Parameters	Return Parameters
HCI_Write_Stored_Link_Key	0x0011	Num_Keys_To_Write, BD_ADDR[i], Link_Key[i]	Status, Num_Keys_Written

Description:

The HCI_Write_Stored_Link_Key command provides the ability to write one or more link keys to be stored in the BR/EDR Controller. The BR/EDR Controller can store a limited number of link keys for other BR/EDR Controllers. If no additional space is available in the BR/EDR Controller then no additional link keys will be stored. If space is limited and if all the link keys to be stored will not fit in the limited space, then the order of the list of link keys without any error will determine which link keys are stored. Link keys at the beginning of the list will be stored first. The Num_Keys_Written parameter will return the number of link keys that were successfully stored. If no additional space is available, then the Host must delete one or more stored link keys before any additional link keys are stored. The link key replacement algorithm is implemented by the Host and not the BR/EDR Controller. Link keys are shared between two BR/EDR Controllers and are used for all security transactions between the two devices. A Host device may have additional storage capabilities, which can be used to save additional link keys to be reloaded to the BR/EDR Controller when needed. See [Section 6.14](#).

Note: Link Keys are only stored by issuing this command.

A Host in Secure Connections Only Mode should not store link keys in the Controller.

Command parameters:

Num_Keys_To_Write:

Size: 1 octet

Value	Parameter Description
0xXX	Number of Link Keys to Write. Range: 0x01 to 0x0B

BD_ADDR [i]:

*Size: 6 octets * Num_Keys_To_Write*

Value	Parameter Description
0XXXXXXXXXXXXX	BD_ADDR for the associated Link Key.

*Link_Key[i]:**Size: 16 octets* Num_Keys_To_Write*

Value	Parameter Description
0xFFFFFFFFXXXXXXX XXXXXXXXXXXX XXXXXXXXXXXX	Link Key for an associated BD_ADDR.

Return parameters:*Status:**Size: 1 octet*

Value	Parameter Description
0x00	HCI_Write_Stored_Link_Key command succeeded.
0x01 to 0xFF	HCI_Write_Stored_Link_Key command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

*Num_Keys_Written:**Size: 1 octet*

Value	Parameter Description
0xFF	Number of Link Keys successfully written. Range: 0x00 to 0xFF

Event(s) generated (unless masked away):

When the HCI_Write_Stored_Link_Key command has completed, an HCI_Command_Complete event shall be generated.



7.3.10 Delete Stored Link Key command

Command	OCF	Command Parameters	Return Parameters
HCI_Delete_Stored_Link_Key	0x0012	BD_ADDR, Delete_All_Flag	Status, Num_Keys_Deleted

Description:

The HCI_Delete_Stored_Link_Key command provides the ability to remove one or more of the link keys stored in the BR/EDR Controller. The BR/EDR Controller can store a limited number of link keys for other BR/EDR devices. Link keys are shared between two BR/EDR devices and are used for all security transactions between the two devices. The Delete_All_Flag parameter is used to indicate if all of the stored Link Keys should be deleted. If the Delete_All_Flag indicates that all Link Keys are to be deleted, then the BD_ADDR command parameter shall be ignored. This command provides the ability to negate all security agreements between two devices. The BD_ADDR command parameter is used to identify which link key to delete. If a link key is currently in use for a connection, then the link key will be deleted when all of the connections are disconnected. See [Section 6.14](#).

Command parameters:

BD_ADDR:

Size: 6 octets

Value	Parameter Description
0xFFFFFFFFXXXX	BD_ADDR for the link key to be deleted.

Delete_All_Flag:

Size: 1 octet

Value	Parameter Description
0x00	Delete only the Link Key for specified BD_ADDR.
0x01	Delete all stored Link Keys.
All other values	Reserved for future use.

Return parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	HCI_Delete_Stored_Link_Key command succeeded.
0x01 to 0xFF	HCI_Delete_Stored_Link_Key command failed. See [Vol 2] Part D, Error Codes for error codes and descriptions.

*Num_Keys_Deleted:**Size: 2 octets*

Value	Parameter Description
0xXXXX	Number of Link Keys Deleted

Event(s) generated (unless masked away):

When the HCI_Delete_Stored_Link_Key command has completed, an HCI_Command_Complete event shall be generated.



7.3.11 Write Local Name command

Command	OCF	Command Parameters	Return Parameters
HCI_Write_Local_Name	0x0013	Local Name	Status

Description:

The HCI_Write_Local_Name command provides the ability to modify the user-friendly name for the BR/EDR Controller. See [Section 6.23](#).

Command parameters:

Local Name:

Size: 248 octets

Value	Parameter Description
	A UTF-8 encoded User-Friendly Descriptive Name for the device. If the name contained in the parameter is shorter than 248 octets, the end of the name is indicated by a NULL octet (0x00), and the following octets (to fill up 248 octets, which is the length of the parameter) do not have valid values.
	Null terminated Zero length String. Default.

Return parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	HCI_Write_Local_Name command succeeded.
0x01 to 0xFF	HCI_Write_Local_Name command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Event(s) generated (unless masked away):

When the HCI_Write_Local_Name command has completed, an HCI_Command_Complete event shall be generated.



7.3.12 Read Local Name command

Command	OCF	Command Parameters	Return Parameters
HCI_Read_Local_Name	0x0014		Status, Local_Name

Description:

The HCI_Read_Local_Name command provides the ability to read the stored user-friendly name for the BR/EDR Controller. See [Section 6.23](#).

Command parameters:

None.

Return parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	HCI_Read_Local_Name command succeeded
0x01 to 0xFF	HCI_Read_Local_Name command failed (see [Vol 2] Part D, Error Codes for list of error codes).

Local_Name:

Size: 248 octets

Value	Parameter Description
	A UTF-8 encoded User Friendly Descriptive Name for the device. If the name contained in the parameter is shorter than 248 octets, the end of the name is indicated by a NULL octet (0x00), and the following octets (to fill up 248 octets, which is the length of the parameter) do not have valid values.

Event(s) generated (unless masked away):

When the HCI_Read_Local_Name command has completed an HCI_Command_Complete event shall be generated.



7.3.13 Read Connection Accept Timeout command

Command	OCF	Command Parameters	Return Parameters
HCI_Read_ Connection_Accept_ Timeout	0x0015		Status, Conn_Accept_Timeout

Description:

This command reads the value for the Connection_Accept_Timeout configuration parameter. See [Section 6.7](#).

Command parameters:

None.

Return parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	HCI_Read_Connection_Accept_Timeout command succeeded.
0x01 to 0xFF	HCI_Read_Connection_Accept_Timeout command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Conn_Accept_Timeout:

Size: 2 octets

Value	Parameter Description
N = 0xXXXX	Connection Accept Timeout measured in number of BR/EDR Baseband slots. Interval Length = $N * 0.625$ ms (1 Baseband slot) Range: 0x0001 to 0xB540 Time Range: 0.625 ms to 29 s

Event(s) generated (unless masked away):

When the HCI_Read_Connection_Timeout command has completed, an HCI_Command_Complete event shall be generated.



7.3.14 Write Connection Accept Timeout command

Command	OCF	Command Parameters	Return Parameters
HCI_Write_ - Connection_Accept_ - Timeout	0x0016	Conn_Accept_Timeout	Status

Description:

This command writes the value for the Connection_Accept_Timeout configuration parameter. See [Section 6.7](#).

Command parameters:

Conn_Accept_Timeout:

Size: 2 octets

Value	Parameter Description
N = 0xXXXX	Connection Accept Timeout measured in number of BR/EDR Baseband slots. Interval Length = $N * 0.625$ ms (1 Baseband slot) Range: 0x0001 to 0xB540 Time Range: 0.625 ms to 29 s Default: 0x1FA0 Time = 5.06 s Mandatory Range for Controller: 0x00A0 to 0xB540

Return parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	HCI_Write_Connection_Accept_Timeout command succeeded.
0x01 to 0xFF	HCI_Write_Connection_Accept_Timeout command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Event(s) generated (unless masked away):

When the HCI_Write_Connection_Accept_Timeout command has completed, an HCI_Command_Complete event shall be generated.



7.3.15 Read Page Timeout command

Command	OCF	Command Parameters	Return Parameters
HCI_Read_Page_Timeout	0x0017		Status, Page_Timeout

Description:

This command reads the value for the Page_Timeout configuration parameter. See [Section 6.6](#).

Command parameters:

None.

Return parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	HCI_Read_Page_Timeout command succeeded.
0x01 to 0xFF	HCI_Read_Page_Timeout command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Page_Timeout:

Size: 2 octets

Value	Parameter Description
N = 0xXXXX	Page Timeout measured in number of Baseband slots. Interval Length = $N * 0.625$ ms (1 Baseband slot) Range: 0x0001 to 0xFFFF Time Range: 0.625 ms to 40.9 s

Event(s) generated (unless masked away):

When the HCI_Read_Page_Timeout command has completed, an HCI_Command_Complete event shall be generated.



7.3.16 Write Page Timeout command

Command	OCF	Command Parameters	Return Parameters
HCI_Write_Page_Timeout	0x0018	Page_Timeout	Status

Description:

This command writes the value for the Page_Timeout configuration parameter. The Page_Timeout configuration parameter defines the maximum time the local Link Manager shall wait for a baseband page response from the remote device at a locally initiated connection attempt. If this time expires and the remote device has not responded to the page at baseband level, the connection attempt will be considered to have failed.

Command parameters:

Page_Timeout:

Size: 2 octets

Value	Parameter Description
N = 0xXXXX	Page Timeout measured in number of Baseband slots. Interval Length = N * 0.625 ms (1 Baseband slot) Range: 0x0001 to 0xFFFF Time Range: 0.625 ms to 40.9 s Default: 0x2000 Time = 5.12 s Mandatory Range for Controller: 0x0016 to 0xFFFF

Return parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	HCI_Write_Page_Timeout command succeeded.
0x01 to 0xFF	HCI_Write_Page_Timeout command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Event(s) generated (unless masked away):

When the HCI_Write_Page_Timeout command has completed, an HCI_Command_Complete event shall be generated.



7.3.17 Read Scan Enable command

Command	OCF	Command Parameters	Return Parameters
HCI_Read_Scan_Enable	0x0019		Status, Scan_Enable

Description:

This command reads the value for the Scan_Enable parameter configuration parameter. See [Section 6.1](#).

Command parameters:

None.

Return parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	HCI_Read_Scan_Enable command succeeded.
0x01 to 0xFF	HCI_Read_Scan_Enable command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Scan_Enable:

Size: 1 octet

Value	Parameter Description
0x00	No Scans enabled.
0x01	Inquiry Scan enabled. Page Scan disabled.
0x02	Inquiry Scan disabled. Page Scan enabled.
0x03	Inquiry Scan enabled. Page Scan enabled.
All other values	Reserved for future use

Event(s) generated (unless masked away):

When the HCI_Read_Scan_Enable command has completed, an HCI_Command_Complete event shall be generated.



7.3.18 Write Scan Enable command

Command	OCF	Command Parameters	Return Parameters
HCI_Write_Scan_Enable	0x001A	Scan_Enable	Status

Description:

This command writes the value for the Scan_Enable configuration parameter. See [Section 6.1](#).

Command parameters:

Scan_Enable:

Size: 1 octet

Value	Parameter Description
0x00	No Scans enabled. Default.
0x01	Inquiry Scan enabled. Page Scan disabled.
0x02	Inquiry Scan disabled. Page Scan enabled.
0x03	Inquiry Scan enabled. Page Scan enabled.
All other values	Reserved for future use

Return parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	HCI_Write_Scan_Enable command succeeded.
0x01 to 0xFF	HCI_Write_Scan_Enable command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Event(s) generated (unless masked away):

When the HCI_Write_Scan_Enable command has completed, an HCI_Command_Complete event shall be generated.



7.3.19 Read Page Scan Activity command

Command	OCF	Command Parameters	Return Parameters
HCI_Read_Page_Scan_Activity	0x001B		Status, Page_Scan_Interval, Page_Scan_Window

Description:

This command reads the value for Page_Scan_Interval and Page_Scan_Window configuration parameters. See [Section 6.8](#) and [Section 6.9](#).

Note: Page Scan is only performed when Page_Scan is enabled (see [6.1](#), [7.3.17](#) and [7.3.18](#)). A changed Page_Scan_Interval could change the local Page Scan Repetition Mode (see [\[Vol 2\] Part B, Section 8.3.1](#)).

Command parameters:

None.

Return parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	HCI_Read_Page_Scan_Activity command succeeded.
0x01 to 0xFF	HCI_Read_Page_Scan_Activity command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Page_Scan_Interval:

Size: 2 octets

Value	Parameter Description
N = 0xFFFF	Size: 2 octets Range: 0x0012 to 0x1000 Time = N * 0.625 ms Range: 11.25 ms to 2560 ms; only even values are valid

Page_Scan_Window:

Size: 2 octets

Value	Parameter Description
N = 0xFFFF	Size: 2 octets Range: 0x0011 to 0x1000 Time = N * 0.625 ms Range: 10.625 ms to 2560 ms

**Event(s) generated (unless masked away):**

When the HCI_Read_Page_Scan_Activity command has completed, an HCI_Command_Complete event shall be generated.



7.3.20 Write Page Scan Activity command

Command	OCF	Command Parameters	Return Parameters
HCI_Write_Page_Scan_Activity	0x001C	Page_Scan_Interval, Page_Scan_Window	Status

Description:

This command writes the values for the Page_Scan_Interval and Page_Scan_Window configuration parameters. The Page_Scan_Window shall be less than or equal to the Page_Scan_Interval. See [Section 6.8](#) and [Section 6.9](#).

Note: Page Scan is only performed when Page_Scan is enabled (see [6.1](#), [7.3.17](#) and [7.3.18](#)). A changed Page_Scan_Interval could change the local Page Scan Repetition Mode (see [\[Vol 2\] Part B, Section 8.3.1](#)).

Command parameters:

Page_Scan_Interval:

Size: 2 octets

Value	Parameter Description
See Section 6.8	

Page_Scan_Window:

Size: 2 octets

Value	Parameter Description
See Section 6.9	

Return parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	HCI_Write_Page_Scan_Activity command succeeded.
0x01 to 0xFF	HCI_Write_Page_Scan_Activity command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Event(s) generated (unless masked away):

When the HCI_Write_Page_Scan_Activity command has completed, an HCI_Command_Complete event shall be generated.



7.3.21 Read Inquiry Scan Activity command

Command	OCF	Command Parameters	Return Parameters
HCI_Read_Inquiry_Scan_Activity	0x001D		Status, Inquiry_Scan_Interval, Inquiry_Scan_Window

Description:

This command reads the value for Inquiry_Scan_Interval and Inquiry_Scan_Window configuration parameter. See [Section 6.2](#) and [Section 6.3](#).

Note: Inquiry Scan is only performed when Inquiry_Scan is enabled see [6.1](#), [7.3.17](#) and [7.3.18](#)).

Command parameters:

None.

Return parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	HCI_Read_Inquiry_Scan_Activity command succeeded.
0x01 to 0xFF	HCI_Read_Inquiry_Scan_Activity command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Inquiry_Scan_Interval:

Size: 2 octets

Value	Parameter Description
N = 0xXXXX	Size: 2 octets Range: 0x0012 to 0x1000 Time = N * 0.625 ms Range: 11.25 to 2560 ms; only even values are valid

Inquiry_Scan_Window:

Size: 2 octets

Value	Parameter Description
N = 0xXXXX	Size: 2 octets Range: 0x0011 to 0x1000 Time = N * 0.625 ms Range: 10.625 ms to 2560 ms

**Event(s) generated (unless masked away):**

When the HCI_Read_Inquiry_Scan_Activity command has completed, an HCI_Command_Complete event shall be generated.



7.3.22 Write Inquiry Scan Activity command

Command	OCF	Command Parameters	Return Parameters
HCI_Write_Inquiry_Scan_Activity	0x001E	Inquiry_Scan_Interval, Inquiry_Scan_Window	Status

Description:

This command writes the values for the Inquiry_Scan_Interval and Inquiry_Scan_Window configuration parameters. The Inquiry_Scan_Window shall be less than or equal to the Inquiry_Scan_Interval. See [Section 6.2](#) and [Section 6.3](#).

Note: Inquiry Scan is only performed when Inquiry_Scan is enabled (see [6.1](#), [7.3.17](#) and [7.3.18](#)).

Command parameters:

Inquiry_Scan_Interval:

Size: 2 octets

Value	Parameter Description
See Section 6.2 .	

Inquiry_Scan_Window:

Size: 2 octets

Value	Parameter Description
See Section 6.3 .	

Return parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	HCI_Write_Inquiry_Scan_Activity command succeeded.
0x01 to 0xFF	HCI_Write_Inquiry_Scan_Activity command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Event(s) generated (unless masked away):

When the HCI_Write_Inquiry_Scan_Activity command has completed, an HCI_Command_Complete event shall be generated.



7.3.23 Read Authentication Enable command

Command	OCF	Command Parameters	Return Parameters
HCI_Read_ Authentication_Enable	0x001F		Status, Authentication_Enable

Description:

This command reads the value for the Authentication_Enable configuration parameter. See [Section 6.16 Authentication Enable](#).

Command parameters:

None.

Return parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	HCI_Read_Authentication_Enable command succeeded.
0x01 to 0xFF	HCI_Read_Authentication_Enable command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Authentication_Enable:

Size: 1 octet

Value	Parameter Description
0x00	Authentication not required.
0x01	Authentication required for all connections.
All other values	Reserved for future use

Event(s) generated (unless masked away):

When the HCI_Read_Authentication_Enable command has completed, an HCI_Command_Complete event shall be generated.



7.3.24 Write Authentication Enable command

Command	OCF	Command Parameters	Return Parameters
HCI_Write_ - Authentication_Enable	0x0020	Authentication_Enable	Status

Description:

This command writes the value for the Authentication_Enable configuration parameter. See [Section 6.16 Authentication Enable](#).

The Authentication_Enable configuration parameter shall only apply to connections (e.g. send an LMP_in_rand or LMP_aud_rand) when the remote device's Host or BR/EDR Controller does not support Secure Simple Pairing or when the local Host does not support Secure Simple Pairing.

Note: Requires LM to read Host features during connection setup.

Command parameters:

Authentication_Enable:

Size: 1 octet

Value	Parameter Description
0x00	Authentication not required. Default.
0x01	Authentication required for all connections.
All other values	Reserved for future use

Return parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	HCI_Write_Authentication_Enable command succeeded.
0x01 to 0xFF	HCI_Write_Authentication_Enable command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Event(s) generated (unless masked away):

When the HCI_Write_Authentication_Enable command has completed, an HCI_Command_Complete event shall be generated.



7.3.25 Read Class of Device command

Command	OCF	Command Parameters	Return Parameters
HCI_Read_Class_of_Device	0x0023		Status, Class_of_Device

Description:

This command reads the value for the Class_of_Device parameter. See [Section 6.26](#).

Command parameters:

None.

Return parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	HCI_Read_Class_of_Device command succeeded.
0x01 to 0xFF	HCI_Read_Class_of_Device command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Class_of_Device:

Size: 3 octets

Value	Parameter Description
0xXXXXXX	Class of Device for the device.

Event(s) generated (unless masked away):

When the HCI_Read_Class_of_Device command has completed, an HCI_Command_Complete event shall be generated.



7.3.26 Write Class of Device command

Command	OCF	Command Parameters	Return Parameters
HCI_Write_Class_of_Device	0x0024	Class_of_Device	Status

Description:

This command writes the value for the Class_of_Device parameter.
See [Section 6.26](#).

Command parameters:

Class_of_Device:

Size: 3 octets

Value	Parameter Description
0xXXXXXX	Class of Device for the device.

Return parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	HCI_Write_Class_of_Device command succeeded.
0x01 to 0xFF	HCI_Write_Class_of_Device command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Event(s) generated (unless masked away):

When the HCI_Write_Class_of_Device command has completed, an HCI_Command_Complete event shall be generated.



7.3.27 Read Voice Setting command

Command	OCF	Command Parameters	Return Parameters
HCI_Read_Voice_Setting	0x0025		Status, Voice_Setting

Description:

This command reads the values for the Voice_Setting configuration parameter. See [Section 6.12](#).

Command parameters:

None.

Return parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	HCI_Read_Voice_Setting command succeeded.
0x01 to 0xFF	HCI_Read_Voice_Setting command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Voice_Setting:

Size: 2 octets (10 bits meaningful)

Value	Parameter Description
See Section 6.12 .	

Event(s) generated (unless masked away):

When the HCI_Read_Voice_Setting command has completed, an HCI_Command_Complete event shall be generated.



7.3.28 Write Voice Setting command

Command	OCF	Command Parameters	Return Parameters
HCI_Write_Voice_Setting	0x0026	Voice_Setting	Status

Description:

This command writes the values for the Voice_Setting configuration parameter. See [Section 6.12](#).

Command parameters:

Voice_Setting: *Size: 2 octets (10 bits meaningful)*

Value	Parameter Description
See Section 6.12 .	

Return parameters:

Status: *Size: 1 octet*

Value	Parameter Description
0x00	HCI_Write_Voice_Setting command succeeded.
0x01 to 0xFF	HCI_Write_Voice_Setting command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Event(s) generated (unless masked away):

When the HCI_Write_Voice_Setting command has completed, an HCI_Command_Complete event shall be generated.



7.3.29 Read Automatic Flush Timeout command

Command	OCF	Command Parameters	Return Parameters
HCI_Read_Automatic_Flush_Timeout	0x0027	Connection_Handle	Status, Connection_Handle, Flush_Timeout

Description:

This command reads the value for the Flush_Timeout parameter for the specified Connection_Handle. See [Section 6.19](#).

Command parameters:

Connection_Handle: *Size: 2 octets (12 bits meaningful)*

Value	Parameter Description
0xFFFF	Connection_Handle Range: 0x0000 to 0x0EFF

Return parameters:

Status: *Size: 1 octet*

Value	Parameter Description
0x00	HCI_Read_Automatic_Flush_Timeout command succeeded.
0x01 to 0xFF	HCI_Read_Automatic_Flush_Timeout command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Connection_Handle: *Size: 2 octets (12 bits meaningful)*

Value	Parameter Description
0xFFFF	Connection_Handle Range: 0x0000 to 0x0EFF

Flush_Timeout: *Size: 2 octets*

Value	Parameter Description
0x0000	Timeout = ∞ ; No Automatic Flush
N = 0xFFFF	Flush Timeout = N * 0.625 ms Size: 11 bits Range: 0x0001 to 0x07FF

**Event(s) generated (unless masked away):**

When the HCI_Read_Automatic_Flush_Timeout command has completed, an HCI_Command_Complete event shall be generated.



7.3.30 Write Automatic Flush Timeout command

Command	OCF	Command Parameters	Return Parameters
HCI_Write_Automatic_Flush_Timeout	0x0028	Connection_Handle, Flush_Timeout	Status, Connection_Handle

Description:

This command writes the value for the Flush_Timeout parameter for the specified Connection_Handle. See [Section 6.19](#).

Command parameters:

Connection_Handle: *Size: 2 octets (12 bits meaningful)*

Value	Parameter Description
0xXXXX	Connection_Handle Range: 0x0000 to 0x0EFF

Flush_Timeout: *Size: 2 octets*

Value	Parameter Description
0x0000	Timeout = ∞ ; No Automatic Flush. Default.
N = 0xXXXX	Flush Timeout = $N * 0.625$ ms Size: 11 bits Range: 0x0001 to 0x07FF Mandatory Range for Controller: 0x0002 to 0x07FF

Return parameters:

Status: *Size: 1 octet*

Value	Parameter Description
0x00	HCI_Write_Automatic_Flush_Timeout command succeeded.
0x01 to 0xFF	HCI_Write_Automatic_Flush_Timeout command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

**Connection_Handle:****Size: 2 octets (12 bits meaningful)**

Value	Parameter Description
0xFFFF	Connection_Handle Range: 0x0000 to 0x0EFF

Event(s) generated (unless masked away):

When the HCI_Write_Automatic_Flush_Timeout command has completed, an HCI_Command_Complete event shall be generated.



7.3.31 Read Num Broadcast Retransmissions command

Command	OCF	Command Parameters	Return Parameters
HCI_Read_Num_Broadcast_Retransmissions	0x0029		Status, Num_Broadcast_Retransmissions

Description:

This command reads the device's parameter value for the Number of Broadcast Retransmissions. See [Section 6.20](#)

Command parameters:

None.

Return parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	HCI_Read_Num_Broadcast_Retransmissions command succeeded.
0x01 to 0xFF	HCI_Read_Num_Broadcast_Retransmissions command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Num_Broadcast_Retransmissions:

Size: 1 octet

Value	Parameter Description
See Section 6.20 .	

Event(s) generated (unless masked away):

When the HCI_Read_Num_Broadcast_Retransmission command has completed, an HCI_Command_Complete event shall be generated.



7.3.32 Write Num Broadcast Retransmissions command

Command	OCF	Command Parameters	Return Parameters
HCI_Write_Num_Broadcast_Retransmissions	0x002A	Num_Broadcast_Re-transmissions	Status

Description:

This command writes the device's parameter value for the Number of Broadcast Retransmissions. See [Section 6.20](#).

Command parameters:

Num_Broadcast_Retransmissions: *Size: 1 octet*

Value	Parameter Description
See Section 6.20 .	

Return parameters:

Status: *Size: 1 octet*

Value	Parameter Description
0x00	HCI_Write_Num_Broadcast_Retransmissions command succeeded.
0x01 to 0xFF	HCI_Write_Num_Broadcast_Retransmissions command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Event(s) generated (unless masked away):

When the HCI_Write_Num_Broadcast_Retransmissions command has completed, an HCI_Command_Complete event shall be generated.



7.3.33 Read Hold Mode Activity command

Command	OCF	Command Parameters	Return Parameters
HCI_Read_Hold_Mode_Activity	0x002B		Status, Hold_Mode_Activity

Description:

This command reads the value for the Hold_Mode_Activity parameter.
See [Section 6.16](#).

Command parameters:

None.

Return parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	HCI_Read_Hold_Mode_Activity command succeeded.
0x01 to 0xFF	HCI_Read_Hold_Mode_Activity command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Hold_Mode_Activity:

Size: 1 octet

Value	Parameter Description
0x00	Maintain current Power State.
0x01	Suspend Page Scan.
0x02	Suspend Inquiry Scan.
0x04	Suspend Periodic Inquiries.
All other values	Reserved for future use.

Event(s) generated (unless masked away):

When the HCI_Read_Hold_Mode_Activity command has completed, an HCI_Command_Complete event shall be generated.



7.3.34 Write Hold Mode Activity command

Command	OCF	Command Parameters	Return Parameters
HCI_Write_Hold_Mode_Activity	0x002C	Hold_Mode_Activity	Status

Description:

This command writes the value for the Hold_Mode_Activity parameter.
See [Section 6.16](#).

Command parameters:

Hold_Mode_Activity:

Size: 1 octet

Value	Parameter Description
0x00	Maintain current Power State. Default.
0x01	Suspend Page Scan.
0x02	Suspend Inquiry Scan.
0x04	Suspend Periodic Inquiries.
All other values	Reserved for future use.

Return parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	HCI_Write_Hold_Mode_Activity command succeeded.
0x01 to 0xFF	HCI_Write_Hold_Mode_Activity command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Event(s) generated (unless masked away):

When the HCI_Write_Hold_Mode_Activity command has completed, an HCI_Command_Complete event shall be generated.



7.3.35 Read Transmit Power Level command

Command	OCF	Command Parameters	Return Parameters
HCI_Read_Transmit_Power_Level	0x002D	Connection_Handle, Type	Status, Connection_Handle, Transmit_Power_Level

Description:

This command reads the values for the Transmit_Power_Level parameter for the specified Connection_Handle. The Connection_Handle shall be a Connection_Handle for an ACL connection.

Command parameters:

Connection_Handle: *Size: 2 octets (12 bits meaningful)*

Value	Parameter Description
0xFFFF	Connection_Handle Range: 0x0000 to 0x0EFF

Type: *Size: 1 octet*

Value	Parameter Description
0x00	Read Current Transmit Power Level.
0x01	Read Maximum Transmit Power Level.
All other values	Reserved for future use

Return parameters:

Status: *Size: 1 octet*

Value	Parameter Description
0x00	HCI_Read_Transmit_Power_Level command succeeded.
0x01 to 0xFF	HCI_Read_Transmit_Power_Level command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Connection_Handle: *Size: 2 octets (12 bits meaningful)*

Value	Parameter Description
0xFFFF	Connection_Handle Range: 0x0000 to 0x0EFF

**Transmit_Power_Level:****Size: 1 octet**

Value	Parameter Description
0xXX	Size: 1 octet (signed integer) Range: -30 to 20 Units: dBm

Event(s) generated (unless masked away):

When the HCI_Read_Transmit_Power_Level command has completed, an HCI_Command_Complete event shall be generated.



7.3.36 Read Synchronous Flow Control Enable command

Command	OCF	Command Parameters	Return Parameters
HCI_Read_- Synchronous_Flow_- Control_Enable	0x002E		Status, Synchronous_Flow_Con- trol_Enable

Description:

The HCI_Read_Synchronous_Flow_Control_Enable command provides the ability to read the Synchronous_Flow_Control_Enable parameter. See [Section 6.22](#).

The Synchronous_Flow_Control_Enable parameter shall only be changed if no connection exists.

Command parameters:

None.

Return parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	HCI_Read_Synchronous_Flow_Control_Enable command succeeded
0x01 to 0xFF	HCI_Read_Synchronous_Flow_Control_Enable command failed (see [Vol 2] Part D, Error Codes for list of error codes).

Synchronous_Flow_Control_Enable:

Size: 1 octet

Value	Parameter Description
0x00	Synchronous Flow Control is disabled. No HCI_Number_Of_Completed_-Packets events will be sent from the Controller for synchronous Connection_Handles.
0x01	Synchronous Flow Control is enabled. HCI_Number_Of_Completed_-Packets events will be sent from the Controller for synchronous Connection_Handles.

Event(s) generated (unless masked away):

When the HCI_Read_Synchronous_Flow_Control_Enable command has completed an HCI_Command_Complete event shall be generated.



7.3.37 Write Synchronous Flow Control Enable command

Command	OCF	Command Parameters	Return Parameters
HCI_Write_ Synchronous_Flow_ Control_Enable	0x002F	Synchronous_Flow_Con- trol_Enable	Status

Description:

The HCI_Write_Synchronous_Flow_Control_Enable command provides the ability to write the Synchronous_Flow_Control_Enable parameter. See [Section 6.22](#).

The Synchronous_Flow_Control_Enable parameter can only be changed if no connections exist.

Command parameters:

Synchronous_Flow_Control_Enable:

Size: 1 octet

Value	Parameter Description
0x00	Synchronous Flow Control is disabled. No HCI_Number_Of_Completed_Packets events shall be sent from the Controller for synchronous Connection_Handles. Default
0x01	Synchronous Flow Control is enabled. HCI_Number_Of_Completed_Packets events shall be sent from the Controller for synchronous Connection_Handles.

Return parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	HCI_Write_Synchronous_Flow_Control_Enable command succeeded
0x01 to 0xFF	HCI_Write_Synchronous_Flow_Control_Enable command failed (see [Vol 2] Part D, Error Codes for list of error codes.)

Event(s) generated (unless masked away):

When the HCI_Write_Synchronous_Flow_Control_Enable command has completed an HCI_Command_Complete event shall be generated.



7.3.38 Set Controller To Host Flow Control command

Command	OCF	Command Parameters	Return Parameters
HCI_Set_Controller_To_Host_Flow_Control	0x0031	Flow_Control_Enable	Status

Description:

This command is used by the Host to turn flow control on or off for data and/or voice sent in the direction from the Controller to the Host. If flow control is turned off, the Host should not send the HCI_Host_Number_Of_Completed_Packets command. That command will be ignored by the Controller if it is sent by the Host and flow control is off. If flow control is turned on for HCI ACL Data packets and off for HCI Synchronous Data packets, HCI_Host_Number_Of_Completed_Packets commands sent by the Host should only contain Connection_Handles for ACL connections. If flow control is turned off for HCI ACL Data packets and on for HCI Synchronous Data packets, HCI_Host_Number_Of_Completed_Packets commands sent by the Host should only contain Connection_Handles for synchronous connections. If flow control is turned on for HCI ACL Data packets and HCI Synchronous Data packets, the Host will send HCI_Host_Number_Of_Completed_Packets commands both for ACL connections and synchronous connections.

The Flow_Control_Enable parameter shall only be changed if no connections exist.

Command parameters:

Flow_Control_Enable:

Size: 1 octet

Value	Parameter Description
0x00	Flow control off in direction from Controller to Host. Default.
0x01	Flow control on for HCI ACL Data packets and off for HCI Synchronous Data packets in direction from Controller to Host.
0x02	Flow control off for HCI ACL Data packets and on for HCI Synchronous Data packets in direction from Controller to Host.
0x03	Flow control on both for HCI ACL Data packets and HCI Synchronous Data packets in direction from Controller to Host.
All other values	Reserved for future use

**Return parameters:***Status:**Size: 1 octet*

Value	Parameter Description
0x00	HCI_Set_Controller_To_Host_Flow_Control command succeeded.
0x01 to 0xFF	HCI_Set_Controller_To_Host_Flow_Control command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Event(s) generated (unless masked away):

When the HCI_Set_Controller_To_Host_Flow_Control command has completed, an HCI_Command_Complete event shall be generated.



7.3.39 Host Buffer Size command

Command	OCF	Command Parameters	Return Parameters
HCI_Host_Buffer_Size	0x0033	Host_ACL_Data_Packet_Length, Host_Synchronous_Data_Packet_Length, Host_Total_Num_ACL_Data_Packets, Host_Total_Num_Synchronous_Data_Packets	Status

Description:

The HCI_Host_Buffer_Size command is used by the Host to notify the Controller about the maximum size of the data portion of HCI ACL and Synchronous Data packets sent from the Controller to the Host. The Controller shall segment the data to be transmitted from the Controller to the Host according to these sizes, so that the HCI Data packets will contain data with up to these sizes. The HCI_Host_Buffer_Size command also notifies the Controller about the total number of HCI ACL and Synchronous Data packets that can be stored in the data buffers of the Host. If flow control from the Controller to the Host is turned off, and the HCI_Host_Buffer_Size command has not been issued by the Host, this means that the Controller will send HCI Data packets to the Host with any lengths the Controller wants to use, and it is assumed that the data buffer sizes of the Host are unlimited. If flow control from the Controller to the Host is turned on, the HCI_Host_Buffer_Size command shall after a power-on or a reset always be sent by the Host before the first HCI_Host_Number_Of_Completed_Packets command is sent.

The [Set Controller To Host Flow Control command](#) is used to turn flow control on or off. The Host_ACL_Data_Packet_Length command parameter will be used to determine the size of the L2CAP segments contained in ACL Data packets, which are transferred from the Controller to the Host. The Host_Synchronous_Data_Packet_Length command parameter is used to determine the maximum size of HCI Synchronous Data packets. Both the Host and the Controller shall support command and event packets, where the data portion (excluding header) contained in the packets is 255 octets in size.

The Host_Total_Num_ACL_Data_Packets command parameter contains the total number of HCI ACL Data packets that can be stored in the data buffers of the Host. The Controller will determine how the buffers are to be divided between different Connection_Handles. The Host_Total_Num_Synchronous_Data_Packets command parameter gives the same information for HCI Synchronous Data packets.

Note: The Host_ACL_Data_Packet_Length and Host_Synchronous_Data_Packet_Length command parameters do not include the length of the HCI Data packet header.

**Command parameters:***Host_ACL_Data_Packet_Length:**Size: 2 octets*

Value	Parameter Description
0xFFFF	Maximum length (in octets) of the data portion of each HCI ACL Data packet that the Host is able to accept.

*Host_Synchronous_Data_Packet_Length:**Size: 1 octet*

Value	Parameter Description
0xFF	Maximum length (in octets) of the data portion of each HCI Synchronous Data packet that the Host is able to accept.

*Host_Total_Num_ACL_Data_Packets:**Size: 2 octets*

Value	Parameter Description
0xFFFF	Total number of HCI ACL Data packets that can be stored in the data buffers of the Host.

*Host_Total_Num_Synchronous_Data_Packets:**Size: 2 octets*

Value	Parameter Description
0xFFFF	Total number of HCI Synchronous Data packets that can be stored in the data buffers of the Host.

Return parameters:*Status:**Size: 1 octet*

Value	Parameter Description
0x00	HCI_Host_Buffer_Size command succeeded.
0x01 to 0xFF	HCI_Host_Buffer_Size command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Event(s) generated (unless masked away):

When the HCI_Host_Buffer_Size command has completed, an HCI_Command_Complete event shall be generated.



7.3.40 Host Number Of Completed Packets command

Command	OCF	Command Parameters	Return Parameters
HCI_Host_Number_Of_Completed_Packets	0x0035	Number_Of_Handles, Connection_Handle[i], Host_Num_Of_Completed_Packets [i]	

Description:

The HCI_Host_Number_Of_Completed_Packets command is used by the Host to indicate to the Controller the number of HCI Data packets that have been completed for each Connection_Handle since the previous HCI_Host_Number_Of_Completed_Packets command was sent to the Controller. This means that the corresponding buffer space has been freed in the Host. Based on this information, and the Host_Total_Num_ACL_Data_Packets and Host_Total_Num_Synchronous_Data_Packets command parameters of the HCI_Host_Buffer_Size command, the Controller can determine for which Connection_Handles the following HCI Data packets should be sent to the Host. The command should only be issued by the Host if flow control in the direction from the Controller to the Host is on and there is at least one connection, or if the Controller is in local loopback mode. Otherwise, the command will be ignored by the Controller. When the Host has completed one or more HCI Data packet(s) it shall send an HCI_Host_Number_Of_Completed_Packets command to the Controller, until it finally reports that all pending HCI Data packets have been completed. The frequency at which this command is sent is manufacturer specific.

The [Set Controller To Host Flow Control command](#) is used to turn flow control on or off. If flow control from the Controller to the Host is turned on, the HCI_Host_Buffer_Size command shall always be sent by the Host after a power-on or a reset before the first HCI_Host_Number_Of_Completed_Packets command is sent.

Note: The HCI_Host_Number_Of_Completed_Packets command is a special command in the sense that no event is normally generated after the command has completed. The command may be sent at any time by the Host when there is at least one connection, or if the Controller is in local loopback mode independent of other commands. The normal flow control for commands is not used for the HCI_Host_Number_Of_Completed_Packets command.

**Command parameters:***Number_Of_Handles:**Size: 1 octet*

Value	Parameter Description
0xXX	The number of Connection_Handles and Host_Num_Of_Completed_Packets parameters pairs contained in this command. Range: 0 to 255

*Connection_Handle[i]: Size: Number_Of_Handles*2 octets (12 bits meaningful)*

Value	Parameter Description
0xFFFF	Connection_Handle Range: 0x0000 to 0x0EFF

*Host_Num_Of_Completed_Packets [i]: Size: Number_Of_Handles * 2 octets*

Value	Parameter Description
0xFFFF	The number of HCI Data packets that have been completed for the associated Connection_Handle since the previous time the event was returned. Range: 0x0000 to 0xFFFF

Return parameters:

None.

Event(s) generated (unless masked away):

Normally, no event is generated after the HCI_Host_Number_Of_Completed_Packets command has completed. However, if the HCI_Host_Number_Of_Completed_Packets command contains one or more invalid parameters, the Controller shall return an HCI_Command_Complete event with a failure status indicating the *Invalid HCI Command Parameters* error code. The Host may send the HCI_Host_Number_Of_Completed_Packets command at any time when there is at least one connection, or if the Controller is in local loopback mode. The normal flow control for commands is not used for this command.



7.3.41 Read Link Supervision Timeout command

Command	OCF	Command Parameters	Return Parameters
HCI_Read_Link_Supervision_Timeout	0x0036	Handle	Status, Handle, Link_Supervision_Timeout

Description:

This command reads the value for the Link_Supervision_Timeout parameter for the Controller.

The Handle used for this command shall be the ACL connection to the appropriate device. See [Section 6.21](#).

Command parameters:

Handle: *Size: 2 octets (12 bits meaningful)*

Value	Parameter Description
0xFFFF	Specifies which Connection_Handle's Link Supervision Timeout value is to be read. The Handle is a Connection_Handle for a BR/EDR Controller. On an AMP, a Physical Link Handle is used as the lower 8 bits of the Handle. The upper 4 bits are reserved and shall be set to 0. Range: 0x0000 to 0x0EFF

Return parameters:

Status: *Size: 1 octet*

Value	Parameter Description
0x00	HCI_Read_Link_Supervision_Timeout command succeeded.
0x01 to 0xFF	HCI_Read_Link_Supervision_Timeout command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Handle: *Size: 2 octets (12 bits meaningful)*

Value	Parameter Description
0xFFFF	Specifies which Connection_Handle's Link Supervision Timeout value was read. The Handle is a Connection_Handle for a BR/EDR Controller. On an AMP a Physical Link Handle is used as the lower 8 bits of the Handle. The upper 4 bits are reserved and shall be set to 0. Range: 0x0000 to 0x0EFF

**Link_Supervision_Timeout:****Size: 2 octets**

Value	Parameter Description
0x0000	No Link_Supervision_Timeout.
N = 0xXXXX	Measured in number of BR/EDR Baseband slots Link_Supervision_Timeout = N * 0.625 ms (1 Baseband slot) Range: 0x0001 to 0xFFFF Time Range: 0.625ms to 40.9 s

Event(s) generated (unless masked away):

When the HCI_Read_Link_Supervision_Timeout command has completed, an HCI_Command_Complete event shall be generated.



7.3.42 Write Link Supervision Timeout command

Command	OCF	Command Parameters	Return Parameters
HCI_Write_Link_Supervision_Timeout	0x0037	Handle, Link_Supervision_Timeout	Status, Handle

Description:

This command writes the value for the Link_Supervision_Timeout parameter for a BR/EDR or AMP Controller. If the Host issues this command with a Connection_Handle which identifies a BR/EDR connection where the Controller is the slave, the Controller shall return the error code *Command Disallowed* (0x0C). The command may be issued to any AMP Controller without restriction to role.

The Handle used for this command shall be the ACL connection to the appropriate device. This command will set the Link_Supervision_Timeout values for other Synchronous Handles to that device.

See [Section 6.21](#).

Command parameters:

Handle:

Size: 2 octets (12 bits meaningful)

Value	Parameter Description
0xXXXX	Specifies which Handle's Link Supervision Timeout value is to be written. The Handle is a Connection_Handle for a BR/EDR Controller. On an AMP a Physical Link Handle is used as the lower 8 bits of the Handle. The upper 4 bits are reserved and shall be set to 0. Range: 0x0000 to 0x0EFF

Link_Supervision_Timeout:

Size: 2 octets

Value	Parameter Description
0x0000	No Link_Supervision_Timeout.
N = 0xXXXX	Measured in number of BR/EDR Baseband slots Link_Supervision_Timeout = N*0.625 ms (1 Baseband slot) Range: 0x0001 to 0xFFFF Time Range: 0.625ms to 40.9 s Default: 0x7D00 Link_Supervision_Timeout = 20 s Mandatory Range for Controller: 0x0190 to 0xFFFF; plus 0 for infinite timeout

**Return parameters:***Status:**Size: 1 octet*

Value	Parameter Description
0x00	HCI_Write_Link_Supervision_Timeout command succeeded.
0x01 to 0xFF	HCI_Write_Link_Supervision_Timeout command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

*Handle:**Size: 2 octets (12 bits meaningful)*

Value	Parameter Description
0xFFFF	Specifies which Handle's Link Supervision Timeout value was written. The Handle is a Connection_Handle for a BR/EDR Controller. On an AMP a Physical Link Handle is used as the lower 8 bits of the Handle. The upper 4 bits (of the 12 meaningful bits) are reserved and shall be set to 0. Range: 0x0000 to 0x0FFF

Event(s) generated (unless masked away):

When the HCI_Write_Link_Supervision_Timeout command has completed, an HCI_Command_Complete event shall be generated.



7.3.43 Read Number Of Supported IAC command

Command	OCF	Command Parameters	Return Parameters
HCI_Read_Number_Of_Supported_IAC	0x0038		Status, Num_Support_IAC

Description:

This command reads the value for the number of Inquiry Access Codes (IAC) that the local BR/EDR Controller can simultaneously listen for during an Inquiry Scan. All BR/EDR Controllers are required to support at least one IAC, the General Inquiry Access Code (the GIAC). Some BR/EDR Controllers support additional IACs.

Command parameters:

None.

Return parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	HCI_Read_Number_Of_Supported_IAC command succeeded.
0x01 to 0xFF	HCI_Read_Number_Of_Supported_IAC command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Num_Support_IAC

Size: 1 octet

Value	Parameter Description
0xXX	Specifies the number of Supported IAC that the local BR/EDR Controller can simultaneously listen for during an Inquiry Scan. Range: 0x01 to 0x40

Event(s) generated (unless masked away):

When the HCI_Read_Number_Of_Supported_IAC command has completed, an HCI_Command_Complete event shall be generated.



7.3.44 Read Current IAC LAP command

Command	OCF	Command Parameters	Return Parameters
HCI_Read_Current_IAC_LAP	0x0039		Status, Num_Current_IAC, IAC_LAP[i]

Description:

This command reads the LAP(s) used to create the Inquiry Access Codes (IAC) that the local BR/EDR Controller is simultaneously scanning for during Inquiry Scans. All BR/EDR Controllers shall support at least one IAC, the General Inquiry Access Code (the GIAC). Some BR/EDR Controllers support additional IACs.

Command parameters:

None.

Return parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	HCI_Read_Current_IAC_LAP command succeeded.
0x01 to 0xFF	HCI_Read_Current_IAC_LAP command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Num_Current_IAC

Size: 1 octet

Value	Parameter Description
0xXX	Specifies the number of IACs which are currently in use by the local BR/EDR Controller to simultaneously listen for during an Inquiry Scan. Range: 0x01 to 0x40

IAC_LAP[i]

*Size: 3 octets * Num_Current_IAC*

Value	Parameter Description
0XXXXXXXX	LAPs used to create the IAC which is currently in use by the local BR/EDR Controller to simultaneously listen for during an Inquiry Scan. Range: 0x9E8B00 to 0x9E8B3F

Event(s) generated (unless masked away):

When the HCI_Read_Current_IAC_LAP command has completed, an HCI_Command_Complete event shall be generated.



7.3.45 Write Current IAC LAP command

Command	OCF	Command Parameters	Return Parameters
HCI_Write_Current_IAC_LAP	0x003A	Num_Current_IAC, IAC_LAP[i]	Status

Description:

This command writes the LAP(s) used to create the Inquiry Access Codes (IAC) that the local BR/EDR Controller is simultaneously scanning for during Inquiry Scans. All BR/EDR Controller shall support at least one IAC, the General Inquiry Access Code (the GIAC). Some BR/EDR Controllers support additional IACs.

This command shall clear any existing IACs and stores Num_Current_IAC and the IAC_LAPs in to the Controller. If Num_Current_IAC is greater than Num_Support_IAC then only the first Num_Support_IAC shall be stored in the Controller, and an HCI_Command_Complete event with error code *Success* (0x00) shall be generated.

Command parameters:

Num_Current_IAC

Size: 1 octet

Value	Parameter Description
0xXX	Specifies the number of IACs that will be used by the local BR/EDR Controller to simultaneously listen for during an Inquiry Scan. Range: 0x01 to 0x40

IAC_LAP[i]

*Size: 3 octets * Num_Current_IAC*

Value	Parameter Description
0XXXXXXXX	LAP(s) that will be used to create the IACs that will be used by the local BR/EDR Controller to simultaneously listen for during an Inquiry Scan. Range: 0x9E8B00 to 0x9E8B3F. The default IAC(s) to be used shall be the GIAC and zero or more other IACs specified by the manufacturer.

Return parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	HCI_Write_Current_IAC_LAP command succeeded.
0x01 to 0xFF	HCI_Write_Current_IAC_LAP command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

**Event(s) generated (unless masked away):**

When the HCI_Write_Current_IAC_LAP command has completed, an HCI_Command_Complete event shall be generated.



7.3.46 Set AFH Host Channel Classification command

Command	OCF	Command Parameters	Return Parameters
HCI_Set_AFH_Host_Channel_Classification	0x003F	AFH_Host_Channel_Classification	Status

Description:

The HCL_Set_AFH_Host_Channel_Classification command allows the Host to specify a channel classification based on its “local information”. This classification persists until overwritten with a subsequent HCL_Set_AFH_Host_Channel_Classification command or until the BR/EDR Controller is reset.

This command shall be supported by a device that declares support for any of the AFH_capable_master, AFH_classification_slave or AFH_classification_master features.

If this command is used, updates should be sent within 10 seconds of the Host knowing that the channel classification has changed. The interval between two successive commands sent shall be at least 1 second.

Command parameters:

AFH Host Channel Classification: *Size: 10 octets (79 bits meaningful)*

Value	Parameter Description
0xFFFFFFFF XXXXXXXXXX XX	This parameter contains 80 1-bit fields. The n^{th} such field (in the range 0 to 78) contains the value for channel n: 0: channel n is bad 1: channel n is unknown The most significant bit (bit 79) is reserved for future use At least N_{min} channels shall be marked as unknown. (See [Vol 2] Part B, Section 2.3.1). If the device supports Synchronizable mode, then the synchronization train channels (see [Vol 2] Part B, Section 2.6.4.8) shall be excluded when checking this requirement.

Return parameters:

Status: *Size: 1 octet*

Value	Parameter Description
0x00	HCI_Set_AFH_Host_Channel_Classification command succeeded.
0x01 to 0xFF	HCI_Set_AFH_Host_Channel_Classification command failed. [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

**Event(s) generated (unless masked away):**

When the HCI_Set_AFH_Host_Channel_Classification command has completed, an HCI_Command_Complete event shall be generated.



7.3.47 Read Inquiry Scan Type command

Command	OCF	Command Parameters	Return Parameters
HCI_Read_Inquiry_Scan_Type	0x0042		Status, Inquiry_Scan_Type

Description:

This command reads the Inquiry_Scan_Type configuration parameter from the local BR/EDR Controller. See [Section 6.4](#). For details, see the Baseband Specification, [\[Vol 2\] Part B, Section 8.4.1](#).

Command parameters:

None.

Return parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	HCI_Read_Inquiry_Scan_Type command succeeded
0x01 to 0xFF	HCI_Read_Inquiry_Scan_Type command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Inquiry_Scan_Type:

Size: 1 octet

Value	Parameter Description
0x00	Standard Scan (default)
0x01	Interlaced Scan
All other values	Reserved for future use

Event(s) generated (unless masked away):

When the HCI_Read_Inquiry_Scan_Type command has completed, an HCI_Command_Complete event shall be generated.



7.3.48 Write Inquiry Scan Type command

Command	OCF	Command Parameters	Return Parameters
HCI_Write_Inquiry_Scan_Type	0x0043	Scan_Type	Status

Description:

This command writes the Inquiry Scan Type configuration parameter of the local BR/EDR Controller. See [Section 6.4](#). For details, see the Baseband Specification, [\[Vol 2\] Part B, Inquiry scan substate](#).

Command parameters:

Scan_Type:

Size: 1 octet

Value	Parameter Description
0x00	Standard Scan (default)
0x01	Interlaced Scan
All other values	Reserved for future use

Return parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	HCI_Write_Inquiry_Scan_Type command succeeded
0x01 to 0xFF	HCI_Write_Inquiry_Scan_Type command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Event(s) generated (unless masked away):

When the HCI_Write_Inquiry_Scan_Type command has completed, an HCI_Command_Complete event shall be generated.



7.3.49 Read Inquiry Mode command

Command	OCF	Command Parameters	Return Parameters
HCI_Read_Inquiry_Mode	0x0044		Status, Inquiry_ Mode

Description:

This command reads the Inquiry_Mode configuration parameter of the local BR/EDR Controller. See [Section 6.5](#).

Command parameters:

None.

Return parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	HCI_Read_Inquiry_Mode command succeeded.
0x01 to 0xFF	HCI_Read_Inquiry_Mode command failed. See [Vol 2] Part D, Error Codes for list of error codes.

Inquiry_Mode:

Size: 1 octet

Value	Parameter Description
0x00	Standard Inquiry Result event format
0x01	Inquiry Result format with RSSI
0x02	Inquiry Result with RSSI format or Extended Inquiry Result format
All other values	Reserved for future use

Event(s) generated (unless masked away):

When the HCI_Read_Inquiry_Mode command has completed, an HCI_Command_Complete event shall be generated.



7.3.50 Write Inquiry Mode command

Command	OCF	Command Parameters	Return Parameters
HCI_Write_Inquiry_Mode	0x0045	Inquiry_Mode	Status

Description:

This command writes the Inquiry_Mode configuration parameter of the local BR/EDR Controller. See [Section 6.5](#).

Command parameters:

Inquiry_Mode:

Size: 1 octet

Value	Parameter Description
0x00	Standard Inquiry Result event format (default)
0x01	Inquiry Result format with RSSI
0x02	Inquiry Result with RSSI format or Extended Inquiry Result format
All other values	Reserved for future use

Return parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	HCI_Write_Inquiry_Mode command succeeded.
0x01 to 0xFF	HCI_Write_Inquiry_Mode command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Event(s) generated (unless masked away):

When the HCI_Write_Inquiry_Mode command has completed, an HCI_Command_Complete event shall be generated.



7.3.51 Read Page Scan Type command

Command	OCF	Command Parameters	Return Parameters
HCI_Read_Page_Scan_Type	0x0046		Status, Page_Scan_Type

Description:

This command reads the Page Scan Type configuration parameter of the local BR/EDR Controller. See [Section 6.11](#). For details, see the Baseband Specification, “[Part B, Page scan substate](#)”.

Command parameters:

None.

Return parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	HCI_Read_Page_Scan_Type command succeeded.
0x01 to 0xFF	HCI_Read_Page_Scan_Type command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Page_Scan_Type:

Size: 1 octet

Value	Parameter Description
0x00	Mandatory: Standard Scan (default)
0x01	Optional: Interlaced Scan
All other values	Reserved for future use

Event(s) generated (unless masked away):

When the HCI_Read_Page_Scan_Type command has completed, an HCI_Command_Complete event shall be generated.



7.3.52 Write Page Scan Type command

Command	OCF	Command Parameters	Return Parameters
HCI_Write_Page_Scan_Type	0x0047	Page_Scan_Type	Status

Description:

This command writes the Page Scan Type configuration parameter of the local BR/EDR Controller. See [Section 6.11](#). For details, see the Baseband Specification, [\[Vol 2\] Part B, Section 8.3.1](#).

Command parameters:

Page_Scan_Type:

Size: 1 octet

Value	Parameter Description
0x00	Mandatory: Standard Scan (default)
0x01	Optional: Interlaced Scan
All other values	Reserved for future use

Return parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	HCI_Write_Page_Scan_Type command succeeded.
0x01 to 0xFF	HCI_Write_Page_Scan_Type command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Event(s) generated (unless masked away):

When the HCI_Write_Page_Scan_Type command has completed, an HCI_Command_Complete event shall be generated.



7.3.53 Read AFH Channel Assessment Mode command

Command	OCF	Command Parameters	Return Parameters
HCI_Read_AFH_Channel_Assessment_Mode	0x0048		Status, AFH_Channel_Assessment_Mode

Description:

The HCI_Read_AFH_Channel_Assessment_Mode command reads the value for the AFH_Channel_Assessment_Mode parameter. The AFH_Channel_Assessment_Mode parameter controls whether the Controller's channel assessment scheme is enabled or disabled.

This command shall be supported by a device that declares support for any of the AFH_capable_master, AFH_classification_slave or AFH_classification_master features.

Command parameters:

None.

Return parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	HCI_Read_AFH_Channel_Assessment_Mode command succeeded.
0x01 to 0xFF	HCI_Read_AFH_Channel_Assessment_Mode command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

AFH_Channel_Assessment_Mode:

Size: 1 octet

Value	Parameter Description
0x00	Controller channel assessment disabled.
0x01	Controller channel assessment enabled.
All other values	Reserved for future use.

Event(s) generated (unless masked away):

When the HCI_Read_AFH_Channel_Assessment_Mode command has completed, an HCI_Command_Complete event shall be generated.



7.3.54 Write AFH Channel Assessment Mode command

Command	OCF	Command Parameters	Return Parameters
HCI_Write_AFH_Channel_Assessment_Mode	0x0049	AFH_Channel_Assessment_Mode	Status

Description:

The HCI_Write_AFH_Channel_Assessment_Mode command writes the value for the AFH_Channel_Assessment_Mode parameter. The AFH_Channel_Assessment_Mode parameter controls whether the Controller's channel assessment scheme is enabled or disabled.

Disabling channel assessment forces all channels to be unknown in the local classification, but does not affect the AFH_reporting_mode or support for the HCI_Set_AFH_Host_Channel_Classification command. A slave in the AFH_reporting_enabled state shall continue to send LMP channel classification messages for any changes to the channel classification caused by either this command (altering the AFH_Channel_Assessment_Mode) or HCI_Set_AFH_Host_Channel_Classification command (providing a new channel classification from the Host).

This command shall be supported by a device that declares support for any of the AFH_capable_master, AFH_classification_slave or AFH_classification_master features.

If the AFH_Channel_Assessment_Mode parameter is enabled and the BR/EDR Controller does not support a channel assessment scheme, other than via the HCI_Set_AFH_Host_Channel_Classification command, then a Status parameter of 'Channel Assessment Not Supported' should be returned. See ["Part D, Error Codes"](#) for a list of error codes and descriptions.

If the BR/EDR Controller supports a channel assessment scheme then the default AFH_Channel_Assessment_Mode is enabled, otherwise the default is disabled.

Command parameters:

AFH_Channel_Assessment_Mode:

Size: 1 octet

Value	Parameter Description
0x00	Controller channel assessment disabled.
0x01	Controller channel assessment enabled.
All other values	Reserved for future use.

**Return parameters:***Status:**Size: 1 octet*

Value	Parameter Description
0x00	HCI_Write_AFH_Channel_Assessment_Mode command succeeded.
0x01 to 0xFF	HCI_Write_AFH_Channel_Assessment_Mode command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Event(s) generated (unless masked away):

When the HCI_Write_AFH_Channel_Assessment_Mode command has completed, an HCI_Command_Complete event shall be generated.



7.3.55 Read Extended Inquiry Response command

Command	OCF	Command Parameters	Return Parameters
HCI_Read_Extended_Inquiry_Response	0x0051		Status, FEC_Required, Extended_Inquiry_Response

Description:

The HCI_Read_Extended_Inquiry_Response command reads the extended inquiry response to be sent during the extended inquiry response procedure. The FEC_Required parameter states if FEC encoding is required.

Command parameters:

None.

Return parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	HCI_Read_Extended_Inquiry_Response command succeeded.
0x01 to 0xFF	HCI_Read_Extended_Inquiry_Response command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

FEC_Required:

Size: 1 octet

Value	Parameter Description
0x00	FEC is not required
0x01	FEC is required
All other values	Reserved for future use

Extended_Inquiry_Response:

Size: 240 octets

Value	Parameter Description
	Extended inquiry response data as defined in [Vol 3] Part C, Section 8 .

Event(s) generated (unless masked away):

When the HCI_Read_Extended_Inquiry_Response command has completed, an HCI_Command_Complete event shall be generated.



7.3.56 Write Extended Inquiry Response command

Command	OCF	Command Parameters	Return Parameters
HCI_Write_Extended_Inquiry_Response	0x0052	FEC_Required, Extended_Inquiry_Response	Status

Description:

The HCI_Write_Extended_Inquiry_Response command writes the extended inquiry response to be sent during the extended inquiry response procedure. The FEC_Required command parameter states if FEC encoding is required. The extended inquiry response data is not preserved over a reset. The initial value of the inquiry response data is all zero octets. The Controller shall not interpret the extended inquiry response data.

Command parameters:

FEC_Required:

Size: 1 octet

Value	Parameter Description
0x00	FEC is not required
0x01	FEC is required
All other values	Reserved for future use

Extended Inquiry Response:

Size: 240 octets

Value	Parameter Description
	Extended inquiry response data as defined in [Vol 3] Part C, Section 8 .
	All octets zero (default).

Return parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	HCI_Write_Extended_Inquiry_Response command succeeded
0x01 to 0xFF	HCI_Write_Extended_Inquiry_Response command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Event(s) generated (unless masked away):

When the HCI_Write_Extended_Inquiry_Response command has completed, an HCI_Command_Complete event shall be generated.



7.3.57 Refresh Encryption Key command

Command	OCF	Command Parameters	Return Parameters
HCI_Refresh_Encryption_Key	0x0053	Connection_Handle	

Description:

This command is used by the Host to cause the BR/EDR Controller to refresh the encryption key on an ACL connection identified by a Connection_Handle by pausing and resuming encryption.

Command parameters:

Connection_Handle: *Size: 2 octets (12 bits meaningful)*

Value	Parameter Description
0xFFFF	Connection_Handle Range: 0x0000 to 0x0EFF

Return parameters:

None.

Event(s) generated (unless masked away):

An HCI_Command_Status event is sent from the BR/EDR Controller to the Host when the Controller has started the Refresh Encryption Key procedure. An HCI_Encryption_Key_Refresh_Complete event shall be generated when the Refresh Encryption Key procedure has completed.



7.3.58 Read Simple Pairing Mode command

Command	OCF	Command Parameters	Return Parameters
HCI_Read_Simple_Pairing_Mode	0x0055		Status, Simple_Pairing_Mode

Description:

This command reads the Simple_Pairing_Mode parameter in the BR/EDR Controller.

Command parameters:

None.

Return parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	HCI_Read_Simple_Pairing_Mode command succeeded.
0x01 to 0xFF	HCI_Read_Simple_Pairing_Mode command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Simple_Pairing_Mode

Size: 1 octet

Value	Parameter Description
0x00	Simple Pairing not set (default)
0x01	Simple pairing enabled
All other values	Reserved for future use

Event(s) generated (unless masked away):

When the HCI_Read_Simple_Pairing_Mode command has completed, an HCI_Command_Complete event shall be generated.



7.3.59 Write Simple Pairing Mode command

Command	OCF	Command Parameters	Return Parameters
HCI_Write_Simple_Pairing_Mode	0x0056	Simple_Pairing_Mode	Status

Description:

This command enables Simple Pairing mode in the BR/EDR Controller. When Simple Pairing Mode is set to 'enabled' the Link Manager shall respond to an LMP_io_capability_req PDU with an LMP_io_capability_res PDU and continue with the subsequent pairing procedure. When Simple Pairing mode is set to 'disabled', the Link Manager shall reject an IO capability request. A Host shall not set the Simple Pairing Mode to 'disabled.'

Until Write_Simple_Pairing_Mode is received by the BR/EDR Controller, it shall not support any Simple Pairing sequences, and shall return the error code *Simple Pairing not Supported by Host* (0x37). This command shall be written before initiating page scan or paging procedures.

The Link Manager Secure Simple Pairing (Host Support) feature bit shall be set to the Simple_Pairing_Mode parameter. The default value for Simple_Pairing_Mode shall be 'disabled.' When Simple_Pairing_Mode is set to 'enabled,' the bit in the LMP features mask indicating support for Secure Simple Pairing (Host Support) shall be set to enabled in subsequent responses to an LMP_features_req from a remote device.

Command parameters:

Simple_Pairing_Mode:

Size: 1 octet

Value	Parameter Description
0x00	Simple Pairing disabled (default)
0x01	Simple pairing enabled
All other values	Reserved for future use

Return parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	HCI_Write_Simple_Pairing_Mode command succeeded.
0x01 to 0xFF	HCI_Write_Simple_Pairing_Mode command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

**Event(s) generated (unless masked away):**

When the HCI_Write_Simple_Pairing_Mode command has completed, an HCI_Command_Complete event shall be generated.



7.3.60 Read Local OOB Data command

Command	OCF	Command Parameters	Return Parameters
HCI_Read_Local_OOB_Data	0x0057		Status, C, R

Description:

This command obtains a Simple Pairing Hash C and Simple Pairing Randomizer R which are intended to be transferred to a remote device using an OOB mechanism. The BR/EDR Controller shall create new values for C and R for each invocation of this command.

Note: Each OOB transfer will have unique C and R values.

After each OOB transfer this command shall be used to obtain a new set of values for the next OOB transfer.

Note: The Controller keeps information used to generate these values for later use in the simple pairing process. If the BR/EDR Controller is powered off or reset then this information is lost and the values obtained before the power off or reset are invalid.

Command parameters:

None.

Return parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	HCI_Read_Local_OOB_Data command succeeded.
0x01 to 0xFF	HCI_Read_Local_OOB_Data command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

C:

Size: 16 octets

Value	Parameter Description
0xFFFFFFFFXX XXXXXXXXXXXX XXXXXXXXXX	Simple Pairing Hash C



R:

Size: 16 octets

Value	Parameter Description
0XXXXXXXXXXXXX XXXXXXXXXXXXX XXXXXXXXXXXXX	Simple Pairing Randomizer R

Event(s) generated (unless masked away):

When the HCI_Read_Local_OOB_Data command has completed, an HCI_Command_Complete event shall be generated.



7.3.61 Read Inquiry Response Transmit Power Level command

Command	OCF	Command Parameters	Return Parameters
HCI_Read_Inquiry_Response_Transmit_Power_Level	0x0058		Status, TX_Power

Description:

This command reads the power level used to transmit the FHS and EIR data packets. This can be used directly in the Tx Power Level EIR data type.

Command parameters:

None.

Return parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	HCI_Read_Inquiry_Response_Transmit_Power_Level command succeeded.
0x01 to 0xFF	HCI_Read_Inquiry_Response_Transmit_Power_Level command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

TX_Power:

Size: 1 octet

Value	Parameter Description
0xXX	Size: 1 octet (signed integer) Range: -70 to 20 Units: dBm

Event(s) generated (unless masked away):

When the HCI_Read_Inquiry_Response_Transmit_Power_Level command has completed, an HCI_Command_Complete event shall be generated.



7.3.62 Write Inquiry Transmit Power Level command

Command	OCF	Command Parameters	Return Parameters
HCI_Write_Inquiry_Transmit_Power_Level	0x0059	TX_Power	Status

Description:

This command writes the inquiry transmit power level used to transmit the inquiry (ID) data packets. The Controller should use the supported TX power level closest to the Tx_Power parameter.

Command parameters:

TX Power:

Size: 1 octet

Value	Parameter Description
0xXX	Size: 1 octet (signed integer) Range: -70 to 20 Units: dBm

Return parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	HCI_Write_Inquiry_Transmit_Power_Level command succeeded
0x01 to 0xFF	HCI_Write_Inquiry_Transmit_Power_Level command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Event(s) generated (unless masked away):

When the HCI_Write_Inquiry_Transmit_Power_Level command has completed, an HCI_Command_Complete event shall be generated.



7.3.63 Send Keypress Notification command

Command	OCF	Command Parameters	Return Parameters
HCI_Send_Keypress_Notification	0x0060	BD_ADDR, Notification_Type	Status, BD_ADDR

Description:

This command is used during the Passkey Entry protocol by a device with KeyboardOnly IO capabilities. It is used by a Host to inform the remote device when keys have been entered or erased.

Command parameters:

BD_ADDR: *Size: 6 octets*

Value	Parameter Description
0xFFFFFFFFXXXX	BD_ADDR of remote device involved in simple pairing process

Notification_Type: *Size: 1 octet*

Value	Parameter Description
0	Passkey entry started
1	Passkey digit entered
2	Passkey digit erased
3	Passkey cleared
4	Passkey entry completed
All other values	Reserved for future use

Return parameters:

Status: *Size: 1 octet*

Value	Parameter Description
0x00	HCI_Send_Keypress_Notification command succeeded
0x01 to 0xFF	HCI_Send_Keypress_Notification command failed

BD_ADDR: *Size: 6 octets*

Value	Parameter Description
0xFFFFFFFFXXXX	BD_ADDR of remote device involved in simple pairing process

**Event(s) generated (unless masked away):**

When the HCI_Send_Keypress_Notification command has completed, an HCI_Command_Complete event shall be generated.



7.3.64 Read Default Erroneous Data Reporting command

Command	OCF	Command Parameters	Return Parameters
HCI_Read_Default_-_Erroneous_Data_Reporting	0x005A		Status, Erroneous_Data_Reporting

Description:

This command reads the Erroneous_Data_Reporting parameter.

Command parameters:

None.

Return parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	HCI_Read_Default_Erroneous_Data_Reporting command succeeded.
0x01 to 0xFF	HCI_Read_Default_Erroneous_Data_Reporting command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Erroneous_Data_Reporting:

Size: 1 octet

Value	Parameter Description
0x00	Erroneous data reporting disabled.
0x01	Erroneous data reporting enabled.
All other values	Reserved for future use.

Event(s) generated (unless masked away):

When the HCI_Read_Default_Erroneous_Data_Reporting command has completed, an HCI_Command_Complete event shall be generated.



7.3.65 Write Default Erroneous Data Reporting command

Command	OCF	Command Parameters	Return Parameters
HCI_Write_Default_Erroneous_Data_Reporting	0x005B	Erroneous_Data_Reporting	Status

Description:

This command writes the Erroneous_Data_Reporting parameter. The BR/EDR Controller shall set the Packet_Status_Flag as defined in [Section 5.4.3 HCI Synchronous Data packets](#), depending on the value of this parameter. The new value for the Erroneous_Data_Reporting parameter shall not apply to existing synchronous connections.

Command parameters:

Erroneous_Data_Reporting:

Size: 1 octet

Value	Parameter Description
0x00	Erroneous Data reporting disabled (default).
0x01	Erroneous data reporting enabled.
All other values	Reserved for future use.

Return parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	HCI_Write_Default_Erroneous_Data_Reporting command succeeded.
0x01 to 0xFF	HCI_Write_Default_Erroneous_Data_Reporting command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Event(s) generated (unless masked away):

When the HCI_Write_Default_Erroneous_Data_Reporting command has completed, an HCI_Command_Complete event shall be generated.



7.3.66 Enhanced Flush command

Command	OCF	Command Parameters	Return Parameters
HCI_Enhanced_Flush	0x005F	Handle, Packet_Type	

Description:

The HCI_Enhanced_Flush command is used to discard all L2CAP packets identified by Packet_Type that are currently pending for transmission in the Controller for the specified Handle, even if there currently are chunks of data that belong to more than one L2CAP packet of the same type in the Controller. The only packet type defined is automatically-flushable. Packets not identified by Packet_Type will not be flushed and will be processed normally by the Controller.

After flushing the packets, all data that is sent to the BR/EDR Controller for the same Handle and packet type shall be discarded by the Controller until an HCI Data packet with the start Packet_Boundary_Flag (0x00 or 0x02) is received. This command allows higher-level software to control how long the baseband should try to retransmit a Baseband packet of a specific type for a Handle before all data of that type currently pending for transmission in the Controller should be flushed. The HCI_Enhanced_Flush command shall be used for ACL-U connections only. On the BR/EDR Controller, the HCI_Flush command can be used to flush all packets (see [Section 7.3.4](#)). In addition to the HCI_Enhanced_Flush and HCI_Flush commands, the automatic flush timers (see [Section 7.3.29](#)) can be used to automatically flush an automatically-flushable L2CAP packet that is currently being transmitted after the specified flush timer has expired.

This command shall be supported by a device that declares support for the Non-Flushable Packet Boundary Flag feature.

On an AMP Controller, in addition to the HCI_Enhanced_Flush command, the automatic flush timer specified in the Flow Specification for this logical link may be used to automatically flush packets on this logical link.

Command parameters:

Handle: *Size: 2 octets (12 bits meaningful)*

Value	Parameter Description
0xFFFF	Handle to be used to identify a connection. The Handle is a Connection_Handle for a BR/EDR Controller and a Logical Link Handle for an AMP Controller. Range: 0x0000 to 0x0EFF

**Packet_Type:****Size: 1 octet**

Value	Parameter Description
0x00	Automatically flushable only.
All other values	Reserved for future use.

Return parameters:

None.

Event(s) generated (unless masked away):

When the Controller receives the HCI_Enhanced_Flush command, the Controller shall send the HCI_Command_Status event to the Host. In addition, when all the packets identified by Packet_Type have been flushed for the specified Handle, the Controller shall send an HCI_Enhanced_Flush_Complete event to the Host. The Controller may send the HCI_Enhanced_Flush_Complete event immediately after flushing all the packets of type Packet_Type for the specified Handle, or it may wait until all packets for the specified Handle, independent of Packet_Type, buffered in the Controller at the time of the receipt of the HCI_Enhanced_Flush command, have been either flushed or transmitted.

Note: No HCI_Command_Complete event will be sent by the Controller to indicate that this command has been completed. Instead, the HCI_Enhanced_Flush_Complete event will indicate that this command has been completed.



7.3.67 Read Logical Link Accept Timeout command

Command	OCF	Command Parameters	Return Parameters
HCI_Read_Logical_Link_Accept_Timeout	0x0061		Status, Logical_Link_Accept_Timeout

Description:

This command reads the value for the Logical_Link_Accept_Timeout configuration parameter.

Command parameters:

None.

Return parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	HCI_Read_Logical_Link_Accept_Timeout command succeeded.
0x01 to 0xFF	HCI_Read_Logical_Link_Accept_Timeout command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Logical_Link_Accept_Timeout:

Size: 2 octets

Value	Parameter Description
N = 0xXXXX	Logical_Link Accept Timeout measured in number of BR/EDR Baseband slots. Interval Length = $N * 0.625 \text{ ms}$ (1 BR/EDR Baseband slot) Range: 0x0001 to 0xB540 Time Range: 0.625 ms to 29 s

Event(s) generated (unless masked away):

When the HCI_Read_Logical_Link_Accept_Timeout command has completed, an HCI_Command_Complete event shall be generated.



7.3.68 Write Logical Link Accept Timeout command

Command	OCF	Command Parameters	Return Parameters
HCI_Write_Logical_Link_Accept_Timeout	0x0062	Logical_Link_Accept_Timeout	Status

Description:

This command writes the value for the Logical_Link_Accept_Timeout configuration parameter. See Logical Link Accept Timeout.

Command parameters:

Logical_Link_Accept_Timeout:

Size: 2 octets

Value	Parameter Description
N = 0xXXXX	Logical_Link Accept Timeout measured in number of BR/EDR Baseband slots. Interval Length = $N * 0.625$ ms (1 BR/EDR Baseband slot) Range: 0x0001 to 0xB540 Time Range: 0.625 ms to 29 s Default: 0x1FA0 Time = 5.06 s Mandatory Range for Controller: 0x00A0 to 0xB540

Return parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	HCI_Write_Logical_Link_Accept_Timeout command succeeded.
0x01 to 0xFF	HCI_Write_Logical_Link_Accept_Timeout command failed. See [Vol 2] Part D, Error Codes or a list of error codes and descriptions.

Event(s) generated (unless masked away):

When the HCI_Write_Logical_Link_Accept_Timeout command has completed, an HCI_Command_Complete event shall be generated.



7.3.69 Set Event Mask Page 2 command

Command	OCF	Command Parameters	Return Parameters
HCI_Set_Event_Mask_Page_2	0x0063	Event_Mask_Page_2	Status

Description:

The HCI_Set_Event_Mask_Page_2 command is used to control which events are generated by the HCI for the Host. The Event_Mask_Page_2 is a logical extension to the Event_Mask parameter of the HCI_Set_Event_Mask command. If the bit in the Event_Mask_Page_2 is set to a one, then the event associated with that bit shall be enabled. The event mask allows the Host to control how much it is interrupted.

The Controller shall ignore those bits which are reserved for future use or represent events which it does not support. If the Host sets any of these bits to 1, the Controller shall act as if they were set to 0.

Command parameters:

Event_Mask_Page_2:

Size: 8 octets

Bit	Parameter Description
0	Physical Link Complete event
1	Channel Selected event
2	Disconnection Physical Link Complete event
3	Physical Link Loss Early Warning event
4	Physical Link Recovery event
5	Logical Link Complete event
6	Disconnection Logical Link Complete event
7	Flow Spec Modify Complete event
8	Number Of Completed Data Blocks event
9	AMP Start Test event
10	AMP Test End event
11	AMP Receiver Report event
12	Short Range Mode Change Complete event
13	AMP Status Change event
14	Triggered Clock Capture event
15	Synchronization Train Complete event
16	Synchronization Train Received event



Bit	Parameter Description
17	Connectionless Slave Broadcast Receive event
18	Connectionless Slave Broadcast Timeout event
19	Truncated Page Complete event
20	Slave Page Response Timeout event
21	Connectionless Slave Broadcast Channel Map Change event
22	Inquiry Response Notification event
23	Authenticated Payload Timeout Expired event
24	SAM Status Change event

The value with all bits set to 0 (which is the default) indicates that no events are specified.

All bits not listed in this table are reserved for future use.

Return parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	HCI_Set_Event_Mask_Page_2 command succeeded.
0x01 to 0xFF	HCI_Set_Event_Mask_Page_2 command failed. See [Vol 2] Part D, Error Codes for error codes and descriptions.

Event(s) generated (unless masked away):

When the HCI_Set_Event_Mask_Page_2 command has completed, an HCI_Command_Complete event shall be generated.



7.3.70 Read Location Data command

Command	OCF	Command Parameters	Return Parameters
HCI_Read_Location_Data	0x0064		Status, Location_Domain_Aware, Location_Domain, Location_Domain Options, Location_Options

Description:

The HCI_Read_Location_Data command provides the ability to read any stored knowledge of environment or regulations or currently in use in the AMP Controller. See [Section 6.29](#) to [Section 6.32](#).

Command parameters:

None.

Return parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	HCI_Read_Location_Data command succeeded.
0x01 to 0xFF	HCI_Read_Location_Data command failed. See [Vol 2] Part D, Error Codes for error codes and descriptions.

See [Section 6.29](#) to [Section 6.32](#) for Location_Domain_Aware, Location_Domain, Location_Domain_Options and Location_Options respectively.

Event(s) generated (unless masked away):

When the HCI_Read_Location_Data command has completed, an HCI_Command_Complete event shall be generated.



7.3.71 Write Location Data command

Command	OCF	Command Parameters	Return Parameters
HCI_Write_Location_Data	0x0065	Location_Domain_Aware, Location_Domain, Location_Domain_Options, Location_Options	Status

Description:

The HCI_Write_Location_Data command writes information about the environment or regulations currently in force, which may affect the operation of the Controller. The Controller may use this information to optimize its operation and to ensure its conformance to currently applicable regulations. The exact way in which this affects AMP operation is AMP type specific.

Command parameters:

See [Section 6.29](#) to [Section 6.32](#) for Location_Domain_Aware, Location_Domain, Location_Domain_Options, Location_Options.

Return parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	HCI_Write_Location_Data command succeeded.
0x01 to 0xFF	HCI_Write_Location_Data command failed. See [Vol 2] Part D, Error Codes for error codes and descriptions.

Event(s) generated (unless masked away):

When the HCI_Write_Location_Data command has completed, an HCI_Command_Complete event shall be generated.



7.3.72 Read Flow Control Mode command

Command	OCF	Command Parameters	Return Parameters
HCI_Read_Flow_Control_Mode	0x0066		Status, Flow_Control_Mode

Description:

This command reads the value for the Flow_Control_Mode configuration parameter. See [Section 6.33](#).

Command parameters:

None.

Return parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	HCI_Read_Flow_Control_Mode command succeeded.
0x01 to 0xFF	HCI_Read_Flow_Control_Mode command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Flow_Control_Mode: See [Section 6.33](#).

Event(s) generated (unless masked away):

When the HCI_Read_Flow_Control_Mode command has completed, an HCI_Command_Complete event shall be generated.



7.3.73 Write Flow Control Mode command

Command	OCF	Command Parameters	Return Parameters
HCI_Write_Flow_Control_Mode	0x0067	Flow_Control_Mode	Status

Description:

This command writes the value for the Flow_Control_Mode configuration parameter. See [Section 6.33](#).

Command parameters:

Flow_Control_Mode:

Size: 1 octet

Value	Parameter Description
0x00	Packet based data flow control mode (default for a BR/EDR Controller)
0x01	Data block based data flow control mode (default for an AMP Controller)
All other values	Reserved for future use

Return parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	HCI_Write_Flow_Control_Mode command succeeded.
0x01 to 0xFF	HCI_Write_Flow_Control_Mode command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Event(s) generated (unless masked away):

When the HCI_Write_Flow_Control_Mode command has completed, an HCI_Command_Complete event shall be generated. If the set fails then the Controller continues using its current mode.



7.3.74 Read Enhanced Transmit Power Level command

Command	OCF	Command Parameters	Return Parameters
HCI_Read_Enhanced_Transmit_Power_Level	0x0068	Connection_Handle, Type	Status, Connection_Handle, Transmit_Power_Level_GFSK, Transmit_Power_Level_DQPSK, Transmit_Power_Level_8DPSK

Description:

This command reads the values for the Enhanced_Transmit_Power_Level parameters for the specified Connection_Handle. The Connection_Handle shall be a Connection_Handle for an ACL connection.

Command parameters:

Connection_Handle:

Size: 2 octets (12 bits meaningful)

Value	Parameter Description
0xXXXX	Connection_Handle Range: 0x0000 to 0x0EFF

Type:

Size: 1 octet

Value	Parameter Description
0x00	Read Current Transmit Power Level
0x01	Read Maximum Transmit Power Level
All other values	Reserved for future use.

Return parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	HCI_Read_Enhanced_Transmit_Power_Level command succeeded.
0x01 to 0xFF	HCI_Read_Enhanced_Transmit_Power_Level command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

**Connection_Handle:****Size: 2 octets (12 bits meaningful)**

Value	Parameter Description
0xXXXX	Connection_Handle Range: 0x0000 to 0x0EFF

Transmit_Power_Level_GFSK:**Size: 1 octet**

Value	Parameter Description
0xXX	Size: 1 octet (signed integer) Range: -100 to 20 Units: dBm

Transmit_Power_Level_DQPSK:**Size: 1 octet**

Value	Parameter Description
0xXX	Size: 1 octet (signed integer) Range: -100 to 20 Units: dBm

Transmit_Power_Level_8DPSK:**Size: 1 octet**

Value	Parameter Description
0xXX	Size: 1 octet (signed integer) Range: -100 to 20 Units: dBm

Event(s) generated (unless masked away):

When the HCI_Read_Enhanced_Transmit_Power_Level command has completed, an HCI_Command_Complete event shall be generated.



7.3.75 Read Best Effort Flush Timeout command

Command	OCF	Command Parameters	Return Parameters
HCI_Read_Best_Effort_Flush_Timeout	0x0069	Logical_Link_Handle	Status, Best_Effort_Flush_Timeout

Description:

This command reads the value of the Best Effort Flush Timeout from the AMP Controller.

Command parameters:

Logical_Link_Handle: *Size: 2 octets (12 bits meaningful)*

Value	Parameter Description
0xFFFF	Logical_Link_Handle to which the HCI_Read_Best_Effort_Flush_Timeout command applies. Range: 0x0000 to 0xFF

Return parameters:

Status: *Size: 1 octet*

Value	Parameter Description
0x00	HCI_Read_Best_Effort_Flush_Timeout command succeeded.
0x01 to 0xFF	HCI_Read_Best_Effort_Flush_Timeout command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Best_Effort_Flush_Timeout: *Size: 4 octets*

Value	Parameter Description
0xFFFFFFFF	0x00000000 to 0xFFFFFFFF: Best Effort Flush Timeout value in microseconds 0xFFFFFFFF: No Best Effort Flush Timeout used (default)

Event(s) generated (unless masked away):

When the HCI_Read_Best_Effort_Flush_Timeout command has completed, an HCI_Command_Complete event shall be generated.



7.3.76 Write Best Effort Flush Timeout command

Command	OCF	Command Parameters	Return Parameters
HCI_Write_Best_Effort_Flush_Timeout	0x006A	Logical_Link_Handle, Best_Effort_Flush_Timeout	Status

Description:

This command writes the value of the Best_Effort_Flush_Timeout into the AMP Controller. The Best_Effort_Flush_Timeout shall be associated with the given logical link, and that Logical_Link_Handle shall specify a Best Effort logical link.

Command parameters:

Logical_Link_Handle: *Size: 2 octets (12 bits meaningful)*

Value	Parameter Description
0xXXXX	Logical_Link_Handle to which the HCI_Write_Best_Effort_Flush_Timeout command applies. Range: 0x0000 to 0xEFF

Best_Effort_Flush_Timeout: *Size: 4 octets*

Value	Parameter Description
0XXXXXXXX	0x00000000 to 0xFFFFFFFF: Best_Effort_Flush_Timeout value in microseconds 0xFFFFFFFF: No Best Effort Flush Timeout used (default)

Return parameters:

Status: *Size: 1 octet*

Value	Parameter Description
0x00	HCI_Write_Best_Effort_Flush_Timeout command succeeded.
0x01 to 0xFF	HCI_Write_Best_Effort_Flush_Timeout command failed. See " [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Event(s) generated (unless masked away):

When the HCI_Write_Best_Effort_Flush_Timeout command has completed, an HCI_Command_Complete event shall be generated.



7.3.77 Short Range Mode command

Command	OCF	Command Parameters	Return Parameters
HCI_Short_Range_Mode	0x006B	Physical_Link_Handle, Short_Range_Mode	

Description:

This command will configure the value of Short_Range_Mode to the AMP Controller and is AMP type specific (see [Volume 5](#)).

Command parameters:

Physical_Link_Handle:

Size: 1 octet

Value	Parameter Description
0xXX	Physical_Link_Handle to which the HCI_Short_Range_Mode command applies.

Short_Range_Mode:

Size: 1 octet

Value	Parameter Description
0x00	Short Range Mode disabled (default)
0x01	Short Range Mode enabled
All other values	Reserved for future use

Return parameters:

None.

Event(s) generated (unless masked away):

An HCI_Command_Status event is sent from the AMP Controller to the Host when the AMP Controller has received the HCI_Short_Range_Mode command. When the HCI_Short_Range_Mode command has completed, an HCI_Short_Range_Mode_Change_Complete event shall be generated.

Note: No HCI_Command_Complete event will be sent by the AMP Controller to indicate that this command has been completed. Instead, the HCI_Short_Range_Mode_Change_Complete event will indicate that this command has been completed.



7.3.78 Read LE Host Support command

Command	OCF	Command Parameters	Return Parameters
HCI_Read_LE_Host_Support	0x006C		Status, LE_Supported_Host, Simultaneous_LE_Host

Description:

The HCI_Read_LE_Host_Support command is used to read the LE Supported (Host) and Simultaneous LE and BR/EDR to Same Device Capable (Host) Link Manager Protocol feature bits. These Link Manager Protocol feature bits are used by the local Host and a remote device (Controller and Host). See [\[Vol 2\] Part C, Section 3.2](#).

The Simultaneous_LE_Host parameter shall be ignored upon receipt.

Command parameters:

None.

Return parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	HCI_Read_LE_Host_Support command succeeded.
0x01 to 0xFF	HCI_Read_LE_Host_Support command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

LE_Supported_Host:

Size: 1 octet

Value	Parameter Description
0xXX	LE_Supported_Host parameter, see Section 6.34 .

Simultaneous_LE_Host:

Size: 1 octet

Value	Parameter Description
0xXX	Simultaneous_LE_Host parameter, see Section 6.35 .

Event(s) generated (unless masked away):

When the HCI_Read_LE_Host_Support command has completed, an HCI_Command_Complete event shall be generated.



7.3.79 Write LE Host Support command

Command	OCF	Command Parameters	Return Parameters
HCI_Write_LE_Host_Support	0x006D	LE_Supported_Host, Simultaneous_LE_Host	Status

Description:

The HCI_Write_LE_Host_Support command is used to set the LE Supported (Host) and Simultaneous LE and BR/EDR to Same Device Capable (Host) Link Manager Protocol feature bits. These Link Manager Protocol feature bits are used by a remote Host. See [\[Vol 2\] Part C, Section 3.2](#).

The default value for these feature bits shall be disabled. When LE_Supported_Host is set to enabled the bit in LMP features mask indicating support for LE Support (Host) shall be set. The Simultaneous_LE_Host parameter shall be set to disabled.

Command parameters:

LE_Supported_Host:

Size: 1 octet

Value	Parameter Description
0xXX	LE_Supported_Host parameter. See Section 6.34

Simultaneous_LE_Host:

Size: 1 octet

Value	Parameter Description
0xXX	Simultaneous_LE_Host parameter. See Section 6.35

Return parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	HCI_Write_LE_Host_Support command succeeded.
0x01 to 0xFF	HCI_Write_LE_Host_Support command failed. See " [Vol 2] Part D, Error Codes " for a list of error codes and descriptions.

Event(s) generated (unless masked away):

When the HCI_Write_LE_Host_Support command has completed, an HCI_Command_Complete event shall be generated.



7.3.80 Set MWS Channel Parameters command

Command	OCF	Command Parameters	Return Parameters
HCI_Set_MWS_Channel_Parameters	0x006E	MWS_Channel_Enable, MWS_RX_Center_Frequency, MWS_TX_Center_Frequency, MWS_RX_Channel_Bandwidth, MWS_TX_Channel_Bandwidth, MWS_Channel_Type	Status

Description:

The HCI_Set_MWS_Channel_Parameters command is used to inform the Controller of the MWS channel parameters.

The MWS_Channel_Enable parameter is used to enable or disable the MWS channel. If it is set to 0x00, the remaining parameters shall be ignored.

The MWS_RX_Center_Frequency and MWS_TX_Center_Frequency parameters are used to indicate the center frequency of the MWS device's uplink (TX) and downlink (RX) channels. The uplink and downlink channel centers may be the same value or different values.

The MWS_RX_Channel_Bandwidth and MWS_TX_Channel_Bandwidth parameters are used to indicate the bandwidth, in kHz, of the MWS device's uplink and downlink channels.

The MWS_Channel_Type parameter describes the type of channel. The types are defined in [Assigned Numbers](#).

Command parameters:

MWS_Channel_Enable:

Size: 1 octet

Value	Parameter Description
0x00	MWS channel is disabled.
0x01	MWS channel is enabled.
All other values	Reserved for future use.

MWS_RX_Center_Frequency:

Size: 2 octets

Value	Parameter Description
0xFFFF	MWS RX center frequency in MHz.

**MWS_TX_Center_Frequency:****Size: 2 octets**

Value	Parameter Description
0xXXXX	MWS TX center frequency in MHz

MWS_RX_Channel_Bandwidth:**Size: 2 octets**

Value	Parameter Description
0xXXXX	MWS RX channel bandwidth in kHz.

MWS_TX_Channel_Bandwidth:**Size: 2 octets**

Value	Parameter Description
0xXXXX	MWS TX channel bandwidth in kHz.

MWS_Channel_Type:**Size: 1 octet**

Value	Parameter Description
0xXX	See Assigned Numbers .

Return parameters:**Status:****Size: 1 octet**

Value	Parameter Description
0x00	HCI_Set_MWS_Channel_Parameters command succeeded.
0x01 to 0xFF	HCI_Set_MWS_Channel_Parameters command failed. See [Vol 2] Part D, Error Codes , for error codes and descriptions

Event(s) generated (unless masked away):

When the HCI_Set_MWS_Channel_Parameters command has completed, an HCI_Command_Complete event shall be generated.



7.3.81 Set External Frame Configuration command

Command	OCF	Command Parameters	Return Parameters
HCI_Set_External_Frame_Configuration	0x006F	Ext_Frame_Duration, Ext_Frame_Sync_Assert_Offset, Ext_Frame_Sync_Assert_Jitter, Ext_Num_Periods, Period_Duration[i], Period_Type[i]	Status

Description:

The HCI_Set_External_Frame_Configuration command allows the Host to specify a frame configuration for an external collocated MWS system. This frame configuration persists until overwritten with a subsequent Set_External_Frame_Configuration or until the Primary Controller is reset.

This command may be supported by a device that declares support for the Coarse Clock Adjustment feature. It may be used to allow the Controller to align the piconet clock with an external frame structure.

When the external frame structure is a multiple of 1.25 ms, it can be aligned in a stable manner with the piconet clock.

The start of the external frame structure is defined as an offset from an external frame synchronization signal. This offset is defined by the Ext_Frame_Sync_Assert_Offset parameter. The offset is represented as the time (in microseconds) from the start of the next MWS frame to the FRAME_SYNC signal.

An external frame consists of downlink periods, uplink periods and guard periods. Downlink means the collocated MWS system is receiving, thus may be interfered with by Bluetooth transmissions. Uplink means the collocated MWS system is transmitting, thus may cause interference to Bluetooth receptions. A guard period may be used by the MWS system to compensate for propagation delays; in this case it should be regarded as split equally between downlink and uplink durations.

The number of specified periods is given by Ext_Num_Periods.

The duration in microseconds of each period is defined by the Period_Duration[i] parameters.

The Period_Type[i] parameter indicates if the specified period is an uplink, downlink, bi-directional or guard period.



The sum of all `Period_Duration[i]` parameters shall be equal to the `Ext_Frame_Duration` parameter.

Upon reception of an `HCI_Set_External_Frame_Configuration` command and a `FRAME_SYNC` signal from the MWS Coexistence Logical Interface, the Controller may compute the type 0 submap for local SAM slot maps. The Controller may then initiate the SAM set type 0 and SAM define map LMP sequences with the remote device.

Command parameters:

Ext_Frame_Duration: *Size: 2 octets*

Value	Parameter Description
0xFFFF	External frame duration in microseconds (unsigned integer)

Ext_Frame_Sync_Assert_Offset: *Size: 2 octets*

Value	Parameter Description
0xFFFF	External frame offset in microseconds (signed integer)

Ext_Frame_Sync_Assert_Jitter: *Size: 2 octets*

Value	Parameter Description
0xFFFF	External frame sync jitter in microseconds (unsigned integer)

Ext_Num_Periods: *Size: 1 octet*

Value	Parameter Description
0xFF	Number of specified periods in an external frame. Valid range: 1 to 32 (unsigned integer)

Period_Duration[i]: *Size: Ext_Num_Periods * 2 octets*

Value	Parameter Description
0xFFFF	Duration of the [i] period in microseconds (unsigned integer)

Period_Type[i]: *Size: Ext_Num_Periods * 1 octet*

Value	Parameter Description
0x00	Downlink
0x01	Uplink
0x02	Bi-Directional
0x03	Guard Period
All other values	Reserved for future use

**Return parameters:***Status:**Size: 1 octet*

Value	Parameter Description
0x00	HCI_Set_External_Frame_Configuration command succeeded.
0x01 to 0xFF	HCI_Set_External_Frame_Configuration command failed. See [Vol 2] Part D, Error Codes , for error codes and descriptions

Event(s) generated (unless masked away):

When the HCI_Set_External_Frame_Configuration command has completed, an HCI_Command_Complete event shall be generated.



7.3.82 Set MWS Signaling command

Command	OCF	Command Parameters	Return Parameters
HCI_Set_MWS_Signaling	0x0070	MWS_RX_Assert_Offset, MWS_RX_Assert_Jitter, MWS_RX_Deassert_Offset, MWS_RX_Deassert_Jitter, MWS_TX_Assert_Offset, MWS_TX_Assert_Jitter, MWS_TX_Deassert_Offset, MWS_TX_Deassert_Jitter, MWS_Pattern_Assert_Offset, MWS_Pattern_Assert_Jitter, MWS_Inactivity_Duration_Assert_Offset, MWS_Inactivity_Duration_Assert_Jitter, MWS_Scan_Frequency_Assert_Offset, MWS_Scan_Frequency_Assert_Jitter, MWS_Priority_Assert_Offset_Request	Status, Bluetooth_Rx_Priority_Assert_Offset, Bluetooth_Rx_Priority_Assert_Jitter, Bluetooth_Rx_Priority_Deassert_Offset, Bluetooth_Rx_Priority_Deassert_Jitter, 802_Rx_Priority_Assert_Offset, 802_Rx_Priority_Assert_Jitter, 802_Rx_Priority_Deassert_Offset, 802_Rx_Priority_Deassert_Jitter, Bluetooth_Tx_On_Assert_Offset, Bluetooth_Tx_On_Assert_Jitter, Bluetooth_Tx_On_Deassert_Offset, Bluetooth_Tx_On_Deassert_Jitter, 802_Tx_On_Assert_Offset, 802_Tx_On_Assert_Jitter, 802_Tx_On_Deassert_Offset, 802_Tx_On_Deassert_Jitter

Description:

The HCI_Set_MWS_Signaling command is used to inform the Bluetooth Controller of the MWS signaling interface logical layer parameters.

All signals are defined in [\[Vol 7\] Part A](#).

Command parameters:

MWS_RX_Assert_Offset:

Size: 2 octets

Value	Parameter Description
0xXXXX	MWS_RX signal assert offset in microseconds (signed integer).

**MWS_RX_Assert_Jitter:****Size: 2 octets**

Value	Parameter Description
0xXXXX	MWS_RX signal assert jitter in microseconds (unsigned integer).

MWS_RX_Deassert_Offset:**Size: 2 octets**

Value	Parameter Description
0xXXXX	MWS_RX signal de-assert offset in microseconds (signed integer).

MWS_RX_Deassert_Jitter:**Size: 2 octets**

Value	Parameter Description
0xXXXX	MWS_RX signal de-assert jitter in microseconds (unsigned integer).

MWS_TX_Assert_Offset:**Size: 2 octets**

Value	Parameter Description
0xXXXX	MWS_TX signal assert offset in microseconds (signed integer).

MWS_TX_Assert_Jitter:**Size: 2 octets**

Value	Parameter Description
0xXXXX	MWS_TX signal assert jitter in microseconds (unsigned integer).

MWS_TX_Deassert_Offset:**Size: 2 octets**

Value	Parameter Description
0xXXXX	MWS_TX signal de-assert offset in microseconds (signed integer).

MWS_TX_Deassert_Jitter:**Size: 2 octets**

Value	Parameter Description
0xXXXX	MWS_TX signal de-assert jitter in microseconds (unsigned integer).

MWS_Pattern_Assert_Offset:**Size: 2 octets**

Value	Parameter Description
0xXXXX	MWS_PATTERN signal assert offset in microseconds (signed integer).

MWS_Pattern_Assert_Jitter:**Size: 2 octets**

Value	Parameter Description
0xXXXX	MWS_PATTERN signal assert jitter in microseconds (unsigned integer).

**MWS_Inactivity_Duration_Assert_Offset:****Size: 2 octets**

Value	Parameter Description
0xFFFF	MWS_INACTIVITY_DURATION signal assert offset in microseconds (signed integer).

MWS_Inactivity_Duration_Assert_Jitter:**Size: 2 octets**

Value	Parameter Description
0xFFFF	MWS_INACTIVITY_DURATION signal assert jitter in microseconds (unsigned integer).

MWS_Scan_Frequency_Assert_Offset:**Size: 2 octets**

Value	Parameter Description
0xFFFF	MWS_SCAN_FREQUENCY signal assert offset in microseconds (signed integer).

MWS_Scan_Frequency_Assert_Jitter:**Size: 2 octets**

Value	Parameter Description
0xFFFF	MWS_SCAN_FREQUENCY signal assert jitter in microseconds (unsigned integer).

MWS_Priority_Assert_Offset_Request:**Size: 2 octets**

Value	Parameter Description
0xFFFF	Minimum advance notification from the beginning of an MWS Uplink period in microseconds (unsigned integer) before which the BLUETOOTH_RX_PRI or 802_RX_PRI signal shall be asserted to be recognized by the MWS.

Return parameters:**Status:****Size: 1 octet**

Value	Parameter Description
0x00	HCI_Set_MWS_Signaling command succeeded.
0x01 to 0xFF	HCI_Set_MWS_Signaling command failed. See [Vol 2] Part D, Error Codes , for error codes and descriptions

Bluetooth_RX_Priority_Assert_Offset:**Size: 2 octets**

Value	Parameter Description
0xFFFF	BLUETOOTH_RX_PRI signal assert offset in microseconds (signed integer).

*Bluetooth_RX_Priority_Assert_Jitter:**Size: 2 octets*

Value	Parameter Description
0xXXXX	BLUETOOTH_RX_PRI signal assert jitter in microseconds (unsigned integer).

*Bluetooth_RX_Priority_Deassert_Offset:**Size: 2 octets*

Value	Parameter Description
0xXXXX	BLUETOOTH_RX_PRI signal de-assert offset in microseconds (signed integer).

*802_RX_Bluetooth_RX_Priority_Deassert_Jitter:**Size: 2 octets*

Value	Parameter Description
0xXXXX	BLUETOOTH_RX_PRI signal de-assert jitter in microseconds (unsigned integer).

*Priority_Assert_Offset:**Size: 2 octets*

Value	Parameter Description
0xXXXX	802_RX_PRI signal assert offset in microseconds (signed integer).

*802_RX_Priority_Assert_Jitter:**Size: 2 octets*

Value	Parameter Description
0xXXXX	802_RX_PRI signal assert jitter in microseconds (unsigned integer).

*802_RX_Priority_Deassert_Offset:**Size: 2 octets*

Value	Parameter Description
0xXXXX	802_RX_PRI signal de-assert offset in microseconds (signed integer).

*802_RX_Priority_Deassert_Jitter:**Size: 2 octets*

Value	Parameter Description
0xXXXX	802_RX_PRI signal de-assert jitter in microseconds (unsigned integer).

*Bluetooth_TX_On_Assert_Offset:**Size: 2 octets*

Value	Parameter Description
0xXXXX	BLUETOOTH_TX_ON signal assert offset in microseconds (signed integer).

*Bluetooth_TX_On_Assert_Jitter:**Size: 2 octets*

Value	Parameter Description
0xXXXX	BLUETOOTH_TX_ON signal assert jitter in microseconds (unsigned integer).

*Bluetooth_TX_On_Deassert_Offset:**Size: 2 octets*

Value	Parameter Description
0xXXXX	BLUETOOTH_TX_ON signal de-assert offset in microseconds (signed integer).

*Bluetooth_TX_On_Deassert_Jitter:**Size: 2 octets*

Value	Parameter Description
0xXXXX	BLUETOOTH_TX_ON signal de-assert jitter in microseconds (unsigned integer).

*802_TX_On_Assert_Offset:**Size: 2 octets*

Value	Parameter Description
0xXXXX	802_TX_ON signal assert offset in microseconds (signed integer).

*802_TX_On_Assert_Jitter:**Size: 2 octets*

Value	Parameter Description
0xXXXX	802_TX_ON signal assert jitter in microseconds (unsigned integer).

*802_TX_On_Deassert_Offset:**Size: 2 octets*

Value	Parameter Description
0xXXXX	802_TX_ON signal de-assert offset in microseconds (signed integer).

*802_TX_On_Deassert_Jitter:**Size: 2 octets*

Value	Parameter Description
0xXXXX	802_TX_ON signal de-assert jitter in microseconds (unsigned integer).

Event(s) generated (unless masked away):

When the HCI_Set_MWS_Signaling command has completed, an HCI_Command_Complete event shall be generated.



7.3.83 Set MWS Transport Layer command

Command	OCF	Command Parameters	Return Parameters
HCI_Set_MWS_Transport_Layer	0x0071	Transport_Layer, To_MWS_Baud_Rate, From_MWS_Baud_Rate	Status

Description:

The HCI_Set_MWS_Transport_Layer command configures the transport layer between the Bluetooth Controller and MWS device.

Command parameters:

Transport_Layer: *Size: 1 octet*

Value	Parameter Description
0xXX	See Assigned Numbers .

To_MWS_Baud_Rate: *Size: 4 octets*

Value	Parameter Description
0XXXXXXXXX	Baud rate in the Bluetooth to MWS direction in Baud.

From_MWS_Baud_Rate: *Size: 4 octets*

Value	Parameter Description
0XXXXXXXXX	Baud rate in the MWS to Bluetooth direction in Baud.

Return parameters:

Status: *Size: 1 octet*

Value	Parameter Description
0x00	HCI_Set_MWS_Transport_Layer command succeeded.
0x01 to 0xFF	HCI_Set_MWS_Transport_Layer command failed. See [Vol 2] Part D, Error Codes , for error codes and descriptions

Event(s) generated (unless masked away):

When the HCI_Set_MWS_Transport_Layer command has completed, an HCI_Command_Complete event shall be generated.



7.3.84 Set MWS Scan Frequency Table command

Command	OCF	Command Parameters	Return Parameters
HCI_Set_MWS_Scan_Frequency_Table	0x0072	Num_Scan_Frequencies, Scan_Frequency_Low[i], Scan_Frequency_High[i]	Status

Description:

The HCI_Set_MWS_Scan_Frequency_Table command configures the MWS scan frequency table in the Controller.

The Num_Scan_Frequencies parameter indicates the number of MWS scan frequencies to be set. A Controller shall support at least 8 table entries.

The Scan_Frequency_Low[i] and Scan_Frequency_High[i] parameters indicate the lower and upper edges for each scan frequency.

Command parameters:

Num_Scan_Frequencies:

Size: 1 octet

Value	Parameter Description
N	Number of MWS scan frequencies to be set in the table.

Scan_Frequency_Low[i]:

*Size: Num_Scan_Frequencies * 2 octets*

Value	Parameter Description
0xXXXX	Lower edge of the MWS scan frequency in MHz (unsigned integer).

Scan_Frequency_High[i]:

*Size: Num_Scan_Frequencies * 2 octets*

Value	Parameter Description
0xXXXX	Upper edge of the MWS scan frequency in MHz (unsigned integer).

Return parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	HCI_Set_MWS_Scan_Frequency_Table command succeeded.
0x01 to 0xFF	HCI_Set_MWS_Scan_Frequency_Table command failed. See [Vol 2] Part D, Error Codes , for error codes and descriptions

**Event(s) generated (unless masked away):**

When the HCI_Set_MWS_Scan_Frequency_Table command has completed, an HCI_Command_Complete event shall be generated.



7.3.85 Set MWS_PATTERN Configuration command

Command	OCF	Command Parameters	Return Parameters
HCI_Set_MWS_PATTERN_Configuration	0x0073	MWS_PATTERN_Index, MWS_PATTERN_NumIntervals, MWS_PATTERN_IntervalDuration[i], MWS_PATTERN_IntervalType[i]	Status

Description:

The HCI_Set_MWS_PATTERN_Configuration command is used by the Host to specify, in conjunction with the HCI_Set_External_Frame_Configuration command, local MWS_PATTERN parameters for an external collocated system.

An MWS_PATTERN configuration shall persist until overwritten by a subsequent Set_MWS_PATTERN_Configuration. All MWS_PATTERN configurations are deleted when an HCI_Set_External_Frame_Configuration command is received or when the Primary Controller is reset.

The sum of the MWS_PATTERN_IntervalDuration parameters shall be an integral multiple of the length of a frame as defined by the most recent HCI_Set_External_Frame_Configuration command.

If any interval with type 4 either has an MWS_PATTERN_IntervalDuration greater than the length of a frame or the sum of the MWS_PATTERN_IntervalDuration parameters for the previous intervals is not a multiple of the length of the frame, the Controller shall return the error code *Invalid HCI Command Parameters* (0x12).

Upon reception of an HCI_Set_MWS_PATTERN_Configuration command, the Controller may compute the local SAM slot map with SAM_Index equal to MWS_PATTERN_Index. If the SAM slot map does not exist, it should be created; if the SAM slot map already exists, its parameters should be replaced. The Controller may then initiate the SAM define map LMP sequence with the remote device.

Upon reception of an MWS_PATTERN signal, with a value other than 3, from the MWS Coexistence Logical Interface (see [\[Vol 7\] Part A](#)), the Controller should check the MWS_PATTERN value against the SAM_Index of those SAM slot maps that have been configured by previous HCI_Set_MWS_PATTERN_Configuration commands. It should then take the following course of action:

1. If MWS_PATTERN does not match any configured SAM slot map, it should take no further action.



2. If MWS_PATTERN matches an available SAM slot map that is already active or is being activated, it should take no further action (i.e. let the current or pending active SAM slot map continue).
3. If MWS_PATTERN matches an available SAM slot map that is neither active nor is being activated, then:
 - a) If the SAM slot map has been activated previously using the LMP_SAM_set_type0 (if relevant) and LMP_SAM_define_map LMP sequences, the Controller should start the SAM switch LMP sequence to activate the matched SAM slot map;
 - b) Otherwise the Controller should start or complete the SAM set type 0 (if relevant), SAM define map, and SAM switch LMP sequences to activate the matched SAM slot map.

Command parameters:

MWS_PATTERN_Index:

Size: 1 octet

Value	Parameter Description
0xXX	Index of the MWS_PATTERN instance to be configured. Range: 0 to 2.

MWS_PATTERN_NumIntervals:

Size: 1 octet

Value	Parameter Description
0xXX	The number of intervals in the following arrays.

MWS_PATTERN_IntervalDuration[i]:

*Size: MWS_PATTERN_NumInterval * 2 octets*

Value	Parameter Description
0xFFFF	The duration of this Bluetooth activity interval in microseconds.

MWS_PATTERN_IntervalType[i]:

*Size: MWS_PATTERN_NumInterval * 1 octet*

Value	Parameter Description
0x00	Neither transmission nor reception is allowed in this interval.
0x01	Transmission is allowed in this interval.
0x02	Reception is allowed in this interval.
0x03	Both transmission and reception are allowed in this interval.
0x04	Interval for the MWS frame as defined by the HCI_Set_External_Frame_Configuration command.
All other values	Reserved for future use

**Return parameters:***Status:**Size: 1 octet*

Value	Parameter Description
0x00	HCI_Set_MWS_PATTERN_Configuration command succeeded
0x01 to 0xFF	HCI_Set_MWS_PATTERN_Configuration command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Event(s) generated (unless masked away):

When the HCI_Set_MWS_PATTERN_Configuration command has completed, an HCI_Command_Complete event shall be generated.



7.3.86 Set Reserved LT_ADDR command

Command	OCF	Command Parameters	Return Parameters
HCI_Set_Reserved_LT_ADDR	0x0074	LT_ADDR	Status, LT_ADDR

Description:

The HCI_Set_Reserved_LT_ADDR command allows the Host to request that the BR/EDR Controller reserve a specific LT_ADDR for Connectionless Slave Broadcast.

If the LT_ADDR indicated in the LT_ADDR parameter is already in use by the BR/EDR Controller, it shall return the *Connection Already Exists* (0x0B) error code. If the LT_ADDR indicated in the LT_ADDR parameter is out of range, the Controller shall return the *Invalid HCI Command Parameters* (0x12) error code. If the command succeeds, then the reserved LT_ADDR shall be used when issuing subsequent HCI_Set_Connectionless_Slave_Broadcast_Data and HCI_Set_Connectionless_Slave_Broadcast commands.

To ensure that the reserved LT_ADDR is not already allocated, it is recommended that this command be issued at some point after HCI_Reset is issued but before page scanning is enabled or paging is initiated.

Command parameters:

LT_ADDR:

Size: 1 octet

Value	Parameter Description
0x01 to 0x07	LT_ADDR to reserve for Connectionless Slave Broadcast
All other values	Reserved for future use

Return parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	HCI_Set_Reserved_LT_ADDR command succeeded.
0x01 to 0xFF	HCI_Set_Reserved_LT_ADDR command failed. See [Vol 2] Part D, Error Codes , for error codes and descriptions.

**LT_ADDR:****Size: 1 octet**

Value	Parameter Description
0x01 to 0x07	LT_ADDR reserved for Connectionless Slave Broadcast. This parameter shall have the same value as the Command Parameter LT_ADDR.
All other values	Reserved for future use

Event(s) generated (unless masked away):

When the HCI_Set_Reserved_LT_ADDR command has completed, an HCI_Command_Complete event shall be generated.



7.3.87 Delete Reserved LT_ADDR command

Command	OCF	Command Parameters	Return Parameters
HCI_Delete_Reserved_LT_ADDR	0x0075	LT_ADDR	Status, LT_ADDR

Description:

The HCI_Delete_Reserved_LT_ADDR command requests that the BR/EDR Controller cancel the reservation for a specific LT_ADDR reserved for the purposes of Connectionless Slave Broadcast.

If the LT_ADDR indicated in the LT_ADDR parameter is not reserved by the BR/EDR Controller, it shall return the *Unknown Connection Identifier* (0x02) error code.

If Connectionless Slave Broadcast mode is still active, then the Controller shall return the *Command Disallowed* (0x0C) error code.

Command parameters:

LT_ADDR:

Size: 1 octet

Value	Parameter Description
0x01 to 0x07	LT_ADDR currently reserved for Connectionless Slave Broadcast and for which reservation is to be cancelled
All other values	Reserved for future use

Return parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	HCI_Delete_Reserved_LT_ADDR command succeeded.
0x01 to 0xFF	HCI_Delete_Reserved_LT_ADDR command failed. See [Vol 2] Part D, Error Codes , for error codes and descriptions.

LT_ADDR:

Size: 1 octet

Value	Parameter Description
0x01 to 0x07	LT_ADDR whose reservation the Host has requested to cancel.
All other values	Reserved for future use

Event(s) generated (unless masked away):

When the HCI_Delete_Reserved_LT_ADDR command has completed, an HCI_Command_Complete event shall be generated.



7.3.88 Set Connectionless Slave Broadcast Data command

Command	OCF	Command Parameters	Return Parameters
HCI_Set_Connectionless_Slave_Broadcast_Data	0x0076	LT_ADDR, Fragment, Data_Length, Data	Status, LT_ADDR

Description:

The HCI_Set_Connectionless_Slave_Broadcast_Data command provides the ability for the Host to set Connectionless Slave Broadcast data in the BR/EDR Controller. This command may be issued at any time after an LT_ADDR has been reserved regardless of whether Connectionless Slave Broadcast mode has been enabled or disabled by the Enable parameter in the HCI_Set_Connectionless_Slave_Broadcast command. If the command is issued without the LT_ADDR reserved, the *Unknown Connection Identifier* (0x02) error code shall be returned.

If Connectionless Slave Broadcast mode is disabled, this data shall be kept by the BR/EDR Controller and used once Connectionless Slave Broadcast mode is enabled. If Connectionless Slave Broadcast mode is enabled, and this command is successful, this data will be sent starting with the next Connectionless Slave Broadcast instant.

The Data_Length field may be zero, in which case no data needs to be provided.

The Host may fragment the data using the Fragment field in the command. If the combined length of the fragments exceeds the capacity of the largest allowed packet size specified in the HCI_Set_Connectionless_Slave_Broadcast command, all fragments associated with the data being assembled shall be discarded and the *Invalid HCI Command Parameters* error code (0x12) shall be returned.

Command parameters:

LT_ADDR:

Size: 1 octet

Value	Parameter Description
0x01 to 0x07	LT_ADDR on which to send Connectionless Slave Broadcast data
All other values	Reserved for future use

**Fragment:****Size: 1 octet**

Value	Parameter Description
0x00	Continuation fragment
0x01	Starting fragment
0x02	Ending fragment
0x03	No fragmentation (single fragment)
All other values	Reserved for future use

Data_Length:**Size: 1 octet**

Value	Parameter Description
0xXX	Length of the Data field

Data:**Size: Data_Length octets**

Value	Parameter Description
Variable	Data to send in future Connectionless Slave Broadcast packets. This data will be repeated in future Connectionless Slave Broadcast instants until new data is provided

Return parameters:**Status:****Size: 1 octet**

Value	Parameter Description
0x00	HCI_Set_Connectionless_Slave_Broadcast_Data command succeeded.
0x01 to 0xFF	HCI_Set_Connectionless_Slave_Broadcast_Data command failed. See [Vol 2] Part D, Error Codes , for error codes and descriptions

LT_ADDR:**Size: 1 octet**

Value	Parameter Description
0x01 to 0x07	LT_ADDR on which Connectionless Slave Broadcast data will be sent
All other values	Reserved for future use

Event(s) generated (unless masked away):

When the HCI_Set_Connectionless_Slave_Broadcast_Data command has completed, an HCI_Command_Complete event shall be generated.



7.3.89 Read Synchronization Train Parameters command

Command	OCF	Command Parameters	Return Parameters
HCI_Read_- Synchronization_Train_- Parameters	0x0077		Status, Sync_Train_Interval, synchroniza- tion_trainTO, Service_Data

Description:

The HCI_Read_Synchronization_Train_Parameters command returns the currently configured values for the Synchronization Train functionality in the master's BR/EDR Controller. This command may be issued at any time.

Command parameters:

None.

Return parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	HCI_Read_Synchronization_Train_Parameters command succeeded.
0x01 to 0xFF	HCI_Read_Synchronization_Train_Parameters command failed. See [Vol 2] Part D, Error Codes , for error codes and descriptions.

Sync_Train_Interval:

Size: 2 octets

Value	Parameter Description
0xFFFF	Interval in slots between consecutive Synchronization Train events on the same channel. Range: 0x0020 to 0xFFFE; only even values are valid

synchronization_trainTO:

Size: 4 octets

Value	Parameter Description
0xFFFFFFFF	Duration in slots to continue sending the synchronization train Range: 0x00000002 to 0x07FFFFFFE; only even values are valid

**Service_Data:****Size: 1 octet**

Value	Parameter Description
0xXX	Host provided value included in Synchronization Train packet, octet 27; see [Vol 2] Part B, Table 8.10 .

Event(s) generated (unless masked away):

When the BR/EDR Controller receives the HCI_Read_Synchronization_Train_Parameters command, it shall send an HCI_Command_Complete event to the Host.



7.3.90 Write Synchronization Train Parameters command

Command	OCF	Command Parameters	Return Parameters
HCI_Write_- Synchronization_- Train_Parameters	0x0078	Interval_Min, Interval_Max, synchronization_trainTO, Service_Data	Status, Sync_Train_Interval

Description:

The HCI_Write_Synchronization_Train_Parameters command configures the Synchronization Train functionality in the BR/EDR Controller. This command may be issued at any time.

Note: The AFH_Channel_Map used in the Synchronization Train packets is configured by the HCI_Set_AFH_Channel_Classification command and the local channel classification in the BR/EDR Controller.

Interval_Min and Interval_Max specify the allowed range of Sync_Train_Interval. Refer to [\[Vol 2\] Part B, Section 2.7.2](#) for a detailed description of Sync_Train_Interval. The BR/EDR Controller shall select an interval from this range and return it in Sync_Train_Interval. If the Controller is unable to select a value from this range, it shall return the *Invalid HCI Command Parameters* (0x12) error code.

Once started (via the HCI_Start_Synchronization_Train command) the Synchronization Train will continue until *synchronization_trainTO* slots have passed or Connectionless Slave Broadcast has been disabled.

Command parameters:

Interval_Min:

Size: 2 octets

Value	Parameter Description
0xXXXX	Minimum value allowed for the interval Sync_Train_Interval in slots. Range: 0x0020 to 0xFFFFE; only even values are valid

Interval_Max:

Size: 2 octets

Value	Parameter Description
0xXXXX	Maximum value allowed for the interval Sync_Train_Interval in slots. Range: 0x0020 to 0xFFFFE; only even values are valid

*synchronization_trainTO:**Size: 4 octets*

Value	Parameter Description
0XXXXXXXX	Duration in slots to continue sending the synchronization train Range: 0x00000002 to 0x07FFFFFFE; only even values are valid

*Service_Data:**Size: 1 octet*

Value	Parameter Description
0xXX	Host provided value to be included in octet 27 of the Synchronization Train packet payload body; see [Vol 2] Part B, Table 8.10.

Return parameters:*Status:**Size: 1 octet*

Value	Parameter Description
0x00	HCI_Write_Synchronization_Train_Parameters command succeeded.
0x01 to 0xFF	HCI_Write_Synchronization_Train_Parameters command failed. See [Vol 2] Part D, Error Codes, for error codes and descriptions.

*Sync_Train_Interval:**Size: 2 octets*

Value	Parameter Description
0XXXXX	Interval in slots between consecutive Synchronization Train packets on the same channel. Range: 0x0020 to 0xFFFFE; only even values are valid

Event(s) generated (unless masked away):

When the BR/EDR Controller receives the HCI_Write_Synchronization_Train_Parameters command, it shall send an HCI_Command_Complete event to the Host.



7.3.91 Read Secure Connections Host Support command

Command	OCF	Command Parameters	Return Parameters
HCI_Read_Secure_Connections_Host_Support	0x0079		Status, Secure_Connections_Host_Support

Description:

This command reads the Secure_Connections_Host_Support parameter in the BR/EDR Controller. When Secure Connections Host Support is set to 'enabled' the Controller uses the enhanced reporting mechanisms for the Encryption_Enabled parameter in the HCI_Encryption_Change event (see [Section 7.7.8](#)) and the Key_Type parameter in the HCI_Link_Key_Notification event (see [Section 7.7.24](#)).

Command parameters:

None.

Return parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	HCI_Read_Secure_Connections_Host_Support command succeeded.
0x01 to 0xFF	HCI_Read_Secure_Connections_Host_Support command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Secure_Connections_Host_Support:

Size: 1 octet

Value	Parameter Description
0x00	Secure_Connections_Host_Support is 'disabled'. Host does not support Secure Connections (default)
0x01	Secure_Connections_Host_Support is 'enabled'. Host supports Secure Connections
All other values	Reserved for future use

Event(s) generated (unless masked away):

When the HCI_Read_Secure_Connections_Host_Support command has completed, an HCI_Command_Complete event shall be generated.



7.3.92 Write Secure Connections Host Support command

Command	OCF	Command Parameters	Return Parameters
HCI_Write_Secure_Connections_Host_Support	0x007A	Secure_Connec-tions_Host_Support	Status

Description:

This command writes the Secure_Connections_Host_Support parameter in the BR/EDR Controller. When Secure Connections Host Support is set to 'enabled' the Controller shall use the enhanced reporting mechanisms for the Encryption_Enabled parameter in the HCI_Encryption_Change event (see [Section 7.7.8](#)) and the Key_Type parameter in the HCI_Link_Key_Notification event (see [Section 7.7.24](#)). If the Host issues this command while the Controller is paging, has page scanning enabled, or has an ACL connection, the Controller shall return the error code *Command Disallowed* (0x0C).

The Link Manager Secure Connections (Host Support) feature bit shall be set to the Secure_Connections_Host_Support parameter. The default value for Secure_Connections_Host_Support shall be 'disabled.' When Secure_Connections_Host_Support is set to 'enabled,' the bit in the LMP features mask indicating support for Secure Connections (Host Support) shall be set to enabled in subsequent responses to an LMP_features_req from a remote device.

Command parameters:

Secure_Connections_Host_Support:

Size: 1 octet

Value	Parameter Description
0x00	Secure_Connections_Host_Support is 'disabled'. Host does not support Secure Connections (default)
0x01	Secure_Connections_Host_Support is 'enabled'. Host supports Secure Connections
All other values	Reserved for future use

Return parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	HCI_Write_Secure_Connections_Host_Support command succeeded.
0x01 to 0xFF	HCI_Write_Secure_Connections_Host_Support command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

**Event(s) generated (unless masked away):**

When the HCI_Write_Secure_Connections_Host_Support command has completed, an HCI_Command_Complete event shall be generated.



7.3.93 Read Authenticated Payload Timeout command

Command	OCF	Command Parameters	Return Parameters
HCI_Read_Authenticated_Payload_Timeout	0x007B	Connection_Handle	Status, Connection_Handle, Authenticated_Payload_Timeout

Description:

This command reads the Authenticated_Payload_Timeout (*authenticatedPayloadTO*, see [Vol 2] Part B, Appendix B for BR/EDR connections and [Vol 6] Part B, Section 5.4 for LE connections) parameter in the Primary Controller on the specified Connection_Handle.

When the Connection_Handle identifies a BR/EDR synchronous connection, the Controller shall return the error code *Command Disallowed* (0x0C).

Command parameters:

Connection_Handle: *Size: 2 octets (12 bits meaningful)*

Value	Parameter Description
0xFFFF	Connection_Handle Range: 0x0000 to 0xFFFF

Return parameters:

Status: *Size: 1 octet*

Value	Parameter Description
0x00	HCI_Read_Authenticated_Payload_Timeout command succeeded.
0x01 to 0xFF	HCI_Read_Authenticated_Payload_Timeout command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Connection_Handle: *Size: 2 octets (12 bits meaningful)*

Value	Parameter Description
0xFFFF	Connection_Handle Range: 0x0000 to 0xFFFF

**Authenticated_Payload_Timeout:****Size: 2 octets**

Value	Parameter Description
N = 0xFFFF	Maximum amount of time specified between packets authenticated by a MIC. Default = 0x0BB8 (30 s) Range: 0x0001 to 0xFFFF Time = N * 10 ms Time Range: 10 ms to 655,350 ms

Event(s) generated (unless masked away):

When the HCI_Read_Authenticated_Payload_Timeout command has completed, an HCI_Command_Complete event shall be generated.



7.3.94 Write Authenticated Payload Timeout command

Command	OCF	Command Parameters	Return Parameters
HCI_Write_Authenticated_Payload_Timeout	0x007C	Connection_Handle, Authenticated_Payload_Timeout	Status, Connection_Handle

Description:

This command writes the `Authenticated_Payload_Timeout` (*authenticatedPayloadTO*, see [Vol 2] Part B, Appendix B and [Vol 6] Part B, Section 5.4 for the LE connection) parameter in the Primary Controller for the specified `Connection_Handle`.

When the `Connection_Handle` identifies a BR/EDR ACL connection:

- If the connection is in Sniff mode, the `Authenticated_Payload_Timeout` shall be equal to or greater than T_{sniff} .
- If the connection is in sniff subrating mode, the `Authenticated_Payload_Timeout` shall be equal to or greater than $(\text{max subrate}) \times T_{\text{sniff}}$.
- If the connection is in Hold mode, the `Authenticated_Payload_Timeout` shall be equal to or greater than the *holdTO* value.

When the `Connection_Handle` identifies a BR/EDR synchronous connection, this command shall be rejected with the error code *Command Disallowed* (0x0C).

When the `Connection_Handle` identifies an LE connection, the `Authenticated_Payload_Timeout` shall be equal to or greater than $\text{connInterval} * (1 + \text{connSlaveLatency})$.

When the `Connection_Handle` is associated with an ACL connection, the Link Manager will use this parameter to determine when to use the LMP ping sequence.

When the `Connection_Handle` is associated with an LE connection, the Link Layer will use this parameter to determine when to use the LE ping sequence.

**Command parameters:***Connection_Handle:**Size: 2 octets (12 bits meaningful)*

Value	Parameter Description
0xXXXX	Connection_Handle Range: 0x0000 to 0x0EFF

*Authenticated_Payload_Timeout:**Size: 2 octets*

Value	Parameter Description
N = 0xXXXX	Maximum amount of time specified between packets authenticated by a valid MIC. Range: 0x0001 to 0xFFFF Time = N * 10 ms Time Range: 10 ms to 655,350 ms

Return parameters:*Status:**Size: 1 octet*

Value	Parameter Description
0x00	HCI_Write_Authenticated_Payload_Timeout command succeeded.
0x01 to 0xFF	HCI_Write_Authenticated_Payload_Timeout command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

*Connection_Handle:**Size: 2 octets (12 bits meaningful)*

Value	Parameter Description
0xXXXX	Connection_Handle Range: 0x0000 to 0x0EFF

Event(s) generated (unless masked away):

When the HCI_Write_Authenticated_Payload_Timeout command has completed, an HCI_Command_Complete event shall be generated.



7.3.95 Read Local OOB Extended Data command

Command	OCF	Command Parameters	Return Parameters
HCI_Read_Local_OOB_Extended_Data	0x007D		Status, C_192, R_192, C_256, R_256

Description:

This command obtains Simple Pairing Hash C_192, Simple Pairing Randomizer R_192, Simple Pairing Hash C_256, and Simple Pairing Randomizer R_256, which are intended to be transferred to a remote device using an OOB mechanism. The BR/EDR Controller shall create new values for C_192, R_192, C_256, and R_256 for each invocation of this command. Each random number (R_192 and R_256) shall be created according to [\[Vol 2\] Part H, Section 2](#).

If the Host issues this command before enabling Secure Connections Host Support, the Controller shall return the error code *Command Disallowed* (0x0C).

Note: Each OOB transfer will have unique C_192, R_192, C_256, and R_256 values.

After each OOB transfer this command shall be used to obtain a new set of values for the next OOB transfer.

Note: The Controller keeps information used to generate these values for later use in the simple pairing process. If the BR/EDR Controller is powered off or reset then this information is lost and the values obtained before the power off or reset are invalid.

Command parameters:

None.

Return parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	HCI_Read_Local_OOB_Extended_Data command succeeded.
0x01 to 0xFF	HCI_Read_Local_OOB_Extended_Data command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

**C_192:****Size: 16 octets**

Value	Parameter Description
0XXXXXXXXXXXXX XXXXXXXXXXXXX XXXXXXXXXXXXX	Simple Pairing Hash C derived from the P-192 public key.

R_192:**Size: 16 octets**

Value	Parameter Description
0XXXXXXXXXXXXX XXXXXXXXXXXXX XXXXXXXXXXXXX	Simple Pairing Randomizer associated with the P-192 public key.

C_256:**Size: 16 octets**

Value	Parameter Description
0XXXXXXXXXXXXX XXXXXXXXXXXXX XXXXXXXXXXXXX	Simple Pairing Hash C derived from the P-256 public key.

R_256:**Size: 16 octets**

Value	Parameter Description
0XXXXXXXXXXXXX XXXXXXXXXXXXX XXXXXXXXXXXXX	Simple Pairing Randomizer associated with the P-256 public key.

Event(s) generated (unless masked away):

When the HCI_Read_Local_OOB_Extended_Data command has completed, an HCI_Command_Complete event shall be generated.



7.3.96 Read Extended Page Timeout command

Command	OCF	Command Parameters	Return Parameters
HCI_Read_Extended_ - Page_Timeout	0x007E		Status, Extended_Page_Timeout

Description:

The HCI_Read_Extended_Page_Timeout command will read the value for the Extended_Page_Timeout configuration parameter. See [Section 6.41](#).

Command parameters:

None.

Return parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	HCI_Read_Extended_Page_Timeout command succeeded.
0x01 to 0x0F	HCI_Read_Extended_Page_Timeout command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Extended_Page_Timeout:

Size: 2 octets

Value	Parameter Description
0xXXXX	Extended Page Timeout measured in number of Baseband slots. Interval Length = $N * 0.625$ ms (1 Baseband slot) Range: 0x0000 (default) to 0xFFFF Time Range: 0 to 40.9 s

Event(s) generated (unless masked away):

When the HCI_Read_Extended_Page_Timeout command has completed, an HCI_Command_Complete event shall be generated.



7.3.97 Write Extended Page Timeout command

Command	OCF	Command Parameters	Return Parameters
HCI_Write_Extended_Page_Timeout	0x007F	Extended_Page_Timeout	Status

Description:

The HCI_Write_Extended_Page_Timeout command will write the value for the Extended_Page_Timeout configuration parameter.

The Extended_Page_Timeout configuration parameter defines the maximum time after the Page_Timeout expires that the local Link Manager may wait for a baseband page response from the remote device at a locally initiated connection attempt. If this time expires and the remote device has not responded to the page at baseband level, the connection attempt will be considered to have failed.

Command parameters:

Extended_Page_Timeout:

Size: 2 octets

Value	Parameter Description
N = 0xXXXX	Extended Page Timeout measured in number of Baseband slots. Interval Length = $N * 0.625$ ms (1 Baseband slot) Range: 0x0000 (default) to 0xFFFF Time Range: 0 to 40.9 s

Return parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	HCI_Write_Extended_Page_Timeout command succeeded.
0x01 to 0x0F	HCI_Write_Extended_Page_Timeout command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Event(s) generated (unless masked away):

When the HCI_Write_Extended_Page_Timeout command has completed, an HCI_Command_Complete event shall be generated.



7.3.98 Read Extended Inquiry Length command

Command	OCF	Command Parameters	Return Parameters
HCI_Read_Extended_Inquiry_Length	0x0080		Status, Extended_Inquiry_Length

Description:

The HCI_Read_Extended_Inquiry_Length command will read the value for the Extended_Inquiry_Length configuration parameter. See [Section 6.42](#).

Command parameters:

None.

Return parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	HCI_Read_Extended_Inquiry_Length command succeeded.
0x01 to 0x0F	HCI_Read_Extended_Inquiry_Length command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Extended_Inquiry_Length:

Size: 2 octets

Value	Parameter Description
0xFFFF	Extended_Inquiry_Length measured in number of Baseband slots. Interval Length = $N * 0.625 \text{ ms}$ (1 Baseband slot) Range: 0x0000 (default) to 0xFFFF Time Range: 0 to 40.9 s

Event(s) generated (unless masked away):

When the HCI_Read_Extended_Inquiry_Length command has completed, an HCI_Command_Complete event shall be generated.



7.3.99 Write Extended Inquiry Length command

Command	OCF	Command Parameters	Return Parameters
HCI_Write_Extended_Inquiry_Length	0x0081	Extended_Inquiry_Length	Status

Description:

The HCI_Write_Extended_Inquiry_Length command will write the value for the Extended_Inquiry_Length configuration parameter.

The Extended_Inquiry_Length configuration parameter defines the maximum time after the Inquiry_Length expires that the local Link Manager may wait for a baseband inquiry response from the remote device at a locally initiated connection attempt. If this time expires and the remote device has not responded to the inquiry at baseband level, the inquiry will be considered to have failed.

Command parameters:

Extended_Inquiry_Length:

Size: 2 octets

Value	Parameter Description
0xXXXX	Extended_Inquiry_Length measured in number of Baseband slots. Interval Length = $N * 0.625 \text{ ms}$ (1 Baseband slot) Range: 0x0000 (default) to 0xFFFF Time Range: 0 to 40.9 s

Return parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	HCI_Write_Extended_Inquiry_Length command succeeded.
0x01 to 0x0F	HCI_Write_Extended_Inquiry_Length command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Event(s) generated (unless masked away):

When the HCI_Write_Extended_Inquiry_Length command has completed, an HCI_Command_Complete event shall be generated.



7.4 INFORMATIONAL PARAMETERS

The Informational Parameters are fixed by the manufacturer of the Bluetooth hardware. These parameters provide information about the BR/EDR Controller and the capabilities of the Link Manager and Baseband in the BR/EDR Controller and PAL in the AMP Controller. The Host device cannot modify any of these parameters.

For Informational Parameters commands, the OGF is defined as 0x04.

7.4.1 Read Local Version Information command

Command	OCF	Command Parameters	Return Parameters
HCI_Read_Local_Version_Information	0x0001		Status, HCI Version, HCI Revision, LMP Version, Manufacturer_Name, LMP Subversion

Description:

This command reads the values for the version information for the local Controller.

The HCI Version information defines the version information of the HCI layer. The LMP/PAL Version information defines the version of the LMP or PAL. The Manufacturer_Name information indicates the manufacturer of the local device.

The HCI Revision and LMP/PAL Subversion are implementation dependent.

Command parameters:

None.

Return parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	HCI_Read_Local_Version_Information command succeeded.
0x01 to 0xFF	HCI_Read_Local_Version_Information command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

**HCI_Version:****Size: 1 octet**

Value	Parameter Description
See Assigned Numbers	

HCI_Revision:**Size: 2 octets**

Value	Parameter Description
0xXXXX	Revision of the Current HCI in the BR/EDR Controller.

LMP/PAL_Version:**Size: 1 octet**

Value	Parameter Description
0xXX	Version of the Current LMP or PAL in the Controller. See Assigned Numbers

Manufacturer_Name:**Size: 2 octets**

Value	Parameter Description
0xXXXX	Manufacturer Name of the BR/EDR Controller. See Assigned Numbers

LMP/PAL_Subversion:**Size: 2 octets**

Value	Parameter Description
0xXXXX	Subversion of the Current LMP or PAL in the Controller. This value is implementation dependent.

Event(s) generated (unless masked away):

When the HCI_Read_Local_Version_Information command has completed, an HCI_Command_Complete event shall be generated.



7.4.2 Read Local Supported Commands command

Command	OCF	Command Parameters	Return Parameters
HCI_Read_Local_Supported_Commands	0x0002		Status, Supported_Commands

Description:

This command reads the list of HCI commands supported for the local Controller.

This command shall return the Supported_Commands configuration parameter. It is implied that if a command is listed as supported, the feature underlying that command is also supported.

See [Section 6.27](#) for more information.

Command parameters:

None.

Return parameters:

Status:

Size: 1 octet

Value	Parameter Description
0	HCI_Read_Local_Supported_Commands command succeeded
0x01 to 0xff	HCI_Read_Local_Supported_Commands command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Supported Commands:

Size: 64 octets

Value	Parameter Description
	Bit mask for each HCI command. If a bit is 1, the Controller supports the corresponding command and the features required for the command. Unsupported or undefined commands shall be set to 0. See Section 6.27 .

Event(s) generated (unless masked away):

When the HCI_Read_Local_Supported_Commands command has completed, an HCI_Command_Complete event shall be generated.



7.4.3 Read Local Supported Features command

Command	OCF	Command Parameters	Return Parameters
HCI_Read_Local_Supported_Features	0x0003		Status, LMP_Features

Description:

This command requests a list of the supported features for the local BR/EDR Controller. This command will return a list of the LMP features. For details see [\[Vol 2\] Part C, Link Manager Protocol Specification](#).

Command parameters:

None.

Return parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	HCI_Read_Local_Supported_Features command succeeded.
0x01 to 0xFF	HCI_Read_Local_Supported_Features command failed. See [Vol 2] Part D, Error Codes .

LMP_Features:

Size: 8 octets

Value	Parameter Description
0xFFFFFFFF XXXXXXXX	Bit Mask List of LMP features. For details see [Vol 2] Part C, Link Manager Protocol Specification .

Event(s) generated (unless masked away):

When the HCI_Read_Local_Supported_Features command has completed, an HCI_Command_Complete event shall be generated.



7.4.4 Read Local Extended Features command

Command	OCF	Command Parameters	Return Parameters
HCI_Read_Local_Extended_Features	0x0004	Page number	Status, Page number, Maximum Page Number, Extended_LMP_Features

Description:

The HCI_Read_Local_Extended_Features command returns the requested page of the extended LMP features.

Command parameters:

Page Number:

Size: 1 octet

Value	Parameter Description
0x00	Requests the normal LMP features as returned by Read_Local_Supported_Features.
0x01 to 0xFF	Return the corresponding page of features.

Return parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	HCI_Read_Local_Extended_Features command succeeded
0x01 to 0xFF	HCI_Read_Local_Extended_Features command failed. See [Vol 2] Part D, Error Codes for list of error codes.

Page Number:

Size: 1 octet

Value	Parameter Description
0x00	The normal LMP features as returned by Read_Local_Supported_Features.
0x01 to 0xFF	The page number of the features returned.

Maximum Page Number:

Size: 1 octet

Value	Parameter Description
0x00 to 0xFF	The highest features page number which contains non-zero bits for the local device.



Extended_LMP_Features:

Size: 8 octets

Value	Parameter Description
0xFFFFFFFFFFFFFFFF	Bit map of requested page of LMP features. See LMP specification for details.

Event(s) generated (unless masked away):

When the HCI_Read_Local_Extended_Features command has completed, an HCI_Command_Complete event shall be generated.



7.4.5 Read Buffer Size command

Command	OCF	Command Parameters	Return Parameters
HCI_Read_Buffer_Size	0x0005		Status, ACL_Data_Packet_Length, Synchronous_Data_Packet_Length, Total_Num_ACL_Data_Packets, Total_Num_Synchronous_Data_Packets

Description:

The HCI_Read_Buffer_Size command is used to read the maximum size of the data portion of HCI ACL and Synchronous Data packets sent from the Host to the Controller. The Host will segment the data to be transmitted from the Host to the Controller according to these sizes, so that the HCI Data packets will contain data with up to these sizes. The HCI_Read_Buffer_Size command also returns the total number of HCI ACL and Synchronous Data packets that can be stored in the data buffers of the Controller. The HCI_Read_Buffer_Size command shall be issued by the Host before it sends any data to the Controller.

For a device supporting BR/EDR and LE, if the HCI_LE_Read_Buffer_Size command returned zero for the number of buffers, then buffers returned by Read_Buffer_Size are shared between BR/EDR and LE.

On a Primary Controller that supports LE only, the HCI_Read_Buffer_Size command shall not be supported (the HCI_LE_Read_Buffer_Size command is to be used in this case).

The ACL_Data_Packet_Length return parameter will be used to determine the size of the L2CAP segments contained in ACL Data packets, which are transferred from the Host to the Controller to be broken up into Baseband packets by the Link Manager. The Synchronous_Data_Packet_Length return parameter is used to determine the maximum size of HCI Synchronous Data packets. The Total_Num_ACL_Data_Packets return parameter contains the total number of HCI ACL Data packets that can be stored in the data buffers of the Controller. The Host will determine how the buffers are to be divided between different Connection_Handles. The Total_Num_Synchronous_Data_Packets return parameter gives the same information but for HCI Synchronous Data packets.

Note: The ACL_Data_Packet_Length and Synchronous_Data_Packet_Length return parameters do not include the length of the HCI Data packet header.

**Command parameters:**

None.

Return parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	HCI_Read_Buffer_Size command succeeded.
0x01 to 0xFF	HCI_Read_Buffer_Size command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

ACL_Data_Packet_Length:

Size: 2 octets

Value	Parameter Description
0xFFFF	Maximum length (in octets) of the data portion of each HCI ACL Data packet that the Controller is able to accept.

Synchronous_Data_Packet_Length:

Size: 1 octet

Value	Parameter Description
0xFF	Maximum length (in octets) of the data portion of each HCI Synchronous Data packet that the Controller is able to accept.

Total_Num_ACL_Data_Packets:

Size: 2 octets

Value	Parameter Description
0xFFFF	Total number of HCI ACL Data packets that can be stored in the data buffers of the Controller.

Total_Num_Synchronous_Data_Packets:

Size: 2 octets

Value	Parameter Description
0xFFFF	Total number of HCI Synchronous Data packets that can be stored in the data buffers of the Controller.

Event(s) generated (unless masked away):

When the HCI_Read_Buffer_Size command has completed, an HCI_Command_Complete event shall be generated.



7.4.6 Read BD_ADDR command

Command	OCF	Command Parameters	Return Parameters
HCI_Read_BD_ADDR	0x0009		Status, BD_ADDR

Description:

On a BR/EDR Controller, this command reads the Bluetooth Controller address (BD_ADDR). See the [\[Vol 2\] Part B, Baseband Specification](#) for details of how BD_ADDR is used (see [\[Vol 3\] Part C, Section 3.2.1](#)).

On an LE Controller, this command shall read the Public Device Address as defined in [\[Vol 6\] Part B, Section 1.3](#). If this Controller does not have a Public Device Address, the value 0x000000000000 shall be returned.

On a BR/EDR/LE Controller, the public address shall be the same as the BD_ADDR.

Command parameters:

None.

Return parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	HCI_Read_BD_ADDR command succeeded.
0x01 to 0xFF	HCI_Read_BD_ADDR command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

BD_ADDR:

Size: 6 octets

Value	Parameter Description
0XXXXXXXXXXXXX	BD_ADDR of the Device

Event(s) generated (unless masked away):

When the HCI_Read_BD_ADDR command has completed, an HCI_Command_Complete event shall be generated.



7.4.7 Read Data Block Size command

Command	OCF	Command Parameters	Return Parameters
HCI_Read_Data_Block_Size	0x000A		Status, Max_ACL_Data_Packet_Length, Data_Block_Length, Total_Num_Data_Blocks

Description:

The HCI_Read_Data_Block_Size command is used to read values regarding the maximum permitted data transfers over the Controller and the data buffering available in the Controller.

The Host uses this information when fragmenting data for transmission, and when performing block-based flow control, based on the HCI_Number_Of_Completed_Data_Blocks event. The HCI_Read_Data_Block_Size command shall be issued by the Host before it sends any data to the Controller.

Command parameters:

None.

Return parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	HCI_Read_Data_Block_Size command succeeded.
0x01 to 0xFF	HCI_Read_Data_Block_Size command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Max_ACL_Data_Packet_Length:

Size: 2 octets

Value	Parameter Description
0xFFFF	Maximum length (in octets) of the data portion of an HCI ACL Data packet that the Controller is able to accept for transmission. For AMP Controllers this always equals to Max_PDU_Size.

Data_Block_Length:

Size: 2 octets

Value	Parameter Description
0xFFFF	Maximum length (in octets) of the data portion of each HCI ACL Data packet that the Controller is able to hold in each of its data block buffers.

**Total_Num_Data_Blocks:****Size: 2 octets**

Value	Parameter Description
0xFFFF	Total number of data block buffers available in the Controller for the storage of data packets scheduled for transmission.

Event(s) generated (unless masked away):

When the HCI_Read_Data_Block_Size command has completed, an HCI_Command_Complete event shall be generated.



7.4.8 Read Local Supported Codecs command

Command	OCF	Command Parameters	Return Parameters
HCI_Read_Local_Supported_Codecs	0x000B		Status, Number_of_Supported_Codecs, Supported_Codecs[i], Number_of_Supported_Vendor_Specific_Codecs, Vendor_Specific_Codecs[k]

The order of the return parameters in an HCI event packet is:

```

Status
Number_of_Supported_Codecs
Supported_Codecs[0]
...
Supported_Codecs[n]
Number_of_Supported_Vendor_Specific_Codecs
Vendor_Specific_Codecs[0]
...
Vendor_Specific_Codecs[m]

```

Description:

This command reads a list of the Bluetooth SIG approved codecs supported by the Controller, as well as vendor specific codecs, which are defined by an individual manufacturer.

Command parameters:

None

Return parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	HCI_Read_Local_Supported_Codecs command succeeded.
0x01 to 0xFF	HCI_Read_Local_Supported_Codecs command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

**Number_of_Supported_Codecs:****Size: 1 octet**

Value	Parameter Description
0xXX	Total number of codecs supported

Supported_Codecs[i]:**Size: 1 * Number_of_Supported_Codecs octets**

Value	Parameter Description
0xXX, 0xXX, ...	An array of codec identifiers. See Assigned Numbers for Codec ID

Number_of_Supported_Vendor_Specific_Codecs:**Size: 1 octet**

Value	Parameter Description
0xXX	Total number of vendor specific codecs supported

Vendor_Specific_Codecs[k]:**Size: 4 * Number_of_Supported_Vendor_Specific_Codecs**

Value	Parameter Description
0XXXXXXXXX	Octets 0 and 1: Company ID, see Assigned Numbers for Company Identifier Octets 2 and 3: Vendor defined codec ID

Event(s) generated (unless masked away):

When the HCI_Read_Local_Supported_Codecs command has completed, an HCI_Command_Complete event shall be generated.



7.4.9 Read Local Simple Pairing Options command

Command	OCF	Command Parameters	Return Parameters
HCI_Read_Local_Simple_Pairing_Options	0x000C		Status, Simple_Pairing_Options, Maximum_Encryption_Key_Size

Description:

The HCI_Read_Local_Simple_Pairing_Options command is used to read the simple pairing options and the maximum encryption key size supported. Bit 0 of the Simple_Pairing_Options return parameter shall be set to 1.

Note: If this command is supported, then the Controller must support remote public key validation (see [\[Vol 2\] Part H, Section 7.6](#)).

Command parameters:

None

Return parameters:

Status: *Size: 1 octet*

Value	Parameter Description
0x00	HCI_Read_Local_Simple_Pairing_Options command succeeded.
0x01 to 0xFF	HCI_Read_Local_Simple_Pairing_Options command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Simple_Pairing_Options: *Size: 1 octet*

Bit Number	Parameter Description
0	Remote public key validation is always performed.
All other bits	Reserved for future use.

Maximum_Encryption_Key_Size: *Size: 1 octet*

Value	Parameter Description
0x07 to 0x10	Maximum encryption key size (in octets) supported.
All other values	Reserved for future use.

Event(s) generated (unless masked away):

When the HCI_Read_Local_Simple_Pairing_Options command has completed, an HCI_Command_Complete event shall be generated.



7.5 STATUS PARAMETERS

The Controller modifies all status parameters. These parameters provide information about the current state of the Link Manager and Baseband in the BR/EDR Controller and the PAL in an AMP Controller. The Host device cannot modify any of these parameters other than to reset certain specific parameters.

For the Status Parameters commands, the OGF is defined as 0x05.

7.5.1 Read Failed Contact Counter command

Command	OCF	Command Parameters	Return Parameters
HCI_Read_Failed_Contact_Counter	0x0001	Handle	Status, Handle, Failed_Contact_Counter

Description:

This command reads the value for the Failed_Contact_Counter parameter for a particular connection to another device. The Handle shall be a Handle for an ACL connection. See [Section 6.15](#).

Command parameters:

Handle:

Size: 2 octets (12 bits meaningful)

Value	Parameter Description
0xFFFF	The Handle for the Connection for which the Failed Contact Counter should be read. The Handle is a Connection_Handle for a BR/EDR Controller and a Logical_Link_Handle for an AMP Controller. Range: 0x0000 to 0x0EFF

Return parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	HCI_Read_Failed_Contact_Counter command succeeded.
0x01 to 0xFF	HCI_Read_Failed_Contact_Counter command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

**Handle:****Size: 2 octets (12 bits meaningful)**

Value	Parameter Description
0xFFFF	The Handle for the connection for which the Failed Contact Counter has been read. The Handle is a Connection_Handle for a BR/EDR Controller and a Logical_Link_Handle for an AMP Controller. Range: 0x0000 to 0x0EFF

Failed_Contact_Counter:**Size: 2 octets**

Value	Parameter Description
0xFFFF	Number of consecutive failed contacts for a connection corresponding to the Handle.

Event(s) generated (unless masked away):

When the HCI_Read_Failed_Contact_Counter command has completed, an HCI_Command_Complete event shall be generated.



7.5.2 Reset Failed Contact Counter command

Command	OCF	Command Parameters	Return Parameters
HCI_Reset_Failed_Contact_Counter	0x0002	Handle	Status, Handle

Description:

This command resets the value for the Failed_Contact_Counter parameter for a particular connection to another device. The Handle shall be a Handle for an ACL connection. See [Section 6.15](#).

Command parameters:

Handle: *Size: 2 octets (12 bits meaningful)*

Value	Parameter Description
0xXXXX	The Handle for the connection for which the Failed Contact Counter should be reset. The Handle is a Connection_Handle for a BR/EDR Controller and a Logical Link Handle for an AMP Controller. Range: 0x0000 to 0x0EFF

Return parameters:

Status: *Size: 1 octet*

Value	Parameter Description
0x00	HCI_Reset_Failed_Contact_Counter command succeeded.
0x01 to 0xFF	HCI_Reset_Failed_Contact_Counter command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Handle: *Size: 2 octets (12 bits meaningful)*

Value	Parameter Description
0xXXXX	The Handle for the connection for which the Failed Contact Counter has been reset. The Handle is a Connection_Handle for a BR/EDR Controller and a Logical Link Handle for an AMP Controller. Range: 0x0000 to 0x0EFF

Event(s) generated (unless masked away):

When the HCI_Reset_Failed_Contact_Counter command has completed, an HCI_Command_Complete event shall be generated.



7.5.3 Read Link Quality command

Command	OCF	Command Parameters	Return Parameters
HCI_Read_Link_Quality	0x0003	Handle	Status, Handle, Link_Quality

Description:

This command returns the value for the Link_Quality for the specified Handle. The Handle shall be a Handle for an ACL connection. This command shall return a Link_Quality value from 0 to 255, which represents the quality of the link between two BR/EDR Controllers. The higher the value, the better the link quality is. Each Bluetooth module vendor will determine how to measure the link quality.

The HCI_Read_Link_Quality command is provided by AMPs. The meaning of the link quality metric is AMP type specific and defined in the AMP PALs (see [Volume 5](#)).

Command parameters:

Handle: *Size: 2 octets (12 bits meaningful)*

Value	Parameter Description
0xFFFF	The Handle for the connection for which link quality parameters are to be read. The Handle is a Connection_Handle for a BR/EDR Controller and a Physical_Link_Handle for an AMP Controller. Range: 0x0000 to 0x0EFF

Return parameters:

Status: *Size: 1 octet*

Value	Parameter Description
0x00	HCI_Read_Link_Quality command succeeded.
0x01 to 0xFF	HCI_Read_Link_Quality command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

**Handle:****Size: 2 octets (12 bits meaningful)**

Value	Parameter Description
0xXXXX	The Handle for the connection for which the link quality parameter has been read. The Handle is a Connection_Handle for a BR/EDR Controller and a Physical_Link_Handle for an AMP Controller. Range: 0x0000 to 0x0EFF

Link_Quality:**Size: 1 octet**

Value	Parameter Description
0xXX	The current quality of the Link connection between the local device and the remote device specified by the Handle. Range: 0x00 to 0xFF The higher the value, the better the link quality is.

Event(s) generated (unless masked away):

When the HCI_Read_Link_Quality command has completed, an HCI_Command_Complete event shall be generated.



7.5.4 Read RSSI command

Command	OCF	Command Parameters	Return Parameters
HCI_Read_RSSI	0x0005	Handle	Status, Handle, RSSI

Description:

This command reads the Received Signal Strength Indication (RSSI) value from a Controller.

For a BR/EDR Controller, a Connection_Handle is used as the Handle command parameter and return parameter. The RSSI parameter returns the difference between the measured Received Signal Strength Indication (RSSI) and the limits of the Golden Receive Power Range for a Connection_Handle to another BR/EDR Controller. The Connection_Handle shall be a Connection_Handle for an ACL connection. Any positive RSSI value returned by the Controller indicates how many dB the RSSI is above the upper limit, any negative value indicates how many dB the RSSI is below the lower limit. The value zero indicates that the RSSI is inside the Golden Receive Power Range.

Note: How accurate the dB values will be depends on the Bluetooth hardware. The only requirements for the hardware are that the BR/EDR Controller is able to tell whether the RSSI is inside, above or below the Golden Device Power Range.

The RSSI measurement compares the received signal power with two threshold levels, which define the Golden Receive Power Range. The lower threshold level corresponds to a received power between -56 dBm and 6 dB above the actual sensitivity of the receiver. The upper threshold level is 20 dB above the lower threshold level to an accuracy of ± 6 dB.

For an AMP Controller, a Physical_Link_Handle is used for the Handle command parameter and return parameter. The meaning of the RSSI metric is AMP type specific and defined in the AMP PALs (see [Volume 5](#)).

For an LE transport, a Connection_Handle is used as the Handle command parameter and return parameter. The meaning of the RSSI metric is an absolute receiver signal strength value in dBm to ± 6 dB accuracy. If the RSSI cannot be read, the RSSI metric shall be set to 127.

**Command parameters:****Handle:****Size: 2 octets (12 bits meaningful)**

Value	Parameter Description
0xXXXX	The Handle for the connection for which the RSSI is to be read. The Handle is a Connection_Handle for a BR/EDR Controller and a Physical_Link_Handle for an AMP Controller. Range: 0x0000 to 0x0EFF

Return parameters:**Status:****Size: 1 octet**

Value	Parameter Description
0x00	HCI_Read_RSSI command succeeded.
0x01 to 0xFF	HCI_Read_RSSI command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Handle:**Size: 2 octets (12 bits meaningful)**

Value	Parameter Description
0xXXXX	The Handle for the connection for which the RSSI has been read. The Handle is a Connection_Handle for a BR/EDR Controller and a Physical_Link_Handle for an AMP Controller. Range: 0x0000 to 0x0EFF

RSSI:**Size: 1 octet**

Value	Parameter Description
N = 0xFF	BR/EDR Range: -128 to 127 (signed integer) Units: dB AMP: Range: AMP type specific (signed integer) Units: dBm LE: Range: -127 to 20, 127 (signed integer) Units: dBm

Event(s) generated (unless masked away):

When the HCI_Read_RSSI command has completed, an HCI_Command_Complete event shall be generated.



7.5.5 Read AFH Channel Map command

Command	OCF	Command Parameters	Return Parameters
HCI_Read_AFH_Channel_Map	0x0006	Connection_Handle	Status, Connection_Handle, AFH_Mode, AFH_Channel_Map

Description:

This command returns the values for the AFH_Mode and AFH_Channel_Map for the specified Connection_Handle. The Connection_Handle shall be a Connection_Handle for an ACL connection.

The returned values indicate the state of the hop sequence specified by the most recent LMP_Set_AFH message for the specified Connection_Handle, regardless of whether the master has received the baseband ACK or whether the AFH_Instant has passed.

This command shall be supported by a device that declares support for either the AFH_capable_slave or AFH_capable_master feature.

Command parameters:

Connection_Handle: *Size: 2 octets (12 bits meaningful)*

Value	Parameter Description
0xFFFF	Connection_Handle Range: 0x0000 to 0x0EFF

Return parameters:

Status: *Size: 1 octet*

Value	Parameter Description
0x00	HCI_Read_AFH_Channel_Map command succeeded.
0x01 to 0xFF	HCI_Read_AFH_Channel_Map command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Connection_Handle: *Size: 2 octets (12 bits meaningful)*

Value	Parameter Description
0xFFFF	Connection_Handle Range: 0x0000 to 0x0EFF

**AFH_Mode:****Size: 1 octet**

Value	Parameter Description
0x00	AFH disabled.
0x01	AFH enabled.
All other values	Reserved for future use.

AFH_Channel_Map:**Size: 10 octets (79 bits meaningful)**

Value	Parameter Description
0XXXXXXXXX XXXXXXXXXX XXX	If AFH_Mode is not AFH enabled then the contents of this parameter are reserved for future use. Otherwise: This parameter contains 80 1-bit fields. The n^{th} such field (in the range 0 to 78) contains the value for channel n: 0: channel n is unused 1: channel n is used The most significant bit (bit 79) is reserved for future use

Event(s) generated (unless masked away):

When the HCI_Read_AFH_Channel_Map command has completed, an HCI_Command_Complete event shall be generated.



7.5.6 Read Clock command

Command	OCF	Command Parameters	Return Parameters
HCI_Read_Clock	0x0007	Connection_Handle, Which_Clock	Status, Connection_Handle, Clock, Accuracy

Description:

This command reads the estimate of the value of the Bluetooth Clock from the BR/EDR Controller.

If the Which_Clock value is 0, then the Connection_Handle shall be ignored, the local Bluetooth Clock value shall be returned and the accuracy parameter shall be set to 0.

If the Which_Clock value is 1, then the Connection_Handle shall be a valid ACL Connection_Handle. If the current role of this ACL connection is Master, then the Bluetooth Clock of this device shall be returned. If the current role is Slave, then an estimate of the Bluetooth Clock of the remote master and the accuracy of this value shall be returned.

The accuracy reflects the clock drift that might have occurred since the slave last received a valid transmission from the master.

Note: The Bluetooth Clock has a minimum accuracy of 250ppm, or about 22 seconds drift in one day.

Note: See [\[Vol 2\] Part B, Section 1.1](#) for more information about the Bluetooth Clock.

Command parameters:

Connection_Handle:

Size: 2 octets (12 bits meaningful)

Value	Parameter Description
0xXXXX	Connection_Handle Range: 0x0000 to 0x0EFF

Which_Clock:

Size 1 octet)

Value	Parameter Description
0xXX	0x00 = Local Clock (Connection_Handle does not have to be valid) 0x01 = Piconet Clock (Connection_Handle shall be valid) 0x02 to 0xFF = Reserved for future use

**Return parameters:***Status:**Size: 1 octet*

Value	Parameter Description
0x00	HCI_Read_Clock command succeeded.
0x01 to 0xFF	HCI_Read_Clock command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

*Connection_Handle:**Size: 2 octets (12 bits meaningful)*

Value	Parameter Description
0xFFFF	The Connection_Handle for the connection for which the master clock has been read. If the Which_Clock parameter was 0, then the Connection_Handle is reserved for future use. Range: 0x0000 to 0x0EFF

*Clock:**Size: 4 octets (28 bits meaningful)*

Value	Parameter Description
0xFFFFFFFF	Bluetooth Clock of the device requested.

*Accuracy:**Size: 2 octets*

Value	Parameter Description
0xFFFF	Maximum (absolute) error in the Bluetooth Clock. Value of 0xFFFF means Unknown. $\text{Accuracy} = \pm N * 0.3125 \text{ ms (1 Bluetooth Clock)}$ Range: 0x0000 to 0xFFFE Time Range: 0 to 20479.375 ms

Event(s) generated (unless masked away):

When the HCI_Read_Clock command has completed, an HCI_Command_Complete event shall be generated.



7.5.7 Read Encryption Key Size command

Command	OCF	Command Parameters	Return Parameters
HCI_Read_Encryption_Key_Size	0x0008	Connection_Handle	Status, Connection_Handle, Key_Size

Description:

This command reads the current encryption key size associated with the Connection_Handle. The Connection_Handle shall be a Connection_Handle for an active ACL connection.

Command parameters:

Connection_Handle: *Size: 2 octets (12 bits meaningful)*

Value	Parameter Description
0xFFFF	Connection_Handle Range: 0x0000 to 0x0EFF

Return parameters:

Status: *Size: 1 octet*

Value	Parameter Description
0x00	Read_Encryption_Key_Size succeeded
0x01 to 0xFF	Read_Encryption_Key_Size failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Connection_Handle: *Size: 2 octets (12 bits meaningful)*

Value	Parameter Description
0xFFFF	Connection_Handle Range: 0x0000 to 0x0EFF

Key_Size: *Size: 1 octet*

Value	Parameter Description
0xFF	Encryption key size. See [Vol 2] Part C, Section 5.2 .

**Event(s) generated (unless masked away):**

When the HCI_Read_Encryption_Key_Size command has completed, an HCI_Command_Complete event shall be generated.

If the ACL connection associated with the Connection_Handle is not encrypted, the Controller shall return an HCI_Command_Complete event with the error code *Insufficient Security* (0x2F).



7.5.8 Read Local AMP Info command

Command	OCF	Command Parameters	Return Parameters
HCI_Read_Local_AMP_Info	0x0009		Status, AMP_Status, Total_Bandwidth, Max_Guaranteed_Bandwidth, Min_Latency, Max_PDU_Size, Controller_Type, PAL_Capabilities, Max_AMP_ASSOC_Length, Max_Flush_Timeout, Best_Effort_Flush_Timeout

Description:

This command returns information about the AMP Controller.

Command parameters:

None.

Return parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	HCI_Read_Local_AMP_Info command succeeded
0x01 to 0xFF	HCI_Read_Local_AMP_Info command failed. See [Vol 2] Part D, Error Codes

AMP_Status:

Size: 1 octet

Value	Parameter Description
0x00	The Controller is available but is currently physically powered down. This value shall be used if the AMP Controller is present and can be powered up by the AMP Manager. This value indicates that there may be a cost of time and power to use this AMP Controller (i.e., the time taken and power required to power up the AMP Controller). These costs are AMP type and AMP implementation dependent.



Value	Parameter Description
0x01	This value indicates that the AMP Controller is only used by Bluetooth technology and will not be shared with other non-Bluetooth technologies. This value shall only be used if the AMP Controller is powered up. This value does not indicate how much bandwidth is currently free on the AMP Controller.
0x02	The AMP Controller has no capacity available for Bluetooth operation This value indicates that all of the AMP Controller's bandwidth is currently allocated to servicing a non Bluetooth technology. A device is permitted to create a Physical Link to an AMP Controller that has this status. This value shall only be used if the AMP Controller is powered up.
0x03	The AMP Controller has low capacity available for Bluetooth operation. This value indicates that the majority of the AMP Controller's bandwidth is currently allocated to servicing a non Bluetooth technology. An AMP Controller with capacity in the approximate range of 0% to 30% should indicate this value. This value does not indicate how much of the capacity available for Bluetooth operation is currently available. This value shall only be used if the AMP Controller is powered up.
0x04	The AMP Controller has medium capacity available for Bluetooth operation. An AMP Controller with capacity in the approximate range of 30% to 70% should indicate this value. This value does not indicate how much of the capacity available for Bluetooth operation is currently available. This value shall only be used if the AMP Controller is powered up.
0x05	The AMP Controller has high capacity available for Bluetooth operation. This value indicates that the majority of the AMP Controller's bandwidth is currently allocated to servicing the Bluetooth technology. An AMP Controller with capacity in the approximate range of 70% to 100% should indicate this value. This value does not indicate how much of the capacity available for Bluetooth operation is currently available. This value shall only be used if the AMP Controller is powered up.



Value	Parameter Description
0x06	The AMP Controller has full capacity available for Bluetooth operation. This value indicates that while currently the AMP is only being used by Bluetooth the device allows a different technology to share the radio. This value shall be used by devices that are not capable of determining the current available capacity of an AMP that is shared by a different technology. This value does not indicate how much of the capacity available for Bluetooth operation is currently available. This value shall only be used if the AMP Controller is powered up.
All other values	Reserved for future use

The AMP Controller to generate HCI_AMP_Status_Change event if this parameter changes after initial Read_Local_AMP_Info request. See [\[Vol 3\] Part E, Section 3.4](#).

Total_Bandwidth:

Size: 4 octets

Value	Parameter Description
	An upper bound on the total data rate that can be achieved by the AMP over HCI. It accounts for the total bandwidth provided by the HCI transport. No sustained combination of transmit and receive operations shall exceed this value. This can be used to help in AMP selection and admission control. Expressed in kb/s.

Max_Guaranteed_Bandwidth:

Size: 4 octets

Value	Parameter Description
	An upper bound on the maximum data the AMP can guarantee for a single logical link over HCI. Any request made by an application above this threshold would be rejected. It accounts for any bandwidth limitations of the HCI transport. No sustained combination of transmit and receive operations shall exceed this value. This can be used to help in AMP selection and admission control. Expressed in kb/s.

Min_Latency:

Size: 4 octets

Value	Parameter Description
	A lower bound on the latency in microseconds that the AMP can guarantee for a logical channel. It accounts for the minimum latency of the HCI transport. This can be used to help in AMP selection and admission control.

*Max_PDU_Size:**Size: 2 octets*

Value	Parameter Description
	An upper bound on the size of L2CAP PDU which may be provided for transmission or reception on this AMP. The Host shall not require the AMP to transport L2CAP PDUs larger than this value. Expressed in octets. The Maximum PDU Size parameter described in [Vol 3] Part A, Section 5.4 for any connection over this AMP should not exceed this value.

*Controller_Type:**Size: 1 octet*

Value	Parameter Description
	See Assigned Numbers .

*PAL_Capabilities:**Size: 2 octets*

Value	Parameter Description
0xXXXX	Bit 0: The PAL supports the Guaranteed service type. All other bits: Reserved for future use. (See [Vol 3] Part A, Section 5.6.)

*Max_AMP_ASSOC_Length:**Size: 2 octets*

Value	Parameter Description
0xXXXX	AMP_ASSOC maximum length for this local AMP Controller. For use with HCI_Read and HCI_Write AMP_ASSOC commands [Section 7.5.8 and Section 7.5.9]. This value is sent to the remote AMP Manager in the AMP Get Info Response (see "[Vol 3] Part E, Section 3.8).

*Max_Flush_Timeout:**Size: 4 octets*

Value	Parameter Description
0xFFFFFFFF	Maximum time period, in microseconds, which the AMP device uses to attempt to transmit a frame on a Guaranteed Logical Link. If the Controller is configured to retry frames for an unbounded time (i.e. there is no flushing at all), then it shall set this value to 0xFFFFFFFF.



Best_Effort_Flush_Timeout

Size: 4 octets

Value	Parameter Description
0xFFFFFFFF	The typical time period, in microseconds, which the AMP device may use to attempt to transmit a frame on a Best Effort Logical Link. The value shall not exceed the value given in Max_Flush_Timeout. If the Controller is configured to retry frames for an unbounded time (i.e. there is no flushing at all), then it shall set this value to 0xFFFFFFFF.

Event(s) generated (unless masked away):

When the HCI_Read_Local_AMP_Info command has completed, an HCI_Command_Complete event shall be generated.



7.5.9 Read Local AMP ASSOC command

Command	OCF	Command Parameters	Return Parameters
HCI_Read_Local_AMP_ASSOC	0x000A	Physical_Link_Handle, Length_So_Far, Max_Remote_AMP_ASSOC_Length	Status, Physical_Link_Handle, AMP_ASSOC_Remaining_Length, AMP_ASSOC_fragment

Description:

This command returns a fragment of the AMP_ASSOC structure. The AMP_ASSOC contains information in an AMP type specific format and is defined in the AMP PALs (see [Volume 5](#)). This may include a combination of address, capabilities and status information. This command can be used when creating an AMP physical link as given by the Physical_Link_Handle parameter, see the HCI_Create_Physical_Link command for more details. If the command is being executed outside of the context of any physical link creation, then the Physical_Link_Handle shall be 0x00.

The Controller shall limit the returned AMP_ASSOC to Max_AMP_ASSOC_Length octets. The Host shall set this parameter to the "AMP_ASSOC_Size" value returned from the remote device in the AMP Get Info Response (see AMP Manager Protocol Specification, [\[Vol 3\] Part E, Section 3.8](#)).

The AMP_ASSOC length may be larger than can fit in a single HCI event. Each time the HCI_Read_Local_AMP_ASSOC command is called, the remaining length of the AMP_ASSOC (including the data returned in the HCI_Command_Complete event) is included in the AMP_ASSOC_Remaining_Length parameter and data from the AMP_ASSOC is contained in the AMP_ASSOC_fragment parameter. In order to obtain the entire AMP_ASSOC, a Host shall continue calling Read_Local_AMP_ASSOC until the AMP_ASSOC_Remaining_Length parameter returned is equal to the size of the AMP_ASSOC_fragment in that event.

The Length_So_Far parameter shall be set to 0 to obtain the first AMP_ASSOC_fragment. Subsequent fragments are obtained by incrementing the Length_So_Far parameter by the length of the previous fragment.

Once the Host begins reading an AMP_ASSOC, all fragments returned in a sequence of HCI_Read_Local_AMP_ASSOC commands shall be from one, static AMP_ASSOC.

**Command parameters:***Physical_Link_Handle:**Size: 1 octet*

Value	Parameter Description
0x00	Used to indicate invalid Physical_Link_Handle
0xXX	Physical_Link_Handle which describes the link to which the AMP_ASSOC belongs.

*Length_So_Far:**Size: 2 octets*

Value
Set Length_So_Far to 0 for the first fragment. Increment parameter by the length of the previous fragment, for each subsequent call

*AMP_ASSOC_Length:**Size: 2 octets*

Value	Parameter Description
0xFFFF	The maximum length, in octets, allowed by the Host for the AMP_ASSOC maximum length for the remote returned by the AMP Controller. For use with Read command to prevent interoperability issues.

Return parameters:*Status:**Size: 1 octet*

Value	Parameter Description
0x00	HCI_Read_Local_AMP_ASSOC command succeeded
0x01 to 0xFF	HCI_Read_Local_AMP_ASSOC command failed. See [Vol 2] Part D, Error Codes for list of error codes.

*Physical_Link_Handle:**Size: 1 octet*

Value	Parameter Description
0xXX	Physical_Link_Handle identifying the physical link to be created with the associated AMP_ASSOC.

*AMP_ASSOC_Remaining_Length:**Size: 2 octets*

Value	Parameter Description
1-size of variable	Length, in octets, of the remainder of the AMP_ASSOC including this fragment.
0	Reserved for future use



AMP_ASSOC_fragment:

Size: 1 to 248 octets

Value
A fragment of the local AMP_ASSOC structure created by a remote AMP of the same Controller type. A fragment, other than the final one, shall be exactly 248 bytes in length.

Event(s) generated (unless masked away):

When the HCI_Read_Local_AMP_ASSOC command has completed, an HCI_Command_Complete event shall be generated.



7.5.10 Write Remote AMP ASSOC command

Command	OCF	Command Parameters	Return Parameters
HCI_Write_Remote_-AMP_ASSOC	0x000B	Physical_Link_Handle, Length_So_Far, AMP_ASSOC_Remain- ing_Length, AMP_ASSOC_fragment	Status, Physical_Link_Handle

Description:

This command writes an AMP_ASSOC fragment to the AMP Controller. The AMP_ASSOC contains information in an AMP type specific format and is defined in the AMP PALs (see [Volume 5](#)). This may include a combination of address, capabilities and status information. This command may be used when creating an AMP physical link. See the HCI_Create_Physical_Link command for more details. See [Section 7.1.37](#).

The AMP_ASSOC length may be larger than can fit in a single HCI command. Each time the HCI_Write_Remote_AMP_ASSOC command is called, the remaining length of the AMP_ASSOC (including the data provided in the command) is included in the AMP_ASSOC_Remaining_Length parameter and data from the AMP_ASSOC is contained in the AMP_ASSOC_fragment parameter. In order to write the entire AMP_ASSOC to the Controller, a Host shall continue calling Write_Remote_AMP_ASSOC until the AMP_ASSOC_Remaining_Length parameter returned is equal to the size of the AMP_ASSOC_fragment in that command.

The HCI_Write_Local_AMP_ASSOC command shall be called immediately after the HCI_Command_Status event (with result of success) for either the HCI_Create_Physical_Link or HCI_Accept_Physical_Link commands.

The Length_So_Far parameter shall be set to 0 to write the first AMP_ASSOC_fragment. Subsequent fragments are written by incrementing the Length_So_Far parameter by the length of the previous fragment.

Command parameters:

Physical_Link_Handle:

Size: 1 octet

Value	Parameter Description
0xXX	Physical_Link_Handle identifying the physical link to be created with the associated AMP_ASSOC.

*Length_So_Far:**Size: 2 octets*

Value
Set Length_So_Far to 0 for the first fragment. Increment it by the length of the previous fragment, for each subsequent call.

*AMP_ASSOC_Remaining_Length:**Size: 2 octets*

Value	Parameter Description
0x0001- Max-AMP_ASSOC_Length	Length, in octets, of the remainder of the AMP_ASSOC, including this fragment. Max_AMP_ASSOC_Length as reported in HCI_Read_Local_AMP_Info command.
All other values	Reserved for future use

*AMP_ASSOC_fragment:**Size: 1 to 248 octets*

Value
A fragment of the AMP_ASSOC structure created by a remote AMP of the same Controller type.
The fragment length shall be 248 bytes for all but the last fragment.

Return parameters:*Status:**Size: 1 octet*

Value	Parameter Description
0x00	HCI_Write_Remote_AMP_ASSOC command succeeded
0x01 to 0xFF	HCI_Write_Remote_AMP_ASSOC command failed. See "[Vol 2] Part D, Error Codes" for list of error codes.

*Physical_Link_Handle:**Size: 1 octet*

Value	Parameter Description
0xXX	Physical_Link_Handle identifying the physical link to be created with the associated AMP_ASSOC.

Event(s) generated (unless masked away):

When the HCI_Write_Remote_AMP_ASSOC command has completed, an HCI_Command_Complete event shall be generated.



7.5.11 Get MWS Transport Layer Configuration command

Command	OCF	Command Parameters	Return Parameters
HCI_Get_MWS_Transport_Layer_Configuration	0x000C		Status, Num_Transports, Transport_Layer[i], Num_Baud_Rates[i], To_MWS_Baud_Rate[k], From_MWS_Baud_Rate[k]

The order of the return parameters in an HCI event packet is:

```

Status
Num_Transports
Transport_Layer[0]
Num_Baud_Rates[0]
...
Transport_Layer[n]
Num_Baud_Rates[n]
To_MWS_Baud_Rate[0]
From_MWS_Baud_Rate[0]
...
To_MWS_Baud_Rate[m]
From_MWS_Baud_Rate[m]

```

Description:

The HCI_Get_MWS_Transport_Layer_Configuration command is used to inform the Host of the Baud rates supported by the Controller for the transport layer.

The Num_Transports parameter is used to indicate the number of MWS coexistence transport interfaces supported by the Controller.
The Num_Baud_Rates parameter indicates the number of supported baud rates for each transport.

The To_MWS_Baud_Rate parameters indicate the supported baud rates in the direction from Bluetooth to MWS for each transport.

The From_MWS_Baud_Rate parameters indicate the supported baud rates in the direction from MWS to Bluetooth for each transport.

If one direction has more supported rates than the other direction, the Controller shall - in the direction with less supported rates - fill with sufficient



zeros to produce the same number of values. The rates for the two directions are not necessarily paired.

Command parameters:

None.

Return parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	HCI_Get_MWS_Transport_Layer_Configuration command succeeded.
0x01 to 0xFF	HCI_Get_MWS_Transport_Layer_Configuration command failed. See [Vol 2] Part D, Error Codes, for error codes and descriptions.

Num_Transports:

Size: 1 octet

Value	Parameter Description
0xXX	Number of supported MWS coexistence transport layers.

Transport_Layer[i]:

*Size: Num_Transports * 1 octet*

Value	Parameter Description
0xXX	See Assigned Numbers .

Num_Baud_Rates[i]:

*Size: Num_Transports * 1 octet*

Value	Parameter Description
0xXX	Number of different baud rates supported for one transport.

To_MWS_Baud_Rate[k]:

*Size: SUM (Num_Baud_Rates [i]) * 4 octets*

Value	Parameter Description
0XXXXXXXX	List of supported Baud rates in the Bluetooth Controller to MWS Device direction in Baud. Each element of the list shall have the format 0XXXXXXXX. The list shall start with the first baud rate for the first transport, followed by the remaining baud rates for the first transport, followed by the baud rates for the second transport (if any), followed by baud rates for subsequent transports (if any).



From_MWS_Baud_Rate[k]: *Size: SUM (Num_Baud_Rates[i]) * 4 octets*

Value	Parameter Description
0XXXXXXXXX	List of supported Baud rates in the Bluetooth Controller for signals in the MWS to Bluetooth Controller Device direction in Baud. Each element of the list shall have the format 0XXXXXXXXX. The list shall start with the first baud rate for the first transport, followed by the remaining baud rates for the first transport, followed by the baud rates for the second transport (if any), followed by baud rates for subsequent transports (if any).

Event(s) generated (unless masked away):

When the HCI_Get_MWS_Transport_Layer_Configuration command has completed, an HCI_Command_Complete event shall be generated.



7.5.12 Set Triggered Clock Capture command

Command	OCF	Command Parameters	Return Parameters
HCI_Set_Triggered_Clock_Capture	0x000D	Connection_Handle, Enable, Which_Clock, LPO_Allowed, Num_Clock_Captures_To_Filter	Status

Description:

The HCI_Set_Triggered_Clock_Capture command configures the BR/EDR Controller for triggered clock capturing.

Triggered clock capturing is enabled or disabled by the Enable parameter. If Enable is set to 0x00, all the other parameters (including Connection_Handle) shall be ignored.

If the Which_Clock value is 0, then the Connection_Handle shall be ignored. If the Which_Clock value is 1, then the Connection_Handle shall be a valid ACL Connection_Handle.

The LPO_Allowed parameter informs the BR/EDR Controller whether it may use a lower accuracy clock or not.

The Num_Clock_Captures_To_Filter parameter is used to filter triggered clock captures between sending HCI_Triggered_Clock_Capture events to the Host. When set to zero, all triggered clock captures shall result in an HCI_Triggered_Clock_Capture event sent to the Host. When set to a non-zero value, after every HCI_Triggered_Clock_Capture event, Num_Clock_Captures_To_Filter triggered clock captures in a row shall not trigger an event to be sent to the Host.

Note: An implementation should ensure that the rate of triggered clock captures does not overwhelm the HCI event queue and processing.

Note: See [\[Vol 2\] Part B, Section 1.1](#) for more information about the Bluetooth Clock.

Command parameters:

Connection_Handle:

Size: 2 octets (12 bits meaningful)

Value	Parameter Description
0xXXXX	Connection_Handle Range: 0x0000 to 0x0EFF

**Enable:****Size: 1 octet**

Value	Parameter Description
0x00	Disable triggered clock capturing on the specified Connection_Handle (Default)
0x01	Enable triggered clock capturing on the specified Connection_Handle
All other values	Reserved for future use

Which_Clock:**Size: 1 octet**

Value	Parameter Description
0xXX	0x00 = Local Clock (Connection_Handle does not have to be valid) 0x01 = Piconet Clock (Connection_Handle shall be valid) 0x02-0xFF = Reserved for future use

LPO_Allowed:**Size: 1 octet**

Value	Parameter Description
0x00	Controller shall not sleep (that is, clock accuracy shall be equal to or better than ± 20 ppm)
0x01	Controller may sleep (that is, clock accuracy shall be equal to or better than ± 250 ppm)
All other values	Reserved for future use

Num_Clock_Captures_To_Filter:**Size: 1 octet**

Value	Parameter Description
0x00	All triggered clock captures result in an HCI_Triggered_Clock_Capture event sent to the Host
0x01 to 0xFF	Number of triggered clock captures filtered between sending an HCI_Triggered_Clock_Capture event to the Host.

Return parameters:**Status:****Size: 1 octet**

Value	Parameter Description
0x00	HCI_Set_Triggered_Clock_Capture command succeeded.
0x01 to 0xFF	HCI_Set_Triggered_Clock_Capture command failed. See [Vol 2] Part D, Error Codes , for error codes and descriptions.

**Event(s) generated (unless masked away):**

When the HCI_Set_Triggered_Clock_Capture command has completed, an HCI_Command_Complete event shall be sent to the Host.

When Triggered Clock Capturing is enabled, HCI_Triggered_Clock_Capture events are returned until Triggered Clock Capturing is disabled.



7.6 TESTING COMMANDS

The Testing commands are used to provide the ability to test various functional capabilities of the Bluetooth hardware. These commands provide the ability to arrange various conditions for testing.

For the Testing commands, the OGF is defined as 0x06.

7.6.1 Read Loopback Mode command

Command	OCF	Command Parameters	Return Parameters
HCI_Read_Loopback_Mode	0x0001		Status, Loopback_Mode

Description:

This command reads the value for the setting of the Controller's Loopback_Mode. The setting of the Loopback_Mode parameter shall determine the path of information. In Non-testing Mode operation, the Loopback_Mode parameter is set to Non-testing Mode and the path of the information is as specified by the Bluetooth specifications. In Local Loopback Mode, every data packet (ACL, SCO and eSCO) and command packet that is sent from the Host to the Controller is sent back with no modifications by the Controller, as shown in [Figure 7.1](#). For details of loopback modes see [Section 7.6.2](#).

Command parameters:

None.

Return parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	HCI_Read_Loopback_Mode command succeeded.
0x01 to 0xFF	HCI_Read_Loopback_Mode command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Loopback_Mode:

Size: 1 octet

Value	Parameter Description
0x00	No Loopback mode enabled (default).
0x01	Enable Local Loopback.
0x02	Enable Remote Loopback. (Not supported by AMPs)
All other values	Reserved for future use.

**Event(s) generated (unless masked away):**

When the HCI_Read_Loopback_Mode command has completed, an HCI_Command_Complete event shall be generated.



7.6.2 Write Loopback Mode command

Command	OCF	Command Parameters	Return Parameters
HCI_Write_Loopback_Mode	0x0002	Loopback_Mode	Status

Description:

This command writes the value for the setting of the BR/EDR Controller's Loopback mode. The setting of the Loopback_Mode parameter shall determine the path of information. In Non-testing mode operation, the Loopback_Mode parameter is set to Non-testing mode and the path of the information as specified by the Bluetooth specifications. In Local Loopback mode, every data packet (ACL, SCO and eSCO) and command packet that is sent from the Host to the BR/EDR Controller is sent back with no modifications by the BR/EDR Controller, as shown in [Figure 7.1](#).

When the BR/EDR Controller enters Local Loopback mode, it shall respond with one to four Connection_Handles, one for an ACL connection and zero to three for synchronous connections. The Host should use these Connection_Handles when sending data in Local Loopback mode. The number of Connection_Handles returned for synchronous connections (between zero and three) is implementation specific. When in Local Loopback mode, the BR/EDR Controller loops back commands and data to the Host. The HCI_Loopback_Command event is used to loop back commands that the Host sends to the Controller.

When an AMP enters Local Loopback mode, all data transmission will be looped back with the received Logical_Link_Handle value, without checking the validity of the Logical Link Handle. This allows local loopback of data without any link creation, and without any data transmission over the air.

There are some commands that are not looped back in Local Loopback mode: HCI_Reset, HCI_Set_Controller_To_Host_Flow_Control, HCI_Host_Buffer_Size, HCI_Host_Number_Of_Completed_Packets, HCI_Read_Buffer_Size, HCI_Read_Loopback_Mode and HCI_Write_Loopback_Mode. These commands should be executed in the way they are normally executed. The commands HCI_Reset and HCI_Write_Loopback_Mode can be used to exit Local Loopback mode.

If HCI_Write_Loopback_Mode is used to exit Local Loopback mode on a BR/EDR Controller, HCI_Disconnection_Complete events corresponding to the HCI_Connection_Complete events that were sent when entering Local Loopback mode should be sent to the Host. Furthermore, no connections are allowed in Local Loopback mode. If there is a connection, and there is an attempt to set the device to Local Loopback mode, the attempt will be refused. When the device is in Local Loopback mode, the Controller will refuse incoming connection attempts. This allows the Host BR/EDR Controller Transport Layer to be tested without any other variables.



If a BR/EDR Controller is set to Remote Loopback mode, it will send back all data (ACL, SCO and eSCO) that comes over the air. It will only allow a maximum of one ACL connection and three synchronous connections, and these shall all be to the same remote device. If there are existing connections to a remote device and there is an attempt to set the local device to Remote Loopback mode, the attempt shall be refused.

If a Host sets Loopback mode to “Remote Loopback” on an AMP Controller, the Controller shall reject the command with the error code *Invalid HCI Command Parameter* (0x12).

See [Figure 7.2](#), where the rightmost device is set to Remote Loopback mode and the leftmost device is set to Non-testing mode. This allows the BR/EDR Air link to be tested without any other variables.

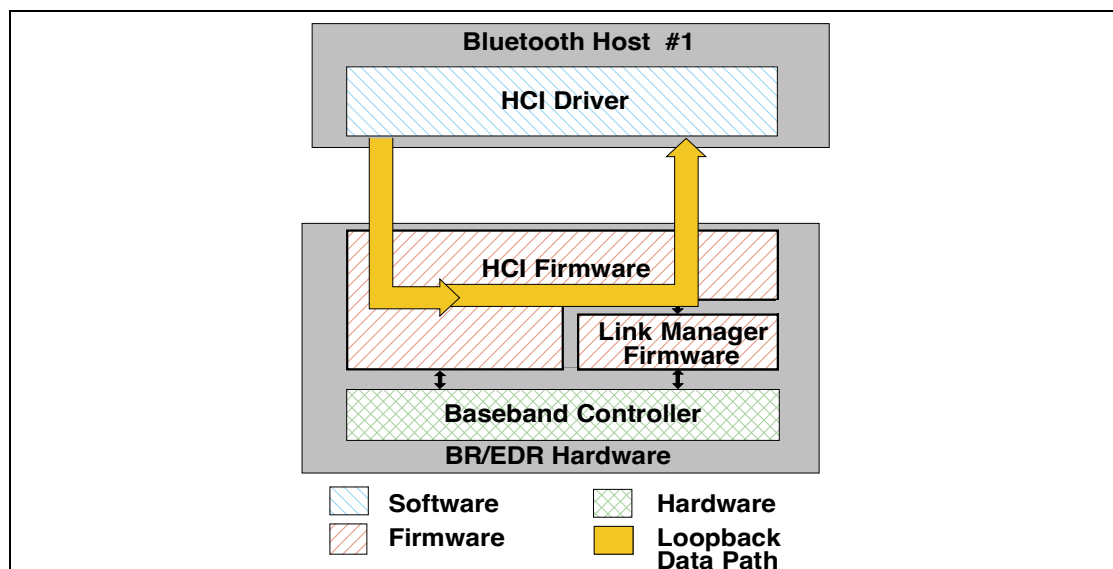


Figure 7.1: Local Loopback mode

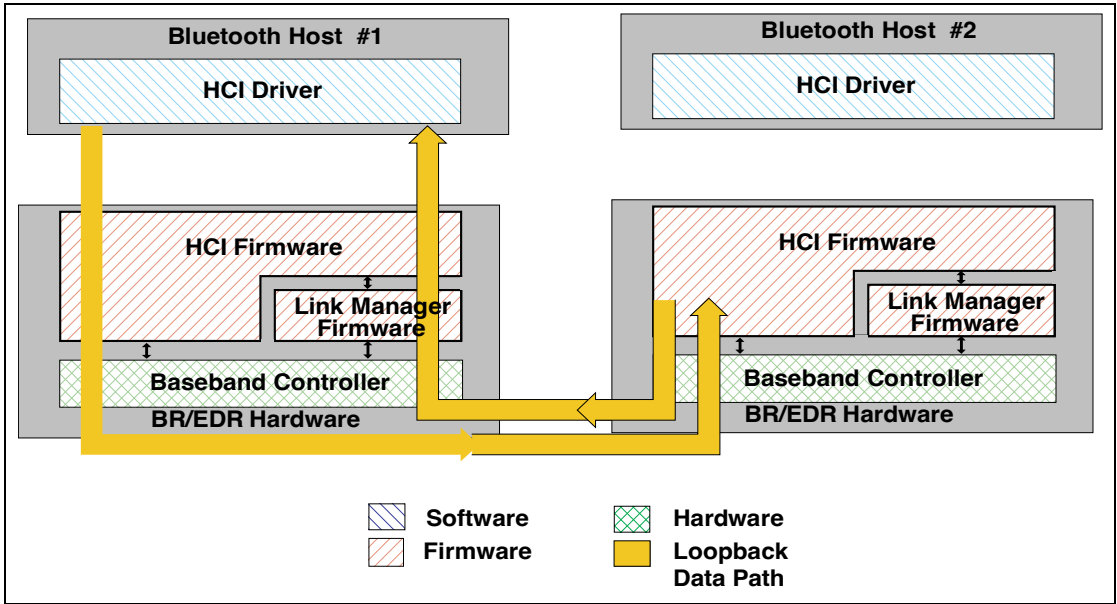


Figure 7.2: Remote Loopback mode

Command parameters:

Loopback_Mode: Size: 1 octet

Value	Parameter Description
0x00	No Loopback mode enabled (default).
0x01	Enable Local Loopback.
0x02	Enable Remote Loopback. (Not Supported by AMPs)
All other values	Reserved for future use.

Return parameters:

Status: Size: 1 octet

Value	Parameter Description
0x00	HCI_Write_Loopback_Mode command succeeded.
0x01 to 0xFF	HCI_Write_Loopback_Mode command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Event(s) generated (unless masked away):

When the HCI_Write_Loopback_Mode command has completed, an HCI_Command_Complete event shall be generated.



7.6.3 Enable Device Under Test Mode command

Command	OCF	Command Parameters	Return Parameters
HCI_Enable_Device_Under_Test_Mode	0x0003		Status

Description:

The HCI_Enable_Device_Under_Test_Mode command allows the local BR/EDR Controller to enter test mode via LMP test commands for BR/EDR Controllers or via the HCI_AMP_Test command for AMP Controllers. For details see [\[Vol 2\] Part C, Link Manager Protocol Specification](#) for BR/EDR Controllers or [Section 7.6.7](#) for AMP Controllers. The Host issues this command when it wants the local device to be the DUT for the Testing scenarios as described in [\[Vol 3\] Part D, Section 1](#). When the BR/EDR Controller receives this command, it shall complete the command with an HCI_Command_Complete event. The BR/EDR Controller functions as normal until the remote tester issues the LMP test command or for AMPs the HCI_AMP_Test command to place the local device into Device Under Test mode. To disable and exit the Device Under Test Mode, the Host may issue the HCI_Reset command. This command prevents remote BR/EDR Controllers from causing the local BR/EDR Controller to enter test mode without first issuing this command.

Command parameters:

None.

Return parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	HCI_Enter_Device_Under_Test_Mode command succeeded.
0x01 to 0xFF	HCI_Enter_Device_Under_Test_Mode command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Event(s) generated (unless masked away):

When the HCI_Enter_Device_Under_Test_Mode command has completed, an HCI_Command_Complete event shall be generated.



7.6.4 Write Simple Pairing Debug Mode command

Command	OCF	Command Parameters	Return Parameters
HCI_Write_Simple_Pairing_Debug_Mode	0x0004	Simple_Pairing_Debug_Mode	Status

Description:

This command configures the BR/EDR Controller to use a predefined Diffie Hellman private key for Simple Pairing to enable debug equipment to monitor the encrypted connection.

Note: Only one side (initiator or responder) needs to set simple pairing debug mode in order for debug equipment to be able to determine the link key and, therefore, be able to monitor the encrypted connection.

When the Simple_Pairing_Debug_Mode parameter is set to enabled the BR/EDR Controller shall use the predefined Diffie Hellman private key. The BR/EDR Controller shall also set the resulting Link_Key type to "Debug Combination Key."

When in Simple Pairing debug mode, the Link Manager shall use the following Diffie Hellman private / public key pairs:

For P-192:

Private key: 07915f86918ddc27005dfd1d6cf0c142b625ed2eff4a518ff

Public key (X): 15207009984421a6586f9fc3fe7e4329d2809ea51125f8ed

Public key (Y): b09d42b81bc5bd009f79e4b59dbbaa857fca856fb9f7ea25

For P-256:

Private key: 3f49f6d4 a3c55f38 74c9b3e3 d2103f50 4aff607b eb40b799 5899b8a6 cd3c1abd

Public key (X): 20b003d2 f297be2c 5e2c83a7 e9f9a5b9 eff49111 acf4fddb cc030148 0e359de6

Public key (Y): dc809c49 652aeb6d 63329abf 5a52155c 766345c2 8fed3024 741c8ed0 1589d28b

Command parameters:

Simple_Pairing_Debug_Mode:

Size: 1 octet

Value	Parameter Description
0x00	Simple Pairing debug mode disabled (default)
0x01	Simple pairing debug mode enabled
All other values	Reserved for future use

**Return parameters:***Status:**Size: 1 octet*

Value	Parameter Description
0x00	HCI_Write_Simple_Pairing_Debug_Mode command succeeded.
0x01 to 0xFF	HCI_Write_Simple_Pairing_Debug_Mode command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Event(s) generated (unless masked away):

When the HCI_Write_Simple_Pairing_Debug_Mode command has completed, an HCI_Command_Complete event shall be generated.



7.6.5 Enable AMP Receiver Reports command

Command	OCF	Command Parameters	Return Parameters
HCI_Enable_AMP_Receiver_Reports	0x0007	Enable, Interval	Status

Description:

This command enables and disables the reporting of frames received. If the Enable parameter is set to 0x00, the Interval parameter shall be ignored. This command shall only be valid in AMP Test Mode.

While reporting is enabled, after multiples of the specified Interval, a report of frames that have been received/not received by the DUT shall be sent to the tester. Optionally, the sums of bits in error are also reported. The values returned are the cumulative totals since the mode was last initiated or reset. The totals are reset by sending the HCI_Enable_AMP_Receiver_Reports, the HCI_AMP_Test command or the HCI_AMP_Test_End command. This event is also generated after an HCI_AMP_Test_End event.

Command parameters:

Enable:

Size: 1 octet

Value	Parameter Description
0x00	Disable
0x01	Enable
All other values	Reserved for future use

Interval:

Size: 1 octet

Value	Parameter Description
0x00	Reserved for future use
0x01 to 0xFF	Number of seconds between event reporting

Return parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	HCI_Enable_AMP_Receiver_Reports command succeeded
0x01 to 0xFF	HCI_Enable_AMP_Receiver_Reports command failed. [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

**Event(s) generated (unless masked away):**

When the HCI_Enable_AMP_Receiver_Reports command has completed, an HCI_Command_Complete event shall be generated.



7.6.6 AMP Test End command

Command	OCF	Command Parameters	Return Parameters
HCI_AMP_Test_End	0x0008		Status

Description:

This command is used to stop any test scenario in progress. This command shall only be used in AMP Test Mode when a test is in progress. This command shall not exit an AMP Controller from AMP Test mode.

Command parameters:

None.

Return parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	HCI_AMP_Test_End command succeeded
0x01 to 0xFF	HCI_AMP_Test_End command failed. [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Event(s) generated (unless masked away):

When the HCI_AMP_Test_End command has completed an HCI_AMP_Test_End event shall be generated and the HCI_AMP_Receiver_Report event shall be generated if the AMP receiver reports are enabled.



7.6.7 AMP Test command

Command	OCF	Command Parameters	Return Parameters
HCI_AMP_Test	0x0009	Test_Parameters	Status

Description:

This command is used to configure and start a test. This command shall be only valid in AMP Test Mode.

When a test scenario has completed or on receiving an HCI_AMP_Test_End command the AMP shall send an HCI_AMP_Test_End event and the AMP returns to an idle state with the RX and TX off.

The details of the Test_Parameters for each Controller Type are detailed in the appropriate AMP PAL specification in [Volume 5](#).

Command parameters:

Test_Parameters:

Size: variable

Value	Parameter Description
0xXX...XX	Depends on the controller type

Return parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	HCI_AMP_Test command succeeded
0x01 to 0xFF	HCI_AMP_Test command failed. See [Vol 2] Part D, Error Codes

Event(s) generated (unless masked away):

When the AMP Controller receives the HCI_AMP_Test command, the AMP Controller shall send the HCI_Command_Status event to the AMP Test Manager which shall be routed to the tester.

The HCI_AMP_Start_Test event shall be generated when the HCI_AMP_Test command has completed and the first data is ready to be sent or received.

HCI_Command_Complete event shall not be sent by the AMP to indicate that this command has been completed. Instead, the HCI_AMP_Start_Test event shall indicate that this command has been completed.

When in a transmitter test scenario and the frames/bursts count have been transmitted the HCI_AMP_Test_End event shall be sent.



7.6.8 Write Secure Connections Test Mode command

Command	OCF	Command Parameters	Return Parameters
HCI_Write_Secure_- Connections_Test_Mode	0x000A	Connection_Handle, DM1_ACL-U_Mode, eSCO_Loopback_Mode	Status, Connection_Handle

Description:

This command configures the BR/EDR Controller to enable and disable the two test modes used for verifying the Secure Connections feature during qualification. The DM1_ACL-U_Mode parameter enables and disables the use of DM1 packets for transmitting ACL-U data. When DM1 ACL-U Mode is disabled, ACL-U traffic may use DM1 packets. When DM1 ACL-U Mode is enabled, ACL-U traffic shall not use DM1 packets unless the Packet_Type parameter only allows DM1 packets (e.g. set to 0x3306 or 0x330E).

The command is used during testing to help make transmit ACL packet selection predictable.

The eSCO_Loopback_Mode parameter enables and disables the loopback of received eSCO payloads. When the eSCO_Loopback_Mode parameter is set to Enabled, the BR/EDR Controller will send back all eSCO data that comes over the air irrespective of whether the CRC check in the received eSCO packet passes or fails. It will only allow one synchronous connection. If there is more than one synchronous connection and there is an attempt to set the local device to eSCO_Loopback_Mode, the attempt shall be refused.

See [Figure 7.3](#), where the rightmost device has the eSCO_Loopback_Mode parameter set to enabled and the leftmost device is in a normal mode of operation. This allows the encryption and decryption of eSCO packets to be tested without any other variables.

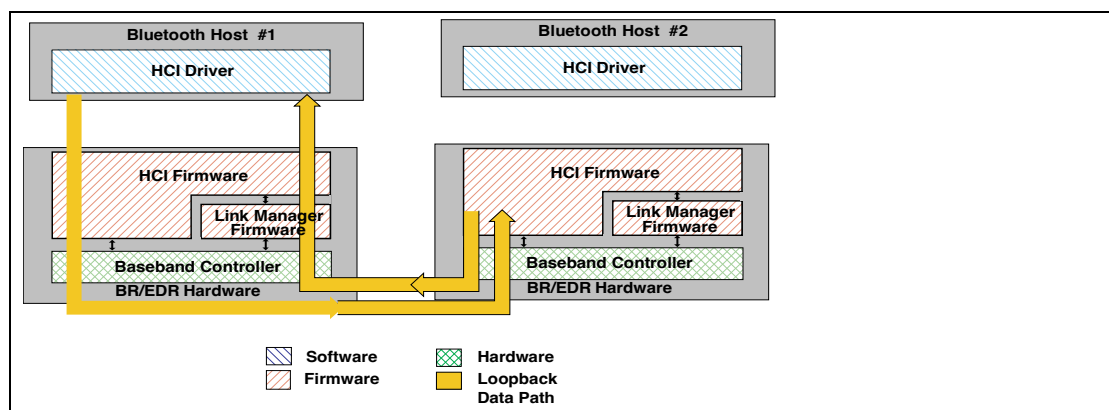


Figure 7.3: Secure Connections eSCO Loopback

The Connection_Handle shall be for an ACL connection.

When the eSCO_Loopback_Mode parameter is set to enabled, received eSCO payloads are looped back as subsequent transmitted eSCO payloads. There may be a delay of 0 or more eSCO intervals before the Controller loops back the payload. This is illustrated in the following three figures.

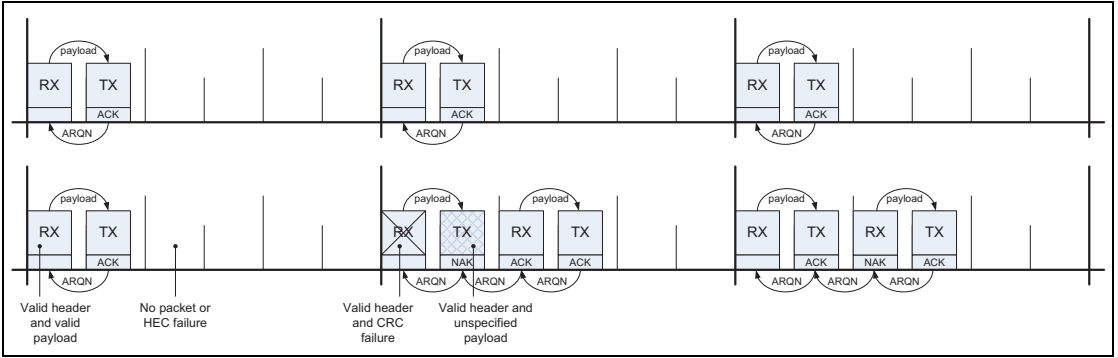


Figure 7.4: Secure Connections eSCO loopback immediate

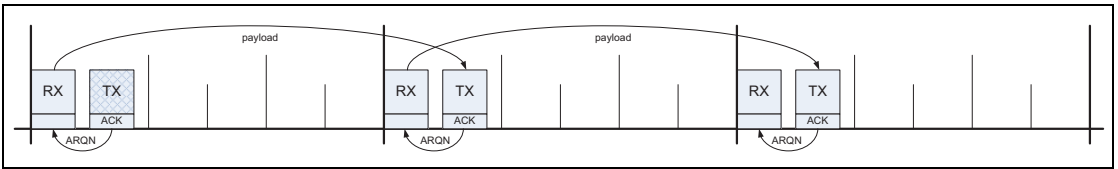


Figure 7.5: Secure Connections eSCO loopback delayed

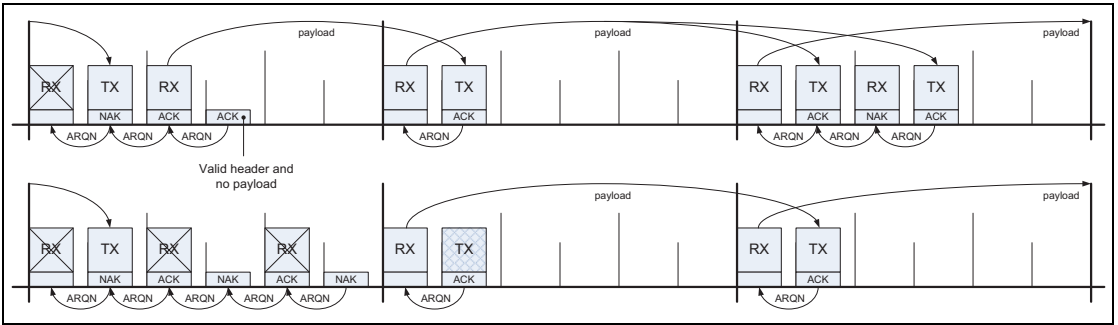


Figure 7.6: Secure Connections eSCO loopback delayed with retransmissions

Command parameters:

Connection_Handle: Size: 2 octets (12 bits meaningful)

Value	Parameter Description
0xXXXX	Connection_Handle Range: 0x0000 to 0x0EFF

**DM1_ACL-U_Mode:****Size: 1 octet**

Value	Parameter Description
0x00	DM1 ACL-U mode disabled (default)
0x01	DM1 ACL-U mode enabled
All other values	Reserved for future use

eSCO_Loopback_Mode:**Size: 1 octet**

Value	Parameter Description
0x00	eSCO loopback mode disabled (default)
0x01	eSCO loopback mode enabled
All other values	Reserved for future use

Return parameters:**Status:****Size: 1 octet**

Value	Parameter Description
0x00	HCI_Write_Secure_Connections_Test_Mode command succeeded.
0x01 to 0xFF	HCI_Write_Secure_Connections_Test_Mode command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Connection_Handle:**Size: 2 octets (12 bits meaningful)**

Value	Parameter Description
0xFFFF	Connection_Handle Range: 0x0000 to 0x0EFF

Events(s) generated (unless masked away):

When the HCI_Write_Secure_Connections_Test_Mode command has completed, an HCI_Command_Complete event shall be generated.



7.7 EVENTS

7.7.1 Inquiry Complete event

Event	Event Code	Event Parameters
HCI_Inquiry_Complete	0x01	Status

Description:

The HCI_Inquiry_Complete event indicates that the Inquiry is finished. This event contains a Status parameter, which is used to indicate if the Inquiry completed successfully or if the Inquiry was not completed.

Event parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	HCI_Inquiry command completed successfully.
0x01 to 0xFF	HCI_Inquiry command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.



7.7.2 Inquiry Result event

Event	Event Code	Event Parameters
HCI_Inquiry_Result	0x02	Num_Responses, BD_ADDR[i], Page_Scan_Repetition_Mode[i], Reserved[i], Reserved[i], Class_of_Device[i] Clock_Offset[i]

Description:

The HCI_Inquiry_Result event indicates that a BR/EDR Controller or multiple BR/EDR Controllers have responded so far during the current Inquiry process. This event will be sent from the BR/EDR Controller to the Host as soon as an Inquiry Response from a remote device is received if the remote device supports only mandatory paging scheme. The BR/EDR Controller may queue these Inquiry Responses and send multiple BR/EDR Controllers information in one HCI_Inquiry_Result event. The event can be used to return one or more Inquiry responses in one event.

This event is only generated if the Inquiry_Mode parameter of the last HCI_Write_Inquiry_Mode command was set to 0x00 (Standard Inquiry Result event format) or if the HCI_Write_Inquiry_Mode command has not been used.

Event parameters:

Num_Responses: *Size: 1 octet*

Value	Parameter Description
0xXX	Number of responses from the Inquiry.

BD_ADDR[i]: *Size: 6 octets * Num_Responses*

Value	Parameter Description
0XXXXXXXXXX XX	BD_ADDR for each device which responded.

Page_Scan_Repetition_Mode[i]: *Size: 1 octet * Num_Responses*

Value	Parameter Description
0x00	R0
0x01	R1
0x02	R2



Value	Parameter Description
All other values	Reserved for future use

*Reserved[ij]:*¹*Size: 1 octet * Num_Responses*

Value	Parameter Description
0xXX	Reserved for future use.

*Reserved[ij]:*²*Size: 1 octet * Num_Responses*

Value	Parameter Description
0xXX	Reserved for future use.

*Class_of_Device[ij]:**Size: 3 octets * Num_Responses*

Value	Parameter Description
0XXXXXXXX	Class of Device for the device

*Clock_Offset[ij]:**Size: 2 octets * Num_Responses*

Bit format	Parameter Description
Bits 14-0	Bits 16-2 of CLKNSlave-CLK
Bit 15	Reserved for future use

1. This was the Page_Scan_Period_Mode parameter in the v1.1 specification. This parameter has no meaning in v1.2 or higher and no default value.

2. This was the Page_Scan_Mode parameter in the v1.1 specification.



7.7.3 Connection Complete event

Event	Event Code	Event Parameters
HCI_Connection_Complete	0x03	Status, Connection_Handle, BD_ADDR, Link_Type, Encryption_Enabled

Description:

The HCI_Connection_Complete event indicates to both of the Hosts forming the connection that a new connection has been established. This event also indicates to the Host which issued the HCI_Create_Connection, HCI_Accept_Connection_Request, or HCI_Reject_Connection_Request command, and then received an HCI_Command_Status event, if the issued command failed or was successful.

Event parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	Connection successfully completed.
0x01 to 0xFF	Connection failed to Complete. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Connection_Handle:

Size: 2 octets (12 bits meaningful)

Value	Parameter Description
0xFFFF	Connection_Handle for the new connection Range: 0x0000 to 0x0EFF

BD_ADDR:

Size: 6 octets

Value	Parameter Description
0XXXXXXXXXXXXX	BD_ADDR of the other connected Device forming the connection.

Link_Type:

Size: 1 octet

Value	Parameter Description
0x00	SCO connection.
0x01	ACL connection (Data Channels).
All other values	Reserved for future use.

Host Controller Interface Functional Specification***Encryption_Enabled:******Size: 1 octet***

Value	Parameter Description
0x00	Link level encryption disabled.
0x01	Link level encryption enabled.
All other values	Reserved for future use.



7.7.4 Connection Request event

Event	Event Code	Event Parameters
HCI_Connection_Request	0x04	BD_ADDR, Class_of_Device, Link_Type

Description:

The HCI_Connection_Request event is used to indicate that a new incoming connection is trying to be established. The connection may either be accepted or rejected. When the Host receives this event and the link type parameter is ACL, it should respond with either an HCI_Accept_Connection_Request or HCI_Reject_Connection_Request command before the timer Conn_Accept_Timeout expires. If the link type is SCO or eSCO, the Host should reply with the HCI_Accept_Synchronous_Connection_Request or the HCI_Reject_Synchronous_Connection_Request command. If the link type is SCO, the Host may respond with HCI_Accept_Connection_Request command. If the event is responded to with HCI_Accept_Connection_Request command, then the default parameter settings of the HCI_Accept_Synchronous_Connection_Request command (see [Section 7.1.27](#)) should be used by the local Link Manager when negotiating the SCO link parameters. In that case, the HCI_Connection_Complete event and not the HCI_Synchronous_Connection_Complete event, shall be returned on completion of the connection.

Note: See [Section 7.3.3](#) for the behavior when the HCI_Connection_Request event is masked or the connection is auto accepted.

Event parameters:

BD_ADDR: Size: 6 octets

Value	Parameter Description
0XXXXXXXXXXXXX	BD_ADDR of the device that requests the connection.

Class_of_Device: Size: 3 octets

Value	Parameter Description
0XXXXXXX	Class of Device for the device, which requests the connection.
0x000000	Unknown Class of Device

Link_Type: Size: 1 octet

Value	Parameter Description
0x00	SCO connection requested
0x01	ACL connection requested



Value	Parameter Description
0x02	eSCO connection requested
All other values	Reserved for future use.



7.7.5 Disconnection Complete event

Event	Event Code	Event Parameters
HCI_Disconnection_Complete	0x05	Status, Connection_Handle, Reason

Description:

The HCI_Disconnection_Complete event occurs when a connection is terminated. The status parameter indicates if the disconnection was successful or not. The reason parameter indicates the reason for the disconnection if the disconnection was successful. If the disconnection was not successful, the value of the reason parameter shall be ignored by the Host. For example, this can be the case if the Host has issued the HCI_Disconnect command and there was a parameter error, or the command was not presently allowed, or a Connection_Handle that didn't correspond to a connection was given.

Note: When a physical link fails, one HCI_Disconnection_Complete event will be returned for each logical channel on the physical link with the corresponding Connection_Handle as a parameter.

Event parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	Disconnection has occurred.
0x01 to 0xFF	Disconnection failed to complete. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Connection_Handle:

Size: 2 octets (12 bits meaningful)

Value	Parameter Description
0xFFFF	Connection_Handle which was disconnected. Range: 0x0000 to 0xFFFF

Reason:

Size: 1 octet

Value	Parameter Description
0xFF	Reason for disconnection. See [Vol 2] Part D, Error Codes for error codes and descriptions.



7.7.6 Authentication Complete event

Event	Event Code	Event Parameters
HCI_Authentication_Complete	0x06	Status, Connection_Handle

Description:

The HCI_Authentication_Complete event occurs when authentication has been completed for the specified connection. The Connection_Handle shall be a Connection_Handle for an ACL connection.

Event parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	Authentication Request successfully completed.
0x01 to 0xFF	Authentication Request failed to complete. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Connection_Handle:

Size: 2 octets (12 bits meaningful)

Value	Parameter Description
0xFFFF	Connection_Handle Range: 0x0000 to 0x0EFF



7.7.7 Remote Name Request Complete event

Event	Event Code	Event Parameters
HCI_Remote_Name_Request_ - Complete	0x07	Status, BD_ADDR, Remote_Name

Description:

The HCI_Remote_Name_Request_Complete event is used to indicate that a remote name request has been completed. The Remote_Name event parameter is a UTF-8 encoded string with up to 248 octets in length. The Remote_Name event parameter will be null-terminated (0x00) if the UTF-8 encoded string is less than 248 octets. The BD_ADDR event parameter is used to identify which device the user-friendly name was obtained from.

Note: The Remote_Name parameter is a string parameter. Endianness does therefore not apply to the Remote_Name parameter. The first octet of the name is received first.

Event parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	HCI_Remote_Name_Request command succeeded.
0x01 to 0xFF	HCI_Remote_Name_Request command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

BD_ADDR:

Size: 6 octets

Value	Parameter Description
0xFFFFFFFFXXXX	BD_ADDR for the device whose name was requested.

Remote_Name:

Size: 248 octets

Value	Parameter Description
Name[248]	A UTF-8 encoded user-friendly descriptive name for the remote device. If the name contained in the parameter is shorter than 248 octets, the end of the name is indicated by a NULL octet (0x00), and the following octets (to fill up 248 octets, which is the length of the parameter) do not have valid values.



7.7.8 Encryption Change event

Event	Event Code	Event Parameters
HCI_Encryption_Change	0x08	Status, Connection_Handle, Encryption_Enabled

Description:

The HCI_Encryption_Change event is used to indicate that the change of the encryption mode has been completed. The Connection_Handle will be a Connection_Handle for an ACL connection and is used to identify the remote device. The Encryption_Enabled event parameter specifies the new Encryption_Enabled parameter for the Connection_Handle specified by the Connection_Handle event parameter. This event will occur on both devices to notify the Hosts when Encryption has changed for all connections between the two devices. This event shall not be generated if encryption is paused or resumed; during a role switch, for example.

The meaning of the Encryption_Enabled parameter depends on whether the Host has indicated support for Secure Connections in the Secure_Connections_Host_Support parameter. When Secure_Connections_Host_Support is 'disabled' or the Connection_Handle refers to an LE link, the Controller shall only use Encryption_Enabled values 0x00 (OFF) and 0x01 (ON). When Secure_Connections_Host_Support is 'enabled' and the Connection_Handle refers to a BR/EDR link, the Controller shall set Encryption_Enabled to 0x00 when encryption is off, to 0x01 when encryption is on and using E0 and to 0x02 when encryption is on and using AES-CCM.

Event parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	An encryption change has occurred.
0x01 to 0xFF	An attempt to change encryption failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Connection_Handle:

Size: 2 octets (12 bits meaningful)

Value	Parameter Description
0xFFFF	Connection_Handle Range: 0x0000 to 0x0EFF

**Encryption_Enabled:****Size: 1 octet**

Value	Parameter Description
0x00	Link Level Encryption is OFF.
0x01	Link Level Encryption is ON with E0 for BR/EDR. Link Level Encryption is ON with AES-CCM for LE.
0x02	Link Level Encryption is ON with AES-CCM for BR/EDR.
All other values	Reserved for future use.



7.7.9 Change Connection Link Key Complete event

Event	Event Code	Event Parameters
HCI_Change_Connection_Link_Key_ - Complete	0x09	Status, Connection_Handle

Description:

The HCI_Change_Connection_Link_Key_Complete event is used to indicate that the change in the Link Key for all connections to a given remote BR/EDR Controller has been completed.

The Connection_Handle will be a Connection_Handle for an ACL connection to the remote Controller. The HCI_Change_Connection_Link_Key_Complete event is sent only to the Host which issued the HCI_Change_Connection_Link_Key command.

Event parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	HCI_Change_Connection_Link_Key command succeeded.
0x01 to 0xFF	HCI_Change_Connection_Link_Key command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Connection_Handle:

Size: 2 octets (12 bits meaningful)

Value	Parameter Description
0xFFFF	Connection_Handle Range: 0x0000 to 0x0EFF



7.7.10 Master Link Key Complete event

Event	Event Code	Event Parameters
HCI_Master_Link_Key_Complete	0x0A	Status, Connection_Handle, Key_Flag

Description:

The HCI_Master_Link_Key_Complete event is used to indicate that the Link Key managed by the master of the piconet has been changed. The Connection_Handle will be a Connection_Handle for an ACL connection within that piconet. The link key used for the connection will be the temporary link key of the master device or the semi-permanent link key indicated by the Key_Flag. The Key_Flag event parameter is used to indicate which Link Key (temporary link key of the Master, or the semi-permanent link keys) is now being used in the piconet.

Note: For a master, the change from a semi-permanent Link Key to temporary Link Key will affect all Connection_Handles related to the piconet. For a slave, this change affects only this particular Connection_Handle. A temporary link key must be used when both broadcast and point-to-point traffic are being encrypted.

Event parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	HCI_Master_Link_Key command succeeded.
0x01 to 0xFF	HCI_Master_Link_Key command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Connection_Handle:

Size: 2 octets (12 bits meaningful)

Value	Parameter Description
0xFFFF	Connection_Handle Range: 0x0000 to 0x0EFF

Key_Flag:

Size: 1 octet

Value	Parameter Description
0x00	Using Semi-permanent Link Key.
0x01	Using Temporary Link Key.



7.7.11 Read Remote Supported Features Complete event

Event	Event Code	Event Parameters
HCI_Read_Remote_Supported_Features_Complete	0x0B	Status, Connection_Handle, LMP_Features

Description:

The HCI_Read_Remote_Supported_Features_Complete event is used to indicate the completion of the process of the Link Manager obtaining the supported features of the remote BR/EDR Controller specified by the Connection_Handle event parameter. The Connection_Handle will be a Connection_Handle for an ACL connection. The event parameters include a list of LMP features. For details see [\[Vol 2\] Part C, Link Manager Protocol Specification](#).

Note: If the features are requested more than once while a connection exists between the two devices, the second and subsequent requests may report a cached copy of the features rather than fetching the feature mask again.

Event parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	HCI_Read_Remote_Supported_Features command succeeded.
0x01 to 0xFF	HCI_Read_Remote_Supported_Features command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Connection_Handle:

Size: 2 octets (12 bits meaningful)

Value	Parameter Description
0xFFFF	Connection_Handle Range: 0x0000 to 0x0EFF

LMP_Features:

Size: 8 octets

Value	Parameter Description
0xFFFFFFFF XXXXXXXX	Bit Mask List of LMP features. See [Vol 2] Part C, Link Manager Protocol Specification .



7.7.12 Read Remote Version Information Complete event

Event	Event Code	Event Parameters
HCI_Read_Remote_Version_Information_Complete	0x0C	Status, Connection_Handle, Version, Manufacturer_Name, Subversion

Description:

The HCI_Read_Remote_Version_Information_Complete event is used to indicate the completion of the process obtaining the version information of the remote Controller specified by the Connection_Handle event parameter. The Connection_Handle shall be for an ACL connection.

The Version event parameter defines the specification version of the BR/EDR or LE Controller. The Manufacturer_Name event parameter indicates the manufacturer of the remote Controller. The Subversion event parameter is controlled by the manufacturer and is implementation dependent. The Subversion event parameter defines the various revisions that each version of the Bluetooth hardware will go through as design processes change and errors are fixed. This allows the software to determine what Bluetooth hardware is being used and, if necessary, to work around various bugs in the hardware.

When the Connection_Handle is associated with a BR/EDR ACL-U logical link, the Version event parameter shall be LMP VersNr parameter, the Manufacturer_Name event parameter shall be the Compld parameter, and the Subversion event parameter shall be the LMP SubVersNr parameter.

When the Connection_Handle is associated with an LE-U logical link, the Version event parameter shall be Link Layer VersNr parameter, the Manufacturer_Name event parameter shall be the Compld parameter, and the Subversion event parameter shall be the SubVersNr parameter.

Event parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	HCI_Read_Remote_Version_Information command succeeded.
0x01 to 0xFF	HCI_Read_Remote_Version_Information command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

**Connection_Handle:****Size: 2 octets (12 bits meaningful)**

Value	Parameter Description
0xFFFF	Connection_Handle Range: 0x0000 to 0x0EFF

Version:**Size: 1 octet**

Value	Parameter Description
0xFF	Version of the Current LMP or Link Layer in the remote Controller. See LMP VersNr and Link LayerVersNr in Assigned Numbers .

Manufacturer_Name:**Size: 2 octets**

Value	Parameter Description
0xFFFF	Manufacturer Name of the remote Controller. See Compld in Assigned Numbers .

Subversion:**Size: 2 octets**

Value	Parameter Description
0xFFFF	Subversion of the LMP or Link Layer in the remote Controller. See [Vol 2] Part C, Table 5.2 and [Vol 6] Part B, Section 2.4.2.13 .



7.7.13 QoS Setup Complete event

Event	Event Code	Event Parameters
HCI_QoS_Setup_Complete	0x0D	Status, Connection_Handle, Unused, Service_Type, Token_Rate, Peak_Bandwidth, Latency, Delay_Variation

Description:

The HCI_QoS_Setup_Complete event is used to indicate the completion of the process of the Link Manager setting up QoS with the remote BR/EDR Controller specified by the Connection_Handle event parameter. The Connection_Handle will be a Connection_Handle for an ACL connection. For more detail see [\[Vol 3\] Part A, Logical Link Control and Adaptation Protocol Specification](#).

The Unused parameter is reserved for future use.

This event is generated following an HCI_QoS_Setup command issued by the local Host.

Note: This event or the HCI_Flow_Specification_Complete event can be generated if the remote device performs an LMP transaction involving the flow parameter values.

Event parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	HCI_QoS_Setup command succeeded.
0x01 to 0xFF	HCI_QoS_Setup command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Connection_Handle:

Size: 2 octets (12 bits meaningful)

Value	Parameter Description
0xFFFF	Connection_Handle Range: 0x0000 to 0x0EFF

Host Controller Interface Functional Specification*Unused:**Size: 1 octet*

Value	Parameter Description
0x00	This value shall be used by the Controller.
All other values	Reserved for future use.

*Service_Type:**Size: 1 octet*

Value	Parameter Description
0x00	No Traffic Available.
0x01	Best Effort Available.
0x02	Guaranteed Available.
All other values	Reserved for future use.

*Token_Rate:**Size: 4 octets*

Value	Parameter Description
0XXXXXXXX	Available Token Rate, in octets per second.

*Peak_Bandwidth:**Size: 4 octets*

Value	Parameter Description
0XXXXXXXX	Available Peak Bandwidth, in octets per second.

*Latency:**Size: 4 octets*

Value	Parameter Description
0XXXXXXXX	Available Latency, in microseconds.

*Delay_Variation:**Size: 4 octets*

Value	Parameter Description
0XXXXXXXX	Available Delay Variation, in microseconds.



7.7.14 Command Complete event

Event	Event Code	Event Parameters
HCI_Command_Complete	0x0E	Num_HCI_Command_Packets, Command_Opcode, Return_Parameters

Description:

The HCI_Command_Complete event is used by the Controller for most commands to transmit return status of a command and the other event parameters that are specified for the issued HCI command.

The Num_HCI_Command_Packets event parameter allows the Controller to indicate the number of HCI command packets the Host can send to the Controller. If the Controller requires the Host to stop sending commands, the Num_HCI_Command_Packets event parameter will be set to zero. To indicate to the Host that the Controller is ready to receive HCI command packets, the Controller generates an HCI_Command_Complete event with the Command_Opcode set to 0x0000 and the Num_HCI_Command_Packets event parameter set to 1 or more. Command_Opcode 0x0000 is a special value indicating that this event is not associated with a command sent by the Host. The Controller can send an HCI_Command_Complete event with Command Opcode 0x0000 at any time to change the number of outstanding HCI command packets that the Host can send before waiting. See each command for the parameters that are returned by this event.

Event parameters:

Num_HCI_Command_Packets:

Size: 1 octet

Value	Parameter Description
0xXX	The Number of HCI Command packets which are allowed to be sent to the Controller from the Host. Range: 0 to 255

Command_Opcode:

Size: 2 octets

Value	Parameter Description
0x0000	No associated command
0xFFFF	(non-zero) Opcode of the command which caused this event.

*Return_Parameter(s):**Size: Depends on command*

Value	Parameter Description
0xXX	This is the return parameter(s) for the command specified in the Command_Opcode event parameter. See each command's definition for the list of return parameters associated with that command.



7.7.15 Command Status event

Event	Event Code	Event Parameters
HCI_Command_Status	0x0F	Status, Num_HCI_Command_Packets, Command_Opcode

Description:

The HCI_Command_Status event is used to indicate that the command described by the Command_Opcode parameter has been received, and that the Controller is currently performing the task for this command. This event is needed to provide mechanisms for asynchronous operation, which makes it possible to prevent the Host from waiting for a command to finish. If the command cannot begin to execute (a parameter error may have occurred, or the command may currently not be allowed), the Status event parameter will contain the corresponding error code, and no complete event will follow since the command was not started. The Num_HCI_Command_Packets event parameter allows the Controller to indicate the number of HCI command packets the Host can send to the Controller. If the Controller requires the Host to stop sending commands, the Num_HCI_Command_Packets event parameter will be set to zero. To indicate to the Host that the Controller is ready to receive HCI command packets, the Controller generates an HCI_Command_Status event with Status 0x00 and Command_Opcode 0x0000 and the Num_HCI_Command_Packets event parameter set to 1 or more. Command_Opcode 0x0000 is a special value indicating that this event is not associated with a command sent by the Host. The Controller can send an HCI_Command_Status event with Command Opcode 0x0000 at any time to change the number of outstanding HCI command packets that the Host can send before waiting.

Event parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	Command currently in pending.
0x01 to 0xFF	Command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Num_HCI_Command_Packets:

Size: 1 octet

Value	Parameter Description
0xXX	The Number of HCI Command packets which are allowed to be sent to the Controller from the Host. Range: 0 to 255



Command_Opcode:

Size: 2 octets

Value	Parameter Description
0x0000	No associated command
0xFFFF	(non-zero) Opcode of the command which caused this event and is pending completion.



7.7.16 Hardware Error event

Event	Event Code	Event Parameters
HCI_Hardware_Error	0x10	Hardware_Code

Description:

The HCI_Hardware_Error event is used to notify the Host that a hardware failure has occurred in the Controller.

Event parameters:

Hardware_Code:

Size: 1 octet

Value	Parameter Description
0x00 to 0xFF	These Hardware_Codes will be implementation-specific, and can be assigned to indicate various hardware problems.



7.7.17 Flush Occurred event

Event	Event Code	Event Parameters
HCI_Flush_Occurred	0x11	Handle

Description:

The HCI_Flush_Occurred event is used to indicate that, for the specified Handle, the current user data to be transmitted has been removed. The Handle shall be a Connection_Handle for a BR/EDR LE ACL connection or a Logical_Link_Handle for an AMP ACL connection. This could result from the HCI_Flush command, or be due to the automatic flush. Multiple blocks of an L2CAP packet could have been pending in the Controller. If one Baseband packet part of an L2CAP PDU is flushed, then the rest of the HCI data packets for the L2CAP PDU shall also be flushed.

Event parameters:

Handle:

Size: 2 octets (12 bits meaningful)

Value	Parameter Description
0xFFFF	Handle that was flushed. The Handle is a Connection_Handle for a Primary Controller or a Logical_Link_Handle for an AMP Controller. Range: 0x0000 to 0x0EFF



7.7.18 Role Change event

Event	Event Code	Event Parameters
HCI_Role_Change	0x12	Status, BD_ADDR, New_Role

Description:

The HCI_Role_Change event is used to indicate that the current role of the BR/EDR Controller related to the particular connection has changed. This event only occurs when both the remote and local BR/EDR Controllers have completed their role change for the BR/EDR Controller associated with the BD_ADDR event parameter. This event allows both affected Hosts to be notified when the Role has been changed.

Event parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	A role change has occurred.
0x01 to 0xFF	A role change failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

BD_ADDR:

Size: 6 octets

Value	Parameter Description
0XXXXXXXXXXXXX	BD_ADDR of the Device for which a role change has completed.

New_Role:

Size: 1 octet

Value	Parameter Description
0x00	Currently the Master for specified BD_ADDR.
0x01	Currently the Slave for specified BD_ADDR.



7.7.19 Number Of Completed Packets event

Event	Event Code	Event Parameters
HCI_Number_Of_Completed_Packets	0x13	Number_of_Handles, Connection_Handle[i], Num_Of_Completed_Packets[i]

Description:

The HCI_Number_Of_Completed_Packets event is used by the Controller to indicate to the Host how many HCI Data packets have been completed (transmitted or flushed) for each Connection_Handle since the previous HCI_Number_Of_Completed_Packets event was sent to the Host. This means that the corresponding buffer space has been freed in the Controller. Based on this information, and the Total_Num_ACL_Data_Packets and Total_Num_Synchronous_Data_Packets return parameter of the HCI_Read_Buffer_Size command, the Host can determine for which Connection_Handles the following HCI Data packets should be sent to the Controller. The HCI_Number_Of_Completed_Packets event shall not specify a given Connection_Handle before the HCI_Connection_Complete event for the corresponding connection or after an event indicating disconnection of the corresponding connection. While the Controller has HCI Data packets in its buffer, it shall keep sending the HCI_Number_Of_Completed_Packets event to the Host at least periodically, until it finally reports that all the pending ACL Data packets have been transmitted or flushed. The rate with which this event is sent is manufacturer specific.

Note: HCI_Number_Of_Completed_Packets events will not report on synchronous Connection_Handles if synchronous Flow Control is disabled. (See [Section 7.3.36](#) and [Section 7.3.37](#).)

Event parameters:

Number_of_Handles:

Size: 1 octet

Value	Parameter Description
0xXX	The number of Connection_Handles and Num_HCI_Data_Packets parameters pairs contained in this event. Range: 0 to 255

*Connection_Handle[i]: Size: Number_of_Handles * 2 octets(12 bits meaningful)*

Value	Parameter Description
0xFFFF	Connection_Handle. Range: 0x0000 to 0x0EFF



Num_Of_Completed_Packets [i]: *Size: Number_of_Handles * 2 octets*

Value	Parameter Description
0xFFFF	The number of HCI Data packets that have been completed (transmitted or flushed) for the associated Connection_Handle since the previous time the event was returned. Range: 0x0000 to 0xFFFF



7.7.20 Mode Change event

Event	Event Code	Event Parameters
HCI_Mode_Change	0x14	Status, Connection_Handle, Current_Mode, Interval

Description:

The HCI_Mode_Change event is used to indicate when the device associated with the Connection_Handle changes between Active mode, Hold mode, and Sniff mode. The Connection_Handle will be a Connection_Handle for an ACL connection. The Connection_Handle event parameter is used to indicate which connection the HCI_Mode_Change event is for. The Current_Mode event parameter is used to indicate which state the connection is currently in. The Interval parameter is used to specify a time amount specific to each state. Each Controller that is associated with the Connection_Handle which has changed modes shall send the HCI_Mode_Change event to its Host.

Event parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	A Mode Change has occurred.
0x01 to 0xFF	HCI_Hold_Mode, HCI_Sniff_Mode, or HCI_Exit_Sniff_Mode command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Connection_Handle:

Size: 2 octets(12 bits meaningful)

Value	Parameter Description
0xFFFF	Connection_Handle. Range: 0x0000 to 0x0EFF

Current_Mode:

Size: 1 octet

Value	Parameter Description
0x00	Active mode.
0x01	Hold mode.
0x02	Sniff mode.
All other values	Reserved for future use.



Interval:

Size: 2 octets

Value	Parameter Description
N = 0xFFFF	Hold: Number of Baseband slots to wait in Hold mode. Hold Interval = N * 0.625 ms (1 Baseband slot) Range: 0x0002 to 0xFFFFE Time Range: 1.25 ms to 40.9 s Sniff: Number of Baseband slots between sniff anchor points. Time between sniff anchor points = N * 0.625 ms (1 Baseband slot) Range: 0x0002 to 0xFFFFE Time Range: 1.25 ms to 40.9 s



7.7.21 Return Link Keys event

Event	Event Code	Event Parameters
HCI_Return_Link_Keys	0x15	Num_Keys, BD_ADDR [i], Link_Key[i]

Description:

The HCI_Return_Link_Keys event is used by the BR/EDR Controller to send the Host the BD_ADDRs associated with one or more stored Link Keys. Zero or more instances of this event will occur after the HCI_Read_Stored_Link_Key command. When there are no link keys stored, no HCI_Return_Link_Keys events shall be returned. When there are link keys stored, the number of link keys returned in each HCI_Return_Link_Keys event is implementation specific. This event shall never return the value of the link keys. The link keys value parameter shall always contain the value of zero.

Event parameters:

Num_Keys:

Size: 1 octet

Value	Parameter Description
0xXX	Number of Link Keys contained in this event. Range: 0x01 to 0x0B

BD_ADDR [i]:

*Size: 6 octets * Num_Keys*

Value	Parameter Description
0XXXXXXXXXXXXX	BD_ADDR for the associated Link Key.

Link_Key[i]:

*Size: 16 octets * Num_Keys*

Value	Parameter Description
0x000000000000 00000000000000 000000	Shall be zero.



7.7.22 PIN Code Request event

Event	Event Code	Event Parameters
HCI_PIN_Code_Request	0x16	BD_ADDR

Description:

The HCI_PIN_Code_Request event is used to indicate that a PIN code is required to create a new link key. The Host shall respond using either the HCI_PIN_Code_Request_Reply or the HCI_PIN_Code_Request_Negative_Reply command, depending on whether the Host can provide the Controller with a PIN code or not.

Note: If the HCI_PIN_Code_Request event is masked away, then the BR/EDR Controller will assume that the Host has no PIN Code.

When the BR/EDR Controller generates an HCI_PIN_Code_Request event in order for the local Link Manager to respond to the request from the remote Link Manager (as a result of an HCI_Create_Connection or HCI_Authentication_Requested command from the remote Host), the local Host shall respond with either an HCI_PIN_Code_Request_Reply or HCI_PIN_Code_Request_Negative_Reply command before the remote Link Manager detects LMP response timeout. (See [\[Vol 2\] Part C, Link Manager Protocol Specification](#).)

Event parameters:

BD_ADDR:

Size: 6 octets

Value	Parameter Description
0xFFFFFFFFXXXX	BD_ADDR of the Device which a new link key is being created for.



7.7.23 Link Key Request event

Event	Event Code	Event Parameters
HCI_Link_Key_Request	0x17	BD_ADDR

Description:

The HCI_Link_Key_Request event is used to indicate that a Link Key is required for the connection with the device specified in BD_ADDR. If the Host has the requested stored Link Key, then the Host shall pass the requested Key to the Controller using the HCI_Link_Key_Request_Reply command. If the Host does not have the requested stored Link Key, or the stored Link Key does not meet the security requirements for the requested service, then the Host shall use the HCI_Link_Key_Request_Negative_Reply command to indicate to the Controller that the Host does not have the requested key.

Note: If the HCI_Link_Key_Request event is masked away, then the BR/EDR Controller will assume that the Host has no additional link keys.

If the Host uses the HCI_Link_Key_Request_Negative_Reply command when the requested service requires an authenticated Link Key and the current Link Key is unauthenticated, the Host should set the Authentication_Requirements parameter one of the MITM Protection Required options.

When the Controller generates an HCI_Link_Key_Request event in order for the local Link Manager to respond to the request from the remote Link Manager (as a result of an HCI_Create_Connection or HCI_Authentication_Requested command from the remote Host), the local Host shall respond with either an HCI_Link_Key_Request_Reply or HCI_Link_Key_Request_Negative_Reply command before the remote Link Manager detects LMP response timeout. (See [\[Vol 2\] Part C, Link Manager Protocol Specification](#).)

Event parameters:

BD_ADDR:

Size: 6 octets

Value	Parameter Description
0xFFFFFFFFXXXX	BD_ADDR of the Device which a stored link key is being requested.



7.7.24 Link Key Notification event

Event	Event Code	Event Parameters
HCI_Link_Key_Notification	0x18	BD_ADDR, Link_Key, Key_Type

Description:

The HCI_Link_Key_Notification event is used to indicate to the Host that a new Link Key has been created for the connection with the device specified in BD_ADDR. The Host can save this new Link Key in its own storage for future use. Also, the Host can decide to store the Link Key in the BR/EDR Controller's Link Key Storage by using the HCI_Write_Stored_Link_Key command. The Key_Type event parameter informs the Host about which key type (combination key, debug combination key, unauthenticated combination key, authenticated combination key or changed combination key) was used during pairing. If the key type is not supported or is reserved for future use, the Host may discard the key or disconnect the link.

The combination key Key_Type is used when standard pairing was used. The debug combination key Key_Type is used when Simple Pairing was used and the debug public key is sent or received. The unauthenticated combination key Key_Type is used when the Just Works Simple Pairing association model was used. The authenticated combination key Key_Type is used when Simple Pairing was used and the Just Works association mode was not used. The changed combination key Key_Type is used when the link key has been changed using the Change Connection Link Key procedure and Simple Pairing Mode is set to enabled.

Note: It is the responsibility of the Host to remember the Key_Type (combination, debug combination, unauthenticated combination, or authenticated combination) prior to changing the link key.

When the unauthenticated or authenticated combination key Key_Type is used, the Controller shall use values 0x04 and 0x05 to indicate keys created with the P-192 elliptic curve and values 0x07 through 0x08 to indicate keys created with the P-256 elliptic curve. The values 0x07 and 0x08 shall only be used when the Host has indicated support for Secure Connections in the Secure_Connections_Host_Support parameter.

Event parameters:

BD_ADDR:

Size: 6 octets

Value	Parameter Description
0xFFFFFFFFXXXX	BD_ADDR of the Device for which the new link key has been generated.

**Link_Key:****Size: 16 octets**

Value	Parameter Description
0xFFFFFFFFXXXXXXX XXXXXXXXXXXX XXXXXXXXXXXX	Link Key for the associated BD_ADDR.

Key_Type:**Size: 1 octet**

Value	Parameter Description
0x00	Combination Key
0x03	Debug Combination Key
0x04	Unauthenticated Combination Key generated from P-192
0x05	Authenticated Combination Key generated from P-192
0x06	Changed Combination Key
0x07	Unauthenticated Combination Key generated from P-256
0x08	Authenticated Combination Key generated from P-256
All other values	Reserved for future use



7.7.25 Loopback Command event

Event	Event Code	Event Parameters
HCI_Loopback_Command	0x19	HCI_Command_Packet

Description:

When in Local Loopback mode, the BR/EDR Controller loops back commands and data to the Host. The HCI_Loopback_Command event is used to loop back all commands that the Host sends to the Controller with some exceptions. See [Section 7.6.1](#) for a description of which commands that are not looped back. The HCI_Command_Packet event parameter contains the entire HCI Command Packet including the header.

Note: The event packet is limited to a maximum of 255 octets in the payload; since an HCI Command packet has 3 octets of header data, only the first 252 octets of the command parameters will be returned.

Event parameters:

HCI_Command_Packet:

Size: Depends on command

Value	Parameter Description
0xXXXXXX	HCI Command packet, including header.



7.7.26 Data Buffer Overflow event

Event	Event Code	Event Parameters
HCI_Data_Buffer_Overflow	0x1A	Link_Type

Description:

This event is used to indicate that the Controller's data buffers have been overflowed. This can occur if the Host has sent more packets than allowed. The Link_Type parameter is used to indicate that the overflow was caused by ACL or synchronous data.

Event parameters:

Link_Type:

Size: 1 octet

Value	Parameter Description
0x00	Synchronous Buffer Overflow (Voice Channels).
0x01	ACL Buffer Overflow (Data Channels).
All other values	Reserved for future use.



7.7.27 Max Slots Change event

Event	Event Code	Event Parameters
HCI_Max_Slots_Change	0x1B	Connection_Handle, LMP_Max_Slots

Description:

This event is used to notify the Host about the LMP_Max_Slots parameter when the value of this parameter changes. It shall be sent each time the maximum allowed length, in number of slots, for Baseband packets transmitted by the local device, changes. The Connection_Handle will be a Connection_Handle for an ACL connection.

Event parameters:

Connection_Handle: *Size: 2 octets (12 bits meaningful)*

Value	Parameter Description
0xFFFF	Connection_Handle. Range: 0x0000 to 0x0EFF

LMP_Max_Slots: *Size: 1 octet*

Value	Parameter Description
0x01, 0x03, 0x05	Maximum number of slots allowed to use for Baseband packets, see [Vol 2] Part C, Section 4.1.10 and Section 5.2 .



7.7.28 Read Clock Offset Complete event

Event	Event Code	Event Parameters
HCI_Read_Clock_Offset_Complete	0x1C	Status, Connection_Handle, Clock_Offset

Description:

The HCI_Read_Clock_Offset_Complete event is used to indicate the completion of the process of the Link Manager obtaining the Clock Offset information of the BR/EDR Controller specified by the Connection_Handle event parameter. The Connection_Handle will be a Connection_Handle for an ACL connection.

Event parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	HCI_Read_Clock_Offset command succeeded.
0x01 to 0xFF	HCI_Read_Clock_Offset command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Connection_Handle:

Size: 2 octets (12 bits meaningful)

Value	Parameter Description
0xFFFF	Connection_Handle Range: 0x0000 to 0x0EFF

Clock_Offset:

Size: 2 octets

Bit format	Parameter Description
Bits 14-0	Bits 16-2 of CLKNSlave-CLK
Bit 15	Reserved for future use.



7.7.29 Connection Packet Type Changed event

Event	Event Code	Event Parameters
HCI_Connection_Packet_Type_Changed	0x1D	Status, Connection_Handle, Packet_Type

Description:

The HCI_Connection_Packet_Type_Changed event is used to indicate that the process has completed of the Link Manager changing which packet types can be used for the connection. This allows current connections to be dynamically modified to support different types of user data. The Packet_Type event parameter specifies which packet types the Link Manager can use for the connection identified by the Connection_Handle event parameter for sending L2CAP data or voice. The Packet_Type event parameter does not decide which packet types the LM is allowed to use for sending LMP PDUs.

Event parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	HCI_Change_Connection_Packet_Type command succeeded.
0x01 to 0xFF	HCI_Change_Connection_Packet_Type command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Connection_Handle:

Size: 2 octets (12 bits meaningful)

Value	Parameter Description
0xFFFF	Connection_Handle. Range: 0x0000 to 0x0EFF

Packet_Type:

Size: 2 octets

For ACL_Link_Type

Bit Number	Parameter Description
1	2-DH1 shall not be used.
2	3-DH1 shall not be used.
3 ¹	DM1 may be used.
4	DH1 may be used.
8	2-DH3 shall not be used.
9	3-DH3 shall not be used.
10	DM3 may be used.



Bit Number	Parameter Description
11	DH3 may be used.
12	2-DH5 shall not be used.
13	3-DH5 shall not be used.
14	DM5 may be used.
15	DH5 may be used.
All other bits	Reserved for future use.

1. This bit will be interpreted as set to 1 by Bluetooth V1.2 or higher Controllers.

For SCO_Link_Type

Bit Number	Parameter Description
5	HV1 may be used.
6	HV2 may be used.
7	HV3 may be used.
All other bits	Reserved for future use.



7.7.30 QoS Violation event

Event	Event Code	Event Parameters
HCI_QoS_Violation	0x1E	Handle

Description:

The HCI_QoS_Violation event is used to indicate the Controller is unable to provide the current QoS requirement for the Connection or Logical Link Handle. This event indicates that the Controller is unable to provide one or more of the agreed QoS parameters.

The Host chooses what action should be done. With a BR/EDR Controller the Host can reissue the HCI_QoS_Setup command to renegotiate the QoS setting for Connection_Handle. With an AMP Controller the Host can reissue the HCI_Flow_Spec_Modify command.

Event parameters:

Handle:

Size: 2 octets (12 bits meaningful)

Value	Parameter Description
0xFFFF	Handle for the link that the Controller cannot provide the current QoS requested. The Handle is a Connection_Handle for a BR/EDR Controller and a Logical_Link_Handle for an AMP Controller. Range: 0x0000 to 0x0EFF



7.7.31 Page Scan Repetition Mode Change event

Event	Event Code	Event Parameters
HCI_Page_Scan_Repetition_Mode_Change	0x20	BD_ADDR, Page_Scan_Repetition_Mode

Description:

The HCI_Page_Scan_Repetition_Mode_Change event indicates that the remote BR/EDR Controller with the specified BD_ADDR has successfully changed the Page Scan Repetition Mode (see [\[Vol 2\] Part B, Section 8.3.1](#)).

Event parameters:

BD_ADDR:

Size: 6 octets

Value	Parameter Description
0XXXXXXXXX XXXX	BD_ADDR of the remote device.

Page_Scan_Repetition_Mode:

Size: 1 octet

Value	Parameter Description
0x00	R0
0x01	R1
0x02	R2
All other values	Reserved for future use



7.7.32 Flow Specification Complete event

Event	Event Code	Event Parameters
HCI_Flow_Specification_Complete	0x21	Status, Connection_Handle, Unused, Flow_direction, Service_Type, Token_Rate, Token_Bucket_Size, Peak_Bandwidth, Access Latency

Description:

The HCI_Flow_Specification_Complete event is used to inform the Host about the Quality of Service for the ACL connection the Controller is able to support. The Connection_Handle will be a Connection_Handle for an ACL connection. The flow parameters refer to the outgoing or incoming traffic of the ACL link, as indicated by the Flow_direction field. The flow parameters are defined in the L2CAP specification [\[Vol 3\] Part A, Section 5.3](#). When the Status parameter indicates a successful completion, the flow parameters specify the agreed values by the Controller. When the Status parameter indicates a failed completion with the error code *QoS Unacceptable Parameters* (0x2C), the flow parameters specify the acceptable values of the Controller. This enables the Host to continue the 'QoS negotiation' with a new HCI_Flow_Specification command with flow parameter values that are acceptable for the Controller. When the Status parameter indicates a failed completion with the error code *QoS Rejected* (0x2D), this indicates a request of the Controller to discontinue the 'QoS negotiation'. When the Status parameter indicates a failed completion, the flow parameter values of the most recently successful completion shall be assumed (or the default values when there was no success completion).

The Unused parameter is reserved for future use.

This event is generated following an HCI_Flow_Specification command issued by the local Host.

Note: This event or the HCI_QoS_Setup_Complete event can be generated if the remote device performs an LMP transaction involving the flow parameter values.

**Event parameters:***Status:**Size: 1 octet*

Value	Parameter Description
0x00	HCI_Flow_Specification command succeeded
0x01 to 0xFF	HCI_Flow_Specification command failed. See [Vol 2] Part D, Error Codes for list of error codes

*Connection_Handle:**Size: 2 octets (12 bits meaningful)*

Value	Parameter Description
0xFFFF	Connection_Handle Range: 0x0000 to 0x0EFF

*Unused:**Size: 1 octet*

Value	Parameter Description
0x00	This value shall be used by the Controller.
All other values	Reserved for future use.

*Flow_direction:**Size: 1 octet*

Value	Parameter Description
0x00	Outgoing Flow i.e., traffic sent over the ACL connection.
0x01	Incoming Flow i.e., traffic received over the ACL connection.
All other values	Reserved for future use.

*Service_Type:**Size: 1 octet*

Value	Parameter Description
0x00	No Traffic
0x01	Best Effort
0x02	Guaranteed
All other values	Reserved for future use

*Token Rate:**Size: 4 octets*

Value	Parameter Description
0xFFFFFFFF	Token Rate in octets per second

**Token Bucket Size:****Size: 4 octets**

Value	Parameter Description
0xFFFFFFFF	Token Bucket Size in octets

Peak_Bandwidth:**Size: 4 octets**

Value	Parameter Description
0xFFFFFFFF	Peak Bandwidth in octets per second

Access Latency:**Size: 4 octets**

Value	Parameter Description
0xFFFFFFFF	Access Latency in microseconds



7.7.33 Inquiry Result with RSSI event

Event	Event Code	Event Parameters
HCI_Inquiry_Result_with_RSSI	0x22	Num_responses, BD_ADDR[i], Page_Scan_Repetition_Mode[i], Reserved[i], Class_of_Device[i], Clock_Offset[i], RSSI[i]

Description:

The HCI_Inquiry_Result_with_RSSI event indicates that a BR/EDR Controller or multiple BR/EDR Controllers have responded so far during the current Inquiry process. This event will be sent from the BR/EDR Controller to the Host as soon as an Inquiry Response from a remote device is received if the remote device supports only mandatory paging scheme. This BR/EDR Controller may queue these Inquiry Responses and send multiple BR/EDR Controllers information in one HCI_Inquiry_Result event. The event can be used to return one or more Inquiry responses in one event. The RSSI parameter is measured during the FHS packet returned by each responding slave.

This event shall only be generated if the Inquiry Mode parameter of the last HCI_Write_Inquiry_Mode command was set to 0x01 (Inquiry Result format with RSSI), or was set to 0x02 (Inquiry Result with RSSI format or Extended Inquiry Result format) and the inquiry response packet had the EIR field set to 0.

Event parameters:

Num_Responses:

Size: 1 octet

Value	Parameter Description
0xXX	Number of responses from the Inquiry.

BD_ADDR[i]:

*Size: 6 octets * Num_Responses*

Value	Parameter Description
0XXXXXXXXXX XX	BD_ADDR for each device which responded.

*Page_Scan_Repetition_Mode[i]:**Size: 1 octet * Num_Responses*

Value	Parameter Description
0x00	R0
0x01	R1
0x02	R2
All other values	Reserved for future use

*Reserved[i]:¹**Size: 1 octet * Num_Responses*

Value	Parameter Description
0xXX	Reserved for future use.

*Class_of_Device[i]:**Size: 3 octets * Num_Responses*

Value	Parameter Description
0XXXXXXX	Class of Device for the device

*Clock_Offset[i]:**Size: 2 octets * Num_Responses*

Bit format	Parameter Description
Bits 14-0	Bits 16-2 of CLKNSlave-CLK
Bit 15	Reserved for future use

*RSSI[i]:**Size: 1 octet * Num_Responses*

Value	Parameter Description
0xXX	Range: -127 to +20 Units: dBm

1. This was the Page_Scan_Period_Mode parameter in the v1.1 specification. This parameter has no meaning in v1.2 or higher and no default value.



7.7.34 Read Remote Extended Features Complete event

Event	Event Code	Event Parameters
HCI_Read_Remote_Extended_Features_Complete	0x23	Status, Connection_Handle, Page_Number, Maximum Page Number, Extended_LMP_Features

Description:

The HCI_Read_Remote_Extended_Features_Complete event is used to indicate the completion of the process of the Link Manager obtaining the remote extended LMP features of the remote device specified by the Connection_Handle event parameter. The Connection_Handle will be a Connection_Handle for an ACL connection. The event parameters include a page of the remote devices extended LMP features. For details see [\[Vol 2\] Part C, Link Manager Protocol Specification](#).

Note: If a feature page is requested more than once while a connection exists between the two devices, the second and subsequent requests may report a cached copy of that page rather than fetching it again.

Event parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	Request for remote extended features succeeded
0x01 to 0xFF	Request for remote extended features failed. See [Vol 2] Part D, Error Codes for error codes.

Connection_Handle:

Size: 2 octets (12 bits meaningful)

Value	Parameter Description
0xFFFF	Connection_Handle Range: 0x0000 to 0x0EFF

Page Number:

Size: 1 octet

Value	Parameter Description
0x00	The normal LMP features as returned by HCI_Read_Remote_Supported_Features command
0x01 to 0xFF	The page number of the features returned



Maximum Page Number:

Size: 1 octet

Value	Parameter Description
0x00 to 0xFF	The highest features page number which contains non-zero bits for the remote device

Extended_LMP_Features:

Size: 8 octets

Value	Parameter Description
0xFFFFFFFFFFFFFFFF	Bit map of requested page of LMP features. See LMP [Vol 2] Part C, Section 3.3 for details.



7.7.35 Synchronous Connection Complete event

Event	Event Code	Event Parameters
HCI_Synchronous_ - Connection_Complete	0x2C	Status, Connection_Handle, BD_ADDR, Link Type, Transmission_Interval, Retransmission Window, Rx_Packet_Length, Tx_Packet_Length, Air Mode

Description:

The HCI_Synchronous_Connection_Complete event is sent to indicate completion of any of the following commands:

- HCI_Setup_Synchronous_Connection
- HCI_Accept_Synchronous_Connection_Request
- HCI_Reject_Synchronous_Connection_Request
- HCI_Enhanced_Setup_Synchronous_Connection
- HCI_Enhanced_Accept_Synchronous_Connection_Request

This event returns the completion status for the command.

When the HCI_Synchronous_Connection_Complete event was triggered by the HCI_Enhanced_Setup_Synchronous_Connection or HCI_Enhanced_Accept_Synchronous_Connection_Request commands, the Controller shall set the Air Mode parameter to the Transmit Air Coding Format parameter of the original command.

Event parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	Connection successfully completed.
0x01 to 0xFF	Connection failed to complete. See [Vol 2] Part D, Error Codes for error codes and descriptions.

**Connection_Handle:****Size: 2 octets (12 bits meaningful)**

Value	Parameter Description
0xXXXX	Connection_Handle Range: 0x0000 to 0x0EFF

BD_ADDR:**Size: 6 octets**

Value	Parameter Description
0XXXXXXXXXXXX	BD_ADDR of the other connected device forming the connection.

Link_Type:**Size: 1 octet**

Value	Parameter Description
0x00	SCO connection
0x01	Reserved for future use
0x02	eSCO connection
All other values	Reserved for future use

Transmission_Interval:**Size: 1 octet**

Value	Parameter Description
0xXX	Time between two consecutive eSCO instants measured in slots. Shall be zero for SCO links.

Retransmission window:**Size: 1 octet**

Value	Parameter Description
0xXX	The size of the retransmission window measured in slots. Shall be zero for SCO links.

Rx_Packet_Length:**Size: 2 octets**

Value	Parameter Description
0XXXX	Length in bytes of the eSCO payload in the receive direction. Shall be zero for SCO links.

Tx_Packet_Length:**Size: 2 octets**

Value	Parameter Description
0XXXX	Length in bytes of the eSCO payload in the transmit direction. Shall be zero for SCO links.

**Air Mode:****Size: 1 octet**

Value	Parameter Description
0x00	μ -law log
0x01	A-law log
0x02	CVSD
0x03	Transparent Data
All other values	Reserved for future use



7.7.36 Synchronous Connection Changed event

Event	Event Code	Event Parameters
HCI_Synchronous_Connection_Changed	0x2D	Status, Connection_Handle, Transmission_Interval, Retransmission Window, Rx_Packet_Length, Tx_Packet_Length

Description:

The HCI_Synchronous_Connection_Changed event indicates to the Host that an existing synchronous connection has been reconfigured. This event also indicates to the initiating Host (if the change was Host initiated) if the issued command failed or was successful.

Command parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	Connection successfully reconfigured.
0x01 to 0xFF	Reconfiguration failed to complete. See [Vol 2] Part D, Error Codes for error codes and descriptions.

Connection_Handle:

Size: 2 octets (12 bits meaningful)

Value	Parameter Description
0xFFFF	Connection_Handle Range: 0x0000 to 0x0EFF

Transmission_Interval:

Size: 1 octet

Value	Parameter Description
0xFF	Time between two consecutive SCO/eSCO instants measured in slots.

Retransmission window:

Size: 1 octet

Value	Parameter Description
0xFF	The size of the retransmission window measured in slots. Shall be zero for SCO links.

***Rx_Packet_Length:******Size: 2 octets***

Value	Parameter Description
0xXXXX	Length in bytes of the SCO/eSCO payload in the receive direction.

Tx_Packet_Length:***Size: 2 octets***

Value	Parameter Description
0xXXXX	Length in bytes of the SCO/eSCO payload in the transmit direction.



7.7.37 Sniff Subrating event

Event	Event Code	Event Parameters
HCI_Sniff_Subrating	0x2E	Status, Connection_Handle, Maximum_Transmit_Latency, Maximum_Receive_Latency, Minimum_Remote_Timeout, Minimum_Local_Timeout

Description:

The HCI_Sniff_Subrating event indicates that the device associated with the Connection_Handle has either enabled sniff subrating or the sniff subrating parameters have been renegotiated by the link manager. The Connection_Handle will be a Connection_Handle for an ACL connection. The Connection_Handle event parameter indicates which connection the HCI_Sniff_Subrating event is for.

Each BR/EDR Controller that is associated with the Connection_Handle that has changed its subrating parameters will send the Sniff Subrating event to its Host.

Event parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	HCI_Sniff_Subrating command succeeded
0x01 to 0xFF	HCI_Sniff_Subrating command failed to complete. See [Vol 2] Part D, Error Codes for error codes and descriptions.

Connection_Handle:

Size: 2 octets (12 bits meaningful)

Value	Parameter Description
0xFFFF	Connection_Handle. Range: 0x0000 to 0x0EFF

**Maximum_Transmit_Latency:****Size: 2 octets**

Value	Parameter Description
N = 0xFFFF	Maximum latency for data being transmitted from the local device to the remote device. Latency = N * 0.625 ms (1 Baseband slot) Range: 0x0000 to 0xFFFFE Time Range: 0 s to 40.9 s

Maximum_Receive_Latency:**Size: 2 octets**

Value	Parameter Description
N = 0xFFFF	Maximum latency for data being received by the local device from the remote device. Latency = N * 0.625 ms (1 Baseband slot) Range: 0x0000 to 0xFFFFE Time Range: 0 s to 40.9 s

Minimum_Remote_Timeout:**Size: 2 octets**

Value	Parameter Description
N = 0xFFFF	The base sniff subrate timeout in Baseband slots that the remote device shall use. Timeout = N * 0.625 ms (1 Baseband slot) Range: 0x0000 to 0xFFFFE Time Range: 0 s to 40.9 s

Minimum_Local_Timeout:**Size: 2 octets**

Value	Parameter Description
N = 0xFFFF	The base sniff subrate timeout in Baseband slots that the local device will use. Timeout = N * 0.625 ms (1 Baseband slot) Range: 0x0000 to 0xFFFFE Time Range: 0 s to 40.9 s



7.7.38 Extended Inquiry Result event

Event	Event Code	Event Parameters
HCI_Extended_Inquiry_ - Result	0x2F	Num_Responses, BD_ADDR, Page_Scan_Repetition_Mode, Reserved, Class_of_Device, Clock_Offset, RSSI, Extended_Inquiry_Response

Description:

The HCI_Extended_Inquiry_Result event indicates that a BR/EDR Controller has responded during the current inquiry process with extended inquiry response data. This event will be sent from the BR/EDR Controller to the Host upon reception of an Extended Inquiry Response from a remote device. One single Extended Inquiry Response is returned per event. This event contains RSSI and inquiry response data for the BR/EDR Controller that responded to the latest inquiry. The RSSI parameter is measured during the FHS packet returned by each responding slave. The Num_Responses parameter shall be set to one.

This event is only generated if the Inquiry_Mode parameter of the last HCI_Write_Inquiry_Mode command was set to 0x02 (Inquiry Result with RSSI format or Extended Inquiry Result format).

Note: This ensures that a Host that does not support Extended Inquiry Results will never receive the HCI_Extended_Inquiry_Result event.

If an inquiry response packet with the EIR field set to zero is received, the HCI_Inquiry_Result_with_RSSI event format shall be used. If the EIR bit is set to one the HCI_Extended_Inquiry_Result event format shall be used. If the EIR bit is set to one but the Controller failed to receive the extended inquiry response packet, the Extended_Inquiry_Response parameter is set to zeros. If an extended inquiry response packet from the same device is correctly received in a later response, another event shall be generated.

Note: The only difference between the HCI_Extended_Inquiry_Result event and the HCI_Inquiry_Result_with_RSSI event is the additional Extended_Inquiry_Response parameter.

Note: The Extended_Inquiry_Response parameter is not interpreted by the Controller. The tagged data set by the other Host should be passed unaltered if it has been correctly received.

**Event parameters:***Num_Responses:**Size: 1 octet*

Value	Parameter Description
0x01	Number of responses from the inquiry. The HCI_Extended_Inquiry_Result event always contains a single response.

*BD_ADDR:**Size: 6 octets*

Value	Parameter Description
0XXXXXXXXX XXXX	BD_ADDR for the device that responded.

*Page_Scan_Repetition_Mode:**Size: 1 octet*

Value	Parameter Description
0x00	R0
0x01	R1
0x02	R2
All other values	Reserved for future use

*Reserved:**Size: 1 octet*

Value	Parameter Description
0xXX	Reserved for future use.

*Class_of_Device:**Size: 3 octets*

Value	Parameter Description
0XXXXXXX	Class of Device for the device that responded.

*Clock_Offset**Size: 2 octets*

Value	Parameter Description
Bits 14-0	Bits 16-2 of CLKNSlave-CLK
Bit 15	Reserved for future use.

*RSSI:**Size: 1 octet*

Value	Parameter Description
0xXX	Range: -127 to +20 Units: dBm



Extended_Inquiry_Response:

Size: 240 octets

Value	Parameter Description
	Extended Inquiry Response data as defined in [Vol 3] Part C, Section 8



7.7.39 Encryption Key Refresh Complete event

Command	Event Code	Event Parameters
HCI_Encryption_Key_Refresh_Complete	0x30	Status, Connection_Handle

Description:

The HCI_Encryption_Key_Refresh_Complete event is used to indicate to the Host that the encryption key was refreshed on the given Connection_Handle any time encryption is paused and then resumed. The BR/EDR Controller shall send this event when the encryption key has been refreshed due to encryption being started or resumed.

If the HCI_Encryption_Key_Refresh_Complete event was generated due to an encryption pause and resume operation embedded within a change connection link key procedure, the HCI_Encryption_Key_Refresh_Complete event shall be sent prior to the HCI_Change_Connection_Link_Key_Complete event.

If the HCI_Encryption_Key_Refresh_Complete event was generated due to an encryption pause and resume operation embedded within a role switch procedure, the HCI_Encryption_Key_Refresh_Complete event shall be sent prior to the HCI_Role_Change event.

Event parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	Encryption key refresh completed successfully
0x01 to 0xFF	Encryption key refresh failed. See [Vol 2] Part D, Error Codes for list of error codes

Connection_Handle:

Size: 2 octets (12 bits meaningful)

Value	Parameter Description
0xFFFF	Connection_Handle Range: 0x0000 to 0x0EFF



7.7.40 IO Capability Request event

Event	Event Code	Event Parameters
HCI_IO_Capability_Request	0x31	BD_ADDR

Description:

The HCI_IO_Capability_Request event is used to indicate that the IO capabilities of the Host are required for a simple pairing process. The Host shall respond with an HCI_IO_Capability_Request_Reply command. This event shall only be generated if simple pairing has been enabled with the HCI_Write_Simple_Pairing_Mode command.

Event parameters:

BD_ADDR:

Size: 6 octets

Value	Parameter Description
0XXXXXXXXXXXXX	BD_ADDR of remote device involved in simple pairing process



7.7.41 IO Capability Response event

Event	Event Code	Event Parameters
HCI_IO_Capability_Response	0x32	BD_ADDR, IO_Capability, OOB_Data_Present, Authentication_Requirements

Description:

The HCI_IO_Capability_Response event is used to indicate to the Host that IO capabilities from a remote device specified by BD_ADDR have been received during a simple pairing process. This event will only be generated if simple pairing has been enabled with the HCI_Write_Simple_Pairing_Mode command.

Event parameters:

BD_ADDR:

Size: 6 octets

Value	Parameter Description
0XXXXXXXXXXXXX	BD_ADDR identifying the device to which the IO capabilities apply.

IO_Capability:

Size: 1 octet

Value	Parameter Description
0x00	DisplayOnly
0x01	DisplayYesNo
0x02	KeyboardOnly
0x03	NoInputNoOutput
All other values	Reserved for future use

OOB_Data_Present:

Size: 1 octet

Value	Parameter Description
0x00	OOB authentication data not present
0x01	OOB authentication data from remote device present
All other values	Reserved for future use

Authentication_Requirements:

Size: 1 octet

Value	Parameter Description
0x00	MITM Protection Not Required – No Bonding. Numeric comparison with automatic accept allowed.



Value	Parameter Description
0x01	MITM Protection Required – No Bonding. Use IO Capabilities to determine authentication procedure
0x02	MITM Protection Not Required – Dedicated Bonding. Numeric comparison with automatic accept allowed.
0x03	MITM Protection Required – Dedicated Bonding. Use IO Capabilities to determine authentication procedure
0x04	MITM Protection Not Required – General Bonding. Numeric Comparison with automatic accept allowed.
0x05	MITM Protection Required – General Bonding. Use IO capabilities to determine authentication procedure.
All other values	Reserved for future use



7.7.42 User Confirmation Request event

Event	Event Code	Event Parameters
HCI_User_Confirmation_Request	0x33	BD_ADDR, Numeric_Value

Description:

The HCI_User_Confirmation_Request event is used to indicate that user confirmation of a numeric value is required. The Host shall reply with either the HCI_User_Confirmation_Request_Reply or the HCI_User_Confirmation_Request_Negative_Reply command. If the Host has output capability (DisplayYesNo or KeyboardOnly), it shall display the Numeric_Value until the HCI_Simple_Pairing_Complete event is received. It shall reply based on the yes/no response from the user. If the Host has no input and no output it shall reply with the HCI_User_Confirmation_Request_Reply command. When the Controller generates an HCI_User_Confirmation_Request event, in order for the local Link Manager to respond to the request from the remote Link Manager, the local Host shall respond with either an HCI_User_Confirmation_Request_Reply or an HCI_User_Confirmation_Request_Negative_Reply command before the remote Link Manager detects LMP response timeout. (See [\[Vol 2\] Part C, Link Manager Protocol Specification](#).)

Event parameters:

BD_ADDR:

Size: 6 octets

Value	Parameter Description
0xFFFFFFFFXXXX	BD_ADDR of device involved in Simple Pairing process.

Numeric_Value:

Size: 4 octets

Value	Parameter Description
0x00000000 to 0x000F423F	Numeric value to be displayed. Valid values are decimal 000000 to 999999.



7.7.43 User Passkey Request event

Event	Event Code	Event Parameters
HCI_User_Passkey_Request	0x34	BD_ADDR

Description:

The HCI_User_Passkey_Request event is used to indicate that a passkey is required as part of a Simple Pairing process. The Host shall respond with either an HCI_User_Passkey_Request_Reply or HCI_User_Passkey_Request_Negative_Reply command. This event will only be generated if simple pairing has been enabled with the HCI_Write_Simple_Pairing_Mode command. When the Controller generates an HCI_User_Passkey_Request event, in order for the local Link Manager to respond to the request from the remote Link Manager, the local Host shall respond with either an HCI_User_Passkey_Request_Reply or HCI_User_Passkey_Request_Negative_Reply command before the remote Link Manager detects LMP response timeout. (See [\[Vol 2\] Part C, Link Manager Protocol Specification](#).)

Event parameters:

BD_ADDR:

Size: 6 octets

Value	Parameter Description
0xFFFFFFFFXXXX	BD_ADDR of the device involved in simple pairing process.



7.7.44 Remote OOB Data Request event

Event	Event Code	Event Parameters
HCI_Remote_OOB_Data_Request	0x35	BD_ADDR

Description:

The HCI_Remote_OOB_Data_Request event is used to indicate that the Simple Pairing Hash C and the Simple Pairing Randomizer R is required for the Secure Simple Pairing process involving the device identified by BD_ADDR. The C and R values were transferred to the Host from the remote device via an OOB mechanism. This event is sent by the Controller because the Host previously set the OOB Data Present parameter to "OOB authentication data from remote device present" in an HCI_IO_Capability_Request_Reply command. If both the Host and Controller support Secure Connections the Host shall respond with the HCI_Remote_OOB_Extended_Data_Request_Reply command. Otherwise, the Host shall respond with the HCI_Remote_OOB_Data_Request_Reply command.

Event parameters:

BD_ADDR:

Size: 6 octets

Value	Parameter Description
0xFFFFFFFFXXXX	BD_ADDR of the device from which the C and R values were received.



7.7.45 Simple Pairing Complete event

Event	Event Code	Event Parameters
HCI_Simple_Pairing_Complete	0x36	Status, BD_ADDR

Description:

The HCI_Simple_Pairing_Complete event is used to indicate that the simple pairing process has completed. A Host that is displaying a numeric value can use this event to change its UI.

When the LMP simple pairing sequences fail for any reason, the HCI_Simple_Pairing_Complete event shall be sent to the Host. When HCI_Simple_Pairing_Complete event is sent in response to the IO capability exchange failing, the Status parameter shall be set to the error code received from the remote device. Otherwise, the Status shall be set to the error code *Authentication Failure* (0x05).

Event parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	Simple Pairing succeeded
0x01 to 0xFF	Simple Pairing failed. See [Vol 2] Part D, Error Codes for a list of error codes.

BD_ADDR:

Size: 6 octets

Value	Parameter Description
0xFFFFFFFFXXXX	BD_ADDR of the device involved in simple pairing process.



7.7.46 Link Supervision Timeout Changed event

Event	Event Code	Event Parameters
HCI_Link_Supervision_Timeout_Changed	0x38	Connection_Handle, Link_Supervision_Timeout

Description:

The HCI_Link_Supervision_Timeout_Changed event is used to notify the slave's Host when the Link_Supervision_Timeout parameter is changed in the slave Controller. This event shall only be sent to the Host by the slave Controller upon receiving an LMP_supervision_timeout PDU from the master.

Note: The Connection_Handle used for this command shall be the ACL connection of the appropriate device.

Event parameters:

Connection_Handle: *Size: 2 octets (12 bits meaningful)*

Value	Parameter Description
0xXXXX	Connection_Handle Range: 0x0000 to 0x0EFF

Link_Supervision_Timeout: *Size: 2 octets*

Value	Parameter Description
N = 0xXXXX	Measured in number of Baseband slots Link_Supervision_Timeout = N*0.625 ms (1 Baseband slot) Range: 0x0001 to 0xFFFF Time Range: 0.625ms to 40.9 s (0 means infinite timeout)



7.7.47 Enhanced Flush Complete event

Event	Event Code	Event Parameters
HCI_Enhanced_Flush_Complete	0x39	Handle

Description:

The HCI_Enhanced_Flush_Complete event is used to indicate that an Enhanced Flush is complete for the specified Handle.

Event parameters:

Handle: *Size: 2 octets (12 bits meaningful)*

Value	Parameter Description
0xXXXX	Handle of the connection for which the Enhanced Flush was done. The Handle is a Connection_Handle for a BR/EDR Controller and a Logical_Link_Handle for an AMP Controller. Range: 0x0000 to 0x0EFF



7.7.48 User Passkey Notification event

Event	Event Code	Event Parameters
HCI_User_Passkey_Notification	0x3B	BD_ADDR, Passkey

Description:

The HCI_User_Passkey_Notification event is used to provide a passkey for the Host to display to the user as required as part of a Simple Pairing process. The Passkey parameter shall be a randomly generated number (see [Vol 2] Part H, Section 2) generated by the Controller modulo 1,000,000.

This event will be generated if the IO capabilities of the local device are DisplayOnly or DisplayYesNo and the IO capabilities of the remote device are KeyboardOnly.

This event shall only be generated if simple pairing has been enabled with the HCI_Write_Simple_Pairing_Mode command.

Event parameters:

BD_ADDR:

Size: 6 octets

Value	Parameter Description
0xFFFFFFFFXXXX	BD_ADDR of the device involved in simple pairing process.

Passkey:

Size: 4 octets

Value	Parameter Description
0x00000000 to 0x000F423F	Passkey to be displayed. Valid values are decimal 000000 to 999999.



7.7.49 Keypress Notification event

Command	Event Code	Event Parameters
HCI_Keypress_ Notification	0x3C	BD_ADDR, Notification_Type

Description:

The HCI_Keypress_Notification event is sent to the Host after a passkey notification has been received by the Link Manager on the given BD_ADDR. The Notification_Type parameter may be used by the Host's user interface.

Event parameters:

BD_ADDR:

Size: 6 octets

Value	Parameter Description
0XXXXXXXXXXXXX	BD_ADDR of remote device involved in simple pairing process

Notification_Type:

Size: 1 octet

Value	Parameter Description
0	Passkey entry started
1	Passkey digit entered
2	Passkey digit erased
3	Passkey cleared
4	Passkey entry completed
All other values	Reserved for future use



7.7.50 Remote Host Supported Features Notification event

Command	Event Code	Event Parameters
HCI_Remote_Host_Supported_Features_Notification	0x3D	BD_ADDR, Host_Supported_Features

Description:

The HCI_Remote_Host_Supported_Features_Notification event is used to return the LMP extended features page containing the Host features. The BD_ADDR shall be the address of the remote device.

This event shall only be generated after the LMP extended features are read from the remote device during a connection initiated by an HCI_Remote_Name_Request command.

Note: This event is not generated during a connection initiated by the HCI_Create_Connection command.

Event parameters:

BD_ADDR:

Size: 6 octets

Value	Parameter Description
0XXXXXXXXXXXXX	BD_ADDR of remote device.

Host_Supported_Features:

Size: 8 octets

Value	Parameter Description
0xFFFFFFFFFFFF FFF	Bit map of Host Supported Features page of LMP extended features. For more information, see [Vol 2] Part C, Link Manager Protocol Specification .



7.7.51 Physical Link Complete event

Event	Event Code	Event Parameters
HCI_Physical_Link_Complete	0x40	Status, Physical_Link_Handle

Description:

The HCI_Physical_Link_Complete event indicates to the Host that a new AMP physical link has been established. This event is used as a response for either HCI_Create_Physical_Link or HCI_Accept_Physical_Link commands and is preceded by an HCI_Command_Status event.

Event parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	Connection successfully completed.
0x01 to 0xFF	Connection failed to Complete. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Physical_Link_Handle:

Size: 1 octet

Value	Parameter Description
0xXX	Physical_Link_Handle identifying the physical link which has been created.



7.7.52 Channel Selected event

Event	Event Code	Event Parameters
HCI_Channel_Selected	0x41	Physical_Link_Handle

Description:

The HCI_Channel_Selected event indicates to the Host originating the physical link that the link information data is available to be read using the HCI_Read_Local_AMP_Assoc command. This allows the originating device to determine link implementation parameters such as an underlying physical channel choice. This structure shall be sent in the AMP Create Physical Link Request (see [\[Vol 3\] Part E, Section 3.11](#)) and be used at the remote end in the HCI_Accept_Physical_Link command.

Event parameters:

Physical_Link_Handle:

Size: 1 octet

Value	Parameter Description
0xXX	Physical_Link_Handle to the established physical link.



7.7.53 Disconnection Physical Link Complete event

Event	Event Code	Event Parameters
HCI_Disconnection_Physical_Link_Complete	0x42	Status, Physical_Link_Handle, Reason

Description:

The HCI_Disconnection_Physical_Link_Complete event occurs when a physical link is terminated. The status parameter indicates if the disconnection was successful or not.

When a physical link fails, one HCI_Disconnection_Logical_Link_Complete event will be returned for each logical channel on the physical link with the corresponding Logical_Link_Handle as a parameter.

Event parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	Disconnection has occurred.
0x01 to 0xFF	Disconnection failed to complete. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Physical_Link_Handle:

Size: 1 octet

Value	Parameter Description
0xXX	Physical_Link_Handle which was disconnected.

Reason:

Size: 1 octet

Value	Parameter Description
0xXX	Reason for disconnection. See [Vol 2] Part D, Error Codes for error codes and descriptions.



7.7.54 Physical Link Loss Early Warning event

Event	Event Code	Event Parameters
HCI_Physical_Link_Loss_Early_Warning	0x43	Physical_Link_Handle, Link_Loss_Reason

Description:

The HCI_Physical_Link_Loss_Early_Warning event occurs when a physical link has early indication that the link may be disrupted. Data transmission and reception may suffer impairment on this link. The Host may use this to indicate to the user that traffic may be disrupted.

The exact low level meaning is dependent on the AMP type. Some AMP types may never generate this event.

Event parameters:

Physical_Link_Handle:

Size: 1 octet

Value	Parameter Description
0xXX	Physical_Link_Handle to which the warning refers.

Link_Loss_Reason:

Size: 1 octet

Value	Parameter Description
0	Unknown
1	Range related
2	Bandwidth related
3	Resolving conflict
4	Interference
All other values	Reserved for future use



7.7.55 Physical Link Recovery event

Event	Event Code	Event Parameters
HCI_Physical_Link_Recovery	0x44	Physical_Link_Handle

Description:

The HCI_Physical_Link_Recovery event indicates that whatever interruption caused an earlier HCI_Physical_Link_Loss_Early_Warning event has now been cleared. The normal data transmission and reception service can be expected. The exact low level meaning is dependent on the AMP type. The Host may use this event to cancel any indication to the user caused by the HCI_Physical_Link_Loss_Early_Warning event.

Event parameters:

Physical_Link_Handle:

Size: 1 octet

Value	Parameter Description
0xXX	Physical_Link_Handle to which the warning referred.



7.7.56 Logical Link Complete event

Event	Event Code	Event Parameters
HCI_Logical_Link_Complete	0x45	Status, Logical_Link_Handle, Physical_Link_Handle, TX_Flow_Spec_ID

Description:

The HCI_Logical_Link_Complete event indicates to both of the Hosts forming the connection whether a new connection has been established successfully.

Event parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	Connection successfully completed.
0x01-0xFF	Connection failed to Complete. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Logical_Link_Handle:

Size: 2 octets (12 bits meaningful)

Value	Parameter Description
0xFFFF	Logical_Link_Handle to be used to identify a connection between two BR/EDR Controllers with communicating AMPs. The Logical_Link_Handle is used as an identifier for transmitting and receiving data. Range: 0x0000 to 0x0EFF

Physical_Link_Handle:

Size: 1 octet

Value	Parameter Description
0xFF	Physical_Link identifying the physical link over which the logical link has been created.

TX_Flow_Spec_ID:

Size: 1 octet

Value	Parameter Description
0xFF	Flow Spec ID identifying the logical link creation that completed. This matches the ID field of the TX_Flow_Spec



7.7.57 Disconnection Logical Link Complete event

Event	Event Code	Event Parameters
HCI_Disconnection_Logical_Link_Complete	0x46	Status, Logical_Link_Handle, Reason

Description:

The HCI_Disconnection_Logical_Link_Complete event occurs when a logical link is terminated on the local Controller. The status parameter indicates if the disconnection was successful or not.

Note: When the local Controller disconnects a logical link, the remote Controller will not be notified.

Note: When a physical link fails, one HCI_Disconnection_Logical_Link_Complete event will be returned for each logical channel on the physical link with the corresponding Logical_Link_Handle as a parameter.

Event parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	Disconnection has occurred.
0x01 to 0xFF	Disconnection failed to complete. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Logical_Link_Handle:

Size: 2 octets (12 bits meaningful)

Value	Parameter Description
0xFFFF	Logical_Link_Handle which was disconnected. Range: 0x0000 to 0xFFFF

Reason:

Size: 1 octet

Value	Parameter Description
0xFF	Reason for disconnection. See [Vol 2] Part D, Error Codes for error codes and descriptions.



7.7.58 Flow Spec Modify Complete event

Event	Event Code	Event Parameters
HCI_Flow_Spec_Modify_Complete	0x47	Status, Handle

Description:

The HCI_Flow_Spec_Modify_Complete event indicates to the Host that an HCI_Flow_Spec_Modify command has been completed. This event also indicates to the Host which issued the HCI_Flow_Spec_Modify command and then received an HCI_Command_Status event if the issued command failed or was successful.

Event parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	HCI_Flow_Spec_Modify command successfully completed.
0x01 to 0xFF	HCI_Flow_Spec_Modify command failed to complete. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Handle:

Size: 2 octets (12 bits meaningful)

Value	Parameter Description
0xFFFF	Connection_Handle if the receiving Controller is a BR/EDR Controller. Logical_Link_Handle, if the receiving Controller is an AMP Controller, identifying the logical link to be changed. Logical_Link_Handle to be used to identify a connection between two BR/EDR Controllers with communicating AMPs. Range: 0x0000 to 0x0EFF



7.7.59 Number Of Completed Data Blocks event

Event	Event Code	Event Parameters
HCI_Number_Of_Completed_Data_Blocks	0x48	Total_Num_Data_Blocks, Number_of_Handles, Handle[i], Num_Of_Completed_Packets[i], Num_Of_Completed_Blocks[i]

Description:

The HCI_Number_Of_Completed_Data_Blocks event is used by the Controller to indicate to the Host how many HCI ACL Data packets have been completed (transmitted or flushed), and how many data block buffers have been freed, for each Handle (a Connection_Handle if the Controller is a BR/EDR Controller or a Logical_Link_Handle if the Controller is an AMP Controller) since the previous HCI_Number_Of_Completed_Data_Blocks event was sent to the Host. This means that the corresponding buffer space has been freed in the Controller. Based on this information, and the Total_Num_Data_Blocks parameter, the Host can determine for which Handles the following HCI ACL Data packets should be sent to the Controller.

The Host should determine the number of blocks occupied by each ACL data packet by dividing the ACL data packet size by the Data_Block_Length parameter of the HCI_Read_Data_Block_Size command.

The Total_Num_Data_Blocks parameter indicates the total number of buffer blocks available in the Controller. Before any HCI_Number_Of_Completed_Data_Blocks event is received, the value of Total_Num_Data_Blocks from the HCI_Read_Data_Block_Size command is used. This allows the value to be updated at any time, which provides the Controller with some flexibility on its buffer allocation.

If the Controller were permitted to reduce its buffer pool in an arbitrary way then there is a potential race condition, in the case where the Host has just started to transmit a new packet. In order to prevent this race condition, the Total_Num_Data_Blocks parameter shall not indicate a reduction in the pool of blocks greater than the sum of the Num_Of_Completed_Blocks values in this event. If a greater reduction in the block pool is required then the value 0 shall be indicated here. The value 0 has a special meaning and indicates that the Host shall re-issue the HCI_Read_Data_Block_Size command in order to find the new buffer pool size. The Host shall wait for any outstanding TX to complete and shall defer further TX until the HCI_Read_Data_Block_Size command has been issued and completed. The Controller shall reduce its allocation only after the HCI_Read_Data_Block_Size command has been issued and completed. This ensures that the race condition described above is avoided.

**Event parameters:***Total_Num_Data_Blocks:**Size: 2 octets*

Value	Parameter Description
0x0000	The size of the buffer pool may have changed. The Host is requested to issue an HCI_Read_Data_Block_Size command in order to determine the new value of Total_Num_Data_Blocks.
0xFFFF	Total number of data block buffers available in the Controller for the storage of data packets scheduled for transmission. This indicates the existing value is unchanged, or increased, or reduced by up to the sum of the Num_Of_Completed_Blocks values in this command.

*Number_of_Handles:**Size: 1 octet*

Value	Parameter Description
0xFF	The number of Handles and Num_Of_Completed_Packets and Num_Of_Completed_Blocks parameter triples contained in this event. Range: 0 to 255

*Handle[i]:**Size: Number_of_Handles * 2 octets(12 bits meaningful)*

Value	Parameter Description
0xFFFF	Handle (Connection_Handle for a BR/EDR Controller or a Logical_Link_Handle for an AMP Controller). Range: 0x0000 to 0xFFFF

*Num_Of_Completed_Packets [i]:**Size: Number_of_Handles * 2 octets*

Value	Parameter Description
0xFFFF	The number of HCI ACL Data packets that have been completed (transmitted or flushed) for the associated Handle since the previous time that an HCI_Number_Of_Completed_Data_Blocks event provided information about this Handle. Range: 0x0000 to 0xFFFF

*Num_Of_Completed_Blocks [i]:**Size: Number_of_Handles * 2 octets*

Value	Parameter Description
0xFFFF	The number of data blocks that have been freed for the associated Handle since the previous time that an HCI_Number_Of_Completed_Data_Blocks event provided information about this Handle. Range: 0x0000 to 0xFFFF



7.7.60 Short Range Mode Change Complete event

Event	Event Code	Event Parameters
HCI_Short_Range_Mode_Change_Complete	0x4C	Status, Physical_Link_Handle, Short_Range_Mode_State

Description:

The HCI_Short_Range_Mode_Change_Complete event occurs after the AMP Controller has been notified to enable or disable Short Range Mode for a given physical link. The status parameter indicates if the configuration change was successful or not.

Event parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	Short Range Mode configuration has occurred.
0x01 to 0xFF	Short Range Mode configuration failed to complete. See [Vol 2 Part D, Error Codes] for a list of error codes and descriptions.

Physical_Link_Handle:

Size: 1 octet

Value	Parameter Description
0xXX	Physical_Link_Handle to which the configuration applies.

Short_Range_Mode_State:

Size: 1 octet

Value	Parameter Description
0x00	Short Range Mode disabled
0x01	Short Range Mode enabled
All other values	Reserved for future use



7.7.61 AMP Status Change event

Event	Event Code	Event Parameters
HCI_AMP_Status_Change	0x4D	Status, AMP_Status

Description:

The HCI_AMP_Status_Change event can occur at any time, after initial HCI_Read_Local_AMP_Info_Request command, and when the AMP Controller detects a change has occurred regarding status and is based on availability thresholds.

Event parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	HCI_AMP_Status_Change has occurred.
0x01 to 0xFF	HCI_AMP_Status_Change failed to complete. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

AMP_Status:

Size: 1 octet

Value	Parameter Description
0x00	The Controller radio is available but is currently physically powered down. This value should be used if the AMP Controller is present and can be powered up by the AMP Manager. This value indicates that there may be a cost of time and power to use this AMP Controller (i.e., the time taken and power required to power up the AMP Controller). These costs are AMP type and AMP implementation dependent.
0x01	This value indicates that the AMP Controller is only used by Bluetooth technology and will not be shared with other non-Bluetooth technologies. This value shall only be used if the AMP Controller is powered up. This value does not indicate how much bandwidth is currently free on the AMP Controller.
0x02	The AMP Controller has no capacity available for Bluetooth operation. This value indicates that all of the AMP Controllers bandwidth is currently allocated to servicing a non Bluetooth technology. A device is permitted to create a Physical Link to an AMP Controller that has this status. This value shall only be used if the AMP Controller is powered up.



Value	Parameter Description
0x03	The AMP Controller has low capacity available for Bluetooth operation. This value indicates that the majority of the AMP Controllers bandwidth is currently allocated to servicing a non Bluetooth technology. An AMP Controller that with capacity in the approximate range of 0 to 30% should indicate this value. This value does not indicate how much of the capacity available for Bluetooth operation is currently available. This value shall only be used if the AMP Controller is powered up.
0x04	The AMP Controller has medium capacity available for Bluetooth operation. An AMP Controller that with capacity in the approximate range of 30% to 70% should indicate this value. This value does not indicate how much of the capacity available for Bluetooth operation is currently available. This value shall only be used if the AMP Controller is powered up
0x05	The AMP Controller has high capacity available for Bluetooth operation. This value indicates that the majority of the AMP Controllers bandwidth is currently allocated to servicing the Bluetooth technology. An AMP Controller that with capacity in the approximate range of 70% to 100% should indicate this value. This value does not indicate how much of the capacity available for Bluetooth operation is currently available. This value shall only be used if the AMP Controller is powered up
0x06	The AMP Controller has full capacity available for Bluetooth operation. This value indicates that while currently the AMP is only being used by Bluetooth the device allows a different technology to share the radio. This value shall be used by devices that are not capable of determining the current available capacity of an AMP that is shared by a different technology. This value does not indicate how much of the capacity available for Bluetooth operation is currently available. This value shall only be used if the AMP Controller is powered up.
All other values	Reserved for future use



7.7.62 AMP Start Test event

Event	Event Code	Event Parameters
HCI_AMP_Start_Test	0x49	Status, Test Scenario

Description:

The HCI_AMP_Start_Test event shall be generated when the HCI_AMP_Test command has completed and the first data is ready to be sent or received.

Event parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	HCI_AMP_Start_Test command succeeded
0x01 to 0xFF	HCI_AMP_Start_Test command failed. See [Vol 2] Part D, Error Codes

Test Scenario:

Size: 1 octet

Value	Parameter Description
0xXX	See the Test commands section of the Protocol Adaptation Layer Functional Specification for the Controller type



7.7.63 AMP Test End event

Event	Event Code	Event Parameters
HCI_AMP_Test_End	0x4A	Status, Test Scenario

Description:

The HCI_AMP_Test_End event shall be generated to indicate that the AMP has transmitted or received the number of frames/bursts configured.

If the Receiver reports are enabled an HCI_AMP_Receiver_Report event shall be generated.

Event parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	HCI_AMP_Test_End command succeeded
0x01 to 0xFF	HCI_AMP_Test_End command failed. See [Vol 2] Part D, Error Codes

Test Scenario:

Size: 1 octet

Value	Parameter Description
0xXX	See the Test commands section of the Protocol Adaptation Layer Functional Specification for the Controller type



7.7.64 AMP Receiver Report event

Event	Event Code	Event Parameters
HCI_AMP_Receiver_Report	0x4B	Controller_Type, Reason, Event_Type, Number_Of_Frames, Number_Of_Error_Frames, Number_Of_Bits, Number_Of_Error_Bits

Description:

The HCI_AMP_Receiver_Report event shall be sent from the AMP to the tester via the AMP Test Manager at the interval configured by the HCI_Enable_AMP_Receiver_Reports command.

The HCI_AMP_Receiver_Report event is generated to indicate the number of frames received and optionally the number of bits in error detected.

Event parameters:

Controller_Type: *Size: 1 octet*

Value	Parameter Description
0xXX	See Assigned Numbers

Reason: *Size: 1 octet*

Value	Parameter Description
0xXX	0x00 Configured Interval report 0x01 Test Ended report 0x02 to 0xFF Reserved for future use

Event_type: *Size: 4 Bytes*

Value	Parameter Description
0xXX	0x00 Frames received report 0x01 Frames received and bits in error report (optional) 0x02 to 0xFF Reserved for future use

Number_Of_Frames: *Size: 2 Bytes*

Value	Parameter Description
0XXXXX	Number of frames received so far

**Number_Of_Error_Frames:****Size: 2 Bytes**

Value	Parameter Description
0xFFFF	Number of frames with errors received so far

Number_Of_Bits:**Size: 4 Bytes**

Value	Parameter Description
0xFFFFFFFF	Number of bits received so far. Shall be set to 0x00000000 if the Event_type is not equal to 0x01.

Number_Of_Error_Bits:**Size: 4 Bytes**

Value	Parameter Description
0xFFFFFFFF	Number of bit errors received so far. Shall be set to 0x00000000 if the Event_type is not equal to 0x01.



7.7.65 LE Meta event

Description:

The LE Meta event is used to encapsulate all LE Controller specific events. The Event Code of all LE Meta events shall be 0x3E. The Subevent_Code is the first octet of the event parameters. The Subevent_Code shall be set to one of the valid Subevent_Codes from an LE specific event. All other Subevent_Parameters are defined in the LE Controller specific events.

7.7.65.1 LE Connection Complete event

Event	Event Code	Event Parameters
HCI_LE_Connection_Complete	0x3E	Subevent_Code, Status, Connection_Handle, Role, Peer_Address_Type, Peer_Address, Conn_Interval, Conn_Latency, Supervision_Timeout, Master_Clock_Accuracy

Description:

The HCI_LE_Connection_Complete event indicates to both of the Hosts forming the connection that a new connection has been created. Upon the creation of the connection a Connection_Handle shall be assigned by the Controller, and passed to the Host in this event. If the connection creation fails this event shall be provided to the Host that had issued the HCI_LE_Create_Connection command.

This event indicates to the Host which issued an HCI_LE_Create_Connection command and received an HCI_Command_Status event if the connection creation failed or was successful.

The Master_Clock_Accuracy parameter is only valid for a slave. On a master, this parameter shall be set to 0x00.

Note: This event is not sent if the HCI_LE_Enhanced_Connection_Complete event (see [Section 7.7.65.10](#)) is unmasked.

**Event parameters:***Subevent_Code:**Size: 1 octet*

Value	Parameter Description
0x01	Subevent code for the HCI_LE_Connection_Complete event

*Status:**Size: 1 octet*

Value	Parameter Description
0x00	Connection successfully completed.
0x01 to 0xFF	Connection failed to complete. [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

*Connection_Handle:**Size: 2 octets (12 bits meaningful)*

Value	Parameter Description
0xFFFF	Connection_Handle Range: 0x0000 to 0x0EFF

*Role:**Size: 1 octet*

Value	Parameter Description
0x00	Connection is master
0x01	Connection is slave
All other values	Reserved for future use

*Peer_Address_Type:**Size: 1 octet*

Value	Parameter Description
0x00	Peer is using a Public Device Address
0x01	Peer is using a Random Device Address
All other values	Reserved for future use

*Peer_Address:**Size: 6 octets*

Value	Parameter Description
0xFFFFFFFFXXXX	Public Device Address or Random Device Address of the peer device

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Value	Parameter Description
0xXXXX	Connection interval used on this connection. Range: 0x0006 to 0x0C80 Time = N * 1.25 ms Time Range: 7.5 ms to 4000 ms.

Conn_Latency:**Size: 2 octets**

Value	Parameter Description
0xXXXX	Slave latency for the connection in number of connection events. Range: 0x0000 to 0x01F3

Supervision_Timeout:**Size: 2 octets**

Value	Parameter Description
0xXXXX	Connection supervision timeout. Range: 0x000A to 0x0C80 Time = N * 10 ms Time Range: 100 ms to 32 s

Master_Clock_Accuracy:**Size: 1 octet**

Value	Parameter Description
0x00	500 ppm
0x01	250 ppm
0x02	150 ppm
0x03	100 ppm
0x04	75 ppm
0x05	50 ppm
0x06	30 ppm
0x07	20 ppm
All other values	Reserved for future use



7.7.65.2 LE Advertising Report event

Event	Event Code	Event Parameters
HCI_LE_Advertising_Report	0x3E	Subevent_Code, Num_Reports, Event_Type[i], Address_Type[i], Address[i], Length[i], Data[i], RSSI[i]

Description:

The HCI_LE_Advertising_Report event indicates that one or more Bluetooth devices have responded to an active scan or have broadcast advertisements that were received during a passive scan. The Controller may queue these advertising reports and send information from multiple devices in one HCI_LE_Advertising_Report event.

This event shall only be generated if scanning was enabled using the HCI_LE_Set_Scan_Enable command. It only reports advertising events that used legacy advertising PDUs.

Event parameters:

Subevent_Code: *Size: 1 octet*

Value	Parameter Description
0x02	Subevent code for the HCI_LE_Advertising_Report event

Num_Reports: *Size: 1 octet*

Value	Parameter Description
0x01 to 0x19	Number of responses in event.
All other values	Reserved for future use

Event_Type[i]: *Size: 1 octet * Num_Reports*

Value	Parameter Description
0x00	Connectable and scannable undirected advertising (ADV_IND)
0x01	Connectable directed advertising (ADV_DIRECT_IND)
0x02	Scannable undirected advertising (ADV_SCAN_IND)
0x03	Non connectable undirected advertising (ADV_NONCONN_IND)



Value	Parameter Description
0x04	Scan Response (SCAN_RSP)
All other values	Reserved for future use

Address_Type[i]: *Size: 1 octet * Num_Reports*

Value	Parameter Description
0x00	Public Device Address
0x01	Random Device Address
0x02	Public Identity Address (Corresponds to Resolved Private Address)
0x03	Random (static) Identity Address (Corresponds to Resolved Private Address)
All other values	Reserved for future use

Address[i]: *Size: 6 octets * Num_Reports*

Value	Parameter Description
0XXXXXXXXXXXXX	Public Device Address, Random Device Address, Public Identity Address or Random (static) Identity Address of the advertising device.

Length_Data[i]: *Size: 1 octets * Num_Reports*

Value	Parameter Description
0x00 to 0x1F	Length of the Data[i] field for each device which responded.
All other values	Reserved for future use.

Data[i]: *Size: SUM (Length_Data[i]) octets*

Value	Parameter Description
	Length_Data[i] octets of advertising or scan response data formatted as defined in [Vol 3] Part C, Section 11 . Note: Each element of this array has a variable length.

RSSI[i]: *Size: 1 octet * Num_Reports*

Value	Parameter Description
0xXX	Size: 1 octet (signed integer) Range: -127 to +20 Units: dBm
0x7F	RSSI is not available



7.7.65.3 LE Connection Update Complete event

Event	Event Code	Event Parameters
HCI_LE_ - Connection_ - Update_Complete	0x3E	Subevent_Code, Status, Connection_Handle, Conn_Interval, Conn_Latency, Supervision_Timeout

Description:

The HCI_LE_Connection_Update_Complete event is used to indicate that the Controller process to update the connection has completed.

This event shall be issued if the HCI_LE_Connection_Update command was issued by the Host or if the connection parameters are updated following a request from the peer device. If no parameters are updated following a request from the peer device then this event shall not be issued.

Note: This event can be issued autonomously by the Master's Controller if it decides to change the connection interval based on the range of allowable connection intervals for that connection.

Note: The parameter values returned in this event may be different from the parameter values provided by the Host through the HCI_LE_Connection_Update command ([Section 7.8.18](#)) or the HCI_LE_Remote_Connection_Parameter_Request_Reply command ([Section 7.8.31](#)).

Event parameters:

Subevent_Code: *Size: 1 octet*

Value	Parameter Description
0x03	Subevent code for the HCI_LE_Connection_Update_Complete event

Status: *Size: 1 octet*

Value	Parameter Description
0x00	HCI_LE_Connection_Update command successfully completed.
0x01 to 0xFF	HCI_LE_Connection_Update command failed to complete. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

**Connection_Handle:****Size: 2 octets (12 bits meaningful)**

Value	Parameter Description
0XXXXX	Connection_Handle Range: 0x0000 to 0x0EFF

Conn_Interval:**Size: 2 octets**

Value	Parameter Description
0XXXXX	Connection interval used on this connection. Range: 0x0006 to 0x0C80 Time = $N * 1.25$ ms Time Range: 7.5 ms to 4000 ms.

Conn_Latency:**Size: 2 octets**

Value	Parameter Description
0XXXXX	Slave latency for the connection in number of connection events. Range: 0x0000 to 0x01F3

Supervision_Timeout:**Size: 2 octets**

Value	Parameter Description
0XXXXX	Supervision timeout for this connection. Range: 0x000A to 0x0C80 Time = $N * 10$ ms Time Range: 100 ms to 32000 ms



7.7.65.4 LE Read Remote Features Complete event

Event	Event Code	Event Parameters
HCI_LE_Read_Remote_Features_Complete	0x3E	Subevent_Code, Status, Connection_Handle, LE_Features

Description:

The HCI_LE_Read_Remote_Features_Complete event is used to indicate the completion of the process of the Controller obtaining the features used on the connection and the features supported by the remote Bluetooth device specified by the Connection_Handle event parameter.

Note: If the features are requested more than once while a connection exists between the two devices, the second and subsequent requests may report a cached copy of the features rather than fetching the feature mask again.

Event parameters:

Subevent_Code: *Size: 1 octet*

Value	Parameter Description
0x04	Subevent code for the HCI_LE_Read_Remote_Features_Complete event.

Status: *Size: 1 octet*

Value	Parameter Description
0x00	HCI_LE_Read_Remote_Features command successfully completed.
0x01 to 0xFF	HCI_LE_Read_Remote_Features command failed to complete. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Connection_Handle: *Size: 2 octets (12 bits meaningful)*

Value	Parameter Description
0xFFFF	Connection_Handle Range: 0x0000 to 0x0EFF

LE_Features: *Size: 8 octets*

Value	Parameter Description
0xFFFFFFFFXXXXXX	Bit Mask List of LE features. See [Vol 6] Part B, Section 4.6 .

**7.7.65.5 LE Long Term Key Request event**

Event	Event Code	Event Parameters
HCI_LE_Long_Term_Key_Request	0x3E	Subevent_Code, Connection_Handle, Random_Number, Encrypted_Diversifier

Description:

The HCI_LE_Long_Term_Key_Request event indicates that the peer device, in the Master role, is attempting to encrypt or re-encrypt the link and is requesting the Long Term Key from the Host. (See [\[Vol 6\] Part B, Section 5.1.3](#)).

This event shall only be generated when the local device's role is Slave.

Event parameters:*Subevent_Code:**Size: 1 octet*

Value	Parameter Description
0x05	Subevent code for the HCI_LE_Long_Term_Key_Request event

*Connection_Handle:**Size: 2 octets (12 bits meaningful)*

Value	Parameter Description
0xFFFF	Connection_Handle Range: 0x0000 to 0x0EFF

*Random_Number:**Size: 8 octets*

Value	Parameter Description
0xFFFFFFFF XXXXXX	64-bit random number.

*Encrypted_Diversifier:**Size: 2 octets*

Value	Parameter Description
0xFFFF	16-bit encrypted diversifier.



7.7.65.6 LE Remote Connection Parameter Request event

Event	Event Code	Event Parameters
HCI_LE_Remote_Connection_Parameter_Request	0x3E	Subevent_Code, Connection_Handle, Interval_Min, Interval_Max, Latency, Timeout

Description:

This event indicates to the master's Host or the slave's Host that the remote device is requesting a change in the connection parameters. The Host replies either with the HCI_LE_Remote_Connection_Parameter_Request_Reply command or the HCI_LE_Remote_Connection_Parameter_Request_Negative_Reply command.

Event parameters:

Subevent_Code:

Size: 1 octet

Value	Parameter Description
0x06	Subevent code for the HCI_LE_Remote_Connection_Parameter_Request event.

Connection_Handle:

Size: 2 octets (12 bits meaningful)

Value	Parameter Description
0xFFFF	Connection_Handle Range 0x0000 to 0x0EFF

Interval_Min:

Size: 2 octets

Value	Parameter Description
N = 0xFFFF	Minimum value of the connection interval requested by the remote device. Range: 0x0006 to 0x0C80 Time = N * 1.25 ms Time Range: 7.5 ms to 4 s

*Interval_Max:**Size: 2 octets*

Value	Parameter Description
N = 0xXXXX	Maximum value of the connection interval requested by the remote device. Range: 0x0006 to 0x0C80 Time = N * 1.25 ms Time Range: 7.5 ms to 4 s

*Latency:**Size: 2 octets*

Value	Parameter Description
0xXXXX	Maximum allowed slave latency for the connection specified as the number of connection events requested by the remote device. Range: 0x0000 to 0x01F3 (499)

*Timeout:**Size: 2 octets*

Value	Parameter Description
N = 0xXXXX	Supervision timeout for the connection requested by the remote device. Range: 0x000A to 0x0C80 Time = N * 10 ms Time Range: 100 ms to 32 s

**7.7.65.7 LE Data Length Change event**

Event	Event Code	Event Parameters
HCI_LE_Data_Length_Change	0x3E	Subevent_Code, Connection_Handle, MaxTxOctets, MaxTxTime, MaxRxOctets, MaxRxTime

Description:

The HCI_LE_Data_Length_Change event notifies the Host of a change to either the maximum LL Data PDU Payload length or the maximum transmission time of packets containing LL Data PDUs in either direction. The values reported are the maximum that will actually be used on the connection following the change. This event shall not be generated if the values have not changed.

Event parameters:*Subevent_Code:**Size: 1 octet*

Value	Parameter Description
0x07	Subevent code for the HCI_LE_Data_Length_Change event

*Connection_Handle:**Size: 2 octets (12 bits meaningful)*

Value	Parameter Description
0xFFFF	Connection_Handle Range 0x0000 to 0x0EFF

*MaxTxOctets:**Size: 2 octets*

Value	Parameter Description
0xFFFF	The maximum number of payload octets in a LLData PDU that the local Controller will send on this connection (<i>connEffectiveMaxTxOctets</i> defined in [Vol 6] Part B, Section 4.5.10). Range 0x001B to 0x00FB

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**MaxTxTime:****Size: 2 octets**

Value	Parameter Description
0xXXXX	The maximum time that the local Controller will take to send a Link Layer packet containing an LL Data PDU on this connection (<i>connEffectiveMaxTxTime</i> defined in [Vol 6] Part B, Section 4.5.10). Range 0x0148 to 0x4290

MaxRxOctets:**Size: 2 octets**

Value	Parameter Description
0xXXXX	The maximum number of payload octets in a Link Layer packet that the local Controller expects to receive on this connection (<i>connEffectiveMaxRxOctets</i> defined in [Vol 6] Part B, Section 4.5.10). Range 0x001B to 0x00FB

MaxRxTime:**Size: 2 octets**

Value	Parameter Description
0xXXXX	The maximum time that the local Controller expects to take to receive a Link Layer packet on this connection (<i>connEffectiveMaxRxTime</i> defined in [Vol 6] Part B, Section 4.5.10). Range 0x0148 to 0x4290



7.7.65.8 LE Read Local P-256 Public Key Complete event

Event	Event Code	Event Parameters
HCI_LE_Read_Local_P-256_Public_Key - Complete	0x3E	Subevent_Code, Status, Local_P-256_Public_Key

Description:

This event is generated when local P-256 key generation is complete.

Event parameters:

Subevent_Code: *Size: 1 octet*

Value	Parameter Description
0x08	Subevent code for the HCI_LE_Read_Local_P-256_Public_Key_Complete event.

Status: *Size: 1 octet*

Value	Parameter Description
0x00	HCI_LE_Read_Local_P-256_Public_Key command completed successfully.
0x01-0xFF	HCI_LE_Read_Local_P-256_Public_Key command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Local_P-256_Public_Key: *Size: 64 octets*

Value	Parameter Description
0xFFFFFFFFFFFFFFFF XXXXXXXXXXXXXXXX XXXXXXXXXXXXXXXX XXXXXXXXXXXXXXXX XXXXXXXXXXXXXXXX XXXXXXXXXXXXXXXX XXXXXXXXXXXXXXXX XXXXXXXXXXXXXXXX XXXXXXXXXXXXXXXX XXXXXXXXXXXXXXXX	Local P-256 public key.



7.7.65.9 LE Generate DHKey Complete event

Event	Event Code	Event Parameters
HCI_LE_Generate_DHKey_Complete	0x3E	Subevent_Code, Status, DHKey

Description:

This event indicates that LE Diffie Hellman key generation has been completed by the Controller.

If the Remote_P-256_Public_Key parameter of the HCI_LE_Generate_DHKey command (see [Section 7.8.93](#)) was invalid (see [\[Vol 3\] Part H, Section 2.3.5.6.1](#)), then all octets of the DHKey event parameter should be set to 0xFF.

Event parameters:

Subevent_Code: *Size: 1 octet*

Value	Parameter Description
0x09	Subevent code for the HCI_LE_Generate_DHKey_Complete event.

Status: *Size: 1 octet*

Value	Parameter Description
0x00	HCI_LE_Generate_DHKey command completed successfully.
0x01-0xFF	HCI_LE_Generate_DHKey command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

DHKey: *Size: 32 octets*

Value	Parameter Description
0xFFFFFFFF XXXXXXXX XXXXXXXX XXXXXXXX XXXXXXXX XXXXXXXX XXXXXXXX XXXXXXXX XXXXXXXX	Diffie Hellman Key.



7.7.65.10 LE Enhanced Connection Complete event

Event	Event Code	Event Parameters
HCI_LE_Enhanced_Connection_Complete	0x3E	Subevent_Code, Status, Connection_Handle, Role, Peer_Address_Type, Peer_Address, Local_Resolvable_Private_Address, Peer_Resolvable_Private_Address Conn_Interval, Conn_Latency, Supervision_Timeout, Master_Clock_Accuracy

Description:

The HCI_LE_Enhanced_Connection_Complete event indicates to both of the Hosts forming the connection that a new connection has been created. Upon the creation of the connection a Connection_Handle shall be assigned by the Controller, and passed to the Host in this event. If the connection creation fails, this event shall be provided to the Host that had issued the HCI_LE_Create_Connection or HCI_LE_Extended_Create_Connection command.

If this event is unmasked and the HCI_LE_Connection_Complete event is unmasked, only the HCI_LE_Enhanced_Connection_Complete event is sent when a new connection has been created.

This event indicates to the Host that issued an HCI_LE_Create_Connection or HCI_LE_Extended_Create_Connection command and received an HCI_Command_Status event if the connection creation failed or was successful.

The Peer_Address, Peer_Resolvable_Private_Address, and Local_Resolvable_Private_Address shall always reflect the most recent packet sent and received on air.

The Master_Clock_Accuracy parameter is only valid for a slave. On a master, this parameter shall be set to 0x00.

**Event parameters:***Subevent Code:**Size: 1 octet*

Value	Parameter Description
0x0A	Subevent code for the HCI_LE_Enhanced_Connection_Complete event

*Status:**Size: 1 octet*

Value	Parameter Description
0x00	Connection successfully completed.
0x01 to 0xFF	Connection failed to complete. See [Vol 2] Part D for a list of error codes and descriptions.

*Connection_Handle:**Size: 2 octets (12 bits meaningful)*

Value	Parameter Description
0xFFFF	Connection_Handle Range: 0x0000 to 0x0EFF

*Role:**Size: 1 octet*

Value	Parameter Description
0x00	Connection is master
0x01	Connection is slave
All other values	Reserved for future use

*Peer_Address_Type:**Size: 1 octet*

Value	Parameter Description
0x00	Public Device Address (default)
0x01	Random Device Address
0x02	Public Identity Address (Corresponds to Resolved Private Address)
0x03	Random (Static) Identity Address (Corresponds to Resolved Private Address)
All other values	Reserved for future use

*Peer_Address:**Size: 6 octets*

Value	Parameter Description
0XXXXXXXXXX XX	Public Device Address, or Random Device Address, Public Identity Address or Random (static) Identity Address of the device to be connected.

**Local_Resolvable_Private_Address:****Size: 6 octets**

Value	Parameter Description
0XXXXXXXXXX XX	Resolvable Private Address being used by the local device for this connection. This is only valid when the Own_Address_Type (from the HCI_LE_Create_Connection, HCI_LE_Set_Advertising_Parameters, HCI_LE_Set_Extended_Advertising_Parameters, or HCI_LE_Extended_Create_Connection commands) is set to 0x02 or 0x03, and the Controller generated a resolvable private address for the local device using a non-zero local IRK. For other Own_Address_Type values, the Controller shall return all zeros.

Peer_Resolvable_Private_Address:**Size: 6 octets**

Value	Parameter Description
0XXXXXXXXXX XX	Resolvable Private Address being used by the peer device for this connection. This is only valid for Peer_Address_Type 0x02 and 0x03. For other Peer_Address_Type values, the Controller shall return all zeros.

Conn_Interval:**Size: 2 octets**

Value	Parameter Description
N = 0XXXXX	Connection interval used on this connection. Range: 0x0006 to 0x0C80 Time = N * 1.25 ms Time Range: 7.5 ms to 4000 ms.

Conn_Latency:**Size: 2 octets**

Value	Parameter Description
0XXXXX	Slave latency for the connection in number of connection events. Range: 0x0000 to 0x01F3

Supervision_Timeout:**Size: 2 octets**

Value	Parameter Description
N = 0XXXXX	Connection supervision timeout. Range: 0x000A to 0x0C80 Time = N * 10 ms Time Range: 100 ms to 32 s

Master_Clock_Accuracy:**Size: 1 octet**

Value	Parameter Description
0x00	500 ppm
0x01	250 ppm



Value	Parameter Description
0x02	150 ppm
0x03	100 ppm
0x04	75 ppm
0x05	50 ppm
0x06	30 ppm
0x07	20 ppm
All other values	Reserved for future use



7.7.65.11 LE Directed Advertising Report event

Event	Event Code	Event Parameters
HCI_LE_Directed_Advertising_Report	0x3E	Subevent_Code, Num_Reports, Event_Type[i], Address_Type[i], Address[i], Direct_Address_Type[i], Direct_Address[i], RSSI[i]

Description:

The HCI_LE_Directed_Advertising_Report event indicates that directed advertisements have been received where the advertiser is using a resolvable private address for the TargetA field of the advertising PDU which the Controller is unable to resolve and the Scanning_Filter_Policy is equal to 0x02 or 0x03, see [Section 7.8.10](#). Direct_Address_Type and Direct_Address specify the address the directed advertisements are being directed to. Address_Type and Address specify the address of the advertiser sending the directed advertisements. The Controller may queue these advertising reports and send information from multiple advertisers in one HCI_LE_Directed_Advertising_Report event.

This event shall only be generated if scanning was enabled using the HCI_LE_Set_Scan_Enable command. It only reports advertising events that used legacy advertising PDUs.

Event parameters:

Subevent Code: *Size: 1 octet*

Value	Parameter Description
0x0B	Subevent code for the HCI_LE_Directed_Advertising_Report event

Num_reports: *Size: 1 octet*

Value	Parameter Description
0x01 to 0x19	Number of responses in event
All other values	Reserved for future use

*Event_Type[i]:**Size: 1 octet*

Value	Parameter Description
0x01	Connectable directed legacy advertising (ADV_DIRECT_IND)
All other values	Reserved for future use

*Address_Type[i]:**Size: 1 octet*

Value	Parameter Description
0x00	Public Device Address (default)
0x01	Random Device Address
0x02	Public Identity Address (Corresponds to Resolved Private Address)
0x03	Random (static) Identity Address (Corresponds to Resolved Private Address)
All other values	Reserved for future use

*Address[i]:**Size: 6 octets*

Value	Parameter Description
0XXXXXXXXX XX	Public Device Address, Random Device Address, Public Identity Address or Random (static) Identity Address of the advertising device.

*Direct_Address_Type[i]:**Size: 1 octet*

Value	Parameter Description
0x01	Random Device Address (default)
All other values	Reserved for future use

*Direct_Address[i]:**Size: 6 octets*

Value	Parameter Description
0XXXXXXXXX XX	Random Device Address

*RSSI[i]:**Size: 1 octet*

Value	Parameter Description
0xFF	Size: 1 octet (signed integer) Range: -127 to +20 Units: dBm
0x7F	RSSI is not available



7.7.65.12 LE PHY Update Complete event

Event	Event Code	Event Parameters
HCI_LE_PHY_Update_Complete	0x3E	Subevent_Code, Status, Connection_Handle, TX_PHY, RX_PHY

Description:

The HCI_LE_PHY_Update_Complete event is used to indicate that the Controller has changed the transmitter PHY or receiver PHY in use.

If the Controller changes the transmitter PHY, the receiver PHY, or both PHYs, this event shall be issued.

If an HCI_LE_Set_PHY command was sent and the Controller determines that neither PHY will change as a result, it issues this event immediately.

Event parameters:

Subevent_Code: *Size: 1 octet*

Value	Parameter Description
0x0C	Subevent code for the HCI_LE_PHY_Update_Complete event

Status: *Size: 1 octet*

Value	Parameter Description
0x00	HCI_LE_Set_PHY command succeeded or autonomous PHY update made by the Controller.
0x01 to 0xFF	HCI_LE_Set_PHY command failed. See [Vol 2] Part D for a list of error codes and descriptions.

Connection_Handle: *Size: 2 octets (12 bits meaningful)*

Value	Parameter Description
0xFFFF	Connection_Handle Range: 0x0000 to 0x0EFF

TX_PHY: *Size: 1 octet*

Value	Parameter Description
0x01	The transmitter PHY for the connection is LE 1M
0x02	The transmitter PHY for the connection is LE 2M



Value	Parameter Description
0x03	The transmitter PHY for the connection is LE Coded
All other values	Reserved for future use

*RX_PHY:**Size: 1 octet*

Value	Parameter Description
0x01	The receiver PHY for the connection is LE 1M
0x02	The receiver PHY for the connection is LE 2M
0x03	The receiver PHY for the connection is LE Coded
All other values	Reserved for future use



7.7.65.13 LE Extended Advertising Report event

Event	Event Code	Event Parameters
HCI_LE_Extended_Advertising_Report	0x3E	Subevent_Code, Num_Reports, Event_Type[i], Address_Type[i], Address[i], Primary_PHY[i], Secondary_PHY[i], Advertising_SID[i], Tx_Power[i], RSSI[i], Periodic_Advertising_Interval[i], Direct_Address_Type[i], Direct_Address[i], Data_Length[i], Data[i]

Description:

The HCI_LE_Extended_Advertising_Report event indicates that one or more Bluetooth devices have responded to an active scan or have broadcast advertisements that were received during a passive scan. The Controller may coalesce multiple advertising reports from the same or different advertisers into a single HCI_LE_Extended_Advertising_Report event, provided all the parameters from all the advertising reports fit in a single HCI event.

This event shall only be generated if scanning was enabled using the HCI_LE_Set_Extended_Scan_Enable command. It reports advertising events using either legacy or extended advertising PDUs.

The Controller may split the data from a single advertisement (whether one PDU or several) into several reports. If so, each report except the last shall have an Event_Type with a data status field of "incomplete, more data to come", while the last shall have the value "complete"; the Address_Type, Address, Advertising_SID, Primary_PHY, and Secondary_PHY fields shall be the same in all the reports.

When a scan response is received, bits 0-2 and 4 of the event type shall indicate the properties of the original advertising event and the Advertising_SID field may be set to the value in the original scannable advertisement.

An Event_Type with a data status field of "incomplete, data truncated" indicates that the Controller attempted to receive an AUX_CHAIN_IND PDU but was not successful.



Where the event being reported used a legacy advertising PDU, the Controller shall set the Event_Type to the value specified in [Table 7.1](#).

PDU Type	Event_Type
ADV_IND	0010011b
ADV_DIRECT_IND	0010101b
ADV_SCAN_IND	0010010b
ADV_NONCONN_IND	0010000b
SCAN_RSP to an ADV_IND	0011011b
SCAN_RSP to an ADV_SCAN_IND	0011010b

Table 7.1: Event_Type values for legacy PDUs

If the Event_Type indicates a legacy PDU (bit 4 = 1), the Primary_PHY parameter shall indicate the LE 1M PHY and the Secondary_PHY parameter shall be set to 0x00. Otherwise, the Primary_PHY parameter indicates the PHY used to send the advertising PDU on the primary advertising physical channel and the Secondary_PHY parameter indicates the PHY used to send the advertising PDU(s), if any, on the secondary advertising physical channel.

The Periodic_Advertising_Interval event parameter is set to zero when no periodic advertising exists as part of the advertising set.

The Direct_Address_Type and Direct_Address parameters indicate the TargetA address in the advertising PDU for directed advertising event types (bit 2 = 1). These parameters shall be ignored for undirected advertising event types (bit 2 = 0). Direct_Address_Type value 0xFE shall be used only when the Scanning_Filter_Policy is equal to 0x02 or 0x03 (see [Section 7.8.64](#)).

When multiple advertising packets are used to complete a single advertising report (e.g., a packet containing an ADV_EXT_IND PDU combined with one containing an AUX_ADV_IND PDU), the RSSI and TxPower event parameters shall be set based on the last packet received.

Event parameters:

Subevent_Code:

Size: 1 octet

Value	Parameter Description
0x0D	Subevent code for the HCI_LE_Extended_Advertising_Report event

Num_Reports:

Size: 1 octet

Value	Parameter Description
0x01 to 0x0A	Number of separate reports in the event
All other values	Reserved for future use

*Event_Type[i]:**Size: 2 octets * Num_Reports*

Bit Number	Parameter Description
0	Connectable advertising
1	Scannable advertising
2	Directed advertising
3	Scan response
4	Legacy advertising PDUs used
5-6	Data status: 00b = Complete 01b = Incomplete, more data to come 10b = Incomplete, data truncated, no more to come 11b = Reserved for future use
All other bits	Reserved for future use

*Address_Type[i]:**Size: 1 octet * Num_Reports*

Value	Parameter Description
0x00	Public Device Address
0x01	Random Device Address
0x02	Public Identity Address (corresponds to Resolved Private Address)
0x03	Random (static) Identity Address (corresponds to Resolved Private Address)
0xFF	No address provided (anonymous advertisement)
All other values	Reserved for future use

*Address[i]:**Size: 6 octets * Num_Reports*

Value	Parameter Description
0XXXXXXXXXXXXX	Public Device Address, Random Device Address, Public Identity Address or Random (static) Identity Address of the advertising device.

*Primary_PHY[i]:**Size: 1 octet * Num_Reports*

Value	Parameter Description
0x01	Advertiser PHY is LE 1M
0x03	Advertiser PHY is LE Coded
All other values	Reserved for future use

*Secondary_PHY[i]:**Size: 1 octet * Num_Reports*

Value	Parameter Description
0x00	No packets on the secondary advertising physical channel
0x01	Advertiser PHY is LE 1M
0x02	Advertiser PHY is LE 2M
0x03	Advertiser PHY is LE Coded
All other values	Reserved for future use

*Advertising_SID[i]:**Size: 1 octet * Num_Reports*

Value	Parameter Description
0x00 to 0x0F	Value of the Advertising SID subfield in the ADI field of the PDU or, for scan responses, in the ADI field of the original scannable advertisement
0xFF	No ADI field provided
All other values	Reserved for future use

*Tx_Power[i]:**Size: 1 octet * Num_Reports*

Value	Parameter Description
0xFF	Size: 1 octet (signed integer) Range: -127 to +20 Units: dBm
0x7F	Tx Power information not available

*RSSI[i]:**Size: 1 octet * Num_Reports*

Value	Parameter Description
0xFF	Size: 1 octet (signed integer) Range: -127 to +20 Units: dBm
0x7F	RSSI is not available

*Periodic_Advertising_Interval[i]:**Size: 2 octets * Num_Reports*

Value	Parameter Description
0x0000	No periodic advertising
N = 0xFFFF	Interval of the periodic advertising Range: 0x0006 to 0xFFFF Time = N * 1.25 ms Time Range: 7.5 ms to 81,918.75 s

*Direct_Address_Type[i]:**Size: 1 octet * Num_Reports*

Value	Parameter Description
0x00	Public Device Address
0x01	Random Device Address
0x02	Public Identity Address (Corresponds to Resolved Private Address)
0x03	Random (static) Identity Address (Corresponds to Resolved Private Address)
0xFE	Random Device Address (Controller unable to resolve)
All other values	Reserved for future use

*Direct_Address[i]:**Size: 6 octets * Num_Reports*

Value	Parameter Description
0XXXXXXXXXXXX	Public Device Address, Random Device Address, Public Identity Address or Random (static) Identity Address of the target device

*Data_Length[i]:**Size: 1 octet * Num_Reports*

Value	Parameter Description
0 to 229	Length of the Data[i] field for each device which responded
All other values	Reserved for future use

*Data[i]:**Size: SUM (Data_Length[i]) octets*

Value	Parameter Description
	Data_Length[i] octets of advertising or scan response data formatted as defined in [Vol 3] Part C, Section 11 . Note: Each element of this array has a variable length.

**7.7.65.14 LE Periodic Advertising Sync Established event**

Event	Event Code	Event Parameters
HCI_LE_Periodic_Advertising_Sync_Established	0x3E	Subevent_Code, Status, Sync_Handle, Advertising_SID, Advertiser_Address_Type, Advertiser_Address, Advertiser_PHY, Periodic_Advertising_Interval, Advertiser_Clock_Accuracy

Description:

The HCI_LE_Periodic_Advertising_Sync_Established event indicates that the Controller has received the first periodic advertising packet from an advertiser after the HCI_LE_Periodic_Advertising_Create_Sync command has been sent to the Controller.

The Sync_Handle parameter identifies the periodic advertising train in subsequent commands and events and shall be assigned by the Controller.

The Advertising_SID parameter is set to the value of the Advertising SID subfield in the ADI field of the advertising PDU referring to the periodic advertising train.

The Advertiser_Address_Type and Advertiser_Address parameters specify the address of the periodic advertiser.

The Advertiser_PHY parameter specifies the PHY used for the periodic advertising.

The Periodic_Advertising_Interval parameter specifies the interval between the periodic advertising events.

The Advertiser_Clock_Accuracy parameter specifies the accuracy of the periodic advertiser's clock.

Event parameters:*Subevent_Code:**Size: 1 octet*

Value	Parameter Description
0x0E	Subevent code for the HCI_LE_Periodic_Advertising_Sync_Established event

**Status:****Size: 1 octet**

Value	Parameter Description
0x00	Periodic advertising sync successful
0x01 to 0xFF	Periodic advertising sync failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions

Sync_Handle:**Size: 2 octets (12 bits meaningful)**

Value	Parameter Description
0xFFFF	Sync_Handle identifying the periodic advertising train Range: 0x0000 to 0x0EFF

Advertising_SID:**Size: 1 octet**

Value	Parameter Description
0x00 to 0x0F	Value of the Advertising SID subfield in the ADI field of the PDU

Advertiser_Address_Type:**Size: 1 octet**

Value	Parameter Description
0x00	Public Device Address or Public Identity Address
0x01	Random Device Address or Random (static) Identity Address
0x02	Public Identity Address (corresponds to Resolved Private Address)
0x03	Random (static) Identity Address (corresponds to Resolved Private Address)
All other values	Reserved for future use

Advertiser_Address:**Size: 6 octets**

Value	Parameter Description
0XXXXXXXXXXXXX	Public Device Address, Random Device Address, Public Identity Address, or Random (static) Identity Address of the advertiser

Advertiser_PHY:**Size: 1 octet**

Value	Parameter Description
0x01	Advertiser PHY is LE 1M
0x02	Advertiser PHY is LE 2M
0x03	Advertiser PHY is LE Coded
All other values	Reserved for future use

Host Controller Interface Functional Specification***Periodic_Advertising_Interval:******Size: 2 octets***

Value	Parameter Description
N = 0xFFFF	Periodic advertising interval Range: 0x0006 to 0xFFFF Time = N * 1.25 ms Time Range: 7.5 ms to 81.91875 s

Advertiser_Clock_Accuracy:***Size: 1 octet***

Value	Parameter Description
0x00	500 ppm
0x01	250 ppm
0x02	150 ppm
0x03	100 ppm
0x04	75 ppm
0x05	50 ppm
0x06	30 ppm
0x07	20 ppm
All other values	Reserved for future use

**7.7.65.15 LE Periodic Advertising Report event**

Event	Event Code	Event Parameters
HCI_LE_Periodic_Advertising_Report	0x3E	Subevent_Code, Sync_Handle, Tx_Power, RSSI, CTE_Type, Data_Status, Data_Length, Data

Description:

The HCI_LE_Periodic_Advertising_Report event indicates that the Controller has received a Periodic Advertising packet.

The Sync_Handle parameter identifies the periodic advertising train that the report relates to.

The RSSI parameter contains the RSSI value for the packet, excluding any Constant Tone Extension. If the Controller supports the Connectionless CTE Receiver feature, RSSI shall not be set to 0x7F.

The Controller may split the data from a single periodic advertisement (whether one PDU or several) into several reports. If so, each report except the last shall have a Data_Status of "incomplete, more data to come", while the last shall have the value "complete".

A Data_Status of "incomplete, data truncated" indicates that the Controller attempted to receive an AUX_CHAIN_IND PDU but was not successful.

The CTE_Type parameter indicates the type of Constant Tone Extension in the periodic advertising packets.

Event parameters:

Subevent_Code:

Size: 1 octet

Value	Parameter Description
0x0F	Subevent code for the HCI_LE_Periodic_Advertising_Report event

**Sync_Handle:****Size: 2 octets (12 bits meaningful)**

Value	Parameter Description
0xFFFF	Sync_Handle identifying the periodic advertising train. Range: 0x0000 to 0x0EFF

Tx_Power:**Size: 1 octet**

Value	Parameter Description
0xFF	Size: 1 octet (signed integer) Range: -127 to +20 Units: dBm
0x7F	Tx Power information not available

RSS[i]:**Size: 1 octet**

Value	Parameter Description
0xFF	Size: 1 octet (signed integer) Range: -127 to +20 Units: dBm
0x7F	RSSI is not available

CTE_Type:**Size: 1 octet**

Value	Parameter Description
0x00	AoA Constant Tone Extension
0x01	AoD Constant Tone Extension with 1 μ s slots
0x02	AoD Constant Tone Extension with 2 μ s slots
0xFF	No Constant Tone Extension
All other values	Reserved for future use

Data_Status:**Size: 1 octet**

Value	Parameter Description
0x00	Data complete
0x01	Data incomplete, more data to come
0x02	Data incomplete, data truncated, no more to come
All other values	Reserved for future use

*Data_Length:**Size: 1 octet*

Value	Parameter Description
0 to 247	Length of the Data field
All other values	Reserved for future use

*Data:**Size: Data_Length octets*

Value	Parameter Description
Variable	Data received from a Periodic Advertising packet

**7.7.65.16 LE Periodic Advertising Sync Lost event**

Event	Event Code	Event Parameters
HCI_LE_Periodic_Advertising_Sync_Lost	0x3E	Subevent_Code, Sync_Handle

Description:

The HCI_LE_Periodic_Advertising_Sync_Lost event indicates that the Controller has not received a Periodic Advertising packet from the train identified by Sync_Handle within the timeout period.

Event parameters:*Subevent_Code:**Size: 1 octet*

Value	Parameter Description
0x10	Subevent code for the HCI_LE_Periodic_Advertising_Sync_Lost event

*Sync_Handle:**Size: 2 octets (12 bits meaningful)*

Value	Parameter Description
0xFFFF	Sync_Handle identifying the periodic advertising train. Range: 0x0000 to 0x0EFF

**7.7.65.17 LE Scan Timeout event**

Event	Event Code	Event Parameters
HCI_LE_Scan_Timeout	0x3E	Subevent_Code

Description:

The HCI_LE_Scan_Timeout event indicates that scanning has ended because the duration has expired.

This event shall only be generated if scanning was enabled using the HCI_LE_Set_Extended_Scan_Enable command.

Event parameters:

Subevent_Code:

Size: 1 octet

Value	Parameter Description
0x11	Subevent code for the HCI_LE_Scan_Timeout event

**7.7.65.18 LE Advertising Set Terminated event**

Event	Event Code	Event Parameters
HCI_LE_- Advertising_Set_- Terminated	0x3E	Subevent_Code, Status, Advertising_Handle, Connection_Handle, Num_Completed_Extended_Advertising_Events

Description:

The HCI_LE_Advertising_Set_Terminated event indicates that the Controller has terminated advertising in the advertising sets specified by the Advertising_Handle parameter.

This event shall be generated every time connectable advertising in an advertising set results in a connection being created or because the advertising duration or the maximum number of extended advertising events has been reached. It shall not be generated if the Host disables the advertising set.

This event shall only be generated if advertising was enabled using the HCI_LE_Set_Extended_Advertising_Enable command.

The Connection_Handle parameter is only valid when advertising ends because a connection was created.

If the Max_Extended_Advertising_Events parameter in the HCI_LE_Set_Extended_Advertising_Enable command was non-zero, the Num_Completed_Extended_Advertising_Events parameter shall be set to the number of completed extended advertising events the Controller had transmitted when either the duration elapsed or the maximum number of extended advertising events was reached; otherwise it shall be set to zero.

If advertising has terminated as a result of the advertising duration elapsing, the Status parameter shall be set to the error code *Advertising Timeout* (0x3C).

If advertising has terminated because the Max_Extended_Advertising_Events was reached, the Status parameter shall be set to the error code *Limit Reached* (0x43).

Event parameters:*Subevent_Code:**Size: 1 octet*

Value	Parameter Description
0x12	Subevent code for the HCI_LE_Advertising_Set_Terminated event

**Status:****Size: 1 octet**

Value	Parameter Description
0x00	Advertising successfully ended with a connection being created
0x01 to 0xFF	Advertising ended for another reason. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Advertising_Handle:**Size: 1 octet**

Value	Parameter Description
0xXX	Advertising_Handle in which advertising has ended Range: 0x00 to 0xEF

Connection_Handle:**Size: 2 octets (12 bits meaningful)**

Value	Parameter Description
0xFFFF	Connection_Handle of the connection whose creation ended the advertising Range: 0x0000 to 0x0EFF

Num_Completed_Extended_Advertising_Events:**Size: 1 octet**

Value	Parameter Description
0xXX	Number of completed extended advertising events transmitted by the Controller

**7.7.65.19 LE Scan Request Received event**

Event	Event Code	Event Parameters
HCI_LE_Scan_Request_Received	0x3E	Subevent_Code, Advertising_Handle, Scanner_Address_Type, Scanner_Address

Description:

The HCI_LE_Scan_Request_Received event indicates that a SCAN_REQ PDU or an AUX_SCAN_REQ PDU has been received by the advertiser. The request contains a device address from a scanner that is allowed by the advertising filter policy. The advertising set is identified by Advertising_Handle.

This event shall only be generated if advertising was enabled using the HCI_LE_Set_Extended_Advertising_Enable command.

The Scanner_Address_Type and Scanner_Address indicates the type of the address and the address of the scanner device.

Event parameters:

Subevent_Code: *Size: 1 octet*

Value	Parameter Description
0x13	Subevent code for the HCI_LE_Scan_Request_Received event

Advertising_Handle: *Size: 1 octet*

Value	Parameter Description
0xXX	Used to identify an advertising set Range: 0x00 to 0xEF

Scanner_Address_Type: *Size: 1 octet*

Value	Parameter Description
0x00	Public Device Address
0x01	Random Device Address
0x02	Public Identity Address (corresponds to Resolved Private Address)
0x03	Random (static) Identity Address (corresponds to Resolved Private Address)
All other values	Reserved for future use



Scanner_Address:

Size: 6 octets

Value	Parameter Description
0XXXXXXXXXXXXX	Public Device Address, Random Device Address, Public Identity Address or Random (static) Identity Address of the advertising device

**7.7.65.20 LE Channel Selection Algorithm event**

Event	Event Code	Event Parameters
HCI_LE_Channel_Selection_Algorithm	0x3E	Subevent_Code, Connection_Handle, Channel_Selection_Algorithm

Description:

The HCI_LE_Channel_Selection_Algorithm event indicates which channel selection algorithm is used on a data physical channel connection (see [\[Vol 6 Part B, Section 4.5.8\]](#)).

Event parameters:

Subevent_Code: *Size: 1 octet*

Value	Parameter Description
0x14	Subevent code for the HCI_LE_Channel_Selection_Algorithm event

Connection_Handle: *Size: 2 octets (12 bits meaningful)*

Value	Parameter Description
0xFFFF	Connection_Handle Range: 0x0000 to 0x0EFF

Channel_Selection_Algorithm: *Size: 1 octet*

Value	Parameter Description
0x00	LE Channel Selection Algorithm #1 is used
0x01	LE Channel Selection Algorithm #2 is used
All other values	Reserved for future use



7.7.65.21 LE Connectionless IQ Report event

Event	Event Code	Event Parameters
HCI_LE_Connectionless_IQ_Report	0x3E	Subevent_Code, Sync_Handle, Channel_Index, RSSI, RSSI_Antenna_ID, CTE_Type, Slot_Durations, Packet_Status, paEventCounter, Sample_Count, I_Sample[i], Q_Sample[i]

Description:

The HCI_LE_Connectionless_IQ_Report event is used by the Controller to report IQ information from the Constant Tone Extension of a received advertising packet forming part of the periodic advertising train identified by Sync_Handle.

The index of the channel on which the packet was received, the RSSI of the packet (excluding the Constant Tone Extension), the ID of the antenna on which this was measured, the type of Constant Tone Extension, the value of *paEventCounter*, and the IQ samples of the Constant Tone Extension of the advertisement are reported in the corresponding parameters. For any given sample, either both or neither of I_Sample[i] and Q_Sample[i] shall equal 0x80.

The Slot_Durations parameter specifies the sampling rate used by the Controller.

The Packet_Status parameter indicates whether the received packet had a valid CRC and, if not, whether the Controller has determined the position and size of the Constant Tone Extension using the Length and CTETime fields.

Note: a Controller is not required to generate this event for packets that have a bad CRC.

The Constant Tone Extension format is defined in [\[Vol 6\] Part B, Section 2.5.1](#).

If the PDU contains AdvData, then the HCI_LE_Periodic_Advertising_Report event shall be generated before this event.

This event is also used by the Controller to report IQ information from the Constant Tone Extension of a received Test Mode packet (see [Section 7.8.50](#)).



The Controller is not required to generate this event for a Constant Tone Extension with a type that it does not support.

This event is also used by the Controller to report that it has insufficient resources to report IQ samples for all received Constant Tone Extensions and has failed to sample at least once. In this case Packet_Status shall be set to 0xFF and Sample_Count to 0x00.

Event parameters:

Subevent_Code: *Size: 1 octet*

Value	Parameter Description
0x15	Subevent code for HCI_LE_Connectionless_IQ_Report event

Sync_Handle: *Size: 2 octets (12 bits meaningful)*

Value	Parameter Description
0xFFFF	Sync_Handle identifying the periodic advertising train. Range: 0x0000 to 0x0EFF
0x0FFF	Receiver Test

Channel_Index: *Size: 1 octet*

Value	Parameter Description
0x00 to 0x24	The index of the channel on which the packet was received.
All other values	Reserved for future use

RSSI: *Size: 2 octets*

Value	Parameter Description
0xFFFF	RSSI of the packet (signed integer) Range: -1270 to +200 Units: 0.1 dBm

RSSI_Antenna_ID: *Size: 1 octet*

Value	Parameter Description
0xFF	Antenna ID

CTE_Type: *Size: 1 octet*

Value	Parameter Description
0x00	AoA Constant Tone Extension
0x01	AoD Constant Tone Extension with 1 μ s slots



Value	Parameter Description
0x02	AoD Constant Tone Extension with 2 μ s slots
All other values	Reserved for future use

*Slot_Durations:**Size: 1 octet*

Value	Parameter Description
0x01	Switching and sampling slots are 1 μ s each
0x02	Switching and sampling slots are 2 μ s each
All other values	Reserved for future use

*Packet_Status:**Size: 1 octet*

Value	Parameter Description
0x00	CRC was correct
0x01	CRC was incorrect and the Length and CTETime fields of the packet were used to determine sampling points
0x02	CRC was incorrect but the Controller has determined the position and length of the Constant Tone Extension in some other way
0xFF	Insufficient resources to sample (Channel_Index, CTE_Type, and Slot_Durations invalid).
All other values	Reserved for future use

*paEventCounter:**Size: 2 octets*

Value	Parameter Description
0xFFFF	The value of <i>paEventCounter</i> (see [Vol 6] Part B, Section 4.4.2.1) for the reported AUX_SYNC_IND PDU

*Sample_Count:**Size: 1 octet*

Value	Parameter Description
0x00	No samples provided (only permitted if Packet_Status is 0xFF).
0x09 to 0x52	Total number of sample pairs (there shall be the same number of I samples and Q samples). Note: This number is dependent on the switch and sample slot durations used.
All other values	Reserved for future use

*I_Sample [i]:**Size: Sample_Count * 1 octet*

Value	Parameter Description
0x80	No valid sample available



Value	Parameter Description
All other values	The list of I samples for the reported packet. Each sample is a signed integer. The list is in the order of the sampling points within the packet.

*Q_Sample [i]:**Size: Sample Count * 1 octet*

Value	Parameter Description
0x80	No valid sample available
All other values	The list of Q samples for the reported packet. Each sample is a signed integer. The list is in the order of the sampling points within the packet.



7.7.65.22 LE Connection IQ Report event

Event	Event Code	Event Parameters
HCI_LE_Connection_IQ_Report	0x3E	Subevent_Code, Connection_Handle, RX_PHY, Data_Channel_Index, RSSI, RSSI_Antenna_ID, CTE_Type, Slot_Durations, Packet_Status, connEventCounter, Sample_Count, I_Sample[i], Q_Sample[i]

Description:

The HCI_LE_Connection_IQ_Report event is used by the Controller to report the IQ samples from the Constant Tone Extension of a received packet (see [\[Vol 6\] Part B, Section 2.4.2.26](#)).

The Connection_Handle parameter identifies the connection that corresponds to the reported information.

The receiver PHY, the index of the data channel, the RSSI value of the packet (excluding the Constant Tone Extension), the ID of the antenna on which this was measured, the type of Constant Tone Extension, the value of *connEventCounter*, and the IQ samples of the Constant Tone Extension of the received packet are reported in the corresponding parameters. For any given sample, either both or neither of I_Sample[i] and Q_Sample[i] shall equal 0x80.

The Slot_Durations parameter specifies the sampling rate used by the Controller.

The Packet_Status parameter indicates whether the received packet had a valid CRC and, if not, whether the Controller has determined the position and size of the Constant Tone Extension using the Length and CTETime fields.

Note: a Controller is not required to generate this event for packets that have a bad CRC.

This event is also used by the Controller to report that it has insufficient resources to report IQ samples for all received Constant Tone Extensions and has failed to sample at least once. In this case Packet_Status shall be set to 0xFF and Sample_Count to 0x00.



The Constant Tone Extension format is defined in [\[Vol 6\] Part B, Section 2.1.5](#).

Event parameters:

Subevent_Code: *Size: 1 octet*

Value	Parameter Description
0x16	Subevent code for HCI_LE_Connection_IQ_Report event

Connection_Handle: *Size: 2 octets (12 bits meaningful)*

Value	Parameter Description
0xFFFF	Connection_Handle Range: 0x0000 to 0x0EFF

RX_PHY: *Size: 1 octet*

Value	Parameter Description
0x01	The receiver PHY for the connection is LE 1M
0x02	The receiver PHY for the connection is LE 2M
All other values	Reserved for future use

Data_Channel_Index: *Size: 1 octet*

Value	Parameter Description
0x00 to 0x24	The index of the data channel on which the Data Physical Channel PDU was received.
All other values	Reserved for future use

RSSI: *Size: 2 octets*

Value	Parameter Description
0xFFFF	RSSI of the packet (signed integer) Range: -1270 to +200 Units: 0.1 dBm

RSSI_Antenna_ID: *Size: 1 octet*

Value	Parameter Description
0xFF	ID of the antenna on which the RSSI is measured

CTE_Type: *Size: 1 octet*

Value	Parameter Description
0x00	AoA Constant Tone Extension



Value	Parameter Description
0x01	AoD Constant Tone Extension with 1 μ s slots
0x02	AoD Constant Tone Extension with 2 μ s slots
All other values	Reserved for future use

*Slot_Durations:**Size: 1 octet*

Value	Parameter Description
0x01	Switching and sampling slots are 1 μ s each
0x02	Switching and sampling slots are 2 μ s each
All other values	Reserved for future use

*Packet_Status:**Size: 1 octet*

Value	Parameter Description
0x00	CRC was correct
0x01	CRC was incorrect and the Length and CTETime fields of the packet were used to determine sampling points
0x02	CRC was incorrect but the Controller has determined the position and length of the Constant Tone Extension in some other way
0xFF	Insufficient resources to sample (Data_Channel_Index, CTE_Type, and Slot_Durations invalid).
All other values	Reserved for future use

*connEventCounter:**Size: 2 octets*

Value	Parameter Description
0xFFFF	The value of <i>connEventCounter</i> (see [Vol 6] Part B, Section 4.5.1) for the reported PDU

*Sample_Count:**Size: 1 octet*

Value	Parameter Description
0x00	No samples provided (only permitted if Packet_Status is 0xFF).
0x09 to 0x52	Total number of sample pairs (there shall be the same number of I samples and Q samples). Note: This number is dependent on the switch and sample slot durations used.
All other values	Reserved for future use

*I_Sample [i]:**Size: Sample_Count * 1 octet*

Value	Parameter Description
0x80	No valid sample available
All other values	The list of I samples for the reported PDU. Each sample is a signed integer. The list is in the order of the sampling points within the PDU.

*Q_Sample [i]:**Size: Sample_Count * 1 octet*

Value	Parameter Description
0x80	No valid sample available
All other values	The list of Q samples for the reported PDU. Each sample is a signed integer. The list is in the order of the sampling points within the PDU.

**7.7.65.23 LE CTE Request Failed event**

Event	Event Code	Event Parameters
HCI_LE_CTE_Request_Failed	0x3E	Subevent_Code, Status, Connection_Handle

Description:

This event is used by the Controller to report an issue following a request to a peer device to reply with a packet containing an LL_CTE_RSP PDU and a Constant Tone Extension. It shall be generated if the packet containing the LL_CTE_RSP PDU sent in response did not contain a Constant Tone Extension or if the peer rejected the request. It shall not be generated if the packet containing the LL_CTE_RSP PDU had a CRC error or if the procedure response timeout timer (see [Vol 6] Part B, Section 5.2) expired.

Event parameters:*Subevent_Code:**Size: 1 octet*

Value	Parameter Description
0x17	Subevent code for HCI_LE_CTE_Request_Failed event

*Status:**Size: 1 octet*

Value	Parameter Description
0x00	LL_CTE_RSP PDU received successfully but without a Constant Tone Extension field.
0x01 to 0xFF	Peer rejected the request. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

*Connection_Handle:**Size: 2 octets (12 bits meaningful)*

Value	Parameter Description
0xFFFF	Connection_Handle Range 0x0000 to 0x0EFF

**7.7.65.24 LE Periodic Advertising Sync Transfer Received event**

Event	Event Code	Event Parameters
HCI_LE_Periodic_Advertising_Sync_Transfer_Received	0x3E	Subevent_Code, Status, Connection_Handle Service_Data, Sync_Handle, Advertising_SID, Advertiser_Address_Type, Advertiser_Address, Advertiser_PHY, Periodic_Advertising_Interval, Advertiser_Clock_Accuracy

Description:

The HCI_LE_Periodic_Advertising_Sync_Transfer_Received event is used by the Controller to report that it has received periodic advertising synchronization information from the device referred to by the Connection_Handle parameter and either successfully synchronized to the periodic advertising train or timed out while attempting to synchronize. The Status will be zero if it successfully synchronized and non-zero otherwise.

The Service_Data value is provided by the Host of the device sending the information.

The Sync_Handle identifies the periodic advertising in subsequent commands and events and shall be assigned by the Controller.

The remaining parameters provide information about the periodic advertising (see [Section 7.7.65.14](#)).

If Status is non-zero, all parameter values are valid except Sync_Handle, which the Host shall ignore.

Note: If the Controller is already synchronized to the periodic advertising train described in the received information, no event will be generated.

Event parameters:

**Subevent_Code:****Size: 1 octet**

Value	Parameter Description
0x18	Subevent code for the HCI_LE_Periodic_Advertising_Sync_Transfer_Received event

Status:**Size: 1 octet**

Value	Parameter Description
0x00	Synchronization to the periodic advertising train succeeded.
0x01–0xFF	Synchronization to the periodic advertising train failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Connection_Handle:**Size: 2 octets (12 bits meaningful)**

Value	Parameter Description
0xFFFF	Connection_Handle Range: 0x0000 to 0x0EFF

Service_Data:**Size: 2 octets**

Value	Parameter Description
0xFFFF	A value provided by the peer device

Sync_Handle:**Size: 2 octets (12 bits meaningful)**

Value	Parameter Description
0xFFFF	Sync_Handle identifying the periodic advertising train. Range: 0x0000 to 0x0EFF

Advertising_SID:**Size: 1 octet**

Value	Parameter Description
0x00–0x0F	Value of the Advertising SID used to advertise the periodic advertising
All other values	Reserved for future use

Advertiser_Address_Type:**Size: 1 octet**

Value	Parameter Description
0x00	Public Device Address or Public Identity Address
0x01	Random Device Address or Random (static) Identity Address
0x02	Public Identity Address (corresponds to Resolved Private Address)
0x03	Random (static) Identity Address (corresponds to Resolved Private Address)



Value	Parameter Description
All other values	Reserved for future use

*Advertiser_Address:**Size: 6 octets*

Value	Parameter Description
0XXXXXXXXXXXXX	Public Device Address, Random Device Address, Public Identity Address, or Random (static) Identity Address of the advertiser

*Advertiser_PHY:**Size: 1 octet*

Value	Parameter Description
0x01	Advertiser PHY is LE 1M
0x02	Advertiser PHY is LE 2M
0x03	Advertiser PHY is LE Coded
All other values	Reserved for future use

*Periodic_Advertising_Interval:**Size: 2 octets*

Value	Parameter Description
N = 0XXXXX	Periodic advertising interval Range: 0x0006 to 0xFFFF Time = N * 1.25 ms Time Range: 7.5 ms to 81.91875 s

*Advertiser_Clock_Accuracy:**Size: 1 octet*

Value	Parameter Description
0x00	500 ppm
0x01	250 ppm
0x02	150 ppm
0x03	100 ppm
0x04	75 ppm
0x05	50 ppm
0x06	30 ppm
0x07	20 ppm
All other values	Reserved for future use



7.7.66 Triggered Clock Capture event

Event	Event Code	Event Parameters
HCI_Triggered_Clock_Capture	0x4E	Connection_Handle, Which_Clock, Clock, Slot_Offset

Description:

The HCI_Triggered_Clock_Capture event is sent to indicate that a triggering event has occurred at the specified clock and offset value. The Which_Clock parameter indicates whether the clock is local or a piconet clock. The Connection_Handle parameter is used when the clock is a piconet clock to indicate which piconet's clock was reported.

The Clock parameter indicates the value of the selected clock at the instant of the triggering event, with bits 1 and 0 set to 0b.

The Slot_Offset parameter indicates the number of microseconds (from 0 to 1249) from the instant at which the selected clock took the value Clock until the triggering event.

Note: What constitutes a triggering event is defined by the Controller implementation. For example, it could be an interrupt signal received by the Controller hardware.

Event parameters:

Connection_Handle: *Size: 2 octets (12 bits meaningful)*

Value	Parameter Description
0xFFFF	Connection_Handle Range: 0x0000 to 0x0EFF

Which_Clock: *Size: 1 octet*

Value	Parameter Description
0xFF	0x00 = Local Clock (Connection_Handle does not have to be valid) 0x01 = Piconet Clock (Connection_Handle shall be valid) 0x02 to 0xFF = Reserved for future use

Clock: *Size: 4 octets (28 bits meaningful)*

Value	Parameter Description
0xFFFFFFFF	Bluetooth clock of the device requested with bits 1 and 0 set to 0b.



Slot_Offset:

Size: 2 octets

Value	Parameter Description
0xFFFF	Number of microseconds from the selected clock attaining the value Clock until the triggering event. Range: 0 to 1249.



7.7.67 Synchronization Train Complete event

Event	Event Code	Event Parameters
HCI_Synchronization_Train_Complete	0x4F	Status

Description:

The HCI_Synchronization_Train_Complete event indicates that the HCI_Start_Synchronization_Train command has completed.

Event parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	HCI_Start_Synchronization_Train command completed successfully.
0x01 to 0xFF	HCI_Start_Synchronization_Train command failed. See [Vol 2] Part D, Error Codes , for error codes and descriptions.



7.7.68 Synchronization Train Received event

Event	Event Code	Event Parameters
HCI_Synchronization_Train_Received	0x50	Status, BD_ADDR, Clock_Offset, AFH_Channel_Map, LT_ADDR, Next_Broadcast_Instant, Connectionless_Slave_Broad- cast_Interval, Service_Data

Description:

The HCI_Synchronization_Train_Received event provides information received from a synchronization train packet transmitted by a Connectionless Slave Broadcast transmitter with the given BD_ADDR. If synchronization was successful, it provides the clock offset, AFH channel map, LT_ADDR, next broadcast instant, broadcast interval, and service data as received from the synchronization train payload. If the command returns a status of 0x01 to 0xFF, then all other parameters are undefined and shall be ignored.

A packet with the Connectionless Slave Broadcast LT_ADDR field in the payload set to zero shall be ignored for the purposes of this event.

Event parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	HCI_Receive_Synchronization_Train command completed successfully.
0x01 to 0xFF	HCI_Receive_Synchronization_Train command failed. See [Vol 2 Part D] , for error codes and descriptions.

BD_ADDR:

Size: 6 octets

Value	Parameter Description
0XXXXXXXXXXXXX	BD_ADDR of the Connectionless Slave Broadcast transmitter.

Clock_Offset:

Size: 4 octets (28 bits meaningful)

Value	Parameter Description
0XXXXXXXX	(CLKNslave - CLK) modulo 2^{28}

**AFH_Channel_Map:****Size: 10 octets (79 bits meaningful)**

Value	Parameter Description
0XXXXXXXXX XXXXXXXXXX XX	This parameter contains 80 1-bit fields. The n^{th} such field (in the range 0 to 78) contains the value for channel n: 0: channel n is unused 1: channel n is used The most significant bit (bit 79) is reserved for future use

LT_ADDR:**Size: 1 octet**

Value	Parameter Description
0x01 to 0x07	LT_ADDR of Connectionless Slave Broadcast channel.
All other values	Reserved for future use

Next_Broadcast_Instant:**Size: 4 octets (28 bits meaningful)**

Value	Parameter Description
0XXXXXXXXX	CLK of a future broadcast on this channel

Connectionless_Slave_Broadcast_Interval:**Size: 2 octets**

Value	Parameter Description
0XXXXX	Interval between Connectionless Slave Broadcast instants in slots. Range: 0x0002 to 0xFFFE; only even values are valid

Service_Data:**Size: 1 octet**

Value	Parameter Description
0XX	Value from octet 27 of the Synchronization Train packet; see [Vol 2] Part B, Table 8.10 .



7.7.69 Connectionless Slave Broadcast Receive event

Event	Event Code	Event Parameters
HCI_Connectionless_Slave_Broadcast_Receive	0x51	BD_ADDR, LT_ADDR, CLK, Offset, Receive_Status, Fragment, Data_Length, Data

Description:

The HCI_Connectionless_Slave_Broadcast_Receive event shall be sent by the BR/EDR Controller every Connectionless Slave Broadcast Instant on which the BR/EDR Controller is scheduled to receive a Connectionless Slave Broadcast packet. If the packet is not received successfully, the event returns a Receive_Status of 0x01. Otherwise, the event returns the payload Data along with the Piconet Clock and the offset from the local CLKN when the packet was received.

The BR/EDR Controller shall send multiple HCI_Connectionless_Slave_Broadcast_Receive events if the length of the received data exceeds the capacity of a single HCI_Connectionless_Slave_Broadcast_Receive event. The fragments shall be marked as starting, continuation, or ending to allow the Host to reassemble the received packet. Only a single event shall be generated for a Connectionless Slave Broadcast instant on which a Connectionless Slave Broadcast packet was scheduled for reception but the BR/EDR Controller failed to successfully receive it.

Event parameters:

BD_ADDR: Size: 6 octets

Value	Parameter Description
0xFFFFFFFFXXXX	BD_ADDR of the broadcasting master device

LT_ADDR: Size: 1 octet

Value	Parameter Description
0x01 to 0x07	LT_ADDR of the Connectionless Slave Broadcast
All other values	Reserved for future use



CLK: *Size: 4 octets (28 bits meaningful)*

Value	Parameter Description
0XXXXXXXX	CLK when Connectionless Slave Broadcast data was received

Offset: *Size: 4 octets (28 bits meaningful)*

Value	Parameter Description
0XXXXXXXX	(CLKNslave – CLK) modulo 2^{28}

Receive Status: *Size: 1 octet*

Value	Parameter Description
0x00	Packet received successfully
0x01	Packet not received successfully (Fragment, Data_Length, and Data fields invalid)
All other values	Reserved for future use

Fragment: *Size: 1 octet*

Value	Parameter Description
0x00	Continuation fragment
0x01	Starting fragment
0x02	Ending fragment
0x03	No fragmentation (single fragment)
All other values	Reserved for future use

Data_Length: *Size: 1 octet*

Value	Parameter Description
0xXX	Length of Data field

Data: *Size: Data_Length octets*

Value	Parameter Description
Variable	Data received from a Connectionless Slave Broadcast packet.



7.7.70 Connectionless Slave Broadcast Timeout event

Event	Event Code	Event Parameters
HCI_Connectionless_Slave_Broadcast_Timeout	0x52	BD_ADDR, LT_ADDR

Description:

On the Connectionless Slave Broadcast Receiver, the HCI_Connectionless_Slave_Broadcast_Timeout event indicates to the Host that the BR/EDR Controller has lost synchronization with the Connectionless Slave Broadcast because no Connectionless Slave Broadcast packets have been received for the timeout interval, *CSB_supervisionTO*, specified in the HCI_Set_Connectionless_Slave_Broadcast_Receive command.

On the Connectionless Slave Broadcast Transmitter, the HCI_Connectionless_Slave_Broadcast_Timeout event indicates to the Host that the BR/EDR Controller has been unable to transmit a Connectionless Slave Broadcast packet for the timeout interval, *CSB_supervisionTO*, specified in the HCI_Set_Connectionless_Slave_Broadcast command.

Event parameters:

BD_ADDR:

Size: 6 octets

Value	Parameter Description
0XXXXXXXXXXXXX	BD_ADDR of the broadcasting master device

LT_ADDR:

Size: 1 octet

Value	Parameter Description
0x01 to 0x07	LT_ADDR of the Connectionless Slave Broadcast
All other values	Reserved for future use



7.7.71 Truncated Page Complete event

Event	Event Code	Event Parameters
HCI_Truncated_Page_Complete	0x53	Status, BD_ADDR

Description:

The HCI_Truncated_Page_Complete event indicates to the Host that an HCI_Truncated_Page command completed. Truncated Paging is considered to be successful when a slave page response ID packet has been received by the local BR/EDR Controller. See [\[Vol 2\] Part B, Section 8.3.3](#) for more information.

An HCI_Truncated_Page_Complete event shall always be sent for each HCI_Truncated_Page command. If the Host issues an HCI_Truncated_Page_Cancel command before the Controller returns the HCI_Truncated_Page_Complete event, then the HCI_Truncated_Page_Complete event shall be sent after the HCI_Command_Complete event for the HCI_Truncated_Page_Cancel command. If the cancellation was successful, the HCI_Truncated_Page_Complete event shall be generated with the error code *Unknown Connection Identifier* (0x02).

Event parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	HCI_Truncated_Page command completed successfully.
0x01 to 0xFF	HCI_Truncated_Page command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

BD_ADDR:

Size: 6 octets

Value	Parameter Description
0xFFFFFFFFXXXX	BD_ADDR of the paged (slave) device



7.7.72 Slave Page Response Timeout event

Event	Event Code	Event Parameters
HCI_Slave_Page_Response_Timeout	0x54	

Description:

The HCI_Slave_Page_Response_Timeout event indicates to the Host that a slave page response timeout has occurred in the BR/EDR Controller.

Note: This event will be generated if the slave BR/EDR Controller responds to a page but does not receive the master FHS packet (see [\[Vol 2\] Part B, Section 8.3.3](#)) within *pagerespTO*.

Event parameters:

None



7.7.73 Connectionless Slave Broadcast Channel Map Change event

Event	Event Code	Event Parameters
HCI_Connectionless_Slave_Broadcast_Channel_Map_Change	0x55	Channel_Map

Description:

The HCI_Connectionless_Slave_Broadcast_Channel_Map_Change event is sent by the master's BR/EDR Controller to the master's Host to indicate that the master's BR/EDR Controller has moved to a new AFH channel map for the PBD logical link.

After an AFH channel map change takes effect for the PBD logical link, the Connectionless Slave Broadcast Transmitter BR/EDR Controller shall send this event to the Host. Upon reception of this event, the Host may restart the synchronization train to allow receivers to obtain the updated AFH channel map.

This event shall also be sent if the Host issues an HCI_Set_AFH_Channel_Classification command which causes the Connectionless Slave Broadcast Channel Map to change.

Event parameters:

Channel_Map: *Size: 10 octets (79 bits meaningful)*

Value	Parameter Description
0xFFFFFFFF XXXXXXXXXX XX	This parameter contains 80 1-bit fields. The n^{th} such field (in the range 0 to 78) contains the value for channel n : 0: channel n is unused 1: channel n is used The most significant bit (bit 79) is reserved for future use



7.7.74 Inquiry Response Notification event

Event	Event Code	Event Parameters
HCI_Inquiry_Response_Notification	0x56	LAP, RSSI

Description:

The HCI_Inquiry_Response_Notification event indicates to the Host that the BR/EDR Controller responded to an Inquiry message. The LAP parameter in the event indicates the LAP used to create the access code received. The parameter may be used by the Host to determine which access code was used in cases where the BR/EDR Controller is performing inquiry scan on multiple inquiry access codes using parallel scanning or sequential scanning. See [\[Vol 3\] Part C, Section 4.1.2.1](#) for details on sequential and parallel scanning.

The LAP parameter returned by the BR/EDR Controller shall be one of the LAPs currently enabled. LAPs are enabled via the HCI_Write_Current_IAC_LAP command.

The RSSI parameter indicates the signal strength of the received ID packet.

Event parameters:

LAP:

Size: 3 octets

Value	Parameter Description
0xXXXXXX	The LAP from which the IAC was derived; see Assigned Numbers . Range: 0x9E8B00 to 0x9E8B3F

RSSI:

Size: 1 octet

Value	Parameter Description
0xXX	Size: 1 octet (signed integer) Range: -100 to 20, +127 indicates unknown RSSI Units: dBm



7.7.75 Authenticated Payload Timeout Expired event

Event	Event Code	Event Parameters
HCI_Authenticated_Payload_Timeout_Expired	0x57	Connection_Handle

Description:

The HCI_Authenticated_Payload_Timeout_Expired event is used to indicate that a packet containing a valid MIC on the Connection_Handle was not received within the *authenticatedPayloadTO* (see [\[Vol 2\] Part B, Appendix B](#) for the BR/EDR and [\[Vol 6\] Part B, Section 5.4](#) for the LE connection).

Note: A Host may choose to disconnect the link when this occurs.

Event parameters:

Connection_Handle:

Size: 2 octet (12 bits meaningful)

Value	Parameter Description
0xFFFF	Connection_Handle Range: 0x0000 to 0x0EFF



7.7.76 SAM Status Change event

Event	Event Code	Event Parameters
HCI_SAM_Status_Change	0x58	Connection_Handle, Local_SAM_Index, Local_SAM_TX_Availability, Local_SAM_RX_Availability, Remote_SAM_Index, Remote_SAM_TX_Availability, Remote_SAM_RX_Availability

Description:

The HCI_SAM_Status_Change event indicates that the Controller has changed the SAM status for the connection identified by the Connection_Handle; i.e., a new SAM slot map has been enabled or the existing one disabled.

Note: A change from one SAM slot map to another only generates one event, not two.

Event parameters:

Connection_Handle: *Size: 2 octets (12 bits meaningful)*

Value	Parameter Description
0xFFFF	Connection_Handle Range: 0x0000 to 0x0EFF

Local_SAM_Index: *Size: 1 octet*

Value	Parameter Description
0xFF	The index of the current SAM slot map used by the local device. 0xFF means SAM is disabled (i.e. all slots are available)

Local_SAM_TX_Availability: *Size: 1 octet*

Value	Parameter Description
0xFF	The proportion of slots available for the local device to transmit. 0 represents "less than 1 in 255" and 255 represents "all"; other proportions shall be linearly scaled. That is, $Local_SAM_TX_Availability = (total_available_TX_slots / local_T_{SAM}) * 255$, rounded to the integer below, where total_available_TX_slots is the total number of slots available for transmission in the current local SAM slot map and local_TSAM is T_{SAM} for the current local SAM slot map.

**Local_SAM_RX_Availability:****Size: 1 octet**

Value	Parameter Description
0xXX	The proportion of slots available for the local device to receive. 0 represents "less than 1 in 255" and 255 represents "all"; other proportions shall be linearly scaled. That is, $\text{Local_SAM_RX_Availability} = (\text{total_available_RX_slots} / \text{local_T}_{\text{SAM}}) * 255$, rounded to the integer below, where $\text{total_available_RX_slots}$ is the total number of slots available for reception in the current local SAM slot map and $\text{local_T}_{\text{SAM}}$ is T_{SAM} for the current local SAM slot map.

Remote_SAM_Index:**Size: 1 octet**

Value	Parameter Description
0xXX	The index of the current SAM slot map used by the remote device. 0xFF means SAM is disabled (i.e. all slots are available)

Remote_SAM_TX_Availability:**Size: 1 octet**

Value	Parameter Description
0xXX	The proportion of slots available for the remote device to transmit. 0 represents "less than 1 in 255" and 255 represents "all"; other proportions shall be linearly scaled. That is, $\text{Remote_SAM_TX_Availability} = (\text{total_available_TX_slots} / \text{remote_T}_{\text{SAM}}) * 255$, rounded to the integer below, where $\text{total_available_TX_slots}$ is the total number of slots available for transmission in the current remote SAM slot map and $\text{remote_T}_{\text{SAM}}$ is T_{SAM} for the current remote SAM slot map.

Remote_SAM_RX_Availability:**Size: 1 octet**

Value	Parameter Description
0xXX	The proportion of slots available for the remote device to receive. 0 represents "less than 1 in 255" and 255 represents "all"; other proportions shall be linearly scaled. That is, $\text{Remote_SAM_RX_Availability} = (\text{total_available_RX_slots} / \text{remote_T}_{\text{SAM}}) * 255$, rounded to the integer below, where $\text{total_available_RX_slots}$ is the total number of slots available for reception in the current remote SAM slot map and $\text{remote_T}_{\text{SAM}}$ is T_{SAM} for the current remote SAM slot map.



7.8 LE CONTROLLER COMMANDS

The LE Controller commands provide access and control to various capabilities of the Bluetooth hardware, as well as methods for the Host to affect how the Link Layer manages the piconet, and controls connections.

For the LE Controller commands, the OGF code is defined as 0x08.

7.8.1 LE Set Event Mask command

Command	OCF	Command Parameters	Return Parameters
HCI_LE_Set_Event_Mask	0x0001	LE_Event_Mask	Status

Description:

The HCI_LE_Set_Event_Mask command is used to control which LE events are generated by the HCI for the Host. If the bit in the LE_Event_Mask is set to a one, then the event associated with that bit will be enabled. The event mask allows the Host to control which events will interrupt it.

The Controller shall ignore those bits which are reserved for future use or represent events which it does not support. If the Host sets any of these bits to 1, the Controller shall act as if they were set to 0.

For LE events to be generated, the LE Meta event bit in the Event_Mask shall also be set. If that bit is not set, then LE events shall not be generated, regardless of how the LE_Event_Mask is set.

Command parameters:

LE_Event_Mask:

Size: 8 octets

Bit	LE Subevent Types
0	HCI_LE_Connection_Complete event
1	HCI_LE_Advertising_Report event
2	HCI_LE_Connection_Update_Complete event
3	HCI_LE_Read_Remote_Features_Complete event
4	HCI_LE_Long_Term_Key_Request event
5	HCI_LE_Remote_Connection_Parameter_Request event
6	HCI_LE_Data_Length_Change event
7	HCI_LE_Read_Local_P-256_Public_Key_Complete event
8	HCI_LE_Generate_DHKey_Complete event
9	HCI_LE_Enhanced_Connection_Complete event



Bit	LE Subevent Types
10	HCI_LE_Directed_Advertising_Report event
11	HCI_LE_PHY_Update_Complete event
12	HCI_LE_Extended_Advertising_Report event
13	HCI_LE_Periodic_Advertising_Sync_Established event
14	HCI_LE_Periodic_Advertising_Report event
15	HCI_LE_Periodic_Advertising_Sync_Lost event
16	HCI_LE_Scan_Timeout event
17	HCI_LE_Advertising_Set_Terminated event
18	HCI_LE_Scan_Request_Received event
19	HCI_LE_Channel_Selection_Algorithm event
20	HCI_LE_Connectionless_IQ_Report event
21	HCI_LE_Connection_IQ_Report event
22	HCI_LE_CTE_Request_Failed event
23	HCI_LE_Periodic_Advertising_Sync_Transfer_Received event

The value with all bits set to 0 indicates that no events are specified. The default is for bits 0 to 4 inclusive (the value 0x0000 0000 0000 001F) to be set.

All bits not listed in this table are reserved for future use.

Return parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	HCI_LE_Set_Event_Mask command succeeded.
0x01 to 0xFF	HCI_LE_Set_Event_Mask command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Event(s) generated (unless masked away):

When the HCI_LE_Set_Event_Mask command has completed, an HCI_Command_Complete event shall be generated.



7.8.2 LE Read Buffer Size command

Command	OCF	Command Parameters	Return Parameters
HCI_LE_Read_Buffer_Size	0x0002		Status, LE_ACL_Data_Packet_Length, Total_Num_LE_ACL_Data_Packets

Description:

The HCI_LE_Read_Buffer_Size command is used to read the maximum size of the data portion of HCI LE ACL Data packets sent from the Host to the Controller. The Host will segment the data transmitted to the Controller according to these values, so that the HCI Data packets will contain data with up to this size. The HCI_LE_Read_Buffer_Size command also returns the total number of HCI LE ACL Data packets that can be stored in the data buffers of the Controller. The HCI_LE_Read_Buffer_Size command shall be issued by the Host before it sends any data to an LE Controller (see [Section 4.1.1](#)).

If the Controller returns a length value of zero, the Host shall use the HCI_Read_Buffer_Size command to determine the size of the data buffers (shared between BR/EDR and LE transports).

Note: Both the HCI_Read_Buffer_Size and HCI_LE_Read_Buffer_Size commands may return buffer length and number of packets parameter values that are non-zero. This allows a Controller to offer different buffers and number of buffers for BR/EDR data packets and LE data packets.

The LE_ACL_Data_Packet_Length return parameter shall be used to determine the size of the L2CAP PDU segments contained in ACL Data packets, which are transferred from the Host to the Controller to be broken up into packets by the Link Layer. The Total_Num_LE_ACL_Data_Packets return parameter contains the total number of HCI ACL Data packets that can be stored in the data buffers of the Controller. The Host determines how the buffers are to be divided between different Connection_Handles.

Note: The LE_ACL_Data_Packet_Length return parameter does not include the length of the HCI Data packet header.

Command parameters:

None.

**Return parameters:***Status:**Size: 1 octet*

Value	Parameter Description
0x00	HCI_LE_Read_Buffer_Size command succeeded.
0x01 to 0xFF	HCI_LE_Read_Buffer_Size command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

*LE_Data_Packet_Length:**Size: 2 octets*

Value	Parameter Description
0x0000	No dedicated LE Buffer – use HCI_Read_Buffer_Size command.
0x0001 to 0x001A	Reserved for future use
0x001B to 0xFFFF	Maximum length (in octets) of the data portion of each HCI ACL Data packet that the Controller is able to accept.

*Total_Num_LE_Data_Packets:**Size: 1 octet*

Value	Parameter Description
0x00	No dedicated LE Buffer – use HCI_Read_Buffer_Size command.
0x01 to 0xFF	Total number of HCI ACL Data packets that can be stored in the data buffers of the Controller.

Event(s) generated (unless masked away):

When the HCI_LE_Read_Buffer_Size command has completed, an HCI_Command_Complete event shall be generated.



7.8.3 LE Read Local Supported Features command

Command	OCF	Command Parameters	Return Parameters
HCI_LE_Read_Local_Supported_Features	0x0003		Status, LE_Features

Description:

This command requests the list of the supported LE features for the Controller.

Command parameters:

None.

Return parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	HCI_LE_Read_Local_Supported_Features command succeeded.
0x01 to 0xFF	HCI_LE_Read_Local_Supported_Features command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

LE_Features:

Size: 8 octets

Value	Parameter Description
0xFFFFFFFFFFFFFFFF	Bit Mask List of supported LE features. See [Vol 6] Part B, Section 4.6.

Event(s) generated (unless masked away):

When the HCI_LE_Read_Local_Supported_Features command has completed, an HCI_Command_Complete event shall be generated.



7.8.4 LE Set Random Address command

Command	OCF	Command Parameters	Return Parameters
HCI_LE_Set_Random_Address	0x0005	Random_Address	Status

Description:

The HCI_LE_Set_Random_Address command is used by the Host to set the LE Random Device Address in the Controller (see [\[Vol 6\] Part B, Section 1.3](#)).

If this command is used to change the address, the new random address shall take effect for advertising no later than the next successful HCI_LE_Set_Advertising_Enable command, for scanning no later than the next successful HCI_LE_Set_Scan_Enable command or HCI_LE_Set_Extended_Scan_Enable command, and for initiating no later than the next successful HCI_LE_Create_Connection command or HCI_LE_Extended_Create_Connection command.

Note: If Extended Advertising is in use, this command only affects the address used for scanning and initiating. The addresses used for advertising are set by the HCI_LE_Set_Advertising_Set_Random_Address command (see [Section 7.8.52](#)).

If the Host issues this command when any of scanning, initiating, or legacy advertising are enabled, the Controller shall return the error code *Command Disallowed* (0x0C).

Command parameters:

Random_Address:

Size: 6 octets

Value	Parameter Description
0xFFFFFFFFXXXX	Random Device Address as defined by [Vol 6] Part B, Section 1.3 .

Return parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	HCI_LE_Set_Random_Address command succeeded.
0x01 to 0xFF	HCI_LE_Set_Random_Address command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

**Event(s) generated (unless masked away):**

When the HCI_LE_Set_Random_Address command has completed, an HCI_Command_Complete event shall be generated.



7.8.5 LE Set Advertising Parameters command

Command	OCF	Command Parameters	Return Parameters
HCI_LE_Set_Advertising_Parameters	0x0006	Advertising_Interval_Min, Advertising_Interval_Max, Advertising_Type, Own_Address_Type, Peer_Address_Type, Peer_Address, Advertising_Channel_Map, Advertising_Filter_Policy	Status

Description:

The HCI_LE_Set_Advertising_Parameters command is used by the Host to set the advertising parameters.

The Advertising_Interval_Min shall be less than or equal to the Advertising_Interval_Max. The Advertising_Interval_Min and Advertising_Interval_Max should not be the same value to enable the Controller to determine the best advertising interval given other activities.

For high duty cycle directed advertising, i.e. when Advertising_Type is 0x01 (ADV_DIRECT_IND, high duty cycle), the Advertising_Interval_Min and Advertising_Interval_Max parameters are not used and shall be ignored.

The Advertising_Type is used to determine the packet type that is used for advertising when advertising is enabled.

Own_Address_Type parameter indicates the type of address being used in the advertising packets.

If Own_Address_Type equals 0x02 or 0x03, the Peer_Address parameter contains the peer's Identity Address and the Peer_Address_Type parameter contains the Peer's Identity Type (i.e. 0x00 or 0x01). These parameters are used to locate the corresponding local IRK in the resolving list; this IRK is used to generate the own address used in the advertisement.

If directed advertising is performed, i.e. when Advertising_Type is set to 0x01 (ADV_DIRECT_IND, high duty cycle) or 0x04 (ADV_DIRECT_IND, low duty cycle mode), then the Peer_Address_Type and Peer_Address shall be valid.

If Own_Address_Type equals 0x02 or 0x03, the Controller generates the peer's Resolvable Private Address using the peer's IRK corresponding to the peer's Identity Address contained in the Peer_Address parameter and peer's Identity Address Type (i.e. 0x00 or 0x01) contained in the Peer_Address_Type parameter.



The Advertising_Channel_Map is a bit field that indicates the advertising channels that shall be used when transmitting advertising packets. At least one channel bit shall be set in the Advertising_Channel_Map parameter.

The Advertising_Filter_Policy parameter shall be ignored when directed advertising is enabled.

The Host shall not issue this command when advertising is enabled in the Controller; if it is the *Command Disallowed* error code shall be used.

If the advertising interval range provided by the Host (Advertising_Interval_Min, Advertising_Interval_Max) is outside the advertising interval range supported by the Controller, then the Controller shall return the *Unsupported Feature or Parameter Value* (0x11) error code.

Command parameters:

Advertising_Interval_Min:

Size: 2 octets

Value	Parameter Description
N = 0xXXXX	Minimum advertising interval for undirected and low duty cycle directed advertising. Range: 0x0020 to 0x4000 Default: 0x0800 (1.28 s) Time = N * 0.625 ms Time Range: 20 ms to 10.24 s

Advertising_Interval_Max:

Size: 2 octets

Value	Parameter Description
N = 0xXXXX	Maximum advertising interval for undirected and low duty cycle directed advertising. Range: 0x0020 to 0x4000 Default: 0x0800 (1.28 s) Time = N * 0.625 ms Time Range: 20 ms to 10.24 s

Advertising_Type:

Size: 1 octet

Value	Parameter Description
0x00	Connectable and scannable undirected advertising (ADV_IND) (default)
0x01	Connectable high duty cycle directed advertising (ADV_DIRECT_IND, high duty cycle)
0x02	Scannable undirected advertising (ADV_SCAN_IND)
0x03	Non connectable undirected advertising (ADV_NONCONN_IND)



Value	Parameter Description
0x04	Connectable low duty cycle directed advertising (ADV_DIRECT_IND, low duty cycle)
All other values	Reserved for future use

*Own_Address_Type:**Size: 1 octet*

Value	Parameter Description
0x00	Public Device Address (default)
0x01	Random Device Address
0x02	Controller generates Resolvable Private Address based on the local IRK from the resolving list. If the resolving list contains no matching entry, use the public address.
0x03	Controller generates Resolvable Private Address based on the local IRK from the resolving list. If the resolving list contains no matching entry, use the random address from LE_Set_Random_Address.
All other values	Reserved for future use

*Peer_Address_Type:**Size: 1 octet*

Value	Parameter Description
0x00	Public Device Address (default) or Public Identity Address
0x01	Random Device Address or Random (static) Identity Address
All other values	Reserved for future use

*Peer_Address:**Size: 6 octets*

Value	Parameter Description
0XXXXXXXXXXXXX	Public Device Address, Random Device Address, Public Identity Address, or Random (static) Identity Address of the device to be connected.

*Advertising_Channel_Map:**Size: 1 octet*

Bit Number	Parameter Description
0	Channel 37 shall be used
1	Channel 38 shall be used
2	Channel 39 shall be used
All other bits	Reserved for future use

The default is 0x07 (all three channels enabled).

**Advertising_Filter_Policy:****Size: 1 octet**

Value	Parameter Description
0x00	Process scan and connection requests from all devices (i.e., the White List is not in use) (default).
0x01	Process connection requests from all devices and scan requests only from devices that are in the White List.
0x02	Process scan requests from all devices and connection requests only from devices that are in the White List.
0x03	Process scan and connection requests only from devices in the White List.
All other values	Reserved for future use.

Return parameters:**Status:****Size: 1 octet**

Value	Parameter Description
0x00	HCI_LE_Set_Advertising_Parameters command succeeded.
0x01 to 0xFF	HCI_LE_Set_Advertising_Parameters command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Event(s) generated (unless masked away):

When the HCI_LE_Set_Advertising_Parameters command has completed, an HCI_Command_Complete event shall be generated.



7.8.6 LE Read Advertising Physical Channel Tx Power command

Command	OCF	Command Parameters	Return Parameters
HCI_LE_Read_Advertising_Physical_Channel_Tx_Power	0x0007		Status, Transmit_Power_Level

Description:

The HCI_LE_Read_Advertising_Physical_Channel_Tx_Power command is used by the Host to read the transmit power level used for LE advertising physical channel packets.

Command parameters:

None.

Return parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	HCI_LE_Read_Advertising_Physical_Channel_Tx_Power command succeeded.
0x01 to 0xFF	HCI_LE_Read_Advertising_Physical_Channel_Tx_Power failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Transmit_Power_Level:

Size: 1 octet

Value	Parameter Description
0xXX	Size: 1 octet (signed integer) Range: -20 to 20 Units: dBm Accuracy: ± 4 dB

Event(s) generated (unless masked away):

When the HCI_LE_Read_Advertising_Physical_Channel_Tx_Power command has completed, an HCI_Command_Complete event shall be generated.



7.8.7 LE Set Advertising Data command

Command	OCF	Command Parameters	Return Parameters
HCI_LE_Set_Advertising_Data	0x0008	Advertising_Data_Length, Advertising_Data	Status

Description:

The HCI_LE_Set_Advertising_Data command is used to set the data used in advertising packets that have a data field.

Only the significant part of the Advertising_Data should be transmitted in the advertising packets, as defined in [\[Vol 3\] Part C, Section 11](#).

If advertising is currently enabled, the Controller shall use the new data in subsequent advertising events. If an advertising event is in progress when this command is issued, the Controller may use the old or new data for that event. If advertising is currently disabled, the data shall be kept by the Controller and used once advertising is enabled.

Command parameters:

Advertising_Data_Length:

Size: 1 octet

Value	Parameter Description
0x00 to 0x1F	The number of significant octets in the Advertising_Data.

Advertising_Data:

Size: 31 octets

Value	Parameter Description
	31 octets of advertising data formatted as defined in [Vol 3] Part C, Section 11 .
	All octets zero (default).

Return parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	HCI_LE_Set_Advertising_Data command succeeded.
0x01 to 0xFF	HCI_LE_Set_Advertising_Data command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Event(s) generated (unless masked away):

When the HCI_LE_Set_Advertising_Data command has completed, an HCI_Command_Complete event shall be generated.



7.8.8 LE Set Scan Response Data command

Command	OCF	Command Parameters	Return Parameters
HCI_LE_Set_Scan_Response_Data	0x0009	Scan_Response_Data_Length, Scan_Response_Data	Status

Description:

This command is used to provide data used in Scanning Packets that have a data field.

Only the significant part of the Scan_Response_Data should be transmitted in the Scanning Packets, as defined in [\[Vol 3\] Part C, Section 11](#).

If advertising is currently enabled, the Controller shall use the new data in subsequent advertising events. If an advertising event is in progress when this command is issued, the Controller may use the old or new data for that event. If advertising is currently disabled, the data shall be kept by the Controller and used once advertising is enabled.

Command parameters:

Scan_Response_Data_Length: *Size: 1 octet*

Value	Parameter Description
0x00 to 0x1F	The number of significant octets in the Scan_Response_Data.

Scan_Response_Data: *Size: 31 octets*

Value	Parameter Description
	31 octets of Scan_Response_Data formatted as defined in [Vol 3] Part C, Section 11 .
	All octets zero (default).

Return parameters:

Status: *Size: 1 octet*

Value	Parameter Description
0x00	HCI_LE_Set_Scan_Response_Data command succeeded.
0x01 to 0xFF	HCI_LE_Set_Scan_Response_Data command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

**Event(s) generated (unless masked away):**

When the HCI_LE_Set_Scan_Response_Data command has completed, an HCI_Command_Complete event shall be generated.



7.8.9 LE Set Advertising Enable command¹

Command	OCF	Command Parameters	Return Parameters
HCI_LE_Set_Advertising_Enable	0x000A	Advertising_Enable	Status

Description:

The HCI_LE_Set_Advertising_Enable command is used to request the Controller to start or stop advertising. The Controller manages the timing of advertisements as per the advertising parameters given in the HCI_LE_Set_Advertising_Parameters command.

The Controller shall continue advertising until the Host issues an HCI_LE_Set_Advertising_Enable command with Advertising_Enable set to 0x00 (Advertising is disabled) or until a connection is created or until the Advertising is timed out due to high duty cycle Directed Advertising. In these cases, advertising is then disabled.

If Advertising_Enable is set to 0x01, the advertising parameters' Own_Address_Type parameter is set to 0x01, and the random address for the device has not been initialized, the Controller shall return the error code *Invalid HCI Command Parameters* (0x12).

If Advertising_Enable is set to 0x01, the advertising parameters' Own_Address_Type parameter is set to 0x03, the controller's resolving list did not contain a matching entry, and the random address for the device has not been initialized, the Controller shall return the error code *Invalid HCI Command Parameters* (0x12).

Note: Enabling advertising when it is already enabled can cause the random address to change. Disabling advertising when it is already disabled has no effect.

Command parameters:

Advertising_Enable:

Size: 1 octet

Value	Parameter Description
0x00	Advertising is disabled (default)
0x01	Advertising is enabled.
All other values	Reserved for future use

1. This command was formerly called "LE Set Advertise Enable".



Return parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	HCI_LE_Set_Advertising_Enable command succeeded.
0x01 to 0xFF	HCI_LE_Set_Advertising_Enable command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Event(s) generated (unless masked away):

When the HCI_LE_Set_Advertising_Enable command has completed, an HCI_Command_Complete event shall be generated.

If the Advertising_Type parameter is 0x01 (ADV_DIRECT_IND, high duty cycle) and the directed advertising fails to create a connection, an HCI_LE_Connection_Complete or HCI_LE_Enhanced_Connection_Complete event shall be generated with the Status code set to Advertising Timeout (0x3C).

If the Advertising_Type parameter is 0x00 (ADV_IND), 0x01 (ADV_DIRECT_IND, high duty cycle), or 0x04 (ADV_DIRECT_IND, low duty cycle) and a connection is created, an HCI_LE_Connection_Complete or HCI_LE_Enhanced_Connection_Complete event shall be generated.

Note: There is a possible race condition if the Advertising_Enable parameter is set to 0x00 (Disable) and the Advertising_Type parameter is 0x00, 0x01, or 0x04. The advertisements might not be stopped before a connection is created, and therefore both the HCI_Command_Complete event and either an HCI_LE_Connection_Complete event or an HCI_LE_Enhanced_Connection_Complete event could be generated. This can also occur when high duty cycle directed advertising is timed out and this command disables advertising.



7.8.10 LE Set Scan Parameters command

Command	OCF	Command Parameters	Return Parameters
HCI_LE_Set_Scan_Parameters	0x000B	LE_Scan_Type, LE_Scan_Interval, LE_Scan_Window, Own_Address_Type, Scanning_Filter_Policy	Status

Description:

The HCI_LE_Set_Scan_Parameters command is used to set the scan parameters.

The LE_Scan_Type parameter controls the type of scan to perform.

The LE_Scan_Interval and LE_Scan_Window parameters are recommendations from the Host on how long (LE_Scan_Window) and how frequently (LE_Scan_Interval) the Controller should scan (See [\[Vol 6\] Part B, Section 4.4.3](#)). The LE_Scan_Window parameter shall always be set to a value smaller or equal to the value set for the LE_Scan_Interval parameter. If they are set to the same value scanning should be run continuously.

Own_Address_Type parameter indicates the type of address being used in the scan request packets.

The Host shall not issue this command when scanning is enabled in the Controller; if it is the *Command Disallowed* error code shall be used.

Command parameters:

LE_Scan_Type:

Size: 1 octet

Value	Parameter Description
0x00	Passive Scanning. No scanning PDUs shall be sent (default)
0x01	Active scanning. Scanning PDUs may be sent.
All other values	Reserved for future use

**LE_Scan_Interval:****Size: 2 octets**

Value	Parameter Description
N = 0xFFFF	This is defined as the time interval from when the Controller started its last LE scan until it begins the subsequent LE scan. Range: 0x0004 to 0x4000 Default: 0x0010 (10 ms) Time = N * 0.625 ms Time Range: 2.5 ms to 10.24 s

LE_Scan_Window:**Size: 2 octets**

Value	Parameter Description
N = 0xFFFF	The duration of the LE scan. LE_Scan_Window shall be less than or equal to LE_Scan_Interval Range: 0x0004 to 0x4000 Default: 0x0010 (10 ms) Time = N * 0.625 ms Time Range: 2.5 ms to 10.24 s

Own_Address_Type:**Size: 1 octet**

Value	Parameter Description
0x00	Public Device Address (default)
0x01	Random Device Address
0x02	Controller generates Resolvable Private Address based on the local IRK from the resolving list. If the resolving list contains no matching entry, use the public address.
0x03	Controller generates Resolvable Private Address based on the local IRK from the resolving list. If the resolving list contains no matching entry, use the random address from LE_Set_Random_Address.
All other values	Reserved for future use.

Scanning_Filter_Policy:**Size: 1 octet**

Value	Parameter Description
0x00	Accept all advertising and scan response PDUs except directed advertising PDUs not addressed to this device (default).
0x01	Accept only advertising and scan response PDUs from devices where the advertiser's address is in the White List. Directed advertising PDUs which are not addressed to this device shall be ignored.



Value	Parameter Description
0x02	Accept all advertising and scan response PDUs except directed advertising PDUs where the initiator's identity address does not address this device. Note: Directed advertising PDUs where the initiator's address is a resolvable private address that cannot be resolved are also accepted.
0x03	Accept all advertising and scan response PDUs except: • advertising and scan response PDUs where the advertiser's identity address is not in the White List; and • directed advertising PDUs where the initiator's identity address does not address this device Note: Directed advertising PDUs where the initiator's address is a resolvable private address that cannot be resolved are also accepted.
All other values	Reserved for future use.

Return parameters:*Status:**Size: 1 octet*

Value	Parameter Description
0x00	HCI_LE_Set_Scan_Parameters command succeeded.
0x01 to 0xFF	HCI_LE_Set_Scan_Parameters command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Event(s) generated (unless masked away):

When the HCI_LE_Set_Scan_Parameters command has completed, an HCI_Command_Complete event shall be generated.



7.8.11 LE Set Scan Enable command

Command	OCF	Command Parameters	Return Parameters
HCI_LE_Set_Scan_Enable	0x000C	LE_Scan_Enable, Filter_Duplicates	Status

Description:

The HCI_LE_Set_Scan_Enable command is used to start and stop scanning. Scanning is used to discover advertising devices nearby.

The Filter_Duplicates parameter controls whether the Link Layer should filter out duplicate advertising reports (Filtering_Enabled) to the Host, or if the Link Layer should generate advertising reports for each packet received (Filtering_Disabled). See [\[Vol 6\] Part B, Section 4.4.3.5](#). If LE_Scan_Enable is set to 0x00 then Filter_Duplicates shall be ignored.

If LE_Scan_Enable is set to 0x01, the scanning parameters' Own_Address_Type parameter is set to 0x01 or 0x03, and the random address for the device has not been initialized, the Controller shall return the error code *Invalid HCI Command Parameters* (0x12).

If the LE_Scan_Enable parameter is set to 0x01 and scanning is already enabled, any change to the Filter_Duplicates setting shall take effect.

Note: Disabling scanning when it is disabled has no effect.

Command parameters:

LE_Scan_Enable:

Size: 1 octet

Value	Parameter Description
0x00	Scanning disabled.
0x01	Scanning enabled.
All other values	Reserved for future use.

Filter_Duplicates:

Size: 1 octet

Value	Parameter Description
0x00	Duplicate filtering disabled.
0x01	Duplicate filtering enabled.
All other values	Reserved for future use.



Return parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	HCI_LE_Set_Scan_Enable command succeeded.
0x01 to 0xFF	HCI_LE_Set_Scan_Enable command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Event(s) generated (unless masked away):

When the HCI_LE_Set_Scan_Enable command has completed, an HCI_Command_Complete event shall be generated.

Zero or more LE Advertising Reports are generated by the Controller based on advertising packets received and the duplicate filtering. More than one advertising packet may be reported in each HCI_LE_Advertising_Report event.

When the Scanning_Filter_Policy is set to 0x02 or 0x03 (see [Section 7.8.10](#)) and a directed advertisement was received where the advertiser used a resolvable private address which the Controller is unable to resolve, an HCI_LE_Directed_Advertising_Report event shall be generated instead of an HCI_LE_Advertising_Report event.



7.8.12 LE Create Connection command

Command	OCF	Command Parameters	Return Parameters
HCI_LE_Create_Connection	0x000D	LE_Scan_Interval, LE_Scan_Window, Initiator_Filter_Policy, Peer_Address_Type, Peer_Address, Own_Address_Type, Conn_Interval_Min, Conn_Interval_Max, Conn_Latency, Supervision_Timeout, Minimum_CE_Length, Maximum_CE_Length	

Description:

The HCI_LE_Create_Connection command is used to create a Link Layer connection to a connectable advertiser.

The LE_Scan_Interval and LE_Scan_Window parameters are recommendations from the Host on how long (LE_Scan_Window) and how frequently (LE_Scan_Interval) the Controller should scan. The LE_Scan_Window parameter shall be set to a value smaller or equal to the value set for the LE_Scan_Interval parameter. If both are set to the same value, scanning should run continuously.

The Initiator_Filter_Policy is used to determine whether the White List is used. If the White List is not used, the Peer_Address_Type and the Peer_Address parameters specify the address type and address of the advertising device to connect to.

Peer_Address_Type parameter indicates the type of address used in the connectable advertisement sent by the peer. The Host shall not set Peer_Address_Type to either 0x02 or 0x03 if both the Host and the Controller support the HCI_LE_Set_Privacy_Mode command. If a Controller that supports the HCI_LE_Set_Privacy_Mode command receives the HCI_LE_Create_Connection command with Peer_Address_Type set to either 0x02 or 0x03, it may use either device privacy mode or network privacy mode for that peer device.

Peer_Address parameter indicates the Peer's Public Device Address, Random (static) Device Address, Non-Resolvable Private Address or Resolvable Private Address depending on the Peer_Address_Type parameter.



Own_Address_Type parameter indicates the type of address being used in the connection request packets.

The Conn_Interval_Min and Conn_Interval_Max parameters define the minimum and maximum allowed connection interval. The Conn_Interval_Min parameter shall not be greater than the Conn_Interval_Max parameter.

The Conn_Latency parameter defines the maximum allowed connection latency (see [Vol 6] Part B, Section 4.5.1).

The Supervision_Timeout parameter defines the link supervision timeout for the connection. The Supervision_Timeout in milliseconds shall be larger than $(1 + \text{Conn_Latency}) * \text{Conn_Interval_Max} * 2$, where Conn_Interval_Max is given in milliseconds. (See [Vol 6] Part B, Section 4.5.2).

The Minimum_CE_Length and Maximum_CE_Length parameters are informative parameters providing the Controller with the expected minimum and maximum length of the connection events. The Minimum_CE_Length parameter shall be less than or equal to the Maximum_CE_Length parameter.

If the Host issues this command when another HCI_LE_Create_Connection command is pending in the Controller, the Controller shall return the error code *Command Disallowed* (0x0C).

If the Own_Address_Type parameter is set to 0x01 and the random address for the device has not been initialized, the Controller shall return the error code *Invalid HCI Command Parameters* (0x12).

If the Own_Address_Type parameter is set to 0x03, the Initiator_Filter_Policy parameter is set to 0x00, the controller's resolving list did not contain a matching entry, and the random address for the device has not been initialized, the Controller shall return the error code *Invalid HCI Command Parameters* (0x12).

If the Own_Address_Type parameter is set to 0x03, the Initiator_Filter_Policy parameter is set to 0x01, and the random address for the device has not been initialized, the Controller shall return the error code *Invalid HCI Command Parameters* (0x12).

Command parameters:

LE_Scan_Interval:

Size: 2 octets

Value	Parameter Description
N = 0xXXXX	This is defined as the time interval from when the Controller started its last LE scan until it begins the subsequent LE scan. Range: 0x0004 to 0x4000 Time = N * 0.625 ms Time Range: 2.5 ms to 10.24 s

*LE_Scan_Window:**Size: 2 octets*

Value	Parameter Description
N = 0xXXXX	Amount of time for the duration of the LE scan. LE_Scan_Window shall be less than or equal to LE_Scan_Interval Range: 0x0004 to 0x4000 Time = N * 0.625 ms Time Range: 2.5 ms to 10.24 s

*Initiator_Filter_Policy:**Size: 1 octet*

Value	Parameter Description
0x00	White List is not used to determine which advertiser to connect to. Peer_Address_Type and Peer_Address shall be used.
0x01	White List is used to determine which advertiser to connect to. Peer_Address_Type and Peer_Address shall be ignored.
All other values	Reserved for future use.

*Peer_Address_Type:**Size: 1 octet*

Value	Parameter Description
0x00	Public Device Address
0x01	Random Device Address
0x02	Public Identity Address (Corresponds to peer's Resolvable Private Address). This value shall only be used by the Host if either the Host or the Controller does not support the HCI_LE_Set_Privacy_Mode command.
0x03	Random (static) Identity Address (Corresponds to peer's Resolvable Private Address). This value shall only be used by a Host if either the Host or the Controller does not support the HCI_LE_Set_Privacy_Mode command.
All other values	Reserved for future use

*Peer_Address:**Size: 6 octets*

Value	Parameter Description
0XXXXXXXXXXXXX	Public Device Address, Random Device Address, Public Identity Address, or Random (static) Identity Address of the device to be connected

*Own_Address_Type:**Size: 1 octet*

Value	Parameter Description
0x00	Public Device Address
0x01	Random Device Address



Value	Parameter Description
0x02	Controller generates Resolvable Private Address based on the local IRK from the resolving list. If the resolving list contains no matching entry, use the public address.
0x03	Controller generates Resolvable Private Address based on the local IRK from the resolving list. If the resolving list contains no matching entry, use the random address from the most recent successful HCI_LE_Set_Random_Address command.
All other values	Reserved for future use

Conn_Interval_Min:**Size: 2 octets**

Value	Parameter Description
N = 0xXXXX	Minimum value for the connection interval. This shall be less than or equal to Conn_Interval_Max. Range: 0x0006 to 0x0C80 Time = N * 1.25 ms Time Range: 7.5 ms to 4 s.

Conn_Interval_Max:**Size: 2 octets**

Value	Parameter Description
N = 0xXXXX	Maximum value for the connection interval. This shall be greater than or equal to Conn_Interval_Min. Range: 0x0006 to 0x0C80 Time = N * 1.25 ms Time Range: 7.5 ms to 4 s.

Conn_Latency:**Size: 2 octets**

Value	Parameter Description
0xXXXX	Slave latency for the connection in number of connection events. Range: 0x0000 to 0x01F3

Supervision_Timeout:**Size: 2 octets**

Value	Parameter Description
N = 0xXXXX	Supervision timeout for the LE Link. (See [Vol 6] Part B, Section 4.5.2) Range: 0x000A to 0x0C80 Time = N * 10 ms Time Range: 100 ms to 32 s

**Minimum_CE_Length:****Size: 2 octets**

Value	Parameter Description
N = 0xXXXX	Information parameter about the minimum length of connection event needed for this LE connection. Range: 0x0000 to 0xFFFF Time = N * 0.625 ms.

Maximum_CE_Length:**Size: 2 octets**

Value	Parameter Description
N = 0xXXXX	Information parameter about the maximum length of connection event needed for this LE connection. Range: 0x0000 to 0xFFFF Time = N * 0.625 ms.

Return parameters:

None.

Event(s) generated (unless masked away):

When the Controller receives the HCI_LE_Create_Connection command, the Controller sends the HCI_Command_Status event to the Host. An HCI_LE_Connection_Complete or HCI_LE_Enhanced_Connection_Complete event shall be generated when a connection is created or the connection creation procedure is cancelled. If a connection is created and the Controller supports the LE Channel Selection Algorithm #2 feature, this event shall be immediately followed by an HCI_LE_Channel_Selection_Algorithm event.

Note: No HCI_Command_Complete event is sent by the Controller to indicate that this command has been completed. Instead, the HCI_LE_Connection_Complete or HCI_LE_Enhanced_Connection_Complete event indicates that this command has been completed.



7.8.13 LE Create Connection Cancel command

Command	OCF	Command Parameters	Return Parameters
HCI_LE_Create_Connection_Cancel	0x000E		Status

Description:

The HCI_LE_Create_Connection_Cancel command is used to cancel the HCI_LE_Create_Connection or HCI_LE_Extended_Create_Connection commands. This command shall only be issued after the HCI_LE_Create_Connection or HCI_LE_Extended_Create_Connection commands have been issued, an HCI_Command_Status event has been received for the HCI_LE_Create_Connection or HCI_LE_Extended_Create_Connection commands, and before the HCI_LE_Connection_Complete or HCI_LE_Enhanced_Connection_Complete events.

Command parameters:

None.

Return parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	HCI_LE_Create_Connection_Cancel command succeeded.
0x01 to 0xFF	HCI_LE_Create_Connection_Cancel command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Event(s) generated (unless masked away):

When the HCI_LE_Create_Connection_Cancel command has completed, an HCI_Command_Complete event shall be generated.

If the HCI_LE_Create_Connection_Cancel command is sent to the Controller without a preceding LE_Create_Connection or HCI_LE_Extended_Create_Connection command, the Controller shall return an HCI_Command_Complete event with the error code *Command Disallowed* (0x0C).

If the cancellation was successful then, after the HCI_Command_Complete event for the HCI_LE_Create_Connection_Cancel command, either an LE Connection Complete or an HCI_LE_Enhanced_Connection_Complete event shall be generated. In either case, the event shall be sent with the error code *Unknown Connection Identifier* (0x02).



7.8.14 LE Read White List Size command

Command	OCF	Command Parameters	Return Parameters
HCI_LE_Read_White_List_Size	0x000F		Status, White_List_Size

Description:

The HCI_LE_Read_White_List_Size command is used to read the total number of White List entries that can be stored in the Controller.

Note: The number of entries that can be stored is not fixed and the Controller can change it at any time (e.g. because the memory used to store the White List can also be used for other purposes).

Command parameters:

None.

Return parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	HCI_LE_Read_White_List_Size command succeeded.
0x01 to 0xFF	HCI_LE_Read_White_List_Size command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

White_List_Size:

Size: 1 octet

Value	Parameter Description
0x01 to 0xFF	Total number of White List entries that can be stored in the Controller.
0x00	Reserved for future use

Event(s) generated (unless masked away):

When the HCI_LE_Read_White_List_Size command has completed, an HCI_Command_Complete event shall be generated.



7.8.15 LE Clear White List command

Command	OCF	Command Parameters	Return Parameters
HCI_LE_Clear_White_List	0x0010		Status

Description:

The HCI_LE_Clear_White_List command is used to clear the White List stored in the Controller.

This command shall not be used when:

- any advertising filter policy uses the White List and advertising is enabled,
- the scanning filter policy uses the White List and scanning is enabled, or
- the initiator filter policy uses the White List and an HCI_LE_Create_Connection or HCI_LE_Extended_Create_Connection command is outstanding.

Command parameters:

None.

Return parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	HCI_LE_Clear_White_List command succeeded.
0x01 to 0xFF	HCI_LE_Clear_White_List command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Event(s) generated (unless masked away):

When the HCI_LE_Clear_White_List command has completed, an HCI_Command_Complete event shall be generated.



7.8.16 LE Add Device To White List command

Command	OCF	Command Parameters	Return Parameters
HCI_LE_Add_Device_To_White_List	0x0011	Address_Type, Address	Status

Description:

The HCI_LE_Add_Device_To_White_List command is used to add a single device to the White List stored in the Controller.

This command shall not be used when:

- any advertising filter policy uses the White List and advertising is enabled,
- the scanning filter policy uses the White List and scanning is enabled, or
- the initiator filter policy uses the White List and an HCI_LE_Create_Connection or HCI_LE_Extended_Create_Connection command is outstanding.

When a Controller cannot add a device to the White List because there is no space available, it shall return the error code *Memory Capacity Exceeded* (0x07).

If the device is already in the White List, the Controller should not add the device to the White List again and should return success.

Address shall be ignored when Address_Type is set to 0xFF.

Command parameters:

Address_Type:

Size: 1 octet

Value	Parameter Description
0x00	Public Device Address
0x01	Random Device Address
0xFF	Devices sending anonymous advertisements
All other values	Reserved for future use.

Address:

Size: 6 octets

Value	Parameter Description
0XXXXXXXXXXXXX	Public Device Address or Random Device Address of the device to be added to the White List.

**Return parameters:***Status:**Size: 1 octet*

Value	Parameter Description
0x00	HCI_LE_Add_Device_To_White_List command succeeded.
0x01 to 0xFF	HCI_LE_Add_Device_To_White_List command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Event(s) generated (unless masked away):

When the HCI_LE_Add_Device_To_White_List command has completed, an HCI_Command_Complete event shall be generated.



7.8.17 LE Remove Device From White List command

Command	OCF	Command Parameters	Return Parameters
HCI_LE_Remove_Device_From_White_List	0x0012	Address_Type, Address	Status

Description:

The HCI_LE_Remove_Device_From_White_List command is used to remove a single device from the White List stored in the Controller.

This command shall not be used when:

- any advertising filter policy uses the White List and advertising is enabled,
- the scanning filter policy uses the White List and scanning is enabled, or
- the initiator filter policy uses the White List and an HCI_LE_Create_Connection or HCI_LE_Extended_Create_Connection command is outstanding.

Address shall be ignored when Address_Type is set to 0xFF.

Command parameters:

Address_Type:

Size: 1 octet

Value	Parameter Description
0x00	Public Device Address
0x01	Random Device Address
0xFF	Devices sending anonymous advertisements
All other values	Reserved for future use.

Address:

Size: 6 octets

Value	Parameter Description
0XXXXXXXXXXXXX	Public Device Address or Random Device Address of the device to be removed from the White List.

Return parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	HCI_LE_Remove_Device_From_White_List command succeeded.



Value	Parameter Description
0x01 to 0xFF	HCI_LE_Remove_Device_From_White_List command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Event(s) generated (unless masked away):

When the HCI_LE_Remove_Device_From_White_List command has completed, an HCI_Command_Complete event shall be generated.



7.8.18 LE Connection Update command

Command	OCF	Command Parameters	Return Parameters
HCI_LE_ - Connection_Update	0x0013	Connection_Handle, Conn_Interval_Min, Conn_Interval_Max, Conn_Latency, Supervision_Timeout, Minimum_CE_Length, Maximum_CE_Length	

Description:

The HCI_LE_Connection_Update command is used to change the Link Layer Connection parameters of a connection. This command may be issued on both the master and slave.

The Conn_Interval_Min and Conn_Interval_Max parameters are used to define the minimum and maximum allowed connection interval. The Conn_Interval_Min parameter shall not be greater than the Conn_Interval_Max parameter.

The Conn_Latency parameter shall define the maximum allowed connection latency.

The Supervision_Timeout parameter shall define the link supervision timeout for the LE link. The Supervision_Timeout in milliseconds shall be larger than $(1 + \text{Conn_Latency}) * \text{Conn_Interval_Max} * 2$, where Conn_Interval_Max is given in milliseconds.

The Minimum_CE_Length and Maximum_CE_Length are information parameters providing the Controller with a hint about the expected minimum and maximum length of the connection events. The Minimum_CE_Length shall be less than or equal to the Maximum_CE_Length.

The actual parameter values selected by the Link Layer may be different from the parameter values provided by the Host through this command.

Command parameters:

Connection_Handle: Size: 2 octets (12 bits meaningful)

Value	Parameter Description
0xFFFF	Connection_Handle Range 0x0000 to 0x0EFF

**Conn_Interval_Min:****Size: 2 octets**

Value	Parameter Description
N = 0xXXXX	Minimum value for the connection interval. This shall be less than or equal to Conn_Interval_Max. Range: 0x0006 to 0x0C80 Time = N * 1.25 ms Time Range: 7.5 ms to 4 s.

Conn_Interval_Max:**Size: 2 octets**

Value	Parameter Description
N = 0xXXXX	Maximum value for the connection interval. This shall be greater than or equal to Conn_Interval_Min. Range: 0x0006 to 0x0C80 Time = N * 1.25 ms Time Range: 7.5 ms to 4 s.

Conn_Latency:**Size: 2 octets**

Value	Parameter Description
0xXXXX	Slave latency for the connection in number of connection events. Range: 0x0000 to 0x01F3

Supervision_Timeout:**Size: 2 octets**

Value	Parameter Description
N = 0xXXXX	Supervision timeout for the LE Link. Range: 0x000A to 0x0C80 Time = N * 10 ms Time Range: 100 ms to 32 s

**Minimum_CE_Length:****Size: 2 octets**

Value	Parameter Description
N = 0xXXXX	Information parameter about the minimum length of connection event needed for this LE connection. How this value is used is outside the scope of this specification. Range: 0x0000 to 0xFFFF Time = N * 0.625 ms.

Maximum_CE_Length:**Size: 2 octets**

Value	Parameter Description
N = 0xXXXX	Information parameter about the maximum length of connection event needed for this LE connection. How this value is used is outside the scope of this specification. Range: 0x0000 to 0xFFFF Time = N * 0.625 ms.

Return parameters:

None.

Event(s) generated (unless masked away):

When the Controller receives the HCI_LE_Connection_Update command, the Controller sends the HCI_Command_Status event to the Host. The HCI_LE_Connection_Update_Complete event shall be generated after the connection parameters have been applied by the Controller or if the command subsequently fails.

Note: An HCI_Command_Complete event is not sent by the Controller to indicate that this command has been completed. Instead, the HCI_LE_Connection_Update_Complete event indicates that this command has been completed.



7.8.19 LE Set Host Channel Classification command

Command	OCF	Command Parameters	Return Parameters
HCI_LE_Set_Host_Channel_Classification	0x0014	Channel_Map	Status

Description:

The HCI_LE_Set_Host_Channel_Classification command allows the Host to specify a channel classification for the data, secondary advertising, and periodic physical channels based on its “local information”. This classification persists until overwritten with a subsequent HCI_LE_Set_Host_Channel_Classification command or until the Controller is reset using the HCI_Reset command (see [\[Vol 6\] Part B, Section 4.5.8.1](#)).

If this command is used, the Host should send it within 10 seconds of knowing that the channel classification has changed. The interval between two successive commands sent shall be at least one second.

This command shall only be used when the local device supports the Master role or supports extended advertising in the Advertising state.

Command parameters:

Channel_Map:

Size: 5 octets (37 bits meaningful)

Value	Parameter Description
0xFFFFFFFF	This parameter contains 37 1-bit fields. The n^{th} such field (in the range 0 to 36) contains the value for the Link Layer channel index n. Channel n is bad = 0. Channel n is unknown = 1. The most significant bits are reserved for future use. At least one channel shall be marked as unknown.

Return parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	HCI_LE_Set_Host_Channel_Classification command succeeded.
0x01 to 0xFF	HCI_LE_Set_Host_Channel_Classification command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

**Event(s) generated (unless masked away):**

When the HCI_LE_Set_Host_Channel_Classification command has completed, an HCI_Command_Complete event shall be generated.



7.8.20 LE Read Channel Map command

Command	OCF	Command Parameters	Return Parameters
HCI_LE_Read_Channel_Map	0x0015	Connection_Handle	Status, Connection_Handle, Channel_Map

Description:

The HCI_LE_Read_Channel_Map command returns the current Channel_Map for the specified Connection_Handle. The returned value indicates the state of the Channel_Map specified by the last transmitted or received Channel_Map (in a CONNECT_IND or LL_CHANNEL_MAP_IND message) for the specified Connection_Handle, regardless of whether the Master has received an acknowledgment.

Command parameters:

Connection_Handle:

Size: 2 octets (12 bits meaningful)

Value	Parameter Description
0xFFFF	Connection_Handle Range 0x0000 to 0x0EFF

Return parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	HCI_LE_Read_Channel_Map command succeeded.
0x01 to 0xFF	HCI_LE_Read_Channel_Map command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Connection_Handle:

Size: 2 octets (12 bits meaningful)

Value	Parameter Description
0xFFFF	Connection_Handle Range 0x0000 to 0x0EFF

**Channel_Map:****Size: 5 octets**

Value	Parameter Description
0xFFFFFFFFXXXX	This parameter contains 37 1-bit fields. The n^{th} such field (in the range 0 to 36) contains the value for the Link Layer channel index n. Channel n is unused = 0. Channel n is used = 1. The most significant bits are reserved for future use.

Event(s) generated (unless masked away):

When the HCI_LE_Read_Channel_Map command has completed, an HCI_Command_Complete event shall be generated.



7.8.21 LE Read Remote Features command

Command	OCF	Command Parameters	Return Parameters
HCI_LE_Read_Remote_Features	0x0016	Connection_Handle	

Description:

This command requests, from the remote device identified by the Connection_Handle, the features used on the connection and the features supported by the remote device. For details see [\[Vol 6\] Part B, Section 4.6](#).

This command may be issued on both the master and slave.

Note: If a connection already exists between the two devices and the features have already been fetched on that connection, the Controller may use a cached copy of the features.

Command parameters:

Connection_Handle:

Size: 2 octets (12 bits meaningful)

Value	Parameter Description
0xXXXX	Connection_Handle Range 0x0000 to 0x0EFF

Return parameters:

None.

Event(s) generated (unless masked away):

When the Controller receives the HCI_LE_Read_Remote_Features command, the Controller shall send the HCI_Command_Status event to the Host. When the Controller has completed the procedure to determine the remote features or has determined that it will be using a cached copy, the Controller shall send an HCI_LE_Read_Remote_Features_Complete event to the Host.

The HCI_LE_Read_Remote_Features_Complete event contains the status of this command and the parameter describing the features used on the connection and the features supported by the remote device.

Note: An HCI_Command_Complete event is not sent by the Controller to indicate that this command has been completed. Instead, the HCI_LE_Read_Remote_Features_Complete event indicates that this command has been completed.



7.8.22 LE Encrypt command

Command	OCF	Command Parameters	Return Parameters
HCI_LE_Encrypt	0x0017	Key, Plaintext_Data	Status, Encrypted_Data

Description:

The HCI_LE_Encrypt command is used to request the Controller to encrypt the Plaintext_Data in the command using the Key given in the command and returns the Encrypted_Data to the Host. The AES-128 bit block cypher is defined in NIST Publication FIPS-197 (<http://csrc.nist.gov/publications/fips/fips197/fips-197.pdf>).

Command parameters:

Key: *Size: 16 octets*

Value	Parameter Description
0xFFFFFFFFXXXX XXXXXXXXXXXXXXX	128 bit key for the encryption of the data given in the command. The most significant octet of the key corresponds to key[0] using the notation specified in FIPS 197.

Plaintext_Data: *Size: 16 octets*

Value	Parameter Description
0xFFFFFFFFXXXX XXXXXXXXXXXXXXX	128 bit data block that is requested to be encrypted. The most significant octet of the PlainText_Data corresponds to in[0] using the notation specified in FIPS 197.

Return parameters:

Status: *Size: 1 octet*

Value	Parameter Description
0x00	HCI_LE_Encrypt command succeeded.
0x01 to 0xFF	HCI_LE_Encrypt command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Encrypted_Data: *Size: 16 octets*

Value	Parameter Description
0xFFFFFFFFXXXX XXXXXXXXXXXXXXX	128 bit encrypted data block. The most significant octet of the Encrypted_Data corresponds to out[0] using the notation specified in FIPS 197.

**Event(s) generated (unless masked away):**

When the HCI_LE_Encrypt command has completed, an HCI_Command_Complete event shall be generated.



7.8.23 LE Rand command

Command	OCF	Command Parameters	Return Parameters
HCI_LE_Rand	0x0018		Status, Random_Number

Description:

The HCI_LE_Rand command is used to request the Controller to generate 8 octets of random data to be sent to the Host. The Random_Number shall be generated according to [\[Vol 2\] Part H, Section 2](#) if the LE Feature (LE Encryption) is supported.

Command parameters:

None.

Return parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	HCI_LE_Rand command succeeded.
0x01 to 0xFF	HCI_LE_Rand command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Random_Number:

Size: 8 octets

Value	Parameter Description
0xFFFFFFFFXX XXXXX	Random Number

Event(s) generated (unless masked away):

When the HCI_LE_Rand command has completed, an HCI_Command_Complete event shall be generated.



7.8.24 LE Start Encryption command

Command	OCF	Command Parameters	Return Parameters
HCI_LE_Start_Encryption	0x0019	Connection_Handle, Random_Number, Encrypted_Diversifier, Long_Term_Key	

Description:

The HCI_LE_Start_Encryption command is used to authenticate the given encryption key associated with the remote device specified by the Connection_Handle, and once authenticated will encrypt the connection. The parameters are as defined in [\[Vol 3\] Part H, Section 2.4.4](#).

If the connection is already encrypted then the Controller shall pause connection encryption before attempting to authenticate the given encryption key, and then re-encrypt the connection. While encryption is paused no user data shall be transmitted.

On an authentication failure, the connection shall be automatically disconnected by the Link Layer. If this command succeeds, then the connection shall be encrypted.

This command shall only be used when the local device's role is Master.

Command parameters:

Connection_Handle: *Size: 2 octets (12 bits meaningful)*

Value	Parameter Description
0xFFFF	Connection_Handle Range 0x0000 to 0x0EFF

Random_Number: *Size: 8 octets*

Value	Parameter Description
0xFFFFFFFFXXXXXX	64 bit random number.

Encrypted_Diversifier: *Size: 2 octets*

Value	Parameter Description
0xFFFF	16 bit encrypted diversifier.



Long_Term_Key:

Size: 16 octets

Value	Parameter Description
0XXXXXXXXXXXXXXXXX XXXXXXXXXXXXXXXXXX XXX	128 bit long term key.

Return parameters:

None.

Event(s) generated (unless masked away):

When the Controller receives the HCI_LE_Start_Encryption command it shall send the HCI_Command_Status event to the Host. If the connection is not encrypted when this command is issued, an HCI_Encryption_Change event shall occur when encryption has been started for the connection. If the connection is encrypted when this command is issued, an HCI_Encryption_Key_Refresh_Complete event shall occur when encryption has been resumed.

Note: An HCI_Command_Complete event is not sent by the Controller to indicate that this command has been completed. Instead, the HCI_Encryption_Change or HCI_Encryption_Key_Refresh_Complete events indicate that this command has been completed.



7.8.25 LE Long Term Key Request Reply command

Command	OCF	Command Parameters	Return Parameters
HCI_LE_Long_Term_Key_Request_Reply	0x001A	Connection_Handle, Long_Term_Key	Status, Connection_Handle

Description:

The HCI_LE_Long_Term_Key_Request_Reply command is used to reply to an HCI_LE_Long_Term_Key_Request event from the Controller, and specifies the Long_Term_Key parameter that shall be used for this Connection_Handle. The Long_Term_Key is used as defined in [\[Vol 6\] Part B, Section 5.1.3](#).

This command shall only be used when the local device's role is Slave.

Command parameters:

Connection_Handle: *Size: 2 octets (12 bits meaningful)*

Value	Parameter Description
0xFFFF	Connection_Handle Range 0x0000 to 0x0EFF

Long Term Key: *Size: 16 octets*

Value	Parameter Description
0xFFFFFFFFFFFFFFFF XXXXXXXXXXXXXXXX XXX	128 bit long term key for the given connection.

Return parameters:

Status: *Size: 1 octet*

Value	Parameter Description
0x00	HCI_LE_Long_Term_Key_Request_Reply command succeeded.
0x01 to 0xFF	HCI_LE_Long_Term_Key_Request_Reply command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Connection_Handle: *Size: 2 octets (12 bits meaningful)*

Value	Parameter Description
0xFFFF	Connection_Handle Range 0x0000 to 0x0EFF

**Event(s) generated (unless masked away):**

When the HCI_LE_Long_Term_Key_Request_Reply command has completed, an HCI_Command_Complete event shall be generated.



7.8.26 LE Long Term Key Request Negative Reply command

Command	OCF	Command Parameters	Return Parameters
HCI_LE_Long_Term_Key_Request_Negative_Reply	0x001B	Connection_Handle	Status, Connection_Handle

Description:

The HCI_LE_Long_Term_Key_Request_Negative_Reply command is used to reply to an HCI_LE_Long_Term_Key_Request event from the Controller if the Host cannot provide a Long Term Key for this Connection_Handle.

This command shall only be used when the local device's role is Slave.

Command parameters:

Connection_Handle: *Size: 2 octets (12 bits meaningful)*

Value	Parameter Description
0xXXXX	Connection_Handle Range 0x0000 to 0x0EFF

Return parameters:

Status: *Size: 1 octet*

Value	Parameter Description
0x00	HCI_LE_Long_Term_Key_Request_Negative_Reply command succeeded.
0x01 to 0xFF	HCI_LE_Long_Term_Key_Request_Negative_Reply command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Connection_Handle: *Size: 2 octets (12 bits meaningful)*

Value	Parameter Description
0xXXXX	Connection_Handle Range 0x0000 to 0x0EFF

Event(s) generated (unless masked away):

When the HCI_LE_Long_Term_Key_Request_Negative_Reply command has completed, an HCI_Command_Complete event shall be generated.



7.8.27 LE Read Supported States command

Command	OCF	Command Parameters	Return Parameters
HCI_LE_Read_Supported_States	0x001C		Status, LE_States

Description:

The HCI_LE_Read_Supported_States command reads the states and state combinations that the Link Layer supports. See [\[Vol 6\] Part B, Section 1.1.1](#).

LE_States is an 8-octet bit field. If a bit is set to 1 then this state or state combination is supported by the Controller. Multiple bits in LE_States may be set to 1 to indicate support for multiple state and state combinations.

Note: this command only provides information about the supported states that can be used with legacy advertising. It does not provide information about states, and combinations of states, that can only be used with the extended advertising commands (see Section 3.1.1).

Command parameters:

None.

Return parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	HCI_LE_Read_Supported_States command succeeded.
0x01 to 0xFF	HCI_LE_Read_Supported_States command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

LE_States:

Size: 8 octets

SUPPORTED STATES & ROLES									
Bit	Scannable Undirected Advertising State	Connectable and Scannable Undirected Advertising State	Non-connectable and Non-Scannable Undirected Advertising State	High Duty Cycle Connectable Directed Advertising State	Low Duty Cycle Connectable Directed Advertising State	Active Scanning State	Passive Scanning State	Initiating State	Connection State (Master Role)
0			X						
1	X								



SUPPORTED STATES & ROLES										
Bit	Scannable Undirected Advertising State	Connectable and Scannable Undirected Advertising State	Non-connectable and Non-Scannable Undirected Advertising State	High Duty Cycle Connectable Directed Advertising State	Low Duty Cycle Connectable Directed Advertising State	Active Scanning State	Passive Scanning State	Initiating State	Connection State (Master Role)	Connection State (Slave Role)
2		X								
3				X						
4							X			
5						X				
6								X		
7										X
8			X				X			
9	X						X			
10		X					X			
11				X			X			
12			X			X				
13	X					X				
14		X				X				
15				X		X				
16			X					X		
17	X							X		
18			X						X	
19	X								X	
20			X							X
21	X									X
22							X	X		
23						X		X		
24							X		X	
25						X			X	
26							X			X
27						X				X
28								X	X	
29					X					
30					X		X			
31					X	X				
32		X						X		
33				X				X		
34					X			X		
35		X							X	
36				X					X	



Bit	SUPPORTED STATES & ROLES								
	Scannable Undirected Advertising State	Connectable and Scannable Undirected Advertising State	Non-connectable and Non-Scannable Undirected Advertising State	High Duty Cycle Connectable Directed Advertising State	Low Duty Cycle Connectable Directed Advertising State	Active Scanning State	Passive Scanning State	Initiating State	Connection State (Slave Role)
37					X				X
38		X							X
39				X					X
40					X				X
41								X	X

All bits not listed in this table, and the value with all bits set to 0, are reserved for future use.

Event(s) generated (unless masked away):

When the HCI_LE_Read_Supported_States command has completed, an HCI_Command_Complete event will be generated.



7.8.28 LE Receiver Test command [v1]

Command	OCF	Command Parameters	Return Parameters
HCI_LE_Receiver_Test [v1]	0x001D	RX_Channel	Status

Description:

This is a previous version of the HCI_LE_Receiver_Test command [v3] (see [Section 7.8.78](#)). It is equivalent to that version of the command with the following command parameter values:

Parameter	Value
PHY	0x01
Modulation_Index	0x00
Expected_CTE_Length	0x00
Expected_CTE_Type	<i>any valid value</i>
Slot_Durations	<i>any valid value</i>
Length_of_Switching_Pattern	<i>any valid value</i>
Antenna_IDs[i]	<i>any valid value</i>



7.8.29 LE Transmitter Test command [v1]

Command	OCF	Command Parameters	Return Parameters
HCI_LE_Transmitter_Test [v1]	0x001E	TX_Channel, Length_Of_Test_Data, Packet_Payload	Status

Description:

This is a previous version of the HCI_LE_Transmitter_Test command [v3] (see [Section 7.8.79](#)). It is equivalent to that version of the command with the following command parameter values:

Parameter	Value
PHY	0x01
CTE_Length	0x00
CTE_Type	<i>any valid value</i>
Length_of_Switching_Pattern	<i>any valid value</i>
Antenna_IDs[i]	<i>any valid value</i>



7.8.30 LE Test End command

Command	OCF	Command Parameters	Return Parameters
HCI_LE_Test_End	0x001F		Status, Number_Of_Packets

Description:

This command is used to stop any test which is in progress. The Number_Of_Packets for a transmitter test shall be reported as 0x0000. The Number_Of_Packets is an unsigned number and contains the number of received packets.

Command parameters:

None.

Return parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	HCI_LE_Test_End command succeeded.
0x01 to 0xFF	HCI_LE_Test_End command failed See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Number_Of_Packets:

Size: 2 octets

Value	Parameter Description
0xXXXX	Number of packets received

Event(s) generated (unless masked away):

When the HCI_LE_Test_End command has completed, an HCI_Command_Complete event shall be generated.



7.8.31 LE Remote Connection Parameter Request Reply command

Command	OCF	Command Parameters	Return Parameters
HCI_LE_Remote_Connection_Parameter_Request_Reply	0x0020	Connection_Handle, Interval_Min, Interval_Max, Latency, Timeout, Minimum_CE_Length, Maximum_CE_Length	Status, Connection_Handle

Description:

Both the master Host and the slave Host use this command to reply to the HCI_LE_Remote_Connection_Parameter_Request event. This indicates that the Host has accepted the remote device's request to change connection parameters.

The Interval_Min and Interval_Max parameters define the minimum and maximum allowed connection interval. The Interval_Min parameter shall not be greater than the Interval_Max parameter.

The Latency parameter shall define the maximum allowed slave latency for the connection in number of connection events.

The Timeout parameter shall define the link supervision timeout for the LE link. The Timeout in milliseconds shall be larger than $(1 + \text{Latency}) * \text{Interval_Max} * 2$, where Interval_Max is given in milliseconds.

The Minimum_CE_Length and Maximum_CE_Length are information parameters providing the Controller with a hint about the expected minimum and maximum length of the connection events. The Minimum_CE_Length shall be less than or equal to the Maximum_CE_Length.

The actual parameter values selected by the Link Layer may be different from the parameter values provided by the Host through this command.

Command parameters:

Connection_Handle:

Size: 2 octets (12 bits meaningful)

Value	Parameter Description
0xXXXX	Connection_Handle Range 0x0000 to 0x0EFF

Host Controller Interface Functional Specification*Interval_Min:**Size: 2 octets*

Value	Parameter Description
N = 0xXXXX	Minimum value of the connection interval. Range: 0x0006 to 0x0C80 Time = N * 1.25 ms Time Range: 7.5 ms to 4 s

*Interval_Max:**Size: 2 octets*

Value	Parameter Description
N = 0xXXXX	Maximum value of the connection interval. Range: 0x0006 to 0x0C80 Time = N * 1.25 ms Time Range: 7.5 ms to 4 s

*Latency:**Size: 2 octets*

Value	Parameter Description
0xXXXX	Maximum allowed slave latency for the connection specified as the number of connection events. Range: 0x0000 to 0x01F3 (499)

*Timeout:**Size: 2 octets*

Value	Parameter Description
N = 0xXXXX	Supervision timeout for the connection. Range: 0x000A to 0x0C80 Time = N * 10 ms Time Range: 100 ms to 32 s

*Minimum_CE_Length:**Size: 2 octets*

Value	Parameter Description
N = 0xXXXX	Information parameter about the minimum length of connection event needed for this LE connection. Range: 0x0000 to 0xFFFF Time = N * 0.625 ms Time Range: 0 ms to 40.9 s

**Maximum_CE_Length:****Size: 2 octets**

Value	Parameter Description
N = 0xXXXX	Information parameter about the maximum length of connection event needed for this LE connection. Range: 0x0000 to 0xFFFF Time = N * 0.625 ms Time Range: 0 ms to 40.9 s

Return parameters:**Status:****Size: 1 octet**

Value	Parameter Description
0x00	HCI_LE_Remote_Connection_Parameter_Request_Reply command succeeded.
0x01 to 0xFF	HCI_LE_Remote_Connection_Parameter_Request_Reply command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Connection_Handle:**Size: 2 octets (12 bits meaningful)**

Value	Parameter Description
0xXXXX	Connection_Handle Range 0x0000 to 0x0EFF

Events(s) generated (unless masked away):

When the HCI_LE_Remote_Connection_Parameter_Request_Reply command has completed, an HCI_Command_Complete event shall be generated.



7.8.32 LE Remote Connection Parameter Request Negative Reply command

Command	OCF	Command Parameters	Return Parameters
HCI_LE_Remote_Connection_Parameter_Request_Negative_Reply	0x0021	Connection_Handle, Reason	Status, Connection_Handle

Description:

Both the master Host and the slave Host use this command to reply to the HCI_LE_Remote_Connection_Parameter_Request event. This indicates that the Host has rejected the remote device's request to change connection parameters. The reason for the rejection is given in the Reason parameter.

Instead of issuing this command, the Host should try to provide alternative connection parameters to the Link Layer via the HCI_LE_Remote_Connection_Parameter_Request_Reply command ([Section 7.8.31](#)).

Command parameters:

Connection_Handle: *Size: 2 octets (12 bits meaningful)*

Value	Parameter Description
0xFFFF	Connection_Handle Range 0x0000 to 0xFFFF

Reason: *Size: 1 octet*

Value	Parameter Description
0xFF	Reason that the connection parameter request was rejected. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Return parameters:

Status: *Size: 1 octet*

Value	Parameter Description
0x00	HCI_LE_Remote_Connection_Parameter_Request_Negative_Reply command succeeded.
0x01 to 0xFF	HCI_LE_Remote_Connection_Parameter_Request_Negative_Reply command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.



Connection_Handle: *Size: 2 octets (12 bits meaningful)*

Value	Parameter Description
0xFFFF	Connection_Handle Range 0x0000 to 0x0EFF

Events(s) generated (unless masked away):

When the
HCI_LE_Remote_Connection_Parameter_Request_Negative_Reply
command has completed, an HCI_Command_Complete event shall be
generated.



7.8.33 LE Set Data Length command

Command	OCF	Command Parameters	Return Parameters
HCI_LE_Set_Data_Length	0x0022	Connection_Handle, TxOctets, TxTime	Status, Connection_Handle

Description:

The HCI_LE_Set_Data_Length command allows the Host to suggest the maximum transmission payload size and maximum packet transmission time (*connMaxTxOctets* and *connMaxTxTime* - see [Vol 6] Part B, Section 4.5.10) to be used for LL Data PDUs on a given connection. The Controller may use smaller or larger values based on local information.

Command parameters:

Connection_Handle: *Size: 2 octets (12 bits meaningful)*

Value	Parameter Description
0xFFFF	Connection_Handle Range 0x0000 to 0x0EFF

TxOctets: *Size: 2 octets*

Value	Parameter Description
0xFFFF	Preferred maximum number of payload octets that the local Controller should include in a single LL Data PDU on this connection. Range 0x001B to 0x00FB

TxTime: *Size: 2 octets*

Value	Parameter Description
0xFFFF	Preferred maximum number of microseconds that the local Controller should use to transmit a single Link Layer packet containing an LL Data PDU on this connection. Range 0x0148 to 0x4290

Return parameters:

Status: *Size: 1 octet*

Value	Parameter Description
0x00	HCI_LE_Set_Data_Length command succeeded.
0x01 to 0xFF	HCI_LE_Set_Data_Length command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

**Connection_Handle:****Size: 2 octets (12 bits meaningful)**

Value	Parameter Description
0xFFFF	Connection_Handle Range 0x0000 to 0x0EFF

Event(s) generated (unless masked away):

When the HCI_LE_Set_Data_Length command has completed, an HCI_Command_Complete event shall be generated.

If the command causes the maximum transmission packet size or maximum packet transmission time to change, an HCI_LE_Data_Length_Change event shall be generated.



7.8.34 LE Read Suggested Default Data Length command

Command	OCF	Command Parameters	Return Parameters
HCI_LE_Read_Suggested_Default_Data_Length	0x0023		Status, SuggestedMaxTxOctets, SuggestedMaxTxTime

Description:

The HCI_LE_Read_Suggested_Default_Data_Length command allows the Host to read the Host's suggested values (SuggestedMaxTxOctets and SuggestedMaxTxTime) for the Controller's maximum transmitted number of payload octets and maximum packet transmission time for packets containing LL Data PDUs to be used for new connections (see [\[Vol 6\] Part B, Section 4.5.10](#)).

Command parameters:

None.

Return parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	HCI_LE_Read_Suggested_Default_Data_Length command succeeded
0x01 to 0xFF	HCI_LE_Read_Suggested_Default_Data_Length command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

SuggestedMaxTxOctets:

Size: 2 octets

Value	Parameter Description
0xFFFF	The Host's suggested value for the Controller's maximum transmitted number of payload octets in LL Data PDUs to be used for new connections. Range 0x001B to 0x00FB Default: 0x001B



SuggestedMaxTxTime:

Size: 2 octets

Value	Parameter Description
0xFFFF	The Host's suggested value for the Controller's maximum packet transmission time for packets containing LL Data PDUs to be used for new connections. Range 0x0148 to 0x4290 Default: 0x0148

Event(s) generated (unless masked away):

When the HCI_LE_Read_Suggested_Default_Data_Length command has completed, an HCI_Command_Complete event shall be generated.



7.8.35 LE Write Suggested Default Data Length command

Command	OCF	Command Parameters	Return Parameters
HCI_LE_Write_Suggested_Default_Data_Length	0x0024	SuggestedMaxTxOctets, SuggestedMaxTxTime	Status

Description:

The HCI_LE_Write_Suggested_Default_Data_Length command allows the Host to specify its suggested values for the Controller's maximum transmission number of payload octets and maximum packet transmission time for packets containing LL Data PDUs to be used for new connections. The Controller may use smaller or larger values for *connInitialMaxTxOctets* and *connInitialMaxTxTime* based on local information. (See [Vol 6] Part B, Section 4.5.10).

Command parameters:

SuggestedMaxTxOctets:

Size: 2 octets

Value	Parameter Description
0xXXXX	The Host's suggested value for the Controller's maximum transmitted number of payload octets in LL Data PDUs to be used for new connections. Range 0x001B to 0x00FB

SuggestedMaxTxTime:

Size: 2 octets

Value	Parameter Description
0xXXXX	The Host's suggested value for the Controller's maximum packet transmission time for packets containing LL Data PDUs to be used for new connections. Range 0x0148 to 0x4290

Return parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	HCI_LE_Write_Suggested_Default_Data_Length command succeeded.
0x01 to 0xFF	HCI_LE_Write_Suggested_Default_Data_Length command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Event(s) generated (unless masked away):



When the HCI_LE_Write_Suggested_Default_Data_Length command has completed, an HCI_Command_Complete event shall be generated.



7.8.36 LE Read Local P-256 Public Key command

Command	OCF	Command Parameters	Return Parameters
HCI_LE_Read_Local_P-256_Public_Key	0x0025		

Description:

The HCI_LE_Read_Local_P-256_Public_Key command is used to return the local P-256 public key from the Controller. The Controller shall generate a new P-256 public/private key pair upon receipt of this command.

The keys returned via this command shall not be used when Secure Connections is used over the BR/EDR transport.

Command parameters:

None.

Return parameters:

None.

Event(s) generated (unless masked away):

When the Controller receives the HCI_LE_Read_Local_P-256_Public_Key command, the Controller shall send the HCI_Command_Status event to the Host. When the local P-256 public key generation finishes, an HCI_LE_Read_Local_P-256_Public_Key_Complete event shall be generated.

No HCI_Command_Complete event is sent by the Controller to indicate that this command has been completed.



7.8.37 LE Generate DHKey command [v1]

Command	OCF	Command Parameters	Return Parameters
HCI_LE_Generate_- DHKey [v1]	0x0026	Remote_P-256_Public_Key	

Description:

This is a previous version of the HCI_LE_Generate_DHKey command [v2] (see [Section 7.8.93](#)). It is equivalent to that version of the command with the following command parameter values:

Parameter	Value
Key_Type	0x00



7.8.38 LE Add Device To Resolving List command

Command	OCF	Command Parameters	Return Parameters
HCI_LE_Add_Device_To_ - Resolving_List	0x0027	Peer_Identity_Address_Type, Peer_Identity_Address, Peer_IRK, Local_IRK	Status

Description:

The HCI_LE_Add_Device_To_Resolving_List command is used to add one device to the resolving list used to generate and resolve Resolvable Private Addresses in the Controller.

This command shall not be used when address resolution is enabled in the Controller and:

- Advertising (other than periodic advertising) is enabled,
- Scanning is enabled, or
- an HCI_LE_Create_Connection, HCI_LE_Extended_Create_Connection, or HCI_LE_Periodic_Advertising_Create_Sync command is outstanding.

This command may be used at any time when address resolution is disabled in the Controller.

The added device shall be set to Network Privacy mode.

When a Controller cannot add a device to the list because there is no space available, it shall return the error code *Memory Capacity Exceeded* (0x07).

Command parameters:

Peer_Identity_Address_Type:

Size: 1 octet

Value	Parameter Description
0x00	Public Identity Address
0x01	Random (static) Identity Address
All other values	Reserved for future use

Peer_Identity_Address:

Size: 6 octets

Value	Parameter Description
0xFFFFFFFFXXXX	Public or Random (static) Identity address of the peer device

**Peer_IRK:****Size: 16 octets**

Value	Parameter Description
0XXXXXXXXXXXXXXXXX XXXXXXXXXXXXXXXXXX	IRK of the peer device

Local_IRK:**Size: 16 octets**

Value	Parameter Description
0XXXXXXXXXXXXXXXXX XXXXXXXXXXXXXXXXXX	IRK of the local device

Return parameters:**Status:****Size: 1 octet**

Value	Parameter Description
0x00	HCI_LE_Add_Device_To_Resolving_List command succeeded
0x01 to 0xFF	HCI_LE_Add_Device_To_Resolving_List command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Event(s) generated (unless masked away):

When the HCI_LE_Add_Device_To_Resolving_List command has completed, an HCI_Command_Complete event shall be generated.



7.8.39 LE Remove Device From Resolving List command

Command	OCF	Command Parameters	Return Parameters
HCI_LE_Remove_Device_From_Resolving_List	0x0028	Peer_Identity_Address_Type, Peer_Identity_Address	Status

Description:

The HCI_LE_Remove_Device_From_Resolving_List command is used to remove one device from the resolving list used to resolve Resolvable Private Addresses in the Controller.

This command shall not be used when address resolution is enabled in the Controller and:

- Advertising (other than periodic advertising) is enabled,
- Scanning is enabled, or
- an HCI_LE_Create_Connection, HCI_LE_Extended_Create_Connection, or HCI_LE_Periodic_Advertising_Create_Sync command is outstanding.

This command may be used at any time when address resolution is disabled in the Controller.

When a Controller cannot remove a device from the resolving list because it is not found, it shall return the error code *Unknown Connection Identifier* (0x02).

Command parameters:

Peer_Identity_Address_Type:

Size: 1 octet

Value	Parameter Description
0x00	Public Identity Address
0x01	Random (static) Identity Address
All other values	Reserved for future use

Peer_Device_Address:

Size: 6 octets

Value	Parameter Description
0xFFFFFFFFXXXX	Public or Random (static) Identity Address of the peer device



Return parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	HCI_LE_Remove_Device_From_Resolving_List command succeeded
0x01 to 0xFF	HCI_LE_Remove_Device_From_Resolving_List command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Event(s) generated (unless masked away):

When the HCI_LE_Remove_Device_From_Resolving_List command has completed, an HCI_Command_Complete event shall be generated.



7.8.40 LE Clear Resolving List command

Command	OCF	Command Parameters	Return Parameters
HCI_LE_Clear_Resolving_List	0x0029		Status

Description:

The HCI_LE_Clear_Resolving_List command is used to remove all devices from the resolving list used to resolve Resolvable Private Addresses in the Controller.

This command shall not be used when address resolution is enabled in the Controller and:

- Advertising (other than periodic advertising) is enabled,
- Scanning is enabled, or
- an HCI_LE_Create_Connection, HCI_LE_Extended_Create_Connection, or HCI_LE_Periodic_Advertising_Create_Sync command is outstanding.

This command may be used at any time when address resolution is disabled in the Controller.

Command parameters:

None

Return parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	HCI_LE_Clear_Resolving_List command succeeded
0x01 to 0xFF	HCI_LE_Clear_Resolving_List command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Event(s) generated (unless masked away):

When the HCI_LE_Clear_Resolving_List command has completed, an HCI_Command_Complete event shall be generated.



7.8.41 LE Read Resolving List Size command

Command	OCF	Command Parameters	Return Parameters
HCI_LE_Read_Resolving_List_Size	0x002A		Status, Resolving_List_Size

Description:

The HCI_LE_Read_Resolving_List_Size command is used to read the total number of entries in the resolving list that can be stored in the Controller.

Note: The number of entries that can be stored is not fixed and the Controller can change it at any time (e.g. because the memory used to store the list can also be used for other purposes).

Command parameters:

None

Return parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	HCI_LE_Read_Resolving_List_Size command succeeded
0x01 to 0xFF	HCI_LE_Read_Resolving_List_Size command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Resolving_List_Size:

Size: 1 octet

Value	Parameter Description
0xXX	Number of entries in the resolving list

Event(s) generated (unless masked away):

When the HCI_LE_Read_Resolving_List_Size command has completed, an HCI_Command_Complete event shall be generated.



7.8.42 LE Read Peer Resolvable Address command

Command	OCF	Command Parameters	Return Parameters
HCI_LE_Read_Peer_Resolvable_Address	0x002B	Peer_Identity_Address_Type, Peer_Identity_Address	Status, Peer_Resolvable_Address

Description:

The HCI_LE_Read_Peer_Resolvable_Address command is used to get the current peer Resolvable Private Address being used for the corresponding peer Public and Random (static) Identity Address. The peer's resolvable address being used may change after the command is called.

This command may be used at any time.

When a Controller cannot find a Resolvable Private Address associated with the Peer Identity Address, or if the Peer Identity Address cannot be found in the resolving list, it shall return the error code *Unknown Connection Identifier* (0x02).

Command parameters:

Peer_Identity_Address_Type

Size: 1 octet

Value	Parameter Description
0x00	Public Identity Address
0x01	Random (static) Identity Address
All other values	Reserved for future use

Peer_Identity_Address:

Size: 6 octets

Value	Parameter Description
0XXXXXXXXXXXXX	Public or Random (static) Identity Address of the peer device

Return parameters:

Status

Size: 1 octet

Value	Parameter Description
0x00	HCI_LE_Read_Peer_Resolvable_Address command succeeded
0x01 to 0xFF	HCI_LE_Read_Peer_Resolvable_Address command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

**Peer_Resolvable_Address:****Size: 6 octets**

Value	Parameter Description
0XXXXXXXXXXXXX	Resolvable Private Address being used by the peer device

Event(s) generated (unless masked away):

When the HCI_LE_Read_Peer_Resolvable_Address command has completed, an HCI_Command_Complete event shall be generated.



7.8.43 LE Read Local Resolvable Address command

Command	OCF	Command Parameters	Return Parameters
HCI_LE_Read_Local_ - Resolvable_Address	0x002C	Peer_Identity_Ad- dress_Type, Peer_Identity_Ad- dress	Status, Local_Resolvable_Ad- dress

Description:

The HCI_LE_Read_Local_Resolvable_Address command is used to get the current local Resolvable Private Address being used for the corresponding peer Identity Address. The local resolvable address being used may change after the command is called.

This command may be used at any time.

When a Controller cannot find a Resolvable Private Address associated with the Peer Identity Address, or if the Peer Identity Address cannot be found in the resolving list, it shall return the error code *Unknown Connection Identifier* (0x02).

Command parameters:

Peer_Identity_Address_Type:

Size: 1 octet

Value	Parameter Description
0x00	Public Identity Address
0x01	Random (static) Identity Address
All other values	Reserved for future use

Peer_Identity_Address:

Size: 6 octets

Value	Parameter Description
0XXXXXXXXXXXXX	Public Identity Address or Random (static) Identity Address of the peer device, 48 bit value.

Return parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	HCI_LE_Read_Local_Resolvable_Address command suc- ceeded



Value	Parameter Description
0x01 to 0xFF	HCI_LE_Read_Local_Resolvable_Address command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

*Local_Resolvable_Address:**Size: 6 octets*

Value	Parameter Description
0XXXXXXXXXXXXX	Resolvable Private Address being used by the local device

Event(s) generated (unless masked away):

When the HCI_LE_Read_Local_Resolvable_Address command has completed, an HCI_Command_Complete event shall be generated.



7.8.44 LE Set Address Resolution Enable command

Command	OCF	Command Parameters	Return Parameters
HCI_LE_Set_Address_Resolution_Enable	0x002D	Address_Resolution_Enable	Status

Description:

The HCI_LE_Set_Address_Resolution_Enable command is used to enable resolution of Resolvable Private Addresses in the Controller. This causes the Controller to use the resolving list whenever the Controller receives a local or peer Resolvable Private Address.

This command shall not be used when:

- Advertising (other than periodic advertising) is enabled,
- Scanning is enabled, or
- an HCI_LE_Create_Connection, HCI_LE_Extended_Create_Connection, or HCI_LE_Periodic_Advertising_Create_Sync command is outstanding.

Note: Enabling address resolution when it is already enabled, or disabling it when it is already disabled, has no effect.

Command parameters:

Address_Resolution_Enable:

Size: 1 octet

Value	Parameter Description
0x00	Address Resolution in Controller disabled (default)
0x01	Address Resolution in Controller enabled
All other values	Reserved for future use

Return parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	HCI_LE_Set_Address_Resolution_Enable command succeeded
0x01 to 0xFF	HCI_LE_Set_Address_Resolution_Enable command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

**Event(s) generated (unless masked away):**

When the HCI_LE_Set_Address_Resolution_Enable command has completed, an HCI_Command_Complete event shall be generated.



7.8.45 LE Set Resolvable Private Address Timeout command

Command	OCF	Command Parameters	Return Parameters
HCI_LE_Set_Resolvable_Private_Address_ - Timeout	0x002E	RPA_Timeout	Status

Description:

The HCI_LE_Set_Resolvable_Private_Address_Timeout command set the length of time the Controller uses a Resolvable Private Address before a new resolvable private address is generated and starts being used.

This timeout applies to all addresses generated by the Controller.

Command parameters:

RPA_Timeout:

Size: 2 octets

Value	Parameter Description
0xXXXX	RPA_Timeout measured in seconds Range: 0x0001 to 0xA1B8 Time range: 1 s to approximately 11.5 hours Default: 0x0384 (900 s or 15 minutes)

Return parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	HCI_LE_Set_Resolvable_Private_Address_Timeout command succeeded
0x01 to 0xFF	HCI_LE_Set_Resolvable_Private_Address_Timeout command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Event(s) generated (unless masked away):

When the HCI_LE_Set_Resolvable_Private_Address_Timeout command has completed, an HCI_Command_Complete event shall be generated.



7.8.46 LE Read Maximum Data Length command

Command	OCF	Command Parameters	Return Parameters
HCI_LE_Read_Maximum_Data_Length	0x002F		Status, supportedMaxTxOctets, supportedMaxTxTime, supportedMaxRxOctets, supportedMaxRxTime

Description:

The HCI_LE_Read_Maximum_Data_Length command allows the Host to read the Controller's maximum supported payload octets and packet duration times for transmission and reception (supportedMaxTxOctets and supportedMaxTxTime, supportedMaxRxOctets, and supportedMaxRxTime, see [\[Vol 6\] Part B, Section 4.5.10](#)).

Command parameters:

None.

Return parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	HCI_LE_Read_Maximum_Data_Length command succeeded.
0x01 to 0xFF	HCI_LE_Read_Maximum_Data_Length command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

supportedMaxTxOctets:

Size: 2 octets

Value	Parameter Description
0xFFFF	Maximum number of payload octets that the local Controller supports for transmission of a single Link Layer packet on a data connection. Range 0x001B to 0x00FB

supportedMaxTxTime:

Size: 2 octets

Value	Parameter Description
0xFFFF	Maximum time, in microseconds, that the local Controller supports for transmission of a single Link Layer packet on a data connection. Range 0x0148 to 0x4290

***supportedMaxRxOctets:******Size: 2 octets***

Value	Parameter Description
0xXXXX	Maximum number of payload octets that the local Controller supports for reception of a single Link Layer packet on a data connection. Range 0x001B to 0x00FB

supportedMaxRxTime:***Size: 2 octets***

Value	Parameter Description
0xXXXX	Maximum time, in microseconds, that the local Controller supports for reception of a single Link Layer packet on a data connection. Range 0x0148 to 0x4290

Event(s) generated (unless masked away):

When the HCI_LE_Read_Maximum_Data_Length command has completed, an HCI_Command_Complete event shall be generated.



7.8.47 LE Read PHY command

Command	OCF	Command Parameters	Return Parameters
HCI_LE_Read_PHY	0x0030	Connection_Handle	Status, Connection_Handle, TX_PHY, RX_PHY

Description:

The HCI_LE_Read_PHY command is used to read the current transmitter PHY and receiver PHY on the connection identified by the Connection_Handle.

Command parameters:

Connection_Handle: *Size: 2 octets (12 bits meaningful)*

Value	Parameter Description
0xFFFF	Connection_Handle Range:0x0000 to 0x0EFF

Return parameters:

Status: *Size: 1 octet*

Value	Parameter Description
0x00	HCI_LE_Read_PHY command succeeded.
0x01 to 0xFF	HCI_LE_Read_PHY command failed. See [Vol 2] Part D for a list of error codes and descriptions.

Connection_Handle: *Size: 2 octets (12 bits meaningful)*

Value	Parameter Description
0xFFFF	Connection_Handle Range:0x0000 to 0x0EFF

TX_PHY: *Size: 1 octet*

Value	Parameter Description
0x01	The transmitter PHY for the connection is LE 1M
0x02	The transmitter PHY for the connection is LE 2M
0x03	The transmitter PHY for the connection is LE Coded
All other values	Reserved for future use



RX_PHY:

Size: 1 octet

Value	Parameter Description
0x01	The receiver PHY for the connection is LE 1M
0x02	The receiver PHY for the connection is LE 2M
0x03	The receiver PHY for the connection is LE Coded
All other values	Reserved for future use

Event(s) generated (unless masked away):

When the HCI_LE_Read_PHY command has completed, an HCI_Command_Complete event shall be generated.



7.8.48 LE Set Default PHY command

Command	OCF	Command Parameters	Return Parameters
HCI_LE_Set_Default_PHY	0x0031	ALL_PHYS, TX_PHYS, RX_PHYS	Status

Description:

The HCI_LE_Set_Default_PHY command allows the Host to specify its preferred values for the transmitter PHY and receiver PHY to be used for all subsequent connections over the LE transport.

The ALL_PHYS parameter is a bit field that allows the Host to specify, for each direction, whether it has no preference among the PHYs that the Controller supports in a given direction or whether it has specified particular PHYs that it prefers in the TX_PHYS or RX_PHYS parameter.

The TX_PHYS parameter is a bit field that indicates the transmitter PHYs that the Host prefers the Controller to use. If the ALL_PHYS parameter specifies that the Host has no preference, the TX_PHYS parameter shall be ignored; otherwise at least one bit shall be set to 1.

The RX_PHYS parameter is a bit field that indicates the receiver PHYs that the Host prefers the Controller to use. If the ALL_PHYS parameter specifies that the Host has no preference, the RX_PHYS parameter shall be ignored; otherwise at least one bit shall be set to 1.

If the Host sets, in the TX_PHYS or RX_PHYS parameter, a bit for a PHY that the Controller does not support, the Controller shall return the error code *Unsupported Feature or Parameter Value* (0x11).

If the Controller does not support asymmetric connections (see [\[Vol 6\] Part B, Section 4.6.9.1](#)) and the Host sets ALL_PHYS to 0x00 and TX_PHYS to a different value than RX_PHYS, the Controller shall return the error code *Unsupported Feature or Parameter Value* (0x11).

Command parameters:

ALL_PHYS:

Size: 1 octet

Bit number	Meaning
0	The Host has no preference among the transmitter PHYs supported by the Controller
1	The Host has no preference among the receiver PHYs supported by the Controller



Bit number	Meaning
All other bits	Reserved for future use

TX_PHYS:*Size: 1 octet*

Bit number	Meaning
0	The Host prefers to use the LE 1M transmitter PHY (possibly among others)
1	The Host prefers to use the LE 2M transmitter PHY (possibly among others)
2	The Host prefers to use the LE Coded transmitter PHY (possibly among others)
All other bits	Reserved for future use

RX_PHYS:*Size: 1 octet*

Bit number	Meaning
0	The Host prefers to use the LE 1M receiver PHY (possibly among others)
1	The Host prefers to use the LE 2M receiver PHY (possibly among others)
2	The Host prefers to use the LE Coded receiver PHY (possibly among others)
All other bits	Reserved for future use

Return parameters:**Status:***Size: 1 octet*

Value	Parameter Description
0x00	HCI_LE_Set_Default_PHY command succeeded.
0x01 to 0xFF	HCI_LE_Set_Default_PHY command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Event(s) generated (unless masked away):

When the HCI_LE_Set_Default_PHY command has completed, an HCI_Command_Complete event shall be generated.



7.8.49 LE Set PHY command

Command	OCF	Command Parameters	Return Parameters
HCI_LE_Set_PHY	0x0032	Connection_Handle, ALL_PHYS, TX_PHYS, RX_PHYS, PHY_options	

Description:

The HCI_LE_Set_PHY command is used to set the PHY preferences for the connection identified by the Connection_Handle. The Controller might not be able to make the change (e.g. because the peer does not support the requested PHY) or may decide that the current PHY is preferable.

The ALL_PHYS parameter is a bit field that allows the Host to specify, for each direction, whether it has no preference among the PHYs that the Controller supports in a given direction or whether it has specified particular PHYs that it prefers in the TX_PHYS or RX_PHYS parameter.

The TX_PHYS parameter is a bit field that indicates the transmitter PHYs that the Host prefers the Controller to use. If the ALL_PHYS parameter specifies that the Host has no preference, the TX_PHYS parameter shall be ignored; otherwise at least one bit shall be set to 1.

The RX_PHYS parameter is a bit field that indicates the receiver PHYs that the Host prefers the Controller to use. If the ALL_PHYS parameter specifies that the Host has no preference, the RX_PHYS parameter shall be ignored; otherwise at least one bit shall be set to 1.

If, for at least one direction, the Host has specified a preference and the current PHY is not one of those preferred, the Controller shall request a change. Otherwise the Controller may, but need not, request a change.

The PHY preferences provided by the HCI_LE_Set_PHY command override those provided via the HCI_LE_Set_Default_PHY command ([Section 7.8.48](#)) or any preferences previously set using the HCI_LE_Set_PHY command on the same connection.

The PHY_options parameter is a bit field that allows the Host to specify options for PHYs. The default value for a new connection shall be all zero bits. The Controller may override any preferred coding for transmitting on the LE Coded PHY.

The Host may specify a preferred coding even if it prefers not to use the LE Coded transmitter PHY since the Controller may override the PHY preference.



If the Host sets, in the TX_PHYS or RX_PHYS parameter, a bit for a PHY that the Controller does not support, the Controller shall return the error code *Unsupported Feature or Parameter Value* (0x11).

If the Controller does not support asymmetric connections (see [Vol 6] Part B, Section 4.6.9.1) and the Host sets ALL_PHYS to 0x00 and TX_PHYS to a different value than RX_PHYS, the Controller shall return the error code *Unsupported Feature or Parameter Value* (0x11).

Command parameters:

Connection_Handle: *Size: 2 octets (12 bits meaningful)*

Value	Parameter Description
0xFFFF	Connection_Handle Range: 0x0000 to 0x0EFF

ALL_PHYS: *Size: 1 octet*

Bit number	Meaning
0	The Host has no preference among the transmitter PHYs supported by the Controller
1	The Host has no preference among the receiver PHYs supported by the Controller
All other bits	Reserved for future use

TX_PHYS: *Size: 1 octet*

Bit number	Meaning
0	The Host prefers to use the LE 1M transmitter PHY (possibly among others)
1	The Host prefers to use the LE 2M transmitter PHY (possibly among others)
2	The Host prefers to use the LE Coded transmitter PHY (possibly among others)
All other bits	Reserved for future use

RX_PHYS: *Size: 1 octet*

Bit number	Meaning
0	The Host prefers to use the LE 1M receiver PHY (possibly among others)
1	The Host prefers to use the LE 2M receiver PHY (possibly among others)
2	The Host prefers to use the LE Coded receiver PHY (possibly among others)
All other bits	Reserved for future use

**PHY_options:****Size: 2 octets**

Bit number	Meaning
0 to 1	0 = the Host has no preferred coding when transmitting on the LE Coded PHY 1 = the Host prefers that S=2 coding be used when transmitting on the LE Coded PHY 2 = the Host prefers that S=8 coding be used when transmitting on the LE Coded PHY 3 = Reserved for future use
All other bits	Reserved for future use

Return parameters:

None.

Event(s) generated (unless masked away):

When the Controller receives the HCI_LE_Set_PHY command, the Controller shall send the HCI_Command_Status event to the Host. The HCI_LE_PHY_Update_Complete event shall be generated either when one or both PHY changes or when the Controller determines that neither PHY will change immediately.

Note: If the peer negotiation resulted in no change to either PHY, this is not an error and the HCI_LE_PHY_Update_Complete event will contain a status indicating success.

Note: An HCI_Command_Complete event is not sent by the Controller to indicate that this command has been completed. Instead, the HCI_LE_PHY_Update_Complete event indicates that this command has been completed. The HCI_LE_PHY_Update_Complete event may also be issued autonomously by the Link Layer.



7.8.50 LE Receiver Test command [v2]

Command	OCF	Command Parameters	Return Parameters
HCI_LE_Receiver_Test [v2]	0x0033	RX_Channel, PHY, Modulation_Index	Status

Description:

This is a previous version of the HCI_LE_Receiver_Test command [v3] (see [Section 7.8.78](#)). It is equivalent to that version of the command with the following command parameter values:

Parameter	Value
Expected_CTE_Length	0x00
Expected_CTE_Type	<i>any valid value</i>
Slot_Durations	<i>any valid value</i>
Length_of_Switching_Pattern	<i>any valid value</i>
Antenna_IDs[i]	<i>any valid value</i>



7.8.51 LE Transmitter Test command [v2]

Command	OCF	Command Parameters	Return Parameters
HCI_LE_Transmitter_Test [v2]	0x0034	TX_Channel, Length_Of_Test_Data, Packet_Payload, PHY	Status

Description:

This is a previous version of the HCI_LE_Transmitter_Test command [v3] (see [Section 7.8.79](#)). It is equivalent to that version of the command with the following command parameter values:

Parameter	Value
CTE_Length	0x00
CTE_Type	<i>any valid value</i>
Length_of_Switching_Pattern	<i>any valid value</i>
Antenna_IDs[i]	<i>any valid value</i>



7.8.52 LE Set Advertising Set Random Address command

Command	OCF	Command Parameters	Return Parameters
HCI_LE_Set_Advertising_Set_Random_Address	0x0035	Advertising_Handle, Random_Address	Status

Description:

The HCI_LE_Set_Advertising_Set_Random_Address command is used by the Host to set the random device address specified by the Random_Address parameter. This address is used in the Controller (see [\[Vol 6\] Part B, Section 1.3.2](#)) for the advertiser's address contained in the advertising PDUs for the advertising set specified by the Advertising_Handle parameter.

If the Host issues this command while an advertising set using connectable advertising is enabled, the Controller shall return the error code *Command Disallowed* (0x0C). The Host may issue this command at any other time.

If this command is used to change the address, the new random address shall take effect for advertising no later than the next successful HCI_LE_Extended_Set_Advertising_Enable command and for periodic advertising no later than the next successful HCI_LE_Periodic_Advertising_Enable command.

Command parameters:

Advertising_Handle: *Size: 1 octet*

Value	Parameter Description
0xXX	Used to identify an advertising set Range: 0x00 to 0xEF

Advertising_Random_Address: *Size: 6 octets*

Value	Parameter Description
0XXXXXXXXXX XX	Random Device Address as defined by [Vol 6] Part B, Section 1.3.2

Return parameters:

Status: *Size: 1 octet*

Value	Parameter Description
0x00	HCI_LE_Set_Advertising_Set_Random_Address command succeeded
0x01 to 0xFF	HCI_LE_Set_Advertising_Set_Random_Address command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions

**Event(s) generated (unless masked away):**

When the HCI_LE_Set_Advertising_Set_Random_Address command has completed, an HCI_Command_Complete event shall be generated.



7.8.53 LE Set Extended Advertising Parameters command

Command	OCF	Command Parameters	Return Parameters
HCI_LE_Set_Extended_Advertising_Parameters	0x0036	Advertising_Handle, Advertising_Event_Properties, Primary_Advertising_Interval_Min, Primary_Advertising_Interval_Max, Primary_Advertising_Channel_Map, Own_Address_Type, Peer_Address_Type, Peer_Address, Advertising_Filter_Policy, Advertising_Tx_Power, Primary_Advertising_PHY, Secondary_Advertising_Max_Skip, Secondary_Advertising_PHY, Advertising_SID, Scan_Request_Notification_Enable	Status, Selected_Tx_Power

Description:

The HCI_LE_Set_Extended_Advertising_Parameters command is used by the Host to set the advertising parameters.

The Advertising_Handle parameter identifies the advertising set whose parameters are being configured.

The Advertising_Event_Properties parameter describes the type of advertising event that is being configured and its basic properties. The type shall be one supported by the Controller. In particular, the following restrictions apply to this parameter:

- If legacy advertising PDU types are being used, then the parameter value shall be one of those specified in [Table 7.2](#). If the advertising set already contains data, the type shall be one that supports advertising data and the amount of data shall not exceed 31 octets.



Event Type	PDU Type	Advertising Event Properties	Advertising Data
Connectable and scannable undirected	ADV_IND	00010011b	Supported
Connectable directed (low duty cycle)	ADV_DIRECT_IND	00010101b	Not allowed
Connectable directed (high duty cycle)	ADV_DIRECT_IND	00011101b	Not allowed
Scannable undirected	ADV_SCAN_IND	00010010b	Supported
Non-connectable and non-scannable undirected	ADV_NONCONN_IND	00010000b	Supported

Table 7.2: Advertising_Event_Properties values for legacy PDUs

- If extended advertising PDU types are being used (bit 4 = 0) then:
 The advertisement shall not be both connectable and scannable.
 High duty cycle directed connectable advertising (≤ 3.75 ms advertising interval) shall not be used (bit 3 = 0).

If the Advertising_Event_Properties parameter does not describe an event type supported by the Controller, contains an invalid bit combination, or specifies a type that does not support advertising data when the advertising set already contains some, the Controller shall return the error code *Invalid HCI Command Parameters* (0x12).

The parameters beginning with “Secondary” are only valid when extended advertising PDU types are being used (bit 4 = 0).

The Own_Address_Type parameter shall be ignored for anonymous advertising (bit 5 = 1).

If Directed advertising is selected, the Peer_Address_Type and Peer_Address shall be valid and the Advertising_Filter_Policy parameter shall be ignored.

The Primary_Advertising_Interval_Min parameter shall be less than or equal to the Primary_Advertising_Interval_Max parameter. The Primary_Advertising_Interval_Min and Primary_Advertising_Interval_Max parameters should not be the same value so that the Controller can choose the best advertising interval given other activities.

For high duty cycle connectable directed advertising event type (ADV_DIRECT_IND), the Primary_Advertising_Interval_Min and Primary_Advertising_Interval_Max parameters are not used and shall be ignored.



If the primary advertising interval range provided by the Host (Primary_Advertising_Interval_Min, Primary_Advertising_Interval_Max) is outside the advertising interval range supported by the Controller, then the Controller shall return the error code *Unsupported Feature or Parameter Value* (0x11).

The Primary_Advertising_Channel_Map is a bit field that indicates the advertising channels that shall be used when transmitting advertising packets. At least one channel bit shall be set in the Primary_Advertising_Channel_Map parameter.

The Own_Address_Type parameter specifies the type of address being used in the advertising packets. For random addresses, the address is specified by the HCI_LE_Set_Advertising_Set_Random_Address command.

If Own_Address_Type equals 0x02 or 0x03, the Peer_Address parameter contains the peer's Identity Address and the Peer_Address_Type parameter contains the peer's Identity Type (i.e., 0x00 or 0x01). These parameters are used to locate the corresponding local IRK in the resolving list; this IRK is used to generate their own address used in the advertisement.

The Advertising_Tx_Power parameter indicates the maximum power level at which the advertising packets are to be transmitted on the advertising physical channels. The Controller shall choose a power level lower than or equal to the one specified by the Host.

The Primary_Advertising_PHY parameter indicates the PHY on which the advertising packets are transmitted on the primary advertising physical channel. If legacy advertising PDUs are being used, the Primary_Advertising_PHY shall indicate the LE 1M PHY. The Secondary_Advertising_PHY parameter indicates the PHY on which the advertising packets are to be transmitted on the secondary advertising physical channel. If the Host specifies a PHY that is not supported by the Controller, it should return the error code *Unsupported Feature or Parameter Value* (0x11). If Constant Tone Extensions are enabled for the advertising set and Secondary_Advertising_PHY specifies a PHY that does not allow Constant Tone Extensions, the Controller shall return the error code *Command Disallowed* (0x0C).

The Secondary_Advertising_Max_Skip parameter is the maximum number of advertising events that can be skipped before the AUX_ADV_IND can be sent.

The Advertising_SID parameter specifies the value to be transmitted in the Advertising SID subfield of the ADI field of the Extended Header of those advertising physical channel PDUs that have an ADI field. If the advertising set only uses PDUs that do not contain an ADI field, Advertising_SID shall be ignored.

The Scan_Request_Notification_Enable parameter indicates whether the Controller shall send notifications upon the receipt of a scan request PDU that



is in response to an advertisement from the specified advertising set that contains its device address and is from a scanner that is allowed by the advertising filter policy.

If the Host issues this command when advertising is enabled for the specified advertising set, the Controller shall return the error code *Command Disallowed* (0x0C).

If periodic advertising is enabled for the advertising set and the Secondary_Advertising_PHY parameter does not specify the PHY currently being used for the periodic advertising, the Controller shall return the error code *Command Disallowed* (0x0C).

If the Advertising_Handle does not identify an existing advertising set and the Controller is unable to support a new advertising set at present, the Controller shall return the error code *Memory Capacity Exceeded* (0x07).

If the advertising set already contains advertising data or scan response data, extended advertising is being used, and the length of the data is greater than the maximum that the Controller can transmit within the longest possible auxiliary advertising segment consistent with the parameters, the Controller shall return the error code *Packet Too Long* (0x45). If advertising on the LE Coded PHY, the S=8 coding shall be assumed.

Command parameters:

Advertising_Handle:

Size: 1 octet

Value	Parameter Description
0xXX	Used to identify an advertising set Range: 0x00 to 0xEF

Advertising_Event_Properties:

Size: 2 octets

Bit number	Parameter Description
0	Connectable advertising
1	Scannable advertising
2	Directed advertising
3	High Duty Cycle Directed Connectable advertising (≤ 3.75 ms Advertising Interval)
4	Use legacy advertising PDUs
5	Omit advertiser's address from all PDUs ("anonymous advertising")
6	Include TxPower in the extended header of at least one advertising PDU
All other bits	Reserved for future use

**Primary_Advertising_Interval_Min:****Size: 3 octets**

Value	Parameter Description
N = 0xXXXXXX	Minimum advertising interval for undirected and low duty cycle directed advertising. Range: 0x000020 to 0xFFFFFFFF Time = N * 0.625 ms Time Range: 20 ms to 10,485.759375 s

Primary_Advertising_Interval_Max:**Size: 3 octets**

Value	Parameter Description
N = 0xXXXXXX	Maximum advertising interval for undirected and low duty cycle directed advertising. Range: 0x000020 to 0xFFFFFFFF Time = N * 0.625 ms Time Range: 20 ms to 10,485.759375 s

Primary_Advertising_Channel_Map:**Size: 1 octet**

Bit number	Parameter Description
0	Channel 37 shall be used
1	Channel 38 shall be used
2	Channel 39 shall be used
All other bits	Reserved for future use

Own_Address_Type:**Size: 1 octet**

Value	Parameter Description
0x00	Public Device Address
0x01	Random Device Address
0x02	Controller generates the Resolvable Private Address based on the local IRK from the resolving list. If the resolving list contains no matching entry, use the public address.
0x03	Controller generates the Resolvable Private Address based on the local IRK from the resolving list. If the resolving list contains no matching entry, use the random address from LE_Set_Advertising_Set_Random_Address.
All other values	Reserved for future use

Peer_Address_Type:**Size: 1 octet**

Value	Parameter Description
0x00	Public Device Address or Public Identity Address



Value	Parameter Description
0x01	Random Device Address or Random (static) Identity Address
All other values	Reserved for future use

*Peer_Address:**Size: 6 octets*

Value	Parameter Description
0XXXXXXXXXX XX	Public Device Address, Random Device Address, Public Identity Address, or Random (static) Identity Address of the device to be connected.

*Advertising_Filter_Policy:**Size: 1 octet*

Value	Parameter Description
0x00	Process scan and connection requests from all devices (i.e., the White List is not in use)
0x01	Process connection requests from all devices and scan requests only from devices that are in the White List.
0x02	Process scan requests from all devices and connection requests only from devices that are in the White List.
0x03	Process scan and connection requests only from devices in the White List.
All other values	Reserved for future use

*Advertising_Tx_Power:**Size: 1 octet*

Value	Parameter Description
0XX	Size: 1 octet (signed integer) Range: -127 to +20 Units: dBm
0x7F	Host has no preference

*Primary_Advertising_PHY:**Size: 1 octet*

Value	Parameter Description
0x01	Primary advertisement PHY is LE 1M
0x03	Primary advertisement PHY is LE Coded
All other values	Reserved for future use

*Secondary_Advertising_Max_Skip:**Size: 1 octet*

Value	Parameter Description
0x00	AUX_ADV_IND shall be sent prior to the next advertising event



Value	Parameter Description
0x01 to 0xFF	Maximum advertising events the Controller can skip before sending the AUX_ADV_IND packets on the secondary advertising physical channel

*Secondary_Advertising_PHY:**Size: 1 octet*

Value	Parameter Description
0x01	Secondary advertisement PHY is LE 1M
0x02	Secondary advertisement PHY is LE 2M
0x03	Secondary advertisement PHY is LE Coded
All other values	Reserved for future use

*Advertising_SID:**Size: 1 octet*

Value	Parameter Description
0x00 to 0x0F	Value of the Advertising SID subfield in the ADI field of the PDU
All other values	Reserved for future use

*Scan_Request_Notification_Enable:**Size: 1 octet*

Value	Parameter Description
0x00	Scan request notifications disabled
0x01	Scan request notifications enabled
All other values	Reserved for future use

Return parameters

*Status:**Size: 1 octet*

Value	Parameter Description
0x00	HCI_LE_Set_Extended_Advertising_Parameters command succeeded
0x01 to 0xFF	HCI_LE_Set_Extended_Advertising_Parameters command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions

*Selected_Tx_Power:**Size: 1 octet*

Value	Parameter Description
0xXX	Size: 1 octet (signed integer) Range: -127 to +20 Units: dBm

Event(s) generated (unless masked away):



When the HCI_LE_Set_Extended_Advertising_Parameters command has completed, an HCI_Command_Complete event shall be generated. The Selected_Tx_Power return parameter indicates the transmit power selected by the Controller. The Controller shall not change the transmit power for this advertising set without being directed to by the Host.



7.8.54 LE Set Extended Advertising Data command

Command	OCF	Command Parameters	Return Parameters
HCI_LE_Set_Extended_Advertising_Data	0x0037	Advertising_Handle, Operation, Fragment_Preference, Advertising_Data_Length, Advertising_Data	Status

Description:

The HCI_LE_Set_Extended_Advertising_Data command is used to set the data used in advertising PDUs that have a data field. This command may be issued at any time after an advertising set identified by the Advertising_Handle parameter has been created using the HCI_LE_Set_Extended_Advertising_Parameters command (see [Section 7.8.53](#)), regardless of whether advertising in that set is enabled or disabled.

If advertising is currently enabled for the specified advertising set, the Controller shall use the new data in subsequent extended advertising events for this advertising set. If an extended advertising event is in progress when this command is issued, the Controller may use the old or new data for that event.

If advertising is currently disabled for the specified advertising set, the data shall be kept by the Controller and used once advertising is enabled for that set. The data shall be discarded when the advertising set is removed.

Only the significant part of the advertising data should be transmitted in the advertising packets as defined in [\[Vol 3\] Part C, Section 11](#).

The Host may set the advertising data in one or more operations using the Operation parameter in the command. If the combined length of the data exceeds the capacity of the advertising set identified by the Advertising_Handle parameter (see [Section 7.8.57](#) LE Read Maximum Advertising Data Length command) or the amount of memory currently available, all the data shall be discarded and the Controller shall return the error code *Memory Capacity Exceeded* (0x07).

If the advertising set uses extended advertising and the combined length of the data is greater than the maximum that the Controller can transmit within the longest possible auxiliary advertising segment consistent with the current parameters of the advertising set (using the current advertising interval if advertising is enabled), all the data shall be discarded and the Controller shall return the error code *Packet Too Long* (0x45). If advertising on the LE Coded PHY, the S=8 coding shall be assumed.



If Operation indicates the start of new data (values 0x01 or 0x03), then any existing partial or complete advertising data shall be discarded.

If the advertising data is discarded by the command or the combined length of the data after the command completes is zero, the advertising set will have no advertising data.

If Operation is 0x04, the behavior is the same as if the current advertising data had been sent again; this can be used to cause the Advertising DID value to be updated (see [Vol 6] Part B, Section 4.4.2.11).

The Fragment_Preference parameter provides a hint to the Controller as to whether advertising data should be fragmented.

If the advertising set specifies a type that does not support advertising data, the Controller shall return the error code *Invalid HCI Command Parameters* (0x12).

If the advertising set uses legacy advertising PDUs that support advertising data and either Operation is not 0x03 or the Advertising_Data_Length parameter exceeds 31 octets, the Controller shall return the error code *Invalid HCI Command Parameters* (0x12).

If Operation is 0x04 and:

- advertising is currently disabled for the advertising set;
- the advertising set contains no data;
- the advertising set uses legacy PDUs; or
- Advertising_Data_Length is not zero;

then the Controller shall return the error code *Invalid HCI Command Parameters* (0x12).

If Operation is not 0x03 or 0x04 and Advertising_Data_Length is zero, the Controller shall return the error code *Invalid HCI Command Parameters* (0x12).

If advertising is currently enabled for the specified advertising set and Operation does not have the value 0x03 or 0x04, the Controller shall return the error code *Command Disallowed* (0x0C).

If advertising is currently enabled for the specified advertising set, the advertising set uses extended advertising, and the length of the data is greater than the maximum that the Controller can transmit within the longest possible auxiliary advertising segment consistent with the current parameters of the advertising set, the Controller shall return the error code *Packet Too Long* (0x45). If advertising on the LE Coded PHY, the S=8 coding shall be assumed.

If the advertising set corresponding to the Advertising_Handle parameter does not exist, then the Controller shall return the error code *Unknown Advertising Identifier* (0x42).

**Command parameters:***Advertising_Handle:**Size: 1 octet*

Value	Parameter Description
0xXX	Used to identify an advertising set Range: 0x00 to 0xEF

*Operation:**Size: 1 octet*

Value	Parameter Description
0x00	Intermediate fragment of fragmented extended advertising data
0x01	First fragment of fragmented extended advertising data
0x02	Last fragment of fragmented extended advertising data
0x03	Complete extended advertising data
0x04	Unchanged data (just update the Advertising DID)
All other values	Reserved for future use

*Fragment_Preference:**Size: 1 octet*

Value	Parameter Description
0x00	The Controller may fragment all Host advertising data
0x01	The Controller should not fragment or should minimize fragmentation of Host advertising data
All other values	Reserved for future use

*Advertising_Data_Length:**Size: 1 octet*

Value	Parameter Description
0 to 251	The number of octets in the Advertising Data parameter
All other values	Reserved for future use

*Advertising_Data:**Size: Advertising_Data_Length octets*

Value	Parameter Description
	Advertising data formatted as defined in [Vol 3] Part C, Section 11 Note: This parameter has a variable length.

Return parameters:*Status:**Size: 1 octet*

Value	Parameter Description
0x00	HCI_LE_Set_Extended_Advertising_Data command succeeded



Value	Parameter Description
0x01 to 0xFF	HCI_LE_Set_Extended_Advertising_Data command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions

Event(s) generated (unless masked away):

When the HCI_LE_Set_Extended_Advertising_Data command has completed, an HCI_Command_Complete event shall be generated.



7.8.55 LE Set Extended Scan Response Data command

Command	OCF	Command Parameters	Return Parameters
HCI_LE_Set_Extended_Scan_Response_Data	0x0038	Advertising_Handle, Operation, Fragment_Preference, Scan_Response_Data_Length, Scan_Response_Data	Status

Description:

The HCI_LE_Set_Extended_Scan_Response_Data command is used to provide scan response data used in scanning response PDUs. This command may be issued at any time after the advertising set identified by the Advertising_Handle parameter has been created using the HCI_LE_Set_Extended_Advertising_Parameters command (see [Section 7.8.53](#)) regardless of whether advertising in that set is enabled or disabled.

If advertising is currently enabled for the specified advertising set, the Controller shall use the new data in subsequent extended advertising events for this advertising set. If an extended advertising event is in progress when this command is issued, the Controller may use the old or new data for that event.

If advertising is currently disabled for the specified advertising set, the data shall be kept by the Controller and used once advertising is enabled for that set. The data shall be discarded when the advertising set is removed.

Only the significant part of the scan response data should be transmitted in the advertising packets as defined in [\[Vol 3\] Part C, Section 11](#).

The Host may set the scan response data in one or more operations using the Operation parameter in the command. If the combined length of the data exceeds the capacity of the advertising set identified by the Advertising_Handle parameter (see [Section 7.8.57](#) LE Read Maximum Advertising Data Length command) or the amount of memory currently available, all the data shall be discarded and the Controller shall return the error code *Memory Capacity Exceeded* (0x07).

If Operation indicates the start of new data (values 0x01 or 0x03), then any existing partial or complete scan response data shall be discarded.

If the scan response data is discarded by the command or the combined length of the data after the command completes is zero, the advertising set will have no scan response data.



The `Fragment_Preference` parameter provides a hint to the Controller as to whether advertising data should be fragmented.

If the advertising set is non-scannable and the Host uses this command other than to discard existing data, the Controller shall return the error code *Invalid HCI Command Parameters* (0x12). If the advertising set uses scannable legacy advertising PDUs and either `Operation` is not 0x03 or the `Scan_Response_Data_Length` parameter exceeds 31 octets, the Controller shall return the error code *Invalid HCI Command Parameters* (0x12).

If `Operation` is not 0x03 and `Scan_Response_Data_Length` is zero, the Controller shall return the error code *Invalid HCI Command Parameters* (0x12).

If advertising is currently enabled for the specified advertising set and `Operation` does not have the value 0x03, the Controller shall return the error code *Command Disallowed* (0x0C).

If the advertising set uses extended advertising and the combined length of the data is greater than the maximum that the Controller can transmit within the longest possible auxiliary advertising segment consistent with the current parameters of the advertising set (using the current advertising interval if advertising is enabled), all the data shall be discarded and the Controller shall return the error code *Packet Too Long* (0x45). If advertising on the LE Coded PHY, the S=8 coding shall be assumed.

If the advertising set uses scannable extended advertising PDUs, advertising is currently enabled for the specified advertising set, and `Scan_Response_Data_Length` is zero, the Controller shall return the error code *Command Disallowed* (0x0C).

If the advertising set corresponding to the `Advertising_Handle` parameter does not exist, then the Controller shall return the error code *Unknown Advertising Identifier* (0x42).

Command parameters:

Advertising_Handle: Size: 1 octet

Value	Parameter Description
0xXX	Used to identify an advertising set Range: 0x00 to 0xEF

Operation: Size: 1 octet

Value	Parameter Description
0x00	Intermediate fragment of fragmented scan response data
0x01	First fragment of fragmented scan response data
0x02	Last fragment of fragmented scan response data



Value	Parameter Description
0x03	Complete scan response data
All other values	Reserved for future use

*Fragment_Preference:**Size: 1 octet*

Value	Parameter Description
0x00	The Controller may fragment all scan response data
0x01	The Controller should not fragment or should minimize fragmentation of scan response data
All other values	Reserved for future use

*Scan_Response_Data_Length:**Size: 1 octet*

Value	Parameter Description
0 to 251	The number of octets in the Scan_Response Data parameter
All other values	Reserved for future use

*Scan_Response_Data:**Size: Scan_Response_Data_Length octets*

Value	Parameter Description
	Scan response data formatted as defined in [Vol 3] Part C, Section 11 Note: This parameter has a variable length.

Return parameters:*Status:**Size: 1 octet*

Value	Parameter Description
0x00	HCI_LE_Set_Extended_Scan_Response_Data command succeeded
0x01 to 0xFF	HCI_LE_Set_Extended_Scan_Response_Data command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions

Event(s) generated (unless masked away):

When the HCI_LE_Set_Extended_Scan_Response_Data command has completed, an HCI_Command_Complete event shall be generated.



7.8.56 LE Set Extended Advertising Enable command

Command	OCF	Command Parameters	Return Parameters
HCI_LE_Set_Extended_Advertising_Enable	0x0039	Enable, Number_of_Sets, Advertising_Handle[i], Duration[i], Max_Extended_Advertising_Events[i]	Status

Description:

The HCI_LE_Set_Extended_Advertising_Enable command is used to request the Controller to enable or disable one or more advertising sets using the advertising sets identified by the Advertising_Handle[i] parameter. The Controller manages the timing of advertisements in accordance with the advertising parameters given in the HCI_LE_Set_Extended_Advertising_Parameters command. The Number_of_Sets parameter is the number of advertising sets contained in the parameter arrays. If Enable and Number_of_Sets are both set to 0x00, then all advertising sets are disabled.

The Controller shall only start an advertising event when the corresponding advertising set is enabled. The Controller shall continue advertising until all advertising sets have been disabled. This can happen when the Host issues an HCI_LE_Set_Extended_Advertising_Enable command with the Enable parameter set to 0x00 (Advertising is disabled), a connection is created, the duration specified in the Duration[i] parameter expires, or the number of extended advertising events transmitted for the set exceeds the Max_Extended_Advertising_Events[i] parameter.

The Duration[i] parameter indicates the duration for which that advertising set is enabled. The duration begins at the start of the first advertising event of this advertising set. The Controller should not start an extended advertising event that it cannot complete within the duration.

If the advertising is high duty cycle connectable directed advertising, then Duration[i] shall be less than or equal to 1.28 seconds and shall not be equal to 0.

The Max_Extended_Advertising_Events[i] parameter, if non-zero, indicates the maximum number of extended advertising events that shall be sent prior to disabling the extended advertising set even if the Duration[i] parameter has not expired.

Duration[i] and Max_Extended_Advertising_Events[i] shall be ignored when Enable is set to 0x00.



If the `HCI_LE_Set_Extended_Advertising_Enable` command is sent again for an advertising set while that set is enabled, the timer used for the duration and the number of events counter are reset and any change to the random address shall take effect.

Disabling the advertising set identified by the `Advertising_Handle[i]` parameter does not disable any periodic advertising associated with that set.

Note: Disabling an advertising set that is already disabled has no effect.

If the same advertising set is identified by more than one entry in the `Advertising_Handle[i]` arrayed parameter, then the Controller shall return the error code *Invalid HCI Command Parameters* (0x12).

If the advertising set corresponding to the `Advertising_Handle[i]` parameter does not exist, then the Controller shall return the error code *Unknown Advertising Identifier* (0x42).

The remainder of this section only applies if `Enable` is set to 0x01.

If `Number_of_Sets` is set to 0x00, the Controller shall return the error code *Invalid HCI Command Parameters* (0x12).

If the advertising data or scan response data in the advertising set is not complete, the Controller shall return the error code *Command Disallowed* (0x0C).

If the advertising set uses scannable extended advertising PDUs and no scan response data is currently provided, the Controller shall return the error code *Command Disallowed* (0x0C).

If the advertising set uses connectable extended advertising PDUs and the advertising data in the advertising set will not fit in the `AUX_ADV_IND` PDU, the Controller shall return the error code *Invalid HCI Command Parameters* (0x12).

Note: The maximum amount of data that will fit in the PDU depends on which options are selected and on the maximum length of PDU that the Controller is able to transmit.

If extended advertising is being used and the length of any advertising data or of any scan response data is greater than the maximum that the Controller can transmit within the longest possible auxiliary advertising segment consistent with the chosen advertising interval, the Controller shall return the error code *Packet Too Long* (0x45). If advertising on the LE Coded PHY, the S=8 coding shall be assumed.

If the advertising set's `Own_Address_Type` parameter is set to 0x01 and the random address for the advertising set has not been initialized, the Controller shall return the error code *Invalid HCI Command Parameters* (0x12).



If the advertising set's `Own_Address_Type` parameter is set to 0x03, the controller's resolving list did not contain a matching entry, and the random address for the advertising set has not been initialized, the Controller shall return the error code *Invalid HCI Command Parameters* (0x12).

Command parameters:

Enable:

Size: 1 octet

Value	Parameter Description
0x00	Advertising is disabled
0x01	Advertising is enabled
All other values	Reserved for future use

Number_of_Sets:

Size: 1 octet

Value	Parameter Description
0x00	Disable all advertising sets
0x01 to 0x3F	Number of advertising sets to enable or disable
All other values	Reserved for future use

Advertising_Handle[i]:

*Size: 1 octet * Number_of_Sets*

Value	Parameter Description
0x00 to 0xEF	Used to identify an advertising set
All other values	Reserved for future use

Duration[i]:

*Size: 2 octets * Number_of_Sets*

Value	Parameter Description
0x0000	No advertising duration. Advertising to continue until the Host disables it.
N = 0xXXXX	Advertising duration Range: 0x0001 to 0xFFFF Time = N * 10 ms Time Range: 10 ms to 655,350 ms

Max_Extended_Advertising_Events[i]:

*Size: 1 octet * Number_of_Sets*

Value	Parameter Description
0xFF	Maximum number of extended advertising events the Controller shall attempt to send prior to terminating the extended advertising
0x00	No maximum number of advertising events.

**Return parameters:***Status:**Size: 1 octet*

Value	Parameter Description
0x00	HCI_LE_Set_Extended_Advertising_Enable command succeeded
0x01 to 0xFF	HCI_LE_Set_Extended_Advertising_Enable command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Event(s) generated (unless masked away):

When the HCI_LE_Set_Extended_Advertising_Enable command has completed, an HCI_Command_Complete event shall be generated.

If the Duration[i] parameter is set to a value other than 0x0000, an HCI_LE_Advertising_Set_Terminated event shall be generated when the duration specified in the Duration[i] parameter expires.

If the Max_Extended_Advertising_Events[i] parameter is set to a value other than 0x00, an HCI_LE_Advertising_Set_Terminated event shall be generated when the maximum number of extended advertising events has been transmitted by the Controller.

If the advertising set is connectable and a connection gets created, an HCI_LE_Connection_Complete event shall be generated followed by an HCI_LE_Advertising_Set_Terminated event with the Status parameter set to 0x00.

If the advertising set is for high duty cycle connectable directed advertising that fails to create a connection before the duration expires, an HCI_LE_Connection_Complete event shall be generated with the Status parameter set to the error code *Advertising Timeout* (0x3C).

Note: If this command is used to disable advertising at about the same time that a connection is established or the advertising duration expires, there is a possible race condition in that it is possible to receive both an HCI_LE_Connection_Complete or HCI_LE_Advertising_Set_Terminated event and the HCI_Command_Complete event for this command.



7.8.57 LE Read Maximum Advertising Data Length command

Command	OCF	Command Parameters	Return Parameters
HCI_LE_Read_Maximum_Advertising_Data_Length	0x003A		Status, Maximum_Advertising_Data_Length

Description:

The HCI_LE_Read_Maximum_Advertising_Data_Length command is used to read the maximum length of data supported by the Controller for use as advertisement data or scan response data in an advertising event or as periodic advertisement data.

Note: The maximum amount may be fragmented across multiple PDUs (see [\[Vol 6\] Part B, Section 2.3.4.9](#)).

Command parameters:

None.

Return parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	HCI_LE_Read_Maximum_Advertising_Data_Length command succeeded
0x01 to 0xFF	HCI_LE_Read_Maximum_Advertising_Data_Length command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Maximum_Advertising_Data_Length:

Size: 2 octets

Value	Parameter Description
0x001F to 0x0672	Maximum supported advertising data length
All other values	Reserved for future use

Event(s) generated (unless masked away):

When the HCI_LE_Read_Maximum_Advertising_Data_Length command has completed, an HCI_Command_Complete event shall be generated.



7.8.58 LE Read Number of Supported Advertising Sets command

Command	OCF	Command Parameters	Return Parameters
HCI_LE_Read_Number_of_Supported_Advertising_Sets	0x003B		Status, Num_Supported_Advertising_Sets

Description:

The HCI_LE_Read_Number_of_Supported_Advertising_Sets command is used to read the maximum number of advertising sets supported by the advertising Controller at the same time.

Note: The number of advertising sets that can be supported is not fixed and the Controller can change it at any time because the memory used to store advertising sets can also be used for other purposes.

Command parameters:

None.

Return parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	HCI_LE_Read_Number_of_Supported_Advertising_Sets command succeeded
0x01 to 0xFF	HCI_LE_Read_Number_of_Supported_Advertising_Sets command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Num_Supported_Advertising_Sets:

Size: 1 octet

Value	Parameter Description
0x01 to 0xF0	Number of advertising sets supported at the same time
All other values	Reserved for future use

Event(s) generated (unless masked away):

When the HCI_LE_Read_Number_of_Supported_Advertising_Sets command has completed, an HCI_Command_Complete event shall be generated.



7.8.59 LE Remove Advertising Set command

Command	OCF	Command Parameters	Return Parameters
HCI_LE_Remove_Advertising_Set	0x003C	Advertising_Handle	Status

Description:

The HCI_LE_Remove_Advertising_Set command is used to remove an advertising set from the Controller.

If the advertising set corresponding to the Advertising_Handle parameter does not exist, then the Controller shall return the error code *Unknown Advertising Identifier* (0x42). If advertising or periodic advertising on the advertising set is enabled, then the Controller shall return the error code *Command Disallowed* (0x0C).

Command parameters:

Advertising_Handle:

Size: 1 octet

Value	Parameter Description
0xXX	Used to identify an advertising set Range: 0x00 to 0xEF

Return parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	HCI_LE_Remove_Advertising_Set command succeeded
0x01 to 0xFF	HCI_LE_Remove_Advertising_Set command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Event(s) generated (unless masked away):

When the HCI_LE_Remove_Advertising_Set command has completed, an HCI_Command_Complete event shall be generated.



7.8.60 LE Clear Advertising Sets command

Command	OCF	Command Parameters	Return Parameters
HCI_LE_Clear_Advertising_Sets	0x003D		Status

Description:

The HCI_LE_Clear_Advertising_Sets command is used to remove all existing advertising sets from the Controller.

If advertising or periodic advertising is enabled on any advertising set, then the Controller shall return the error code *Command Disallowed* (0x0C).

Note: All advertising sets are cleared on HCI reset.

Command parameters:

None.

Return parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	HCI_LE_Clear_Advertising_Sets command succeeded
0x01 to 0xFF	HCI_LE_Clear_Advertising_Sets command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Event(s) generated (unless masked away):

When the HCI_LE_Clear_Advertising_Sets command has completed, an HCI_Command_Complete event shall be generated.



7.8.61 LE Set Periodic Advertising Parameters command

Command	OCF	Command Parameters	Return Parameters
HCI_LE_Set_Periodic_Advertising_Parameters	0x003E	Advertising_Handle, Periodic_Advertising_Interval_Min, Periodic_Advertising_Interval_Max, Periodic_Advertising_Properties	Status

Description:

The HCI_LE_Set_Periodic_Advertising_Parameters command is used by the Host to set the parameters for periodic advertising.

The Advertising_Handle parameter identifies the advertising set whose periodic advertising parameters are being configured. If the corresponding advertising set does not already exist, then the Controller shall return the error code *Unknown Advertising Identifier* (0x42).

The Periodic_Advertising_Interval_Min parameter shall be less than or equal to the Periodic_Advertising_Interval_Max parameter. The Periodic_Advertising_Interval_Min and Periodic_Advertising_Interval_Max parameters should not be the same value to enable the Controller to determine the best advertising interval given other activities.

The Periodic_Advertising_Properties parameter indicates which fields should be included in the advertising packet.

If the advertising set identified by the Advertising_Handle specified anonymous advertising, the Controller shall return the error code *Invalid HCI Command Parameters* (0x12).

If the Host issues this command when periodic advertising is enabled for the specified advertising set, the Controller shall return the error code *Command Disallowed* (0x0C).

If the Advertising_Handle does not identify an advertising set that is already configured for periodic advertising and the Controller is unable to support more periodic advertising at present, the Controller shall return the error code *Memory Capacity Exceeded* (0x07).

If the advertising set already contains periodic advertising data and the length of the data is greater than the maximum that the Controller can transmit within a periodic advertising interval of Periodic_Advertising_Interval_Max, the Controller shall return the error code *Packet Too Long* (0x45). If advertising on the LE Coded PHY, the S=8 coding shall be assumed.

**Command parameters:***Advertising_Handle:**Size: 1 octet*

Value	Parameter Description
0xXX	Used to identify a periodic advertising train Range: 0x00 to 0xEF

*Periodic_Advertising_Interval_Min:**Size: 2 octets*

Value	Parameter Description
N = 0xXXXX	Minimum advertising interval for periodic advertising. Range: 0x0006 to 0xFFFF Time = N * 1.25 ms Time Range: 7.5ms to 81.91875 s

*Periodic_Advertising_Interval_Max:**Size: 2 octets*

Value	Parameter Description
N = 0xXXXX	Maximum advertising interval for periodic advertising. Range: 0x0006 to 0xFFFF Time = N * 1.25 ms Time Range: 7.5ms to 81.91875 s

*Periodic_Advertising_Properties:**Size: 2 octets*

Bit Number	Parameter Description
6	Include TxPower in the advertising PDU
All other bits	Reserved for future use

Return parameters:*Status:**Size: 1 octet*

Value	Parameter Description
0x00	HCI_LE_Set_Periodic_Advertising_Parameters command succeeded
0x01 to 0xFF	HCI_LE_Set_Periodic_Advertising_Parameters command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Event(s) generated (unless masked away):

When the HCI_LE_Set_Periodic_Advertising_Parameters command has completed, an HCI_Command_Complete event shall be generated.



7.8.62 LE Set Periodic Advertising Data command

Command	OCF	Command Parameters	Return Parameters
HCI_LE_Set_Periodic_Advertising_Data	0x003F	Advertising_Handle, Operation, Advertising_Data_Length, Advertising_Data	Status

Description:

The HCI_LE_Set_Periodic_Advertising_Data command is used to set the data used in periodic advertising PDUs. This command may be issued at any time after the advertising set identified by the Advertising_Handle parameter has been configured for periodic advertising using the HCI_LE_Set_Periodic_Advertising_Parameters command (see [Section 7.8.61](#)), regardless of whether periodic advertising in that set is enabled or disabled. If the advertising set has not been configured for periodic advertising, then the Controller shall return the error code *Command Disallowed* (0x0C).

If periodic advertising is currently enabled for the specified advertising set, the Controller shall use the new data in subsequent periodic advertising events for this advertising set. If a periodic advertising event is in progress when this command is issued, the Controller may use the old or new data for that event.

If periodic advertising is currently disabled for the specified advertising set, the data shall be kept by the Controller and used once periodic advertising is enabled for that set. The data shall be discarded when the advertising set is removed.

Only the significant part of the periodic advertising data should be transmitted in the advertising packets as defined in [\[Vol 3\] Part C, Section 11](#).

The Host may set the periodic advertising data in one or more operations using the Operation parameter in the command. If the combined length of the data exceeds the capacity of the advertising set identified by the Advertising_Handle parameter (see [Section 7.8.57](#) LE Read Maximum Advertising Data Length command) or the amount of memory currently available, all the data shall be discarded and the Controller shall return the error code *Memory Capacity Exceeded* (0x07).

If the combined length of the data is greater than the maximum that the Controller can transmit within the current periodic advertising interval (if periodic advertising is currently enabled) or the Periodic_Advertising_Interval_Max for the advertising set (if currently disabled), all the data shall be discarded and the Controller shall return the error code *Packet Too Long* (0x45). If advertising on the LE Coded PHY, the S=8 coding shall be assumed.



If Operation indicates the start of new data (values 0x01 or 0x03), then any existing partial or complete data shall be discarded.

If the periodic advertising data is discarded by the command or the combined length of the data after the command completes is zero, the advertising set will have no periodic advertising data.

If periodic advertising is currently enabled for the specified advertising set and Operation does not have the value 0x03, the Controller shall return the error code *Command Disallowed* (0x0C).

If the advertising set corresponding to the Advertising_Handle parameter does not exist, then the Controller shall return the error code *Unknown Advertising Identifier* (0x42).

Command parameters:

Advertising_Handle: *Size: 1 octet*

Value	Parameter Description
0xXX	Used to identify an advertising set Range: 0x00 to 0xEF

Operation: *Size: 1 octet*

Value	Parameter Description
0x00	Intermediate fragment of fragmented periodic advertising data
0x01	First fragment of fragmented periodic advertising data
0x02	Last fragment of fragmented periodic advertising data
0x03	Complete periodic advertising data
All other values	Reserved for future use

Advertising_Data_Length: *Size: 1 octet*

Value	Parameter Description
0 to 252	The number of octets in the Advertising Data parameter
All other values	Reserved for future use

Advertising_Data: *Size: Advertising_Data_Length octets*

Value	Parameter Description
	Periodic advertising data formatted as defined in [Vol 3] Part C, Section 11 . Note: This parameter has a variable length.

**Return parameters:***Status:**Size: 1 octet*

Value	Parameter Description
0x00	HCI_LE_Set_Periodic_Advertising_Data command succeeded
0x01 to 0xFF	HCI_LE_Set_Periodic_Advertising_Data command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Event(s) generated (unless masked away):

When the HCI_LE_Set_Periodic_Advertising_Data command has completed, an HCI_Command_Complete event shall be generated.



7.8.63 LE Set Periodic Advertising Enable command

Command	OCF	Command Parameters	Return Parameters
HCI_LE_Set_Periodic_Advertising_Enable	0x0040	Enable, Advertising_Handle	Status

Description:

The HCI_LE_Set_Periodic_Advertising_Enable command is used to request the Controller to enable or disable the periodic advertising for the advertising set specified by the Advertising_Handle parameter (ordinary advertising is not affected).

If the advertising set is not currently enabled (see the HCI_LE_Set_Extended_Advertising_Enable command), the periodic advertising is not started until the advertising set is enabled. Once the advertising set has been enabled, the Controller shall continue periodic advertising until the Host issues an HCI_LE_Set_Periodic_Advertising_Enable command with Enable set to 0x00 (periodic advertising is disabled). Disabling the advertising set has no effect on the periodic advertising once the advertising set has been enabled.

The Controller manages the timing of advertisements in accordance with the advertising parameters given in the HCI_LE_Set_Periodic_Advertising_Parameters command.

If the advertising set corresponding to the Advertising_Handle parameter does not exist, the Controller shall return the error code *Unknown Advertising Identifier* (0x42).

If Enable is set to 0x01 and the periodic advertising data in the advertising set is not complete, the Controller shall return the error code *Command Disallowed* (0x0C).

If Enable is set to 0x01 (Advertising is enabled) and the length of the periodic advertising data is greater than the maximum that the Controller can transmit within the chosen periodic advertising interval, the Controller shall return the error code *Packet Too Long* (0x45). If advertising on the LE Coded PHY, the S=8 coding shall be assumed.

Note: Enabling periodic advertising when it is already enabled can cause the random address to change. Disabling periodic advertising when it is already disabled has no effect.

Command parameters:

**Enable:****Size: 1 octet**

Value	Parameter Description
0x00	Periodic advertising is disabled (default)
0x01	Periodic advertising is enabled
All other values	Reserved for future use

Advertising_Handle:**Size: 1 octet**

Value	Parameter Description
0xXX	Used to identify an advertising set Range: 0x00 to 0xEF

Return parameters:**Status:****Size: 1 octet**

Value	Parameter Description
0x00	HCI_LE_Set_Periodic_Advertising_Enable command succeeded
0x01 to 0xFF	HCI_LE_Set_Periodic_Advertising_Enable command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Event(s) generated (unless masked away):

When the HCI_LE_Set_Periodic_Advertising_Enable command has completed, an HCI_Command_Complete event shall be generated.



7.8.64 LE Set Extended Scan Parameters command

Command	OCF	Command Parameters	Return Parameters
HCI_LE_Set_Extended_Scan_Parameters	0x0041	Own_Address_Type Scanning_Filter_Policy Scanning_PHYs, Scan_Type[i], Scan_Interval[i], Scan_Window[i]	Status

Description:

The HCI_LE_Set_Extended_Scan_Parameters command is used to set the extended scan parameters to be used on the advertising physical channels.

The Scanning_PHYs parameter indicates the PHY(s) on which the advertising packets should be received on the primary advertising physical channel. The Host may enable one or more scanning PHYs. If the Host specifies a PHY that is not supported by the Controller, it should return the error code *Unsupported Feature or Parameter Value* (0x11). The Scan_Type[i], Scan_Interval[i], and Scan_Window[i] parameters array elements are ordered in the same order as the set bits in the Scanning_PHY parameter, starting from bit 0. The number of array elements is determined by the number of bits set in the Scanning_PHY parameter.

The Scan_Type[i] parameter specifies the type of scan to perform.

The Scan_Interval[i] and Scan_Window[i] parameters are recommendations from the Host on how long (Scan_Window[i]) and how frequently (Scan_Interval[i]) the Controller should scan (see [\[Vol 6\] Part B, Section 4.4.3](#)); however the frequency and length of the scan is implementation specific. If the requested scan cannot be supported by the implementation, the Controller shall return the error code *Invalid HCI Command Parameters* (0x12).

The Own_Address_Type parameter indicates the type of address being used in the scan request packets.

If the Host issues this command when scanning is enabled in the Controller, the Controller shall return the error code *Command Disallowed* (0x0C).

**Command parameters:***Own_Address_Type:**Size: 1 octet*

Value	Parameter Description
0x00	Public Device Address
0x01	Random Device Address
0x02	Controller generates the Resolvable Private Address based on the local IRK from the resolving list. If the resolving list contains no matching entry, then use the public address.
0x03	Controller generates the Resolvable Private Address based on the local IRK from the resolving list. If the resolving list contains no matching entry, then use the random address from LE_Set_Random_Address.
All other values	Reserved for future use

*Scanning_Filter_Policy:**Size: 1 octet*

Value	Parameter Description
0x00	Accept all advertising and scan response PDUs except directed advertising PDUs not addressed to this device
0x01	Accept only advertising and scan response PDUs from devices where the advertiser's address is in the White List. Directed advertising PDUs which are not addressed to this device shall be ignored.
0x02	Accept all advertising and scan response PDUs except directed advertising PDUs where the initiator's identity address does not address this device. Note: Directed advertising PDUs where the initiator's address is a resolvable private address that cannot be resolved are also accepted.
0x03	Accept all advertising and scan response PDUs except: • advertising and scan response PDUs where the advertiser's identity address is not in the White List; and • directed advertising PDUs where the initiator's identity address does not address this device Note: Directed advertising PDUs where the initiator's address is a resolvable private address that cannot be resolved are also accepted.
All other values	Reserved for future use

*Scanning_PHYs:**Size: 1 octet*

Bit number	Parameter Description
0	Scan advertisements on the LE 1M PHY
2	Scan advertisements on the LE Coded PHY
All other bits	Reserved for future use



Scan_Type[i]: *Size: 1 octet * bits set in Scanning_PHYs*

Value	Parameter Description
0x00	Passive Scanning. No scan request PDUs shall be sent.
0x01	Active Scanning. Scan request PDUs may be sent.
All other values	Reserved for future use

Scan_Interval[i]: *Size: 2 octets * bits set in Scanning_PHYs*

Value	Parameter Description
N = 0xFFFF	Time interval from when the Controller started its last scan until it begins the subsequent scan on the primary advertising physical channel. Range: 0x0004 to 0xFFFF Time = N * 0.625 ms Time Range: 2.5 ms to 40.959375 s

Scan_Window[i]: *Size: 2 octets * bits set in Scanning_PHYs*

Value	Parameter Description
N = 0xFFFF	Duration of the scan on the primary advertising physical channel. Range: 0x0004 to 0xFFFF Time = N * 0.625 ms Time Range: 2.5 ms to 40.959375 s

Return parameters:

Status: *Size: 1 octet*

Value	Parameter Description
0x00	HCI_LE_Set_Extended_Scan_Parameters command succeeded
0x01 to 0xFF	HCI_LE_Set_Extended_Scan_Parameters command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Event(s) generated (unless masked away):

When the HCI_LE_Set_Extended_Scan_Parameters command has completed, an HCI_Command_Complete event shall be generated.



7.8.65 LE Set Extended Scan Enable command

Command	OCF	Command Parameters	Return Parameters
HCI_LE_Set_Extended_Scan_Enable	0x0042	Enable, Filter_Duplicates, Duration, Period	Status

Description:

The HCI_LE_Set_Extended_Scan_Enable command is used to enable or disable scanning.

The Enable parameter determines whether scanning is enabled or disabled. If it is set to 0x00, the remaining parameters shall be ignored.

The Filter_Duplicates parameter controls whether the Link Layer should filter out duplicate advertising reports (filtering duplicates enabled) to the Host or if the Link Layer should generate advertising reports for each packet received (filtering duplicates disabled). See [\[Vol 6\] Part B, Section 4.4.3.5](#).

If the Filter_Duplicates parameter is set to 0x00, all advertisements received from advertisers shall be sent to the Host in advertising report events.

If the Filter_Duplicates parameter is set to 0x01, duplicate advertisements should not be sent to the Host in advertising report events until scanning is disabled.

If the Filter_Duplicates parameter is set to 0x02, duplicate advertisements in a single scan period should not be sent to the Host in advertising report events; this setting shall only be used if both Period and Duration are non-zero. If Filter_Duplicates is set to 0x02 and either Period or Duration to zero, the Controller shall return the Invalid error code *HCI Command Parameters* (0x12).

If the Duration parameter is zero or both the Duration parameter and Period parameter are non-zero, the Controller shall continue scanning until scanning is disabled by the Host issuing an HCI_LE_Set_Extended_Scan_Enable command with the Enable parameter set to 0x00 (Scanning is disabled). The Period parameter shall be ignored when the Duration parameter is zero.

If the Duration parameter is non-zero and the Period parameter is zero, the Controller shall continue scanning until the duration specified in the Duration parameter has expired.

If both the Duration and Period parameters are non-zero and the Duration is greater than or equal to the Period, the Controller shall return the error code *Invalid HCI Command Parameters* (0x12).



When the Duration and Period parameters are non-zero, the Controller shall scan for the duration of the Duration parameter within a scan period specified by the Period parameter. After the scan period has expired, a new scan period shall begin and scanning shall begin again for the duration specified. The scan periods continue until the Host disables scanning.

If the HCI_LE_Set_Extended_Scan_Enable command with Enable set to 0x01 is sent while scanning is already enabled, the timers used for duration and period are reset to the new parameter values and a new scan period is started. Any change to the Filter_Duplicates setting or the random address shall take effect.

Note: Disabling scanning when it is disabled has no effect.

Note: The duration of a scan period refers to the time spent scanning on both the primary and secondary advertising physical channels. However, expiry of the duration does not prevent the Link Layer from scanning for and receiving auxiliary packets of received advertisements.

If Enable is set to 0x01, the scanning parameters' Own_Address_Type parameter is set to 0x01 or 0x03, and the random address for the device has not been initialized, the Controller shall return the error code *Invalid HCI Command Parameters* (0x12).

Command parameters:

Enable:

Size: 1 octet

Value	Parameter Description
0x00	Scanning disabled
0x01	Scanning enabled
All other values	Reserved for future use

Filter_Duplicates:

Size: 1 octet

Value	Parameter Description
0x00	Duplicate filtering disabled
0x01	Duplicate filtering enabled
0x02	Duplicate filtering enabled, reset for each scan period
All other values	Reserved for future use

**Duration:****Size: 2 octets**

Value	Parameter Description
0x0000	Scan continuously until explicitly disable
N = 0xXXXX	Scan duration Range: 0x0001 to 0xFFFF Time = N * 10 ms Time Range: 10 ms to 655.35 s

Period:**Size: 2 octets**

Value	Parameter Description
0x0000	Scan continuously
N = 0xXXXX	Time interval from when the Controller started its last Scan_Duration until it begins the subsequent Scan_Duration. Range: 0x0001 to 0xFFFF Time = N * 1.28 sec Time Range: 1.28 s to 83,884.8 s

Return parameters:**Status:****Size: 1 octet**

Value	Parameter Description
0x00	HCI_LE_Set_Extended_Scan_Enable command succeeded
0x01 to 0xFF	HCI_LE_Set_Extended_Scan_Enable command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Event(s) generated (unless masked away):

When the HCI_LE_Set_Extended_Scan_Enable command has completed, an HCI_Command_Complete event shall be generated.

Zero or more LE Extended Advertising Reports are generated by the Controller based on any advertising packets received and the duplicate filtering in effect. More than one advertising packet may be reported in each HCI_LE_Extended_Advertising_Report event.

At the end of a single scan (Duration non-zero but Period zero), an HCI_LE_Scan_Timeout event shall be generated.



7.8.66 LE Extended Create Connection command

Command	OCF	Command Parameters	Return Parameters
HCI_LE_Extended_Create_Connection	0x0043	Initiator_Filter_Policy, Own_Address_Type, Peer_Address_Type, Peer_Address, Initiating_PHYs, Scan_Interval[i], Scan_Window[i], Conn_Interval_Min[i], Conn_Interval_Max[i], Conn_Latency[i], Supervision_Timeout[i], Minimum_CE_Length[i], Maximum_CE_Length[i]	

Description:

The HCI_LE_Extended_Create_Connection command is used to create a Link Layer connection to a connectable advertiser.

The Initiator_Filter_Policy parameter is used to determine whether the White List is used. If the White List is not used, the Peer_Address_Type and the Peer_Address parameters specify the address type and address of the advertising device to connect to.

The Own_Address_Type parameter indicates the type of address being used in the connection request packets.

The Peer_Address_Type parameter indicates the type of address used in the connectable advertisement sent by the peer.

The Peer_Address parameter indicates the Peer's Public Device Address, Random (static) Device Address, Non-Resolvable Private Address, or Resolvable Private Address depending on the Peer_Address_Type parameter.

The Initiating_PHYs parameter indicates the PHY(s) on which the advertising packets should be received on the primary advertising physical channel and the PHYs for which connection parameters have been specified. The Host may enable one or more initiating PHYs. If the Host specifies a PHY that is not supported by the Controller, the latter should return the error code *Unsupported Feature or Parameter Value* (0x11). The array elements of the arrayed parameters are ordered in the same order as the set bits in the Initiating_PHYs parameter, starting from bit 0. The number of array elements is determined by the number of bits set in the Initiating_PHYs parameter. When a



connectable advertisement is received and a connection request is sent on one PHY, scanning on any other PHYs is terminated.

The `Scan_Interval[i]` and `Scan_Window[i]` parameters are recommendations from the Host on how long (`Scan_Window[i]`) and how frequently (`Scan_Interval[i]`) the Controller should scan (see [Vol 6] Part B, Section 4.5.3); however the frequency and length of the scan is implementation specific. If the requested scan cannot be supported by the implementation, the Controller shall return the error code *Invalid HCI Command Parameters* (0x12). If bit 1 is set in `Initiating_PHYs`, the values for the LE 2M PHY shall be ignored.

The `Conn_Interval_Min[i]` and `Conn_Interval_Max[i]` parameters define the minimum and maximum allowed connection interval. The `Conn_Interval_Min[i]` parameter shall not be greater than the `Conn_Interval_Max[i]` parameter.

The `Conn_Latency[i]` parameter defines the maximum allowed connection latency (see [Vol 6] Part B, Section 4.5.1).

The `Supervision_Timeout[i]` parameter defines the link supervision timeout for the connection. The `Supervision_Timeout[i]` in milliseconds shall be larger than $(1 + \text{Conn_Latency}[i]) * \text{Conn_Interval_Max}[i] * 2$, where `Conn_Interval_Max[i]` is given in milliseconds (see [Vol 6] Part B, Section 4.5.2).

The `Minimum_CE_Length[i]` and `Maximum_CE_Length[i]` parameters are informative parameters providing the Controller with the expected minimum and maximum length of the connection events. The `Minimum_CE_Length[i]` parameter shall be less than or equal to the `Maximum_CE_Length[i]` parameter.

Where the connection is made on a PHY whose bit is not set in the `Initiating_PHYs` parameter, the Controller shall use the `Conn_Interval_Min[i]`, `Conn_Interval_Max[i]`, `Conn_Latency[i]`, `Supervision_Timeout[i]`, `Minimum_CE_Length[i]`, and `Maximum_CE_Length[i]` parameters for an implementation-chosen PHY whose bit is set in the `Initiating_PHYs` parameter.

If the Host issues this command when another `HCI_LE_Extended_Create_Connection` command is pending in the Controller, the Controller shall return the error code *Command Disallowed* (0x0C).

If the `Own_Address_Type` parameter is set to 0x01 and the random address for the device has not been initialized, the Controller shall return the error code *Invalid HCI Command Parameters* (0x12).

If the `Own_Address_Type` parameter is set to 0x03, the `Initiator_Filter_Policy` parameter is set to 0x00, the controller's resolving list did not contain a matching entry, and the random address for the device has not been initialized, the Controller shall return the error code *Invalid HCI Command Parameters* (0x12).



If the `Own_Address_Type` parameter is set to 0x03, the `Initiator_Filter_Policy` parameter is set to 0x01, and the random address for the device has not been initialized, the Controller shall return the error code *Invalid HCI Command Parameters* (0x12).

If the `Initiating_PHYs` parameter does not have at least one bit set for a PHY allowed for scanning on the primary advertising physical channel, the Controller shall return the error code *Invalid HCI Command Parameters* (0x12).

Command parameters:

Initiating_Filter_Policy:

Size: 1 octet

Value	Parameter Description
0x00	White List is not used to determine which advertiser to connect to. <code>Peer_Address_Type</code> and <code>Peer_Address</code> shall be used.
0x01	White List is used to determine which advertiser to connect to. <code>Peer_Address_Type</code> and <code>Peer_Address</code> shall be ignored.
All other values	Reserved for future use

Own_Address_Type:

Size: 1 octet

Value	Parameter Description
0x00	Public Device Address
0x01	Random Device Address
0x02	Controller generates the Resolvable Private Address based on the local IRK from the resolving list. If the resolving list contains no matching entry, then use the public address.
0x03	Controller generates the Resolvable Private Address based on the local IRK from the resolving list. If the resolving list contains no matching entry, then use the random address from the most recent successful <code>HCI_LE_Set_Random_Address</code> command.
All other values	Reserved for future use

Peer_Address_Type:

Size: 1 octet

Value	Parameter Description
0x00	Public Device Address or Public Identity Address
0x01	Random Device Address or Random (static) Identity Address
All other values	Reserved for future use

**Peer_Address:****Size: 6 octets**

Value	Parameter Description
0XXXXXXXXXX XX	Public Device Address, Random Device Address, Public Identity Address, or Random (static) Identity Address of the device to be connected.

Initiating_PHYs:**Size: 1 octet**

Bit number	Parameter Description
0	Scan connectable advertisements on the LE 1M PHY. Connection parameters for the LE 1M PHY are provided.
1	Connection parameters for the LE 2M PHY are provided.
2	Scan connectable advertisements on the LE Coded PHY. Connection parameters for the LE Coded PHY are provided.
All other bits	Reserved for future use

Scan_Interval[i]:**Size: 2 octets * bits set in Initiating_PHYs**

Value	Parameter Description
N = 0XXXXX	Time interval from when the Controller started its last scan until it begins the subsequent scan on the primary advertising physical channel. Range: 0x0004 to 0xFFFF Time = N * 0.625 ms Time Range: 2.5 ms to 40.959375 s

Scan_Window[i]:**Size: 2 octets * bits set in Initiating_PHYs**

Value	Parameter Description
N = 0XXXXX	Duration of the scan on the primary advertising physical channel. Range: 0x0004 to 0xFFFF Time = N * 0.625 ms Time Range: 2.5 ms to 40.959375 s

Conn_Interval_Min[i]:**Size: 2 octets * bits set in Initiating_PHYs**

Value	Parameter Description
N = 0XXXXX	Minimum value for the connection interval. This shall be less than or equal to Conn_Interval_Max[i]. Range: 0x0006 to 0x0C80 Time = N * 1.25 ms Time Range: 7.5 ms to 4 s
All other values	Reserved for future use



Conn_Interval_Max[i]: *Size: 2 octets * bits set in Initiating_PHYs*

Value	Parameter Description
N = 0xFFFF	Maximum value for the connection interval. This shall be greater than or equal to Conn_Interval_Min[i]. Range: 0x0006 to 0x0C80 Time = N * 1.25 ms Time Range: 7.5 ms to 4 s
All other values	Reserved for future use

Conn_Latency[i]: *Size: 2 octets * bits set in Initiating_PHYs*

Value	Parameter Description
0xFFFF	Slave latency for the connection in number of connection events. Range: 0x0000 to 0x01F3
All other values	Reserved for future use

Supervision_Timeout[i]: *Size: 2 octets * bits set in Initiating_PHYs*

Value	Parameter Description
N = 0xFFFF	Supervision timeout for the LE Link. (See [Vol 6] Part B, Section 4.5.2) Range: 0x000A to 0x0C80 Time = N * 10 ms Time Range: 100 ms to 32 s
All other values	Reserved for future use

Minimum_CE_Length[i]: *Size: 2 octets * bits set in Initiating_PHYs*

Value	Parameter Description
N = 0xFFFF	Informative parameter recommending the minimum length of connection event needed for this LE connection. Range: 0x0000 to 0xFFFF Time = N * 0.625 ms

Maximum_CE_Length[i]: *Size: 2 octets * bits set in Initiating_PHYs*

Value	Parameter Description
N = 0xFFFF	Informative parameter recommending the maximum length of connection event needed for this LE connection. Range: 0x0000 to 0xFFFF Time = N * 0.625 ms

Return parameters:

None.

**Event(s) generated (unless masked away):**

When the Controller receives the HCI_LE_Extended_Create_Connection command, the Controller sends the HCI_Command_Status event to the Host. An HCI_LE_Enhanced_Connection_Complete event shall be generated when a connection is created or the connection creation procedure is cancelled. If a connection is created, this event shall be immediately followed by an HCI_LE_Channel_Selection_Algorithm event.

Note: No HCI_Command_Complete event is sent by the Controller to indicate that this command has been completed. Instead, the HCI_LE_Enhanced_Connection_Complete event indicates that this command has been completed.



7.8.67 LE Periodic Advertising Create Sync command

Command	OCF	Command Parameters	Return Parameters
HCI_LE_Periodic_Advertising_Create_Sync	0x0044	Options, Advertising_SID, Advertiser_Address_Type, Advertiser_Address, Skip, Sync_Timeout, Sync_CTE_Type	

Description:

The HCI_LE_Periodic_Advertising_Create_Sync command is used to synchronize with a periodic advertising train from an advertiser and begin receiving periodic advertising packets.

This command may be issued whether or not scanning is enabled and scanning may be enabled and disabled (see the [LE Set Extended Scan Enable command](#)) while this command is pending. However, synchronization can only occur when scanning is enabled. While scanning is disabled, no attempt to synchronize will take place.

The Options parameter is used to determine whether the Periodic Advertiser List is used and whether HCI_Periodic_Advertising_Report events for this periodic advertising train are initially enabled or disabled. If the Periodic Advertiser List is not used, the Advertising_SID, Advertiser Address_Type, and Advertiser Address parameters specify the periodic advertising device to listen to; otherwise they shall be ignored.

The Advertising_SID parameter, if used, specifies the value that must match the Advertising SID subfield in the ADI field of the received advertisement for it to be used to synchronize.

The Skip parameter specifies the maximum number of consecutive periodic advertising events that the receiver may skip after successfully receiving a periodic advertising packet.

The Sync_Timeout parameter specifies the maximum permitted time between successful receives. If this time is exceeded, synchronization is lost.

The Sync_CTE_Type parameter specifies whether to only synchronize to periodic advertising with certain types of Constant Tone Extension (a value of 0 indicates that the presence or absence of a Constant Tone Extension is irrelevant). If the periodic advertising has the wrong type of Constant Tone Extension then:



- If bit 0 of Options is set, the Controller shall ignore this address and SID and continue to search for other periodic advertisements.
- Otherwise, the Controller shall cancel the synchronization with the error code *Unsupported Remote Feature/Unsupported LMP Feature* (0x1A).

If the periodic advertiser changes the type of Constant Tone Extension after the scanner has synchronized with the periodic advertising, the scanner's Link Layer shall remain synchronized.

If the Host sets all the non-reserved bits of the Sync_CTE_Type parameter to 1, the Controller shall return the error code *Command Disallowed* (0x0C).

Irrespective of the value of the Skip parameter, the Controller should stop skipping packets before the Sync_Timeout would be exceeded.

If the Host issues this command when another HCI_LE_Periodic_Advertising_Create_Sync command is pending (see page 1454), the Controller shall return the error code *Command Disallowed* (0x0C).

If the Host issues this command with bit 0 of Options not set and with Advertising_SID, Advertising_Address_Type, and Advertiser_Address the same as those of a periodic advertising train that the Controller is already synchronized to, the Controller shall return the error code *Connection Already Exists* (0x0B).

If the Host issues this command and the Controller has insufficient resources to handle any more periodic advertising trains, the Controller shall return the error code *Memory Capacity Exceeded* (0x07).

If bit 1 of the Options parameter is set to 1 and the Controller does not support the HCI_LE_Set_Periodic_Advertising_Receive_Enable command, the Controller shall return the error code *Connection Failed to be Established / Synchronization Timeout* (0x3E).

Command parameters:

Options:

Size: 1 octet

Bit Number	Parameter Description
0	0: Use the Advertising_SID, Advertiser_Address_Type, and Advertiser_Address parameters to determine which advertiser to listen to. 1: Use the Periodic Advertiser List to determine which advertiser to listen to.
1	0: Reporting initially enabled 1: Reporting initially disabled
All other bits	Reserved for future use

**Advertising_SID:****Size: 1 octet**

Value	Parameter Description
0x00 to 0x0F	Advertising SID subfield in the ADI field used to identify the Periodic Advertising
All other values	Reserved for future use

Advertiser_Address_Type:**Size: 1 octet**

Value	Parameter Description
0x00	Public Device Address
0x01	Random Device Address
All other values	Reserved for future use

Advertiser_Address:**Size: 6 octets**

Value	Parameter Description
0XXXXXXXXXX XX	Public Device Address, Random Device Address, Public Identity Address, or Random (static) Identity Address of the advertiser

Skip:**Size: 2 octets**

Value	Parameter Description
0XXXXX	The maximum number of periodic advertising events that can be skipped after a successful receive Range: 0x0000 to 0x01F3

Sync_Timeout:**Size: 2 octets**

Value	Parameter Description
N = 0XXXXX	Synchronization timeout for the periodic advertising train Range: 0x000A to 0x4000 Time = N*10 ms Time Range: 100 ms to 163.84 s

Sync_CTE_Type:**Size: 1 octet**

Bit Number	Parameter Description
0	Do not sync to packets with an AoA Constant Tone Extension
1	Do not sync to packets with an AoD Constant Tone Extension with 1 μ s slots
2	Do not sync to packets with an AoD Constant Tone Extension with 2 μ s slots
3	Do not sync to packets with a type 3 Constant Tone Extension (currently reserved for future use)



Bit Number	Parameter Description
4	Do not sync to packets without a Constant Tone Extension
All other bits	Reserved for future use

Return parameters:

None.

Event(s) generated (unless masked away):

When the HCI_LE_Periodic_Advertising_Create_Sync command has been received, the Controller sends the HCI_Command_Status event to the Host. An HCI_LE_Periodic_Advertising_Sync_Established event shall be generated when the Controller starts receiving periodic advertising packets; until the event is generated, the command is considered to be pending.

When the Controller receives periodic advertising packets then, if reporting is enabled, it sends HCI_LE_Periodic_Advertising_Report events to the Host.

Note: No HCI_Command_Complete event is sent by the Controller to indicate that this command has been completed. Instead, the HCI_LE_Periodic_Advertising_Sync_Established event indicates that this command has been completed.



7.8.68 LE Periodic Advertising Create Sync Cancel command

Command	OCF	Command Parameters	Return Parameters
HCI_LE_Periodic_Advertising_Create_Sync_Cancel	0x0045		Status

Description:

The HCI_LE_Periodic_Advertising_Create_Sync_Cancel command is used to cancel the HCI_LE_Periodic_Advertising_Create_Sync command while it is pending.

If the Host issues this command while no HCI_LE_Periodic_Advertising_Create_Sync command is pending, the Controller shall return the error code *Command Disallowed* (0x0C).

Command parameters:

None.

Return parameters:

Status: Size: 1 octet

Value	Parameter Description
0x00	HCI_LE_Periodic_Advertising_Create_Sync_Cancel command succeeded
0x01 to 0xFF	HCI_LE_Periodic_Advertising_Create_Sync_Cancel command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Event(s) generated (unless masked away):

When the HCI_LE_Periodic_Advertising_Create_Sync_Cancel command has completed, the Controller sends an HCI_Command_Complete event to the Host.

After the HCI_Command_Complete is sent and if the cancellation was successful, the Controller sends an HCI_LE_Periodic_Advertising_Sync_Established event to the Host with the error code *Operation Cancelled by Host* (0x44).



7.8.69 LE Periodic Advertising Terminate Sync command

Command	OCF	Command Parameters	Return Parameters
HCI_LE_Periodic_Advertising_Terminate_Sync	0x0046	Sync_Handle	Status

Description:

The HCI_LE_Periodic_Advertising_Terminate_Sync command is used to stop reception of the periodic advertising train identified by the Sync_Handle parameter.

If the periodic advertising train corresponding to the Sync_Handle parameter does not exist, then the Controller shall return the error code *Unknown Advertising Identifier* (0x42).

Following successful completion of this command the Sync_Handle is destroyed.

Command parameters:

Sync_Handle: *Size: 2 octets (12 bits meaningful)*

Value	Parameter Description
0xXXXX	Sync_Handle identifying the periodic advertising train Range: 0x0000 to 0x0EFF

Return parameters:

Status: *Size: 1 octet*

Value	Parameter Description
0x00	HCI_LE_Periodic_Advertising_Terminate_Sync command succeeded
0x01 to 0xFF	HCI_LE_Periodic_Advertising_Terminate_Sync command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Event(s) generated (unless masked away):

When the HCI_LE_Periodic_Advertising_Terminate_Sync command has completed, an HCI_Command_Complete event shall be generated.



7.8.70 LE Add Device To Periodic Advertiser List command

Command	OCF	Command Parameters	Return Parameters
HCI_LE_Add_Device_To_Periodic_Advertiser_List	0x0047	Advertiser_Address_Type, Advertiser_Address, Advertising_SID	Status

Description:

The HCI_LE_Add_Device_To_Periodic_Advertiser_List command is used to add an entry, consisting of a single device address and SID, to the Periodic Advertiser list stored in the Controller. Any additions to the Periodic Advertiser list take effect immediately. If the entry is already on the list, the Controller shall return the error code *Invalid HCI Command Parameters* (0x12).

If the Host issues this command when an HCI_LE_Periodic_Advertising_Create_Sync command is pending, the Controller shall return the error code *Command Disallowed* (0x0C).

When a Controller cannot add an entry to the Periodic Advertiser list because the list is full, the Controller shall return the error code *Memory Capacity Exceeded* (0x07).

Command parameters:

Advertiser_Address_Type: *Size: 1 octet*

Value	Parameter Description
0x00	Public Device Address or Public Identity Address
0x01	Random Device Address or Random (static) Identity Address
All other values	Reserved for future use

Advertiser_Address: *Size: 6 octets*

Value	Parameter Description
0XXXXXXXXXX XX	Public Device Address, Random Device Address, Public Identity Address, or Random (static) Identity Address of the advertiser

Advertising_SID: *Size: 1 octet*

Value	Parameter Description
0x00 to 0x0F	Advertising SID subfield in the ADI field used to identify the Periodic Advertising
All other values	Reserved for future use

**Return parameters:***Status:**Size: 1 octet*

Value	Parameter Description
0x00	HCI_LE_Add_Device_To_Periodic_Advertiser_List command succeeded
0x01 to 0xFF	HCI_LE_Add_Device_To_Periodic_Advertiser_List command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Event(s) generated (unless masked away):

When the HCI_LE_Add_Device_To_Periodic_Advertiser_List command has completed, an HCI_Command_Complete event shall be generated.



7.8.71 LE Remove Device From Periodic Advertiser List command

Command	OCF	Command Parameters	Return Parameters
HCI_LE_Remove_Device_From_Periodic_Advertiser_List	0x0048	Advertiser_Address_Type, Advertiser_Address, Advertising_SID	Status

Description:

The HCI_LE_Remove_Device_From_Periodic_Advertiser_List command is used to remove one entry from the list of Periodic Advertisers stored in the Controller. Removals from the Periodic Advertisers List take effect immediately.

If the Host issues this command when an HCI_LE_Periodic_Advertising_Create_Sync command is pending, the Controller shall return the error code *Command Disallowed* (0x0C).

When a Controller cannot remove an entry from the Periodic Advertiser list because it is not found, the Controller shall return the error code *Unknown Advertising Identifier* (0x42).

Command parameters:

Advertiser_Address_Type:

Size: 1 octet

Value	Parameter Description
0x00	Public Device Address or Public Identity Address
0x01	Random Device Address or Random (static) Identity Address
All other values	Reserved for future use

Advertiser_Address:

Size: 6 octets

Value	Parameter Description
0xFFFFFFFFXX	Public Device Address, Random Device Address, Public Identity Address, or Random (static) Identity Address of the advertiser

Advertising_SID:

Size: 1 octet

Value	Parameter Description
0x00 to 0x0F	Advertising SID subfield in the ADI field used to identify the Periodic Advertising
All other values	Reserved for future use

**Return parameters:***Status:**Size: 1 octet*

Value	Parameter Description
0x00	HCI_LE_Remove_Device_From_Periodic_Advertiser_List command succeeded
0x01 to 0xFF	HCI_LE_Remove_Device_From_Periodic_Advertiser_List command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Event(s) generated (unless masked away):

When the HCI_LE_Remove_Device_From_Periodic_Advertiser_List command has completed, an HCI_Command_Complete event shall be generated.



7.8.72 LE Clear Periodic Advertiser List command

Command	OCF	Command Parameters	Return Parameters
HCI_LE_Clear_Periodic_Advertiser_List	0x0049		Status

Description:

The HCI_LE_Clear_Periodic_Advertiser_List command is used to remove all entries from the list of Periodic Advertisers in the Controller.

If this command is used when an HCI_LE_Periodic_Advertising_Create_Sync command is pending, the Controller shall return the error code *Command Disallowed* (0x0C).

Command parameters:

None.

Return parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	HCI_LE_Clear_Periodic_Advertiser_List command succeeded
0x01 to 0xFF	HCI_LE_Clear_Periodic_Advertiser_List command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Event(s) generated (unless masked away):

When the HCI_LE_Clear_Periodic_Advertiser_List command has completed, an HCI_Command_Complete event shall be generated.



7.8.73 LE Read Periodic Advertiser List Size command

Command	OCF	Command Parameters	Return Parameters
HCI_LE_Read_Periodic_Advertiser_List_Size	0x004A		Status, Periodic_Advertiser_List_Size

Description:

The HCI_LE_Read_Periodic_Advertiser_List_Size command is used to read the total number of Periodic Advertiser list entries that can be stored in the Controller.

Note: The number of entries that can be stored is not fixed and the Controller can change it at any time (e.g., because the memory used to store the list can also be used for other purposes).

Command parameters:

None.

Return parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	HCI_LE_Read_Periodic_Advertiser_List_Size command succeeded
0x01 to 0xFF	HCI_LE_Read_Periodic_Advertiser_List_Size command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Periodic_Advertiser_List_Size:

Size: 1 octet

Value	Parameter Description
0x01 to 0xFF	Total number of Periodic Advertiser list entries that can be stored in the Controller
0x00	Reserved for future use

Event(s) generated (unless masked away):

When the HCI_LE_Read_Periodic_Advertiser_List_Size command has completed, an HCI_Command_Complete event shall be generated.



7.8.74 LE Read Transmit Power command

Command	OCF	Command Parameters	Return Parameters
HCI_LE_Read_Transmit_Power	0x004B		Status, Min_Tx_Power, Max_Tx_Power

Description:

The HCI_LE_Read_Transmit_Power command is used to read the minimum and maximum transmit powers supported by the Controller.

Command parameters:

None.

Return parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	HCI_LE_Read_Transmit_Power command succeeded
0x01 to 0xFF	HCI_LE_Read_Transmit_Power command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Min_Tx_Power:

Size: 1 octet

Value	Parameter Description
0xXX	Size: 1 octet (signed integer) Range: -127 to +20 Units: dBm

Max_Tx_Power:

Size: 1 octet

Value	Parameter Description
0xXX	Size: 1 octet (signed integer) Range: -127 to +20 Units: dBm

Event(s) generated (unless masked away):

When the HCI_LE_Read_Transmit_Power command has completed, an HCI_Command_Complete event shall be generated.



7.8.75 LE Read RF Path Compensation command

Command	OCF	Command Parameters	Return Parameters
HCI_LE_Read_RF_Path_Compensation	0x004C		Status, RF_Tx_Path_Compensation_Value, RF_Rx_Path_Compensation_Value

Description:

The HCI_LE_Read_RF_Path_Compensation command is used to read the RF Path Compensation Values parameter used in the Tx Power Level and RSSI calculation.

Command parameters:

None.

Return parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	HCI_LE_Read_RF_Path_Compensation command succeeded
0x01 – 0xFF	HCI_LE_Read_RF_Path_Compensation command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

RF_Tx_Path_Compensation_Value:

Size: 2 octets

Value	Parameter Description
0xFFFF	Size: 2 octets (signed integer) Range: -128.0 dB (0xFB00) to 128.0 dB (0x0500) Units: 0.1 dB

RF_Rx_Path_Compensation_Value:

Size: 2 octets

Value	Parameter Description
0xFFFF	Size: 2 octets (signed integer) Range: -128.0 dB (0xFB00) to 128.0 dB (0x0500) Units: 0.1 dB

Event(s) generated (unless masked away):

When the HCI_LE_Read_RF_Path_Compensation command has completed, an HCI_Command_Complete event shall be generated.



7.8.76 LE Write RF Path Compensation command

Command	OCF	Command Parameters	Return Parameters
HCI_LE_Write_RF_Path_Compensation	0x004D	RF_Tx_Path_Compensation_Value, RF_Rx_Path_Compensation_Value	Status

Description:

The HCI_LE_Write_RF_Path_Compensation command is used to indicate the RF path gain or loss between the RF transceiver and the antenna contributed by intermediate components. A positive value means a net RF path gain and a negative value means a net RF path loss. The RF Tx Path Compensation Value parameter shall be used by the Controller to calculate radiative Tx Power Level used in the TxPower field in the Extended Header using the following equation:

Radiative Tx Power Level = Tx Power Level at RF transceiver output + RF Tx Path Compensation Value

For example, if the Tx Power Level is +4 (dBm) at RF transceiver output and the RF Path Compensation Value is -1.5 (dB), the radiative Tx Power Level is $+4+(-1.5) = 2.5$ (dBm).

The RF Rx Path Compensation Value parameter shall be used by the Controller to calculate the RSSI value reported to the Host.

Command parameters:

RF_Tx_Path_Compensation_Value:

Size: 2 octets

Value	Parameter Description
0xXXXX	Size: 2 octets (signed integer) Range: -128.0 dB (0xFB00) to 128.0 dB (0x0500) Units: 0.1 dB

RF_Rx_Path_Compensation_Value:

Size: 2 octets

Value	Parameter Description
0xXXXX	Size: 2 octets (signed integer) Range: -128.0 dB (0xFB00) to 128.0 dB (0x0500) Units: 0.1 dB

**Return parameters:***Status:**Size: 1 octet*

Value	Parameter Description
0x00	HCI_LE_Write_RF_Path_Compensation command succeeded
0x01 to 0xFF	HCI_LE_Write_RF_Path_Compensation command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Event(s) generated (unless masked away):

When the HCI_LE_Write_RF_Path_Compensation command has completed, an HCI_Command_Complete event shall be generated.



7.8.77 LE Set Privacy Mode command

Command	OCF	Command Parameters	Return Parameters
HCI_LE_Set_Privacy_Mode	0x004E	Peer_Identity_Address_Type, Peer_Identity_Address, Privacy_Mode	Status

Description:

The HCI_LE_Set_Privacy_Mode command is used to allow the Host to specify the privacy mode to be used for a given entry on the resolving list. The effect of this setting is specified in [\[Vol 6\] Part B, Section 4.7](#).

When an entry on the resolving list is removed, the mode associated with that entry shall also be removed.

This command shall not be used when address resolution is enabled in the Controller and:

- Advertising (other than periodic advertising) is enabled,
- Scanning is enabled, or
- an HCI_LE_Create_Connection, HCI_LE_Extended_Create_Connection, or HCI_LE_Periodic_Advertising_Create_Sync command is outstanding.

This command may be used at any time when address resolution is disabled in the Controller.

If the device is not on the resolving list, the Controller shall return the error code *Unknown Connection Identifier* (0x02).

Command parameters:

Peer_Identity_Address_Type:

Size: 1 octet

Value	Parameter Description
0x00	Public Identity Address
0x01	Random (static) Identity Address
All other values	Reserved for future use

Peer Identity_Address:

Size: 6 octets

Value	Parameter Description
0xFFFFFFFFXX	Public Identity Address or Random (static) Identity Address of the advertiser

**Privacy_Mode:****Size: 1 octet**

Value	Parameter Description
0x00	Use Network Privacy Mode for this peer device (default)
0x01	Use Device Privacy Mode for this peer device
All other values	Reserved for future use

Return parameters:**Status:****Size: 1 octet**

Value	Parameter Description
0x00	HCI_LE_Set_Privacy_Mode command succeeded
0x01 to 0xFF	HCI_LE_Set_Privacy_Mode command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Event(s) generated (unless masked away):

When the HCI_LE_Set_Privacy_Mode command has completed, an HCI_Command_Complete event shall be generated.



7.8.78 LE Receiver Test command [v3]

Command	OCF	Command Parameters	Return Parameters
HCI_LE_Receiver_Test [v3]	0x004F	RX_Channel, PHY, Modulation_Index, Expected_CTE_Length, Expected_CTE_Type, Slot_Durations, Length_of_Switching_Pattern, Antenna_IDs[i]	Status

For version v1 of this command, see [Section 7.8.28](#).

For version v2 of this command, see [Section 7.8.50](#).

Description:

This command is used to start a test where the DUT receives test reference packets at a fixed interval. The tester generates the test reference packets.

The RX_Channel and PHY parameters specify the RF channel and PHY to be used by the receiver.

The Modulation_Index parameter specifies whether or not the Controller should assume the receiver has a stable modulation index.

The Expected_CTE_Length and Expected_CTE_Type parameters specify the expected length and type of the Constant Tone Extensions in received test reference packets. When receiving on a PHY that allows Constant Tone Extensions, if the Constant Tone Extension in a received test reference packet does not match both of these, the DUT shall discard that packet. If Expected_CTE_Length is not zero and PHY specifies a PHY that does not allow Constant Tone Extensions, the Controller shall return the error code *Command Disallowed* (0x0C).

If the Slot_Durations parameter is set to 0x01 and the Controller does not support 1 μ s switching and sampling, the Controller shall return the error code *Unsupported Feature or Parameter Value* (0x11).

Slot_Durations, Length_of_Switching_Pattern, and Antenna_IDs[i] are only used when expecting an AoA Constant Tone Extension and shall be ignored when expecting an AoD Constant Tone Extension.

If the Controller determines that any of the Antenna_IDs[i] values do not identify an antenna in the device's antenna array, it shall return the error code *Unsupported Feature or Parameter Value* (0x11).



Note: Some Controllers may be unable to determine which values do or do not identify an antenna.

Command parameters:

RX_Channel:

Size: 1 octet

Value	Parameter Description
N = 0xXX	N = (F-2402) / 2 Range: 0x00 to 0x27. Frequency Range: 2402 MHz to 2480 MHz

PHY:

Size: 1 octet

Value	Parameter Description
0x01	Receiver set to use the LE 1M PHY
0x02	Receiver set to use the LE 2M PHY
0x03	Receiver set to use the LE Coded PHY
All other values	Reserved for future use

Modulation_Index:

Size: 1 octet

Value	Parameter Description
0x00	Assume transmitter will have a standard modulation index
0x01	Assume transmitter will have a stable modulation index
All other values	Reserved for future use

Expected_CTE_Length:

Size: 1 octet

Value	Parameter Description
0x00	No Constant Tone Extension expected (default)
0x02 to 0x14	Expected length of the Constant Tone Extension in 8 μ s units
All other values	Reserved for future use

Expected_CTE_Type:

Size: 1 octet

Value	Parameter Description
0x00	Expect AoA Constant Tone Extension
0x01	Expect AoD Constant Tone Extension with 1 μ s slots
0x02	Expect AoD Constant Tone Extension with 2 μ s slots
All other values	Reserved for future use

**Slot_Durations:****Size: 1 octet**

Value	Parameter Description
0x01	Switching and sampling slots are 1 μ s each
0x02	Switching and sampling slots are 2 μ s each
All other values	Reserved for future use

Length_of_Switching_Pattern:**Size: 1 octet**

Value	Parameter Description
0x02 to 0x4B	The number of Antenna IDs in the pattern
All other values	Reserved for future use

Antenna_IDs[i]:**Size: Length_of_Switching_Pattern * 1 octet**

Value	Parameter Description
0xXX	List of Antenna IDs in the pattern

Return parameters:**Status:****Size: 1 octet**

Value	Parameter Description
0x00	HCI_LE_Receiver_Test command succeeded.
0x01 to 0xFF	HCI_LE_Receiver_Test command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Event(s) generated (unless masked away):

When the HCI_LE_Receiver_Test command has completed, an HCI_Command_Complete event shall be generated.

If the Expected_CTE_Length parameter is not set to zero, then HCI_LE_Connectionless_IQ_Report events may be generated by the Controller.



7.8.79 LE Transmitter Test command [v3]

Command	OCF	Command Parameters	Return Parameters
HCI_LE_Transmitter_Test [v3]	0x0050	TX_Channel, Length_Of_Test_Data, Packet_Payload, PHY, CTE_Length, CTE_Type, Length_of_Switching_Pattern, Antenna_IDs[i]	Status

For version v1 of this command, see [Section 7.8.29](#).

For version v2 of this command, see [Section 7.8.51](#).

Description:

This command is used to start a test where the DUT generates test reference packets at a fixed interval. The Controller shall transmit at maximum power.

The TX_Channel and PHY parameters specify the RF channel and PHY to be used by the transmitter.

The Length_of_Test_Data and Packet_Payload parameters specify the length and contents of the Payload of the test reference packets. An LE Controller supporting the HCI_LE_Transmitter_Test command shall support Packet_Payload values 0x00, 0x01 and 0x02. An LE Controller supporting the LE Coded PHY shall also support Packet_Payload value 0x04. An LE Controller may support other values of Packet_Payload.

The CTE_Length and CTE_Type parameters specify the length and type of the Constant Tone Extension in the test reference packets. If the CTE_Type parameter is set to 0x01 and the Controller does not support 1 μ s switching, the Controller shall return the error code *Unsupported Feature or Parameter Value* (0x11). If CTE_Length is not zero and PHY specifies a PHY that does not allow Constant Tone Extensions, the Controller shall return the error code *Command Disallowed* (0x0C).

The Length_of_Switching_Pattern and Antenna_IDs[i] parameters specify the antenna switching pattern. They are only used when transmitting an AoD Constant Tone Extension and shall be ignored when transmitting an AoA Constant Tone Extension.

If the Controller determines that any of the Antenna_IDs[i] values do not identify an antenna in the device's antenna array, it shall return the error code *Unsupported Feature or Parameter Value* (0x11).



Note: Some Controllers may be unable to determine which values do or do not identify an antenna.

Command parameters:

TX_Channel:

Size: 1 octet

Value	Parameter Description
N = 0xXX	N = (F-2402) / 2 Range: 0x00 to 0x27 Frequency Range: 2402 MHz to 2480 MHz

Length_Of_Test_Data:

Size: 1 octet

Value	Parameter Description
0x00 to 0xFF	Length in bytes of payload data in each packet

Packet_Payload:

Size: 1 octet

Value	Parameter Description
0x00	PRBS9 sequence '1111111100000111101...' (in transmission order) as described in [Vol 6] Part F, Section 4.1.5
0x01	Repeated '11110000' (in transmission order) sequence as described in [Vol 6] Part F, Section 4.1.5
0x02	Repeated '10101010' (in transmission order) sequence as described in [Vol 6] Part F, Section 4.1.5
0x03	PRBS15 sequence as described in [Vol 6] Part F, Section 4.1.5
0x04	Repeated '11111111' (in transmission order) sequence
0x05	Repeated '00000000' (in transmission order) sequence
0x06	Repeated '00001111' (in transmission order) sequence
0x07	Repeated '01010101' (in transmission order) sequence
All other values	Reserved for future use

PHY:

Size: 1 octet

Value	Parameter Description
0x01	Transmitter set to use the LE 1M PHY
0x02	Transmitter set to use the LE 2M PHY
0x03	Transmitter set to use the LE Coded PHY with S=8 data coding
0x04	Transmitter set to use the LE Coded PHY with S=2 data coding
All other values	Reserved for future use

**CTE_Length:****Size: 1 octet**

Value	Parameter Description
0x00	Do not transmit a Constant Tone Extension
0x02 to 0x14	Length of the Constant Tone Extension in 8 μ s units
All other values	Reserved for future use

CTE_Type:**Size: 1 octet**

Value	Parameter Description
0x00	AoA Constant Tone Extension
0x01	AoD Constant Tone Extension with 1 μ s slots
0x02	AoD Constant Tone Extension with 2 μ s slots
All other values	Reserved for future use

Length_of_Switching_Pattern:**Size: 1 octet**

Value	Parameter Description
0x02 to 0x4B	The number of Antenna IDs in the pattern
All other values	Reserved for future use

Antenna_IDs[i]:**Size: Length_of_Switching_Pattern * 1 octet**

Value	Parameter Description
0xXX	List of Antenna IDs in the pattern

Return parameters:**Status:****Size: 1 octet**

Value	Parameter Description
0x00	HCI_LE_Transmitter_Test command succeeded.
0x01 to 0xFF	HCI_LE_Transmitter_Test command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Event(s) generated (unless masked away):

When the HCI_LE_Transmitter_Test command has completed, an HCI_Command_Complete event shall be generated.



7.8.80 LE Set Connectionless CTE Transmit Parameters command

Command	OCF	Command Parameters	Return Parameters
HCI_LE_Set_Connectionless_CTE_Transmit_Parameters	0x0051	Advertising_Handle, CTE_Length, CTE_Type, CTE_Count, Length_of_Switching_Pattern, Antenna_IDs[i]	Status

Description:

The HCI_LE_Set_Connectionless_CTE_Transmit_Parameters command is used to set the type, length, and antenna switching pattern for the transmission of Constant Tone Extensions in any periodic advertising on the advertising set identified by the Advertising_Handle parameter.

The CTE_Count parameter specifies how many packets with a Constant Tone Extension are to be transmitted in each periodic advertising event. If the number of packets that would otherwise be transmitted is less than this, the Controller shall transmit sufficient AUX_CHAIN_IND PDUs with no AdvData to make up the number. However, if a change in circumstances since this command was issued means that the Controller can no longer schedule all of these packets, it should transmit as many as possible.

If the Host issues this command when Constant Tone Extensions have been enabled in the advertising set, the Controller shall return the error code *Command Disallowed* (0x0C).

The Length_of_Switching_Pattern and Antenna_IDs[i] parameters are only used when transmitting an AoD Constant Tone Extension and shall be ignored when transmitting an AoA Constant Tone Extension.

If the CTE_Length parameter is greater than the maximum length of Constant Tone Extension supported, the Controller shall return the error code *Unsupported Feature or Parameter Value* (0x11).

If the Host requests a type of Constant Tone Extension that the Controller does not support, the Controller shall return the error code *Unsupported Feature or Parameter Value* (0x11).

If the Controller is unable to schedule CTE_Count packets in each event, the Controller shall return the error code *Unsupported Feature or Parameter Value* (0x11).



If the advertising set corresponding to the Advertising_Handle parameter does not exist, the Controller shall return the error code *Unknown Advertising Identifier* (0x42).

If Length_of_Switching_Pattern is greater than the maximum length of switching pattern supported by the Controller (see [Section 7.8.87](#)), the Controller shall return the error code *Unsupported Feature or Parameter Value* (0x11).

If the Controller determines that any of the Antenna_IDs[i] values do not identify an antenna in the device's antenna array, it shall return the error code *Unsupported Feature or Parameter Value* (0x11).

Note: Some Controllers may be unable to determine which values do or do not identify an antenna.

Command parameters:

Advertising_Handle: *Size: 1 octet*

Value	Parameter Description
0xXX	Used to identify an advertising set Range: 0x00 to 0xEF

CTE_Length: *Size: 1 octet*

Value	Parameter Description
0x02 to 0x14	Constant Tone Extension length in 8 μ s units
All other values	Reserved for future use

CTE_Type: *Size: 1 octet*

Value	Parameter Description
0x00	AoA Constant Tone Extension
0x01	AoD Constant Tone Extension with 1 μ s slots
0x02	AoD Constant Tone Extension with 2 μ s slots
All other values	Reserved for future use

CTE_Count: *Size: 1 octet*

Value	Parameter Description
0xXX	The number of Constant Tone Extensions to transmit in each periodic advertising interval Range: 0x01 to 0x10

**Length_of_Switching_Pattern:****Size: 1 octet**

Value	Parameter Description
0x02 to 0x4B	The number of Antenna IDs in the pattern
All other values	Reserved for future use

Antenna_IDs[i]:**Size: Length_of_Switching_Pattern * 1 octet**

Value	Parameter Description
0xXX	List of Antenna IDs in the pattern.

Return parameters:**Status:****Size: 1 octet**

Value	Parameter Description
0x00	HCI_LE_Set_Connectionless_CTE_Transmit_Parameters command succeeded.
0x01 to 0xFF	HCI_LE_Set_Connectionless_CTE_Transmit_Parameters command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Event(s) generated (unless masked away):

When the HCI_LE_Set_Connectionless_CTE_Transmit_Parameters command has completed, an HCI_Command_Complete event shall be generated.



7.8.81 LE Set Connectionless CTE Transmit Enable command

Command	OCF	Command Parameters	Return Parameters
HCI_LE_Set_Connectionless_CTE_Transmit_Enable	0x0052	Advertising_Handle, CTE_Enable	Status

Description:

The HCI_LE_Set_Connectionless_CTE_Transmit_Enable command is used to request that the Controller enables or disables the use of Constant Tone Extensions in any periodic advertising on the advertising set identified by Advertising_Handle.

In order to start sending periodic advertisements containing a Constant Tone Extension, the Host must also enable periodic advertising using the HCI_LE_Set_Periodic_Advertising_Enable command (see [Section 7.8.63](#)).

Note: Periodic advertising can only be enabled when advertising is enabled on the same advertising set, but can continue after advertising has been disabled.

If the Host issues this command before it has issued the HCI_LE_Set_Periodic_Advertising_Parameters command (see [Section 7.8.61](#)) for the advertising set, the Controller shall return the error code *Command Disallowed* (0x0C).

Once enabled, the Controller shall continue advertising with Constant Tone Extensions until either one of the following occurs:

- The Host issues an HCI_LE_Set_Connectionless_CTE_Transmit_Enable command with CTE_Enable set to 0x00 (disabling Constant Tone Extensions but allowing periodic advertising to continue).
- The Host issues an HCI_LE_Set_Periodic_Advertising_Enable command (see [Section 7.8.63](#)) with Enable set to 0x00 (disabling periodic advertising). If periodic advertising is re-enabled then it shall continue to contain Constant Tone Extensions.

If the Host issues this command before it has issued the HCI_LE_Set_Connectionless_CTE_Transmit_Parameters command for the advertising set, the Controller shall return the error code *Command Disallowed* (0x0C).

If the periodic advertising is on a PHY that does not allow Constant Tone Extensions, the Controller shall return the error code *Command Disallowed* (0x0C).



If the advertising set corresponding to the Advertising_Handle parameter does not exist, the Controller shall return the error code *Unknown Advertising Identifier* (0x42).

The Host may issue this command when advertising or periodic advertising is enabled in the advertising set.

Command parameters:*Advertising_Handle:**Size: 1 octet*

Value	Parameter Description
0xXX	Identifier for the advertising set in which Constant Tone Extension is being enabled or disabled Range: 0x00 to 0xEF

*CTE_Enable:**Size: 1 octet*

Value	Parameter Description
0x00	Advertising with Constant Tone Extension is disabled (default)
0x01	Advertising with Constant Tone Extension is enabled
All other values	Reserved for future use

Return parameters:*Status:**Size: 1 octet*

Value	Parameter Description
0x00	HCI_LE_Set_Connectionless_CTE_Transmit_Enable command succeeded.
0x01 to 0xFF	HCI_LE_Set_Connectionless_CTE_Transmit_Enable command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Event(s) generated (unless masked away):

When the HCI_LE_Set_Connectionless_CTE_Transmit_Enable command has completed, an HCI_Command_Complete event shall be generated.



7.8.82 LE Set Connectionless IQ Sampling Enable command

Command	OCF	Command Parameters	Return Parameters
HCI_LE_Set_Connectionless_IQ_Sampling_Enable	0x0053	Sync_Handle, Sampling_Enable, Slot_Durations, Max_Sampled_CTEs, Length_of_Switching_Pattern, Antenna_IDs[i]	Status, Sync_Handle

Description:

The HCI_LE_Set_Connectionless_IQ_Sampling_Enable command is used to request that the Controller enables or disables capturing IQ samples from the Constant Tone Extension of periodic advertising packets in the periodic advertising train identified by the Sync_Handle parameter. If that periodic advertising train does not exist, then the Controller shall return the error code *Unknown Advertising Identifier* (0x42).

The Max_Sampled_CTEs parameter specifies the maximum number of Constant Tone Extensions in each periodic advertising event that the Controller should collect and report IQ samples from. The Controller should sample all Constant Tone Extensions up to this number.

If the Sampling_Enable parameter is set to 0x01 (sampling is enabled), the Controller starts attempting to capture IQ samples from the periodic advertisements.

Once sampling has been enabled, the Controller shall continue taking IQ samples until the Host issues an HCI_LE_Set_Connectionless_IQ_Enable command with Sampling_Enable set to 0x00 (sampling is disabled) or synchronization with the periodic advertising train is lost.

If Sampling_Enable is set to 0x00, Slot_Durations, Max_Sampled_CTEs, Length_of_Switching_Pattern, and Antenna_IDs shall be ignored.

The command is also used to set the antenna switching pattern and switching and sampling slot durations to be used while receiving the Constant Tone Extension.

If Slot_Durations is set to 0x01 and the Controller does not support 1 μ s switching and sampling, the Controller shall return the error code *Unsupported Feature or Parameter Value* (0x11).



The Slot_Durations, Length_of_Switching_Pattern, and Antenna_IDs parameters are only used when receiving an AoA Constant Tone Extension and shall be ignored when receiving an AoD Constant Tone Extension.

If Length_of_Switching_Pattern is greater than the maximum length of switching pattern supported by the Controller, the Controller shall return the error code *Unsupported Feature or Parameter Value* (0x11).

If the Controller determines that any of the Antenna_IDs[i] values do not identify an antenna in the device's antenna array, it shall return the error code *Unsupported Feature or Parameter Value* (0x11).

Note: Some Controllers may be unable to determine which values do or do not identify an antenna.

If Sampling_Enable is set to 0x01 and the periodic advertising is on a PHY that does not allow Constant Tone Extensions, the Controller shall return the error code *Command Disallowed* (0x0C).

Command parameters:

Sync_Handle: *Size: 2 octets (12 bits meaningful)*

Value	Parameter Description
0xXXXX	Sync_Handle identifying the periodic advertising train. Range: 0x0000 to 0x0EFF

Sampling_Enable: *Size: 1 octet*

Value	Parameter Description
0x00	Connectionless IQ sampling is disabled (default)
0x01	Connectionless IQ sampling is enabled
All other values	Reserved for future use

Slot_Durations: *Size: 1 octet*

Value	Parameter Description
0x01	Switching and sampling slots are 1 μ s each
0x02	Switching and sampling slots are 2 μ s each
All other values	Reserved for future use

Max_Sampled_CTEs: *Size: 1 octet*

Value	Parameter Description
0x00	Sample and report all available Constant Tone Extensions



Value	Parameter Description
0xXX	The maximum number of Constant Tone Extensions to sample and report in each periodic advertising interval Range: 0x01 to 0x10

*Length_of_Switching_Pattern:**Size: 1 octet*

Value	Parameter Description
0x02 to 0x4B	The number of Antenna IDs in the pattern
All other values	Reserved for future use

*Antenna_IDs[i]:**Size: Length_of_Switching_Pattern * 1 octet*

Value	Parameter Description
0xXX	List of Antenna IDs in the pattern.

Return parameters:*Status:**Size: 1 octet*

Value	Parameter Description
0x00	HCI_LE_Set_Connectionless_IQ_Sampling_Enable command succeeded.
0x01 to 0xFF	HCI_LE_Set_Connectionless_IQ_Sampling_Enable command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

*Sync_Handle:**Size: 2 octets (12 bits meaningful)*

Value	Parameter Description
0xFFFF	Sync_Handle identifying the periodic advertising. Range: 0x0000 to 0x0EFF

Event(s) generated (unless masked away):

When the HCI_LE_Set_Connectionless_IQ_Sampling_Enable command has completed, an HCI_Command_Complete event shall be generated.

HCI_LE_Connectionless_IQ_Report events are generated by the Controller based on the advertising packets received.



7.8.83 LE Set Connection CTE Receive Parameters command

Command	OCF	Command Parameters	Return Parameters
HCI_LE_Set_Connection_-_CTE_Receive_Parameters	0x0054	Connection_Handle, Sampling_Enable, Slot_Durations, Length_of_Switching_Pattern, Antenna_IDs[i]	Status, Connection_Handle

Description:

The HCI_LE_Set_Connection_CTE_Receive_Parameters command is used to enable or disable sampling received Constant Tone Extension fields on the connection identified by the Connection_Handle parameter and to set the antenna switching pattern and switching and sampling slot durations to be used.

If the Sampling_Enable parameter is set to 0x01, the Controller shall sample Constant Tone Extensions on the specified connection and report the samples to the Host. If it is set to 0x00, the Controller shall cease sampling on the specified connection; the remaining parameters shall be ignored.

If Slot_Durations is set to 0x01 and the Controller does not support 1 μ s switching and sampling, the Controller shall return the error code *Unsupported Feature or Parameter Value* (0x11).

The Slot_Durations, Length_of_Switching_Pattern, and Antenna_IDs parameters are only used when receiving an AoA Constant Tone Extension and shall be ignored when receiving an AoD Constant Tone Extension.

If Length_of_Switching_Pattern is greater than the maximum length of switching pattern supported by the Controller, the Controller shall return the error code *Unsupported Feature or Parameter Value* (0x11).

If the Controller determines that any of the Antenna_IDs[i] values do not identify an antenna in the device's antenna array, it shall return the error code *Unsupported Feature or Parameter Value* (0x11).

Note: Some Controllers may be unable to determine which values do or do not identify an antenna.

Command parameters:

**Connection_Handle:****Size: 2 octets (12 bits meaningful)**

Value	Parameter Description
0xFFFF	Connection_Handle Range: 0x0000 to 0x0EFF

Sampling_Enable:**Size: 1 octet**

Value	Parameter Description
0x00	Connection IQ sampling is disabled (default)
0x01	Connection IQ sampling is enabled
All other values	Reserved for future use

Slot_Durations:**Size: 1 octet**

Value	Parameter Description
0x01	Switching and sampling slots are 1 μ s each
0x02	Switching and sampling slots are 2 μ s each
All other values	Reserved for future use

Length_of_Switching_Pattern:**Size: 1 octet**

Value	Parameter Description
0x02 to 0x4B	The number of Antenna IDs in the pattern
All other values	Reserved for future use

Antenna_IDs[i]:**Size: Length_of_Switching_Pattern * 1 octet**

Value	Parameter Description
0xFF	List of Antenna IDs in the pattern.

Return parameters:**Status:****Size: 1 octet**

Value	Parameter Description
0x00	HCI_LE_Set_Connection_CTE_Receive_Parameters command succeeded.
0x01 to 0xFF	HCI_LE_Set_Connection_CTE_Receive_Parameters command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.



Connection_Handle: *Size: 2 octets (12 bits meaningful)*

Value	Parameter Description
0xFFFF	Connection_Handle Range: 0x0000 to 0x0EFF

Event(s) generated (unless masked away):

When the HCI_LE_Set_Connection_CTE_Receive_Parameters command has completed, an HCI_Command_Complete event shall be generated.



7.8.84 LE Set Connection CTE Transmit Parameters command

Command	OCF	Command Parameters	Return Parameters
HCI_LE_Set_Connection_-_CTE_Transmit_Parameters	0x0055	Connection_Handle, CTE_Types, Length_of_Switching_Pattern, Antenna_IDs[i]	Status, Connection_Handle

Description:

The HCI_LE_Set_Connection_CTE_Transmit_Parameters command is used to set the antenna switching pattern and permitted Constant Tone Extension types used for transmitting Constant Tone Extensions requested by the peer device on the connection identified by the Connection_Handle parameter.

If the Host issues this command when Constant Tone Extension responses have been enabled on the connection, the Controller shall return the error code *Command Disallowed* (0x0C).

If the CTE_Types parameter has a bit set for a type of Constant Tone Extension that the Controller does not support, the Controller shall return the error code *Unsupported Feature or Parameter Value* (0x11).

The Length_of_Switching_Pattern and Antenna_IDs[i] parameters are only used when transmitting an AoD Constant Tone Extension and shall be ignored when transmitting an AoA Constant Tone Extension.

If Length_of_Switching_Pattern is greater than the maximum length of switching pattern supported by the Controller, the Controller shall return the error code *Unsupported Feature or Parameter Value* (0x11).

If the Controller determines that any of the Antenna_IDs[i] values do not identify an antenna in the device's antenna array, it shall return the error code *Unsupported Feature or Parameter Value* (0x11).

Note: Some Controllers may be unable to determine which values do or do not identify an antenna.

Command parameters:

Connection_Handle:

Size: 2 octets (12 bits meaningful)

Value	Parameter Description
0xFFFF	Connection_Handle Range: 0x0000 to 0x0EFF

**CTE_Types:***Size: 1 octet*

Bit Number	Parameter Description
0	Allow AoA Constant Tone Extension Response
1	Allow AoD Constant Tone Extension Response with 1 μ s slots
2	Allow AoD Constant Tone Extension Response with 2 μ s slots
All other values	Reserved for future use

Length_of_Switching_Pattern:*Size: 1 octet*

Value	Parameter Description
0x02 to 0x4B	The number of Antenna IDs in the pattern
All other values	Reserved for future use

Antenna_IDs[i]:*Size: Length_of_Switching_Pattern * 1 octet*

Value	Parameter Description
0xXX	List of Antenna IDs in the pattern.

Return parameters:**Status:***Size: 1 octet*

Value	Parameter Description
0x00	HCI_LE_Set_Connection_CTE_Transmit_Parameters command succeeded.
0x01 to 0xFF	HCI_LE_Set_Connection_CTE_Transmit_Parameters command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Connection_Handle:*Size: 2 octets (12 bits meaningful)*

Value	Parameter Description
0xFFFF	Connection_Handle Range: 0x0000 to 0x0EFF

Event(s) generated (unless masked away):

When the HCI_LE_Set_Connection_CTE_Transmit_Parameters command has completed, an HCI_Command_Complete event shall be generated.



7.8.85 LE Connection CTE Request Enable command

Command	OCF	Command Parameters	Return Parameters
HCI_LE_Connection_CTE_Request_Enable	0x0056	Connection_Handle, Enable, CTE_Request_Interval, Requested_CTE_Length, Requested_CTE_Type	Status, Connection_Handle

Description:

The HCI_LE_Connection_CTE_Request_Enable command is used to request the Controller to start or stop initiating the Constant Tone Extension Request procedure (see [Vol 6] Part B, Section 5.1.12) on a connection identified by the Connection_Handle parameter.

If the Host issues this command when the Controller is aware (e.g. through a previous feature exchange) that the peer device's Link Layer does not support the Connection CTE Response feature, the Controller shall return the error code *Unsupported Remote Feature / Unsupported LMP Feature* (0x1A).

If Enable is set to 0x00, the remaining parameters shall be ignored.

The CTE_Request_Interval parameter defines whether the Constant Tone Extension Request procedure is initiated only once or periodically. In the case of periodic operation, the procedure is initiated every CTE_Request_Interval. However, the Controller may delay initiating the procedure beyond the requested interval (e.g., in order to prioritize other activities).

The Requested_CTE_Length parameter indicates the minimum length of the Constant Tone Extension and the Requested_CTE_Type parameter indicates the type of Constant Tone Extension that the Controller shall request from the remote device.

A request is active on a connection from when the Host issues a successful command with Enable set to 0x01 until the single procedure has been performed, the period specified by CTE_Request_Interval has ended, or a command with Enable set to 0x00 has succeeded, whichever happens first.

If the Host issues this command with Enable set to 0x01 while a request is active for the specified connection, the Controller shall return the error code *Command Disallowed* (0x0C).

Note: The failed command will not affect the behavior of the Link Layer in respect of the currently-active request.



If the Host issues this command before issuing the HCI_LE_Set_Connection_CTE_Receive_Parameters command at least once on the connection, the Controller shall return the error code *Command Disallowed* (0x0C).

If the Host issues this command when the receiver PHY for the connection is not a PHY that allows Constant Tone Extensions, the Controller shall return the error code *Command Disallowed* (0x0C).

If the Host sets CTE_Request_Interval to a non-zero value less than or equal to connSlaveLatency, the Controller shall return the error code *Command Disallowed* (0x0C).

If Enable is set to 0x01 and the receiver PHY for the connection changes to a PHY that does not allow Constant Tone Extensions, then the Controller shall automatically disable Constant Tone Extension requests.

Command parameters:

Connection_Handle: *Size: 2 octets (12 bits meaningful)*

Value	Parameter Description
0xFFFF	Connection_Handle Range: 0x0000 to 0xFFFF

Enable: *Size: 1 octet*

Value	Parameter Description
0x00	Disable Constant Tone Extension Request for the connection (default)
0x01	Enable Constant Tone Extension Request for the connection
All other values	Reserved for future use

CTE_Request_Interval: *Size: 2 octets*

Value	Parameter Description
0x0000	Initiate the Constant Tone Extension Request procedure once, at the earliest practical opportunity
0x0001 to 0xFFFF	Requested interval for initiating the Constant Tone Extension Request procedure in number of connection events.

Requested_CTE_Length: *Size: 1 octet*

Value	Parameter Description
0x02 to 0x14	Minimum length of the Constant Tone Extension being requested in 8 μ s units
All other values	Reserved for future use

**Requested_CTE_Type:****Size: 1 octet**

Value	Parameter Description
0x00	AoA Constant Tone Extension
0x01	AoD Constant Tone Extension with 1 μ s slots
0x02	AoD Constant Tone Extension with 2 μ s slots
All other values	Reserved for future use

Return parameters:**Status:****Size: 1 octet**

Value	Parameter Description
0x00	HCI_LE_Connection_CTE_Request_Enable command succeeded.
0x01 to 0xFF	HCI_LE_Connection_CTE_Request_Enable command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Connection_Handle:**Size: 2 octets (12 bits meaningful)**

Value	Parameter Description
0xFFFF	Connection_Handle Range: 0x0000 to 0x0EFF

Event(s) generated (unless masked away):

When the HCI_LE_Connection_CTE_Request_Enable command has completed, an HCI_Command_Complete event shall be generated.

HCI_LE_Connection_IQ_Report events are generated by the Controller based on Constant Tone Extensions received, whether in packets containing an LL_CTE_RSP PDU or otherwise. If a packet is received containing an LL_CTE_RSP PDU but no Constant Tone Extension, or if the peer device rejects the request, an HCI_LE_CTE_Request_Failed event shall be generated.



7.8.86 LE Connection CTE Response Enable command

Command	OCF	Command Parameters	Return Parameters
HCI_LE_Connection_CTE_Response_Enable	0x0057	Connection_Handle, Enable	Status, Connection_Handle

Description:

The HCI_LE_Connection_CTE_Response_Enable command is used to request the Controller to respond to LL_CTE_REQ PDUs with LL_CTE_RSP PDUs on the specified connection.

If the Host issues this command before issuing the HCI_LE_Set_Connection_CTE_Transmit_Parameters command at least once on the connection, the Controller shall return the error code *Command Disallowed* (0x0C).

If the Host issues this command when the transmitter PHY for the connection is not a PHY that allows Constant Tone Extensions, the Controller shall return the error code *Command Disallowed* (0x0C).

If the transmitter PHY for the connection changes to a PHY that does not allow Constant Tone Extensions, then the Controller shall automatically disable Constant Tone Extension responses.

Command parameters:

Connection_Handle: *Size: 2 octets (12 bits meaningful)*

Value	Parameter Description
0xFFFF	Connection_Handle Range: 0x0000 to 0x0EFF

Enable: *Size: 1 octet*

Value	Parameter Description
0x00	Disable Constant Tone Extension Response for the connection (default)
0x01	Enable Constant Tone Extension Response for the connection
All other values	Reserved for future use

Return parameters:



Status:

Size: 1 octet

Value	Parameter Description
0x00	HCI_LE_Connection_CTE_Response_Enable command succeeded.
0x01 to 0xFF	HCI_LE_Connection_CTE_Response_Enable command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Connection_Handle:

Size: 2 octets (12 bits meaningful)

Value	Parameter Description
0xFFFF	Connection_Handle Range: 0x0000 to 0x0EFF

Event(s) generated (unless masked away):

When the HCI_LE_Connection_CTE_Response_Enable command has completed, an HCI_Command_Complete event shall be generated.



7.8.87 LE Read Antenna Information command

Command	OCF	Command Parameters	Return Parameters
HCI_LE_Read_Antenna_Information	0x0058		Status, Supported_Switching_Sampling_Rates, Number_of_Antennae, Max_Length_of_Switching_Pattern, Max_CTE_Length

Description:

The HCI_LE_Read_Antenna_Information command allows the Host to read the switching rates, the sampling rates, the number of antennae, and the maximum length of a transmitted Constant Tone Extension supported by the Controller.

Command parameters:

None.

Return parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	HCI_LE_Read_Antenna_Information command succeeded.
0x01 to 0xFF	HCI_LE_Read_Antenna_Information command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Supported_Switching_Sampling_Rates:

Size: 1 octet

Bit Number	Meaning
0	1 μ s switching supported for AoD transmission
1	1 μ s sampling supported for AoD reception
2	1 μ s switching and sampling supported for AoA reception
3 to 7	Reserved for future use

Number_of_Antennae:

Size: 1 octet

Value	Parameter Description
0x01 to 0x4B	The number of antennae supported by the Controller
All other values	Reserved for future use

**Max_Length_of_Switching_Pattern:****Size: 1 octet**

Value	Parameter Description
0x02 to 0x4B	Maximum length of antenna switching pattern supported by the Controller
All other values	Reserved for future use

Max_CTE_Length:**Size: 1 octet**

Value	Parameter Description
0x02 to 0x14	Maximum length of a transmitted Constant Tone Extension supported in 8 μ s units
All other values	Reserved for future use

Event(s) generated (unless masked away):

When the HCI_LE_Read_Antenna_Information command has completed, an HCI_Command_Complete event shall be generated.



7.8.88 LE Set Periodic Advertising Receive Enable command

Command	OCF	Command Parameters	Return Parameters
HCI_LE_Set_Periodic_Advertising_Receive_Enable	0x0059	Sync_Handle, Enable	Status

Description:

The HCI_LE_Set_Periodic_Advertising_Receive_Enable command will enable or disable reports for the periodic advertising train identified by the Sync_Handle parameter.

The Enable parameter determines whether reporting is enabled or disabled. If the value is the same as the current state, the command has no effect.

If the periodic advertising train corresponding to the Sync_Handle parameter does not exist, the Controller shall return the error code *Unknown Advertising Identifier* (0x42).

Command parameters:

Sync_Handle: *Size: 2 octets (12 bits meaningful)*

Value	Parameter Description
0xXXXX	Sync_Handle identifying the periodic advertising train. Range: 0x0000 to 0x0EFF

Enable: *Size: 1 octet*

Value	Parameter Description
0x00	Reporting disabled
0x01	Reporting enabled
All other values	Reserved for future use

Return parameters:

Status: *Size: 1 octet*

Value	Parameter Description
0x00	HCI_LE_Set_Periodic_Advertising_Receive_Enable command succeeded.
0x01–0xFF	HCI_LE_Set_Periodic_Advertising_Receive_Enable command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

**Event(s) generated (unless masked away):**

When the HCI_LE_Set_Periodic_Advertising_Receive_Enable command has completed, an HCI_Command_Complete event shall be generated.



7.8.89 LE Periodic Advertising Sync Transfer command

Command	OCF	Command Parameters	Return Parameters
HCI_LE_Periodic_Advertising_Sync_Transfer	0x005A	Connection_Handle, Service_Data, Sync_Handle	Status, Connection_Handle

Description:

The HCI_LE_Periodic_Advertising_Sync_Transfer command is used to instruct the Controller to send synchronization information about the periodic advertising train identified by the Sync_Handle parameter to a connected device.

The Service_Data parameter is a value provided by the Host for use by the Host of the peer device. It is not used by the Controller.

The connected device is identified by the Connection_Handle parameter.

If the periodic advertising train corresponding to the Sync_Handle parameter does not exist, the Controller shall return the error code *Unknown Advertising Identifier* (0x42).

If the Connection_Handle parameter does not identify a current connection, the Controller shall return the error code *Unknown Connection Identifier* (0x02).

If the remote device has not indicated support for the Periodic Advertising Sync Transfer - Recipient feature, the Controller shall return the error code *Unsupported Remote Feature / Unsupported LMP Feature* (0x1A).

Note: This command may complete before the periodic advertising synchronization information is sent. No indication is given as to how the recipient handled the information.

Command parameters:

Connection_Handle: *Size: 2 octets (12 bits meaningful)*

Value	Parameter Description
0xXXXX	Connection_Handle Range: 0x0000 to 0x0EFF

Service_Data: *Size: 2 octets*

Value	Parameter Description
0xXXXX	A value provided by the Host

**Sync_Handle:****Size: 2 octets (12 bits meaningful)**

Value	Parameter Description
0xXXXX	Sync_Handle identifying the periodic advertising train. Range: 0x0000 to 0x0EFF

Return parameters:**Status:****Size: 1 octet**

Value	Parameter Description
0x00	HCI_LE_Periodic_Advertising_Sync_Transfer command succeeded.
0x01–0xFF	HCI_LE_Periodic_Advertising_Sync_Transfer command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Connection_Handle:**Size: 2 octets (12 bits meaningful)**

Value	Parameter Description
0xXXXX	Connection_Handle Range: 0x0000 to 0x0EFF

Event(s) generated (unless masked away):

When the HCI_LE_Periodic_Advertising_Sync_Transfer command has completed, an HCI_Command_Complete event shall be generated.



7.8.90 LE Periodic Advertising Set Info Transfer command

Command	OCF	Command Parameters	Return Parameters
HCI_LE_Periodic_Advertising_Set_Info_Transfer	0x005B	Connection_Handle, Service_Data, Advertising_Handle	Status, Connection_Handle

Description:

The HCI_LE_Periodic_Advertising_Set_Info_Transfer command is used to instruct the Controller to send synchronization information about the periodic advertising in an advertising set to a connected device.

The Advertising_Handle parameter identifies the advertising set. If the parameters in the advertising set have changed since the periodic advertising was first enabled, the current parameters – not the original ones – are sent.

The Service_Data parameter is a value provided by the Host to identify the periodic advertising train to the peer device. It is not used by the Controller.

The connected device is identified by the Connection_Handle parameter.

If the advertising set corresponding to the Advertising_Handle parameter does not exist, the Controller shall return the error code *Unknown Advertising Identifier* (0x42).

If periodic advertising is not currently in progress for the advertising set, the Controller shall return the error code *Command Disallowed* (0x0C).

If the Connection_Handle parameter does not identify a current connection, the Controller shall return the error code *Unknown Connection Identifier* (0x02).

If the remote device has not indicated support for the Periodic Advertising Sync Transfer - Recipient feature, the Controller shall return the error code *Unsupported Remote Feature / Unsupported LMP Feature* (0x1A).

Note: This command may complete before the periodic advertising synchronization information is sent. No indication is given as to how the recipient handled the information.

Command parameters:

Connection_Handle:

Size: 2 octets (12 bits meaningful)

Value	Parameter Description
0xFFFF	Connection_Handle Range: 0x0000 to 0x0EFF

**Service_Data:****Size: 2 octets**

Value	Parameter Description
0xXXXX	A value provided by the Host

Advertising_Handle:**Size: 1 octet**

Value	Parameter Description
0x00–0xEF	Used to identify an advertising set
All other values	Reserved for future use

Return parameters:**Status:****Size: 1 octet**

Value	Parameter Description
0x00	HCI_LE_Periodic_Advertising_Set_Info_Transfer command succeeded.
0x01–0xFF	HCI_LE_Periodic_Advertising_Set_Info_Transfer command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Connection_Handle:**Size: 2 octets (12 bits meaningful)**

Value	Parameter Description
0xXXXX	Connection_Handle Range: 0x0000 to 0x0EFF

Event(s) generated (unless masked away):

When the HCI_LE_Periodic_Advertising_Set_Info_Transfer command has completed, an HCI_Command_Complete event shall be generated.



7.8.91 LE Set Periodic Advertising Sync Transfer Parameters command

Command	OCF	Command Parameters	Return Parameters
HCI_LE_Set_Periodic_Advertising_Sync_Transfer_Parameters	0x005C	Connection_Handle, Mode, Skip, Sync_Timeout, CTE_Type	Status, Connection_Handle

Description:

The HCI_LE_Set_Periodic_Advertising_Sync_Transfer_Parameters command is used to specify how the Controller will process periodic advertising synchronization information received from the device identified by the Connection_Handle parameter (the "transfer mode").

The Mode parameter specifies the action to be taken when periodic advertising synchronization information is received. If Mode is 0x00, the Controller will ignore the information. Otherwise it will notify the Host and synchronize to the periodic advertising. Mode also specifies whether periodic advertising reports are initially enabled or disabled.

The Skip parameter specifies the number of consecutive periodic advertising packets that the receiver may skip after successfully receiving a periodic advertising packet.

The Sync_Timeout parameter specifies the maximum permitted time between successful receives. If this time is exceeded, synchronization is lost.

Irrespective of the value of the Skip parameter, the Controller should stop skipping packets before the Sync_Timeout would be exceeded.

The CTE_Type parameter specifies whether to only synchronize to periodic advertising with certain types of Constant Tone Extension. If the periodic advertiser changes the type of the Constant Tone Extension after the Controller has synchronized with the periodic advertising, it shall remain synchronized.

Note: A value of 0 (i.e. all bits clear) indicates that the presence or absence of a Constant Tone Extension is irrelevant.

This command does not affect any processing of any periodic advertising synchronization information already received from the peer device, whether or not the Controller has yet synchronized to the periodic advertising train it describes.

The parameter values provided by this command override those provided via the HCI_LE_Set_Default_Periodic_Advertising_Sync_Transfer_Parameters



command ([Section 7.8.92](#)) or any preferences previously set using the HCI_LE_Set_Periodic_Advertising_Sync_Transfer_Parameters command on the same connection.

If the Connection_Handle parameter does not identify a current connection, the Controller shall return the error code *Unknown Connection Identifier* (0x02).

Command parameters:**Connection_Handle:****Size: 2 octets (12 bits meaningful)**

Value	Parameter Description
0xFFFF	Connection_Handle Range: 0x0000 to 0x0EFF

Mode:**Size: 1 octet**

Value	Parameter Description
0x00	No attempt is made to synchronize to the periodic advertising and no HCI_LE_Periodic_Advertising_Sync_Transfer_Received event is sent to the Host.
0x01	An HCI_LE_Periodic_Advertising_Sync_Transfer_Received event is sent to the Host. HCI_LE_Periodic_Advertising_Report events will be disabled.
0x02	An HCI_LE_Periodic_Advertising_Sync_Transfer_Received event is sent to the Host. HCI_LE_Periodic_Advertising_Report events will be enabled.
All other values	Reserved for future use

Skip:**Size: 2 octets**

Value	Parameter Description
0xFFFF	The number of periodic advertising packets that can be skipped after a successful receive Range: 0x0000 to 0x01F3

Sync_Timeout:**Size: 2 octets**

Value	Parameter Description
N=0xFFFF	Synchronization timeout for the periodic advertising train Range: 0x000A to 0x4000 Time = N*10 ms Time Range: 100 ms to 163.84 s

**CTE_Type:****Size: 1 octet**

Bit Number	Parameter Description
0	Do not sync to packets with an AoA Constant Tone Extension
1	Do not sync to packets with an AoD Constant Tone Extension with 1 μ s slots
2	Do not sync to packets with an AoD Constant Tone Extension with 2 μ s slots
4	Do not sync to packets without a Constant Tone Extension
All other values	Reserved for future use

Return parameters:**Status:****Size: 1 octet**

Value	Parameter Description
0x00	HCI_LE_Set_Periodic_Advertising_Sync_Transfer_Parameters command succeeded.
0x01–0xFF	HCI_LE_Set_Periodic_Advertising_Sync_Transfer_Parameters command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Connection_Handle:**Size: 2 octets (12 bits meaningful)**

Value	Parameter Description
0xFFFF	Connection_Handle Range: 0x0000 to 0x0EFF

Event(s) generated (unless masked away):

When the HCI_LE_Set_Periodic_Advertising_Sync_Transfer_Parameters command has completed, an HCI_Command_Complete event shall be generated.



7.8.92 LE Set Default Periodic Advertising Sync Transfer Parameters command

Command	OCF	Command Parameters	Return Parameters
HCI_LE_Set_Default_Periodic_Advertising_Sync_Transfer_Parameters	0x005D	Mode, Skip, Sync_Timeout, CTE_Type	Status

Description:

The HCI_LE_Set_Default_Periodic_Advertising_Sync_Transfer_Parameters command is used to specify the initial value for the mode, skip, timeout, and Constant Tone Extension type (set by the HCI_LE_Set_Periodic_Advertising_Sync_Transfer_Parameters command; see [Section 7.8.91](#)) to be used for all subsequent connections over the LE transport.

The Mode parameter specifies the initial action to be taken. If Mode is 0x00, the Controller will ignore the information. Otherwise it will notify the Host and synchronize to the periodic advertising. Mode also specifies whether periodic advertising reports are initially enabled or disabled.

The Skip parameter specifies the number of consecutive periodic advertising packets that the receiver may skip after successfully receiving a periodic advertising packet.

The Sync_Timeout parameter specifies the maximum permitted time between successful receives. If this time is exceeded, synchronization is lost.

The CTE_Type parameter specifies whether to only synchronize to periodic advertising with certain types of Constant Tone Extension. If the periodic advertiser changes the type of the Constant Tone Extension after the Controller has synchronized with the periodic advertising, it shall remain synchronized.

Note: A value of 0 (i.e. all bits clear) indicates that the presence or absence of a Constant Tone Extension is irrelevant.

This command does not affect any existing connection.

Command parameters:

**Mode:****Size: 1 octet**

Value	Parameter Description
0x00	No attempt is made to synchronize to the periodic advertising and no HCI_LE_Periodic_Advertising_Sync_Transfer_Received event is sent to the Host (default).
0x01	An HCI_LE_Periodic_Advertising_Sync_Transfer_Received event is sent to the Host. HCI_LE_Periodic_Advertising_Report events will be disabled.
0x02	An HCI_LE_Periodic_Advertising_Sync_Transfer_Received event is sent to the Host. HCI_LE_Periodic_Advertising_Report events will be enabled.
All other values	Reserved for future use

Skip:**Size: 2 octets**

Value	Parameter Description
0xFFFF	The number of periodic advertising packets that can be skipped after a successful receive Range: 0x0000 to 0x01F3

Sync_Timeout:**Size: 2 octets**

Value	Parameter Description
N=0xFFFF	Synchronization timeout for the periodic advertising train Range: 0x000A to 0x4000 Time = N*10 ms Time Range: 100 ms to 163.84 s

CTE_Type:**Size: 1 octet**

Bit Number	Parameter Description
0	Do not sync to packets with an AoA Constant Tone Extension
1	Do not sync to packets with an AoD Constant Tone Extension with 1 μ s slots
2	Do not sync to packets with an AoD Constant Tone Extension with 2 μ s slots
4	Do not sync to packets without a Constant Tone Extension
All other values	Reserved for future use

Return parameters:



Status:

Size: 1 octet

Value	Parameter Description
0x00	HCI_LE_Set_Default_Periodic_Advertising_Sync_Transfer_ - Parameters command succeeded.
0x01–0xFF	HCI_LE_Set_Default_Periodic_Advertising_Sync_Transfer_ - Parameters command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Event(s) generated (unless masked away):

When the HCI_LE_Set_Default_Periodic_Advertising_Sync_Transfer_Parameters command has completed, an HCI_Command_Complete event shall be generated.



7.8.93 LE Generate DHKey command [v2]

Command	OCF	Command Parameters	Return Parameters
HCI_LE_Generate_- DHKey [v2]	0x005E	Remote_P-256_Public_Key, Key_Type	

For version v1 of this command, see [Section 7.8.37](#).

Description:

The HCI_LE_Generate_DHKey command is used to initiate generation of a Diffie-Hellman key in the Controller for use over the LE transport. This command takes the remote P-256 public key as input. The Diffie-Hellman key generation uses the private key generated by the HCI_LE_Read_Local_P-256_Public_Key command or the private debug key (see [\[Vol 3\] Part H, Section 2.3.5.6.1](#)).

The Diffie-Hellman key returned via this command shall not be generated using any keys used for Secure Connections over the BR/EDR transport.

If the remote P-256 public key is invalid (see [\[Vol 3\] Part H, Section 2.3.5.6.1](#)), the Controller shall return an error and should use the error code *Invalid HCI Command Parameters* (0x12).

Command parameters:

Remote_P-256_Public_Key:

Size: 64 octets

Value	Parameter Description
0xFFFFFFFFFFFFFFFF XXXXXXXXXXXXXXXX XXXXXXXXXXXXXXXX XXXXXXXXXXXXXXXX XXXXXXXXXXXXXXXX XXXXXXXXXXXXXXXX XXXXXXXXXXXXXXXX XXXXXXXXXXXXXXXX XXXXXXXXXXXXXXXX	The remote P-256 public key: X, Y format Octets 31-0: X co-ordinate Octets 63-32: Y co-ordinate Little-endian Format

Key_Type:

Size: 1 octet

Value	Parameter Description
0x00	Use the generated private key
0x01	Use the debug private key
All other values	Reserved for future use

Return parameters:



None.

Event(s) generated (unless masked away):

When the Controller receives the HCI_LE_Generate_DHKey command, the Controller shall send the HCI_Command_Status event to the Host. When the DHKey generation finishes, an HCI_LE_Generate_DHKey_Complete event shall be generated.

No HCI_Command_Complete event is sent by the Controller to indicate that this command has been completed.



7.8.94 LE Modify Sleep Clock Accuracy command

Command	OCF	Command Parameters	Return Parameters
HCI_LE_Modify_Sleep_Clock_Accuracy	0x005F	Action	Status

Description:

This command is used to request that the Controller changes its sleep clock accuracy for testing purposes. It should not be used under other circumstances.

The Action parameter specifies whether the sleep clock should be changed to one that is more accurate or one that is less accurate.

If Action is 0x00 and the Controller is already using its most accurate clock, or Action is 0x01 and the Controller is already using its least accurate clock, it shall return the error code *Limit Reached* (0x43).

If the Controller is unable to switch to a different clock accuracy because some other activity requires the current accuracy, it shall return the error code *Controller Busy* (0x3A).

If the Controller is unable to switch to a different sleep clock for any other reason, it shall return the error code *Command Disallowed* (0x0C).

Command parameters:

Action:

Size: 1 octet

Value	Parameter Description
0x00	Switch to a more accurate clock
0x01	Switch to a less accurate clock
All other values	Reserved for future use

Return parameters:

Status:

Size: 1 octet

Value	Parameter Description
0x00	HCI_LE_Modify_Sleep_Clock_Accuracy command succeeded
0x01–0xFF	HCI_LE_Modify_Sleep_Clock_Accuracy command failed. See [Vol 2] Part D, Error Codes for a list of error codes and descriptions.

Event(s) generated (unless masked away):



When the HCI_LE_Modify_Sleep_Clock_Accuracy command has completed, an HCI_Command_Complete event shall be generated.

Note: The Controller might not have changed the clock when it returns the HCI_Command_Complete event indicating success, for example because it still needs to notify peers of the pending change.